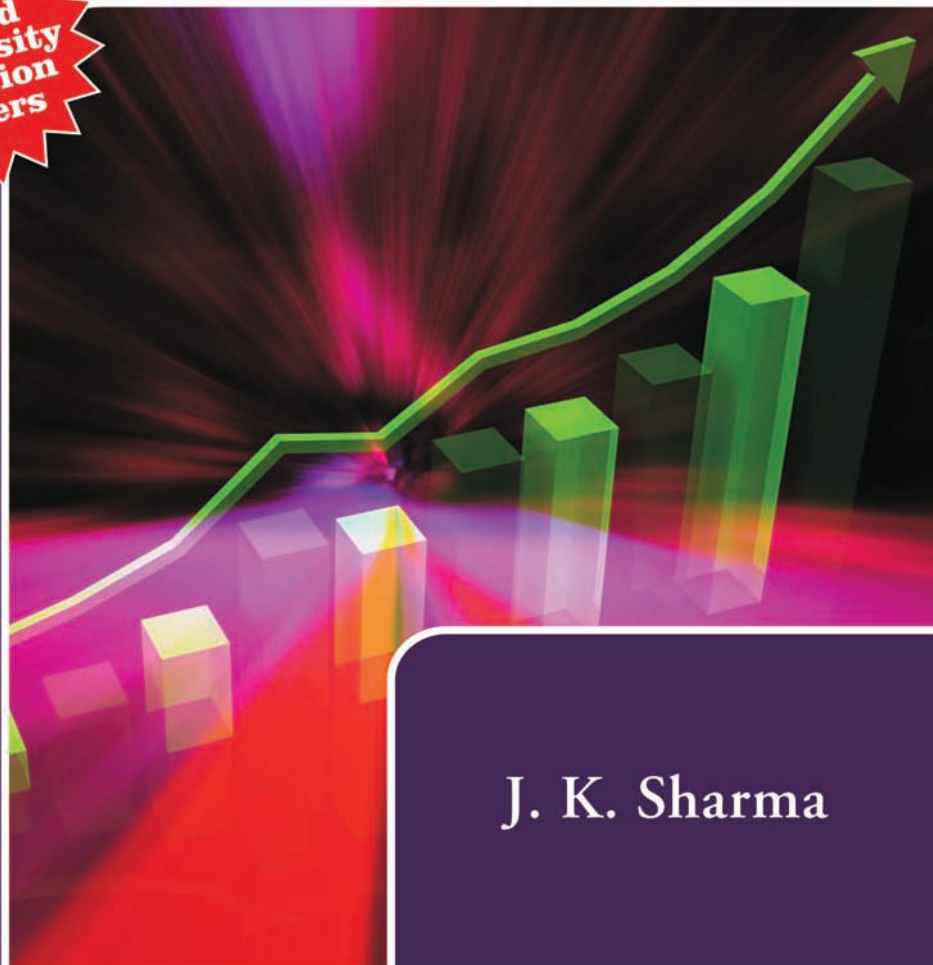


BUSINESS STATISTICS

With
solved
university
question
papers



J. K. Sharma

FOR UNIVERSITY OF DELHI

ALWAYS LEARNING

PEARSON

Business Statistics

This page is intentionally left blank.

Business Statistics

For University of Delhi

As per the syllabus of B.Com. (Hons.) course

J. K. Sharma

Professor, Amity Business School
Amity University, Noida

PEARSON

Delhi • Chennai • Chandigarh

Copyright © 2011 Dorling Kindersley (India) Pvt. Ltd.

Licensees of Pearson Education in South Asia

No part of this eBook may be used or reproduced in any manner whatsoever without the publisher's prior written consent.

This eBook may or may not include all assets that were part of the print version. The publisher reserves the right to remove any material present in this eBook at any time.

ISBN 9788131759684

eISBN 9789332506770

Head Office: A-8(A), Sector 62, Knowledge Boulevard, 7th Floor, NOIDA 201 309, India
Registered Office: 11 Local Shopping Centre, Panchsheel Park, New Delhi 110 017, India

Contents

Syllabus
Preface
About the Author

xv
xvii
xix

UNIT 1: STATISTICAL DATA AND DESCRIPTIVE STATISTICS

CHAPTER 1 STATISTICAL DATA AND THEIR SOURCES

3-17

- 1.1 Reasons For Learning Statistics 3
- 1.2 Statistics Defined 3
- 1.3 Types of Statistical Methods 5
- 1.4 Importance and Scope of Statistics 5
 - 1.4.1 Statistics in Economics 6
 - 1.4.2 Statistics in Business Management 6
 - 1.4.3 Statistics in Social Sciences 6
 - 1.4.4 Statistics In Medical Sciences 7
 - *Conceptual Questions 1A* 7
- 1.5 Need For Data 7
 - 1.5.1 Types of Data 8
- 1.6 Principles of Measurement 9
- 1.7 Sources of Data 12
 - 1.7.1 Primary Data Sources 13
 - 1.7.2 Secondary Data Sources 14
 - *Conceptual Questions 1B* 16
 - *True or False Concepts Quiz* 17

CHAPTER 2 DATA CLASSIFICATION AND THEIR PRESENTATION

18-61

- 2.1 Introduction 18
- 2.2 Classification of Data 18
 - 2.2.1 Requisites of Ideal Classification 19
 - 2.2.2 Basis of Classification 19

- 2.3 Organizing Data Using Data Array 21
 - 2.3.1 Frequency Distribution 22
 - 2.3.2 Methods of Data Classification 25
 - 2.3.3 Bivariate Frequency Distribution 31
 - 2.3.4 Types of Frequency Distributions 33
 - *Conceptual Questions 2A* 35
 - *Self-Practice Problem 2A* 36
 - *Hints and Answers* 37
- 2.4 Graphical Presentation of Data 38
 - 2.4.1 Functions of a Graph 38
 - 2.4.2 General Rules For Drawing Diagrams 39
- 2.5 Types of Diagrams 40
 - 2.5.1 One-dimensional Diagrams 40
 - 2.5.2 Two-dimensional Diagrams 53
 - 2.5.3 Three-dimensional Diagrams 56
 - 2.5.4 Pictograms or Ideographs 56
 - *Conceptual Questions 2B* 57
 - *Self-Practice Problem 2B* 57
 - *Formulae Used* 58
 - *True or False Concepts Quiz* 59
 - *Review Self-Practice Problems* 59
 - *Glossary of Terms* 61

CHAPTER 3 MEASURES OF CENTRAL TENDENCY

62-121

- 3.1 Introduction 62
- 3.2 Requisites of a Measure of Central Tendency 63
- 3.3 Measures of Central Tendency 64
- 3.4 Mathematical Averages 64
 - 3.4.1 Arithmetic Mean of Ungrouped (or Raw) Data 64
 - 3.4.2 Arithmetic Mean of Grouped (or Classified) Data 68
 - 3.4.3 Some Special Types of Problems and Their Solutions 72
 - 3.4.4 Advantages and Disadvantages of Arithmetic Mean 78
 - 3.4.5 Weighted Arithmetic Mean 80
 - *Conceptual Questions 3A* 84
 - *Self-Practice Problem 3A* 84
 - *Hints and Answers* 85
- 3.5 Geometric Mean 85
 - 3.5.1 Advantages, Disadvantages and Applications of G.M. 87
 - 3.5.2 Combined Geometric Mean 90
 - 3.5.3 Weighted Geometric Mean 91
 - *Conceptual Questions 3B* 93
 - *Self-Practice Problem 3B* 93
 - *Hints and Answers* 94
- 3.6 Harmonic Mean 94
- 3.7 Relationship Among A.M., G.M. And H.M. 98
 - *Self-Practice Problem 3C* 98
 - *Hints and Answers* 98

- 3.8 Averages of Position 99
 - 3.8.1 Median 99
 - 3.8.2 Advantages, Disadvantages and Applications of Median 102
- 3.9 Partition Values—Quartiles, Deciles and Percentiles 102
 - 3.9.1 Graphical Method for Calculating Partition Values 103
 - *Conceptual Questions 3C* 108
 - *Self-Practice Problem 3D* 108
 - *Hints and Answers* 109
- 3.10 Mode 109
 - 3.10.1 Graphical Method for Calculating Mode Value 112
 - 3.10.2 Advantages and Disadvantages of Mode Value 113
- 3.11 Relationship Between Mean, Median and Mode 113
- 3.12 Comparison Between Measures of Central Tendency 114
 - *Conceptual Questions 3D* 115
 - *Self-Practice Problem 3E* 116
 - *Hints and Answers* 117
 - *Formulae Used* 117
 - *True or False Concepts Quiz* 117
 - *Review Self-Practice Problems* 118
 - *Hints and Answers* 120
 - *Glossary of Terms* 121

CHAPTER 4 MEASURES OF VARIATION: ABSOLUTE AND RELATIVE

122-162

- 4.1 Introduction 122
- 4.2 Significance of Measuring Dispersion 123
- 4.3 Classification of Measures of Dispersion 124
- 4.4 Distance Measures 124
 - 4.4.1 Range 125
 - 4.4.2 Interquartile Range or Deviation 126
 - *Conceptual Questions 4A* 129
 - *Self-Practice Problem 4A* 130
 - *Hints and Answers* 131
- 4.5 Average Deviation Measures 131
 - 4.5.1 Mean Absolute Deviation 131
 - 4.5.2 Variance and Standard Deviation 138
 - 4.5.3 Mathematical Properties of Standard Deviation 145
 - 4.5.4 Coefficient of Variation 150
 - *Conceptual Questions 4B* 154
 - *Self-Practice Problem 4B* 154
 - *Hints and Answers* 156
 - *Formulae Used* 157
 - *True or False Concepts Quiz* 158
 - *Review Self-Practice Problems* 158
 - *Hints and Answers* 160
 - *Glossary of Terms* 162

CHAPTER 5 SKEWNESS, MOMENTS AND KURTOSIS**163-194**

- 5.1 Introduction 163
- 5.2 Measures of Skewness 164
 - 5.2.1 Relative Measures of Skewness 165
 - *Conceptual Questions 5A* 174
 - *Self-Practice Problem 5A* 174
 - *Hints and Answers* 175
- 5.3 Moments 177
 - 5.3.1 Moments About Mean 178
 - 5.3.2 Moments About Arbitrary Point 178
 - 5.3.3 Moments About Zero or Origin 179
 - 5.3.4 Relationship Between Moments About Mean and Moments About an Arbitrary Point 179
 - 5.3.5 Moments in Standard Units 179
 - 5.3.6 Sheppard's Corrections For Moments 180
- 5.4 Kurtosis 182
 - 5.4.1 Measures of Kurtosis 183
 - *Conceptual Questions 5B* 191
 - *Self-Practice Problem 5B* 191
 - *Hints and Answers* 192
 - *Formulae Used* 192
 - *Review Self-Practice Problems* 192
 - *Hints and Answers* 193
 - *Glossary of Terms* 194

UNIT 2: PROBABILITY, PROBABILITY DISTRIBUTIONS AND DECISION THEORY**CHAPTER 6 FUNDAMENTALS OF PROBABILITY****197-235**

- 6.1 Introduction 197
- 6.2 Concepts of Probability 197
- 6.3 Definition of Probability 200
- 6.4 Fundamental Rules of Probability 202
- 6.5 Counting Rules for Determining the Number of Outcomes 203
 - *Conceptual Questions 6A* 206
 - *Self-Practice Problem 6A* 207
 - *Hints and Answers* 207
- 6.6 Rules of Probability and Algebra of Events 209
 - 6.6.1 Rules of Addition 209
 - 6.6.2 Rules of Multiplication 213
 - *Self-Practice Problem 6B* 218
 - *Hints and Answers* 219

- 6.7 Bayes' Theorem 221
 - *Self-Practice Problem 6C* 224
 - *Hints and Answers* 225
- 6.8 Expected Value and Variance of a Random Variable 225
 - 6.8.1 Properties of Expected Value and Variance 226
 - *Conceptual Questions 6B* 228
 - *Self-Practice Problem 6D* 228
 - *Hints and Answers* 229
 - *Formulae Used* 230
 - *True or False Concepts Quiz* 230
 - *Review Self-Practice Problems* 231
 - *Hints and Answers* 233

CHAPTER 7 PROBABILITY DISTRIBUTIONS

236-263

- 7.1 Introduction 236
- 7.2 Discrete Probability Distributions 237
 - *Conceptual Questions 7A* 242
 - *Self-Practice Problem 7A* 242
 - *Hints and Answers* 243
 - *Conceptual Questions 7B* 246
 - *Self-Practice Problem 7B* 247
 - *Hints and Answers* 247
- 7.3 Continuous Probability Distributions 248
 - 7.3.1 Normal Probability Distribution Function 249
 - 7.3.2 Approximation of Binomial and Poisson Distributions to Normal Distribution 251
 - *Conceptual Questions 7C* 257
 - *Self-Practice Problem 7C* 257
 - *Hints and Answers* 258
 - *Formulae Used* 259
 - *True or False Concepts Quiz* 260
 - *Review Self-Practice Problems* 260
 - *Hints and Answers* 262

CHAPTER 8 STATISTICAL DECISION THEORY

264-304

- 8.1 Introduction 264
- 8.2 Steps in Decision Theory Approach 265
- 8.3 Types of Decision-Making Environments 266
- 8.4 Decision-Making Under Uncertainty 267
 - 8.4.1 Criterion of Optimism (Maximum or Minimum) 267
 - 8.4.2 Criterion of Pessimism (Maximum or Minimum) 268
 - 8.4.3 Equally Likely Decision (Laplace) Criterion 268
 - 8.4.4 Criterion of Realism (Hurwicz Criterion) 268
 - 8.4.5 Criterion of Regret 269
 - *Self-Practice Problem 8A* 271
 - *Hints and Answers* 273

- 8.5 Decision Making Under Risk 273
 - 8.5.1 Expected Monetary Value (EMV) 273
 - 8.5.2 Expected Opportunity Loss (EOL) 277
 - 8.5.3 Expected Value of Perfect Information (EVPI) 279
- 8.6 Decision Tree Analysis 287
 - *Conceptual Questions 8A* 297
 - *Self-Practice Problem 8B* 297
 - *Hints and Answers* 299
 - *Formulae Used* 299
 - *True or False Concepts Quiz* 299
 - *Review Self-Practice Problems* 300
 - *Hints and Answers* 303

UNIT 3: SIMPLE CORRELATION AND REGRESSION ANALYSIS

CHAPTER 9 CORRELATION ANALYSIS

307-342

- 9.1 Introduction 307
- 9.2 Significance of Measuring Correlation 308
- 9.3 Correlation and Causation 308
- 9.4 Types of Correlations 309
 - 9.4.1 Positive and Negative Correlation 309
 - 9.4.2 Linear and Non-linear Correlation 310
 - 9.4.3 Simple, Partial and Multiple Correlation 310
- 9.5 Methods of Correlation Analysis 311
 - 9.5.1 Scatter Diagram Method 311
 - 9.5.2 Karl Pearson's Correlation Coefficient 314
 - 9.5.3 Probable Error and Standard Error of Coefficient of Correlation 315
 - 9.5.4 The Coefficient of Determination 318
 - *Conceptual Questions 9A* 326
 - *Hints and Answers* 327
 - 9.5.5 Spearman's Rank Correlation Coefficient 327
 - *Self-Practice Problem 9A* 337
 - *Hints and Answers* 339
 - *Conceptual Questions 9B* 339
 - *Formulae Used* 340
 - *True or False Concepts Quiz* 340
 - *Review Self-Practice Problems* 341
 - *Hints and Answers* 342

CHAPTER 10 REGRESSION ANALYSIS**343-372**

- 10.1 Introduction 343
- 10.2 Advantages of Regression Analysis 344
- 10.3 Parameters of Simple Linear Regression Model 345
 - 10.3.1 Regression Coefficients 345
- 10.4 Methods to Determine Regression Coefficients 346
 - 10.4.1 Least-squares Normal Equations 346
 - 10.4.2 Deviations Method 349
 - *Conceptual Questions 10A* 362
 - *Hints and Answers* 365
 - *Conceptual Questions* 368
 - *Formulae Used* 368
 - *True or False Concepts Quiz* 369
 - *Review Self-Practice Problems* 369
 - *Hints and Answers* 371

UNIT 4: INDEX NUMBERS**CHAPTER 11 INDEX NUMBERS****375-436**

- 11.1 Introduction 375
- 11.2 Index Number 376
- 11.3 Types of Index Numbers 376
- 11.4 Characteristics and Uses of Index Numbers 378
 - 11.4.1 Characteristics of Index Numbers 378
 - 11.4.2 Uses of Index Numbers 378
 - *Conceptual Questions 11A* 380
- 11.5 Methods For Construction of Price Indexes 380
- 11.6 Unweighted Price Indexes 380
 - 11.6.1 Single Price Index 380
 - 11.6.2 Aggregate Price Index 381
 - 11.6.3 Average Price Relative Index 383
 - *Self-Practice Problem 11A* 385
 - *Hints and Answers* 386
- 11.7 Weighted Price Indexes 388
 - 11.7.1 Weighted Aggregate Price Index 388
 - 11.7.2 Weighted Average of Price Relative Index 396
- 11.8 Quantity or Volume Indexes 398
- 11.9 Value Indexes 402
 - *Self-Practice Problem 11B* 402
 - *Hints and Answers* 404

- 11.10 Tests of Adequacy of Indexes 405
 - 11.10.1 Time Reversal Test 406
 - 11.10.2 Factor Reversal Test 406
 - 11.10.3 Circular Test 407
- 11.11 Chain Indexes 412
 - *Self-Practice Problem 11C* 420
 - *Hints and Answers* 422
- 11.13 Consumer Price Indexes 424
 - 11.13.1 Uses of Consumer Price Index (CPI) Number 425
 - 11.13.2 Construction of a Consumer Price Index 425
 - *Conceptual Questions 11B* 431
 - *Formulae Used* 432
 - *True or False Concepts Quiz* 433
 - *Review Self-Practice Problems* 433
 - *Hints and Answers* 435

UNIT 5: TIME SERIES ANALYSIS

CHAPTER 12 TIME SERIES ANALYSIS

439-491

- 12.1 Introduction 439
- 12.2 Forecasting Methods 440
 - 12.2.1 Quantitative Forecasting Methods 440
 - 12.2.2 Qualitative Forecasting Methods 441
- 12.3 Time Series Analysis 441
 - 12.3.1 Objectives of Time Series Analysis 442
 - 12.3.2 Time Series Patterns 442
 - 12.3.3 Components of a Time Series 442
- 12.4 Time Series Decomposition Models 443
 - 12.4.1 Multiplicative Model 443
 - 12.4.2 Additive Model 444
 - *Conceptual Questions 12A* 444
- 12.5 Quantitative Forecasting Methods 445
 - 12.5.1 Freehand (or Graphical) Method 445
 - 12.5.2 Smoothing Methods 446
 - *Self-Practice Problem 12A* 454
 - *Hints and Answers* 455
- 12.6 Trend Projection Methods 457
 - 12.6.1 Linear Trend Model 457
 - 12.6.2 Exponential Trend Model 463
 - 12.6.3 Changing the Origin and Scale of Equations 465
 - *Self-Practice Problem 12B* 466
 - *Hints and Answers* 467

12.7	Measurement of Seasonal Effects	468
12.7.1	Seasonal Index	469
12.7.2	Method of Simple Averages	470
12.7.3	Ratio-to-Trend Method	474
12.7.4	Ratio-to-Moving Average Method	478
12.7.5	Link Relative Method	480
12.8	Measurement of Cyclical Variations—Residual Method	484
12.9	Measurement of Irregular Variations	485
	• <i>Conceptual Questions 12B</i>	485
	• <i>Self-Practice Problem 12C</i>	486
	• <i>Hints and Answers</i>	487
	• <i>Formulae Used</i>	488
	• <i>Review Self-Practice Problems</i>	489
	• <i>Hints and Answers</i>	490

Solved B.Com. (Hons.) Questions Papers (2008–2010)

492-532

This page is intentionally left blank.

Syllabus

B.Com. (Hons.)
Paper – CH 2.1: Semester - II
BUSINESS STATISTICS

Duration: 3 hrs
Max. Marks: 100

Lectures: 75

Objective: The objective of this course is to familiarize students with the basic statistical tools used to summarize and analyse quantitative information for decision making.

Expected Learning Outcomes: The student is expected to be equipped with the tools of processing and description of statistical data. In addition, the student would develop competence to use computer for statistical calculations especially for comparatively large-sized problems.

Unit 1

(No. of lectures: 23)

Statistical Data and Descriptive Statistics

1.1 Measures of Central Tendency

- (a) Mathematical averages including arithmetic mean, geometric mean and harmonic mean. Properties and applications.
- (b) Positional Averages
 - Mode
 - Median (and other partition values including quartiles, deciles and percentiles).

12 Lectures

1.2 Measures of Variation: absolute and relative.

- Range, quartile deviation, mean deviation, standard deviation and their co-efficients, Properties of standard deviation/variance.
- 5 Lectures**

1.3 Moments: calculation (including Sheppard's corrections) and significance. Skewness, Kurtosis and Moments.

6 Lectures

Unit 2**(No. of lectures: 18)****Probability, Probability Distributions and Decision Theory**

- 2.1 Theory of Probability. Approaches to the calculation of probability (2 lectures)
- 2.2 Calculation of event probabilities. Addition and multiplication laws of probability. (3 lectures)
- 2.3 Conditional probability and Bayes' Theorem. Expectation and variance of a random variable. (3 lectures)
- 2.4 Probability distributions: Binomial, Poisson and Normal. (5 lectures)
- 2.5 The decision environment. Pay-off and regret matrices. Criteria of decision making: Laplace, Maximin/Minimax, Maximax/Minimin, Savage; and Expectation. (Excluding Bayesian analysis) EVPI and its calculation. (3 lectures)
- 2.6 Decision Trees (2 lectures)

Unit 3**(No. of lectures: 10)****Simple Correlation and Regression Analysis**

- 3.1 Correlation Analysis. Meaning of Correlation simple, multiple and partial; linear and non-linear; Causation and correlation, Scatter diagram, Pearson's co-efficient of correlation; calculation and properties (proofs not required). Probable and standard errors, Rank Correlation. (5 lectures)
- 3.2 Regression Analysis. Principle of least squares and regression lines. Regression equations and estimation. Standard Error of Estimates. (5 lectures)

Unit 4**(No. of lectures: 12)****Index Numbers**

- 4.1 Meaning and uses of index numbers. Construction of index numbers: fixed and chain base: univariate and composite. Aggregative and average of relatives – simple and weighted. (5 lectures)
- 4.2 Tests of adequacy of index numbers, Base shifting, splicing and deflating. Problems in the construction of index numbers. (3 lectures)
- 4.3 Construction of consumer price indices. Important share price indices including BSE SENSEX and NSE NIFTY. (4 lectures)

Unit 5**(No. of lectures: 12)****Time Series Analysis**

- 5.1 Components of time series. Additive and multiplicative models. (2 lectures)
- 5.2 Trend analysis. Fitting of trend line using principle of least squares – linear, second degree parabola and exponential. Conversion of annual linear trend equation to quarterly/monthly basis and vice-versa. Moving averages. (6 lectures)
- 5.3 Seasonal variations- calculation and uses. Simple averages, ratio-to-trend, ratio-to-moving averages and link-relatives methods. (4 lectures)

The students will be familiarized with software and the statistical and other functions contained therein related to formation of frequency distributions and calculation of averages, measures of variation, correlation and regression co-efficients. All of this shall be done through practicals in paper CH 2.2 (Part C).

Preface

The primary objective of writing this book is to explain the concepts of *Business Statistics* lucidly along with solution procedures and analysis of results. It enables students to understand fundamental concepts of *Business Statistics* and apply formulae directly to business problems. Numerous solved business-oriented examples have been presented throughout the book. *Unsolved Self-Practice Problems* with Hints and Answers and *Review Questions* have been added in each chapter to strengthen the conceptual as well as practical knowledge of readers.

This book is designed to be self-contained. It comprises 12 chapters. The presentation and sequence of chapters have made the text interesting and lucid. It is intended to serve as a core primary text for students of B.Com. (Hons.) and BA (Eco. Hons.) courses of University of Delhi and also a few similar other undergraduate courses.

In writing this book, I have benefitted by referring to many books and publications. I express my gratitude and acknowledgment to such authors, publishers and institutions.

I am indebted to my family members, friends and students for their valuable suggestions and encouragement during the developing stage of the manuscript. I am thankful to the editorial team of Pearson Education for their cooperation and the manner in which this project was managed.

Suggestions and comments to improve the book in content and style are always welcome, and will be appreciated and acknowledged. Readers can directly mail me at my e-mail ID: jks_sharma@yahoo.com.

J. K. Sharma

This page is intentionally left blank.

About the Author

J.K. Sharma is a professor at Amity Business School, Amity University, Noida. He has also served as a professor at the Faculty of Management Studies, University of Delhi. He has also served as Director of G L Bajaj Institute of Management and Research, Greater Noida; Institute of Management Studies, Noida and Maharishi Institute of Management, Noida.

Professor Sharma has more than 33 years of teaching experience in Operations Research, Business Statistics, Mathematics and Service Operations Management. His research interest lies in the application of operations research in the functional areas of management and hospital administration. He is a gold medalist in Mathematics and has taught as a visiting professor at Group ESSEC (A Graduate School of Management) in France. He has authored 20 books on Operations Research, Business Statistics, Business Mathematics, Management of Systems and Discrete Mathematics. He has also written than 90 research papers and 20 case studies. He has supervised 15 Ph.D. students and is also a Ph.D. thesis examiner of several Indian universities. He is also a research paper reviewer of several. National and International research journals.

UNIT 1

STATISTICAL DATA AND DESCRIPTIVE STATISTICS

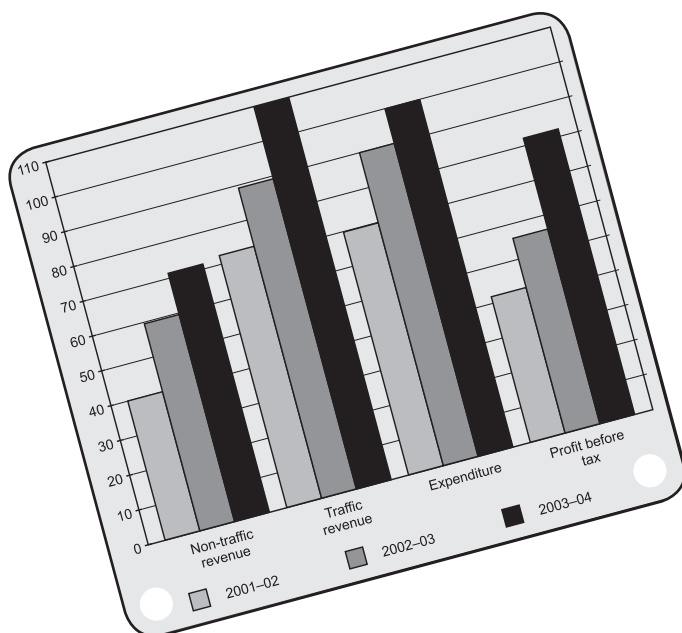
Chapter 1: Statistical Data and Their Sources

Chapter 2: Data Classification and Their Presentation

Chapter 3: Measures of Central Tendency

Chapter 4: Measures of Variation: Absolute and Relative

Chapter 5: Skewness, Moments and Kurtosis



This page is intentionally left blank.

1

Statistical Data and Their Sources

LEARNING OBJECTIVES

After studying this chapter, you should be able to

- present a broad overview of statistics as a subject.
- bring out applications of statistics and its usefulness in managerial decision making.
- describe the data collection process.
- understand basic concepts of questionnaire design and measurement scales.

1.1 REASONS FOR LEARNING STATISTICS

H. G. Wells' statement that "statistical thinking will one day be as necessary as the ability to read and write" is valid in the context of today's competitive business environment where many organizations find themselves data rich but information poor. Thus, for decision makers, it is important to develop the ability to extract meaningful information from raw data to make better decisions. It is possible only through the careful analysis of data guided by statistical thinking.

The reason for analysis of *data* is an understanding of *variation and its causes* in any phenomenon. Since variation is present in all phenomena, therefore knowledge of it leads to better decisions about a phenomenon that produced the data. From this perspective, the learners of statistics enables the decision maker to understand how to:

- present and describe information (data) so as to improve decisions.
- draw conclusions about the large *population* based upon information obtained from samples.
- seek out relationship between pair of variables to improve processes.
- obtain reliable forecasts of statistical variables of interest.

Thus, a statistical study might be a simple exploration enabling us to gain insight into a virtually unknown situation or it might be a sophisticated analysis to produce numerical confirmation or a reflection of some widely held belief.

1.2 STATISTICS DEFINED

As Statistical Data The word statistics refers to a special discipline or a collection of procedures and principles useful as an aid in gathering and analysing numerical information for the purpose of drawing conclusions and making decisions. Since any numerical figure, or figures, cannot be

called statistics owing to many considerations which decide its use, statistical data or mere data is a more appropriate expression to indicate numerical facts.

A few definitions which describe the characteristics of statistics are as follows:

- *The classified facts respecting the condition of the people in a state . . . especially those facts which can be stated in numbers or in tables of numbers or in any tabular or classified arrangement.*
—Webster

This definition is quite narrow as it confines the scope of statistics only to such facts and figures which are related to the conditions of the people in a state.

- *By statistics we mean **quantitative data** affected to a marked extent by multiplicity of causes.*
—Yule and Kendall
- *By statistics we mean **aggregates of facts** affected to a marked extent by multiplicity of causes numerically expressed, enumerated, or estimated according to reasonable standards of accuracy, collected in a systematic manner for predetermined purpose and placed in relation to each other.*
—Horace Secrist

As Statistical Methods Methods used in the collection and analysis of numerical information or statistical data for the purpose of drawing conclusions and making decisions are called *statistical methods*.

Statistical methods, also called statistical techniques, are sometimes loosely referred to cover 'statistics' as a subject in whole. There are two branches of statistics: (i) *Mathematical statistics* and (ii) *Applied statistics*. Mathematical statistics is a branch of mathematics and is theoretical. It deals with the basic theory about how a particular statistical method is developed. Applied statistics, on the other hand, uses statistical theory in formulating and solving problems in other subject areas such as economics, sociology, medicine, business/industry, education and psychology.

The field of applied statistics is not easy because the rules necessary to solve a particular problem are not always obvious although the guiding principles that underlie the various methods are identical regardless of the field of their application. As a matter of fact, experience and judgment are otherwise more necessary to execute a given statistical investigation.

A few relevant definitions of statistical methods are given below:

- *Statistics is the science which deals with the methods of collecting, classifying, presenting, comparing and interpreting numerical data collected to throw some light on any sphere of enquiry.*
—Seligman
- *The science of statistics is the method of judging, collecting natural or social phenomenon from the results obtained from the analysis or enumeration or collection of estimates.*
—King

A. L. Bowley has given the following three definitions keeping in mind various aspects of statistics as a science:

- *Statistics may be called the science of counting.*
- *Statistics may be called the science of average.*
- *Statistics is the science of the measurement of social organism regarded as a whole in all its manifestations.*

Another definition of statistics given by Croxton and Cowden is as follows:

- *Statistics may be defined as a science of collection, presentation, analysis and interpretation of numerical data.*
—Croxton and Cowden

1.3 TYPES OF STATISTICAL METHODS

Statistical methods, broadly, fall into the following two categories:

- (i) Descriptive statistics
- (ii) Inferential statistics

Descriptive statistics (also referred to as *Exploratory Data Analysis*) includes statistical methods involving the collection, presentation and characterization of a set of data in order to describe the various features of that set of data.

In general, methods of descriptive statistics include graphic methods and numeric measures. Bar charts, line graphs and pie charts comprise the graphic methods, whereas numeric measures include measures of central tendency, dispersion, skewness and kurtosis.

Inferential statistics includes statistical methods which facilitate estimating the characteristics of a population or making decisions concerning a population on the basis of sample results. Sample and population are two relative terms. The larger group of units about which inferences are to be made is called the population or universe, while a sample is a fraction, subset or portion of that universe.

Inferential statistics can be categorized as *parametric* or *non-parametric*. The use of parametric statistics is based on the assumption that the population from which the sample is drawn is normally distributed. Parametric statistics can be used only when data are collected on an interval or a ratio scale. Non-parametric statistics makes no explicit assumption regarding the normality of distribution in the population and is used when the data are collected on a nominal or an ordinal scale.

The need for sampling arises because in many situations data are sought for a large group of elements such as individuals, companies, voters, households, products, customers and so on to make inferences about the population that the sample represents. Thus, due to time, cost and other considerations data are collected from only a small portion of the population called *sample*. The concepts derived from probability theory help to ascertain the likelihood that the analysis of the characteristics based on a sample do reflect the characteristics of the population from which the sample is drawn. This helps the decision maker to draw conclusions about the characteristics of a large population under study.

1.4 IMPORTANCE AND SCOPE OF STATISTICS

The scope of application of statistics has assumed unprecedented dimensions these days. Statistical methods are applicable in diverse fields such as economics, trade, industry, commerce, agriculture, bio-sciences, physical sciences, education, astronomy, insurance, accountancy and auditing, sociology, psychology, meteorology and so on. Bringing out its wide applications, Carroll D. Wright (1887), United States Commissioner of the Bureau of Labour, has explained the importance of statistics by saying:

To a very striking degree, our culture has become a statistical culture. Even a person who may never have heard of an index number is affected by those index numbers which describe the cost of living. It is impossible to understand Psychology, Sociology, Economics or a Physical Science without some general idea of the meaning of an average, of variation, of concomitance of sampling, of how to interpret charts and tables.

According to the statistician Bowley—‘*A knowledge of statistics is like a knowledge of foreign language or of algebra, it may prove of use at any time under any circumstances*’.

1.4.1 Statistics in Economics

Statistical methods are extensively used in all branches of economics. For example:

- (i) Time-series analysis is used for studying the behaviour of prices, production and consumption of commodities, money in circulation, and bank deposits and clearings.
- (ii) Index numbers are useful in economic planning as they indicate the changes over a specified period of time in (a) prices of commodities, (b) imports and exports, (c) industrial/agricultural production, (d) cost of living, and the like.
- (iii) Demand analysis is used to study the relationship between the price of a commodity and its output (supply).
- (iv) Forecasting techniques are used for curve fitting by the principle of least squares and exponential smoothing to predict inflation rate, unemployment rate or manufacturing capacity utilization.

1.4.2 Statistics in Business Management

According to Wallis and Roberts—‘*Statistics may be regarded as a body of methods for making wise decisions in the face of uncertainty.*’ Ya-Lin-Chou gave a modified definition over this, saying that ‘*Statistics is a method of decision making in the face of uncertainty on the basis of numerical data and calculated risks.*’ These definitions reflect the applications of statistics in the development of general principles for dealing with uncertainty.

Statistical reports provide a summary of business activities which improves the capability of making more effective decisions regarding future activities. Certain activities are discussed below of a typical organization where statistics plays an important role in their efficient execution.

1.4.3 Statistics in Social Sciences

The following definitions reflect the importance of statistics in social sciences.

- *Statistics is the science of the measurement of social organism, regarded as a whole in all its manifestations.* —Bowley
- *The science of statistics is the method of judging collective, natural or social phenomenon from the results obtained from the analysis, enumeration or collection of estimates.* —W. I. King

Some specific areas of applications of statistics in social sciences are as listed below:

- (i) Regression and correlation analysis techniques are used to study and isolate all those factors associated with each social phenomenon which bring out the changes in data with respect to time, place and object.
- (ii) Sampling techniques and estimation theory are indispensable methods for conducting any social survey pertaining to any strata of society and drawing valid inferences.
- (iii) In sociology, statistical methods are used to study mortality (death) rates, fertility (birth rates) trends, population growth and other aspects of vital statistics.

1.4.4 Statistics in Medical Sciences

The knowledge of statistical techniques in all natural sciences—zoology, botany, meteorology, and medicine—is of great importance. For example, for proper diagnosis of a disease, the doctor needs and relies heavily on factual data relating to pulse rate, body temperature, blood pressure, heart beats and body weight.

An important application of statistics lies in using the *test of significance* for testing the efficacy of a particular drug or injection meant to cure a specific disease. Comparative studies for effectiveness of a particular drug/injection manufactured by different companies can also be made by using statistical techniques such as the *t*-test and *F*-test.

To study plant life, a botanist has to rely on data about the effect of temperature, type of environment rainfall, and so on.

Conceptual Questions 1A

1. What is statistics? How do you think that the knowledge of statistics is essential in management decisions. Give examples.
2. (a) Explain how statistics plays an important role in management planning and decision making?
(b) Define statistics and statistical methods. Explain the uses of statistical methods in modern business.
3. Statistics affects everybody and touches life at many points. It is both a science and an art. Explain this statement with suitable examples.
4. With the help of few examples, explain the role of statistics as a managerial tool.
5. 'Statistics is a body of methods for making wise decisions in the face of uncertainty'. Comment on the statement bringing out clearly, how statistics helps in business decision making.
6. One can say that statistical inference includes an interest in statistical description as well, since the ultimate purpose of statistical inference is to describe population data. Does statistical inference differ from statistical description? Discuss.
7. 'Modern statistical tools and techniques are important for improving the quality of managerial decisions'. Explain this statement.
8. 'Statistics are numerical statements of facts but all facts numerically stated are not statistics'. Comment upon the statement.
9. (a) Define statistics. Why do some people look at this science with an eye of distrust?
(b) 'The science of statistics is the most useful servant but only of great value to those who understand its proper use'. Discuss.
10. Bring out the applications of statistics in economics and business administration as a scientific tool.
11. Discuss the differences between statistics as numerical facts and as a discipline or field of study.

1.5 NEED FOR DATA

Statistical data are the basic material needed to make an effective decision in a particular situation. The main reasons for collecting data are listed below:

- (i) To provide necessary inputs to a given phenomenon or situation under study.
- (ii) To measure performance in an ongoing process such as production, service and so on.
- (iii) To enhance the quality of decision making by enumerating alternative courses of action in a decision making process and selecting an appropriate one.
- (iv) To satisfy the desire to understand an unknown phenomenon.
- (v) To assist in guessing the causes and probable effects of certain characteristics in given situations.

For any statistical analysis to be useful, the collection and use of input data is extremely important. One can collect an enormous amount of data on a subject of interest in a compact and usable form from the internet. However, the reliability of such data is always doubtful. Thus, before relying on any interpreted data, either from a computer, internet or other source, we should study answers to the following questions:

- (i) Have data come from an unbiased source, i.e., source should not have an interest in supplying the data that lead to a misleading conclusion?
- (ii) Do data represent the entire population under study, i.e., how many observations should represent the population?
- (iii) Do the data support other evidences already available? Is any evidence missing that may cause to arrive at a different conclusion?
- (iv) Do data support the logical conclusions drawn? Have we made conclusions which are not supported by data?

A useful acronym in statistics is GIGO, meaning *garbage in, garbage out*. If the data quality is poor, irrespective of the sophistication of the statistical analysis used, the results may be meaningless. This can result in low user confidence in statistical findings. *Data quality* is influenced by three important factors:

- The *type of data* used in any analysis.
- The *source* from which that data comes.
- The *data collection methods* used to record the data.

The choice of statistical method is to a large extent determined by the *type of data* available. Certain statistical methods are valid for certain data types only. The incorrect use of a statistical method for a certain data type can render the statistical findings invalid.

The other two factors, namely the *source of data* and choice of a *data collection method*, can influence user confidence in the accuracy and reliability of the data. This again impacts on the validity of statistical findings.

1.5.1 Types of Data

Statistical data are the outcome of a continuous process of measuring, counting and/or observing. These may pertain to several aspects of a phenomenon (or a problem) which are measurable, quantifiable, countable or classifiable. While conducting a survey or making a study, an investigator develops a method to ask several questions to deal with the variety of characteristics of the given population or universe. These characteristics which one intends to investigate and analyse are termed as *variables*. The data, which are the observed outcomes of these variables, may vary from response to response. Consumer behaviour (attitude), profit/loss to a company, job satisfaction, drinking and/or smoking habits, leadership ability, class affiliation or status are the examples of a variable.

Data type is determined by the nature of variable under study. A variable is either *qualitative* or *quantitative* in nature. *Qualitative variables* produce *categorical (non-numeric)* data. Such data consists of a *coded value* assigned to each response category of the variable. These coded values are *arbitrarily* assigned. Normal arithmetic operations can not therefore, be performed on these coded values as they have no numerical meaning.

Quantitative variables are variables that produce *numeric* responses. The *data* consists of *numbers* which have arithmetic properties. Such data can therefore, be manipulated using conventional arithmetic operations (add, subtract, multiply and divide operations) and produce meaningful results.

Table 1.1 summarizes the types of variables which can be studied to yield the observed outcomes in relation to the nature of data, information and measurement.

Table 1.1 Nature of Data, Information and Measurement

<i>Data Type</i>	<i>Information Type</i>	<i>Measurement Type</i>
Categorical	→ Do you practice Yoga?	Yes ☐ No ☐
Numerical	→ How many books do you have in your library?	Number
	→ What is your height?	Centimetres or Inches

1.6 PRINCIPLES OF MEASUREMENT

There are four different ways in which data can be classified according to its *type*. This classification provides way to distinguish between different types of data that can be collected in any research study.

In general, the principles of measurement (scaling) has three characteristics:

- (i) Numbers are ordered. One number is less than, equal to or greater than another number.
- (ii) Difference between numbers are ordered. The difference between any pair of numbers is greater than, equal or less than to the difference between any other pair of numbers.
- (iii) The number series has a unique origin indicated by the number zero.

Each variable is associated with only one of the four possible types. The combinations of these characteristics of *order*, *distance* and *origin* provide the following widely used classification of measurement scales:

<i>Classification</i>	<i>Characteristics</i>	<i>Basic Empirical Operation</i>
Nominal Scale	No order, distance or unique origin	Determination of categorical information. Numbers only identify groups which cannot be ordered.
Ordinal Scale	Order but no distance or unique origin	Determination of greater or lesser values. Numbers allow ranking but no arithmetic.
Interval Scale	Both order and distance but not unique	Determination of equality of intervals or differences. Intervals between numbers are meaningful.
Ratio Scale	Order, distance and unique origin	Determination of equality of ratios. Intervals between numbers are meaningful and also their ratio as the lowest value is a meaningful zero.

Nominal Scale This scales is associated with *qualitative* variables which have fixed response categories, all of equal importance. Here it is assumed that no response category is more or less important than any other category. In nominal scaling, the numerical values are either named or

categorized in such a way that these values are mutually exclusive and collectively exhaustive. For example, shirt numbers in a football or cricket match are measured at a nominal level. A player wearing a shirt number 24 is not more of anything than a player wearing a shirt number 12 and is certainly not twice the number 12. In other words, if we use numbers to identify categories, they are recognized as levels only.

Statistically, nominal scaled data is the weakest form of data for analysis purposes. Since the numerical values recorded are only arbitrary *codes* assigned to the various response categories, they possess no numerical properties. At most, these coded response values can be *counted* and *grouped*. Thus, only limited statistical analysis can be applied to nominal scaled data. Few examples of nominal scaling are gender, blood group, religion, marital status, place of work and brand name.

Ordinal Scale This scaling is also associated with *qualitative* variables. It differs from nominal in that there is an *implied ranking* between the response categories. Each consecutive response category possesses either *more* or *less* of a given characteristic. The implied ranking between consecutive categories could apply to, for example, increasing (or decreasing) age, size, responsibility, authority, accountability, strength, preference, economic value or weight.

Ordinal scaled data is also generated when respondents are asked to *rank objects* (products, brands, companies) or product attributes (e.g. taste, packaging, price and size) from *most preferred* to *least preferred* or *most important* to *least important*. The most preferred (or most important) item in a list is assigned a rank value of 1; the second most preferred (or important) item a rank value of 2 and so on.

Few examples of qualitative variables with ordinal scaled response categories are shown below:

<i>Variable</i>	<i>Rank Order Response Categories (Showing Ranking Characteristic)</i>	<i>Data Codes</i>
Income groups	Upper, middle, lower (decreasing purchasing power)	1, 2, 3
Management positions	Junior, middle, senior, director (increasing responsibility)	1, 2, 3, 4
Stress	Low, moderate, high (increasing anxiety)	1, 2, 3
Examination grades	First, second, third, fail (decreasing knowledge)	1, 2, 3, 4

From an arithmetic view point, the *rank values* assigned to consecutive categories are not true numeric values. These numbers cannot be assumed to measure the absolute difference between the categories. The distances between the ranks are not assumed to be *equal*, hence difference between ranks are not a constant value. They only indicate the *order* in which the categories appear.

Statistically, ordinal scaled data is also numerically weak data. However, unlike nominal scaled data, it has the property of order giving it modest numeric properties.

Interval Scaling This scaling is associated with *quantitative* variables. Interval scaled data is designed to possess the arithmetic properties of both *order* (i.e. implied ranking) and *distance* between the data values, but with an arbitrary value for its origin. Thus on a 1–5 rating scale, for example, a 4 is considered to reflect a stronger perception (or preference) than a 2. This is the property of *order*. Also, the differences between the ratings are assumed to reflect equal differences between perceptions or expressed preferences. It is possible to conclude that the difference in perception

(or preference) between ratings of 3 and 4 is the same as between ratings of 1 and 2. This is the property of *distance*. However, because there is no absolute origin of zero (the origin is arbitrarily set to 1), it is not possible to meaningfully compare the ratio of interval scaled values to one another. For example, it is not valid to conclude that a rating of 4 is twice as important as a rating of 2, nor that a rating of 3 is three times more important than a rating of 1. Only absolute differences make sense between interval scaled data values.

These properties of *order* and *distance* give interval scaled data sufficient numeric properties to allow arithmetic operations to be meaningfully performed on such data. It should be noted that the wider the rating scale (e.g. 1–10 as opposed to 1–5), the more valid it is to perform numeric operations on the interval scaled data.

Few examples of quantitative variables with interval scaled response values are as follows:

- (i) *A semantic differential rating scale:*

Question: Rate your *taste* of banana

Bitter *Sweet*
 1 2 3 4 5 6 7

The variable is *taste*, with interval response values of 1, 2, 3, 4, 5, 6 or 7.

- (ii) *A likert rating scale:*

Question: Comment on the statement: Airplane season tickets are overpriced.

Strongly disagree *Disagree* *Unsure* *Agree* *Strongly agree*
 1 2 3 4 5

The variable is the *statement* with response values of 1, 2, 3, 4 or 5.

Ratio Scaling This scaling is also associated with *quantitative* variables. Ratio scaled data values are derived from direct *measurement* where there is an absolute origin of zero.

Ratio scaled data values can be any number in an interval. For example, the *age of a car owner* can be any value between, say, 18 and 90 years. Such data is the strongest type of statistical data possible, as it possesses all arithmetic properties, namely *order*, *distance* and an *absolute origin of zero*. The zero origin indicates the absence of the attribute being measured and permits relative comparisons through ratios (e.g. 5 is half of 10; 4 is a quarter of 16, etc.). Such data can, therefore, be manipulated using all arithmetic operations (add, subtract, multiply and divide).

Few examples of quantitative variables with ratio scaled data values are as follows:

<i>Example</i>	<i>Variable Data Values</i>
Age of car driver	36 years
Distance to work place	2,4 km
Flight time from Delhi to Mumbai	120 min
Employee salary per month	₹ 4265
Milk price (per litre)	₹ 249
Number of children in household	4

Statistically, ratio scaled data is the strongest and most desirable data to gather as it lends itself to the widest range of statistical methods. Almost all statistical methods are valid for ratio scaled data.

If ratio scaled data is grouped into categories, the gathered data type becomes *ordinal scaled*, since only a coded data value for the chosen category is recorded instead of the actual ratio-scaled data value.

The few examples showing the difference in data type depending on the question format are:

<i>Quantitative Variable</i>	<i>Question Format</i>	
	<i>Ratio Data</i>	<i>Ordinal Data</i>
Shopper's age	How old are you? (years)	Indicate the age group to which you belong (Tick only one block.) • 16–25 • 26–35 • 36–45 • 46–55 • 56–65 • Over 65
Shopper's purchasing frequency	How many times a month do you shop for groceries? (times)	How often did you visit your local market last month to buy groceries? (Tick only one block.) • Daily • 4–5 times/week • 2–3 times/week • weekly • 2–3 times/month • monthly

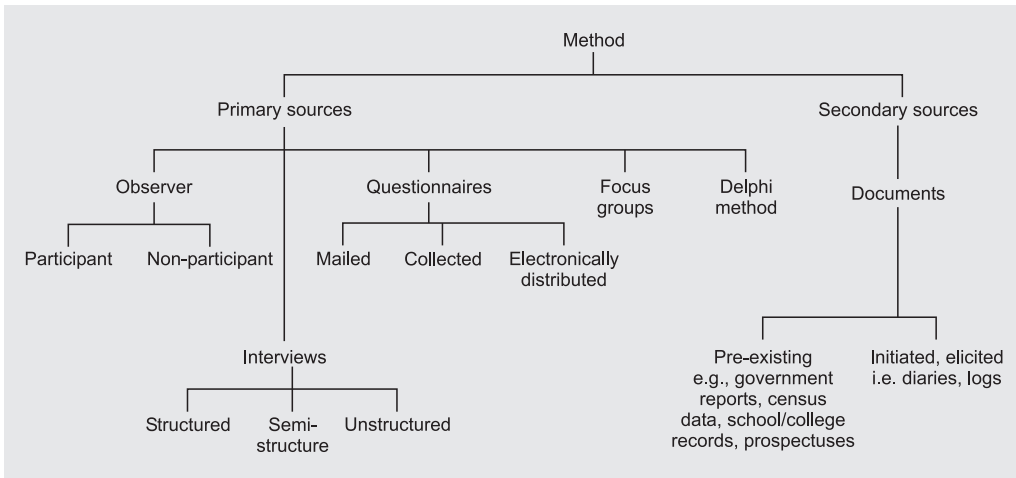
1.7 SOURCES OF DATA

The choice of a data collection method from a particular source depends on the facilities available, the extent of accuracy required in analyses, the expertise of the investigator, the time span of the study and the amount of money and other resources required for data collection. When the data to be collected are very voluminous and require huge amounts of money, manpower, and time, reasonably accurate conclusions can be drawn by observing even a small part of the population provided the concept of sampling is used objectively.

While data is collected from several sources, these sources can be classified in one of two ways:

- Secondary versus primary data sources
- Internal versus external data sources.

Secondary data is data which already exists in processed form. *Primary* data, on the other hand, originates only when it is collected. Various methods of data collection are summarized as follows:



Internal data is generated from within the organization during its normal activities. *External* data, on the other hand, exists in the micro- and macro-environments in which an organization operates.

1.7.1 Primary Data Sources

Individuals, focus groups and/or panels of respondents specifically decided upon and set up by the investigator for data collection are examples of **primary data sources**.

Advantages

- *Direct relevance* to the problem being researched.
- Greater accuracy and reliability of data resulting from greater control over the data collection process.

Disadvantages

- It could be time consuming to collect.
- It is generally *more expensive* to collect.

The methods which may be used for primary data collection are briefly discussed below.

Observation In observational studies, the investigator does not ask questions to seek clarifications on certain issues. Instead, he records the behaviour, as it occurs, of an event in which he is interested. Sometimes mechanical devices are also used to record the desired data.

Studies based on observations are best suited for researches requiring non-self report descriptive data. That is, when respondents' behaviours are to be understood without asking them to part with the needed information. Diverse opinions in the diagnosis of a particular disease could be an example of an observational study.

Certain difficulties do arise during the collection of such data on account of (i) the observer's training, philosophy, opinions, and expectations, (ii) the interdependence of observations and inferences, and (iii) the inadequacies of the sense organs causing significant variations in the observations of the same phenomenon.

Interviewing Interviews can be conducted either face-to-face or over telephone. Such interviews provide an opportunity to establish a rapport with the interviewer and help extract valuable information. Direct interviews are expensive and time consuming, if a big sample of respondents is to be personally interviewed. Interviewers' biases also come in the way. Such interviews should be conducted at the exploratory stages of research to handle concepts and situational factors.

Telephonic interviews help establish contact with interviewers spread over distantly separated geographic locations and obtain responses quickly. This method is effective only when the interviewer has specific questions to ask and the needs responses promptly. Since the interviewer in this case cannot observe the non-verbal responses at the other end, the respondent can unilaterally terminate the interview without warning or explanation.

Questionnaire It is a formalized set of questions for extracting information from the target respondents. The form of the questions should correspond to the form of the required information. The three general forms of questions are: *dichotomous* (yes/no response type), *multiple choice*, and *open-ended*. A questionnaire can be administered personally or mailed to the respondents. It is an efficient method of collecting primary data when the investigator knows what exactly is required and how to measure such variables of interest as:

- Behaviour—past, present, or intended.
- Demographic characteristics—age, sex, income, and occupation.
- Level of knowledge.
- Attitudes and opinions.

As such there are no set principles that must be used to design a questionnaire. However, general principles of questionnaire design based on numerous studies and experiences of survey researchers. A good questionnaire does, however, require the application of common sense, concern for the respondent, a clear concept of the information needed, and a thorough pre-testing of the questionnaire.

1.7.2 Secondary Data Sources

A secondary data source contains *processed* data. Such data has already been gathered, summarized, interpreted and reported in documents such as academic, technical and commercial publications and in-company reports. These documents are generally prepared by different objectives from those of the present study.

Secondary data sources must be carefully evaluated, however, before the data is included in the current research study in order to determine the data's *relevance* and *usefulness* to the problem. There are a number of benefits and drawbacks associated with the use of secondary data. Some of these are noted below.

Advantages

- The data is already in existence.
- Access time is relatively short (data is generally readily accessible).
- The cost of acquiring secondary data is generally low.

It is recommended that relevant secondary sources always be identified and consulted as they generally constitute a *more economical* source of data both *financially* and *time wise*.

Disadvantages

Secondary data does have certain *drawbacks*. Secondary data can differ in format, quality and quantity from the data desired for the current study. In short, a researcher should consider secondary data's *reliability* and *compatibility* in the context of the current study. Few disadvantages include.

- The data may not be problem specific (relevancy).
- The data may be dated and therefore inappropriate.
- It may be difficult to assess data accuracy and reliability.
- The data may not lend itself to further manipulation (the format and the extent of data aggregation may not be appropriate).
- Combining data from various sources can lead to data incompatibility (i.e. errors of collation and possible bias).

External Secondary Data Sources

- Government publications which include the following:
 - (i) The National Accounts Statistics published by the Central Statistical Organization (CSO). It contains estimates of national income for several years, growth rate and rate on major economic activities such as agriculture, industry, trade, transport and so on.
 - (ii) Wholesale Price Index published by the office of the Economic Advisor, Ministry of Commerce and Industry.
 - (iii) Consumer Price Index.
 - (iv) Reserve Bank of India Bulletins.
 - (v) Economic Survey.
- Non-Government publications include publications of various industrial and trade associations such as
 - (i) The Indian Cotton Mills Association
 - (ii) The various Chambers of Commerce
 - (iii) The Bombay Stock Exchange, which publishes a directory containing financial accounts, key profitability and other relevant data.
- Various syndicate services such as Operations Research Group (ORG). The Indian Market Research Bureau (IMRB) also collects and tabulates abundant marketing information to suit the requirements of individual firms, making the same available at regular intervals.
- International organizations which publish data are as follows:
 - (i) The International Labour Organization (ILO)—which publishes data on the total and active population, employment, unemployment, wages and consumer prices.
 - (ii) The Organization for Economic Cooperation and Development (OECD)—which publishes data on foreign trade, industry, food, transport and science and technology.
 - (iii) The International Monetary Fund (IMF)—which publishes reports on national and international foreign exchange regulations and other trade barriers, foreign trade and economic developments.

Internal Secondary Data Sources The data generated within an organization in the process of routine business activities are referred to as internal secondary data. Financial accounts, production, quality control and sales records are examples of such data. However, data originating from one department of an organization may not be useful for another department in its original form. It is, therefore, desirable to condense such data into a form needed by the other.

Conceptual Questions 1B

12. A manager of a large corporation has recommended that ₹ 1000 raise be given to keep a valued subordinate from moving to another company. What internal and external sources of data might be used to decide whether such a salary increase is appropriate?
 13. In a recent study of causes of death in men 60 years of age and older, a sample of 120 men indicated that 48 died as a result of some form of heart disease.
 - (a) Develop a descriptive statistic that can be used as an estimate of the percentage of men 60 years of age or older who die from some form of heart disease
 - (b) Are the data on the causes of death qualitative or quantitative?
 14. Determine whether each of the following random variables is categorical or numerical. If it is numerical, determine whether the phenomenon of interest is discrete or continuous.
 - (a) Amount of time the personal computer is used per week.
 - (b) Number of persons in the household who use the personal computer.
 - (c) Amount of money spent on clothing in the last month.
 - (d) Favourite shopping centre.
 - (e) Amount of time spent on shopping for clothing in the last month.
 15. State whether each of the following variables is qualitative or quantitative and indicate the measurement scale that is appropriate for each.
 - (a) Age
 - (b) Gender
 - (c) Class rank
 - (d) Make of automobile
 - (e) Annual sales
 - (f) Soft-drink size (small, medium, large)
 - (g) Earnings per share
 - (h) Method of payment (cash, check, credit card)
 16. A firm is interested in testing the advertising effectiveness of a new television commercial. As part of the test, the commercial is shown on 6:30 p.m. local news programme in Delhi. Two days later, a market research firm conducts a telephone survey to obtain information on recall rates (percentage of viewers who recall seeing the commercial) and impressions of the commercial.
 - (a) What is the population for this study?
 - (b) What is the sample for this study?
 - (c) Why would a sample be used in this situation? Explain.
 17. Suppose the following information is obtained from a person on his application for a home loan from a bank:
 - (a) Place of residence: GK II, New Delhi
 - (b) Type of residence: Single family home
 - (c) Date of birth: 14 August 1975
 - (d) Monthly income: ₹ 25,000
 - (e) Occupation: Systems Engineer
 - (f) Employer: Telecom company
 - (g) Number of years at job: 5
 - (h) Other income: ₹ 30,000 per year
 - (i) Marital status: Married
 - (j) Number of children: 1
 - (k) Loan requested: ₹ 5,00,000
 - (l) Term of Loan: 10 years
 - (m) Other loan: Car
 - (n) Amount of other loan: ₹ 1,00,000
- Classify each of the 14 responses by type of data and level of measurement.

True or False Chapter Concepts Quiz

1. The scale of measurement of a variable is a nominal scale when data are labels to identify an attribute of the element.
2. The statistical method used to summarize data depends upon whether the data are qualitative or quantitative.
3. Statistical studies can be classified as either experimental or observational.
4. Learning statistics does not help to improve processes.
5. Statistics cannot be misused.
6. All facts numerically stated are not statistics.
7. Statistical thinking focuses on ways to understand, manage and reduce variation.
8. An average value computed from the set of all observations in the population is called a statistic.
9. Inferential statistics help in generalizing the results of a sample to the entire population.
10. Descriptive statistical methods are used for presenting and characterizing data.
11. A statistic is a summary measure that describes the characteristic of a population.
12. A descriptive measure computed from a sample of the population is called a parameter.
13. Enumerative studies involve decision making regarding a population and/or its characteristics.
14. Analytical studies involve taking some action on a process to improve performance in the future.
15. Data are needed to satisfy our curiosity.
16. A continuous variable can also be used for quantitative data when every value within some interval is a possible result.
17. A summary measure computed from sample data is called statistic.
18. The summary numbers for either a population or a sample are called descriptive statistics.

Concepts Quiz Answers

1. T	2. T	3. T	4. F	5. F	6. T	7. T	8. F	9. T
10. T	11. T	12. T	13. F	14. F	15. T	16. T	17. F	18. T

2

Data Classification and Their Presentation

LEARNING OBJECTIVES

After studying this chapter, you should be able to

- understand types of data and the basis of their classification.
- use techniques of organizing data in tabular and graphical form in order to enhance data analysis and interpretation.

2.1 INTRODUCTION

In Chapter 1, we learned how to collect data through primary and/or secondary sources. Whenever a set of data which we have collected contains a large number of observations, the best way to examine such data is to present it in some compact and orderly form to become suitable for *analysis* and *interpretation*. For this reason, the data set is organized and summarized in such a way that patterns are revealed and more easily interpreted. Such an arrangement of data is known as *distribution* of the data. Distribution is important because it reveals the pattern of variation and helps in a better understanding of the phenomenon the data present.

2.2 CLASSIFICATION OF DATA

Classification of data is the process of arranging data in groups/classes on the basis of certain properties. The classification of statistical data serves the following purposes:

- (i) It condenses the raw data into a form suitable for statistical analysis.
- (ii) It removes complexities and highlights the features of the data.
- (iii) It facilitates comparisons and drawing inferences from the data. For example, if university students in a particular course are divided according to sex, their results can be compared.
- (iv) It provides information about the mutual relationships among elements of a data set. For example, based on literacy and criminal tendency of a group of people, it can be established whether literacy has any impact or not on criminal tendency.
- (v) It helps in statistical analysis by separating elements of the data set into homogeneous groups and hence brings out the points of similarity and dissimilarity.

2.2.1 Requisites of Ideal Classification

The classification of data is decided after taking into consideration the nature, scope, and purpose of the investigation. However, an ideal classification should have following characteristics:

It Should be Unambiguous It is necessary that the various classes should be so defined that there is no room for confusion. There must be only one class for each element of the data set. For example, if the population of the country is divided into two classes, say literates and illiterates, then an exhaustive definition of the terms used would be essential.

Classes Should be Exhaustive and Mutually Exclusive Each element of the data set must belong to a class. For this, an extra class can be created with the title ‘others’ so as to accommodate all the remaining elements of the data set.

Each class should be mutually exclusive so that each element must belong to only one class. For example, classification of students according to the age: below 25 years and more than 20 years, is not correct because students of age 20 to 25 may belong to both the classes.

It Should be Stable The classification of a data set into various classes must be done in such a manner that if each time an investigation is conducted, it remains unchanged and hence the results of one investigation may be compared with that of another. For example, classification of the country’s population by a census survey based on occupation suffers from this defect because various occupations are defined in different ways in successive censuses and, as such, these figures are not strictly comparable.

It Should be Flexible A classification should be flexible so that suitable adjustments can be made in new situations and circumstances. However, flexibility does not mean instability. The data should be divided into few major classes which must be further subdivided. Ordinarily there would not be many changes in the major classes. Only small sub-classes may need a change and the classification can, thus, retain the merit of stability and yet have flexibility.

2.2.2 Basis of Classification

Statistical data are classified after taking into account the nature, scope and purpose of an investigation. Generally, data are classified on the basis of the following four bases:

Geographical Classification In geographical classification, data are classified on the basis of geographical or locational differences such as—cities, districts, or villages between various elements of the data set. The following is an example of a geographical distribution:

City	:	Mumbai	Kolkata	Delhi	Chennai
Population density (per square km)	:	654	685	423	205

Such a classification is also known as *spatial classification*. Geographical classifications are generally listed in alphabetical order. Elements in the data set are also listed by the frequency size to emphasize the importance of various geographical regions as in ranking the metropolitan cities by population density. The first approach is followed in case of reference tables while the second approach is followed in the case of summary tables.

Chronological Classification When data are classified on the basis of time, the classification is known as chronological classification. Such classifications are also called *time series* because data are usually listed in chronological order starting with the earliest period. The following example would give an idea of chronological classification:

Year	:	1941	1951	1961	1971	1981	1991	2001
Population	:	31.9	36.9	43.9	54.7	75.6	85.9	98.6
(crore)								

Qualitative Classification In qualitative classification, data are classified on the basis of descriptive characteristics or on the basis of attributes like sex, literacy, region, caste or education, which cannot be quantified. This is done in the following two ways:

- (i) *Simple classification*: In this type of classification, each class is subdivided into two sub-classes and only one attribute is studied such as: male and female; blind and not blind, educated and uneducated and so on.
- (ii) *Manifold classification*: In this type of classification, a class is subdivided into more than two sub classes which may be sub divided further. An example of this form of classification is shown in the box.

Quantitative Classification In this classification, data are classified on the basis of some characteristics which can be measured such as height, weight, income, expenditure, production or sales.

Quantitative variables can be divided into the following two types. The term variable refers to any quantity or attribute whose value varies from one investigation to another.

- (i) *Continuous variable* is the one that can take any value within the range of numbers. Thus, the height or weight of individuals can be of any value within the limits. In such a case, data are obtained by measurement.
- (ii) *Discrete (also called discontinuous) variable* is the one whose values change by steps or jumps and cannot assume a fractional value. The number of children in a family, number of workers (or employees), and number of students in a class are few examples of a discrete variable. In such a case, data are obtained by counting.

The following are examples of continuous and discrete variables in a data set (Table 2.1).

Table 2.1 Continuous and Discrete Variables in a Data Set

Discrete Series		Continuous Series	
Number of Children	Number of Families	Weight (kg)	Number of Persons
0	10	100 to 110	10
1	30	110 to 120	20
2	60	120 to 130	25
3	90	130 to 140	35
4	110	140 to 150	50
—	—	—	—
	300		140

2.3 ORGANIZING DATA USING DATA ARRAY

The best way to examine a large set of numerical data is to first organize and present it in an appropriate tabular and graphical format.

Table 2.2 presents the total number of overtime hours worked for 30 consecutive weeks by machinists in a machine shop. The data displayed here are in *raw form*, that is, the numerical observations are not arranged in any particular order or sequence.

Table 2.2 Raw Data Pertaining to Total Time Hours Worked by Machinists

94	89	88	89	90	94	92	88	87	85
88	93	94	93	94	93	92	88	94	90
93	84	93	84	91	93	85	91	89	95

These raw data are not amenable even to a simple reading and do not highlight any characteristic/trend, such as the highest, the lowest and the average weekly hours. Even a careful look at these data do not easily reveal any significant trend regarding the nature and pattern of variations therein. As such no meaningful inference can be drawn, unless these data are reorganized to make them more useful. For example, if we are to ascertain a value around which most of the overtime hours cluster, such a value is difficult to obtain from the raw data.

Moreover, as the number of observations gets large, it becomes more and more difficult to focus on the specific features in a set of data. Thus, we need to organize the observation so that we can better understand the information that the data are revealing.

The raw data can be reorganized in a data array and frequency distribution. Such an arrangement enables us to see quickly some of the characteristics of the data we have collected.

When a raw data set is arranged in rank order, from the smallest to the largest observation or vice versa, the ordered sequence obtained is called an *ordered array*. Table 2.3 reorganizes data given in Table 2.2 in the ascending order.

Table 2.3 Ordered Array of Total Overtime Hours Worked by Machinists

84	84	85	85	87	88	88	88
88	89	89	89	90	90	91	91
92	92	93	93	93	93	93	93
94	94	94	94	94	95		

It may be observed that an ordered array does not summarize the data in any way as the number of observations in the array remains the same. However, a few advantages of ordered arrays are as follows:

Advantages

- (i) It provides a quick look at the highest and lowest observations in the data within which individual values vary.
- (ii) It helps in dividing the data into various sections or parts.
- (iii) It enables us to know the degree of concentration around a particular observation.
- (iv) It helps to identify whether any values appear more than once in the array.

Disadvantages

In spite of various advantages of converting a set of raw data into an ordered array, an array is a cumbersome form of presentation which is tiresome to construct. It neither summarizes nor organizes the data to present them in a more meaningful way. It also fails to highlight the salient characteristics of the data which may be crucial in terms of their relevance to decision making.

This task cannot be accomplished unless the observations are appropriately condensed. The best way to do so is to display them into a convenient number of groupings with the number of observations falling in different groups indicated against each. Such tabular summary presentation showing the number (frequency) of observations in each of several non-overlapping classes or groups is known as *frequency distribution* (also referred to as *grouped data*).

2.3.1 Frequency Distribution

A **frequency distribution** divides observations in the data set into conveniently established, numerically ordered classes (groups or categories). The number of observations in each class is referred to as *frequency* denoted as f .

Few examples of instances where frequency distributions would be useful are when (i) a marketing manager wants to know how many units (and what proportions or percentage) of each product sells in a particular region during a given period, (ii) a tax consultant desires to keep count of the number of times different size of firms are audited and (iii) a financial analyst wants to keep track of the number of times the shares of manufacturing and service companies to be or gain order a period of time.

Advantages

The following are a few advantages of grouping and summarizing raw data in this compact form

- (i) The data are expressed in a more compact form. One can get a deeper insight into the salient characteristics of the data at the very first glance.
- (ii) One can quickly note the pattern of distribution of observations falling in various classes.
- (iii) It permits the use of more complex statistical techniques which help to reveal certain other obscure and hidden characteristics of the data.

Disadvantages

A frequency distribution suffers from some disadvantages as stated below:

- (i) In the process of grouping, individual observations lose their identity. It becomes difficult to notice how the observations contained in each class are distributed. This applies more to a frequency distribution which uses the tally method in its construction.
- (ii) A serious limitation inherent in this kind of grouping is that there will be too much clustering of observations in various classes in case the number of classes is too small. This will cause some of the essential information to remain unexposed.

Hence, it is important that summarizing data should not be at the cost of losing essential details. The purpose should be to seek an appropriate compromise between having too much of details or very little. To be able to achieve this compromise, certain criteria are discussed for constructing a frequency distribution.

The frequency distribution of the number of hours of overtime given in Table 2.2 is shown in Table 2.4.

Table 2.4 Array and Tallies

<i>Number of Overtime Hours</i>	<i>Tally</i>	<i>Number of Weeks (Frequency)</i>
84		2
85		2
86	—	0
87		1
88		4
89		3
90		2
91		2
92		2
93		6
94		5
95		1
		30

Constructing a Frequency Distribution As the number of observations obtained gets larger, the method discussed above to condense the data becomes quite difficult and time consuming. Thus to further condense the data into frequency distribution tables, the following steps should be taken:

- (i) Select an appropriate number of non-overlapping class intervals.
- (ii) Determine the width of the class intervals.
- (iii) Determine class limits (or boundaries) for each class interval to avoid overlapping.

1. Decide the Number of Class Intervals The decision on the number of class groupings depends largely on the judgment of the individual investigator and/or the range that will be used to group the data, although there are certain guidelines that can be used. As a general rule, a frequency distribution should have at least five class intervals (groups), but not more than 15. The following two rules are often used to decide approximate number of classes in a frequency distribution:

- (i) If k represents the number of classes and N is the total number of observations, then the value of k will be the smallest exponent of the number 2, so that $2^k \geq N$.

In Table 2.3, we have $N = 30$ observations. If we apply this rule, then we shall have

$$2^3 = 8 (< 30); \quad 2^4 = 16 (< 30); \quad 2^5 = 32 (> 30)$$

Thus, we may choose $k = 5$ as the number of classes.

- (ii) According to Sturge's rule, the number of classes can be determined by the formula

$$k = 1 + 3.222 \log_e N$$

where k is the number of classes and $\log_e N$ is the logarithm of the total number of observations.

Applying this rule to the data given in Table 2.3, we get

$$k = 1 + 3.222 \log 30 = 1 + 3.222(1.4771) = 5.759 \cong 5$$

2. Determine the Width of Class Intervals When constructing the frequency distribution, it is desirable that the width of each class interval should be equal in size. The size (or width) of each class interval can be determined by first taking the difference between the largest and smallest numerical values in the data set and then dividing it by the number of class intervals desired.

$$\text{Width of class interval } (h) = \frac{\text{Largest numerical value} - \text{Smallest numerical value}}{\text{Number of classes desired}}$$

The value obtained from this formula can be rounded off to a more convenient value based on the investigator's preference.

From the ordered array in Table 2.3, the range is: $95 - 84 = 11$ hours. Using the above formula with desired five classes, the width of the class intervals is approximated as

$$\text{Width of class interval} = \frac{11}{5} = 2.2 \text{ hours}$$

For convenience, the selected width (or interval) of each class is rounded to 3 hours.

3. Determine Class Limits (Boundaries) The limits of each class interval should be clearly defined so that each observation (element) of the data set belongs to one and only one class.

Each class has two limits—a *lower limit* and an *upper limit*. The usual practice is to let the lower limit of the first class be a convenient number slightly below or equal to the lowest value in the data set. In Table 2.3, we may take the lower class limit of the first class as 82 and the upper class limit as 85. Thus, the class would be written as 82–85. This class interval includes all overtime hours ranging from 82 upto but not including 85 hours. The various other classes can be written as

<i>Overtime Hours (Class intervals)</i>	<i>Tallies</i>	<i>Frequency</i>
82 but less than 85		2
85 but less than 88		3
88 but less than 91		9
91 but less than 94		10
94 but less than 97		6
		30

Mid-point of Class Intervals The main advantage of using the above summary table is that the major data characteristics become clear to the decision maker. However, it is difficult to know how the individual values are distributed within a particular class interval without access to the original data. The *class mid-point* is the point halfway between the boundaries (both upper and lower class limits) of each class and is representative of all the observations contained in that class.

Arriving at the correct class mid-points is important, for these are used as representative of all the observations contained in their respective class while computing many important statistical measures. A mid-point is obtained by dividing the sum of the upper and lower class limits by two. Problems in computing mid-points arise when the class limits are ambiguous and not clearly defined.

The width of the class interval should, as far as possible, be equal for all the classes. If this is not possible to maintain, the interpretation of the distribution becomes difficult. For example, it will be difficult to say whether the difference between the frequencies of the two classes is due to the difference in the concentration of observations in the two classes or due to the width of the class intervals being different.

The width of the class intervals should preferably be not only the same throughout, but should also be a convenient number such as 5, 10 or 15. A width given by integers 7, 13 or 19 should be avoided.

2.3.2 Methods of Data Classification

There are two ways in which observations in the data set are classified on the basis of class intervals, namely

- (i) Exclusive method
- (ii) Inclusive method

Exclusive Method When the data are classified in such a way that the upper limit of a class interval is the lower limit of the succeeding class interval (i.e. no data point falls into more than one class interval), then it is said to be the exclusive method of classifying data. This method is illustrated in Table 2.5.

Table 2.5 Exclusive Method of Data Classification

<i>Dividends Declared in percent (Class Intervals)</i>	<i>Number of Companies (Frequencies)</i>
0–10	5
10–20	7
20–30	15
30–40	10

Such classification ensures continuity of data because the upper limit of one class is the lower limit of succeeding class. As shown in Table 2.5, five companies declared dividends ranging from 0 to 10 percent, this means a company which declared exactly 10 percent dividend would not be included in the class 0–10 but would be included in the next class 10–20. Since this point is not always clear, therefore to avoid confusion data are displayed in a slightly different manner as given in Table 2.6.

Table 2.6 Exclusive Method of Data Classification (Different Method)

<i>Dividends Declared in percent (Class Intervals)</i>	<i>Number of Companies (Frequencies)</i>
0 but less than 10	5
10 but less than 20	7
20 but less than 30	15
30 but less than 40	10

Inclusive Method When the data are classified in such a way that both lower and upper limits of a class interval are included in the interval itself, then it is said to be the inclusive method of classifying data. This method is shown in Table 2.7.

Table 2.7 Inclusive Method of Data Classification

<i>Number of Accidents (Class Intervals)</i>	<i>Number of Weeks (Frequencies)</i>
0–4	5
5–9	22
10–14	13
15–19	8
20–24	2

Remarks: 1. An exclusive method should be used to classify a set of data involving continuous variables and an inclusive method should be used to classify a set of data involving discrete variables.

2. If a continuous variable is classified according to the inclusive method, then certain adjustment in the class interval is needed to obtain continuity as shown in Table 2.8.

Table 2.8 Data Classification

<i>Class Intervals</i>	<i>Frequency</i>
30 – 44	28
45 – 59	32
60 – 74	45
75 – 89	50
90 – 104	35

To ensure continuity, first calculate correction factor as

$$x = \frac{\text{Upper limit of a class} - \text{Lower limit of the next higher class}}{2}$$

and then subtract it from the lower limits of all the classes and add it to the upper limits of all the classes.

From Table 2.8, we have $x = (45 - 44) \div 2 = 0.5$. Subtract 0.5 from the lower limits of all the classes and add 0.5 to the upper limits. The adjusted classes would then be as shown in Table 2.9.

Table 2.9 Inclusive Method of Data Classification (Different format)

<i>Class Intervals</i>	<i>Frequency</i>
29.5 – 44.5	28
44.5 – 59.5	32
59.5 – 74.5	45
74.5 – 89.5	50
89.5 – 104.5	35

3. Class intervals should be of equal size to make meaningful comparison between classes. In a few cases, extreme values in the data set may require the inclusion of *open-ended classes* and

this distribution is known as an *open-ended distribution*. Such open-ended classes do not pose any problem in data analysis as long as only a few frequencies (or values) lie in these classes. However, an open-ended distribution is not fit for further mathematical calculations because *mid-value* which is used to represent the class cannot be determined for an open-ended class. An example of an open-ended distribution is given in Table 2.10.

Table 2.10 Open-ended Distribution

<i>Age (Years)</i>	<i>Population (Millions)</i>
Under 5	17.8
5–17	44.7
18–24	29.9
25–44	69.6
45–64	44.6
65 and above	27.4
	<u>234.0</u>

Table 2.11 provides a tentative guide to determine an adequate number of classes.

Table 2.11 Guide to Determine the Number of Classes to Use

<i>Number of Observations, N</i>	<i>Suggested Number of Classes</i>
20	5
50	7
100	8
200	9
500	10
1000	11

Example 2.1: The following set of numbers represents mutual fund prices reported at the end of a week for selected 40 nationally sold funds.

10	17	15	22	11	16	19	24	29	18
25	26	32	14	17	20	23	27	30	12
15	18	24	36	18	15	21	28	33	38
34	13	10	16	20	22	29	29	23	31

Arrange these prices into a frequency distribution having a suitable number of classes.

Solution: Since the number of observations are 40, it seems reasonable to choose 6 ($2^6 > 40$) class intervals to summarize values in the data set. Again, since the smallest value is 10 and the largest is 38, therefore the class interval is given by

$$h = \frac{\text{Range}}{\text{Number of classes}} = \frac{38 - 10}{6} = \frac{28}{6} = 4.66 \sim 5$$

Now performing the actual tally and counting the number of values in each class, we get the frequency distribution by exclusive method as shown in Table 2.12:

Table 2.12 Frequency Distribution

<i>Class Interval (Mutual Fund Prices, ₹)</i>	<i>Tally</i>	<i>Frequency (Number of Mutual Funds)</i>
10 – 15		6
15 – 20		11
20 – 25		9
25 – 30		7
30 – 35		5
35 – 40		2
		40

Example 2.2: The take-home salary (in ₹) of 40 unskilled workers from a company for a particular month was.

2482	2392	2499	2412	2440	2444
2446	2540	2394	2365	2412	2458
2482	2394	2450	2444	2440	2494
2460	2425	2500	2390	2414	2365
2390	2460	2422	2500	2470	2428

Construct a frequency distribution having a suitable number of classes.

Solution: Since the number of observations are 30, we choose $5(2^5 > 30)$ class intervals to summarize values in the data set. In the data set, the smallest value is 2365 and the largest is 2500, so the width of each class interval will be

$$h = \frac{\text{Range}}{\text{Number of classes}} = \frac{2540 - 2365}{5} = \frac{175}{5} = 35$$

Sorting the data values into classes and counting the number of values in each class, we get the frequency distribution by exclusive method as shown in Table 2.13.

Table 2.13 Frequency Distribution

<i>Class Interval (Salary, ₹)</i>	<i>Tally</i>	<i>Frequency (Number of Workers)</i>
2365 – 2400		6
2400 – 2435		7
2435 – 2470		10
2470 – 2505		6
2505 – 2540		1
		30

Example 2.3: A computer company received a rush order for as many home computers as could be shipped during a six-week period. Company records provide the following daily shipments:

22	65	65	67	55	50	65
77	73	30	62	54	48	65
79	60	63	45	51	68	79
83	33	41	49	28	55	61
65	75	55	75	39	87	45
50	66	65	59	25	35	53

Group these daily shipments figures into a frequency distribution having the suitable number of classes.

Solution: Since the number of observations are 42, it seems reasonable to choose 6 ($2^6 > 42$) classes. Again, since the smallest value is 22 and the largest is 87, therefore the class interval is given by

$$h = \frac{\text{Range}}{\text{Number of classes}} = \frac{87 - 22}{6} = \frac{65}{6} = 10.833 \text{ or } 11$$

Now performing the actual tally and counting the number of values in each class, we get the following frequency distribution by inclusive method as shown in Table 2.14.

Table 2.14 Frequency Distribution

<i>Class Interval (Number of Computers)</i>	<i>Tally</i>	<i>Frequency (Number of Days)</i>
22–32		4
33–43		4
44–54		9
55–65		14
66–76		6
77–87		5
		<u>42</u>

Example 2.4: Following is the increase of D.A. in the salaries of employees of a firm at the following rates:

₹ 250 for the salary range up to ₹ 4749

₹ 260 for the salary range from ₹ 4750

₹ 270 for the salary range from ₹ 4950

₹ 280 for the salary range from ₹ 5150

₹ 290 for the salary range from ₹ 5350

No increase of D.A. for salary of ₹ 5500 or more. What will be the additional amount required to be paid by the firm in a year which has 32 employees with the following salaries (in ₹)?

5422	4714	5182	5342	4835	4719	5234	5035
5085	5482	4673	5335	4888	4769	5092	4735
5542	5058	4730	4930	4978	4822	4686	4730
5429	5545	5345	5250	5375	5542	5585	4749

Solution: Performing the actual tally and counting the number of employees in each salary range (or class), we get the following frequency distribution as shown in Table 2.15.

Table 2.15 Frequency Distribution

<i>Class Interval (Pay Range)</i>	<i>Tally</i>	<i>Frequency, f (Number of Employees)</i>	<i>Rate of D.A. (₹ x)</i>	<i>Total Amount Paid (₹ fx)</i>
upto 4749		8	250	2000
4750–4949		5	260	1300
4950–5149		5	270	1350
5150–5349		6	280	1680
5350–5549		8	290	2320
		<u>32</u>		<u>8650</u>

Hence additional amount required by the firm for payment of D.A. is ₹ 8650.

Example 2.5: Following are the number of items of similar type produced in a factory during the last 50 days.

21	22	17	23	27	15	16	22	15	23
24	25	36	19	14	21	24	25	14	18
20	31	22	19	18	20	21	20	36	18
21	20	31	22	19	18	20	20	24	35
25	26	19	32	22	26	25	26	27	22

Arrange these observations into a frequency distribution with both inclusive and exclusive class intervals choosing a suitable number of classes.

Solution: Since the number of observations are 50, it seems reasonable to choose 6 ($2^6 > 50$) or less classes. Since smallest value is 14, and the largest is 36 therefore the class interval is given by

$$h = \frac{\text{Range}}{\text{Number of classes}} = \frac{36 - 14}{6} = \frac{22}{6} = 3.66 \text{ or } 4$$

Performing the actual tally and counting the number of observations in each class, we get the following frequency distribution with inclusive class intervals as shown in Table 2.16.

Table 2.16 Frequency Distribution With Inclusive Class Intervals

<i>Class Intervals</i>	<i>Tally</i>	<i>Frequency (Number of Items Produced)</i>
14–17		6
18–21		18
22–25		15
26–29		5
30–33		3
34–37		3
		<u>50</u>

Converting the class intervals shown in Table 2.16 into exclusive class intervals is shown in Table 2.17.

Table 2.17 Frequency Distribution With Exclusive Class Intervals

<i>Class Intervals</i>	<i>Mid-Value of Class Intervals</i>	<i>Frequency (Number of Items Produced)</i>
13.5–17.5	15.5	6
17.5–21.5	19.5	18
21.5–25.5	23.5	15
25.5–29.5	27.5	5
29.5–33.5	31.5	3
33.5–37.5	35.5	3

2.3.3 Bivariate Frequency Distribution

The frequency distributions discussed so far involved only one variable and therefore, called *univariate frequency distributions*. In case the data involve two variables (such as profit and expenditure on advertisements of a group of companies, income and expenditure of a group of individuals, supply and demand of a commodity etc.), then frequency distribution so obtained as a result of cross classification is called *bivariate frequency distribution*. It can be summarized in the form of a *two-way (bivariate) frequency table* and the values of each variable are grouped into various classes (not necessarily same for each variable) in the same way as for univariate distributions.

If the data corresponding to one variable, say x , is grouped into m classes and the data corresponding to another variable, say y , is grouped into n classes, then bivariate frequency table will have $m \times n$ cells.

Frequency distribution of variable x for a given value of y is obtained by the values of x and vice versa. Such frequencies in each cell are called *conditional frequencies*. The frequencies of the values of variables x and y together with their frequency totals are called the *marginal frequencies*.

Example 2.6: The following figures indicate income (x) and percentage expenditure on food (y) of 25 families. Construct a bivariate frequency table classifying x into intervals 200 – 300, 300 – 400, ... and y into 10 – 15, 15 – 20, ...

Write the marginal distribution of x and y and the conditional distribution of x when y lies between 15 and 20.

x	y	x	y	x	y	x	y	x	y
550	12	225	25	680	13	202	29	689	11
623	14	310	26	300	25	255	27	523	12
310	18	640	20	425	16	492	18	317	18
420	16	512	18	555	15	587	21	384	17
600	15	690	12	325	23	643	19	400	19

Solution: The two-way frequency table showing income (in ₹) and percentage expenditure on food is shown in Table 2.18.

Table 2.18 Two-way Frequency

<i>Expenditure (y)</i> (Percentage)	<i>Income (x)</i>					<i>Marginal</i> <i>Frequencies, f_y</i>
	200–300	300–400	400–500	500–600	600–700	
10–15				(2)	(4)	6
15–20		(3)	(4)	(2)	(2)	11
20–25		(1)		(1)	(1)	3
25–30	(3)	(2)				5
<i>Marginal</i> <i>Frequencies, f_x</i>	3	6	4	5	7	25

The conditional distribution of x when y lies between 15 and 20 per cent is as follows:

Income (x) :	200–300	300–400	400–500	500–600	600–700
15%–20% :	0	3	4	2	2

Example 2.7: The following data give the points scored in a tennis match by two players X and Y at the end of 20 games:

(10, 12) (7, 11) (7, 9) (15, 19) (17, 21) (12, 8) (16, 10) (14, 14) (22, 18) (16, 7)
 (15, 16) (22, 20) (19, 15) (7, 18) (11, 11) (12, 18) (10, 10) (5, 13) (11, 7) (10, 10)

Taking class intervals as: 5–9, 10–14, 15–19 ..., for both X and Y, construct

- Bivariate frequency table.
- Conditional frequency distribution for Y given $X > 15$.

Solution: (i) The two-way frequency distribution is shown in Table 2.19.

Table 2.19 Bivariate Frequency Table

<i>Player Y</i>	<i>Player</i>				<i>Marginal</i> <i>Frequencies, f_y</i>
	5–9	10–14	15–19	20–24	
5–9	(1)	(2)	(1)	—	4
10–14	(2)	(5)	(1)	—	8
15–19	(1)	(1)	(3)	(1)	6
20–24	—	—	(1)	(1)	2
<i>Marginal</i> <i>Frequencies, f_x</i>	4	8	6	2	20

- Conditional frequency distribution for Y given $X > 15$.

<i>Player Y</i>	<i>Player X</i>	
	15–19	20–24
5–9	1	—
10–14	1	—
15–19	3	1
20–24	1	1
	<u>6</u>	<u>2</u>

2.3.4 Types of Frequency Distributions

Cumulative Frequency Distribution Sometimes it is preferable to present data in a **cumulative frequency (cf) distribution** or simply a distribution which shows the cumulative number of observations below the upper boundary (limit) of each class in the given frequency distribution. A cumulative frequency distribution is of two types: (i) *more than* type and (ii) *less than* type.

In a *less than* cumulative frequency distribution, the frequencies of each class interval are added successively from top to bottom and represent the cumulative number of observations less than or equal to the class frequency to which it relates. But in the *more than* cumulative frequency distribution, the frequencies of each class interval are added successively from bottom to top and represent the cumulative number of observations greater than or equal to the class frequency to which it relates.

The frequency distribution given in Table 2.20 illustrates the concept of cumulative frequency distribution.

Table 2.20 Cumulative Frequency Distribution

<i>Number of Accidents</i>	<i>Number of Weeks (Frequency)</i>	<i>Cumulative Frequency (Less than)</i>	<i>Cumulative Frequency (More than)</i>
0–4	5	5	50
5–9	22	$5 + 22 = 27$	$50 - 5 = 45$
10–14	13	$27 + 13 = 40$	$45 - 22 = 23$
15–19	8	$40 + 8 = 48$	$23 - 13 = 10$
20–24	2	$48 + 2 = 50$	$10 - 8 = 2$
	50		

From Table 2.20, it may be noted that cumulative frequencies are corresponding to the lower limit and upper limit of class intervals. The ‘less than’ cumulative frequencies are corresponding to the upper limit of class intervals and ‘more than’ cumulative frequencies are corresponding to the lower limit of class intervals shown in Table 2.21(a) and (b).

Table 2.21(a)

<i>Upper Limits</i>	<i>Cumulative Frequency (Less than)</i>
less than 4	5
less than 9	27
less than 14	40
less than 19	48
less than 24	50

Table 2.21(b)

<i>Lower Limits</i>	<i>Cumulative Frequency (More than)</i>
0 and more	50
5 and more	45
10 and more	23
15 and more	10
20 and more	2

Relative Frequency Distribution To enrich data analysis, it is sometimes important to show what percentage of observations fall within each class of a distribution instead of showing the actual class frequencies. To convert a frequency distribution into a corresponding **relative frequency distribution**, we divide each class frequency by the total number of observations in the entire distribution. Each relative frequency is thus a proportion as shown in Table 2.22.

Table 2.22 Relative and Percentage Frequency Distributions

<i>Number of Accidents</i>	<i>Number of Weeks (Frequency)</i>	<i>Relative Frequency</i>	<i>Percentage Frequency</i>
0–4	5	$\frac{5}{50} = 0.10$	$\frac{5}{50} \times 100 = 10$
5–9	22	$\frac{22}{50} = 0.44$	$\frac{22}{50} \times 100 = 44$
10–14	13	$\frac{13}{50} = 0.26$	$\frac{13}{50} \times 100 = 26$
15–19	8	$\frac{8}{50} = 0.16$	$\frac{8}{50} \times 100 = 16$
20–24	2	$\frac{2}{50} = 0.04$	$\frac{2}{50} \times 100 = 4$
	<u>50</u>	<u>1.00</u>	<u>100</u>

Percentage Frequency Distribution A **percentage frequency distribution** is one in which the number of observations for each class interval is converted into a percentage frequency by dividing it by the total number of observations in the entire distribution. The quotient so obtained is then multiplied by 100 as shown in Table 2.22.

Example 2.8: Following are the number of two wheelers sold by a dealer during eight weeks of six working days each.

13	19	22	14	13	16	19	21
23	11	27	25	17	17	13	20
23	17	26	20	24	15	20	21
23	17	29	17	19	14	20	20
10	22	18	25	16	23	19	20
21	17	18	24	21	20	19	26

- (a) Group these figures into a table having the classes 10–12, 13–15, 16–18, ... and 28–30.
 (b) Convert the distribution of part (i) into a corresponding percentage frequency distribution and (ii) also a percentage cumulative frequency distribution.

Solution: (a) Frequency distribution of the given data is shown in Table 2.23.

Table 2.23 Frequency Distribution

<i>Number of Automobiles Sold (Class Intervals)</i>	<i>Tally</i>	<i>Number of Days (Frequency)</i>
10–12		2
13–15		6
16–18		10
19–21		16
22–24		8
25–27		5
28–30		1
		48

(b) Percentage and more than Cumulative Percentage Distribution (Table 2.24).

Table 2.24 Percentage Cumulative Frequency Distribution

<i>Number of Automobiles Sold (Class Intervals)</i>	<i>Number of Days (Frequency)</i>	<i>Cumulative Frequency</i>	<i>Percentage Frequency</i>	<i>Percentage Cumulative Frequency</i>
10–12	2	2	4.17	4.17
13–15	6	8	12.50	16.67
16–18	10	18	20.83	37.50
19–21	16	34	33.34	70.84
22–24	8	42	16.67	87.51
25–27	5	47	10.41	97.92
28–30	1	48	2.08	100.00
	48		100.00	

Conceptual Questions 2A

1. Explain the characteristics of a frequency distribution.
2. Illustrate two methods of classifying data in class intervals.
3. What are the advantages of using a frequency distribution to describe a body of raw data? What are the disadvantages?
4. When constructing a grouped frequency distribution, should equal intervals always be used? Under what circumstances should unequal intervals be used instead?
5. What are the advantages and disadvantages of using open-end intervals when constructing a group frequency distribution?

6. When constructing a group frequency distribution, is it necessary that the resulting distribution be symmetric? Explain.
7. Why is it necessary to summarize data? Explain the approaches available to summarize data distributions.
8. What are the objections to unequal class and open class intervals? State the conditions under which the use of unequal class intervals and open class intervals are desirable and necessary.
9. (a) What do you understand by cumulative frequency distribution?
(b) What do you understand by bivariate or two-way frequency distribution?

Self-practice Problems 2A

- 2.1 A portfolio contains 51 stocks whose prices are given below:

67 34 36 48 49 31 61 34
43 45 38 32 27 61 29 47
36 50 46 30 40 32 30 33
45 49 48 41 53 36 37 47
47 30 50 28 35 35 38 36
46 43 34 62 69 50 28 44
43 60 39

Summarize these stock prices in the form of a frequency distribution.

- 2.2 Construct a frequency distribution of the data given below, where class interval is 4 and the mid-value of one of the classes is zero.

-8 -7 10 12 6 4 3 0 7
-4 -3 -2 2 3 4 7 5 6
10 12 9 13 11 -10 -7 1 0
53 2 6 10 -6 -4

- 2.3 Form a frequency distribution of the following data. Use an equal class interval of 4 where the lower limit of the first class is 10.

10 17 15 22 11 16 19 24 29
18 25 26 32 14 17 20 23 27
30 12 15 18 24 36 18 15 21
28 33 38 34 13 10 16 20 22
29 29 23 31

- 2.4 If class mid-points in a frequency distribution of the ages of a group of persons are: 25, 32, 39, 46, 53 and 60, find:

- (a) the size of the class-interval
- (b) the class boundaries
- (c) the class limits, assuming that the age quoted is the age completed on the last birthdays

- 2.5 The distribution of ages of 500 readers of a nationally distributed magazine is given below:

<i>Age (in Years)</i>	<i>Number of Readers</i>
Below 14	20
15-19	125
20-24	25
25-29	35
30-34	80
35-39	140
40-44	30
45 and above	45

Find the relative and cumulative frequency distributions for this distribution.

- 2.6 The distribution of inventory to sales ratio of 200 retail outlets is given below:

<i>Inventory to Sales Ratio</i>	<i>Number of Retail Outlets</i>
1.0-1.2	20
1.2-1.4	30
1.4-1.6	60
1.6-1.8	40
1.8-2.0	30
2.0-2.2	15
2.2-2.4	5

Find the relative and cumulative frequency distributions for this distribution.

- 2.7 A wholesaler's daily shipments of a particular item varied from 1152 to 9888 units per day. Indicate the limits of nine classes into which these shipments might be grouped.

2.8 The class marks of distribution of the number of electric light bulbs replaced daily in an office building are 5, 10, 15 and 20. Find (a) the class boundaries and (b) class limits.

2.9 Classify the following data by taking class intervals such that their mid-values are 17, 22, 27, 32 and so on:

30 42 30 54 40 48 15 17 51
 42 25 41 30 27 42 36 28 26
 37 54 44 31 36 40 36 22 30
 31 19 48 16 42 32 21 22 46
 33 41 21

2.10 In degree colleges of a city, no teacher is less than 30 years or more than 60 years in age. Their cumulative frequencies are as follows:

Less than	:	60	55	50	45
		40	35	30	25
Total frequency	:	980	925	810	675
		535	380	220	75

Find the frequencies in the class intervals 25–30, 30–35, ...

Hints and Answers

2.3 The classes for preparing frequency distribution by inclusive method will be 10–13, 14–17, 18–21, ..., 34–37, 38–41

2.4 (a) Size of the class interval = Difference between the mid-values of any two consecutive classes = 7

(b) The class boundaries for different classes are obtained by adding (for upper class boundaries or limits) and subtracting (for lower class boundaries or limits) half the magnitude of the class interval, that is, $7 \div 2 = 3.5$ from the mid-values.

Class Intervals : 21.5–28.5 28.5–35.5 35.5–42.5

Mid-Values : 25 32 39

Class Intervals : 42.5–49.5 49.5–56.5 56.5–63.5

Mid-Values : 46 53 60

(c) The distribution can be expressed in inclusive class intervals with width of 7 as 22–28, 29–35, ..., 57–63.

2.7 One possibility is 1000–1999, 2000–2999, 3000–3999, ..., 9000–9999 units of the item.

2.10	Age (year)	Cumulative Frequency	Age	Frequency
	Less than 25	75	20–25	75
	Less than 30	220	25–30	$220 - 75 = 145$
	Less than 35	380	30–35	$380 - 220 = 160$
	Less than 40	535	35–40	$535 - 380 = 155$
	Less than 45	675	40–45	$675 - 535 = 140$
	Less than 50	810	45–50	$810 - 675 = 135$
	Less than 55	925	50–55	$925 - 810 = 115$
	Less than 60	980	55–60	$980 - 925 = 55$

2.4 GRAPHICAL PRESENTATION OF DATA

It has already been discussed that one of the important functions of statistics is to present complex and unorganized (raw) data in such a manner that they would easily be understandable. According to King—‘*One of the chief aims of statistical science is to render the meaning of masses of figures clear and comprehensible at a glance.*’ This is often best accomplished by presenting the data in a pictorial (or graphical) form.

The graphical (diagrammatical) presentation of data has many advantages. The following persons rightly observed that

- *With but few exceptions, memory depends upon the faculty of our brains possess of forming visual images and it is this power of forming visual images which lies at the root of the utility of diagrammatic presentation.*

—R. L. A. Holmes

- *Cold figures are uninspiring to most people. Diagrams help us to see the pattern and shape of any complex situation. Just as a map gives us a bird’s eye-view of the wide stretch of a country, so diagrams help as visualise the whole meaning of the numerical complex at a single glance.*

—M. J. Moroney

According to Calvin F. Schmid—‘*Charts and graphs represent an extremely useful and flexible medium for explaining, interpreting and analysing numerical facts largely by means of points, lines, areas and other geometric forms and symbols. They make possible the presentation of quantitative data in a simple, clear, and effective manner and facilitate comparison of values, trends and relationships.*’

2.4.1 Functions of a Graph

Graphic presentation of frequency distributions facilitate easy understanding of data presentation and interpretation. The shape of the graph offers easy answers to several questions. The same information can also be obtained from tabular presentation of a frequency distribution, but the same is not as effective in highlighting the essential characteristics as explicitly as is possible in the case of graphic presentation.

The shape of the graph gives an exact idea of the variations of the distribution trends. Graphic presentation, therefore, serves as an easy technique for quick and effective comparison between two or more frequency distributions. When the graph of one frequency distribution is superimposed on the other, the points of contrast regarding the type of distribution and the pattern of variation become quite obvious. All these advantages necessitate a clear understanding of the various forms of graphic representation of a frequency distribution.

According to P. Maslov—‘*Diagrams are drawn for two purposes (i) to permit the investigator to graph the essence of the phenomenon he is observing and (ii) to permit others to see the results at a glance, i.e. for the purpose of popularisation.*’

Advantages

- Diagrams give an attractive and elegant presentation:* Diagrams have greater attraction and effective impression. People, in general, avoid figures, but are always impressed by diagrams. Since people see pictures carefully, their effect on the mind is more stable. Thus, diagrams give delight to the eye and add the spark of interest.

- (ii) *Diagrams leave good visual impact:* Diagrams have the merit of rendering any idea readily. The impression created by a diagram is likely to last longer in the minds of people than the effect created by figures. Thus, diagrams have greater memorizing value than figures.
- (iii) *Diagrams facilitate comparison:* With the help of diagrams, comparisons of groups and series of figures can be made easily. While comparing absolute figures, the significance is not clear but when these are presented by diagrams, the comparison is easy. The technique of diagrammatic representation should not be used when comparison is either not possible or is not necessary.
- (iv) *Diagrams save time:* Diagrams present the set of data in such a way that their significance is known without loss of much time. Moreover, diagrams save time and effort which are otherwise needed in drawing inferences from a set of figures.
- (v) *Diagrams simplify complexity and depict the characteristics of the data:* Diagrams, besides being attractive and interesting, also highlight the characteristics of the data. Large data can easily be represented by diagrams and thus, without straining one's mind, the basic features of the data can be understood and inferences can be drawn in a very short time.

Limitations We often find tabular and graphical presentations of data in annual reports, newspapers, magazines, bulletins, and so on. But inspite their usefulness, diagrams can also be misused. A few limitations of these as a tool for statistical analysis are as under:

- (i) They provide only an approximate picture of the data.
- (ii) They cannot be used as alternative to tabulation of data.
- (iii) They can be used only for comparative study.
- (iv) They are capable of representing only homogeneous and comparable data.

2.4.2 General Rules For Drawing Diagrams

To draw useful inferences from graphical presentation of data, it is important to understand how they are prepared and how they should be interpreted. When we say that 'one picture is worth a thousand words', it neither proves (nor disproves) a particular fact, nor is it suitable for further analysis of data. However, if diagrams are properly drawn, they highlight the different characteristics of data. The following general guidelines are taken into consideration while preparing diagrams.

Title: Each diagram should have a suitable title. It may be given either at the top of the diagram or below it. The title must convey the main theme which the diagram intends to portray.

Size: The size and portion of each component of a diagram should be such that all the relevant characteristics of the data are properly displayed and can be easily understood.

Proportion of length and breadth: An appropriate proportion between the length and breadth of the diagram should be maintained. As such there are no fixed rules about the ratio of length to width. However, a ratio of $\sqrt{2} : 1$ or 1.414 (long side) : 1 (short side) suggested by Lutz in his book *Graphic Presentation* may be adopted as a general rule.

Proper scale: There are again no fixed rules for selection of scale. The diagram should neither be too small nor too large. The scale for the diagram should be decided after taking into consideration the magnitude of data and the size of the paper on which it is to be drawn. The scale showing the values as far as possible should be in even numbers or in multiples of 5, 10, 20 and so on. The scale should specify the size of the unit and the nature of data it represents, for example, 'millions

of tonnes', in ₹ thousand, and so on. The scale adopted should be indicated on both vertical and horizontal axes if different scales are used. Otherwise, it can be indicated at some suitable place on the graph paper.

Footnotes and Source note: To clarify or elucidate any points which need further explanation but cannot be shown in the graph, footnotes are given at the bottom of the diagrams.

Index: A brief index explaining the different types of lines, shades, designs or colours used in the construction of the diagram should be given to understand its contents.

Simplicity: Diagrams should be prepared in such a way that they can be understood easily. To keep it simple, too much information should not be loaded in a single diagram as it may create confusion. Thus if the data are large, then it is advisable to prepare more than one diagram, each depicting some identified characteristic of the same data.

2.5 TYPES OF DIAGRAMS

There are a variety of diagrams used to represent statistical data. Different types of diagrams, used to describe sets of data, are divided into the following categories:

- **Dimensional diagrams**
 - (i) One dimensional diagrams such as histograms, frequency polygons and pie charts.
 - (ii) Two-dimensional diagrams such as rectangles, squares and circles.
 - (iii) Three dimensional diagrams such as cylinders and cubes.
- **Pictograms or Ideographs**
- **Cartographs or Statistical maps**

2.5.1 One-dimensional Diagrams

These diagrams are most useful, simple and popular in the diagrammatic presentation of frequency distributions. These diagrams provide a useful and quick understanding of the *shape* of the distribution and its characteristics. According to Calvin F. Schmid—*'The simple bar chart with many variations is particularly appropriate for comparing the magnitude (or size) of coordinate items or of parts of a total. The basis of comparison in the bar is linear or one-dimensional.'*

These diagrams are called one-dimensional diagrams because only the length (height) of the bar (not the width) is taken into consideration. Of course, width or thickness of the bar has no effect on the diagram, even then the thickness should not be too much otherwise the diagram would appear like a two-dimensional diagram.

Tips for Constructing a Diagram

- (i) The width of all the bars drawn should be same.
- (ii) The gap between one bar and another must be uniform.
- (iii) There should be a common base to all the bars.
- (iv) It is desirable to write the value of the variable represented by the bar at the top end so that the reader can understand the value without looking at the scale.
- (v) The frequency, relative frequency or percent frequency of each class interval is shown by drawing a rectangle whose base is the class interval on the horizontal axis and whose height is the corresponding frequency, relative frequency or percent frequency.

- (vi) The value of variables (or class boundaries in case of grouped data) under study are scaled along the horizontal axis, and the number of observations (frequencies, relative frequencies or percentage frequencies) are scaled along the vertical axis.

The one-dimensional diagrams (charts) used for graphical presentation of data sets are as follows:

- Histogram
- Frequency polygon
- Frequency curve
- Cumulative frequency distribution (Ogive)
- Pie diagram

Histograms (Bar Diagrams) These diagrams are used to graph both ungrouped and grouped data. In the case of an ungrouped data, values of the variable (the characteristic to be measured) are scaled along the horizontal axis and the number of observations (or frequencies) along the vertical axis of the graph. The plotted points are then connected by straight lines to enhance the shape of the distribution. The height of such boxes (rectangles) measures the number of observations in each of the classes.

Listed below are the various types of histograms:

- | | |
|-----------------------------------|-------------------------------------|
| (i) Simple bar charts | (v) Paired bar charts |
| (ii) Grouped (or multiple) charts | (vi) Sliding bar charts |
| (iii) Deviation bar charts | (vii) Relative frequency bar charts |
| (iv) Subdivided bar charts | (viii) Percentage bar charts |

For plotting a histogram of a grouped frequency distribution, the end points of class intervals are specified on the horizontal axis and the number of observations (or frequencies) are specified on the vertical axis of the graph. Often class mid-points are posted on the horizontal axis rather than the end points of class intervals. In either case, the width of each bar indicates the class interval while the height indicates the frequency of observations in that class. Figure 2.1 is a histogram for the frequency distribution given in Table 2.12 of Example 2.1.

Remarks: Bar diagrams are not suitable to represent long period time series.

Simple Bar Charts The graphic techniques described earlier are used for group frequency distributions. The graphic techniques presented in this section can also be used for displaying values of categorical variables. Such data is first tallied into summary tables and then graphically displayed as either *bar charts* or *pie charts*.

Bar charts are used to represent only one characteristic of data and there will be as many bars as number of observations. For example, the data obtained on the production of oil seeds in a particular year can be represented by such bars. Each bar would represent the yield of a particular oil seed in that year. Since the bars are of the same width and only the length varies, the relationship among them can be easily established.

Sometimes only lines are drawn for comparison of given variable values. Such lines are not thick and their number is sufficiently large. The different measurements to be shown should not have too much difference, so that the lines may not show too much dissimilarity in their heights.

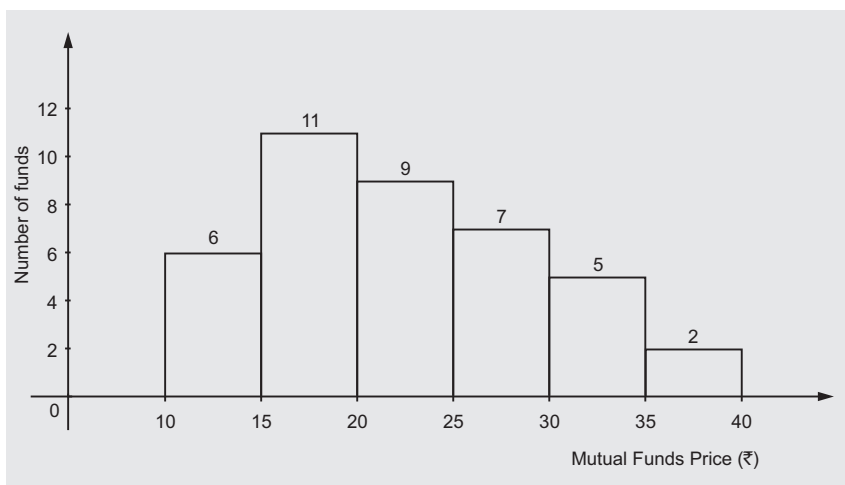


Figure 2.1 Histogram for Mutual Funds

Such charts are used to economize space, specially when observations are large. The lines may be either vertical or horizontal depending upon the type of variable—numerical or categorical.

Example 2.9: The data on the production of oil seeds in a particular year is presented in Table 2.25.

Table 2.25 Production of Oil Seeds

<i>Oil Seed</i>	<i>Yield (Million tonnes)</i>	<i>Percentage Production (Million tonnes)</i>
Ground nut	5.80	43.03
Rapeseed	3.30	24.48
Coconut	1.18	8.75
Cotton	2.20	16.32
Soyabean	1.00	7.42
	<u>13.48</u>	<u>100.00</u>

Represent this data by a suitable bar chart.

Solution: The information provided in Table 2.25 is expressed graphically as the frequency bar chart as shown in Fig. 2.2. In this figure, each type of seed is depicted by a bar, the length of which represents the frequency (or percentage) of observations falling into that category.

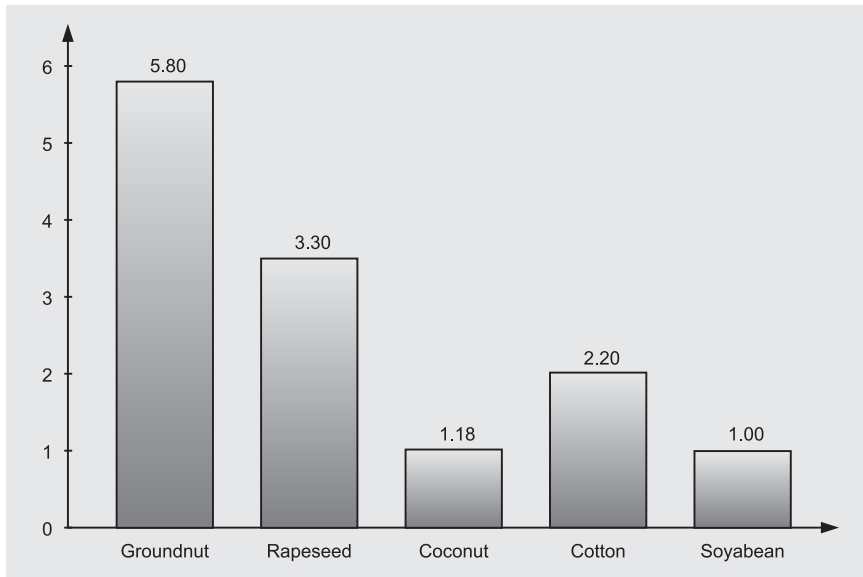


Figure 2.2 Bar Chart Pertaining to Production of Oil Seeds

Remark: The bars should be constructed vertically (as shown in Fig. 2.2) when categorized observations are the outcome of a numerical variable. But if observations are the outcome of a categorical variable, then the bars should be constructed horizontally.

Example 2.10: An advertising company kept an account of response letters received each day over a period of 50 days. The observations were:

0	2	1	1	1	2	0	0	1	0	1	0	0	1	0	1	1	0
2	0	0	2	0	1	0	1	0	1	0	3	1	0	1	0	1	0
2	5	1	2	0	0	0	0	5	0	1	1	2	0				

Construct a frequency table and draw a line chart (or diagram) to present the data.

Solution: The observations are tallied into the summary table as shown in Table 2.26.

Figure 2.3 depicts a frequency bar chart for the number of letters received during a period of 50 days presented in Table 2.26.

Table 2.26 Frequency Distribution of Letters Received

Number of Letters Received	Tally	Number of Days (Frequency)
0		23
1		17
2		7
3		2
4	—	0
5		1
		50

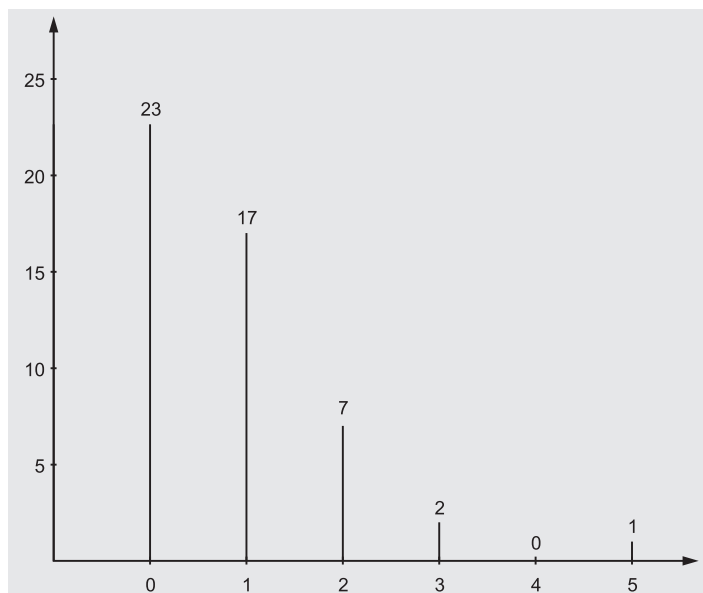


Figure 2.3 Number of Letters Received

Multiple Bar Charts A multiple bar chart is also known as grouped (or compound) bar chart. Such charts are useful for direct comparison between two or more sets of data. The technique of drawing such a chart is same as that of a single bar chart with a difference that each set of data is represented in different shades or colours on the same scale. An index explaining shades or colours must be given.

Example 2.11: The data on fund flow (in ₹ crore) of an International Airport Authority during financial years 2001–02 to 2003–04 are given below:

	2001–02	2002–03	2003–04
Non-traffic revenue	40.00	50.75	70.25
Traffic revenue	70.25	80.75	110.00
Profit before tax	40.15	50.50	80.25

Represent this data by a suitable bar chart.

Solution: The multiple bar chart of the given data is shown in Fig. 2.4.

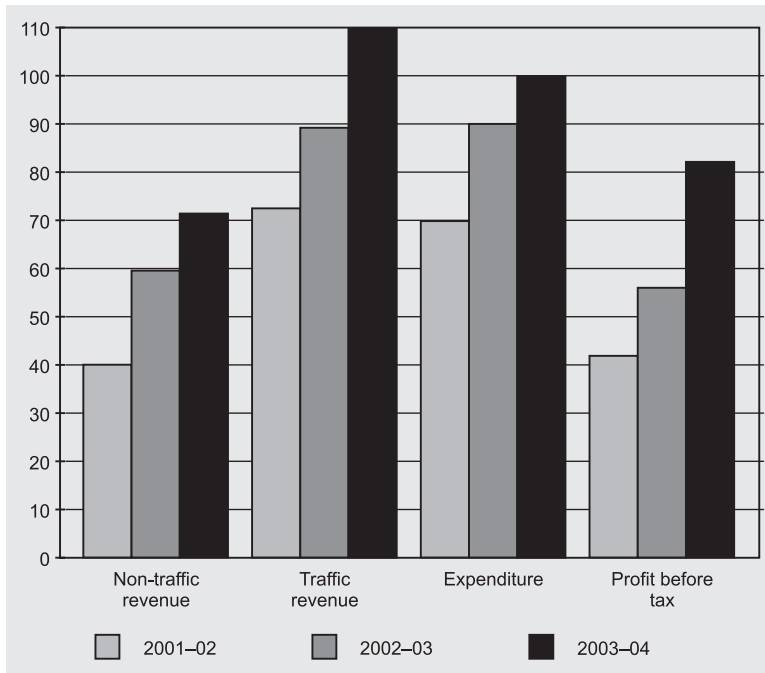


Figure 2.4 Multiple Bar Chart Pertaining to Performance of an International Airport Authority

Deviation Bar Charts Deviation bar charts are suitable for presentation of net quantities in excess or deficit such as profit, loss, import or exports. The excess (or positive) values and deficit (or negative) values are shown above and below the base line.

Example 2.12: The following are the figures of sales and net profits of a company over the last three years.

(per cent change over previous year)		
<i>Year</i>	<i>Sales Growth</i>	<i>Net Profit</i>
2002-2003	15	30
2003-2004	12	53
2004-2005	18	-72

Present this data by a suitable bar chart.

Solution: Figure 2.5 depicts deviation bar charts for sales and percent change in sales over previous year's data.

Subdivided Bar Chart Subdivided bar charts are suitable for expressing information in terms of ratios or percentages. For example, net per capita availability of food grains, results of a college faculty-wise in last few years, and so on. While constructing these charts the various components in each bar should be in the same order to avoid confusion. Different shades must be used to

represent various ratio values but the shade of each component should remain the same in all the other bars. An index of the shades should be given with the diagram.

A common arrangement while making these charts is that of presenting each bar in order of magnitude from the largest component at the base of the bar to the smallest at the end.

Since the different components of the bars do not start on the same scale, the individual bars are to be studied properly for their mutual comparisons.

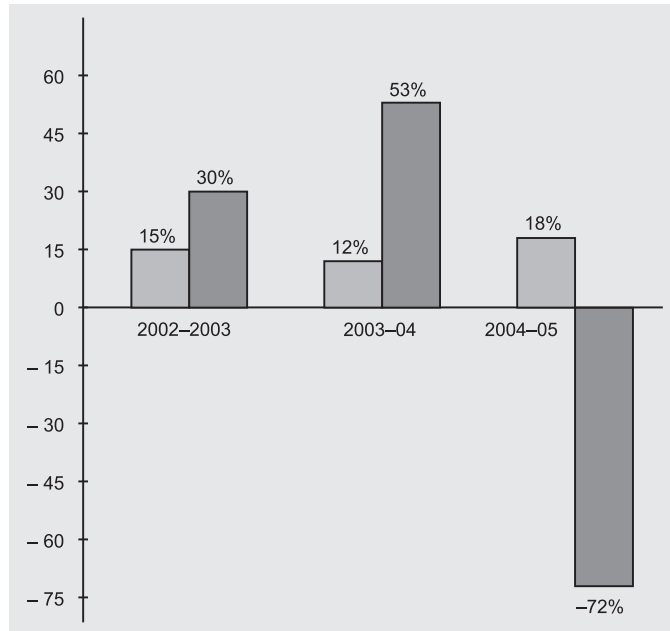


Figure 2.5 Deviation Bar Chart Pertaining to Sales and Profits

Example 2.13: The data on sales (₹ in million) of a company are given below:

	2005	2006	2007
Export	1.4	1.8	2.29
Home	1.6	2.7	2.9
Total	<u>3.0</u>	<u>4.5</u>	<u>5.19</u>

Solution: Figure 2.6 depicts a subdivided bar chart for the given data.

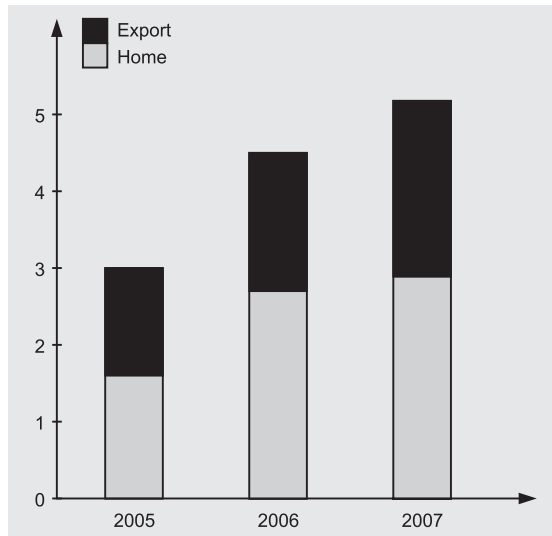


Figure 2.6 Subdivided Bar Chart Pertaining to Sales

Percentage Bar Charts When the relative proportions of components of a bar are more important than their absolute values, then each bar can be constructed with same size to represent 100 per cent. The component values are then expressed in terms of percentage of the total to obtain the necessary length for each of these in the full length of the bars. The other rules regarding the shades, index and thickness are the same as mentioned earlier.

Example 2.14: The following table shows the data on cost, profit or loss per unit of a good produced by a company during the year 2006–07.

Particulars	2006			2007		
	Amount (₹)	Percentage	Cumulative Percentage	Amount (₹)	Percentage	Cumulative Percentage
Cost per unit						
(a) Labour	25	41.67	41.67	34	40.00	40.00
(b) Material	20	33.33	75.00	30	35.30	75.30
(c) Miscellaneous	15	25.00	100.00	21	24.70	100.00
Total cost	60	100		85	100	
Sales proceeds per unit	80	110		80	88	
Profit (+) or loss (-) per item	+ 20	+ 10		-5	-12	

Represent diagrammatically the data given above on percentage basis.

Solution: The cost, sales and profit/loss data expressed in terms of percentages have been represented in the bar chart as shown in Fig. 2.7.

Frequency Polygon As shown in Fig. 2.8, the frequency polygon is formed by marking the mid-point at the top of horizontal bars and then joining these dots by a series of straight lines. The frequency polygons are formed as a closed figure with the horizontal axis, therefore a series of straight lines is drawn from the mid-point of the top base of the first and the last rectangles to the mid-point falling on the horizontal axis of the next outlying interval with zero frequency. The frequency polygon is sometimes jagged in appearance.

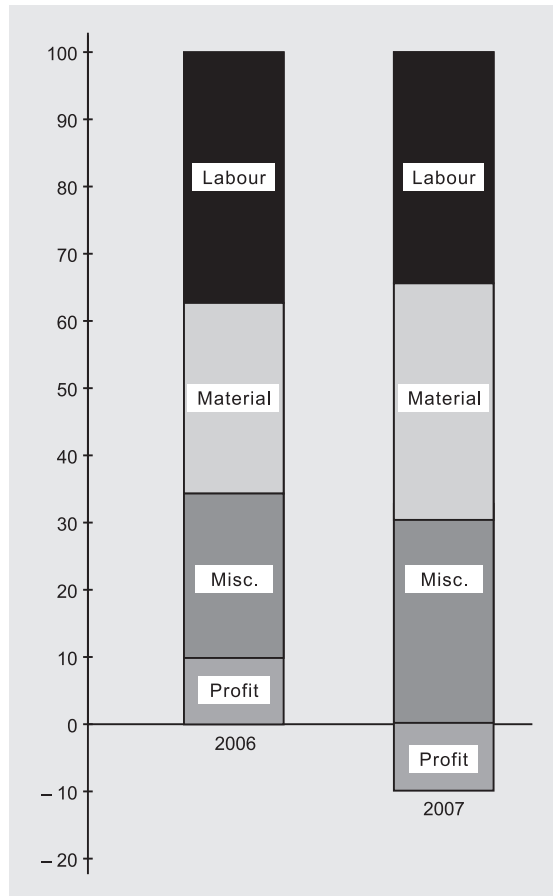


Figure 2.7 Percentage Bar Chart Pertaining to Cost, Sales and Profit/Loss

A frequency polygon can also be converted back into a histogram by drawing vertical lines from the bounds of the classes shown on the horizontal axis, and then connecting them with horizontal lines at the heights of the polygon at each mid-point.

Drawing a frequency polygon does not necessarily require constructing a histogram first. A frequency polygon can be obtained directly on plotting points above each class mid-point at heights equal to the corresponding class frequency. The points so drawn are then joined by a series of straight lines and the polygon is closed as explained earlier. In this case, horizontal x -axis measures the successive class mid-points and not the lower class limits. Figure 2.8 shows the frequency polygon for the frequency distribution presented by histogram in Fig. 2.1.

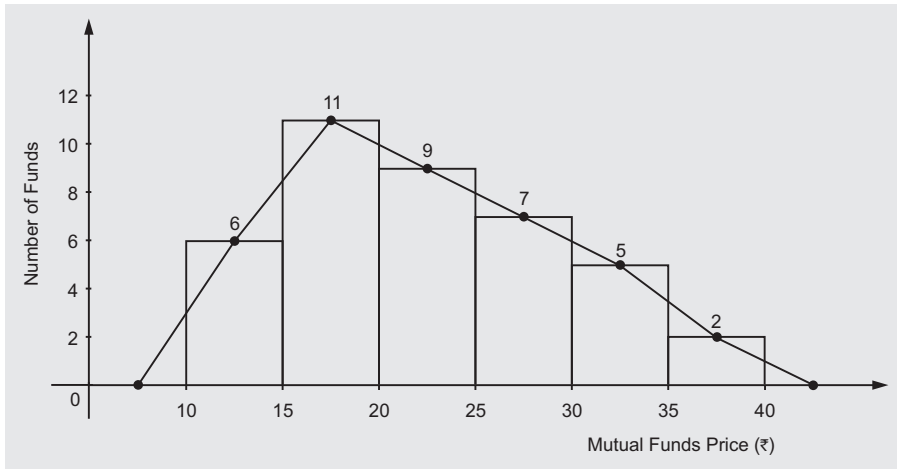


Figure 2.8 Frequency Polygon for Mutual Fund

Frequency Curve It is described as a smooth frequency polygon as shown in Fig. 2.9. A frequency curve is described in terms of its (i) symmetry (skewness) and (ii) its degree of peakedness (kurtosis). The concepts of skewness and kurtosis describing a frequency distribution will be discussed in Chapter 5.

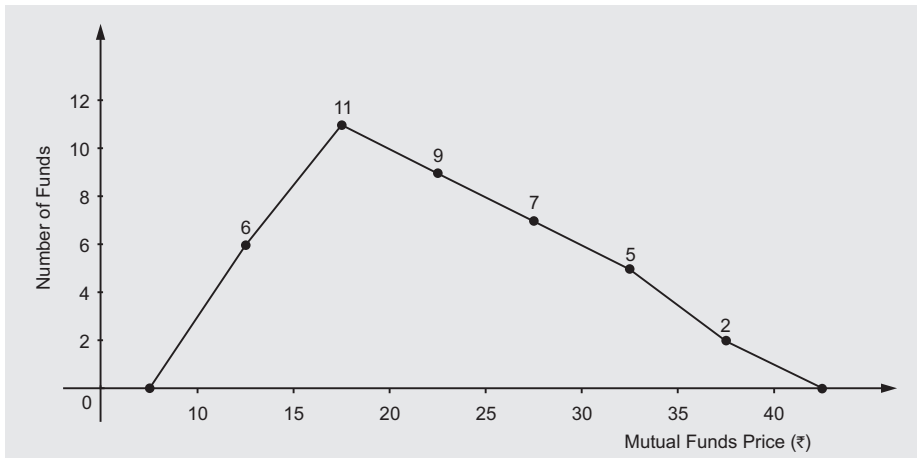


Figure 2.9 Frequency Curve

Two frequency distributions can also be compared by superimposing two or more frequency curves provided the width of their class intervals and the total number of frequencies are equal for the given distributions. Even if the distributions to be compared differ in terms of total frequencies, they still can be compared by drawing percent frequency curves where the vertical axis measures the percent class frequencies and not the absolute frequencies.

Cumulative Frequency Distribution (Ogive) It enables us to see how many observations lie above or below certain values rather than merely recording the number of observations within intervals. Cumulative frequency distribution is another method of data presentation that helps in data

analysis and interpretation. Table 2.27 shows the cumulative number of observations below and above the upper boundary of each class in the distribution.

A cumulative frequency curve popularly known as *Ogive* is another form of graphic presentation of a cumulative frequency distribution. The ogive for the cumulative frequency distribution given in Table 2.27 is presented in Fig. 2.10.

Once cumulative frequencies are obtained, the remaining procedure for drawing curve called ogive is as usual. The only difference being that the y-axis now has to be so scaled that it accommodates the total frequencies. The x-axis is labelled with the upper class limits in the case of less than ogive, and the lower class limits in case of more than ogive.

Table 2.27 Calculation of Cumulative Frequencies

Mutual Funds Price (₹)	Upper Class Boundary	Number of Funds (f)	Cumulative Frequency	
			Less than	More than
10–15	15	6	6	40
15–20	20	11	6 + 11 = 17	40 – 6 = 34
20–25	25	9	17 + 9 = 26	34 – 11 = 23
25–30	30	7	26 + 7 = 33	23 – 9 = 14
30–35	35	5	33 + 5 = 38	14 – 7 = 7
35–40	40	2	38 + 2 = 40	7 – 5 = 2

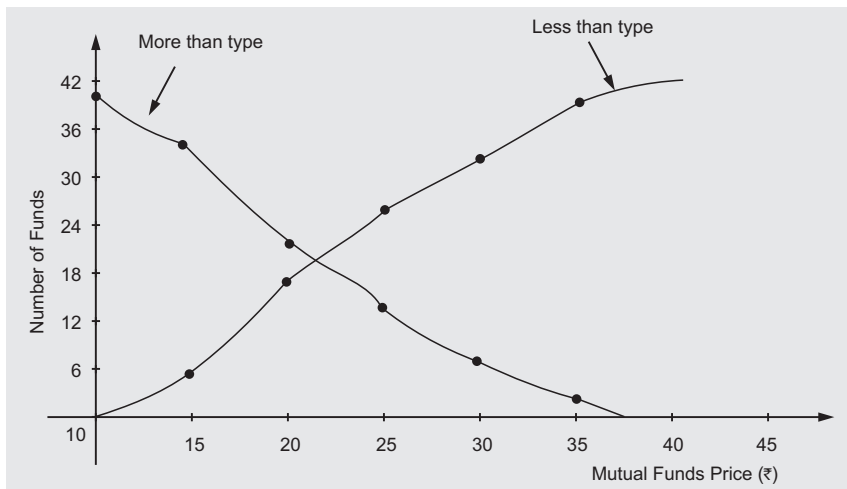


Figure 2.10 Ogive for Mutual Funds Prices

To draw a cumulative 'less than ogive', points are plotted against each successive upper class limit and a corresponding less than cumulative frequency value. These points are then joined by a series of straight lines and the resultant curve is closed at the bottom by extending it so as to meet the horizontal axis at the real lower limit of the first class interval.

To draw a cumulative 'more than ogive', points are plotted against each successive lower class limit and the corresponding more than cumulative frequency. These points are joined by a series of straight lines and the curve is closed at the bottom by extending it to meet the horizontal axis at the upper limit of the last class interval. Both the types of ogives so drawn are shown in Fig. 2.10.

It may be mentioned that a line drawn parallel to the vertical axis through the point of intersection of the two types of ogives will meet the x -axis at its middle point, and the value corresponding to this point will be the median of the distribution. Similarly, the perpendicular drawn from the point of intersection of the two curves on the vertical axis will divide the total frequencies into two equal parts.

Two ogives, whether *less than* or *more than* type, can be readily compared by drawing them on the same graph paper. The presence of unequal class intervals poses no problem in their comparison, as it does in the case of comparison of two frequency polygons. If the total frequencies are not the same in the two distributions, they can be first converted into per cent frequency distributions and then ogives drawn on a single graph paper to facilitate comparison.

Pie Diagram These diagrams are normally used to show the total number of observations of different types in the data set on a percentage basis rather than on an absolute basis through a circle. Usually the largest percentage portion of data in a pie diagram is shown first at 12 o'clock position on the circle, whereas the other observations (in per cent) are shown in clockwise succession in descending order of magnitude. The steps to draw a pie diagram are summarized below:

- (i) Convert the various observations (in per cent) in the data set into corresponding degrees in the circle by multiplying each by 3.6 ($360 \div 100$).
- (ii) Draw a circle of appropriate size with a compass.
- (iii) Draw points on the circle according to the size of each portion of the data with the help of a protractor and join each of these points to the center of the circle.

The pie chart has two distinct advantages: (i) it is aesthetically pleasing and (ii) it shows that the total for all categories or slices of the pie adds to 100 per cent.

Example 2.15: The data shows market share (in per cent) by revenue of the following companies in a particular year:

Batata–BPL	30	Escorts–First Pacific	5
Hutchison–Essar	26	Reliance	3
Bharti–Sing Tel	19	RPG	2
Modi Dista Com	12	Srinivas	2
		Shyam	1

Draw a pie diagram for the above data.

Solution: Converting percentage figures into angle outlay by multiplying each of them by 3.6 as shown in Table 2.28.

Table 2.28 Calculation of Market Share of Different Companies

<i>Company</i>	<i>Market Share (Percent)</i>	<i>Angle Outlay (Degree)</i>
Batata–BPL	30	108.0
Hutchison–Essar	26	93.6
Bharti–Sing Tel	19	68.4
Modi Dista Com	12	43.2
Escorts—First Pacific	5	18.0
Reliance	3	10.8
RPG	2	7.2
Srinivas	2	7.2
Shyam	1	3.6
Total	100	360.0

Using the data given in Table 2.28, construct the pie chart displayed in Fig. 2.11 by dividing the circle into nine parts according to degrees of angle at the centre.

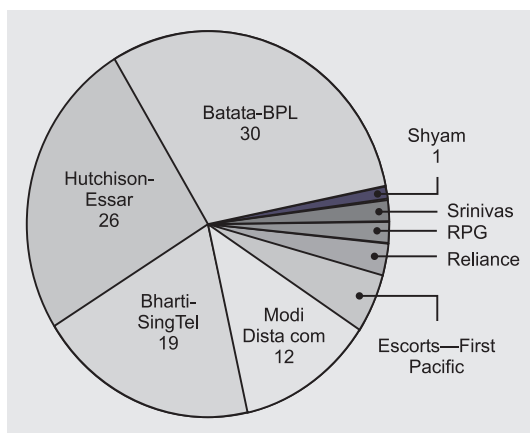


Figure 2.11 Percentage Pie Chart

Example 2.16: The following data relate to area in millions of square kilometer of oceans of the world.

<i>Ocean</i>	<i>Area (Million sq. km)</i>
Pacific	70.8
Atlantic	41.2
Indian	28.5
Antarctic	7.6
Arctic	4.8

Solution: Converting given areas into angle outlay as shown in Table 2.29.

Table 2.29 Calculation of Area

<i>Ocean</i>	<i>Area (Million sqkm)</i>	<i>Angle Outlay (Degrees)</i>
Pacific	70.8	$\frac{70.8}{152.9} \times 360 = 166.70$
Atlantic	41.2	$\frac{41.2}{152.9} \times 360 = 97.00$
Indian	28.5	67.10
Antarctic	7.6	17.89
Arctic	4.8	11.31
Total	152.9	360.00

Pie diagram is shown in Fig. 2.12.

2.5.2 Two-dimensional Diagrams

In one-dimensional diagrams or charts, only the length of the bar is taken into consideration. But in two-dimensional diagrams, both its height and width are taken into account for presenting the data. These diagrams, also known as *surface diagrams* or *area diagrams*, are:

- Rectangles
- Squares
- Circles

Rectangles Since area of a rectangle is equal to the product of its length and width, therefore while making such type of diagrams both length and width are considered.

Rectangles are suitable in cases where two or more quantities are to be compared and each quantity is sub-divided into several components.

Example 2.17: The following data represent the income of two families A and B. Construct a rectangular diagram.

<i>Item of Expenditure</i>	<i>Family A</i> (Monthly Income ₹ 30,000)	<i>Family B</i> (Monthly Income ₹ 40,000)
Food	5550	7280
Clothing	5100	6880
House rent	4800	6480
Fuel and light	4740	6320
Education	4950	6640
Miscellaneous	4860	6400
Total	30,000	40,000

Solution: Converting individual values into percentages taking total income as equal to 100 as shown in Table 2.30.

Table 2.30 Percentage Summary Table Pertaining to Expenses Incurred by Two Families

<i>Item of Expenditure</i>	<i>Family A (Monthly Income ₹ 30,000)</i>			<i>Family B (Monthly Income ₹ 40,000)</i>		
	<i>Actual Expenses</i>	<i>Percentage of Expenses</i>	<i>Cumulative Percentage</i>	<i>Actual Expenses</i>	<i>Percentage of Expenses</i>	<i>Cumulative Percentage</i>
Food	5550	18.50	18.50	7280	18.20	18.20
Clothing	5100	17.00	35.50	6880	17.20	35.40
House rent	4800	16.00	51.50	6480	16.20	51.60
Fuel and light	4740	15.80	67.30	6320	15.80	67.40
Education	4950	16.50	83.80	6640	16.60	84.00
Miscellaneous	4860	16.20	100.00	6400	16.00	100.00
Total	30,000	100.00		40,000	100.00	

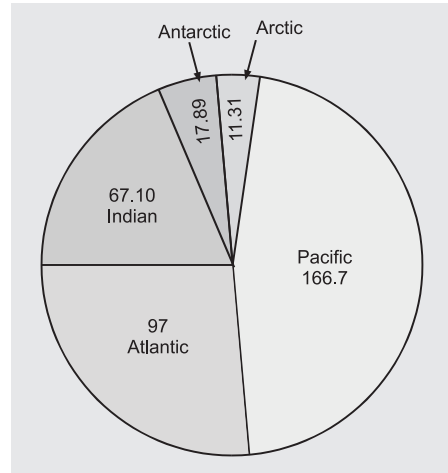


Figure 2.12 Per cent Pie Diagram

The height of the rectangles shown in Fig. 2.13 is equal to 100. The difference in the total income is represented by the difference on the base line which is in the ratio of 3:4.

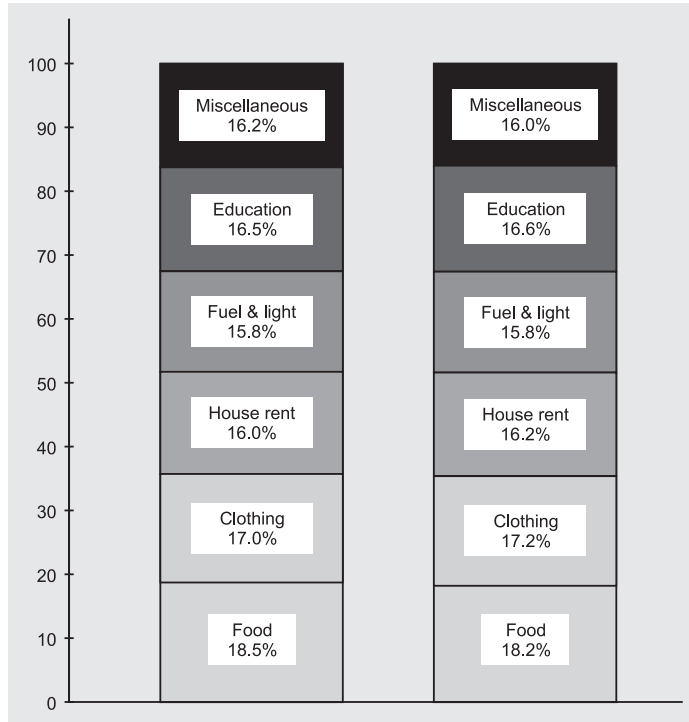


Figure 2.13 Percentage of Expenditure by Two Families

Squares Squares give a better comparison than rectangular bars when the difference of totals to be compared is large. For example, if in Example 2.17 the total expenses of families A and B are ₹2000 and 20,000, respectively, then the width of the rectangles would be in the ratio 1 : 10. If such a ratio is taken, the diagram would look very unwieldy. Thus to overcome this difficulty, squares are constructed to make use of their areas to represent given data for comparison.

To construct a square diagram, first take the square-root of the values of various figures to be represented and then these values are divided either by the lowest figure or by some other common figure to obtain proportions of the sides of the squares. The squares constructed on these proportionate lengths must have either the base or the centre on a straight line. The scale is attached with the diagram to show the variable value represented by one square unit area of the squares.

Example 2.18: The following data represent the production (in million tonnes) of coal by different countries in a particular year.

Country	Production
USA	130.1
USSR	44.0
UK	16.4
India	3.3

Represent the data graphically by constructing a suitable diagram.

Solution: The given data can be represented graphically by square diagrams. For constructing the sides of the squares, the necessary calculations are shown in Table 2.31.

Table 2.31 Side of a Square Pertaining to Production of Coal

<i>Country</i>	<i>Production (Million tonnes)</i>	<i>Square Root of Production Amount</i>	<i>Side of a Square (One square inch)</i>
USA	130.1	11.406	1.267
USSR	44.0	6.633	0.737
UK	16.4	4.049	0.449
India	3.3	1.816	0.201

The squares representing the amount of coal production by various countries are shown in Fig. 2.14.

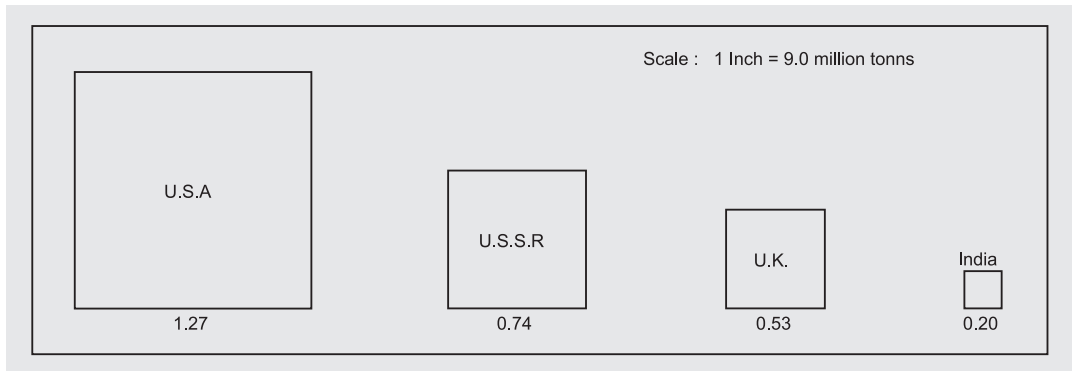


Figure 2.14 Coal Production in Different Countries

Circles Circles are alternatives to squares to represent data graphically. The circles are also drawn such that their areas are in proportion to the figures represented by them. The circles are constructed in such a way that their centres lie on the same horizontal line and the distance between the circles is equal.

Since the area of a circle is directly proportional to the square of its radius, therefore the radii of the circles are obtained in proportion to the square root of the figures under representation. Thus, the lengths which were used as the sides of the square can also be used as the radii of circles.

Example 2.19: The following data represent the land area in different countries. Represent this data graphically using suitable diagram.

<i>Country</i>	<i>Land Area (crore acres)</i>
USSR	590.4
China	320.5
USA	190.5
India	81.3

Solution: The data can be represented graphically using circles. The calculations for constructing radii of circles are shown in Table 2.32.

Table 2.32 Radii of Circles Pertaining to Land Area of Countries

Country	Land Area (crore acres)	Square Root of Land Area	Radius of Circles (Inches)
USSR	590.4	24.3	0.81
China	320.5	17.9	0.60
USA	190.5	13.8	0.46
India	81.3	9.0	0.30

The various circles representing the land area of respective countries are shown in Fig. 2.15.

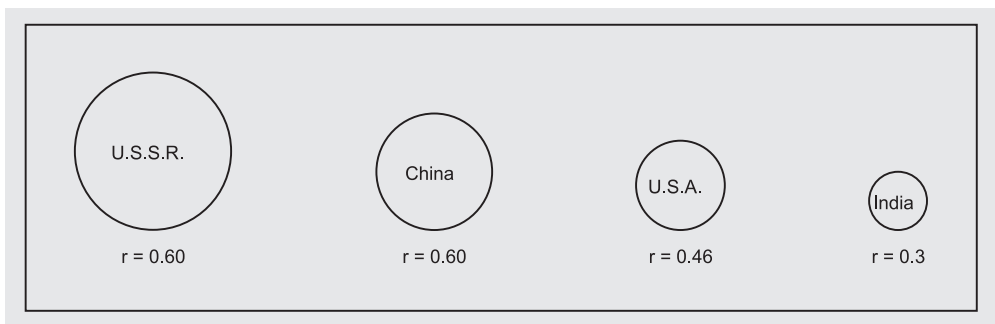


Figure 2.15 Land Area of Different Countries

2.5.3 Three-dimensional Diagrams

Cylinders, spheres cubes and so on are known as three-dimensional diagrams because three dimensions—length, breadth and depth are taken into consideration for representing figures. These diagrams are used when only one point is to be compared and the ratio between the highest and the lowest measurements is more than 100 : 1. For constructing these diagrams, the cube root of various measurements is calculated and the side of each cube is taken in proportion to the cube roots.

Amongst the three-dimensional diagrams, cubes are the easiest and should be used only in those cases where the figures cannot be adequately presented through bar, square or circle diagrams.

2.5.4 Pictograms or Ideographs

A pictogram is another form of pictorial bar chart. Such charts are useful in presenting data to people who cannot understand charts. Small symbols or simplified pictures are used to represent the size of the data. To construct pictograms or ideographs, the following suggestions are made:

- The symbols must be simple and clear.
- The quantity represented by a symbol should be given.
- Larger quantities are shown by increasing the number of symbols, and not by increasing the size of the symbols. A part of a symbol can be used to represent a quantity smaller than the whole symbol.

Conceptual Questions 2B

10. What are the different types of charts known to you? What are their uses?
11. Point out the role of diagrammatic presentation of data. Explain briefly the different types of bar diagrams known to you.
12. Charts are more effective in attracting attention than other methods of presenting data. Do you agree? Give reasons for your answer.
13. Discuss the utility and limitations (if any) of diagrammatic presentation of statistical data.
14. Diagrams are meant for a rapid view of the relation of different data and their comparisons. Discuss.
15. Write short notes on pictographic and cartographic representations of statistical data.
16. What are the advantages of using a graph to describe a frequency distribution?
17. When constructing a graph of a grouped frequency distribution, is it necessary that the resulting distribution be symmetric? Explain.
18. Explain what is meant by a frequency polygon, a histogram and a frequency curve.
19. Define the terms relative frequency and cumulative frequency. How are these related to a frequency distribution?
20. The distribution of heights of all students in the commerce department of the university has two peaks or is bimodal. The distribution of the IQs of the same students, however, has only one peak. How is this possible since the same students are considered in both the cases? Explain.

Self-practice Problems 2B

- 2.11** The following data represent the gross income, expenditure (in ₹ lakh) and net profit (in ₹ lakh) during the years 1999 to 2002.

	1999– 2000	2000– 2001	2001– 2002
Gross income	570	592	632
Gross expenditure	510	560	610
Net income	60	32	22

Construct a diagram or chart you prefer to use here.

- 2.12** Which of the charts would you prefer to represent the following data pertaining to the monthly income of two families and the expenditure incurred by them.

<i>Expenditure</i>	<i>Family A</i> (<i>Income</i> ₹ 17,000)	<i>Family B</i> (<i>Income</i> ₹ 10,000)
Food	4000	5400
Clothing	2800	3600

House rent	3000	3500
Education	2300	2800
Miscellaneous	3000	5000
Saving or deficits	+1900	–300

- 2.13** The following data represent the outlays (₹ crore) by heads of development.

<i>Heads of Development</i>	<i>Centre</i>	<i>States</i>
Agriculture	4765	7039
Irrigation and Flood control	6635	11,395
Energy	9995	8293
Industry and Minerals	12,770	2985
Transport and Communication	12,200	5120
Social services	8216	1420
	<u>54,581</u>	<u>36,252</u>

Represent the data by a suitable diagram and write a report on the data bringing out the salient features.

- 2.14** Make a diagrammatic representation of the following textile production and imports.

	<i>Value (in crore)</i>	<i>Length (in hundred yards)</i>
Mill production	116.4	426.9
Handloom production	106.8	192.8
Imports	319.7	64.7

What conclusions do you draw from the diagram?

- 2.15** Make a diagrammatic representation of the following data:

<i>Country</i>	<i>Production of Sugar in a Certain Year in Quintals (10,00,000)</i>
Cuba	32
Australia	30
India	20
Japan	5
Java	1
Egypt	1

- 2.16** The following data represent the estimated gross area under different cereal crops during a particular year.

<i>Crop</i>	<i>Gross Area (‘000 hectares)</i>	<i>Crop</i>	<i>Gross Areas (‘000 hectares)</i>
Paddy	34,321	Ragi	2656
Wheat	18,287	Maize	6749

Jowar	22,381	Barley	4422
Bajra	15,859	Small millets	6258

Draw a suitable chart to represent the data.

- 2.17** The following data indicate the rupee sales (in '000) of three products according to region.

<i>Product Group</i>	<i>Sales (in ₹ '000)</i>			<i>Total Sales (₹ '000)</i>
	<i>North</i>	<i>South</i>	<i>East</i>	
A	70	75	90	235
B	90	60	100	250
C	50	60	40	150
	<u>210</u>	<u>195</u>	<u>230</u>	<u>635</u>

- Using vertical bars, construct a bar chart depicting total sales region-wise.
- Construct a component chart to illustrate the product breakdown of sales region-wise by horizontal bars.
- Construct a pie chart illustrating total sales.

Formulae Used

- Class interval for a class in a frequency distribution

$$h = \text{Upper limit} - \text{Lower limit}$$
- Mid-point of a class in a frequency distribution

$$m = \frac{\text{Upper limit} + \text{Lower limit}}{2}$$
- Approximate interval size to be used in constructing a frequency distribution

$$h = \frac{\text{Largest data value} - \text{Smallest data value}}{\text{Number of class intervals}}$$

- Approximate number of class intervals for constructing a frequency distribution: $2^k \geq N$, where k and N represent the number of classes and total number of observations, respectively.

True or False Concepts Quiz

1. The mid-point of a class interval is h .
2. Frequency distribution of continuous data can be presented graphically as histograms or frequency polygons.
3. A frequency polygon is appropriate for graphing continuously distributed variables.
4. The percentile rank of a score is equal to the frequency of the scores falling up to and including the score.
5. Simple bar diagram is used only for one-dimensional comparisons.
6. Pie diagram is a circle divided into sections with areas equal to the corresponding component.
7. A pie diagram is inappropriate for representing nominally scaled data.
8. The height of a bar represents the frequency rather than the value of a variable.
9. The wider the class interval, the more specific information is lost about the actual data.
10. We cannot construct frequency distribution tables for nominally or ordinally scaled data.
11. The frequency distribution represents data in a compressed form.
12. The classes in any frequency distribution are all inclusive and mutually exclusive.
13. A frequency polygon can always be used to construct a histogram.
14. A histogram shows each separate class in the distribution more clearly than a frequency polygon.
15. The data array does not allow us to locate the highest and lowest values in the data set.

Concepts Quiz Answers

1. F	2. T	3. F	4. T	5. T	6. T	7. T	8. F
9. F	10. F	11. T	12. T	13. F	14. T	15. F	

Review Self-practice Problems

- 2.18** If the price of a two-bed room flat in Gurgaon varies from ₹ 9,00,000 to ₹ 12,00,000, then

- (a) Indicate the class boundaries of 10 classes into which these values can be grouped.
- (b) What class interval width did you choose?
- (c) What are the 10 class mid-points?

- 2.19** The raw data displayed here are the scores (out of 100 marks) of a market survey regarding the acceptability of a new

product launch by a company for a random sample of 50 respondents

40 45 41 45 45 30 39 8 48
 25 26 9 23 24 26 29 8 40
 41 42 39 35 18 25 35 40 42
 43 44 36 27 32 28 27 25 26
 38 37 36 35 32 28 40 41 43
 44 45 40 39 41

- (a) Form a frequency distribution having 9 class intervals
- (b) Form a percentage distribution from the frequency distribution in part (a)

- (c) Form a histogram of the frequency distribution in part (a)

2.20 State whether each of the following variables is qualitative or quantitative and indicate the measurement scale that is appropriate for each:

- (a) Age (b) Gender
(c) Class rank (d) Annual sales
(e) Mode of payment
(f) Earnings per share

2.21 The following data represent the sales of car tyres of various brands by a retail showroom of tyres during the year 2001–02.

Brand of Tyre	Tyres Sold
Dunlop	136
Modi	221
Firestone	138
Ceat	84
Goodyear	101
JK	120

- (a) Construct a bar chart and pie chart.
(b) Which of these charts do you prefer to use? Why?

2.22 The following data represent the expenditure incurred on following heads by a company during the year 2002

Expenditure Head	Amount (₹ in lakh)
Raw materials	1,689
Taxes	582
Manufacturing expenses	543
Employees salary	470
Depreciation	94
Dividend	75
Misc. expenses	286
Retained income	51

- (a) Construct a bar chart and pie chart.
(b) Which of these charts do you prefer to use? Why?

2.23 Draw an ogive by less than method and determine the number of companies earning profits between ₹ 45 crore and ₹ 75 crore:

Profit (₹ in crore)	Number of Companies
10–20	8
20–30	12
30–40	20
40–50	24
50–60	15
60–70	10
70–80	7
80–90	3
90–100	1

2.24 The following data represent the hottest career options in marketing:

Career Option	Percentage
Product Manager	23
Market Research Executive	10
Direct Marketing Manager	20
Manager Events and Productions	10
VP Marketing	16
Other Marketing Careers	21

Develop the appropriate display(s) and thoroughly analyse the data.

2.25 The data represent the closing prices of 40 common stocks.

29	34	43	8	37	8	7	30	35
19	9	16	38	53	16	1	48	18
9	9	10	37	18	8	28	24	21
18	33	31	32	29	79	11	38	11
52	14	9	33					

- (a) Construct frequency and relative frequency distributions for the data.
(b) Construct cumulative frequency and cumulative relative frequency distributions of the data.

2.26 The distribution of disability adjusted life year (DALY) loss by certain causes in certain year (in percentage) is given below:

Cause	India	China	World
• Communicable diseases	50.50	25.30	45.80
• Non-communicable diseases	40.40	58.00	42.20
• Injuries	9.10	16.70	12.00

Depict this data by pie chart and bar chart.

- 2.27** A government hospital has the following data representing weight in kg at birth of 200 premature babies:

<i>Weight</i>	<i>Number of Babies</i>
0.5–0.7	14
0.8–1.0	16
1.1–1.3	25
1.4–1.6	26
1.7–1.9	28
2.0–2.2	36
2.3–2.5	37
2.6–2.8	18

- Develop an appropriate display(s).
- Calculate the approximate middle value in the data set.
- If a baby below 2 kg is kept in the ICU as a precaution, then what percentage of premature babies need extra care in the ICU?

- 2.28** The medical superintendent of a hospital is concerned about the amount of waiting time for a patient before being treated in the OPD. The following data of waiting time (in minutes) were collected during a typical day:

<i>Waiting Time</i>	<i>Number of Patients</i>
30–40	125
40–50	195
50–60	305
60–70	185
70–80	120
80–90	

- Use the data to construct 'more than' and 'less than' frequency distributions and ogive.
- Use the ogive to estimate how long 75 percent of the patients should expect to wait.

Glossary of Terms

Frequency distribution: A tabular summary of data showing the number (frequency) of observations in each of several non-overlapping class intervals.

Class mid-point: The point in each class that is halfway between the lower and upper class limits.

Cumulative frequency distribution: The cumulative number of observations less than or equal to the upper class limit of each class.

Bar graph: A graphical device for depicting data that have been summarized in a frequency distribution, relative frequency distribution or per cent frequency distribution.

3

Measures of Central Tendency

LEARNING OBJECTIVES

After studying this chapter, you should be able to

- understand the role of descriptive statistics in summarization, description and interpretation of the data.
- understand the importance of summary measures to describe characteristics of a data set.
- use several numerical methods belonging to measures of central tendency to describe the characteristics of a data set.

3.1 INTRODUCTION

In Chapter 2, we discussed how raw data can be organized in terms of charts and frequency distributions in order to be easily understood and analysed. Although frequency distributions and corresponding graphical representations make raw data more meaningful, yet they fail to identify three major properties that describe a set of quantitative data. These three major properties are as follows:

- (i) The numerical value of an observation (also called *central value*) around which most numerical values of other observations in the data set show a tendency to cluster or group is called the *central tendency*.
- (ii) The extent to which numerical values are dispersed around the central value is called *variation*.
- (iii) The extent of departure of numerical values from symmetrical (normal) distribution around the central value is called *skewness*.

These three properties—*central tendency*, *variation* and *shape* of the frequency distribution—may be used to extract and summarize major features of the data set by the application of certain statistical methods called *descriptive measures* or *summary measures*. There are three types of summary measures:

- (i) Measures of central tendency
- (ii) Measures of dispersion or variation
- (iii) Measure of symmetry—skewness

These measures can also be used for comparing two or more populations in terms of the properties mentioned in the previous page to draw useful inferences.

The term 'central tendency' was coined because observations (numerical values) in most data sets show a distinct tendency to a group or cluster around a value of an observation located somewhere in the middle of all observations. It is necessary to identify or calculate this typical *central value* (also called *average*) to describe or project the characteristic of the entire data set. This descriptive value is the measure of the *central tendency* or *location* and methods of computing this central value are called *measures of central tendency*.

If the descriptive summary measures are computed using data of samples, then these are called **sample statistic** or simply *statistic* but if these measures are computed using data of the population, they are called **population parameters** or simply *parameters*. The population parameter is represented by the Greek letter μ (read : mu) and sample statistic is represented by the Roman letter $\bar{x}$ (read : x bar).

3.2 REQUISITES OF A MEASURE OF CENTRAL TENDENCY

The following are the few requirements to be satisfied by an average or a measure of central tendency:

It Should be Rigidly Defined The definition of an average should be clear and rigid so that there must be uniformity in its interpretation by different decision makers or investigators. There should not be any chance for applying discretion; rather it should be defined by an algebraic formula.

It Should be Based on All the Observations To ensure that it should represent the entire data set, its value should be calculated by taking into consideration the entire data set.

It Should be Easy to Understand and Calculate The value of an average should be computed by using a simple method without reducing its accuracy and other advantages.

It Should Have Sampling Stability The value of an average calculated from various independent random samples of the same size from a given population should not vary much from another. The least amount of difference (if any) in the values is considered to be the sampling error.

It Should be Capable of Further Algebraic Treatment The nature of the average should be such that it could be used for statistical analysis of the data set. For example, it should be possible to determine the average production in a particular year by the use of average production in each month of that year.

It Should Not be Unduly Affected by Extreme Observations The value of an average should not be unduly affected by very small or very large observations in the given data. Otherwise the average value may not truly represent characteristics of the entire set of data.

3.3 MEASURES OF CENTRAL TENDENCY

The various measures of central tendency or averages commonly used can be broadly classified in the following categories:

(i) Mathematical Averages

- (a) Arithmetic Mean (commonly called the mean or average)
 - Simple
 - Weighted
- (b) Geometric Mean
- (c) Harmonic Mean

(ii) Averages of Position

- (a) Median
- (b) Quartiles
- (c) Deciles
- (d) Percentiles
- (e) Mode

Notations

- m_i = mid-point for the i th class in the data set
- f_i = number of observations (or frequency) in the i th class; ($i = 1, 2, \dots, N$)
- N = total number of observations in the population
- n = number of observations in the sample (sample size)
- l = lower limit of any class interval
- h = width (or size) of the class interval
- cf = cumulative frequency
- Σ = summation (read: sigma) of all values of observations

3.4 MATHEMATICAL AVERAGES

Various methods of calculating mathematical averages of a data set are classified in accordance of the nature of data available, that is, ungrouped (unclassified or raw) or grouped (classified) data.

3.4.1 Arithmetic Mean of Ungrouped (or Raw) Data

There are two methods for calculating **arithmetic mean (A.M.)** for ungrouped or unclassified data:

- (i) Direct method
- (ii) Indirect (or Short-cut) method

Direct Method

In this method, A.M. is calculated by adding the values of all observations and dividing the total by the number of observations. Thus, if $x_1, x_2, \dots, x_N$ represent the values of N observations, then A.M. for a population of N observations is:

$$\text{Population mean, } \mu = \frac{x_1 + x_2 + \dots + x_N}{N} = \frac{1}{N} \sum_{i=1}^N x_i \quad (3-1a)$$

However, for a sample containing n observations $x_1, x_2, \dots, x_n$, the sample A.M. can be written as

$$\text{Sample mean, } \bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n} = \frac{1}{n} \sum_{i=1}^n x_i \quad (3-1b)$$

The denominator of above two formulae is different because in statistical analysis the uppercase letter N is used to indicate the number of observations in the population, while the lower case letter n is used to indicate the number of observations in the sample.

Example 3.1: In a survey of 5 cement companies, the profit (in ₹ lakh) earned during a year was 15, 20, 10, 35 and 32. Find the arithmetic mean of the profit earned.

Solution: Applying the formula (3-1b), we have

$$\bar{x} = \frac{1}{n} \sum_{i=1}^5 x_i = \frac{1}{5} (15 + 20 + 10 + 35 + 32) = 22.4$$

Thus the arithmetic mean of the profit earned by these companies during a year was ₹ 22.4 lakh.

Alternative Formula In general, when observations x_i ($i = 1, 2, \dots, n$) are grouped as a frequency distribution, then A.M. formula (3-1b) should be modified as

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n f_i x_i \quad (3-2)$$

where f_i represents the frequency (number of observations) with which variable x_i occurs in the

given data set, i.e. $n = \sum_{i=1}^n f_i$.

Example 3.2: If A, B, C and D are four chemicals costing ₹ 15, ₹ 12, ₹ 8 and ₹ 5 per 100g, respectively, and are contained in a given compound in the ratio of 1, 2, 3 and 4 parts, respectively, then what should be the price of the resultant compound.

Solution: Using the formula (3-2), the sample arithmetic mean is

$$\bar{x} = \frac{1}{n} \sum_{i=1}^4 f_i x_i = \frac{1 \times 15 + 2 \times 12 + 3 \times 8 + 4 \times 5}{1 + 2 + 3 + 4} = ₹ 8.30.$$

Thus, the average price of the resultant compound should be ₹ 8.30 per 100g.

Example 3.3: The number of new orders received by a company over the last 25 working days were recorded as follows: 3, 0, 1, 4, 4, 4, 2, 5, 3, 6, 4, 5, 1, 4, 2, 3, 0, 2, 0, 5, 4, 2, 3, 3, 1. Calculate the arithmetic mean for the number of orders received over all similar working days (Table 3.1).

Solution: Applying the formula (3-1b), the arithmetic mean is

$$\begin{aligned} \bar{x} &= \frac{1}{n} \sum_{i=1}^{25} x_i = \frac{1}{25} [3 + 0 + 1 + 4 + 4 + 4 + 2 + 5 + 3 + 6 + 4 \\ &\quad + 5 + 1 + 4 + 2 + 3 + 0 + 2 + 0 + 5 + 4 + 2 + 3 + 3 + 1] \\ &= \frac{1}{25} (71) = 2.84 \cong 3 \text{ orders (approx.).} \end{aligned}$$

Alternative approach: Use of formula (3-2)

Table 3.1 Calculations of Mean ($\bar{x}$) Value

Number of Orders (x_i)	Frequency (f_i)	$f_i x_i$
0	3	0
1	3	3
2	4	8
3	5	15
4	6	24
5	3	15
6	1	6
	<u>25</u>	<u>71</u>

$$\text{Arithmetic mean, } \bar{x} = \frac{1}{n} \sum f_i x_i = \frac{71}{25} = 2.8 \cong 3 \text{ orders (approx.)}$$

Example 3.4: From the following information on the number of defective components in 1000 boxes

Number of defective components	:	0	1	2	3	4	5	6
Number of boxes	:	25	306	402	200	51	10	6

Calculate the arithmetic mean of defective components for the whole of the production line.

Solution: The calculations of mean defective components for the whole production line are shown in Table 3.2.

Table 3.2 Calculations of $\bar{x}$ for Ungrouped Data

Number of Defective Components (x_i)	Number of Boxes (f_i)	$f_i x_i$
0	25	0
1	306	306
2	402	804
3	200	600
4	51	204
5	10	50
6	6	36
	<u>1000</u>	<u>2000</u>

Applying the formula (3-2), the arithmetic mean is

$$\bar{x} = \frac{1}{n} \sum_{i=0}^6 f_i x_i = \frac{1}{1000} (2000) = 2 \text{ defective components.}$$

Indirect or Short-Cut Method (Ungrouped Data)

In this method, an arbitrary *assumed mean* is used as a basis for calculating deviations from individual values in the data set. Let A be the arbitrary assumed A.M. and let

$$d_i = x_i - A \quad \text{or} \quad x_i = A + d_i$$

Substituting the value of x_i in formula (3-1b), we have

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i = \frac{1}{n} \sum_{i=1}^n (A + d_i) = A + \frac{1}{n} \sum_{i=1}^n f_i d_i \quad (3-3)$$

If frequencies of the numerical values are also taken into consideration, then the formula (3-3) becomes

$$\bar{x} = A + \frac{1}{n} \sum_{i=1}^n f_i d_i \quad (3-4)$$

where $n = \sum_{i=1}^n f_i$ = total number of observations in the sample.

Example 3.5: The daily earnings (in rupees) of employees working on a daily basis in a firm are

Daily earnings (₹)	:	100	120	140	160	180	200	220
Number of employees	:	3	6	10	15	24	42	75

Calculate the average daily earning for all employees.

Solution: The calculations of average daily earning for employees are shown in Table 3.3.

Table 3.3 Calculations of $\bar{x}$ for Ungrouped Data

Daily Earnings (in ₹) (x_i)	Number of Employees (f_i)	$d_i = x_i - A$ $= x_i - 160$	$f_i d_i$
100	3	-60	-180
120	6	-40	-240
140	10	-20	-200
160 ← A	15	0	0
180	24	20	480
200	42	40	1680
220	75	60	4500
	175		6040

Here, $A = 160$ is taken as assumed mean. The required A.M. using the formula (3-4) is given by

$$\bar{x} = A + \frac{1}{n} \sum_{i=1}^7 f_i d_i = 160 + \frac{6040}{175} = ₹ 194.51$$

Example 3.6: The human resource manager at a city hospital began a study of the overtime hours of the registered nurses. 25 nurses were selected at random, and following overtime hours during a month were recorded:

13	13	12	15	7	15	5	12	6	7	12	10	9	13	12
5	9	6	10	5	6	9	6	9	12					

Calculate the arithmetic mean of overtime hours during the month.

Solution: Calculations of arithmetic mean of overtime hours are shown in Table 3.4.

Table 3.4 Calculations of $\bar{x}$ for Ungrouped Data

Overtime Hours (x_i)	Number of Number (f_i)	$d_i = x_i - A$ $= x_i - 10$	$f_i d_i$
5	3	-5	-15
6	4	-4	-16
7	2	-3	-6
9	4	-1	-4
10 ← A	2	0	0
12	5	2	10
13	3	3	9
15	2	5	10
	<u>25</u>		<u>-12</u>

Here, $A=10$ is taken as assumed mean. The required arithmetic mean of overtime using the formula (3-4) is as follows

$$\bar{x} = A + \frac{1}{n} \sum_{i=1}^{25} f_i d_i = 10 - \frac{12}{25} = 9.52 \text{ hours}$$

3.4.2 Arithmetic Mean of Grouped (or Classified) Data

Arithmetic mean for grouped data can also be calculated by applying any of the following methods:

- (i) Direct method
- (ii) Indirect (or Step-deviation) method

For calculating arithmetic mean for a grouped data set, the following assumptions are made:

- (i) The class intervals must be closed.
- (ii) The width of each class interval should be equal.
- (iii) The values of the observations in each class interval must be uniformly distributed between its lower and upper limits.
- (iv) The mid-value of each class interval must represent the average of all values in that class, that is, it is assumed that all values of observations are evenly distributed between the lower and upper class limits.

Direct Method

The formula used in this method is same as formula (3-2) except that x_i is replaced with the mid-point value m_i of class intervals. The new formula becomes

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n f_i m_i \quad (3-5)$$

where m_i = mid-value of i th class interval.
 f_i = frequency of i th class interval.
 $n = \sum f_i$, sum of all frequencies

Example 3.7: A company is planning to improve plant safety. For this, accident data for the last 50 weeks was compiled. These data are grouped into the frequency distribution as shown below. Calculate the A.M. of the number of accidents per week.

Number of accidents :	0–4	5–9	10–14	15–19	20–24
Number of weeks :	5	22	13	8	2

Solution: The calculations of A.M. are shown in Table 3.5 using formula (3-5).

Table 3.5 Arithmetic Mean of Accidents

Number of Accidents	Mid-value (m_i)	Number of Weeks (f_i)	$f_i m_i$
0–4	2	5	10
5–9	7	22	154
10–14	12	13	156
15–19	17	8	136
20–24	22	2	44
		50	500

The A.M. of the number of accidents per week is

$$\bar{x} = \frac{1}{n} \sum_{i=1}^5 f_i m_i = \frac{500}{50} = 10 \text{ accidents per week}$$

Indirect or Step-deviation Method

The formula (3-5) for calculating A.M. can be improved as formula (3-6). This improved formula is also known as the *step-deviation method*:

$$\bar{x} = A + \frac{\sum f_i d_i}{n} \times h \quad (3-6)$$

where A = assumed value for the A.M.

h = width of the class intervals

$$d_i = \frac{m_i - A}{h}, \text{ deviation from the assumed mean}$$

The formula (3-6) is very useful in those cases where mid-values (m_i) and/or frequencies (f_i) are in three or more digits. The calculation of d_i from m_i involves reducing each m_i by an amount A (called assumed arithmetic mean) and then dividing the reduced values by h (width of class intervals). This procedure is usually referred to as *change of location and scale* or *coding*.

Example 3.8: Calculate the arithmetic mean of accidents per week by the short-cut method using the data of Example 3.7.

Solution: The calculations of the average number of accidents are shown in the Table 3.6.

Table 3.6 Arithmetic Mean of Accidents

Number of Accidents	Mid-value (m_i)	$d_i = (m_i - A)/h = (m_i - 12)/5$	Number of Weeks (f_i)	$f_i d_i$
0–4	2	–2	5	–10
5–9	7	–1	22	–22
10–14	(12) ← A	0	13	0
15–19	17	1	8	8
20–24	22	2	2	4
			<u>50</u>	<u>–20</u>

$$\begin{aligned}\text{Arithmetic mean, } \bar{x} &= A + \left(\frac{1}{n} \sum f_i d_i \right) h \\ &= 12 - \frac{20}{50} \times 5 = 10 \text{ accidents per week}\end{aligned}$$

Example 3.9: The following distribution gives the pattern of overtime work done by 100 employees of a company. Calculate the average overtime work done per employee.

Overtime hours	: 10–15	15–20	20–25	25–30	30–35	35–40
Number of employees	: 11	20	35	20	8	6

Solution: The calculations of the average overtime work done per employee with assumed mean $A = 22.5$ and $h = 5$ are given in Table 3.7.

Table 3.7 Calculations of Average Overtime

Overtime (hrs) x_i	Number of Employees, f_i	Mid-value (m_i)	$d_i = (m_i - 22.5)/5$	$f_i d_i$
10–15	11	12.5	–2	–22
15–20	20	17.5	–1	–20
20–25	35	(22.5) ← A	0	0
25–30	20	27.5	1	20
30–35	8	32.5	2	16
35–40	6	37.5	3	18
	<u>100</u>			<u>12</u>

The required A.M. is $\bar{x} = A + \frac{\sum f_i d_i}{n} \times h = 22.5 + \frac{12}{100} \times 5 = 23.1$ hours

Example 3.10: The following is the age distribution of 1000 persons working in an organization:

<i>Age Group</i>	<i>Number of Persons</i>	<i>Age Group</i>	<i>Number of Persons</i>
20–25	30	45–50	105
25–30	160	50–55	70
30–35	210	55–60	60
35–40	180	60–65	40
40–45	145		

Due to continuous losses, it is desired to bring down the manpower strength to 70 per cent of the present number according to the following scheme:

- Retrench the first 15 per cent from the lower age group.
- Absorb the next 45 per cent in other branches.
- Make 10 per cent from the highest age group retire permanently, if necessary.

Calculate the age limits of the persons retained and those to be transferred to other departments. Also find the average age of those retained.

Solution: (a) The first 15 per cent persons to be retrenched from the lower age groups are $(15/100) \times 1000 = 150$. But the lowest age group 20–25 has only 30 persons and therefore the remaining $150 - 30 = 120$ will be taken from next higher age group, that is, 25–30, which has 160 persons.

(b) The next 45 per cent, that is, $(45/100) \times 1000 = 450$ persons who are to be absorbed in other branches, belong to the following age groups:

<i>Age Groups</i>	<i>Number of Persons</i>
25–30	$(160 - 120) = 40$
30–35	210
35–40	180
40–45	$(450 - 40 - 210 - 180) = 20$

(c) Those who are likely to be retired are 10 per cent, that is, $(10/100) \times 1000 = 100$ persons and belong to the following highest age groups:

<i>Age Group</i>	<i>Number of Persons</i>
55–60	$(100 - 40) = 60$
60–65	40

Hence, the calculations of the average age of those retained and/or to be transferred to other departments are shown in Table 3.8.

Table 3.8 Calculations of Average Age

Age Group (x_i)	Mid-value (m_i)	Number of Persons (f_i)	$d_i = (m - 47.5)/5$	$f_i d_i$
40–45	42.5	$145 - 20 = 125$	-1	-125
45–50	47.5 ← A	105	0	0
50–55	52.5	70	1	70
		<u>300</u>		<u>-55</u>

The required average age is $\bar{x} = A + \frac{\sum f_i d_i}{n} \times h = 47.5 - \frac{55}{300} \times 5 = 46.58 = 47$ years (approx.).

3.4.3 Some Special Types of Problems and Their Solutions

Case 1: Frequencies are given in cumulative form, that is, either ‘more than type’ or ‘less than type’

As we know that the ‘more than type’ cumulative frequencies are calculated by adding frequencies from bottom to top, so that the first class interval has the highest cumulative frequency and it goes on decreasing in subsequent classes. But in case of ‘less than cumulative frequencies’, the cumulation is done downward so that the first class interval has the lowest cumulative frequency and it goes on increasing in the subsequent classes.

In both of these cases, data are first converted into inclusive class intervals or exclusive class intervals. Then the calculations for $\bar{x}$ are done in the usual manner, as discussed earlier.

Example 3.11: Following is the cumulative frequency distribution of the preferred length of kitchen slabs obtained from the preference study on housewives:

Length (in metres) more than	:	1.0	1.5	2.0	2.5	3.0	3.5
Preference of housewives	:	50	48	42	40	10	5

A manufacturer has to take a decision on what length of slabs to manufacture. What length would you recommend and why?

Solution: The given data are converted into exclusive class intervals as shown in Table 3.9. The frequency of each class has been found out by deducting the given cumulative frequency from the cumulative frequency of the previous class:

Table 3.9 Conversion into Exclusive Class Intervals

Length (in metres)	Preference of Housewives (More than)	Class Interval	Frequency
1.0	50	1.0–1.5	$(50 - 48) = 2$
1.5	48	1.5–2.0	$(48 - 42) = 6$
2.0	42	2.0–2.5	$(42 - 40) = 2$
2.5	40	2.5–3.0	$(40 - 10) = 30$
3.0	10	3.0–3.5	$(10 - 5) = 5$
3.5	5		

The calculations for mean length of slab are shown in Table 3.10.

Table 3.10 Calculations of Mean Length of Slab

<i>Class Interval</i>	<i>Mid-value (m_i)</i>	<i>Preference of Housewives (f_i)</i>	$d_i = \frac{m_i - 2.25}{0.5}$	$f_i d_i$
1.0–1.5	1.25	2	–2	–4
1.5–2.0	1.75	6	–1	–6
2.0–2.5	(2.25) ← A	2	0	0
2.5–3.0	2.75	30	1	30
3.0–3.5	3.25	5	2	10
		45		30

The mean length of the slab is $\bar{x} = A + \frac{\sum f_i d_i}{n} \times h = 2.25 + \frac{30}{45} \times 0.5 = 2.58$ metres.

Example 3.12: In an examination of 675 candidates, the examiner supplied the following information:

<i>Marks Obtained (Percentage)</i>	<i>Number of Candidates</i>	<i>Marks Obtained (Percentage)</i>	<i>Number of Candidates</i>
Less than 10	7	Less than 50	381
Less than 20	39	Less than 60	545
Less than 30	95	Less than 70	631
Less than 40	201	Less than 80	675

Calculate the mean percentage of marks obtained.

Solution: Arranging the given data into inclusive class intervals as shown in Table 3.11.

Table 3.11 Calculations of Mean Percentage of Marks

<i>Marks Obtained (Percentage)</i>	<i>Cumulative Frequency</i>	<i>Class-intervals (Less than)</i>	<i>Frequency</i>
Less than 10	7	0–10	7
Less than 20	39	10–20	(39 – 7) = 32
Less than 30	95	20–30	(95 – 39) = 56
Less than 40	201	30–40	(201 – 95) = 106
Less than 50	381	40–50	(381 – 201) = 180
Less than 60	545	50–60	(545 – 381) = 164
Less than 70	631	60–70	(631 – 545) = 86
Less than 80	675	70–80	(675 – 631) = 44

The calculations for mean percentage of marks obtained by the candidates are shown in Table 3.12.

Table 3.12 Calculations of Mean Percentage of Marks

Class Intervals	Mid-value (m_i)	Number of Candidates (f_i)	$d_i = \frac{m_i - 35}{10}$	$f_i d_i$
0-10	5	7	-3	-21
10-20	15	32	-2	-64
20-30	25	56	-1	-56
30-40	(35) ← A	106	0	0
40-50	45	180	1	180
50-60	55	164	2	328
60-70	65	86	3	258
70-80	75	44	4	176
		<u>675</u>		<u>801</u>

The mean percentage of marks obtained is

$$\bar{x} = A + \frac{\sum f_i d_i}{n} \times h = 35 + \frac{801}{675} \times 10 = 46.86 \text{ marks}$$

Case 2: Frequencies are not given but have to be calculated from the given data.

Example 3.13: 168 handloom factories have the following distribution of average number of workers in various income groups:

Income groups	:	800-1000	1000-1200	1200-1400	1400-1600	1600-1800
Number of firms	:	40	32	26	28	42
Average number of workers	:	8	12	8	8	4

Find the mean salary paid to the workers.

Solution: Since the total number of workers (i.e. frequencies) working in different income groups are not given, therefore these have to be determined as shown in Table 3.13.

Table 3.13 Calculation of Mean Salary

Income Group (x_i)	Mid-value (m_i)	$d_i = \frac{m_i - A}{h}$ $= \frac{m_i - 1300}{200}$	Number of Firms	Average Number of Workers	Frequencies (f_i) (5) = (3) × (4)	$m_i f_i$
(1)	(2)		(3)	(4)		
800-1000	900	-2	40	8	320	-640
1000-1200	1100	-1	32	12	384	-384
1200-1400	(1300) ← A	0	26	8	208	0
1400-1600	1500	1	28	8	224	224
1600-1800	1700	2	42	4	168	336
			<u>168</u>	<u>40</u>	<u>1304</u>	<u>-464</u>

The required A.M. is given by

$$\bar{x} = A + \frac{\Sigma m_i d_i}{n} \times h = 1300 - \frac{464}{1304} \times 200 = 1228.84$$

Example 3.14: Find the missing frequencies in the following frequency distribution. The A.M. of the given data is 11.09.

<i>Class Interval</i>	<i>Frequency</i>	<i>Class</i>	<i>Frequency</i>
9.3–9.7	2	11.3–11.7	14
9.8–10.2	5	11.8–12.2	6
10.3–10.7	f_3	12.3–12.7	3
10.8–11.2	f_4	12.8–13.2	1
			60

Solution: The calculations for A.M. are shown in Table 3.14.

Table 3.14 Calculation of A.M.

<i>Class Interval</i>	<i>Frequency</i> (f_i)	<i>Mid-value</i> (m_i)	$d_i = \frac{m_i - 11.0}{0.5}$	$f_i d_i$
9.3–9.7	2	9.5	–3	–6
9.8–10.2	5	10.0	–2	–10
10.3–10.7	f_3	10.5	–1	$-f_3$
10.8–11.2	f_4	11.0 ← A	0	0
11.3–11.7	14	11.5	1	14
11.8–12.2	6	12.0	2	12
12.3–12.7	3	12.5	3	9
12.8–13.2	1	13.0	4	4
	60			$23 - f_3$

where the assumed mean is $A = 11$. Applying the formula

$$\bar{x} = A + \frac{\Sigma f_i d_i}{n} \times h$$

we get $11.09 = 11.0 + \frac{23 - f_3}{60} \times 0.5$

or $0.09 = \frac{23 - f_3}{120}$ or $f_3 = 23 - 0.09 \times 120 = 12.2$

Since the total of the frequencies is 60 and $f_3 = 12.2$, therefore

$$f_4 = 60 - (2 + 5 + 12.2 + 14 + 6 + 3 + 1) = 16.8$$

Case 3: Complete data are not given

Example 3.15: The pass result of 50 students who took a class test is given below:

Marks	:	40	50	60	70	80	90
Number of students	:	8	10	9	6	4	3

If the mean marks for all the students were 51.6, find out the mean marks of the students who failed.

Solution: The marks obtained by 40 students who passed are given in Table 3.15.

Table 3.15 Calculation of Mean

Marks	Frequency (f_i)	$f_i x_i$
40	8	320
50	10	500
60	9	540
70	6	420
80	4	320
90	3	270
	<u>40</u>	<u>2370</u>

Total marks of all the students = $50 \times 51.6 = 2580$

Total marks of 40 students who passed = $\sum f_i x_i = 2370$

Thus, marks of the remaining 10 students = $2580 - 2370 = 210$. Hence, the average marks of 10 students who failed are $210/10 = 21$ marks.

Case 4: Incorrect values have been used for the calculation of Arithmetic Mean

Example 3.16: (a) The average dividend declared by a group of 10 chemical companies was 18 percent. Later on, it was discovered that one correct figure, 12, was misread as 22. Find the correct average dividend.

(b) The mean of 200 observations was 50. Later on, it was found that two observations were misread as 92 and 8 instead of 192 and 88. Find the correct mean.

Solution: (a) Given $n = 10$ And $\bar{x} = 18$ percent. We know that

$$\bar{x} = \frac{\sum x}{n} \quad \text{or} \quad \sum x = n\bar{x} = 10 \times 18 = 180$$

Since one numerical value 12 was misread as 22, therefore after subtracting the incorrect value and then adding the correct value in the total $n\bar{x}$, we have $180 - 22 + 12 = 170$. Hence, correct mean is $\bar{x} = \sum x/n = 170/10 = 17$ percent.

(b) Given that $n = 200$, $\bar{x} = 50$.

We know that $\bar{x} = \frac{\sum x}{n}$ or $\sum x = n\bar{x} = 200 \times 50 = 10,000$

Since two observations were misread, therefore the correct total $\sum x = n\bar{x}$ can be obtained as

$$\sum x = 10,000 - (92 + 8) + (192 + 88) = 10,180.$$

Hence, correct mean is $\bar{x} = \frac{\sum x}{n} = \frac{10,180}{200} = 50.9$.

Case 5: Frequency distributions have open-ended class intervals

Example 3.17: The annual salaries (in ₹ thousands) of employees in an organization are given below. The total salary of 10 employees in the class over ₹ 40,000 is ₹ 9,00,000. Compute the mean salary. Every employee belonging to the top 25 per cent of earners has to pay 5 per cent of his salary to the workers' relief fund. Estimate the contribution to this fund.

Salary (₹ '000)	Number of Employees
below 10	4
10–20	6
20–30	10
30–40	20
40 and above	10

Solution: Since class intervals are uniform, therefore we can take some width for open-end class intervals also. Calculations of mean are shown in Table 3.16.

Table 3.16 Calculation of Mean Salary

Salary (₹ '000)	Mid-value (m_i)	Number of Employees (f_i)	$d_i = \frac{m_i - 25}{10}$	$f_i d_i$
0–10	5	4	–2	–8
10–20	15	6	–1	–6
20–30	25 ← A	10	0	0
30–40	35	20	1	20
40 and above	45 (given)	10	2	20
		50		26

where mid-value 25 is considered as the assumed mean. Applying the formula, we get

$$\bar{x} = A + \frac{\sum f_i d_i}{n} \times h = 25 + \frac{26}{50} \times 10 = ₹ 30.2$$

The number of employees belonging to the top 25 per cent of the earners are $0.25 \times 50 = 13$ employees and the distribution of these top earners would be as follows

Salary (₹ '000)	Number of Employees
40 and above	10
30–40	3

This calculation implies that 3 employees have been selected from the salary range 30–40. Under the assumption that frequencies are equally distributed between lower and upper limits of a class interval, the calculations would be as follows:

Since 20 employees have salary in the range 30–40 = 10 or ₹ 10,000, therefore 3 employees will have income in the range $(10/20) \times 3 = 1.5$ or ₹ 1,500. But we are interested in the top 3 earners in the range 30–40, their salaries will range from $(40 - 1.5)$ to 40, i.e., 38.5 to 40. Thus, the distribution of salaries of the top 25 persons is as follows:

Salary (₹ '000)	Mid-value (m_i)	Number of Employees (f_i)	Total Salary ($m_i f_i$)
40 and above	—	10	9,00,000 (given)
30–40	35	3	1,05,000
		13	10,05,000

This shows that the total income of the top 25 percent of earners is ₹ 10,05,000. Hence 5 percent contribution to the fund is $0.05 \times 10,05,000 = ₹ 50,250$.

Remark: If the width of class intervals is not same, then in accordance with the magnitude of change in the width, fix the width of last class interval.

3.4.4 Advantages and Disadvantages of Arithmetic Mean

Advantages

1. The calculation of arithmetic mean is simple and it is unique, that is, every data set has one and only one mean.
2. The calculation of arithmetic mean is based on all values given in the data set.
3. The arithmetic mean is reliable single value that reflects all values in the data set.
4. The arithmetic mean is least affected by fluctuations in the sample size. In other words, its value, determined from various samples drawn from a population, varies by the least possible amount.
5. It can be readily put to algebraic treatment. Some of the algebraic properties of arithmetic mean are as follows:
 - (a) *The algebraic sum of deviations of all the observations x_i ($i = 1, 2, \dots, n$) from the A.M. is always zero, that is,*

$$\sum_{i=1}^n (x_i - \bar{x}) = \sum_{i=1}^n x_i - n\bar{x} = \sum_{i=1}^n x_i - n\left(\frac{1}{n}\right) \sum_{i=1}^n x_i = 0 = 0$$

Here, the difference $x_i - \bar{x}$ ($i = 1, 2, \dots, n$) is usually referred to as *deviation from the arithmetic mean*. This result is also true for grouped data.

Due to this property, the mean is characterized as a *point of balance*, i.e., sum of the positive deviations from mean is equal to the sum of the negative deviations from mean.

- (b) *The sum of the squares of the deviations of all the observations from the A.M. is less than the sum of the squares of all the observations from any other quantity.*

Let x_i ($i = 1, 2, \dots, n$) be the given observations and $\bar{x}$ be their arithmetic mean. Then this property implies that

$$\sum_{i=1}^n (x_i - \bar{x})^2 \leq \sum_{i=1}^n (x_i - a)^2$$

where 'a' is any constant quantity.

This property of A.M. is also known as the *least square property* and shall be quite helpful in defining the concept of standard deviation.

- (c) *It is possible to calculate the combined (or pooled) arithmetic mean of two or more than two sets of data of the same nature.*

Let $\bar{x}_1$ and $\bar{x}_2$ be arithmetic means of two sets of data of the same nature of size n_1 and n_2 respectively. Then their *combined A.M.* can be calculated as

$$\bar{x}_{12} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2} \quad (3-7)$$

The result (3-7) can also be generalized in the same way for more than two sets of data of different sizes having different arithmetic means.

- (d) While compiling the data for calculating A.M., it is possible that we may wrongly read and/or write certain number of observations. In such a case, the correct value of A.M. can be calculated first by subtracting the sum of observations wrongly recorded from Σx_i (total of all observations) and then adding the sum of the correct observations to it. The result is then divided by the total number of observations.

Disadvantages

1. The value of A.M. cannot be calculated accurately for unequal and open-ended class intervals either at the beginning or end of the given frequency distribution.
2. The A.M. is reliable and reflects all the values in the data set. However, it is very much affected by the extreme observations (or outliers) which are not representative of the rest of the data set. Outliers at the high end will increase the mean, while outliers at the lower end will decrease it. For example, if monthly income of four persons is 50, 70, 80 and 1000, then their A.M. will be 300, which does not represent the data.
3. The calculation of A.M. sometime becomes difficult because every data element is used in the calculation (unless the short-cut method for grouped data is used to calculate the mean). Moreover, the value so obtained may not be among the observations included in the data.
4. The mean cannot be calculated for qualitative characteristics such as intelligence, honesty, beauty or loyalty.
5. The mean cannot be calculated for a data set that has open-ended classes at either the high or low end of the scale.

Example 3.18: The mean salary paid to 1500 employees of an organization was found to be ₹ 12,500. Later on, after disbursement of salary, it was discovered that the salary of two employees was wrongly entered as ₹ 15,760 and 9590. Their correct salaries were ₹ 17,760 and 8590. Calculate correct mean.

Solution: Let x_i ($i = 1, 2, \dots, 1500$) be the salary of i th employee. Then, we are given that

$$\bar{x} = \frac{1}{1500} \sum_{i=1}^{1500} x_i = 12,500$$

$$\text{or} \quad \sum_{i=1}^{1500} x_i = 12,500 \times 1500 = ₹ 1,87,50,000$$

This gives the total salary disbursed to all 1500 employees. Now after adding the correct salary figures of two employees and subtracting the wrong salary figures posted against two employees, we have

$$\begin{aligned} \Sigma x_i &= 1,87,50,000 + (\text{Sum of correct salaries figures}) \\ &\quad - (\text{Sum of wrong salaries figures}) \\ &= 1,87,50,000 + (17,760 + 8590) - (15,760 + 9590) \\ &= 1,87,50,000 + 26,350 - 25,350 = 1,87,51,000 \end{aligned}$$

Thus, the correct mean salary is given by

$$\bar{x} = 1,87,51,000 \div 1500 = ₹ 12500.67$$

Example 3.19: There are two units of an automobile company in two different cities employing 760 and 800 persons, respectively. The arithmetic means of monthly salaries paid to persons in these two units are ₹ 18,750 and ₹ 16,950 respectively. Find the combined arithmetic mean of salaries of the employees in both the units.

Solution: Let n_1 and n_2 be the number of persons working in unit 1 and 2, respectively, and $\bar{x}_1$ and $\bar{x}_2$ be the arithmetic mean of salaries paid to these persons, respectively. We are given that:

$$\text{Unit 1: } n_1 = 760 ; \quad \bar{x}_1 = ₹ 18,750$$

$$\text{Unit 2: } n_2 = 800 ; \quad \bar{x}_2 = ₹ 16,950$$

Thus, the combined mean of salaries paid by the company is:

$$\begin{aligned} \bar{x}_{12} &= \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2} = \frac{760 \times 18,750 + 800 \times 16,950}{760 + 800} \\ &= ₹ 17,826.92 \text{ per month} \end{aligned}$$

Example 3.20: The mean yearly salary paid to all employees in a company was ₹ 24,00,000. The mean yearly salaries paid to male and female employees were ₹ 25,00,000 and ₹ 19,00,000, respectively. Find out the percentage of male to female employees in the company.

Solution: Let n_1 and n_2 be the number of employees as male and female, respectively. We are given that

Characteristics	Groups		Combined Group (Total Employees)
	Male	Female	
Number of employees	$n_1 = ?$	$n_2 = ?$	$n = n_1 + n_2$
Mean salary (₹)	$\bar{x}_1 = 25,00,000$	$\bar{x}_2 = 19,00,000$	$\bar{x}_{12} = 24,00,000$

Applying the formula for mean of combined group

$$\bar{x}_{12} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2}$$

$$\text{or} \quad (n_1 + n_2) \bar{x}_{12} = n_1 \bar{x}_1 + n_2 \bar{x}_2$$

$$(n_1 + n_2) 24,00,000 = 25,00,000 n_1 + 19,00,000 n_2$$

$$1000 n_1 = 5000 n_2$$

$$\frac{n_1}{n_2} = \frac{5000}{1000} = \frac{5}{1} \quad \text{or} \quad n_1 : n_2 = 5 : 1$$

Hence, male employees in the company are $\{5 \div (5 + 1)\} \times 100 = 83.33$ per cent and female employees are $\{1 \div (5 + 1)\} \times 100 = 16.67$ per cent.

3.4.5 Weighted Arithmetic Mean

The arithmetic mean, as discussed earlier, gives equal importance (or weight) to each observation in the data set. However, there are situations in which values of individual observations in the data

set are not of equal importance. If values occur with different frequencies, then computing A.M. of values (as opposed to the A.M. of observations) may not be a true representative of the data set characteristic and thus may be misleading. Under these circumstances, we may attach to each observation value a 'weight' $w_1, w_2, \dots, w_N$ as an indicator of their importance perhaps because of size or importance and compute a weighted mean or average denoted by $\bar{x}_w$ as

$$\mu_w \text{ or } \bar{x}_w = \frac{\sum x_i w_i}{\sum w_i}$$

This is similar to the method for dealing with frequency data when the value is multiplied by the frequency, within each class, totalled and divided by the total number of values.

Remark: The **weighted arithmetic mean** should be used

- (i) when the importance of all the numerical values in the given data set is not equal.
- (ii) when the frequencies of various classes are widely varying.
- (iii) where there is a change either in the proportion of numerical values or in the proportion of their frequencies.
- (iv) when ratios, percentages or rates are being averaged.

Example 3.21: An examination was held to decide the awarding of a scholarship. The weights of various subjects were different. The marks obtained by three candidates (out of 100 in each subject) are given below.

Subject	Weight	Students		
		A	B	C
Mathematics	4	60	57	62
Physics	3	62	61	67
Chemistry	2	55	53	60
English	1	67	77	49

Calculate the weighted A.M. to award the scholarship.

Solution: The calculations of the weighted arithmetic mean are shown in Table 3.17

Table 3.17 Calculations of Weighted Arithmetic Mean

Subject	Weight (w_i)	Students					
		Student A		Student B		Student C	
		Marks (x_i)	$x_i w_i$	Marks (x_i)	$x_i w_i$	Marks (x_i)	$x_i w_i$
Mathematics	4	60	240	57	228	62	248
Physics	3	62	186	61	183	67	201
Chemistry	2	55	110	53	106	60	120
English	1	67	67	77	77	49	49
	10	244	603	248	594	238	618

Applying the formula for weighted mean, we get

$$\begin{aligned}\bar{x}_{wA} &= \frac{603}{10} = 60.3 ; \quad \bar{x}_A = \frac{244}{4} = 61 \\ \bar{x}_{wB} &= \frac{594}{10} = 59.4 ; \quad \bar{x}_B = \frac{248}{4} = 62 \\ \bar{x}_{wC} &= \frac{618}{10} = 61.8 ; \quad \bar{x}_C = \frac{238}{4} = 59.5\end{aligned}$$

From the above calculations, it may be noted that student B should get the scholarship as per simple A.M. values, but according to weighted A.M., student C should get the scholarship because all the subjects of examination are not of equal importance.

Example 3.22: The owner of a general store was interested in knowing the mean contribution (sales price minus variable cost) of his stock of 5 items. The data is given below.

<i>Product</i>	<i>Contribution per Unit</i>	<i>Quantity Sold</i>
1	6	160
2	11	60
3	8	260
4	4	460
5	14	110

Solution: If the owner ignores the values of the individual products and gives equal importance to each product, then the mean contribution per unit sold will be

$$\bar{x} = \frac{1}{5} \{6 + 11 + 8 + 4 + 14\} = ₹ 8.60$$

This value ₹ 8.60 may not necessarily be the mean contribution per unit of different quantities of the products sold. In this case, the owner has to take into consideration the number of units of each product sold as different weights. Computing weighted A.M. by multiplying units sold (w) of a product by its contribution (x). That is,

$$\bar{x}_w = \frac{6(160) + 11(60) + 8(260) + 4(460) + 14(110)}{160 + 60 + 260 + 460 + 110} = \frac{7,080}{1,050} = ₹ 6.74$$

This value, ₹ 6.74, is different from the earlier value, ₹ 8.60. The owner must use the value ₹ 6.74 for decision making purpose.

Example 3.23: A management consulting firm has four types of professionals on its staff: managing consultants, senior associates, field staff and office staff. Average rates charged to consulting clients for the work of each of these professional categories are ₹ 3150 per hour, ₹ 1680 per hour, ₹ 1260 per hour and 630 per hour, respectively. Office records indicate the following number of hours billed last year in each category: 8000, 14,000, 24,000 and 35,000 respectively. If the firm is trying to come up with an average billing rate for estimating client charges for next year, what would you suggest they do and what do you think is an appropriate rate?

Solution: The data given in the problem are as follows:

<i>Staff</i>	<i>Consulting Charges</i> (₹ per hour) x_i	<i>Hours Billed</i> w_i
Managing consultants	3150	8000
Senior associates	1680	14,000
Field staff	1260	24,000
Office staff	630	35,000

Applying the formula for weighted mean, we get

$$\begin{aligned}
 \bar{x}_w &= \frac{\sum x_i w_i}{\sum w_i} = \frac{3150(8000) + 1680(14,000) + 1260(24,000) + 630(35,000)}{8000 + 14,000 + 24,000 + 35,000} \\
 &= \frac{2,52,00,000 + 2,35,20,000 + 3,02,40,000 + 2,20,50,000}{81,000} \\
 &= ₹ 1247.037 \text{ per hour}
 \end{aligned}$$

However, the firm should cite this as an average rate for clients who use the four professional categories for approximately 10 per cent, 17 per cent, 30 per cent and 43 per cent of the total hours billed.

Example 3.24: Calculate arithmetic mean from the data given below :

Marks :	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50
No. of Students :	7	6	15	12	10

[GJ Univ., BBA, 2005]

Solution: Calculations required to find arithmetic mean (A.M.) from a classified data are shown below:

<i>Marks</i>	<i>Mid-value (m)</i>	<i>No. of Students (f)</i>	<i>fm</i>
0 – 10	5	7	35
10 – 20	15	6	90
20 – 30	25	15	375
30 – 40	35	12	420
40 – 50	45	10	450
		$\Sigma f = 50$	$\Sigma fm = 1370$

$$\text{Required A.M. } \bar{x} = \frac{\Sigma fm}{\Sigma f} = \frac{1370}{50} = 27.4$$

Conceptual Questions 3A

1. Explain the term *average*. What are the merits of a good average? Explain with examples.
2. What are the measures of central tendency? Why are they called measures of central tendency?
3. What are the different measures of central tendency? Mention the advantages and disadvantages of arithmetic mean.
4. What are the different measures of central tendency? Discuss the essentials of an ideal average.
5. Give a brief description of the various measures of central tendency. Why is arithmetic mean so popular?
6. What information about a body of data is provided by an average? How are averages useful as a descriptive measure?
7. It is said that the weighted mean is commonly referred to as a 'weighted average'. How is the use of this phrase inconsistent with the definition of an average?
8. How is an average considered as a representative measure or a measure of central tendency? How is the ability of an average to measure central tendency related to other characteristics of data?
9. What is statistical average? What are the desirable properties for an average to possess? Mention the different types of averages and state why arithmetic mean is the most commonly used amongst them.

Self-practice Problems 3A

- 3.1 An investor buys ₹ 12,000 worth of shares of a company each month. During the first 5 months, he bought the shares at a price of ₹ 100, ₹ 120, ₹ 150, ₹ 200 and ₹ 240 per share respectively. After 5 months what is the average price paid for the shares by him?
- 3.2 A company wants to pay bonus to members of the staff. The bonus is to be paid as under:

Monthly Salary (in ₹)	Bonus
3000–4000	1000
4000–5000	1200
5000–6000	1400
6000–7000	1600
7000–8000	1800
8000–9000	2200
9000–10,000	2200
10,000–11,000	2400

Actual amount of salary drawn by the employees is given below:

3250	3780	4200	4550	6200
6600	6800	7250	3630	8320
9420	9520	8000	10,020	10,280
11,000	6100	6250	7630	3820
5400	4630	5780	7230	6900

How much would the company need to pay by way of bonus? What shall be the average bonus paid per member of the staff?

- 3.3 Calculate the simple and weighted arithmetic mean price per tonne of coal purchased by a company for the half year. Account for difference between the two:

Month	Price/ tonne	Tonnes Pur- chased	Month	Price/ tonne	Tonnes Pur- chased
January	4205	25	April	5200	52
February	5125	30	May	4425	10
March	5000	40	June	5400	45

- 3.4 Salary paid by a company to its employees is as follows:

Designation	Monthly Salary (in ₹)	Number of Persons
Senior Manager	35,000	1
Manager	30,000	20
Executives	25,000	70
Jr. Executives	20,000	10
Supervisors	15,000	150

Calculate the simple and weighted arithmetic mean of salary paid.

- 3.5 The mean monthly salary paid to all employees in a company is ₹ 16,000. The mean monthly salaries paid to technical and non-technical employees are ₹ 18,000 and ₹ 12,000, respectively. Determine the percentage of technical and non-technical employees in the company.
- 3.6 The mean marks in statistics of 100 students in a class were 72 percent. The mean marks of

boys were 75 percent, while their number was 70 percent. Find out the mean marks of girls in the class.

- 3.7 The arithmetic mean height of 50 students of a college is 5'8". The height of 30 of these is given in the frequency distribution below. Find the arithmetic mean height of the remaining 20 students.

Height in inches : 5'4" 5'6" 5'8" 5'10" 6'0"

Frequency : 4 12 4 8 2

- 3.8 An appliances manufacturing company is forecasting regional sales for next year. The Delhi branch, with current yearly sales of ₹ 387.6 million, is expected to achieve a sales growth of 7.25 per cent; the Kolkata branch, with current sales of ₹ 158.6 million, is expected to grow by 8.20 per cent; and the Mumbai branch, with sales of ₹ 115 million, is expected to increase sales by 7.15 per cent. What is the average rate of growth forecasted for next year?

Hints and Answers

3.1 $x = ₹ 146.30$

3.2 ₹ 42,000 to pay bonus; Average bonus paid per member = $42,000/25 = ₹ 1,680$

3.3 $\bar{x} = ₹ 4892.5$ tonne; $\bar{x}_w = 5032.30$

3.4 $\bar{x} = ₹ 25,000$; $\bar{x}_w = ₹ 19,262.94$

3.5 Percentage of technical personnel = 66.67 percent ; Non-technical = 33.33 percent

3.6 Mean marks of girls, $\bar{x}_2 = 65$ percent

3.7 $\bar{x}_1$ (mean height of $n_1 = 30$ students) = 5'6".

Given $n_2 = 50 - 30 = 20$, $\bar{x}_2$ (mean height of 50 students) = 6'8". Thus

$$\bar{x} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2} = 5'9''$$

$$\begin{aligned} 3.8 \quad \bar{x}_w &= \frac{\sum x_i w_i}{\sum w_i} \\ &= \frac{387.6 \times 7.25 + 158.6 \times 8.20 + 115 \times 7.15}{387.6 + 158.6 + 115} \\ &= \frac{2810.10 + 1300.52 + 822.25}{661.20} \\ &= \frac{4932.87}{661.20} = 7.46 \text{ per cent} \end{aligned}$$

3.5 GEOMETRIC MEAN

In many business and economics problems, we deal with quantities (variables) that change over a period of time. In such cases, the aim is to know an average percentage change rather than simple average value to represent the average growth or declining rate in the variable value over a period of time. Thus, we need to calculate another measure of central tendency called **geometric mean** (G.M.). The specific application of G.M. is to show multiplicative effects over time in compound interest and inflation calculations.

Consider, for example, the annual rate of growth of output of a company in the last five years.

Year	Growth Rate (Percent)	Output at the End of the Year
1998	5.0	105
1999	7.5	112.87
2000	2.5	115.69
2001	5.0	121.47
2002	10.0	133.61

The simple arithmetic mean of the growth rate is

$$\bar{x} = \frac{1}{5} (5 + 7.5 + 2.5 + 5 + 10) = 6.$$

This value of 'mean' implies that if 6 per cent is the growth rate, then output at the end of 2002 should be 133.81, which is slightly more than the actual value, 133.61. Thus, the correct growth rate should be slightly less than 6.

To find the correct growth rate, we apply the formula of geometric mean

$$\begin{aligned} \text{G.M.} &= \sqrt[n]{\text{Product of all the } n \text{ values}} \\ &= \sqrt[n]{x_1 \cdot x_2 \cdots x_n} = (x_1 \cdot x_2 \cdots x_n)^{1/n} \end{aligned} \quad (3-8)$$

In other words, *G.M. of a set of n observations is the n th root of their product.*

For the above example, substituting the values of growth rate in the given formula, we have

$$\begin{aligned} \text{G.M.} &= \sqrt[5]{5 \times 7.5 \times 2.5 \times 5 \times 10} = \sqrt[5]{4687.5} \\ &= 5.9 \text{ per cent average growth.} \end{aligned}$$

Calculation of G.M. When the number of observations are more than three, the G.M. can be calculated by taking logarithm on both sides of the equation. The formula (3-8) for G.M. for ungrouped data can be expressed in terms of logarithm as shown below:

$$\begin{aligned} \log (\text{G.M.}) &= \frac{1}{n} \log (x_1 x_2 \cdots x_n) \\ &= \frac{1}{n} \{ \log x_1 + \log x_2 + \cdots + \log x_n \} = \frac{1}{n} \sum_{i=1}^n \log x_i \end{aligned}$$

and therefore

$$\text{G.M.} = \text{Antilog} \left\{ \frac{1}{n} \sum \log x_i \right\} \quad (3-9)$$

If the observations $x_1, x_2, \dots, x_n$ occur with frequencies $f_1, f_2, \dots, f_n$, respectively, and the total of frequencies is $n = \sum f_i$, then the G.M. for such data is given by

$$\begin{aligned} \text{G.M.} &= \left(x_1^{f_1} x_2^{f_2} \cdots x_n^{f_n} \right)^{1/n} \\ \text{or} \quad \log (\text{G.M.}) &= \frac{1}{n} \{ f_1 \log x_1 + f_2 \log x_2 + \cdots + f_n \log x_n \} = \frac{1}{n} \sum_{i=1}^n f_i \log x_i \\ \text{or} \quad \text{G.M.} &= \text{Antilog} \left\{ \frac{1}{n} \sum f_i \log x_i \right\} \end{aligned} \quad (3-10)$$

3.5.1 Advantages, Disadvantages and Applications of G.M.

Advantages

- (i) The value of G.M. is not much affected by extreme observations and is computed by taking all the observations into account.
- (ii) It is useful for averaging ratio and percentage as well as in determining rate of increase and decrease.
- (iii) In the calculation of G.M., more weight is given to smaller values and less weight to higher values. For example, it is useful in the study of price fluctuations where the lower limit can touch zero, whereas the upper limit may go upto any number.
- (iv) It is suitable for algebraic manipulations. The calculation of weighted G.M. and combined G.M. are two examples of algebraic manipulations of the original formula of geometric mean.

Disadvantages

- (i) The calculation of G.M. as compared to A.M. is more difficult and intricate.
- (ii) The value of G.M. cannot be calculated when any of the observations in the data set is either negative or zero.
- (iii) While calculating weighted geometric, mean equal importance (or weight) is not given to each observation in the data set.

Applications

- (i) The concept of G.M. is used in the construction of index numbers.
- (ii) Since $G.M. \leq A.M.$, therefore G.M. is useful in those cases where smaller observations are to be given importance. Such cases usually occur in social and economic areas of study.
- (iii) The G.M. of a data set is useful in estimating the average rate of growth in the initial value of an observation per unit per period. For example, it is useful in finding the percentage increase in sales, profit, production, population and so on. It is also useful in calculating the amount of money accumulated at the end of n periods with an original principal amount of P_0 . The formula is as follows

$$P_n = P_0 (1 + r)^n \quad \text{or} \quad r = \left(\frac{P_n}{P_0} \right)^{\frac{1}{n}} - 1$$

where r = interest rate (rate of growth) per unit period
 n = number of years or length of the period

Example 3.25: The rate of increase in population of a country during the last three decades is 5 percent, 8 percent and 12 percent. Find the average rate of growth during the last three decades.

Solution: Since the data is given in terms of percentage, therefore geometric mean is a more appropriate measure. The calculations of geometric mean are shown in Table 3.18.

Table 3.18 Calculations of G.M.

<i>Decade</i>	<i>Rate of Increase in Population (%)</i>	<i>Population at the End of Decade (x) Taking Preceding Decade as 100</i>	<i>$\log_{10} x$</i>
1	5	105	2.0212
2	8	108	2.0334
3	12	112	2.0492
			<u>6.1038</u>

Using the formula (3-10), we have

$$\begin{aligned}\text{G.M.} &= \text{Antilog} \left\{ \frac{1}{n} \sum \log x \right\} = \text{Antilog} \left\{ \frac{1}{3} (6.1038) \right\} \\ &= \text{Antilog} (2.0346) = 108.2\end{aligned}$$

Hence the average rate of increase in population over the last three decades is

$$108.2 - 100 = 8.2 \text{ per cent.}$$

Example 3.26: A given machine is assumed to depreciate 40 per cent in value in the first year, 25 per cent in the second year and 10 per cent per year for the next three years, each percentage being calculated on the diminishing value. What is the average depreciation recorded on the diminishing value for the period of five years?

Solution: The calculation of geometric mean is shown in Table 3.19.

Table 3.19 Calculation of G.M.

<i>Rate of Depreciation (x_i) (in percentage)</i>	<i>Number of Years (f_i)</i>	<i>$\log_{10} x_i$</i>	<i>$f_i \log_{10} x_i$</i>
40	1	1.6021	1.6021
25	1	1.3979	1.3979
10	3	1.0000	3.0000
			<u>6.0000</u>

Using formula (3-11), we have

$$\begin{aligned}\text{G.M.} &= \text{Antilog} \left\{ \frac{1}{n} \sum f \log x \right\} = \text{Antilog} \left\{ \frac{1}{5} (6.0000) \right\} \\ &= \text{Antilog} (1.2) = 15.85\end{aligned}$$

Hence, the average rate of depreciation for first five years is 15.85 per cent.

Example 3.27: A candidate obtained the following percentage of marks in different subjects in the half-yearly examination:

English : 46%, Statistics : 67%, Cost Accountancy : 72%,
Economics : 58%, Income Tax : 53%

It is agreed to give double weights to marks in English and Statistics as compared to other subjects. What is the simple and weighted arithmetic mean?

Solution: The calculations for simple and weighted mean are shown in the Table 3.20:

Table 3.18 Calculation of Simple and Weighted Mean

<i>Examination</i>	<i>Percentage of Marks (x)</i>	<i>Weights age (w)</i>	<i>w x</i>
English	46	2	92
Statistics	67	2	134
Cost Accountancy	72	1	72
Economics	58	1	58
Income Tax	53	1	53
$N = 5$	$\Sigma x = 296$	$\Sigma w = 7$	$\Sigma wx = 409$

$$\text{Simple Mean, } \bar{x} = \frac{\Sigma x}{N} = \frac{296}{5} = 59.2 \text{ per cent}$$

$$\text{Weighted Mean, } \bar{x}_w = \frac{\Sigma wx}{\Sigma w} = \frac{409}{7} = 58.43 \text{ per cent (approx.)}$$

Example 3.28: The annual rate of growth of output of a factory in five years is 5.0, 6.5, 4.5, 8.5 and 7.5 per cent, respectively? What is the compound rate of growth of output per annum for the period?

Solution: Since the question involves the average rate of growth in percentage, so we need to calculate G.M. Let the output be 100 initially. Then,

<i>Year</i>	<i>Percentage</i>	<i>Output (x)</i>	<i>log x</i>
1	5.0	105.0	2.0212
2	6.5	106.5	2.0273
3	4.5	104.5	2.0191
4	8.5	108.5	2.0354
5	7.5	107.5	2.0314
			<u>10.1344</u>

$$\text{G.M.} = \text{Antilog} \left(\frac{\Sigma \log x}{N} \right) = \text{Antilog} \left(\frac{10.1344}{5} \right) = \text{Antilog} (2.0269) = 106.38.$$

Hence, per cent growth (increase) of output is $106.38 - 100 = 6.38$ per cent

Example 3.29: The number of divorces per 1000 marriages in a big city in India increased from 96 in 1980 to 120 in 1990. Find the annual rate of increase of the divorce rate for the period 1980 to 1990. [Delhi Univ., BCom.(Hons), 1994]

Solution: Let the rate of growth in divorce be r . Then

$$P_n = P_0 \left(1 + \frac{r}{100} \right)^n \quad \text{or} \quad 120 = 96 \left(1 + \frac{r}{100} \right)^{10}$$

$$\frac{120}{96} = \left(1 + \frac{r}{100} \right)^{10}$$

Taking log on both sides, we get

$$\log 120 - \log 96 = 10 \log \left(1 + \frac{r}{100} \right)$$

$$10 \log \left(1 + \frac{r}{100} \right) = 2.0792 - 1.9823 = 0.0969$$

$$1 + \frac{r}{100} = \text{Antilog } (0.00969) = 1.0226$$

$$\frac{r}{100} = 0.0226 \quad \text{or} \quad r = 0.0226 \times 100 = 2.26 \text{ per cent.}$$

Example 3.30: A machinery depreciates by 40 per cent in the first year, 25 per cent in second year and 10 per cent per year for the next three years, each percentage being calculated as a diminishing value. What is the average of depreciation for the entire period ?

Solution: The calculations of average percentage of depreciation are shown below in the table.

Year	Depreciation (d) (Per cent)	$x = 100 - d$	$\log x$
1	40	60	1.7782
2	25	75	1.8751
3	10	90	1.9542
4	10	90	1.9542
5	10	90	1.9542
			9.5159

$$\text{Geometric Mean} = \text{Antilog} \left(\frac{\sum \log x}{n} \right) = \text{Antilog} \left(\frac{9.5159}{5} \right) = \text{Antilog } (1.90318) = 80$$

Thus, the average depreciation is $100 - 80 = 20$ per cent.

Example 3.31: The G.M. of four values was calculated as 16. It was later discovered that one of the values was recorded wrongly as 32 when in fact it was 162. Calculate correct G.M.

[Delhi Univ., B.Com., 2004]

Solution: Since $\text{G.M.} = (x_1 x_2 x_3 x_4)^{1/4}$, therefore

$$16 = (x_1 x_2 x_3 x_4)^{1/4} \quad \text{or} \quad x_1 x_2 x_3 x_4 = (16)^4 = 65,536 \text{ (given)}$$

$$\text{Thus} \quad \text{Correct product} = \frac{65,536 \times \text{Correct value}}{\text{Wrong value}} = \frac{65,536 \times 162}{32} = 3,31,776$$

$$\text{Hence, correct Geometric Mean (G.M.)} = (3,31,776)^{1/4} = 24.$$

3.5.2 Combined Geometric Mean

The combined geometric mean of observations formed by pooling the geometric means of different sets of data is defined as

$$\log \text{G.M.} = \frac{\sum_{i=1}^n n_i \log G_i}{\sum_{i=1}^n n_i} \quad (3-11)$$

where G_i is the geometric mean of the i th data set having n_i number of observations.

3.5.3 Weighted Geometric Mean

If different observations x_i ($i = 1, 2, \dots, n$) are given different weights (importance), say w_i ($i = 1, 2, \dots, n$) respectively, then their weighted geometric mean is defined as

$$\text{G.M.}(w) = \text{Antilog} \left[\left(\frac{1}{n} \right) \sum w \log x \right] = \text{Antilog} \left[\left(\frac{1}{\sum w} \right) \sum w \log x \right] \quad (3-12)$$

Example 3.32: Three sets of data contain 8, 7, and 5 observations and their geometric means are 8.52, 10.12 and 7.75, respectively. Find the combined geometric mean of 20 observations.

Solution: Applying the formula (3-12), the combined geometric mean can be obtained as follows

$$\begin{aligned} \text{G.M.} &= \text{Antilog} \left[\frac{n_1 \log G_1 + n_2 \log G_2 + n_3 \log G_3}{n_1 + n_2 + n_3} \right] \\ &= \text{Antilog} \left[\frac{8 \log (8.52) + 7 \log (10.12) + 5 \log (7.75)}{8 + 7 + 5} \right] \\ &= \text{Antilog} \left[\frac{(8 \times 0.9304) + (7 \times 1.0051) + (5 \times 0.8893)}{20} \right] \\ &= \text{Antilog} \left(\frac{18.9254}{20} \right) = \text{Antilog} (0.94627) = 8.835 \end{aligned}$$

Hence, the combined G.M. of 20 observations is 8.835.

Example 3.33: The weighted geometric mean of four numbers 8, 25, 17 and 30 is 15.3. If the weights of the first three numbers are 5, 3 and 4, respectively, find the weight of fourth number.

Solution: Let weight of fourth number be w . Then the weighted geometric mean of four numbers can be calculated as shown in Table 3.21.

Table 3.21 Calculations of Weighted G.M.

Numbers (x)	Weight of Each Number (w)	$\log_{10} x$	$w \log_{10} x$
8	5	0.9031	4.5155
25	3	1.3979	4.1937
17	4	1.2304	4.9216
30	w	1.4771	1.4771 w
	$12 + w$		$13.6308 + 1.4771w$

Thus, the weighted G.M. is

$$\log \{ \text{G.M. } (w) \} = \left[\left(\frac{1}{\sum w} \right) \sum w \log x \right]$$

$$\text{or} \quad \log (15.3) = \left[\left(\frac{1}{12 + w} \right) (13.6308 + 1.4771w) \right]$$

$$(1.1847) (12 + w) = 13.6308 + 1.4771w$$

$$14.2164 + 1.1847w = 13.6308 + 1.4771w$$

$$0.5856 = 0.2924 w \quad \text{or} \quad w = \frac{0.5856}{0.2924} = 2 \text{ (approx.)}$$

Thus, the weight of fourth number is 2.

Example 3.34: A distribution consists of three components with total frequencies of 200, 250 and 300 having means 25, 10 and 15, respectively. Find mean of combined distribution.

[Delhi Univ., B. Com., 2006]

Solution: Given that $n_1 = 200$, $\bar{x}_1 = 25$, $n_2 = 250$, $\bar{x}_2 = 10$ and $n_3 = 300$, $\bar{x}_3 = 15$. The combined mean is given by

$$\begin{aligned} \text{Combined mean, } \bar{x} &= \frac{n_1\bar{x}_1 + n_2\bar{x}_2 + n_3\bar{x}_3}{n_1 + n_2 + n_3} = \frac{200(25) + 250(10) + 300(15)}{200 + 250 + 300} \\ &= \frac{5000 + 2500 + 4500}{750} = \frac{12000}{750} = 16 \end{aligned}$$

Example 3.35: The weighted G.M. of 5 numbers 10, 15, 25, 12 and 20 is 18.15. If the weights of the first four numbers are 2, 3, 5, and 2, respectively, find the weight of the 5th number.

Solution: Let the weight of 5th number be x . Then other calculations are as follows:

x	w	$\log x$	$w \log x$
10	2	1.0000	2.0000
15	3	1.1761	3.5283
25	5	1.3979	6.9895
12	2	1.0792	2.1584
20	x	1.3010	1.3010x
	$12 + x$		$14.6762 + 1.3010x$

$$\text{Weighted G.M.} = \text{Antilog} \left(\frac{\sum w \log x}{\sum w} \right)$$

$$18.15 = \text{Antilog} \left(\frac{14.6762 + 1.3010x}{12 + x} \right)$$

$$\log 18.15 = \frac{14.6762 + 1.3010x}{12 + x}$$

$$1.2343 = \frac{14.6762 + 1.3010x}{12 + x}$$

$$\begin{aligned}
 1.2343(12 + x) &= 14.6751 + 1.3010x \\
 14.8116 + 1.2343x &= 14.6762 + 1.3010x \\
 1.2343x - 1.3010x &= 14.6762 - 14.8116 \\
 -0.0667x &= -0.1354 \text{ or } x = 2.029
 \end{aligned}$$

Thus, the weight of 5th number is 2.029.

Conceptual Questions 3B

10. Define simple and weighted geometric mean of a given distribution. Under what circumstances would you recommend its use?
11. Discuss the advantages, disadvantages and uses of geometric mean.

Self-practice Problems 3B

- 3.9** Find the geometric mean of the following distribution of data:

Dividend : 0–10 10–20 20–30 30–40 40–45
declared (%)

Number of : 5 7 15 25 8
companies

- 3.10** The weighted geometric mean of the four numbers 20, 18, 12 and 4 is 11.75. If the weights of the first three numbers are 1, 3, and 4, respectively, find the weight of the fourth number.

- 3.11** A machinery is assumed to depreciate 44 per cent in value in the first year, 15 per cent in the second year and 10 per cent per year for the next three years. Each percentage being calculated on diminishing value. What is the average percentage of depreciation for the entire period?

- 3.12** A manufacturer of electrical circuit boards, has manufactured the following number of units over the past 5 years:

2000	2001	2002	2003	2004
14,300	15,150	16,110	17,540	19,430

Calculate the average percentage increase in units produced over this time period, and use this to estimate production for 2006.

- 3.13** The owner of a warehouse is calculating the average growth factor for his warehouse over the last 6 years. Using a geometric

mean, he comes up with an answer of 1.42. Individual growth factors for the first 5 years were 1.91, 1.53, 1.32, 1.91 and 1.40, but he lost the records for the sixth year, after he calculated the mean. What was it?

- 3.14** Industrial Gas Supplier keeps records on the cost of processing a purchase order. Over the last 5 years, this cost has been ₹ 355, 358, 361, 365 and 366. What has supplier's average percentage increase been over this period? If this average rate stays the same for 3 more years, what will cost supplier to process a purchase order at that time?

- 3.15** A sociologist has been studying the yearly changes in the number of convicts assigned to the largest correctional facility in the state. His data are expressed in terms of the percentage increase in the number of prisoners (a negative number indicates a percentage decrease). The sociologist's most recent data are as follows:

1999	2000	2001	2002	2003	2004
5%	6%	9%	4%	7%	6%

- (a) Calculate the average percentage increase using only the 1999–2002 data.
- (b) A new penal code was passed in 1998. Previously, the prison population grew at a rate of about 2 per cent per year. What seems to be the effect of the new penal code?

Hints and Answers

3.9 G.M. = 25.64 percent.

3.10 Apply log G.M.

$$\frac{\sum w \log w}{\sum w} = \log 11.75$$

$$\frac{9.4974 + 0.6021w_4}{8 + w_4} = \log 11.75$$

$$\text{or } w_4 = 0.850 \text{ (approx.)}$$

3.11 Depreciation rate : 44 15 10 10 10

Diminishing value

taking 100 as base (x) : 56 85 90 90 90

$$\log x: 1.7582 + 1.9294 + 1.9542 + 1.9542 + 1.9542 \\ = 9.5502$$

$$\text{G.M.} = \text{Antilog } (\Sigma \log x/N)$$

$$= \text{Antilog } (9.5502/5)$$

$$= \text{Antilog } (1.91004) = 81.28$$

The diminishing value is ₹ 81.28 and average depreciation is 18.72 percent.

3.12 G.M.: $\sqrt[4]{19430/14300} = \sqrt[4]{1.3587} = 1.07964$. So the average increase is 7.96 per cent per year. In 2006, the estimated

production will be $19430 (1.0796)^2 = 22,646$ units (approx.)

3.13 Since G.M.

$$1.42 = x \times \sqrt[6]{1.91 \times 1.53 \times 1.32 \times 1.91 \times 1.40} \\ x = (1.42)^6 / (1.91 \times 1.53 \\ \times 1.32 \times 1.91 \times 1.40) \\ = 8.195/10.988 = 0.7458$$

3.14 G.M. = $\sqrt[4]{366/355} = \sqrt[4]{1.0309} = 1.00765$.

So the average increase is 0.765 per cent per year. In three more years, the estimated cost will be $366 (1.00765)^3 = ₹ 757.600$

$$3.15 \text{ (a) G.M.: } \sqrt[4]{0.95 \times 1.06 \times 1.09 \times 1.04} \\ = \sqrt[4]{2.5132} = 1.03364.$$

So the average rate of increase from 1999–2002 was 3.364 per cent per year.

(b) G.M.:

$$\sqrt[6]{0.95 \times 1.06 \times 1.09 \times 1.04 \times 1.07 \times 0.94} \\ = \sqrt[6]{1.148156} = 1.01741. \text{ So the new code appears to have slight effect on the rate of growth of convicts, which has decrease from 2 per cent to 1.741 per cent per year.}$$

3.6 HARMONIC MEAN

The **harmonic mean** (H.M.) of a set of observations is defined as the reciprocal of the arithmetic mean of the reciprocal of the individual observations, that is,

$$\frac{1}{\text{H.M.}} = \frac{1}{n} \sum_{i=1}^n \frac{1}{x_i} \\ \text{or } \text{H.M.} = \frac{n}{\sum_{i=1}^n \left(\frac{1}{x_i} \right)} \quad \text{(For ungrouped data)} \quad (3-13)$$

If $f_1, f_2, \dots, f_n$ are the frequencies of observations $x_1, x_2, \dots, x_n$, then the harmonic mean is defined as

$$\text{H.M.} = \frac{n}{\sum_{i=1}^n f_i \left(\frac{1}{x_i} \right)} \quad \text{(For grouped data)} \quad (3-14)$$

$$\text{where } n = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2}$$

Advantages

- (i) The H.M. of the given data set is also computed based on its every element.
- (ii) While calculating H.M., more weightage is given to smaller values in a data set because in this case, the reciprocal of given values is taken for the calculation of H.M.
- (iii) The original formula of H.M. can be extended to accommodate further analysis of data by certain algebraic manipulations.

Disadvantages

- (i) The H.M. is not often used for analysing business problems.
- (ii) The H.M. of any data set cannot be calculated if it has negative and/or zero elements.
- (iii) The calculation of H.M. involves complicated calculations. For calculating the H.M. of a data set, the largest weight is given to smaller values of elements, therefore it does not represent the true characteristic of the data set.

Applications

The harmonic mean is particularly useful for computation of average rates and ratios. Such rates and ratios are generally used to express relations between two different types of measuring units that can be expressed reciprocally. For example, distance (in km) and time (in hours).

Example 3.36: An investor buys ₹ 20,000 worth of shares of a company each month. During the first 3 months he bought the shares at a price of ₹ 120, ₹ 160 and ₹ 210. After 3 months, what is the average price paid by him for the shares?

Solution: Since the value of shares is changing after every one month, therefore the required average price per share is the harmonic mean of the prices paid in first three months.

$$\begin{aligned} \text{H.M.} &= \frac{3}{(1 \div 120) + (1 \div 160) + (1 \div 210)} = \frac{3}{0.008 + 0.006 + 0.004} \\ &= 3/0.018 = ₹ 166.66 \end{aligned}$$

Example 3.37: Find the harmonic mean of the following distribution of data:

Dividend yield (percent)	: 2–6	6–10	10–14
Number of companies	: 10	12	18

Solution: The calculation of harmonic mean is shown in Table 3.21.

Table 3.22 Calculation of H.M.

Class Intervals (Dividend Yield)	Mid-value (m_i)	Number of Companies (Frequency, f_i)	Reciprocal $\left(\frac{f_i}{m_i}\right)$
2–6	4	10	$10/4 = 2.5$
6–10	8	12	$12/8 = 1.5$
10–14	12	18	$18/12 = 1.5$
		$N = 40$	5.5

$$\text{The harmonic mean is H.M.} = \frac{n}{\sum_{i=1}^3 f_i \left(\frac{1}{m_i} \right)} = \frac{40}{5.5} = 7.27$$

Hence the average dividend yield of 40 companies is 7.27 per cent.

Example 3.38: A certain store's profits were ₹ 50,000, ₹ 1,00,000 and ₹ 8,00,000 in the year 2002, 2003 and 2004, respectively. Determine the average rate of growth of the store's point.

[Delhi Univ., B.Com.(Hons), 2005]

Solution: Using compound interest formula of geometric mean, we have

$$P_n = P_0 (1 + r)^n$$

Since, given that $P_n = 8,00,000$, $P_0 = 50,000$, $r = ?$, $n = 2$, therefore

$$\therefore 8,00,000 = 50,000 (1 + r)^2 \quad \text{or} \quad \frac{8,00,000}{50,000} = (1 + r)^2$$

$$16 = (1 + r)^2 \quad \text{or} \quad 4 = 1 + r, \text{ i.e. } r = 3 \text{ per cent}$$

Example 3.39: A piece of property was purchased for ₹ 20 lakh and sold for ₹ 32.60 lakh after 10 years. What is the average annual rate of return on the original investment?

[Delhi Univ., B. Com.(Pass), 2004]

Solution: Using compound interest formula to calculate average rate of return as follows:

$$P_n = P_0 (1 + r)^n$$

where (given) $P_0 = 20$, $P_n = 32.60$, $n = 10$ and $r = ?$. Thus

$$32.60 = 20 (1 + r)^{10}$$

$$\text{or} \quad \frac{32.60}{20} = (1 + r)^{10}$$

Taking logarithm on both sides, we get

$$\log 32.60 - \log 20 = 10 \log (1 + r)$$

$$\log (1 + r) = \frac{\log 32.60 - \log 20}{10} = \frac{1.5132 - 1.3010}{10} = \frac{0.2122}{10} = 0.02122$$

$$1 + r = \text{Antilog } (0.02122)$$

$$1 + r = 1.050 \quad \text{or} \quad r = 0.05 = 5 \text{ per cent.}$$

Example 3.40: A certain store made weekly profits of ₹ 5000, ₹ 10,000 and ₹ 18,000 in the year 1998, 1999 and 2000, respectively. Determine the average rate of growth of this store's profit.

[GJ Univ., BBA, 2004]

Solution: To determine the average rate of growth, apply the following formula:

$$P_n = P_0 (1 + r)^n$$

Then,

$$18,000 = 5000 (1 + r)^2; \text{ since } P_n = 18,000, \quad P_0 = 5000, \quad n = 2$$

$$(1 + r)^2 = \frac{18,000}{5000} = 3.6$$

$$1 + r = \sqrt{3.6} = 1.897, \text{ i.e. } r = 0.897$$

Thus, average rate of growth is 0.897 per cent.

Example 3.41: The per piece price of banana is ₹ 1.50 and that of orange is ₹ 3. Mr. Ram spends ₹ 10 each on banana and orange. The average price per piece is not ₹ 2.25 but ₹ 2.00. Comment.

Solution: Average price of fruits is $(1.50 + 3)/2$ ₹ 2.25, but it is not true. Since this question involves the ratio of two quantities, we need to applying the formula of H.M. as follows:

$$\text{Average price} = \frac{2}{\frac{1}{x_1} + \frac{1}{x_2}} = \frac{2}{\frac{1}{1.50} + \frac{1}{3}} = ₹ 2$$

Example 3.42: Three men take 12, 8 and 6 hours, respectively, to husk an acre of corn. Determine the average number of hours to husk an acre of corn. [GJ Univ., BBA, 2004]

Solution: The H.M. of digits: 12, 8, 6 is

$$\text{H.M.} = \frac{n}{\Sigma(1/x_i)} = \frac{3}{\frac{1}{12} + \frac{1}{8} + \frac{1}{6}} = \frac{3 \times 576}{216} = 8 \text{ hours.}$$

Thus, the average number of hours to husk an acre is 8 hours.

Example 3.43: If X travels 8 km at 4 km/hr, 6 km at 3 km/hr and 4 km at 2 km/hr, what would be the average rate per hour at which he travelled? Verify your answer.

Solution: Since this question involves the ratio of two quantities—speed and distance, we calculate weighted harmonic mean as shown below:

Speed (x)	Distance (w)	w/x
4	8	$8/4 = 2$
3	6	$6/3 = 2$
2	4	$4/2 = 2$

$$\text{Harmonic Mean} = \frac{\Sigma w}{\Sigma(w/x)} = \frac{18}{6} = 3 \text{ km/hr}$$

Example 3.44: An investor buys ₹ 1200 worth of shares of a company each month. During the first five months he bought the shares at a price of ₹10, ₹12, ₹ 15 ₹ 20 and ₹ 24 per share. After five months, what is the average price paid for the shares by him.

Solution: Since shares are bought with varying prices by investing the same amount of money each time, the average price per share is given by harmonic mean of 10, 12, 15, 20, 24. The formula for H.M. is as follows

$$\text{H.M.} = \frac{n}{\Sigma(1/x_i)} = \frac{5}{\frac{1}{10} + \frac{1}{12} + \frac{1}{15} + \frac{1}{20} + \frac{1}{24}} = \frac{5 \times 120}{41} = ₹ 14.63.$$

Example 3.45: The rate of a certain commodity in the first week is 0.4 kg/rupee. It is 0.6 kg/rupee in second week and is 0.5 kg/rupee in the third week. Is it correct to say that the average price is 0.5 kg/rupee? Justify.

Solution: Since it involves the ratio of two quantities, i.e. kg and rupees, we need to calculate H.M. as shown below:

$$\text{H.M.} = \frac{n}{\Sigma(1/x_i)} = \frac{3}{\frac{1}{0.4} + \frac{1}{0.6} + \frac{1}{0.5}} = \frac{3}{10\left(\frac{1}{4} + \frac{1}{6} + \frac{1}{5}\right)} = 0.482.$$

Thus, average price is 0.482 kg/rupee rather than 0.5 kg/rupee.

Example 3.46: Cities A , B and C are equidistant from each other. A motorist travels from A to B at 30 km/hr, B to C at 40 km/hr and C to A at 50 km/hr. Determine the average speed for the entire trip. [Delhi Univ., B.Com. (Hons.), 2006]

Solution: Average speed is calculated using formula of harmonic mean as follows:

$$\text{H.M.} = \frac{n}{\Sigma(1/x_i)} = \frac{3}{\frac{1}{30} + \frac{1}{40} + \frac{1}{50}} = 38.46 \text{ km/hr}$$

3.7 RELATIONSHIP AMONG A.M., G.M. AND H.M.

For any set of observations, its A.M., G.M. and H.M. are related to each other in the relationship $\text{A.M.} \geq \text{G.M.} \geq \text{H.M.}$. The sign of '=' holds if and only if all the observations are identical.

If observations in a data set take the values $a, ar, ar^2, \dots, ar^{n-1}$, each with single frequency, then $(\text{G.M.})^2 = \text{A.M.} \times \text{H.M.}$

Self-practice Problems 3C

- 3.16** In a certain factory, a unit of work is completed by A in 4 minutes, B in 5 minutes, C in 6 minutes, D in 10 minutes, and E in 12 minute. (a) What is the average rate of completing the work? (b) What is the average number of units of work completed per minute? (c) At this rate, how many units will they complete in a 6-hour day?
- 3.17** An investor buys ₹ 12,000 worth of shares of a company each month. During the first 5 months, he bought the shares at a price of ₹ 100, ₹ 120, ₹ 150, ₹ 200 and ₹ 240 per share, respectively. After 5 months what is the average price paid by him for the shares?
- 3.18** Calculate the A.M., G.M. and H.M. of the following observations and show that $\text{A.M.} > \text{G.M.} > \text{H.M.}$
- | | | | | | | |
|----|----|----|----|----|----|----|
| 32 | 35 | 36 | 37 | 39 | 41 | 43 |
|----|----|----|----|----|----|----|
- 3.19** The profit earned by 18 companies is given below:
- | | | | | | | |
|--------------------|------|----|----|----|----|----|
| Profit (in ₹ lakh) | : 20 | 21 | 22 | 23 | 24 | 25 |
| No. of companies | : 4 | 2 | 7 | 1 | 3 | 1 |
- Calculate the harmonic mean of profit earned.
- 3.20** Find the harmonic mean for the following distribution of data:
- | | | | | |
|----------------|--------|-------|-------|-------|
| Class interval | : 0–10 | 10–20 | 20–30 | 30–40 |
| Frequency | : 5 | 8 | 3 | 4 |

Hints and Answers

- 3.16** (a) Average rate of completing the work per minute = 6.25
 (b) Average units/minute = $1 \div 6.25 = 0.16$
 (c) Units completed in 6-hours (360 minutes) day by all 5 workers = $360 \times 0.16 = 288$ units

3.17 Average price for shares = ₹ 146.30

3.18 A.M. = 37.56; G.M. = 37.52

H.M. = 37.25

3.19 H.M. = ₹ 21.9 lakh

3.20 H.M. = 9.09

3.8 AVERAGES OF POSITION

Different from mathematical averages—arithmetic mean, geometric mean and harmonic mean, which are mathematical in nature and deal with those characteristics of a data set which can be directly measured quantitatively, such as income, profit, level of production and rate of growth. However, in cases where we want to guard against the influence of a few outlying observations (called outliers), and/or we need to measure qualitative characteristics of a data set, such as honesty, intelligence, beauty, consumer acceptance and so on, other measures of central tendency namely *median*, *quartiles*, *deciles*, *percentiles* and *mode* are used. These measures are also called *positional averages*. The term ‘position’ refers to the place of the value of an observation in the data set. These measures help in identifying the value of an observation of interest rather than computing it.

3.8.1 Median

Median may be defined as the *middle value* in the data set when its elements are arranged in a sequential order, that is, in either ascending or descending order of magnitude. It is called a middle value in an ordered sequence of data in the sense that half of the observations are smaller and half of them are larger than this value. The **median** is thus a measure of the *location* or *centrality* of the observations.

The median can be calculated for both ungrouped and grouped data sets.

Ungrouped Data

In this case, the data is arranged in either ascending or descending order of magnitude.

- (i) If the number of observations (n) is an *odd number*, then the median (Med) is represented by the numerical value corresponding to the positioning point of $(n + 1)/2$ ordered observation. That is,

$$\text{Med} = \text{Size or value of } \left(\frac{n+1}{2} \right) \text{th observation in the data array}$$

- (ii) If the number of observations (n) is an *even number*, then the median is defined as the arithmetic mean of the numerical values of $n/2$ th and $(n/2 + 1)$ th observations in the data array. That is,

$$\text{Med} = \frac{\frac{n}{2} \text{th observation} + \left(\frac{n}{2} + 1 \right) \text{th observation}}{2}$$

Example 3.47: Calculate the median of the following data that relates to the service time (in minutes) per customer for 7 customers at a railway reservation counter: 3.5, 4.5, 3, 3.8, 5.0, 5.5, 4

Solution: The data are arranged in ascending order as follows:

Observations in the data array :	1	2	3	4	5	6	7
Service time (in minutes) :	3	3.5	3.8	4	4.5	5	5.5

The median for this data would be

$$\begin{aligned}\text{Med} &= \text{value of } (n + 1)/2\text{th observation in the data array} \\ &= \{(7 + 1) \div 2\}\text{th} = 4\text{th observation in the data array} = 4\end{aligned}$$

Thus, the median service time is 4 minutes per customer.

Example 3.48: Calculate the median of the following data that relates to the number of patients examined per hour in the outpatient ward (OPD) in a hospital: 10, 12, 15, 20, 13, 24, 17, 18

Solution: The data are arranged in ascending order as follows:

Observations in the data array :	1	2	3	4	5	6	7	8
Number of patients :	10	12	13	15	17	18	20	24

Since the number of observations in the data array are even, the average of $(n/2)\text{th} = 4\text{th}$ observation, i.e. 15 and $(n/2) + 1 = 5\text{th}$ observation, i.e., 17 will give the median, that is,

$$\text{Med} = (15 + 17) \div 2 = 16$$

Thus, median number of patients examined per hour in OPD in a hospital is 16.

Grouped Data

To find the median value for grouped data, first identify the class interval which contains the median value or $(n/2)\text{th}$ observation of the data set. To identify such class interval, find the cumulative frequency of each class until the class for which the cumulative frequency is equal to or greater than the value of $(n/2)\text{th}$ observation. The value of the median within that class is found by using interpolation. That is, it is assumed that the observation values are evenly spaced over the entire class interval. The following formula is used to determine the median of grouped data:

$$\text{Med} = l + \frac{(n/2) - cf}{f} \times h$$

where l = lower class limit (or boundary) of the median class interval

cf = cumulative frequency of the class prior to the median class interval, that is, the sum of all the class frequencies upto, but not including the median class interval

f = frequency of the median class

h = width of the median class interval

n = total number of observations in the distribution.

Example 3.49: A survey was conducted to determine the age (in years) of 120 automobiles. The result of such a survey is as follows:

Age of auto :	0-4	4-8	8-12	12-16	16-20
Number of autos :	13	29	48	22	8

What is the median age for the autos?

Solution: Finding the cumulative frequencies to locate the median class as shown in Table 3.23.

Table 3.22 Calculations for Median Value

Age of Auto (in years)	Number of Autos (<i>f</i>)	Cumulative Frequency (<i>cf</i>)
0–4	13	13
4–8	29	42
8–12	48	90 ← Median class
12–16	22	112
16–20	8	120
	<u><i>n</i> = 120</u>	

Here, the total number of observations (frequencies) are $n = 120$. Median is the size of $(n/2)$ th = $120 \div 2 = 60$ th observation in the data set. This observation lies in the class interval 8–12. Applying the formula (3-16), we have

$$\text{Med} = l + \frac{(n/2) - cf}{n} \times h = 8 + \frac{(120 \div 2) - 42}{48} \times 4 = 8 + 1.5 = 9.5$$

Example 3.50: In a factory employing 3000 persons, 5 per cent earn less than ₹ 150 per day, 580 earn from ₹ 151 to ₹ 200 per day, 30 per cent earn from ₹ 201 to ₹ 250 per day, 500 earn from ₹ 251 to ₹ 300 per day, 20 per cent earn from ₹ 301 to ₹ 350 per day, and the rest earn ₹ 351 or more per day. What is the median wage?

Solution: Calculation of median wage per day is shown in Table 3.24.

Table 3.24 Calculation of Median Wage

Earnings (₹)	Percentage of Workers (per cent)	Number of Persons (<i>f</i>)	Cumulative Frequency (<i>cf</i>)
Less than 150	5	150	150
151–200	—	580	730
201–250	30	900	1630 ← Median class
251–300	—	500	2130
301–350	20	600	2730
351 and above	—	270	3000
		<u>3000</u>	

Median observation = $(n/2)$ th = $(3000) \div 2 = 1500$ th observation. This observation lies in the class interval 201–250.

Now applying the formula (3-16), we have

$$\begin{aligned} \text{Med} &= l + \frac{(n/2) - cf}{f} \times h \\ &= 201 + \frac{1500 - 730}{900} \times 50 = 201 + 42.78 = ₹ 243.78 \end{aligned}$$

Hence, the median wage is ₹ 243.78 per day.

3.8.2 Advantages, Disadvantages and Applications of Median

Advantages

- (i) Median is unique, i.e. like mean, there is only one median for a set of data.
- (ii) The value of median is easy to understand and may be calculated from any type of data. The median in many situations can be located simply by inspection.
- (iii) The sum of the absolute differences of all observations in the data set from median value is minimum. In other words, the absolute difference of observations from the median is less than from any other value in the distribution. That is, $\sum |x - \text{Med}| = \text{a minimum value}$.
- (iv) The extreme values in the data set does not affect the calculation of the median value and therefore it is the useful measure of central tendency when such values do occur.
- (v) The median is considered the best statistical technique for studying the qualitative attribute of an observation in the data set.
- (vi) The median value may be calculated for an open-ended distribution of data set.

Disadvantages

- (i) The median is not capable of algebraic treatment. For example, the median of two or more sets of data cannot be determined.
- (ii) The value of median is affected more by sampling variations, that is, it is affected by the number of observations rather than the values of the observations. Any observation selected at random is just as likely to exceed the median as it is to be exceeded by it.
- (iii) Since median is an average of position, therefore arranging the data in ascending or descending order of magnitude is time consuming in case of a large number of observations.
- (iv) The calculation of median in case of grouped data is based on the assumption that values of observations are evenly spaced over the entire class interval.

Applications

The median is helpful in understanding the characteristic of a data set when

- (i) observations are qualitative in nature.
- (ii) extreme values are present in the data set.
- (iii) a quick estimate of an average is desired.

3.9 PARTITION VALUES—QUARTILES, DECILES AND PERCENTILES

The basic purpose of all the measures of central tendency discussed so far was to know more and more about the characteristics of a data set. Another method to analyse a data set is by arranging all the observations in either ascending or descending order of their magnitude and then dividing this ordered series into two equal parts by applying the concept of median. However, to have more knowledge about the data set, we may decompose it into more parts of equal size. The measures of central tendency which are used for dividing the data into several equal parts are called *partition values*.

In this section, we shall discuss data analysis by dividing it into *four*, *ten* and *hundred* parts of equal size and the corresponding partition values are called *quartiles*, *deciles* and *percentiles*. All these values can be determined in the same way as median. The only difference is in their location.

Quartiles The values of observations in a data set, when arranged in an ordered sequence, can be divided into four equal parts, or quarters, using three quartiles namely Q_1 , Q_2 and Q_3 . The first quartile Q_1 divides a distribution in such a way that 25 percent ($=n/4$) of observations have a value less than Q_1 and 75 percent ($= 3n/4$) have a value more than Q_1 , i.e. Q_1 is the median of the ordered values that are below the median.

The second quartile Q_2 has the same number of observations above and below it. It is therefore same as median value.

The quartile Q_3 divides the data set in such a way that 75 percent of the observations have a value less than Q_3 and 25 percent have a value more than Q_3 , i.e. Q_3 is the median of the order values that are above the median.

The generalized formula for calculating quartiles in case of grouped data is

$$Q_i = l + \left[\frac{i(n/4) - cf}{f} \right] \times h; \quad i = 1, 2, 3 \quad (3-15)$$

where cf = cumulative frequency prior to the quartile class interval

l = lower limit of the quartile class interval

f = frequency of the quartile class interval

h = width of the class interval

Deciles The values of observations in a data set when arranged in an ordered sequence can be divided into ten equal parts, using nine deciles, D_i ($i = 1, 2, \dots, 9$). The generalized formula for calculating deciles in case of grouped data is

$$D_i = l + \left[\frac{i(n/10) - cf}{f} \right] \times h; \quad i = 1, 2, \dots, 9 \quad (3-16)$$

where the symbols have their usual meaning and interpretation.

Percentiles The values of observations in a data when arranged in an ordered sequence can be divided into hundred equal parts using ninety nine percentiles, P_i ($i = 1, 2, \dots, 99$). In general, the i th percentile is a number that has i per cent of the data values at or below it and $(100 - i)$ per cent of the data values at or above it. The lower quartile (Q_1), median and upper quartile (Q_3) are also the 25th percentile, 50th percentile and 75th percentile, respectively. For example, if you are told that you scored 90th percentile in a test (like the CAT), it indicates that 90 per cent of the scores were at or below your score, while 10 per cent were at or above your score. The generalized formula for calculating percentiles in case of grouped data is

$$P_i = l + \left[\frac{i(n/100) - cf}{f} \right] \times h; \quad i = 1, 2, \dots, 99 \quad (3-17)$$

where the symbols have their usual meaning and interpretation.

3.9.1 Graphical Method for Calculating Partition Values

The graphical method of determining various partition values can be summarized into following steps:

- (i) Draw an ogive (cumulative frequency curve) by 'less than' method.
- (ii) Take the values of observations or class intervals along the horizontal scale (i.e. x -axis) and cumulative frequency along vertical scale (i.e., y -axis).
- (iii) Determine the median value, that is, value of $(n/2)$ th observation, where n is the total number of observations in the data set.
- (iv) Locate this value on the y -axis and from this point draw a line parallel to the x -axis meeting the ogive at a point, P. Draw a perpendicular on x -axis from P and it meets the x -axis at a point, M.

The other partition values such as quartiles, deciles and percentiles can also be obtained by drawing lines parallel to the x -axis to the distance $i(n/4)$ ($i = 1, 2, 3$); $i(n/10)$ ($i = 1, 2, \dots, 9$) and $i(n/100)$ ($i = 1, 2, \dots, 99$), respectively.

Example 3.51: The following is the distribution of weekly wages of 600 workers in a factory:

<i>Weekly Wages (in ₹)</i>	<i>Number of Workers</i>	<i>Weekly Wages (in ₹)</i>	<i>Number of Workers</i>
Below 875	69	1100 – 1175	58
875 – 950	167	1175 – 1250	24
950 – 1025	207	1250 – 1325	10
1025 – 1100	65	Total	600

- (a) Draw an ogive for the above data and hence obtain the median value. Check it against the calculated value.
- (b) Obtain the limits of weekly wages of central 50 per cent of the workers.
- (c) Estimate graphically the percentage of workers who earned weekly wages between 950 and 1250.

Solution: (a) The calculations of median value are shown in Table 3.25.

Table 3.25 Calculations of Median Value

<i>Weekly Wages (in ₹)</i>	<i>Number of Workers (f)</i>	<i>Cumulative Frequency (Less than type)</i>	<i>Per cent Cumulative Frequency</i>
Less than 875	69	69	11.50
Less than 950	167	236 $\leftarrow Q_1$ class	39.33
Less than 1025	207	443 $\leftarrow$ Median class	73.83
Less than 1100	65	508 $\leftarrow Q_3$ class	84.66
Less than 1175	58	566	94.33
Less than 1250	24	590	98.33
Less than 1325	10	600	100.00

Since a median observation in the data set is the $(n/2)$ th observation = $(600 \div 2)$ th observation, that is, 300th observation. This observation lies in the class interval 950–1025. Applying the formula (3-16) to calculate median wage value, we have

$$\text{Med} = l + \frac{(n/2) - cf}{f} \times h = 950 + \frac{300 - 236}{207} \times 75 = 950 + 23.2 = ₹ 973.2 \text{ per week}$$

The median wage value can also be obtained by applying the graphical method as shown in Fig. 3.1.

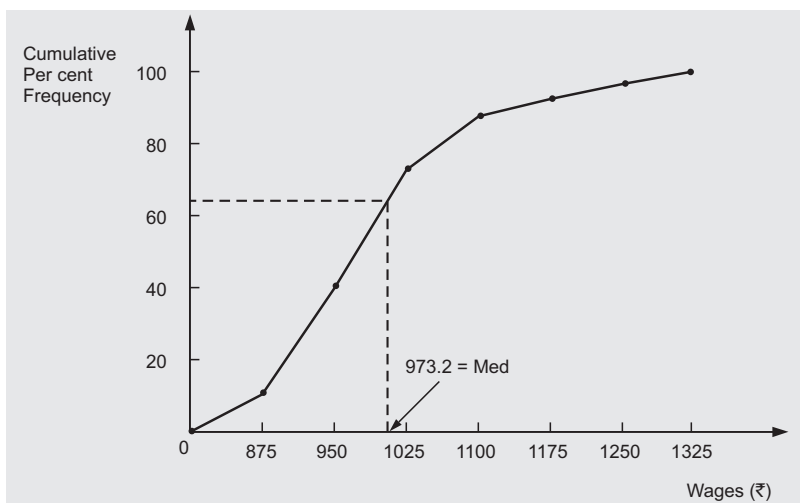


Figure 3.1 Cumulative Frequency Curve

Q_1 = value of $(n/4)$ th observation
 = value of $(600/4)$ th = 150th observation

(b) The limits of weekly wages of central 50 percent of the workers can be calculated by taking the difference of Q_1 and Q_3 . This implies that Q_1 lies in the class interval 875–950. Thus,

$$\begin{aligned}
 Q_1 &= l + \frac{(n/4) - cf}{f} \times h \\
 &= 875 + \frac{150 - 69}{167} \times 75 = 875 + 36.38 = ₹ 911.38 \text{ per week}
 \end{aligned}$$

Similarly, Q_3 = Value of $(3n/4)$ th observation
 = Value of $(3 \times 600/4)$ th = 450th observation

This value of Q_3 lies in the class interval 1025–1100. Thus,

$$\begin{aligned}
 Q_3 &= l + \frac{(3n/4) - cf}{f} \times h \\
 &= 1025 + \frac{450 - 443}{65} \times 75 = 1025 + 8.08 = ₹ 1033.08 \text{ per week}
 \end{aligned}$$

Hence, the limits of weekly wages of central 50 percent workers are ₹ 911.38 and ₹ 1033.08.

(c) The percentage of workers who earned weekly wages less than or equal to ₹ 950 is 39.33 and who earned weekly wages less than or equal to ₹ 1250 is 98.33. Thus, the percentage of workers who earned weekly wages between ₹ 950 and ₹ 1250 is $(98.33 - 39.33) = 59$.

Example 3.52: You are working for the transport manager of a 'call centre' which hires cars for the staff. You are interested in the weekly distances covered by these cars. Kilometers recorded for a sample of hired cars during a given week yielded the following data:

<i>Kilometers Covered</i>	<i>Number of Cars</i>	<i>Kilometers Covered</i>	<i>Number of Cars</i>
100–110	4	150–160	8
110–120	0	160–170	5
120–130	3	170–180	0
130–140	7	180–190	2
140–150	11		<u>40</u>

- (a) Form a cumulative frequency distribution and draw a cumulative frequency ogive.
 (b) Estimate graphically the number of cars which covered less than 165 km in the week.
 (c) Calculate Q_1 , Q_2 , Q_3 and P_{75} .

Solution: (a) The calculations to obtain a cumulative frequency distribution and to draw ogive are shown in Table 3.26.

Table 3.26

<i>Kilometers Covered Less than</i>	<i>Number of Cars</i>	<i>Cumulative Frequency</i>	<i>Per cent Cumulative Frequency</i>
110	4	4	10.0
120	0	4	10.0
130	3	7	17.5
140	7	14 $\leftarrow Q_1$	35.0
150	11	25 $\leftarrow Me = Q_2$	62.5
160	8	33 $\leftarrow Q_3$	82.5 $\leftarrow P_{75}$
170	5	38	95.0
180	0	38	95.0
190	2	40	100.0

Plotting cumulative frequency values on the graph paper, frequency polygon is as shown in Fig. 3.2.

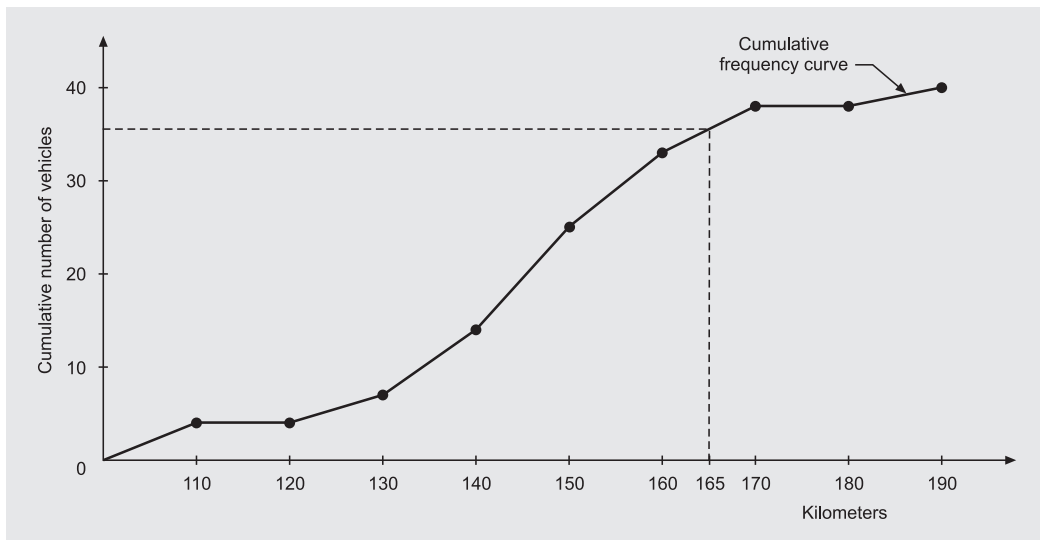


Figure 3.2 Cumulative Frequency Curve

(b) The number of cars which covered less than 165 km in the week are 35 as shown in the Fig. 3.2.

(c) Since there are 40 observations in the data set, we can take 10th, 20th and 30th cumulative values to corresponds to Q_1 , Q_2 and Q_3 , respectively. These values from the graph gives $Q_1 = 134$, $Q_2 = 146$ and $Q_3 = 156$.

$$P_{75} = \frac{(75n/100) - cf}{f} \times h = 150 + \frac{30 - 25}{8} \times 10 = 156.25$$

This implies that 75 per cent of cars covered less than or equal to 156.25 kilometers.

Example 3.53: The following distribution gives the pattern of overtime work per week done by 100 employees of a company. Calculate median, first quartile and seventh decile.

Overtime hours	10–15	15–20	20–25	25–30	30–35	35–40
No. of employees	11	20	35	20	8	6

Calculate Q_1 , D_7 and P_{60} .

Solution: The calculations of median, first quartile (Q_1) and seventh decile (D_7) are shown in Table 3.27.

Table 3.27

Overtime Hours	Number of Employees	Cumulative Frequency (Less than type)
10–15	11	11
15–20	20	31 $\leftarrow Q_1$ class
20–25	35	66 $\leftarrow$ Median and P_{60} class
25–30	20	86 $\leftarrow D_7$ class
30–35	8	94
35–40	6	100
	<u>100</u>	

Since the number of observations in this data set are 100, the median value is $(n/2)$ th = $(100 \div 2)$ th = 50th observation. This observation lies in the class interval 20–25. Applying the formula (3-16) to get median overtime hours value, we have

$$\begin{aligned} \text{Med} &= l + \frac{(n/2) - cf}{f} \times h \\ &= 20 + \frac{50 - 31}{35} \times 5 = 20 + 2.714 = 22.714 \text{ hours} \end{aligned}$$

Q_1 = value of $(n/4)$ th observation = value of $(100/4)$ th = 25th observation

$$= l + \frac{(n/4) - cf}{f} \times h = 15 + \frac{25 - 11}{20} \times 5 = 15 + 3.5 = 18.5 \text{ hours}$$

Also D_7 = value of $(7n/10)$ th observation = value of $(7 \times 100)/10 = 70$ th observation

$$= l + \frac{(7n/100) - cf}{f} \times h = 25 + \frac{70 - 66}{20} \times 5 = 25 + 1 = 26 \text{ hours}$$

P_{60} = Value of $(60n/100)$ th observation = $60 \times (100/100) = 60$ th observation

$$= l + \frac{(60 \times n/100) - cf}{f} \times h = 20 + \frac{60 - 31}{35} \times 5 = 24.14 \text{ hours}$$

Conceptual Questions 3C

12. Define median and discuss its advantages and disadvantages.
13. When is the use of median considered more appropriate than mean?
14. Write a short criticism of the following statement: 'Median is more representative than mean because it is relatively less affected by extreme values'.
15. What are quartiles of a distribution? Explain their uses.
16. It has been said that the same percentage of frequencies falls between the first and ninth decile for symmetric and skewed distributions. Criticize or explain this statement. Generalize your answer to other percentiles.
17. Describe the similarities and differences among median, quartiles and percentiles as descriptive measures of position.

Self-practice Problems 3D

- 3.21 On a university campus 200 teachers are asked to express their views on how they feel about the performance of their Union's president. The views are classified into the following categories:

Disapprove strongly	= 94
Disapprove	= 52
Approve	= 43
Approve strongly	= 11

What is the median view?

- 3.22 The following are the profit figures earned by 50 companies in the country:

Profit (in ₹ lakh)	Number of Companies
10 or less	4
20 or less	10
30 or less	30
40 or less	40
50 or less	47
60 or less	50

Calculate

- (a) the median
 - (b) the range of profit earned by the middle 80 per cent of the companies. Also verify your results by graphical method.
- 3.23 A number of particular items has been classified according to their weights. After drying for two weeks the same items have

again been weighted and similarly classified. It is known that the median weight in the first weighing was 20.83 g, while in the second weighing it was 17.35 g. Some frequencies, a and b , in the first weighing and x and y in the second weighing are missing. It is known that $a = x/3$ and $b = y/2$. Find out the values of the missing frequencies.

Class	Frequencies		Class	Frequencies	
	I	II		I	II
0–5	a	x	15–20	52	50
5–10	b	y	20–25	75	30
10–15	11	40	25–30	22	28

- 3.24 The length of time taken by each of 18 workers to complete a specific job was observed to be the following:

Time (in min):	5–9	10–14	15–19	20–24	25–29
Number of workers	: 3	8	4	2	1

(a) Calculate the median time

(b) Calculate Q_1 and Q_3

- 3.25 The following distribution is with regard to weight (in g) of mangoes of a given variety. If mangoes less than 443 g in weight be considered unsuitable for the foreign market, what is the percentage of total yield

suitable for it? Assume the given frequency distribution to be typical of the variety.

Weight (in g)	Number of Mangoes	Weight (in g)	Number of Mangoes
410–419	10	450–459	45
420–429	20	460–469	18
430–439	42	470–479	7
440–449	54		

Draw an ogive of ‘more than’ type of the above data and deduce how many mangoes will be more than 443 g.

3.26 Given the following frequency distribution with some missing frequencies:

Class	Frequency	Class	Frequency
10–20	185	50–60	136
20–30	—	60–70	—
30–40	34	70–80	50
40–50	180		

If the total frequency is 685 and median is 42.6, find out the missing frequencies.

Hints and Answers

3.21 Disapprove

3.22 $\text{Med} = 27.5$; $P_{90} - P_{10} = 47.14 - 11.67 = 35.47$

3.23 $a = 3, b = 6; x = 9, y = 12$

3.24 (a) 13.25 (b) $Q_3 = 17.6, Q_1 = 10.4$

3.25 52.25 per cent; 103

3.26 20–30(77)

3.10 MODE

The **mode** is that value of an observation which occurs most frequently in the data set, that is, the point (or class mark) with the highest frequency.

The concept of mode is of great use to large scale manufacturers of consumable items such as ready-made garments, shoe-makers, and so on. In all such cases, it is important to know the size that fits most persons rather than ‘mean’ size.

The mode is a poor measure of central tendency when most frequently occurring values of an observation do not appear close to the centre of the data. The mode need not even be a unique value. Consider the frequency distributions shown in Fig. 3.3(a) and (b). The distribution in Fig. 3.3(a) has its mode at the lowest class and certainly cannot be considered representative of central location. The distribution shown in Fig. 3.3(b) has two modes. Obviously neither of these values appear to be representative of the central location of the data. For these reasons, the mode has limited use as a measure of central tendency for decision-making. However, for descriptive analysis, mode is a useful measure of central tendency.

Calculation of Mode It is always preferable to calculate mode from grouped data. Table 3.28, for example, shows the sales per day of an item for 20 days period. The mode of this data is 71 since this value occurs more frequently (four times than any other value). However, it fails to reveal the fact that most of the values are under 70.

Table 3.28 Sales During 20 Days Period
(Data arranged in ascending order)

53,	56,	57,	58,	58,	60,	61,	63,	63,	64
64,	65,	65,	67,	68,	71,	71,	71,	71,	74

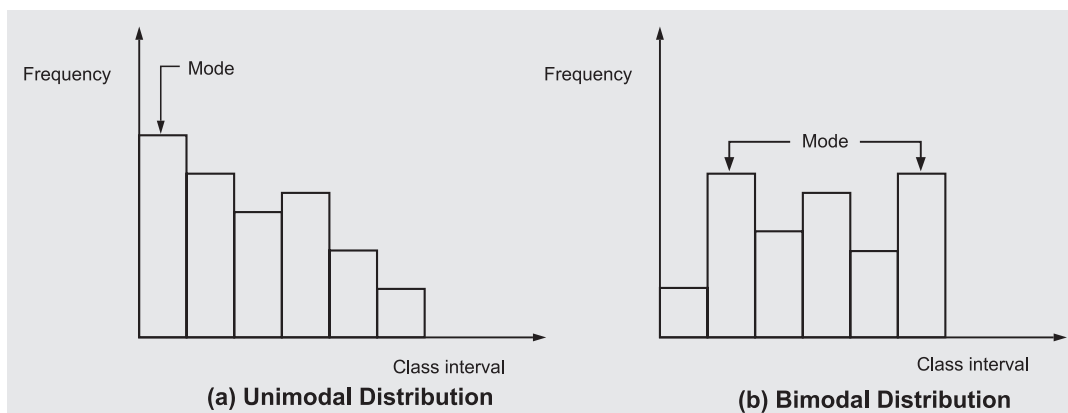


Figure 3.3 Frequency Distribution

Converting this data into a frequency distribution as shown in Table 3.29:

Table 3.29 Frequency Distribution of Sales Per Day

<i>Sales volume</i> (Class interval)	:	53–56	57–60	61–64	65–68	69–72	72 and above
<i>Number of days</i> (Frequency)	:	2	4	5	4	4	1

Table 3.29 shows that a sale of 61–64 units of the item was achieved on 5 days. Thus, this class is more representative of the sales per day.

In the case of grouped data, the following formula is used for calculating mode:

$$\text{Mode} = l + \frac{f_m - f_{m-1}}{2f_m - f_{m-1} - f_{m+1}} \times h$$

where

- l = lower limit of the modal class interval
- f_{m-1} = frequency of the class preceding the modal class interval
- f_{m+1} = frequency of the class following the modal class interval
- h = width of the modal class interval

Example 3.54: Find the missing frequency for the following incomplete distribution by using appropriate formula when mode is 36.

Marks	No. of Students
0 – 10	5
10 – 20	7
20 – 30	–
30 – 40	–
40 – 50	10
50 – 60	6
	<u>50</u>

Solution: Let the missing frequency corresponding to class 20 – 30 be x . Then missing frequency corresponding to class 30 – 40 will be $50 - (5 + 7 + x + 10 + 6) = 22 - x$

Since mode is 36, therefore modal class is 30 – 40. Then

$$\text{Mode} = l + \frac{f_m - f_{m-1}}{2f_m - f_{m-1} - f_{m+1}} \times h = 30 + \frac{22 - x - x}{2(22 - x) - x - 10} \times 10$$

$$36 = 30 + \frac{22 - 2x}{34 - 3x} \times 10$$

$$6(34 - 3x) = (22 - 2x)(10)$$

$$204 - 18x = 220 - 20x$$

$$2x = 16, \text{ i.e., } x = 8$$

Missing frequency is $x = 8$ and $22 - x = 22 - 8 = 14$.

Example 3.55: Median and mode are not affected by extreme values. Give examples in support of your answer. [Delhi Univ., B.Com. (Hons.), 2006]

Solution: For calculating median consider the following five figures: 2500, 3000, 4000, 4500 and 11,500

$$\text{Median value} = \text{Size of } \left(\frac{n+1}{2} \right)^{\text{th}} \text{ item} = \text{Size of 3rd item} = 4000$$

If values of the first and last figure are changed to 1500 and 9500, then median value again be 3500. This is because position of the third observation remains unchanged.

For modal value, consider an example : 115, 120, 125, 130, 135, 140, 130, 130, 142 and 132. The value 130 is occurring more number of times than others. Thus, mode value does not change with the change in extreme values:

Example 3.56: Using the data of Table 3.28, calculate the mode of sales distribution of the units of item during the 20 days period.

Solution: Since the largest frequency corresponds to the class interval 61–64, therefore it is the mode class. Then, we have, $l = 61, f_m = 5, f_{m-1} = 4, f_{m+1} = 4$ and $h = 3$. Thus,

$$\begin{aligned} M_0 &= l + \frac{f_m - f_{m-1}}{2f_m - f_{m-1} - f_{m+1}} \times h \\ &= 61 + \frac{5 - 4}{10 - 4 - 4} \times 3 = 61 + 1.5 = 62.5 \end{aligned}$$

Hence, the modal sale is of 62.5 units.

Example 3.57: In 500 small-scale industrial units, the return on investment ranged from 0 to 30 percent; no unit sustaining loss. 5 percent of the units had returns ranging from 0 percent to (and including) 5 percent, and 15 percent of the units earned returns exceeding 5 percent but not exceeding 10 percent. The median rate of return was 15 percent and the upper quartile 2 percent. The uppermost layer of returns exceeding 25 percent was earned by 50 units.

(a) Present the information in the form of a frequency table as follows:

Exceeding 0 percent but not exceeding 5 percent

Exceeding 5 percent but not exceeding 10 percent

Exceeding 10 percent but not exceeding 15 percent

and so on.

(b) Find the rate of return around which there is maximum concentration of units.

Solution: (a) The given information is summarized in the form of a frequency distribution as shown in Table 3.30.

Table 3.30

<i>Rate of Return</i>	<i>Industrial Units</i>
Exceeding 0 percent but not exceeding 5 percent	$500 \times \frac{5}{100} = 25$
Exceeding 5 percent but not exceeding 10 percent	$500 \times \frac{5}{100} = 75$
Exceeding 10 percent but not exceeding 15 percent	$250 - 100 = 150$
Exceeding 15 percent but not exceeding 20 percent	$375 - 250 = 125$
Exceeding 20 percent but not exceeding 25 percent	$500 - 375 - 50 = 75$
Exceeding 25 percent but not exceeding 30 percent	50

(b) Calculating mode to find out the rate of return around which there is maximum concentration of the units. The mode lies in the class interval 10–15. Thus 0

$$\begin{aligned}
 M_0 &= l + \frac{f_m - f_{m-1}}{2f_m - f_{m-1} - f_{m+1}} \\
 &= 10 + \frac{150 - 75}{2 \times 150 - 75 - 125} \times 5 = 10 + 3.75 = 13.75 \text{ percent}
 \end{aligned}$$

3.10.1 Graphical Method for Calculating Mode Value

The procedure of calculating mode using the graphical method is summarized below:

- Draw a histogram of the data, the tallest rectangle will represent the modal class.
- Draw two diagonal lines from the top right corner and left corner of the tallest rectangle to the top right corner and left corner of the adjacent rectangles.
- Draw a perpendicular line from the point of intersection of the two diagonal lines on the x -axis. The value on the x -axis marked by the line will represent the modal value.

Example 3.58: Calculate the mode using the graphical method for the following distribution of data:

Sales (in units)	:	53–56	57–60	61–64	65–68	69–72	73–76
Number of days	:	2	4	5	4	4	1

Solution: Construct a histogram of the data shown in Fig. 3.4 and draw other lines for the calculation of mode value.

The mode value from Fig. 3.4 is 62.5 which is same as calculated in Example 3.37.

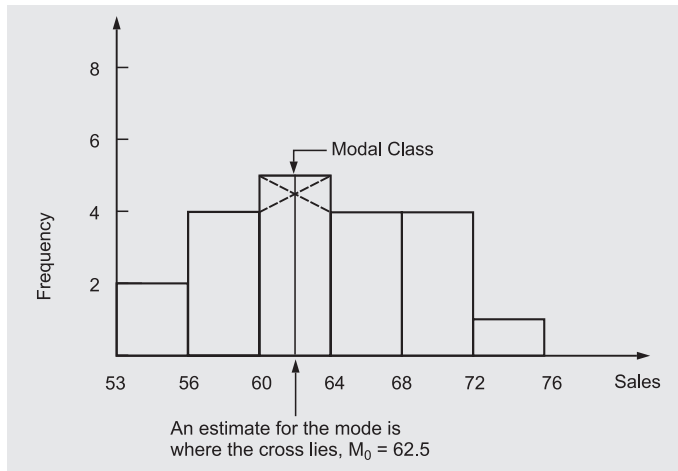


Figure 3.4 Graph For Modal Value

3.10.2 Advantages and Disadvantages of Mode Value

Advantages

- (i) Mode value is easy to understand and calculate. Mode class can also be located by inspection.
- (ii) The mode is not affected by the extreme values in the distribution and can also be calculated for open-ended frequency distributions.
- (iii) The mode can be used to describe quantitative as well as qualitative data. For example, its value is used for comparing consumer preferences for various types of products, for example cigarettes, soaps, toothpastes or other products.

Disadvantages

- (i) Mode is not a rigidly defined measure as there are several methods for calculating its value.
- (ii) It is difficult to locate modal class in the case of multi-modal frequency distributions.
- (iii) Mode is not suitable for algebraic manipulations.
- (iv) When data sets contain more than one modes, such values are difficult to interpret and compare.

3.11 RELATIONSHIP BETWEEN MEAN, MEDIAN AND MODE

In an *unimodal* and a symmetrical distribution, the values of mean, median and mode are equal as indicated in Fig. 3.5. In other words, when all these three values are not equal to each other, the distribution is not symmetrical.

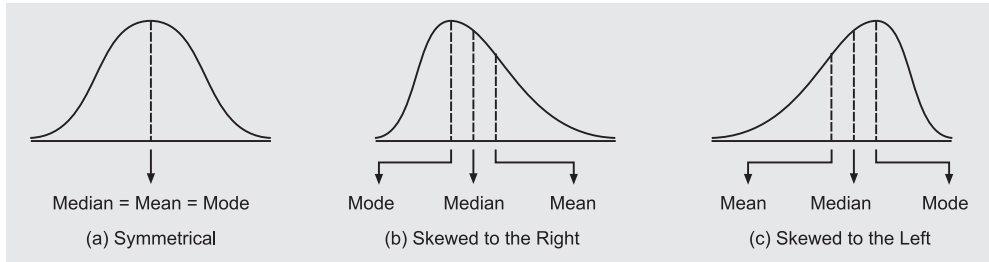


Figure 3.5 A Comparison of Mean, Median and Mode for Three Distributional Shapes

A distribution that is not symmetrical, but rather has most of its values either to the right or the left of the mode is said to be *skewed*. For such asymmetrical distribution, Karl Pearson has suggested a relationship between these three measures of central tendency as

$$\text{Mean} - \text{Mode} = 3 (\text{Mean} - \text{Median}) \quad (3-18)$$

or

$$\text{Mode} = 3 \text{ Median} - 2 \text{ Mean}$$

This implies that the value of any of these three measures can be calculated provided we know any two values out of three. The relationship (3-18) is shown in Figs. 3.5(b) and (c).

If most of the values of observations in a distribution fall to the right of the mode as shown in Fig. 3.5(b), then it is said to be skewed to the right or *positively skewed*. Distributions that are skewed right contain a few unusually large values of observations. In this case, mode remains under the peak (i.e., representing highest frequency) but the median (value that depends on the number of observations) and mean move to the right (value that is affected by extreme values). The order of magnitude of these measures will be

$$\text{Mean} > \text{Median} > \text{Mode}$$

But if the distribution is skewed to the left or *negatively skewed* (i.e., values of lower magnitude are concentrated more to the left of the mode) then mode is again under the peak whereas median and mean move to the left of mode. The order of magnitude of these measures will be

$$\text{Mean} < \text{Median} < \text{Mode}$$

In both these cases, the difference between mean and mode is three times the difference between mean and median.

In general, for a single-peaked skewed distribution (non-symmetrical), the median is preferred to the mean for measuring location because it is neither influenced by the frequency of occurrence of a single observation value as mode nor it is affected by extreme values.

3.12 COMPARISON BETWEEN MEASURES OF CENTRAL TENDENCY

In this chapter, we have already presented three methods to understand the characteristics of a data set. However, the choice of which method to use for describing a distribution of values of observations in a data set is not always easy. The choice to use any one of these three is mainly guided by their characteristics. The characteristics of these three differ from each other with regard to following three factors:

1. **The Presence of Outlier Data Values:** The data values that differ in a big way from the other values in a data set are known as *outliers* (either very small or very high values). As mentioned earlier, the median is not sensitive to outlier values because its value depends only on the number of observations and the value always lies in the middle of the ordered set of values, whereas mean, which is calculated using all data values, is sensitive to the outlier values in a data set. Obviously, smaller the number of observations in a data set, greater the influence of any outliers on the mean. The median is said to be *resistant* to the presence of outlier data values, but the mean is not.
2. **Shape of Frequency Distribution:** The effect of the shape of frequency distribution on mean, median and mode is shown in Fig. 3.5. In general, the median is preferred to the mean as a way of measuring location for single peaked, skewed distributions. One of the reasons is that it satisfies the criterion that the *sum of absolute* difference (i.e., absolute error of judgment) of median from values in the data set is minimum, that is, $\sum |x - \text{Med}| = \text{Min}$. In other words, the smallest sum of the absolute errors is associated with the median value is the data set as compared to either mean or mode. When data is multi-modal, there is no single measure of central location and the mode can vary dramatically from one sample to another, particularly when dealing with small samples.
3. **The Status of Theoretical Development:** Although the three measures of central tendency, mean, median, and mode, satisfy different mathematical criteria but the objective of any statistical analysis in *inferential statistics* is always to minimize the *sum of squared deviations (errors)* taken from these measures to every value in the data set. The criterion of the sum of squared deviations is also called *least squares criterion*. Since A.M. satisfies the least squares criterion, it is mathematically consistent with several techniques of statistical inference.

As with the median, it can not be used to develop theoretical concepts and models and so is only used for basic descriptive purposes.

Conceptual Questions 3D

18. Give a brief description of the different measures of central tendency. Why is arithmetic mean so popular?
19. What are the advantages and disadvantages of the three common averages: mean, median and mode?
20. Identify the mathematical criteria associated with mean, median and mode and briefly explain the meaning of each criterion.
21. What is a statistical average? What are the desirable properties for an average to possess?
22. Mention the different types of averages and state why arithmetic mean is most commonly used amongst them.
22. What is the relationship between mean, median and mode? Under what circumstances are they equal?
23. Which measure of central tendency is usually preferred if the distribution is known to be single peaked and skewed? Why?

Self-practice Problems 3E

- 3.27** Given below is the distribution of profits (in '000 rupees) earned by 94 per cent of the retail grocery shops in a city.

<i>Profits</i>	<i>Number of Shops</i>	<i>Profit</i>	<i>Number of Shops</i>
0–10	0	50–60	68
10–20	5	60–70	83
20–30	14	70–80	91
30–40	27	80–90	94
40–50	48		

Calculate the modal value.

- 3.28** Compute mode value from the following data relating to dividend paid by companies in a particular financial year.

<i>Dividend (in percent)</i>	<i>Number of Companies (in percent)</i>	<i>Dividend</i>	<i>Number of Companies</i>
5.0–7.5	182	15.0–17.5	280
7.5–10.0	75	17.5–20.0	236
10.0–12.5	59	20.0–22.5	378
12.5–15.0	127	22.5–25.0	331

- 3.29** Following is the cumulative frequency distribution of the preferred length of kitchen slabs obtained from the preference study on 50 housewives:

<i>Length (metres) (more than)</i>	<i>Number of Housewives</i>
1.0	50
1.5	46
2.0	40
2.5	42
3.0	10
3.5	3

A manufacturer has to take a decision on what length of slabs to manufacture. What length would you recommend and why?

- 3.30** The number of solar heating systems available to the public is quite large, and their heat storage capacities are quite varied. Here is a distribution of heat storage capacity (in days) of 28 systems that were tested recently by a testing agency

<i>Days</i>	<i>Frequency</i>	<i>Days</i>	<i>Frequency</i>
0–0.99	2	4–4.99	5
1–1.99	4	5–5.99	3
2–2.99	6	6–6.99	1
3–3.99	7		

The agency knows that its report on the tests will be widely circulated and used as the basis for solar heat allowances.

Compute the mean, median and mode of these data.

- 3.31** Pandey does statistical analysis for an automobile racing team. The data on fuel consumption (in km per litre) for the team's cars in recent races are as follows:

14.77 16.11 15.05 15.99 14.91
15.27 16.01 15.75 14.89 16.05 15.22
16.02 15.24 16.11 15.02

- Calculate the mean and median fuel consumption.
- Group the data into five equally sized classes. What is the fuel consumption value of the modal class?
- Which of the three measures of central tendency is best to use? Explain.

- 3.32** An agriculture farm sells grab bags of flower bulbs. The bags are sold by weight; thus the number of bulbs in each bag can vary depending on the varieties included. Below are the number of bulbs in each of the 20 bags sampled:

21 33 37 56 47 25 33 32 47 34
36 23 26 33 37 26 37 37 43 45

- What are the mean and median number of bulbs per bag?
- Based on your answer, what can you conclude about the shape of the distribution of number of bulbs per bag?

- 3.33** The table below is the frequency distribution of ages to the nearest birthday for a random sample of 50 employees in a large company.

Age to nearest birthday	20-29	30-39	40-49	50-59	60-69
Number of employees	5	12	13	8	12

Compute the mean, median and mode for these data.

Hints and Answers

3.27 $M_0 = 3 \text{ Med} - 2\bar{x} = ₹ 50.04 \text{ thousand}$

3.28 $M_0 = 21.87$ (percent of the share value)

Formulae Used

1. Summation of n numbers

$$\sum_{i=1}^n x_i = x_1 + x_2 + \dots + x_n$$

Simplified expression for the summation of n numbers

$$\sum x_i = x_1 + x_2 + \dots + x_n$$

2. Sample mean, $\bar{x} = \frac{\sum x_i}{n}$

$$\text{Population mean, } \mu = \frac{\sum x_i}{N}$$

Sample mean for grouped data,

$$\bar{x} = \frac{\sum f_i m_i}{n}$$

where $n = \sum f_i$ and m_i = mid-value of class intervals

3. Weighted mean for a population or a sample

$$\sum f_i \text{ or } \bar{x}_w = \frac{\sum w_i x_i}{\sum w_i}$$

where w_i = weight for observation i

4. Position of the median in an ordered set of observation belong to a population or a sample is $\text{Med} = x_{(n/2) + (1/2)}$

Median for grouped data,

$$\text{Med} = l + \left[\frac{(n/2) - cf}{f} \right] h$$

5. Quartile for a grouped data

$$Q_i = l + \left[\frac{i(n/4) - cf}{f} \right] h; \quad i = 1, 2, 3$$

Decile for a grouped data

$$D_i = l + \left[\frac{i(n/10) - cf}{f} \right] h; \quad i = 1, 2, \dots, 9$$

Percentile for a grouped data

$$P_i = l + \left[\frac{i(n/100) - cf}{f} \right] h; \quad i = 1, 2, \dots, 99$$

6. Mode for a grouped data

$$M_0 = l + \left[\frac{f_m - f_{m-1}}{2f_m - f_{m-1} - f_{m+1}} \right] h$$

Mode for a multimode frequency distribution

$$M_0 = 3 \text{ Median} - 2 \text{ Mean}$$

True or False Concepts Quiz

- Inferential statistics are used to describe specific characteristics of the data.
- With nominal data, the mean should be used as a measure of central tendency.
- With ordinal data, we can use both the mode and the mean as a measure of central tendency.
- When the data are interval or ratio, we can use the mean as a measure of central tendency.

5. Harmonic mean is the reciprocal of arithmetic mean.
6. Sum of absolute deviations from median is minimum.
7. The mean of a data set remains unaffected if an observation equal to mean is included in it.
8. With continuous data, the median is the most appropriate measure of central tendency.
9. Weighted mean is useful in problems relating to the construction of index numbers and standardized birth and death rates.
10. It is possible to have data with three different values for measures of central tendency.
11. The median is less affected than the mean by extreme values of observations in a distribution.
12. The sum of deviations from mean is zero.
13. If the number of observations is even, the median is in the middle of the distribution.
14. The mode is always found at the highest point of a graph of a frequency distribution.
15. For grouped data, it is possible to calculate an approximate mean by assuming that each value in a given class is equal to its mid-point.

Concepts Quiz Answers

1. F	2. F	3. F	4. T	5. F	6. T	7. T	8. F	9. T
10. T	11. T	12. T	13. F	14. T	15. T			

Review Self-practice Problems

- 3.35** The following is the data on profit margin (in percent) of three products and their corresponding sales (in ₹) during a particular period.

Product	Profit Margin (percent)	Sales (₹ in thousand)
A	12.5	2,000
B	10.3	6,000
C	6.4	10,000

- (a) Determine the mean profit margin.
 - (b) Determine the weighted mean considering the rupee sales as weight for each product.
 - (c) Which of the means calculated in part (a) and (b) is the correct one?
- 3.36** The number of cars sold by each of the 10 car dealers during a particular month, arranged in ascending order, is 12, 14, 17, 20, 20, 20, 22, 22, 24, 25. Considering this scale to be the statistical population of interest, determine the mean, median, and mode for the number of cars sold.
- (a) Which value calculated above best describes the 'typical' sales volume per dealer?
 - (b) For the given data, determine the values at the (i) quartile Q_1 and (ii) percentile P_{30} for these sales amounts.
- 3.37** A quality control inspector tested nine samples of each of three designs A, B and C of certain bearing for a new electrical winch. The following data are the number of hours, it took for each bearing to fail when the winch motor was run continuously at maximum output, with a load on the winch equivalent to 1.9 times the intended capacity.
- A : 16 16 53 15 31 17 14 30 20
 B : 18 27 23 21 22 26 39 17 28
 C : 31 16 42 20 18 17 16 15 19
- Calculate the mean and median for each group and suggest which design is best and why?
- 3.38** The following are the weekly wages in rupees of 30 workers of a firm:
- 140 139 126 114 100 88 62 77 99
 103 108 129 144 148 134 63 69 148
 132 118 142 116 123 104 95 80 85
 106 123 133

The firm gave bonus of ₹ 10, 15, 20, 25, 30, and 35 for individuals in the respective salary slabs: exceeding 60 but not exceeding 75; exceeding 75 but not exceeding 90; and so on up to exceeding 135 and not exceeding 150. Find the average bonus paid.

- 3.39** The mean monthly salaries paid to 100 employees of a company was ₹ 5,000. The mean monthly salaries paid to male and female employees were ₹ 5,200 and ₹ 4,200, respectively. Determine the percentage of males and females employed by the company.

- 3.40** A factory pays workers on piece rate basis and also a bonus to each worker on the basis of individual output in each quarter. The rate of bonus payable is as follows:

Output (in units)	Bonus (₹)	Output (in units)	Bonus (₹)
70–74	40	90–94	70
75–79	45	95–99	80
80–84	50	100–104	100
85–89	60		

The individual output of a batch of 50 workers is given below:

94 83 78 76 88 86 93 80 91 82
89 97 92 84 92 80 85 83 98 103
87 88 88 81 95 86 99 81 87 90
84 97 80 75 93 101 82 82 89 72
85 83 75 72 83 98 77 87 71 80

By suitable classification you are required to find:

- Average bonus per worker for the quarter,
- Average output per worker.

- 3.41** An economy grows at the rate of 2 per cent in the first year, 2.5 per cent in the second year, 3 per cent in the third year, 4 per cent

in the fourth year ... and 10 per cent in the tenth year. What is the average rate of growth of the company?

- 3.42** A man travelled by car for 3 days. He covered 480 km each day. On the first day he drove for 10 hours at 48 km an hour, on the second day he drove for 12 hours at 40 km an hour, and on the last day he drove for 15 hours at 32 km per hour. What was his average speed?

[Bangalore Univ., B.Com., 1996]

- 3.43** The price of a certain commodity in the first week of January is 400 g per rupee; it is 600 g per rupee in the second week and 500 g per rupee in the third week. Is it correct to say that the average price is 500 g per rupee? Verify.

- 3.44** Find the missing information in the following table:

	A	B	C	Combined
Number	10	8	—	24
Mean	20	—	6	15
Geometric Mean	10	7	—	8.397

[Delhi Univ., B.Com.(Hons.), 1998]

- 3.45** During a period of decline in stock market prices, a stock is sold at ₹ 50 per share on one day, ₹ 40 on the next day, and ₹ 25 on the third day.

- If an investor bought 100, 120, and 180 shares on the respective three days, find the average price paid per share.
- If the investor bought ₹ 1000 worth of shares on each of the three days, find the average price paid per share.

[Delhi Univ., BA (Hons Eco.), 1998]

Hints and Answers

$$3.35 \text{ (a) } \mu = \frac{\sum x_i}{N} = \frac{29.2}{3} = 9.73 \text{ percent}$$

$$(b) \mu_w = \frac{\sum w_i x_i}{\sum w_i} = \frac{1,50,800}{18,000} = 8.37 \text{ percent}$$

- (c) The weighted mean of 8.37 percent considering sales (in ₹) as weights is the correct mean profit margin. Percentages should never be averaged without being weighted.

$$3.36 \mu = \frac{\sum x_i}{N} = \frac{196}{10} = 19.6$$

$$\text{Med} = \frac{\left(\frac{n}{2}\right) + \left(\frac{n}{2} + 1\right)}{2} = \frac{5\text{th} + 6\text{th}}{2} = 20.0$$

M_0 = most frequent value = 20

- (a) Median is best used as the 'typical' value because of the skewness in the distribution of values

$$(b) Q_1 = x_{(n/4) + (1/2)} = x_{(10/4) + (1/2)} = x_{3.0} = 17$$

$$P_{30} = x_{(3n/10) + (1/2)} = x_{3.5} = 17 + 0.5(20 - 17) = 18.5$$

- 3.37 Listing the data in ascending order:

A : 14 15 16 16 17 20 30 31 53

B : 17 18 21 22 23 26 27 28 39

C : 15 16 16 17 18 19 20 31 42

$$\bar{x}_A = 212/9 = 23.56; \text{ Med (A)} = 17$$

$$\bar{x}_B = 221/9 = 24.56; \text{ Med (B)} = 23$$

$$\bar{x}_C = 194/9 = 21.56; \text{ Med (C)} = 18$$

Since medians are the fifth observation in each data set, therefore design B is best because both the mean and median are highest.

- 3.38 Prepare a frequency distribution as follows:

Weekly Wages (₹)	Frequency (f)	Bonus Paid (x)	Weekly Wages (₹)	Frequency (f)	Bonus Paid (x)
61-75	3	10	106-120	5	25
76-90	4	15	121-135	7	30
91-105	5	20	136-150	6	35

$$\text{Average bonus paid} = \frac{\sum fx}{n} = \frac{375}{30} = ₹ 24.5$$

$$3.39 \text{ Given } n_1 + n_2 = 100, \bar{x}_{12} = 5000, \bar{x}_1 = 5200 \text{ and } \bar{x}_2 = 4200$$

$$\bar{x}_{12} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2}$$

$$\text{or } 5000 = \frac{n_1 (5200) + n_2 (4200)}{n_1 + n_2} = \frac{n_1 (5200) + (100 - n_1) (4200)}{100}$$

$$1000n_1 = 80,000 \text{ or } n_1 = 80 \text{ and } n_2 = 100 - n_1 = 20$$

Output (in units)	Frequency (f)	Bonus (₹)	Output (in units)	Frequency (f)	Bonus (₹)
70-74	3	40	90-94	7	70
75-79	5	45	95-99	6	80
80-84	15	50	100-104	2	100
85-89	12	60			

- (a) Average bonus/worker for quarter,

$$\bar{x} = \sum fx/n = 2,985/50 = ₹ 59.7$$

- (b) Total quarterly bonus paid

$$= ₹ 59.7 \times 50 = ₹ 2,985$$

- (c) Average output/worker, $\bar{x} = 86.1$ units

$$3.41 \text{ Year : } 1 \ 2 \ 3 \ 4 \ 5 \ 6 \ 7 \ 8 \ 9 \ 10$$

Growth

$$\text{rate : } 2 \ 2.5 \ 3 \ 4 \ 5 \ 6 \ 7 \ 8 \ 9 \ 10$$

Value at the

$$\text{end of year } x : 102 \ 102.5 \ 103 \ 104 \ 105$$

$$106 \ 107 \ 108 \ 109 \ 110$$

$$\text{G.M.} = \text{Antilog} (\Sigma \log x/n)$$

$$= \text{Antilog} (20.237/10) = 105.6$$

$$\text{Average growth rate} = 105.6 - 100$$

$$= 5.6 \text{ percent}$$

$$\begin{aligned}
 3.42 \text{ H.M.} &= n / \left(\frac{1}{x_1} + \frac{1}{x_2} + \frac{1}{x_3} \right) \\
 &= 3 / \left(\frac{1}{48} + \frac{1}{40} + \frac{1}{32} \right) \\
 &= 38.98 \text{ km per hour}
 \end{aligned}$$

$$\begin{aligned}
 3.43 \text{ Harmonic mean} &= n / \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right) \\
 &= 3 / \left(\frac{1}{400} + \frac{1}{600} + \frac{1}{500} \right) = 0.486
 \end{aligned}$$

$$\begin{aligned}
 3.44 \text{ Mean : Let } x &\text{ be the mean of B. Then} \\
 (20 \times 10) + (8 \times x) + (6 \times 6) &= (15 \times 24) \\
 8x &= 124 \quad \text{or} \quad x = 15.5
 \end{aligned}$$

Hence mean of B = 15.5

Geometric mean: Let x be the geometric mean of C. Then

$$(10)^{10} \times (7)^8 \times x^6 = (8.397)^{24}$$

$$10 \log 10 + 8 \log 7 + 6 \log x = 24 \log 8.397$$

$$10 + (8 \times 0.8451) + 6 \log x = 24 (0.9241)$$

$$6 \log x = 5.4176 \quad \text{or} \quad x = \text{Antilog } 0.9029 = 7.997$$

Hence, geometric mean of C is 7.997.

3.45 Average price paid per share

$$= \frac{\sum wx}{\sum w} = \frac{14,300}{400} = 35.75.$$

Glossary of Terms

Population parameter: A numerical value used as a summary measure using data of the population.

Sample statistic: A numerical value used as a summary measure using data of the sample for estimation or hypothesis testing.

Mean: The sum of all the data values divided by their number.

Mean value: A measure of central location (tendency) for a data set such that the observations in the data set tend to cluster around it.

Weighted arithmetic mean: The mean for a data set obtained by assigning each observation, a weight that reflects its importance within the data set.

Geometric mean: A value that represents n th root of the product of a set of n numbers.

Harmonic mean: A value that is the reciprocal of the mean of the reciprocals of a set of numbers.

Median: A measure of central location such that one half of the observations in the data set is less than or equal to the given value.

Quartiles: The values which divide an ordered data set into 4 equal parts. The 2nd quartile is the median.

Deciles: The values which divide an ordered data set into 10 equal parts. The 5th decile is the median.

Percentiles: The values which divide an ordered data set into 100 equal parts. The 50th percentile is the median.

Mode value: A measure of location recognised by the location of the most frequently occurring value of a set of data.

Outlier: A very small or very large value in the data set.

4

Measures of Variation : Absolute and Relative

LEARNING OBJECTIVES

After studying this chapter, you should be able to

- provide the importance of the concept of variability (dispersion)
- measure the spread or dispersion, understand it, and identify its causes to provide a basis for action

4.1 INTRODUCTION

The measures of central tendency describe that the major part of values in the data set appears to concentrate (cluster) around a central value called *average* with the remaining values scattered (spread or distributed) on either sides of that value. But these measures do not reveal how these values are dispersed (spread or scattered) on each side of the central value. *The dispersion of values is indicated by the extent to which these values tend to spread over an interval rather than cluster closely around an average.*

The statistical techniques to measure such dispersion are of two types:

- (i) Techniques that are used to measure the extent of variation or the deviation (also called degree of variation) of each value in the data set from a measure of central tendency, usually the mean or median. Such statistical techniques are called *measures of dispersion* (or *variation*).
- (ii) Techniques that are used to measure the direction (away from uniformity or symmetry) of variation in the distribution of values in the data set. Such statistical techniques are called *measures of skewness*.

A small dispersion among values in the data set indicates that data are clustered closely around the mean. The mean is therefore considered representative of the data, i.e. mean is a reliable average. Conversely, a large dispersion among values in the data set indicates that the mean is not reliable, i.e. it is not representative of the data.

Illustration Suppose over the 6-year period the net profits (in percentage) of two firms are as follows:

Firm 1	:	5.2	4.5	3.9	4.7	5.1	5.4
Firm 2	:	7.8	7.1	5.3	14.3	11.0	16.1

Since average amount of profit is 4.8 percent for both firms, therefore operating results of both the firms are equally good and that a choice between them for investment purposes must depend on other considerations. However, the difference among the values is greater in Firm, 2, that is, profit is varying from 5.3 to 16.1 percent, while the net profit values of Firm 1 was varying from 3.9 to 5.4 percent. This shows that the values in data set 2 are spread more than those in data set 1. This implies that Firm 1 has a consistent performance while Firm 2 has a highly inconsistent performance. Thus for investment purposes, a comparison of the average (mean) profit values alone should not be sufficient.

4.2 SIGNIFICANCE OF MEASURING DISPERSION

Following are some of the purposes for which measures of variation are needed.

Test the Reliability of An Average Measures of variation are used to test to what extent an average represents the characteristic of a data set. If the variation is small, that is, extent of dispersion or scatter is less on each side of an average, then it indicates high uniformity of values in the distribution and the average represents an individual value in the data set. On the other hand, if the variation is large, then it indicates a lower degree of uniformity in values in the data set, and the average may be unreliable. No variation indicates perfect uniformity and, therefore, values in the data set are identical.

Control The Variability Measuring variation helps to identify the nature and causes of variation. Such information is useful in controlling the variations. According to Spurr and Bonini, *'In matters of health, variations in, body temperature, pulse beat and blood pressure are the basic guides to diagnosis. Prescribed treatment is designed to control their variation. In industrial production, efficient operation requires control of quality variation, the causes of which are sought through inspection and quality control programmes.'* In social science, the measurement of 'inequality' of distribution of income and wealth requires the measurement of variability.

Compare Two or More Sets of Data with Respect to Their Variability Measures of variation help in the comparison of the spread in two or more sets of data with respect to their uniformity or consistency. For example, (i) the measurement of variation in share prices and their comparison with respect to different companies over a period of time requires the measurement of variation, (ii) the measurement of variation in the length of stay of patients in a hospital every month may be used to set staffing levels, number of beds, number of doctors, and other trained staff, patient admission rates, and so on.

Facilitate The Use of Other Statistical Techniques Measures of variation facilitate the use of other statistical techniques such as correlation and regression analysis, hypothesis testing, forecasting, quality control and so on.

4.3 CLASSIFICATION OF MEASURES OF DISPERSION

The various measures of dispersion (variation) can be classified into two categories:

- (i) Absolute measures
- (ii) Relative measures

Absolute measures are described by a number or value to represent the amount of variation or differences among values in a data set. Such a number or value is expressed in the same unit of measurement as the set of values in the data such as rupees, inches, feet, kilograms or tonnes. Such measures help in comparing two or more sets of data in terms of absolute magnitude of variation, provided the variable values are expressed in the same unit of measurement and have almost the same average value.

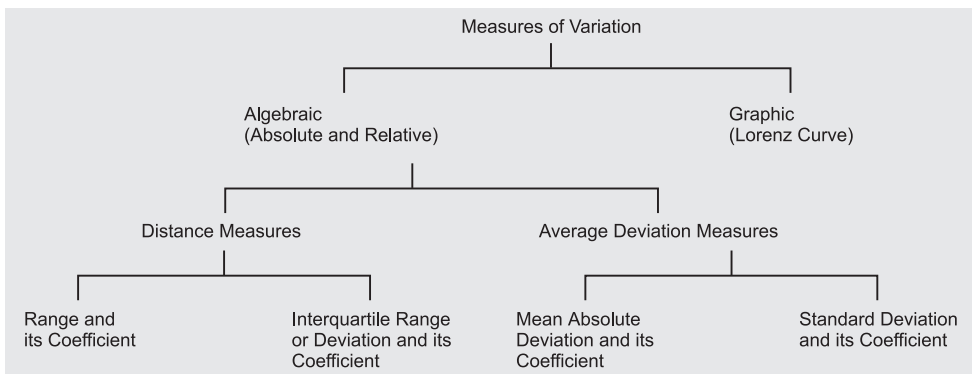
The **relative measures** are described as the ratio of a measure of absolute variation to an average and is termed as *coefficient of variation*. The word 'coefficient' means a number that is independent of any unit of measurement. While computing the relative variation, the average value used as base should be the same from which the absolute deviations were calculated.

Another classification of the measures of variation is based on the method employed for their calculations:

- (i) Distance measures
- (ii) Average deviation measures

The **distance measures** describe the spread or dispersion of values of a variable in terms of difference among values in the data set. The **average deviation measures** describe the average deviation for a given measure of central tendency.

The above-mentioned classification of various measures of dispersion (variation) may be summarized as shown below:



4.4 DISTANCE MEASURES

As mentioned above, two distance measures discussed in this section are namely:

- (i) Range
- (ii) Interquartile deviation

4.4.1 Range

The range is the most simple measure of dispersion and is based on the location of the largest and the smallest values in the data. Thus, the **range** is defined to be the difference between the largest and lowest observed values in a data set. In other words, it is the length of an interval which covers the highest and the lowest observed values in a data set and thus measures the dispersion or spread within the interval in the most direct possible way.

$$\begin{aligned}\text{Range (R)} &= \text{Highest value of an observation} - \text{Lowest value of an observation} \\ &= H - L\end{aligned}\quad (4-1)$$

For example, if the smallest value of an observation in the data set is 160 and the largest value is 250, then the range is $250 - 160 = 90$.

For grouped frequency distributions of values in the data set, the range is the difference between the upper class limit of the last class and the lower class limit of first class. In this case, the range obtained may be higher than as compared to ungrouped data because of the fact that the class limits are extended slightly beyond the extreme values in the data set.

Advantages

- (i) It is independent of the measure of central tendency and easy to calculate and understand.
- (ii) It is quite useful in cases where the purpose is only to find out the extent of extreme variation, such as industrial quality control, temperature, and rainfall.

Disadvantages

- (i) The calculation of range is based only on two values—largest and smallest in the data set and fail to take account of any other observations.
- (ii) It is largely influenced by two extreme values and completely independent of the other values. For example, range of two data sets {1, 2, 3, 7, 12} and {1, 1, 1, 12, 12} is 11, but the two data sets differ in terms of overall dispersion of values.
- (iii) Its value is sensitive to change in sampling, that is, different samples of the same size from the same population may have widely different ranges.
- (iv) It cannot be computed in case of open-ended frequency distributions because no highest or lowest value exists in open-ended class.
- (v) It does not describe the variation among values in the data between two extremes. For example,

Set 1	:	9	21	21	21	21	21	21	21
Set 2	:	9	9	9	9	21	21	21	21
Set 3	:	9	10	12	14	15	19	20	21

has a range of $21 - 9 = 12$, but the variation of values is quite different in each case (each of the following sets of data) between the highest and the lowest values.

Applications

- (i) *Fluctuation in share prices*: The range is useful in the study of small variations among values in a data set, such as variation in share prices and other commodities that are very sensitive to price changes from one period to another.

- (ii) *Quality control*: It is widely used in industrial quality control. Quality control is exercised by preparing suitable *control charts*. These charts are based on setting an upper control limit (range) and a lower control limit (range) within which produced items shall be accepted. The variation in the quality beyond these ranges requires necessary correction in the production process or system.
- (iii) *Weather forecasts*: The concept of range is used to determine the difference between maximum and minimum temperature or rainfall by meteorological departments to announce for the knowledge of the general public.

Coefficient of Range

The relative measure of range, also known as the coefficient of range, is obtained by applying the following formula:

$$\text{Coefficient of range} = \frac{H - L}{H + L} \quad (4-2)$$

Example 4.1: The following are the sales figures of a firm for the last 12 months.

Months	:	1	2	3	4	5	6	7	8	9	10	11	12
Sales													
(₹ '000)	:	80	82	82	84	84	86	86	88	88	90	90	92

Calculate the range and coefficient of range for sales.

Solution: Given that $H = 92$ and $L = 80$. Therefore,

$$\text{Range} = H - L = 92 - 80 = 12$$

$$\text{and Coefficient of range} = \frac{H - L}{H + L} = \frac{92 - 80}{92 + 80} = \frac{12}{172} = 0.069$$

Example 4.2: The following data show the waiting time (to the nearest 100th of a minute) of telephone calls to be matured:

<i>Waiting Time</i>	<i>Frequency</i> (Minutes)	<i>Waiting Time</i>	<i>Frequency</i> (Minutes)
0.10–0.35	6	0.88–1.13	8
0.36–0.61	10	1.14–1.39	4
0.62–0.87	8		

Calculate the range and coefficient of range.

Solution: Given that $H = 1.39$ and $L = 0.10$. Therefore,

$$\text{Range} = H - L = 1.39 - 0.10 = 1.29 \text{ minutes}$$

$$\text{and Coefficient of Range} = \frac{H - L}{H + L} = \frac{1.39 - 0.10}{1.39 + 0.10} = \frac{1.29}{1.49} = 0.865$$

4.4.2 Interquartile Range or Deviation

The limitations or disadvantages of the range can partially be overcome by using another measure of variation which measures the spread over the middle half of the values in the data set so as to minimize the influence of outliers (extreme values) in the calculation of range. Since a large

number of values in the data set lie in the central part of the frequency distribution, therefore it is necessary to study the **Interquartile Range** (also called midspread). To compute this value, the entire data set is divided into four parts each of which contains 25 percent of the observed values. The quartiles are the highest values in each of these four parts. The *interquartile range is a measure of dispersion or spread of values in the data set between the third quartile, Q_3 and the first quartile, Q_1* . In other words, the *interquartile range or deviation (IQR)* is the range for the middle 50 percent of the data. The concept of IQR is shown in Fig. 4.1 as

$$\text{Interquartile range (IQR)} = Q_3 - Q_1 \quad (4-3)$$

Half the distance between Q_1 and Q_3 is called the *semi-interquartile range* or the *quartile deviation (Q.D.)*.

$$\text{Quartile deviation (QD)} = \frac{Q_3 - Q_1}{2} \quad (4-4)$$

The median is not necessarily midway between Q_1 and Q_3 , although this will be so for a symmetrical distribution. The median and quartiles divide the data into equal numbers of values but do not necessarily divide the data into equally wide intervals.

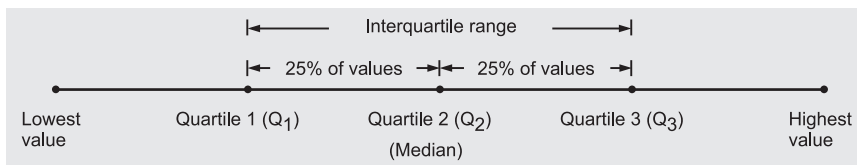


Figure 4.1 Interquartile Range

In a non-symmetrical distribution, the two quartiles Q_1 and Q_3 are at equal distance from the median, that is, $\text{Median} - Q_1 = Q_3 - \text{Median}$. Thus, $\text{Median} \pm \text{Quartile Deviation}$ covers exactly 50 percent of the observed values in the data set.

A smaller value of quartile deviation indicates high uniformity or less variation among the middle 50 percent observed values around the median value. On the other hand, a high value of quartile deviation indicates large variation among the middle 50 percent observed values.

Coefficient of Quartile Deviation

Since quartile deviation is an absolute measure of variation, therefore its value gets affected by the size and number of observed values in the data set. Thus, the Q.D. of two or more than two sets of data may differ. Due to this reason, to compare the degree of variation in different sets of data, we compute the relative measure corresponding to Q.D., called the *coefficient of Q.D.*, and it is calculated as follows:

$$\text{Coefficient of QD} = \frac{Q_3 - Q_1}{Q_3 + Q_1} \quad (4-5)$$

Example 4.3: Use an appropriate measure to evaluate the variation in the following data:

<i>Farm Size (acre)</i>	<i>No. of Farms</i>	<i>Farm Size (acre)</i>	<i>No. of Farms</i>
Below 40	394	161–200	169
41–80	461	201–240	113
81–120	391	241 and above	148
121–160	334		

Solution: Since the frequency distribution has open-end class intervals on the two extreme sides, therefore Q.D. would be an appropriate measure of variation. The computation of Q.D. is shown in Table 4.1.

Table 4.1 Calculations of Quartile Deviation

<i>Farm Size (acre)</i>	<i>No. of Farms</i>	<i>Cumulative Frequency (cf)</i> <i>(Less than)</i>
Below 40	394	394
41–80	461	855 ← Q_1 class
81–120	391	1246
121–160	334	1580 ← Q_3 class
161–200	169	1749
201–240	113	1862
241 and above	148	2010
	<u>2010</u>	

Q_1 = Value of $(n/4)$ th observation = $2010 \div 4$ or 502.5th observation

This observation lies in the class 41–80. Therefore,

$$\begin{aligned}
 Q_1 &= l + \frac{(n/4) - cf}{f} \times h \\
 &= 41 + \frac{502.5 - 394}{461} \times 40 = 41 + 9.41 = 50.41 \text{ acres}
 \end{aligned}$$

Q_3 = Value of $(3n/4)$ th observation = $(3 \times 2010) \div 4$ or 1507.5th observation

This observation lies in the class 121–160. Therefore,

$$\begin{aligned}
 Q_3 &= l + \frac{(3n/4) - cf}{f} \times h \\
 &= 121 + \frac{1507.5 - 1246}{334} \times 40 = 121 + 31.31 = 152.31 \text{ acres}
 \end{aligned}$$

Thus, the quartile deviation is given by

$$Q.D. = \frac{Q_3 - Q_1}{2} = \frac{152.31 - 50.41}{2} = 50.95 \text{ acres}$$

and

$$\text{Coefficient of Q.D.} = \frac{Q_3 - Q_1}{Q_3 + Q_1} = \frac{50.95}{202.72} = 0.251$$

Example 4.4: The Mean and S.D. of a normal distribution are ₹ 60 and ₹ 5 respectively. Find interquartile range and mean deviation of the distribution.

Solution: Given, $\bar{x} = 60$, $\sigma = 5$. Thus,

$$\text{Mean deviation (M.D.)} = \frac{4}{5} \sigma = \frac{4}{5} \times 5 = 4.$$

$$\text{Also, Quartile deviation (Q.D.)} = \frac{2}{3} \sigma = \frac{2}{3} \times 5 = \frac{10}{3}.$$

Thus,

$$\text{Coefficient of Q.D.} = \frac{Q_3 - Q_1}{Q_3 + Q_1} = \frac{10}{3} \quad \text{or} \quad Q_3 - Q_1 = \frac{20}{3}$$

$$\text{Interquartile Range} = Q_3 - Q_1 = 20/3.$$

Advantages

- (i) It is not difficult to calculate but can only be used to evaluate variation among observed values within the middle of the data set. Its value is not affected by the extreme (highest and lowest) values in the data set.
- (ii) It is an appropriate measure of variation for a data set summarized in open-ended class intervals.
- (iii) Since it is a positional measure of variation, therefore it is useful in case of erratic or highly skewed distributions, where other measures of variation get affected by extreme values in the data set.

Disadvantages

- (i) The value of Q.D. is based on the middle 50 percent observed values in the data set, therefore it cannot be considered as a good measure of variation as it is not based on all the observations.
- (ii) The value of Q.D. is very much affected by sampling fluctuations.
- (iii) The Q.D. has no relationship with any particular value or an average in the data set for measuring the variation. Its value is not affected by the distribution of the individual values within the interval of the middle 50 percent observed values.

Conceptual Questions 4A

1. Explain the term variation. What does a measure of variation serve? In the light of these, comment on some of the well-known measures of variation.
2. What are the requisites of a good measure of variation?
3. Explain how measures of central tendency and measures of variation are complementary to each other in the context of analysis of data.
4. Distinguish between absolute and relative measures of variation. Give a broad classification of the measures of variation.
5. (a) Critically examine the different methods of measuring variation.
(b) Explain with suitable examples the term 'variation'. Mention some common measures of variation and describe the one which you think is the most important.
6. What do you understand by 'coefficient of variation'? Discuss its importance in business problems.

Self-practice Problems 4A

- 4.1** The following are the prices of shares of a company from Monday to Saturday:

<i>Days</i>	<i>Price (₹)</i>	<i>Days</i>	<i>Price (₹)</i>
Monday	200	Thursday	160
Tuesday	210	Friday	220
Wednesday	208	Saturday	250

Calculate the range and its coefficient.

- 4.2** The day's sales figures (in ₹) for the last 15 days at Nirula's ice-cream counter, arranged in ascending order of magnitude, are recorded as follows: 2000, 2000, 2500, 2500, 2500, 3500, 4000, 5300, 9000, 12,500, 13,500, 24,500, 27,100, 30,900, and 41,000. Determine the range and middle 50 per cent range for this sample data.

- 4.3** The following distribution shows the sales of the 50 largest companies for a recent year:

<i>Sales (Million of rupees)</i>	<i>Number of Companies</i>
0–9	18
10–19	19
20–29	6
30–39	2
40–49	5

Calculate the coefficient of range.

- 4.4** You are given the frequency distribution of 292 workers of a factory according to their average weekly income.

<i>Weekly Income (₹)</i>	<i>No. of Workers</i>	<i>Weekly Income (₹)</i>	<i>No. of Workers</i>
Below 1350	8	1450–1470	22
1350–1370	16	1470–1490	15
1370–1390	39	1490–1510	15
1390–1410	58	1510–1530	9
1410–1430	60	1530 and above	10
1430–1450	40		

Calculate the quartile deviation and its coefficient from the above mentioned data.

- 4.5** You are given the data pertaining to kilowatt hours of electricity consumed by 100 persons in a city.

<i>Consumption (kilowatt hour)</i>	<i>No. of Users</i>
0–10	6
10–20	25
20–30	36
30–40	20
40–50	13

Calculate the range within which the middle 50 per cent of the consumers fall.

- 4.6** The following sample shows the weekly number of road accidents in a city during a two-year period:

<i>Number of Accidents</i>	<i>Frequency</i>	<i>Number of Accidents</i>	<i>Frequency</i>
0–4	5	25–29	9
5–9	12	30–34	4
10–14	32	35–39	3
15–19	27	40–44	1
20–24	11		

Find the interquartile range and coefficient of quartile deviation.

- 4.7** A city development authority subdivided the available land for housing into the following building lot sizes:

<i>Lot Size (Square meters)</i>	<i>Frequency</i>
Below 69.44	19
69.44–104.15	25
104.16–208.32	42
208.33–312.49	12
312.50–416.65	5
416.66 and above	17

Find the interquartile range and quartile deviation.

- 4.8 The cholera cases reported in different hospitals of a city in a rainy season are given below:

Calculate the quartile deviation for the given distribution and comment upon the meaning of your result.

Age Group (Years)	Frequency	Age Group (Years)	Frequency
Less than 1	15	25–35	132
1–5	113	35–45	65
5–10	122	45–65	46
10–15	91	65 and above	15
15–25	229		

Hints and Answers

- 4.1 Range = ₹ 90,

Coefficient of range = 0.219

- 4.2 Range = ₹ 39,000;

Middle 50 per cent, $R = P_{75} - P_{25}$

$$= x_{(75n/100) + (1/2)} - x_{(25n/100) + (1/2)}$$

$$= x_{(11.25 + 0.50)} - x_{(3.75 + 0.50)}$$

$$= x_{11.75} - x_{4.25}$$

$$= (13,500 + 8250) - (2500 + 00)$$

$$= 19,250$$

Here $x_{11.75}$ is the interpolated value for the 75 per cent of the distance between 11th and 12th ordered sales amount. Similarly, $x_{4.25}$ is

the interpolated value for the 25 per cent of the distance between 4th and 5th order sales amount.

- 4.3 Coefficient of range = 1

- 4.4 Quartile deviation = 27.76; Coeff. of Q.D. = 0.020; $Q_1 = 1393.48$; $Q_3 = 1449$

- 4.5 $Q_3 - Q_1 = 34 - 17.6 = 16.4$

- 4.6 $Q_3 - Q_1 = 30.06$;

Coefficient of Q.D. = 0.561

- 4.7 $Q_3 - Q_1 = 540.26$; Q.D. = 270.13

- 4.8 Q.D. = 10 years

4.5 AVERAGE DEVIATION MEASURES

In this section, we shall discuss two more measures of dispersion to measure the mean (or average) amount by which all values in a data set (population or sample) vary from their mean. These measures deal with the average deviation from some measure of central tendency—usually mean or median. These measures are:

- Mean Absolute Deviation or Average Deviation
- Variance and Standard Deviation

4.5.1 Mean Absolute Deviation

Since two measures of variation, range and quartile deviation discussed earlier do not show how values in a data set are scattered about a central value or disperse themselves throughout the range, therefore it is quite reasonable to measure the variation as a degree (amount) to which values within a data set deviate from either mean or median.

The mean of deviations of individual values in the data set from their actual mean is always zero, so such a measure (zero) would be useless as an indicator of variation. This problem can be solved in two ways:

- Ignore the signs of the deviations by taking their absolute value, or
- Square the deviations because the square of a negative number is positive.

Since the absolute difference between a value x_i of an observation from A.M. is always a positive number, whether it is less than or more than the A.M., therefore we take the absolute value of each such deviation from the A.M. (or median). Taking the average of these deviations from the A.M., we get a measure of variation called the *mean absolute deviation* (MAD). In general, the mean absolute deviation is given by

$$\text{MAD} = \frac{1}{N} \sum_{i=1}^N |x - \mu|, \text{ for a population} \quad (4-6)$$

$$\text{MAD} = \frac{1}{n} \sum_{i=1}^n |x - \bar{x}|, \text{ for a sample}$$

where $| |$ indicates the absolute value. That is, the signs of deviations from the mean are disregarded.

For a grouped frequency distribution, MAD is given by

$$\text{MAD} = \frac{\sum_{i=1}^n f_i |x_i - \bar{x}|}{\sum f_i} \quad (4-7)$$

Formulae (4-6) and (4-7), in different contexts, indicate that the MAD provides a useful method of comparing the relative tendency of values in the distribution to scatter around a central value or to disperse themselves throughout the range.

While calculating the mean absolute deviation, the median is also considered for computing because the sum of the absolute values of the deviations from the median is smaller than that from any other value. However, in general, arithmetic mean is used for this purpose.

If a frequency distribution is symmetrical, then A.M. and median values coincide and the same MAD value is obtained. In such a case $\bar{x} \pm \text{MAD}$ provides a range in which 57.5 per cent of the observations are included. Even if the frequency distribution is moderately skewed, the interval $\bar{x} \pm \text{MAD}$ includes the same percentage of observations. This shows that more than half of the observations are scattered within one unit of the MAD around the arithmetic mean.

Coefficient of MAD

The relative measure of mean absolute deviation (MAD), called the *coefficient of MAD*, is obtained by dividing the MAD by a measure of central tendency (arithmetic mean or median) used for calculating the MAD. Thus,

$$\text{Coefficient of MAD} = \frac{\text{Mean absolute deviation}}{\bar{x} \text{ or Me}} \quad (4-8)$$

If the value of relative measure is desired in percentage, then

$$\text{Coefficient of MAD} = \frac{\text{MAD}}{\bar{x} \text{ or Me}} \times 100$$

Example 4.5: The number of patients seen in the emergency ward of a hospital for a sample of 5 days in the last month were: 153, 147, 151, 156 and 153. Determine the mean deviation and interpret.

Solution: The mean number of patients is $\bar{x} = (153 + 147 + 151 + 156 + 153)/5 = 152$. Below are the details of the calculations of MAD using formula (4-6).

<i>Number of Patients (x)</i>	$x - \bar{x}$	<i>Absolute Deviation</i> $ x - \bar{x} $
153	$153 - 152 = 1$	1
147	$147 - 152 = -5$	5
151	$151 - 152 = -1$	1
156	$156 - 152 = 4$	4
153	$153 - 152 = 1$	1
		12

$$\text{MAD} = \frac{1}{n} \sum |x - \bar{x}| = \frac{12}{5} = 2.4 \cong 3 \text{ patients (approx.)}$$

The mean absolute deviation is 3 patients per day. The number of patients deviate on the average by 3 patients from the mean of 152 patients per day.

Example 4.6: Calculate mean deviation from median from the data given below:

Wages (₹) (mid Point)	:	125	175	225	275
No. of persons	:	3	8	21	6

Solution: Calculations for mean deviation are shown below:

<i>Mid Point</i>	<i>Class Interval</i>	<i>Frequency</i>	<i>Cumulative Frequency (cf)</i>	$ d = x - \text{Med} $ $= x - 221.43 $	$f d $
125	100 — 150	3	3	96.43	289.29
175	150 — 200	8	11	46.43	371.44
225	200 — 250	21	32	3.57	74.97
275	250 — 300	6	38	53.57	321.42
325	300 — 350	2	40	103.57	207.14
		40			1264.26

Median = Size of $\left(\frac{n}{2}\right)^{\text{th}} = \frac{40}{2} = 20\text{th item}$. Thus, Median class is 200 – 250.

$$\text{Med} = l + \frac{\frac{n}{2} - cf}{f} \times h = 200 + \frac{20 - 11}{21} \times 50 = 221.43$$

$$\text{Mean Deviation (M.D.)} = \frac{\sum f |d|}{\sum f} = \frac{1264.26}{40} = 31.6.$$

Example 4.7: The income of a person in a particular week is ₹ 20 per day. Find the mean deviation of his income for 5 days.

Solution: The income for 7 days is as follows:

<i>Days</i>	<i>Income</i>	$ d = x - 20 $
Monday	20	0
Tuesday	20	0
Wednesday	20	0
Thursday	20	0
Friday	20	0
Saturday	20	0
Sunday	20	0

Since number of observations are odd, median is 4th observation, i.e. = 20. Hence

$$\text{Mean deviation (M.D.)} = \frac{\Sigma |d|}{n} = 0.$$

Example 4.8: Mean and S.D. of the following continuous series are 31 and 15.9, respectively. The distribution after taking step deviation is as follows:

$dx :$	-3	-2	-1	0	1	2	3
$f :$	10	15	25	25	10	10	5

Determine the actual class intervals.

Solution: Calculations to determine actual class intervals are shown below:

d	f	d^2	fd	fd^2
-3	10	9	-30	90
-2	15	4	-30	60
-1	25	1	-25	25
0	25	0	0	0
1	10	1	10	10
2	10	4	20	40
3	5	9	15	45
	<u>100</u>		<u>-40</u>	<u>270</u>

$$\text{Mean} = A + \frac{\Sigma fd}{N} \times h \quad \text{or} \quad 31 = A + \frac{(-40)}{100} \times h = A - 0.40h$$

$$\begin{aligned} \text{Also} \quad \text{S.D. } (\sigma) &= \sqrt{\frac{\Sigma fd^2}{n} - \left(\frac{\Sigma fd}{n}\right)^2} \times h \\ 15.9 &= \sqrt{\frac{270}{100} - \left(\frac{-40}{100}\right)^2} \times h = \sqrt{2.7 - 0.16} \times h = \sqrt{2.54} \times h \\ \text{i.e.} \quad h &= \frac{15.9}{\sqrt{2.54}} = \frac{15.9}{1.59} = 10 \end{aligned}$$

Now for $h = 10$, we get $31 = A - 0.4 \times 10 = A - 4$, i.e. $A = 35$. Since we know that $d = \frac{m - A}{h}$ or $-30 = m - 35$, i.e. $m = 5$.

Similarly, we can find other mid-values as: 5, 15, 25, 35, 45, 55, 65. Since class size $h = 10$. Thus, class interval corresponding to $m = 5$ is 0–10.

Similarly the other class intervals are: 0–10, 10–20, 20–30, 30–40, 40–50, 50–60, 60–70.

Example 4.9: Calculate the mean absolute deviation and its coefficient from median for the following data

Year	Sales (₹ in thousand)	
	Product A	Product B
1996	23	36
1997	41	39
1998	29	36
1999	53	31
2000	38	47

Solution: The median sales (Me) of the two products A and B are 38 and 36, respectively. The calculations of MAD in both the cases are shown in Table 4.2.

Table 4.2 Calculations of MAD

Product A		Product B	
Sales (x)	$ x - \text{Me} = x - 38 $	Sales (x)	$ x - \text{Me} = x - 36 $
23	15	31	5
29	9	36	0
38	0	36	0
41	3	39	3
53	15	47	11
$n = 5$	$ x - \text{Me} = 42$	$n = 5$	$ x - \text{Me} = 19$

$$\text{Product A:} \quad \text{MAD} = \frac{1}{n} \sum |x - \text{Me}| = \frac{42}{5} = 8.4$$

$$\text{Coefficient of MAD} = \frac{\text{MAD}}{\text{Me}} = \frac{8.4}{38} = 0.221$$

$$\text{Product B:} \quad \text{MAD} = \frac{1}{n} \sum |x - \text{Me}| = \frac{19}{5} = 3.8$$

$$\text{Coefficient of MAD} = \frac{\text{MAD}}{\text{Me}} = \frac{3.8}{36} = 0.106$$

Example 4.10: Find the mean absolute deviation from mean for the following frequency distribution of sales (₹ in thousand) in a co-operative store.

Sales	:	50–100	100–150	150–200	200–250	250–300	300–350
Number of days	:	11	23	44	19	8	7

Solution: The mean absolute deviation can be calculated by using the formula (4-6) for mean. The calculations for MAD are shown in Table 4.3. Let the assumed mean be $A = 175$.

Table 4.3 Calculations for MAD

Sales (₹)	Mid-Value (m)	Frequency (f)	$(m - 175)/50$ (= d)	fd	$\frac{ x - \bar{x} }{ x - \bar{x} }$	f x - $\bar{x}$
50–100	75	11	–2	–22	104.91	1154.01
100–150	125	23	–1	–23	54.91	1262.93
150–200	175 ← A	44	0	0	4.91	216.04
200–250	225	19	1	19	45.09	856.71
250–300	275	8	2	16	95.09	760.72
300–350	325	7	3	21	145.09	1015.63
		112		11		5266.04

$$\bar{x} = A + \frac{\sum fd}{\sum f} \times h = 175 + \frac{11}{112} \times 50 = ₹ 179.91 \text{ per day}$$

$$\text{MAD} = \frac{\sum f|x - \bar{x}|}{\sum f} = \frac{5266.04}{112} = ₹ 47.01$$

Thus, the average sales is ₹ 179.91 thousand per day and the mean absolute deviation of sales is ₹. 47.01 thousand per day.

Example 4.11: A welfare organization introduced an education scholarship scheme for school going children of a backward village. The rates of scholarship were fixed as given below:

Age Group (Years)	Amount of Scholarship per Month (₹)
5–7	300
8–10	400
11–13	500
14–16	600
17–19	700

The ages of 30 school children are noted as 11, 8, 10, 5, 7, 12, 7, 17, 5, 13, 9, 8, 10, 15, 7, 12, 6, 7, 8, 11, 14, 18, 6, 13, 9, 10, 6, 15, 3, 5 yzears, respectively. Calculate mean and standard deviation of monthly scholarship. Find out the total monthly scholarship amount being paid to the students.

Solution: The number of students in the age group from 5–7 to 17–19 are calculated as shown in Table 4.4:

Table 4.4

Age Group (Years)	Tally Bars	Number of Students
5–7		10
8–10		8
11–13		7
14–16		3
17–19		2
		<u>30</u>

The calculations for mean and standard deviation are shown in Table 4.5.

Table 4.5 Calculations for Mean and Standard Deviation

Age Group (Years)	Number of Students (<i>f</i>)	Mid-value (<i>m</i>)	$d = \frac{m - A}{h} = \frac{m - 12}{3}$	<i>fd</i>	<i>fd</i> ²
5–7	10	6	–2	–20	40
8–10	8	9	–1	–8	8
11–13	7	A → 12	0	0	0
14–16	3	15	1	3	3
17–19	2	18	2	4	8
	30			–21	59

$$\text{Mean, } \bar{x} = A + \frac{\Sigma fd}{\Sigma f} \times h = 12 - \frac{21}{30} \times 3 = 12 - 2.1 = 9.9$$

$$\begin{aligned} \text{Standard deviation, } \sigma &= \sqrt{\frac{\Sigma fd^2}{N} - \left(\frac{\Sigma fd}{N}\right)^2} \times h = \sqrt{\frac{59}{30} - \left(\frac{-21}{30}\right)^2} \times 3 \\ &= \sqrt{1.967 - 0.49} \times 3 = 1.2153 \times 3 = 3.6459 \end{aligned}$$

Calculations for monthly scholarship paid to 30 students are shown in Table 4.6.

Table 4.6 Calculations for Monthly Scholarship

Number of Students	Amount of Scholarship per Month (₹)	Total Monthly Scholarship (₹)
10	300	3000
8	400	3200
7	500	3500
3	600	1800
2	700	1400
		12,900

Advantages

- The calculation of MAD is based on all observations in the distribution and shows the dispersion of values around the measure of central tendency.
- The value of MAD is easy to compute and therefore makes it popular among those users who are not even familiar with statistical methods.
- While calculating MAD, equal weightage is given to each observed value and thus it indicates how far each observation lies from either the mean or median.
- Average deviation from mean is always zero in any data set. The MAD avoids this problem by using absolute values to eliminate the negative signs.

Disadvantages

- The algebraic signs are ignored while calculating MAD. If the signs are not ignored, then the sum of the deviations taken from arithmetic mean will be zero and close to zero when deviations are taken from median.

- (ii) The value of MAD is considered to be best when deviations are taken from median. However, median does not provide a satisfactory result in case of a high degree of variability in a data set. Moreover, the sum of the deviations from mean (ignoring signs) is greater than the sum of the deviations from median (ignoring signs). In such a situation, computations of MAD by taking deviations from mean are also not desirable.
- (iii) The MAD is generally unwieldy in mathematical discussions.

4.5.2 Variance and Standard Deviation

Another way to disregard the negative sign of deviations from mean is to square them. Instead of computing the absolute value of each deviation from mean, we square the deviations from mean. Then the sum of all such squared deviations is divided by the number of observations in the data set. This value is called **population variance** and is denoted by σ^2 (a lower-case Greek letter sigma). It is usually referred to as 'sigma squared'. Symbolically, it is written as

$$\begin{aligned}\text{Population variance, } \sigma^2 &= \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2 \\ &= \frac{1}{N} \sum_{i=1}^N x_i^2 - (\mu)^2 \quad (\text{Deviation is taken from actual population A.M.})\end{aligned}\quad (4-9)$$

Since σ^2 is the average or mean of squared deviations from arithmetic mean, it is also called the *mean square average*.

If the *sum of the squared* deviations about a sample mean $\bar{x}$ in Eq. (4-9) is divided by n (sample size), then invariably value of σ^2 tends to be lower than its actual value. This condition is called *bias*. However, this *bias* in the estimation of population variance from a sample can be removed by dividing the sum of the squared deviations between the sample mean and each element in the population by $n-1$ rather than by n . Thus, the *unbiased* sample variance denoted by s^2 is defined as follows:

$$\text{Sample variance, } s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 = \frac{\sum x_i}{n-1} - \frac{n \bar{x}^2}{n-1} = \frac{\sum x_i}{n-1} - \frac{(\sum x_i)^2}{n(n-1)} \quad (4-10)$$

The numerator $\sum (x_i - \bar{x})^2$ in Eq. (4-10) is called the *total sum of squares*. This quantity measures the total variation among values in a data set (whereas the variance measures only the *average variation*). The larger the value of $\sum (x_i - \bar{x})^2$, the greater the variation among the values in a data set.

Standard Deviation

The numerical value of population or a sample variance is difficult to interpret because it is expressed in square units. To reach at interpretable measure of variance expressed in the units of original data, we take a positive square root of the variance, which is known as the **standard deviation** or *root-mean square deviation*. The standard deviation of population and sample is denoted by σ and s , respectively.

(a) *Ungrouped Data*

Population standard deviation,

$$\sigma = \sqrt{\sigma^2} = \sqrt{\frac{1}{N} \sum (x - \mu)^2} = \sqrt{\frac{1}{N} \sum x^2 - (\mu)^2} = \sqrt{\frac{\sum d^2}{N} - \left(\frac{\sum d}{N}\right)^2}$$

Sample standard deviation $s = \sqrt{\frac{\sum x^2}{n-1} - \frac{n\bar{x}^2}{n-1}}$; where n = sample size(b) *Grouped Data*

$$\text{Population standard deviation, } \sigma = \sqrt{\frac{\sum fd^2}{N} - \left(\frac{\sum fd}{N}\right)^2} \times h$$

where

 h = width of class interval $d = \frac{m - A}{h}$, where A is any constant (also called assumed A.M.)

$$\text{Sample standard deviation, } s = \sqrt{s^2} = \sqrt{\frac{\sum f(x - \bar{x})^2}{n-1}} = \sqrt{\frac{\sum fx^2}{n-1} - \frac{(\sum fx)^2}{n(n-1)}} \quad (4-11)$$

Remarks: 1. For any data set, MAD is always less than the σ because MAD is less sensitive to the extreme observations. Thus, when a data contains few very large observations, the MAD provides a more realistic measure of variation than σ .

2. When sample size (n) becomes very large, $(n - 1)$ becomes indistinguishable and irrelevant.

Advantages

- (i) The value of standard deviation is based on every observation in a set of data. It is the only measure of variation capable of algebraic treatment and less affected by fluctuations of sampling as compared to other measures of variation.
- (ii) It is possible to calculate the combined standard deviation of two or more sets of data.
- (iii) Standard deviation has a definite relationship with the area under the symmetric curve of a frequency distribution. Due to this reason, standard deviation is called a *standard* measure of variation.
- (iv) Standard deviation is useful in further statistical investigations. For example, standard deviation plays a vital role in comparing skewness, correlation, and so on, and also widely used in sampling theory.

Disadvantages

- (i) As compared to other measures of variation, calculations of standard deviation are difficult.
- (ii) While calculating standard deviation, more weight is given to extreme values and less to those near mean. Since for calculating S.D., the deviations from the mean are squared, therefore large deviations when squared are proportionately more than small deviations. For example, the deviations 2 and 10 are in the ratio of 1 : 5 but their squares 4 and 100 are in the ratio of 1 : 25.

Example 4.12: The wholesale prices of a commodity for seven consecutive days in a month is as follows:

Days	:	1	2	3	4	5	6	7
Commodity price/quintal	:	240	260	270	245	255	286	264

Calculate the variance and standard deviation.

Solution: The computations for variance and standard deviation are shown in Table 4.7.

Table 4.7 Computations of Variance and Standard Deviation by Actual Mean Method

Observation (x)	$x - \bar{x} = x - 260$	$(x - \bar{x})^2$
240	-20	400
260	0	0
270	10	100
245	-15	225
255	-5	25
286	26	676
264	4	16
<u>1820</u>		<u>1442</u>

$$\bar{x} = \frac{\sum x}{N} = \frac{1820}{7} = 260$$

$$\text{Variance, } \sigma^2 = \frac{\sum (x - \bar{x})^2}{N} = \frac{1442}{7} = 206$$

$$\text{Standard deviation, } \sigma = \sqrt{\sigma^2} = \sqrt{206} = 14.352$$

If we take deviation from an assumed A.M. = 255 instead of actual A.M. = 260. The calculations for standard deviation will be as shown in Table 4.8.

Table 4.8 Computation of Standard Deviation by Assumed Mean Method

Observation (x)	$d = x - A = x - 255$	d^2
240	-15	225
260	5	25
270	15	225
245	-10	100
(255) ← A	0	0
286	31	961
264	9	81
	<u>35</u>	<u>1617</u>

$$\begin{aligned} \text{Standard deviation } \sigma &= \sqrt{\frac{\sum d^2}{N} - \left(\frac{\sum d}{N}\right)^2} = \sqrt{\frac{1617}{7} - \left(\frac{35}{7}\right)^2} \\ &= \sqrt{231 - 25} = \sqrt{206} = 14.352 \end{aligned}$$

This result is same as obtained earlier in Table 4.7.

Remark: When actual A.M. is not a whole number, assumed A.M. method should be used to reduce the computation time.

Example 4.13: A study of 100 engineering companies gives the following information:

Profit (₹ in crore)	:	0–10	10–20	20–30	30–40	40–50	50–60
Number of companies	:	8	12	20	30	20	10

Calculate the standard deviation of the profit earned.

Solution: Let assumed mean, A be 35 and the value of h be 10. Calculations for standard deviation are shown in Table 4.9.

Table 4.9 Calculations of Standard Deviation

Profit (₹ in crore)	Mid-value (m)	$d = \frac{m - A}{h} = \frac{m - 35}{10}$	f	fd	fd^2
0–10	5	–3	8	–24	72
10–20	15	–2	12	–24	48
20–30	25	–1	20	–20	20
30–40	(35) ← A	0	30	0	0
40–50	45	1	20	20	20
50–60	55	2	10	20	40
				–28	200

$$\begin{aligned}\text{Standard deviation, } \sigma &= \sqrt{\frac{\sum fd^2}{N} - \left(\frac{\sum fd}{N}\right)^2} \times h \\ &= \sqrt{\frac{200}{100} - \left(\frac{-28}{100}\right)^2} \times 10 = \sqrt{2 - 0.078} \times 10 = 13.864\end{aligned}$$

Example 4.14: Mr. Gupta, a retired government servant is considering, investing his money in two proposals. He wants to choose the one that has higher average net present value and lower standard deviation. The relevant data are given below. Can you help him in choosing the proposal?

Proposal A:	Net Present Value (NPV)	Chance of the Possible Outcome of NPV
	1559	0.30
	5662	0.40
	9175	0.30

<i>Proposal B:</i>	<i>Net Present Value (NPV)</i>	<i>Chance of the Possible Outcome of NPV</i>
	-10,050	0.30
	5,812	0.40
	20,584	0.30

Solution: To suggest to Mr. Gupta a proposal for high average net present value, first calculate the expected (average) net present value for both the proposals.

$$\begin{aligned}\text{Proposal A: Expected NPV} &= 1559 \times 0.30 + 5662 \times 0.40 + 9175 \times 0.30 \\ &= 467.7 + 2264.8 + 2752.5 = ₹ 5485\end{aligned}$$

$$\begin{aligned}\text{Proposal B: Expected NPV} &= -10,050 \times 0.30 + 5812 \times 0.40 + 20,584 \times 0.30 \\ &= -3015 + 2324.8 + 6175.2 = ₹ 5485\end{aligned}$$

Since the expected NPV in both the cases is same, he would like to choose the less risky proposal. For this we have to calculate the standard deviation in both the cases.

Standard deviation for Proposal A:

$NPV(x_i)$	$Expected\ NPV(\bar{x})$	$x - \bar{x}$	f	$f(x - \bar{x})^2$
1559	5485	-3926	0.30	46,24,042.8
5662	5485	177	0.40	12,531.6
9175	5485	3690	0.30	40,84,830.0
			1.00	87,21,404.4

$$s_A = \sqrt{\frac{\sum f(x - \bar{x})^2}{\sum f}} = \sqrt{87,21,404.4} = ₹ 2953.20$$

Standard deviation for Proposal B:

$NPV(x_i)$	$Expected\ NPV(\bar{x})$	$x - \bar{x}$	f	$f(x - \bar{x})^2$
-10,050	5485	-15,535	0.30	7,24,00,867.5
5812	5485	327	0.40	42,771.6
20,584	5485	15,099	0.30	6,83,93,940.3
			1.00	14,08,37,579.4

$$s_B = \sqrt{\frac{\sum f(x - \bar{x})^2}{\sum f}} = \sqrt{14,08,37,579} = ₹ 11,867.50$$

The $s_A < s_B$ indicates uniform net profit for Proposal A. Thus Proposal A may be chosen.

Example 4.15: The number of employees, wages per employee and the variance of the wages of employees for two factories are given below :

	Factory A	Factory B
No. of employees	100	100
Average wages per employee per month (₹)	1200	850
Variance of the wages (₹)	81	256

- (a) In which factory is there greater variation in the distribution of wages per employee?
 (b) Suppose in factory B, the wages of an employee were wrongly noted as ₹ 900 instead of ₹ 910.

Solution: (a) Coefficient of variation, $CV = \frac{\sigma}{\bar{x}} \times 100$. Thus,

$$CV \text{ for factory, } A = \frac{9}{1200} \times 100 = 0.75 \text{ per cent}$$

$$CV \text{ for factory, } B = \frac{16}{850} \times 100 = 1.88 \text{ per cent}$$

Since co-efficient of variation is more for firm B, it shows greater variation in the distribution of wages per employee in the firm B.

(b) For factory B, $\Sigma x = \bar{x} \times N = 850 \times 100 = 85000$

$$\text{Correct } \Sigma x = 85000 - 900 + 910 = 85,010$$

$$\text{Correct, } \bar{x} = \frac{85010}{100} = 850.10$$

$$\text{Correct variance} = \frac{\Sigma x^2}{N} - (\bar{x})^2$$

$$256 = \frac{\Sigma x^2}{100} - (850)^2 \quad \text{or} \quad 25600 = \Sigma x^2 - 7,22,50,000$$

$$\Sigma x^2 = 7,22,50,000 + 25,600 = 7,22,75,600$$

$$\begin{aligned} \text{Correct } \Sigma x^2 &= 7,22,75,600 - (900)^2 + (910)^2 \\ &= 7,22,75,600 - 8,10,000 + 8,28,100 = 7,22,93,700 \end{aligned}$$

$$\begin{aligned} \text{Correct variance} &= \frac{\text{Correct } \Sigma x^2}{N} - \text{Correct } (\bar{x})^2 = \frac{7,22,93,700}{100} - (850.10)^2 \\ &= 7,22,937 - 7,22,670.01 = 266.99 \end{aligned}$$

Example 4.16: In a series of 5 observations, the means and variance are 4.4 and 8.24. If three observations are 1, 2, 6, find other two. [Delhi Univ., B.Com. (Hons.), 1993]

Solution: Let the other two observations be x, y . Then

$$\frac{x + y + 1 + 2 + 6}{5} = 4.4 \quad \text{or} \quad x + y + 9 = 4.4 \times 5 = 22$$

$$x + y = 13, \text{ i.e. } y = 13 - x$$

$$\text{Var } (\sigma^2) = \frac{1}{n} \Sigma x^2 - (\bar{x})^2$$

$$8.24 = \frac{1}{5} \Sigma x^2 - (4.4)^2 = \frac{1}{5} \Sigma x^2 - 19.36$$

$$8.24 + 19.36 = \frac{1}{5} \Sigma x^2, \text{ i.e. } \Sigma x^2 = 138$$

$$x^2 + y^2 + 1 + 4 + 36 = 138$$

$$x^2 + y^2 + 41 = 138$$

$$x^2 + y^2 = 97$$

$$\begin{aligned}
 x^2 + (13 - x)^2 &= 97 \quad (\text{since } y = 13 - x) \\
 x^2 + 169 + x^2 - 26x &= 97 \\
 2x^2 - 26x + 72 &= 0 \\
 x^2 - 13x + 36 &= 0 \\
 (x - 9)(x - 4) &= 0 \quad \text{or } x = 9, 4 \quad \text{and } y = 4, 9
 \end{aligned}$$

Hence, two numbers are 4 and 9.

Example 4.17: Mean and S.D. of 100 items are found by a student as 50 and 0.1. If at the time of calculation, two items are wrongly taken as 40 and 50 instead of 60 and 30, find correct mean and S.D. [Delhi Univ., B.Com. 1996]

Solution: Given, $\bar{x} = 50$, $\sigma = 0.1$ and $N = 100$.

Thus,
$$\bar{x} = \frac{\Sigma x}{n} \quad \text{or } \Sigma x = n \times \bar{x} = 100 \times 50 = 5000$$

$$\begin{aligned}
 \text{Correct } \Sigma x &= 5000 - \text{wrong items} + \text{correct items} \\
 &= 5000 - 40 - 50 + 60 + 30 = 5000
 \end{aligned}$$

$$\text{Correct Mean, } \bar{x} = \frac{5000}{100} = 50$$

Now
$$\sigma = \sqrt{\frac{\Sigma x^2}{n} - (\bar{x})^2} \quad \text{or } \sigma^2 = \frac{\Sigma x^2}{n} - (\bar{x})^2$$

Substituting the values as: $\sigma = 0.1$, $\bar{x} = 50$, $N = 100$, we get

$$(0.1)^2 = \frac{\Sigma x^2}{100} - (50)^2$$

$$0.01 = \frac{\Sigma x^2}{100} - 2500 \quad \text{or } \Sigma x^2 = 2,50,001$$

$$\begin{aligned}
 \text{Correct, } \Sigma x^2 &= 250001 - (40)^2 - (50)^2 + (60)^2 + (30)^2 \\
 &= 250001 - 1600 - 2500 + 3600 + 900 = 2,50,401
 \end{aligned}$$

$$\text{Correct, S.D.} = \sqrt{\frac{\Sigma x^2}{n} - (\bar{x})^2} = \sqrt{\frac{250401}{100} - (50)^2} = 2.0025.$$

Example 4.18: A charitable organization decided to give old age pension to people over sixty years of age. The scales of pension were fixed as follows:

Age Group (years)	Amount of Pension (₹ per month)
60–65	20
65–70	25
70–75	30
75–80	35
80–85	40

The ages of 25 people who secured the pension right are given as: 74, 62, 84, 72, 61, 83, 72, 81, 64, 71, 63, 61, 67, 74, 64, 79, 73, 75, 76, 69, 68, 78, 66, 67, 64. Calculate the monthly average pension payable per pensioner and the S.D. [Delhi Univ., B.Com.(Hons.), 2005]

Solution: The calculations for monthly average pension and S.D. are shown below in the table:

Age Group (Years)	Tally	No. of Persons (f)	Amount of Pension (₹) (x)	fx	fx ²
60 – 65		7	20	140	2800
65 – 70		5	25	125	3125
70 – 75		6	30	180	5400
75 – 80		4	35	140	4900
80 – 85		3	40	120	4800
		<u>25</u>		<u>705</u>	<u>21025</u>

$$\text{Mean} = \frac{\Sigma fx}{\Sigma f} = \frac{705}{25} = ₹ 28.20$$

$$\text{S.D.} = \sqrt{\frac{\Sigma fx^2}{n} - (\bar{x})^2} = \sqrt{\frac{21025}{25} - (28.20)^2} = \sqrt{45.76} = 6.77.$$

4.5.3 Mathematical Properties of Standard Deviation

- 1. Combined standard deviation:** The combined standard deviation of two sets of data containing n_1 and n_2 observations with means $\bar{x}_1$ and $\bar{x}_2$ and standard deviations σ_1 and σ_2 , respectively is given by

$$\sigma_{12} = \sqrt{\frac{n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2)}{n_1 + n_2}}$$

where σ_{12} = combined standard deviation; $d_1 = \bar{x}_{12} - \bar{x}_1$; $d_2 = \bar{x}_{12} - \bar{x}_2$

and

$$\bar{x}_{12} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2} \text{ (combined arithmetic mean)}$$

This formula for combined standard deviation of two sets of data can be extended to compute the standard deviation of more than two sets of data on the same lines.

- 2. Standard deviation of natural numbers:** The standard deviation of the first n natural numbers is given by

$$\sigma = \sqrt{\frac{1}{12}(n^2 - 1)}$$

For example, the standard deviation of the first 100 (i.e., from 1 to 100) natural numbers will be

$$\sigma = \sqrt{\frac{1}{12}(100^2 - 1)} = \sqrt{\frac{1}{12}(9999)} = \sqrt{833.25} = 28.87$$

- 3. Standard deviation is independent of change of origin but not of scale.**

Example 4.19: For a group of 50 male workers, the mean and standard deviation of their monthly wages are ₹ 6300 and ₹ 900 respectively. For a group of 40 female workers, these are ₹ 5400 and ₹ 600 respectively. Find the standard deviation of monthly wages for the combined group of workers.

Solution: Given that

$$n_1 = 50, \bar{x}_1 = 6300, \sigma_1 = 900$$

$$n_2 = 40, \bar{x}_2 = 5400, \sigma_2 = 600$$

$$\text{Then, Combined Mean } \bar{x}_{12} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2} = \frac{50 \times 6300 + 40 \times 5400}{50 + 40} = 5,900$$

and Combined Standard Deviation

$$\begin{aligned} \sigma_{12} &= \sqrt{\frac{n_1 (\sigma_1^2 + d_1^2) + n_2 (\sigma_2^2 + d_2^2)}{n_1 + n_2}} \\ &= \sqrt{\frac{50(8,10,000 + 1,60,000) + 40(3,60,000 + 2,50,000)}{50 + 40}} = ₹ 900 \end{aligned}$$

$$\text{where } d_1 = \bar{x}_{12} - \bar{x}_1 = 5900 - 6300 = -400$$

$$d_2 = \bar{x}_{12} - \bar{x}_2 = 5900 - 5400 = 500$$

Example 4.20: A study of the age of 100 persons grouped into intervals 20–22, 22–24, 24–26,... revealed the mean age and standard deviation to be 32.02 and 13.18, respectively. While checking, it was discovered that the observation 57 was misread as 27. Calculate the correct mean age and standard deviation.

Solution: From the data given in the problem, we have $\bar{x} = 32.02$, $\sigma = 13.18$ and $N = 100$. We know that

$$\bar{x} = \frac{\Sigma fx}{N} \text{ or } \Sigma fx = N \times \bar{x} = 100 \times 32.02 = 3202$$

$$\begin{aligned} \text{and } \sigma^2 &= \frac{\Sigma fx^2}{N} - (\bar{x})^2 \text{ or } \Sigma fx^2 = N\sigma^2 + (\bar{x})^2 = 100[(13.18)^2 + (32.02)^2] \\ &= 100(173.71 + 1025.28) = 100 \times 1198.99 \\ &= 1,19,899 \end{aligned}$$

On re-placing the correct observation, we get

$$\Sigma fx = 3202 - 27 + 57 = 3232.$$

$$\text{Also } \Sigma fx^2 = 1,19,899 - (27)^2 + (57)^2 = 1,19,899 - 729 + 3249 = 1,22,419$$

$$\text{Thus, correct A.M. is } \bar{x} = \frac{\Sigma fx}{N} = \frac{3232}{100} = 32.32.$$

$$\begin{aligned} \text{and correct variance is } \sigma^2 &= \frac{\Sigma fx^2}{N} - (\bar{x})^2 = \frac{1,22,419}{100} - (32.32)^2 \\ &= 1224.19 - 1044.58 = 179.61 \end{aligned}$$

$$\text{or correct standard deviation is } \sigma = \sqrt{\sigma^2} = \sqrt{179.61} = 13.402.$$

Example 4.21: For a group containing 100 items, the A.M. and S.D. are 8 and $\sqrt{10.5}$. For 50 observations selected from these 100 observations, A.M. and S.D. are 10 and 2, respectively. Find A.M. and S.D. of the remaining 50 observations. [Delhi Univ., B.Com. (Hons.), 1994]

Solution: Let the two subgroups be denoted by 1 and 2. Then given

$$\text{Group 1 : } n_1 = 50, \bar{x}_1 = 10, \sigma_1 = 2, \bar{x}_{12} = 8$$

$$\text{Group 2 : } n_2 = 50, \bar{x}_2 = ?, \sigma_2 = ?, \sigma_{12} = \sqrt{10.5}$$

Combining A.M. of both groups, we get

$$\bar{x}_{12} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2}$$

$$8 = \frac{50 \times 10 + 50 \bar{x}_2}{100}, \text{ i.e. } 800 = 500 + 50 \bar{x}_2 \quad \text{or} \quad \bar{x}_2 = 6$$

Further $d_1 = |\bar{x}_{12} - \bar{x}_1| = |8 - 10| = 2$ and $d_2 = |\bar{x}_{12} - \bar{x}_2| = |8 - 6| = 2$

Combined

$$\sigma_{12} = \sqrt{\frac{n_1 \sigma_1^2 + n_2 \sigma_2^2 + n_1 d_1^2 + n_2 d_2^2}{n_1 + n_2}}$$

$$\sqrt{10.5} = \sqrt{\frac{50 \times 4 + 50 \sigma_2^2 + 50 \times 4 + 50 \times 4}{100}}$$

$$10.5 = \frac{200 + 50 \sigma_2^2 + 200 + 200}{100}$$

$$10.5 \times 100 = 600 + 50 \sigma_2^2, \text{ i.e. } \sigma_2^2 = 9 \quad \text{or} \quad \sigma_2 = 3$$

Example 4.22 : The A.M. of two samples of size 50 and 100 are 54.4 and 50.3 and their S.D. are 8 and 7, respectively. Obtain the combined S.D. of 150 items. [Delhi Univ., B.Com.(Hons.), 2005]

Solution: Given, $\bar{x}_1 = 54.4$, $\bar{x}_2 = 50.3$; $n_1 = 50$, $n_2 = 100$; $\sigma_1 = 8$, $\sigma_2 = 7$. Thus

$$\text{Combined, } \bar{x}_{12} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2} = \frac{50(54.4) + 100(50.3)}{50 + 100} = \frac{2720 + 5030}{150} = 51.67$$

Also $d_1 = |\bar{x}_{12} - \bar{x}_1| = |51.67 - 54.4| = 2.73$; $d_2 = |\bar{x}_{12} - \bar{x}_2| = |51.67 - 50.3| = 1.37$

$$\begin{aligned} \text{Combined } \sigma_{12} &= \sqrt{\frac{n_1 \sigma_1^2 + n_2 \sigma_2^2 + n_1 d_1^2 + n_2 d_2^2}{n_1 + n_2}} \\ &= \sqrt{\frac{50(8)^2 + 100(7)^2 + 50(2.73)^2 + 100(1.37)^2}{50 + 100}} \\ &= \sqrt{\frac{50 \times 64 + 100 \times 49 + 50(7.4529) + 100(1.8769)}{150}} \\ &= \sqrt{\frac{3200 + 4900 + 372.645 + 187.69}{150}} = \sqrt{57.736} = 7.598. \end{aligned}$$

Example 4.23: For two groups of observations, the following results are available:

Group I : $\Sigma (x - 5) = 8$, $\Sigma (x - 5)^2 = 40$, $n_1 = 20$

Group II : $\Sigma (x - 8) = -10$, $\Sigma (x - 8)^2 = 70$, $n_2 = 25$

Find A.M. and S.D. of both groups taken together.

[Delhi Univ., B.Com.(Hons.), 2006]

Solution: From the data of the problem, we have

$$\bar{x}_1 = A + \frac{\Sigma d}{n} = 5 + \frac{8}{20} = 5 + 0.4 = 5.4$$

$$\bar{x}_2 = A + \frac{\Sigma d}{n} = 8 + \frac{(-10)}{25} = 8 - 0.4 = 7.6$$

$$\text{Combined mean, } \bar{x}_{12} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2} = \frac{(20)(5.4) + (25)(7.6)}{20 + 25} = 6.6$$

$$\text{S.D. } (\sigma_1) = \sqrt{\frac{\Sigma d^2}{n} - \left(\frac{\Sigma d}{n}\right)^2} = \sqrt{\frac{40}{20} - \left(\frac{8}{20}\right)^2} = \sqrt{2 - 0.16} = \sqrt{1.84} = 1.356$$

$$\text{S.D. } (\sigma_2) = \sqrt{\frac{\Sigma d^2}{n} - \left(\frac{\Sigma d}{n}\right)^2} = \sqrt{\frac{70}{25} - \left(\frac{-10}{25}\right)^2} = \sqrt{2.8 - 0.16} = \sqrt{2.64} = 1.624$$

$$\begin{aligned} \text{Combined S.D. } (\sigma_{12}) &= \sqrt{\frac{n_1\sigma_1^2 + n_2\sigma_2^2 + n_1d_1^2 + n_2d_2^2}{n_1 + n_2}} \\ &= \sqrt{\frac{20(1.356)^2 + 25(1.624)^2 + 20(1.2)^2 + 25(1.0)^2}{20 + 25}} \\ &= \sqrt{\frac{36.774 + 65.934 + 28.8 + 25}{45}} = \sqrt{3.477} = 1.865. \end{aligned}$$

where $d_1 = |\bar{x}_{12} - \bar{x}_1| = |6.6 - 5.4| = 1.2$; $d_2 = |\bar{x}_{12} - \bar{x}_2| = |6.6 - 7.6| = 1.0$

Example 4.24: The mean of 5 observations is 15 and the variance is 9. If two more observations having values -3 and 10 are combined with these 5 observations, what will be the new mean and variance of 7 observations?

Solution: From the data of the problem, we have $\bar{x} = 15$, $\sigma^2 = 9$ and $n = 5$. We know that

$$\bar{x} = \frac{1}{n} \Sigma x \quad \text{or} \quad \Sigma x = n \times \bar{x} = 5 \times 15 = 75$$

If two more observations having values -3 and 10 are added to the existing 5 observations, then after adding these 6th and 7th observations, we get

$$\Sigma x = 75 - 3 + 10 = 82$$

$$\text{Thus, the new A.M. is } \bar{x} = \frac{\Sigma x}{n} = \frac{82}{7} = 11.71$$

$$\text{Variance, } \sigma^2 = \frac{\Sigma x^2}{n} - (\bar{x})^2$$

$$9 = \frac{\Sigma x^2}{5} - (15)^2 \quad \text{or} \quad \Sigma x^2 = 1170$$

On adding two more observations, i.e., -3 and 10 , we get

$$\Sigma x^2 = 1170 + (-3)^2 + (10)^2 = 1279$$

$$\text{Variance, } \sigma^2 = \frac{\Sigma x^2}{n} - (\bar{x})^2 = \frac{1279}{7} - (11.71)^2 = 45.59$$

Hence, the new mean and variance of 7 observations is 11.71 and 45.59 , respectively.

Relationship Between Different Measures of Variation

$$(a) \text{ Quartile deviation (Q.D.)} = \frac{2}{3} \sigma$$

$$\text{Mean absolute deviation (MAD)} = \frac{4}{5} \sigma$$

$$(b) \text{ Quartile deviation} = \frac{5}{6} \text{ MAD}$$

$$\text{Standard deviation} = \frac{5}{4} \text{ MAD} \quad \text{or} \quad \frac{3}{2} \text{ QD}$$

$$(c) \text{ Mean absolute deviation} = \frac{6}{5} \text{ QD}$$

These relationships are applicable only to symmetrical distributions.

Example 4.25: A collar manufacturer is considering the production of a new collar to attract young men. Thus following statistics of neck circumference are available based on measurement of a typical group of the college students:

Mid-value (in inches):	12.0	12.5	13.0	13.5	14.0	14.5	15.0	15.5	16.0
Number of students :	2	16	36	60	76	37	18	3	2

Compute the standard deviation and use the criterion $\bar{x} \pm 3\sigma$, where σ is the standard deviation and $\bar{x}$ is the arithmetic mean to determine the largest and smallest size of the collar he should make in order to meet the needs of practically all the customers bearing in mind that collar are worn on average half inch longer than neck size.

Solution: Calculations for mean and standard deviation in order to determine the range of collar size to meet the needs of customers are shown in Table 4.10.

Table 4.10 Calculations for Mean and Standard Deviation

Mid-value (in inches)	Number of Students (f)	$d = \frac{x - A}{h} = \frac{x - 14}{0.5}$	fd	fd^2
12.0	2	-4	-8	32
12.5	16	-3	-48	144
13.0	36	-2	-72	144
13.5	60	-1	-60	60
14.0 ← A	76	0	0	0
14.5	37	1	37	37
15.0	18	2	36	72
15.5	3	3	9	27
16.0	2	4	8	32
	250		-98	548

$$\text{Mean, } \bar{x} = A + \frac{\sum fd}{N} \times h = 14.0 - \frac{98}{250} \times 0.5 = 14.0 - 0.196 = 13.804$$

$$\begin{aligned} \text{Standard deviation, } \sigma &= \sqrt{\frac{\sum fd^2}{N} - \left(\frac{\sum fd}{N}\right)^2} \times h = \sqrt{\frac{548}{250} - \left(\frac{-98}{250}\right)^2} \times 0.5 \\ &= 4 \times 0.5 = 1.427 \times 0.5 = 0.7135 \end{aligned}$$

$$\text{Largest and smallest neck size} = \bar{x} \pm 3\sigma = 13.804 \pm 3 \times 0.713 = 11.666 \text{ and } 15.944.$$

Since all the customers are to wear collar half inch longer than their neck size 0.5 is to be added to the neck size range given above. The new range then becomes
(11.666 + 0.5) and (15.944 + 0.5) or 12.165 and 16.444, i.e. 12.2 and 16.4 inches.

4.5.4 Coefficient of Variation

Standard deviation is an absolute measure of variation and expresses variation in the same unit of measurement as the arithmetic mean or the original data. A relative measure called the **coefficient of variation** (CV) developed by Karl Pearson is very useful measure for (i) comparing two or more data sets expressed in different units of measurement, and (ii) comparing data sets that are in same unit of measurement but the mean values of data sets in a comparable field are widely dissimilar (such as mean wages received per month by the top management personnel and labour class personnel of a large organization).

Thus in view of this limitation, we need to convert absolute measure of variation, that is, S.D. into a relative measure, which can be helpful in comparing the variability of two or more sets of data. The new measure, coefficient of variation (CV), measures the standard deviation relative to the mean in percentages. In other words, CV indicates how large the standard deviation is in relation to the mean and is computed as follows:

$$\text{Coefficient of variation (CV)} = \frac{\text{Standard deviation}}{\text{Mean}} \times 100 = \frac{\sigma}{\bar{x}} \times 100$$

Multiplying by 100 converts the decimal to a percent.

The set of data for which the coefficient of variation is low is said to be more uniform (consistent) or more homogeneous (stable).

Example 4.26: The weekly sales of two products A and B were recorded as given below:

Product A	:	59	75	27	63	27	28	56
Product B	:	150	200	125	310	330	250	225

Find out which of the two shows greater fluctuation in sales.

Solution: For comparing the fluctuation in sales of two products, we will prefer to calculate coefficient of variation for both the products (Table 4.11).

Product A: Let A = 56 be the assumed mean of sales for product A.

Table 4.11 Calculations of Mean and Standard Deviation

Sales (x)	Frequency (f)	$d = x - A$	fd	fd^2
27	2	-29	-58	1682
28	1	-28	-28	784
56 ← A	1	0	0	0
59	1	3	3	9
63	1	7	7	49
75	1	19	19	361
	<u>7</u>		<u>-57</u>	<u>2885</u>

$$\bar{x} = A + \frac{\Sigma fd}{\Sigma f} = 56 - \frac{57}{7} = 47.86$$

$$\sigma_A^2 = \frac{\Sigma fd^2}{\Sigma f} - \left(\frac{\Sigma fd}{\Sigma f} \right)^2 = \frac{2885}{7} - \left(-\frac{57}{7} \right)^2 = 412.14 - 66.30 = 345.84$$

$$s_A = \sqrt{345.84} = 18.60$$

Then $CV(A) = \frac{\sigma_A}{\bar{x}} \times 100 = \frac{18.60}{47.86} \times 100 = 38.86 \text{ percent}$

Product B: Let $A = 225$ be the assumed mean of sales for product B (Table 4.12).

Table 4.12 Calculations of Mean and Standard Deviation

Sales (x)	Frequency (f)	$d = x - A$	fd	fd^2
125	1	-100	-100	10,000
150	1	-75	-75	5625
200	1	-25	-25	625
(225) ← A	1	0	0	0
250	1	25	25	625
310	1	85	85	7225
330	1	105	105	11,025
	<u>7</u>		<u>15</u>	<u>35,125</u>

$$\sigma_A = \sqrt{345.84} = 18.60$$

$$\bar{x} = A + \frac{\Sigma fd}{\Sigma f} = 225 + \frac{15}{7} = 227.14$$

$$\sigma_A^2 = \frac{\Sigma fd^2}{\Sigma f} - \left(\frac{\Sigma fd}{\Sigma f} \right)^2 = \frac{35,125}{7} - \left(\frac{15}{7} \right)^2 = 5017.86 - 4.59 = 5013.27$$

or $\sigma_B = \sqrt{5013.27} = 70.80$

Then $CV(B) = \frac{\sigma_B}{\bar{x}} \times 100 = \frac{70.80}{227.14} \times 100 = 31.17 \text{ percent}$

Since the coefficient variation for product A is more than that of product B, therefore the sales fluctuation in case of product A is higher.

Example 4.27: Verify the correctness of the following statement: A batsman scored at an average of 60 runs in an inning against Pakistan. The S.D. of the scores by him was 12. A year later against Australia, his average came down to 50 runs an inning and the S.D. of the runs scored fell down to 9. Therefore, it is correct to say that his performance was worse against Australia and that there was lesser consistency in his batting against Australia.

Solution: Against Pakistan: $\bar{x}_p = 60$; $\sigma_p = 12$

$$CV = \frac{\sigma_p}{\bar{x}_p} \times 100 = \frac{12}{60} \times 100 = 20 \text{ per cent}$$

Against Australia: $\bar{x}_A = 50$; $\sigma_A = 9$

$$CV = \frac{\sigma_A}{\bar{x}_A} \times 100 = \frac{9}{50} \times 100 = 18 \text{ per cent}$$

Since CV for Australia is less than that of Pakistan, there is more consistency in his batting against Australia. Hence the statement is incorrect. However, the average runs scored by him were more against Pakistan.

Example 4.28: After settlement the average weekly wages in a factory has increased from ₹ 8000 to ₹ 12000 and the S.D. has increased from ₹ 100 to ₹ 150. After settlement, the wages has become higher and more uniform. Do you agree? [Delhi Univ., B.Com.(Hons.), 2005]

Solution: Before settlement, $CV = \frac{\text{S.D.}}{\text{Mean}} \times 100 = \frac{100}{8000} \times 100 = 1.25 \text{ per cent}$

$$\text{After settlement, } CV = \frac{150}{12000} \times 100 = 1.25 \text{ per cent}$$

Since there is no change in CV, there is no improvement in uniformity.

Example 4.29: The following is the record of goals scored by team A in a football session :

No. of goals scored : 0, 1, 2, 3, 4

No. of matches : 1, 9, 7, 5, 3

From team B, the average number of goals scored per match are 2.5 with a standard deviation of 1.25 goals. Find which team may be considered as more consistent. [Delhi Univ., B.Com.(Pass), 2006]

Solution: Calculating coefficient of variation of two teams to comment on consistency as follows:

No. of Goals (x)	No. of Matches (f)	$d = x - A$ $= x - 2$	fd	fd ²
0	1	-2	-2	4
1	9	-1	-9	9
2	7	0	0	0
3	5	1	5	5
4	3	2	6	12
	25		0	30

$$\bar{x} = A + \frac{\Sigma fd}{n} = 2 + \frac{0}{25} = 2$$

$$\sigma = \sqrt{\frac{\Sigma fd^2}{n} - \left(\frac{\Sigma fd}{n}\right)^2} = \sqrt{\frac{30}{25} - \left(\frac{0}{25}\right)^2} = 1.09$$

$$CV \text{ for team A} = \frac{\sigma_A}{\bar{x}_A} \times 100 = \frac{1.09}{2} \times 100 = 54.5 \text{ per cent}$$

$$CV \text{ for team B} = \frac{\sigma_B}{\bar{x}_B} \times 100 = \frac{1.25}{2.5} \times 100 = 50 \text{ per cent.}$$

Since CV of team B is lesser than that of A, team B is more consistent.

Example 4.30: From the analysis of monthly wages paid to employees in two service organizations X and Y, the following results were obtained:

	Organization X	Organization Y
Number of wage-earners	550	650
Average monthly wages	5000	4500
Variance of the distribution of wages	900	1600

- (a) Which organization pays a larger amount as monthly wages?
 (b) In which organization is there greater variability in individual wages of all the wage earners taken together?

Solution: (a) For finding out which organization, X or Y pays larger amount of monthly wages, we have to compare the total wages:

Total wage bill paid monthly by X and Y is

$$X : n_1 \times \bar{x}_1 = 550 \times 5000 = ₹ 27,50,000$$

$$Y : n_2 \times \bar{x}_2 = 650 \times 4500 = ₹ 29,25,000$$

Organization Y pays a larger amount as monthly wages as compared to organization X.

- (b) For calculating the combined variation, we will first calculate the combined mean as follows:

$$\bar{x}_{12} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2} = \frac{27,50,000 + 29,25,000}{1200} = ₹ 4729.166$$

$$\begin{aligned} \sigma_{12} &= \sqrt{\frac{n_1 (\sigma_1^2 + d_1^2) + n_2 (\sigma_2^2 + d_2^2)}{n_1 + n_2}} \\ &= \sqrt{\frac{550(900 + 73,351.05) + 650(1600 + 52,517.05)}{550 + 650}} \\ &= \sqrt{\frac{4,08,38,077.55 + 3,51,76,082.50}{1200}} = \sqrt{63345.13} = 251.68 \end{aligned}$$

where $d_1 = |\bar{x}_{12} - \bar{x}_1| = 4729.166 - 5000 = 270.834$

$$d_2 = |\bar{x}_{12} - \bar{x}_2| = 4729.166 - 4500 = 229.166$$

Calculating coefficient of variance of both organization to comment on variability in wages as follows:

$$CV(X) = \frac{\sigma}{\bar{x}} \times 100 = \frac{30}{5000} \times 100 = 0.60$$

$$CV(Y) = \frac{40}{4500} \times 100 = 0.89$$

Since $CV(Y) > CV(X)$, there is greater variability in individual wages of organization Y.

Conceptual Questions 4B

7. What purpose does a measure of variation serve? In the light of these, comment on some of the well-known measures of variation.
8. What do you understand by 'coefficient of variation'? Discuss its importance in business problems.
9. Explain and illustrate how the measures of variation afford a supplement to the information about frequency distribution furnished by averages.
10. Explain the advantages of standard deviation as a measure of variation over range and the average deviation. Under what circumstances will the variance of a variable be zero?
11. Comment on the comparative merits and demerits of measures of variation.
12. Describe the various methods of measuring variation along with their respective merits and demerits.
13. It has been said that the lesser the variability that exists, the more an average is representative of a set of data. Comment.
14. What information is provided by variance or standard deviation?
15. What advantages are associated with variance and standard deviation relative to range as the measure of variability?

Self-practice Problems 4B

- 4.9 Find the average deviation from mean for the following distribution:
Quantity demanded (in units):
60 61 62 63 64 65 66 67 68
Frequency :
2 0 15 29 25 12 10 4 3
- 4.10 Find the average deviation from mean for the following distribution:
Dividend yield:
0-3 3-6 6-9 9-12 12-15 15-18 18-21
Number of companies:
2 7 10 12 9 6 4
- 4.11 Find the average deviation from median for the following distribution:
Sales (₹ '000):
1-3 3-5 5-7 7-9 9-11 11-13 13-15 15-17
Number of shops:
6 53 85 56 21 26 4 4
- 4.12 In a survey of 48 engineering companies, following data was collected:
Level of profit (₹ in lakh):
10 11 12 13 14
Number of companies:
3 12 18 12 3
- 4.13 Calculate the variance and standard deviation for the distribution.
A manufacturer of T-shirts approaches you with the following information
Length of shoulder (in inches):
12.0 12.5 13.0 13.5 14 14.5 15 15.5 16
Frequency:
5 20 30 43 60 56 37 16 3
Calculate the standard deviation and advice the manufacturer as to the largest and the smallest shoulder size T-shirts, he should make in order to meet the needs of his customers.
- 4.14 A charitable organization decided to give old-age pension to people over 60 years of age. The scales of pension were fixed as follows:

<i>Age Group</i>	<i>Pension/month (₹)</i>
60-65	200
65-70	250
70-75	300
75-80	350
80-85	400

The ages of 25 persons who secured the pension are as given below:

74 62 84 72 61 83 72 81
64 71 63 61 60 67 74 64
79 73 75 76 69 68 78 66
67

Calculate the monthly average pension payable per person and the standard deviation.

- 4.15** Two automatic filling machines A and B are used to fill tea in 500 g cartons. A random sample of 100 cartons on each machine showed the following:

Tea Contents (in g)	Machine A	Machine B
485–490	12	10
490–495	18	15
495–500	20	24
500–505	22	20
505–510	24	18
510–515	4	13

Comment on the performance of the two machines on the basis of average filling and dispersion.

- 4.16** An analysis of production rejects resulted in the following observations

No. of Rejects per Operator	No. of Operators	No. of Rejects per Operator	No. of Operators
21–25	5	41–45	15
26–30	15	46–50	12
31–35	28	51–55	3
36–40	42		

Calculate the mean and standard deviation.

- 4.17** Blood serum cholestrol levels of 10 persons are as under:

240 260 290 245 255 288 272 263 277 250

Calculate the standard deviation with the help of assumed mean.

- 4.18** 32 trials of a process to finish a certain job revealed the following information:

Mean time taken to complete the job = 80 minutes

Standard deviation = 16 minutes

Another set of 8 trials gave mean time as 100 minutes and standard deviation equalled to 25 minutes.

Find the combined mean and standard deviation.

- 4.19** From the analysis of monthly wages paid to workers in two organizations X and Y, the following results were obtained:

	X	Y
Number of wage-earners	550	600
Average monthly wages (₹)	1260	1348.5
Variance of distribution of wages (₹)	100	841

Obtain the average wages and the variability in individual wages of all the workers in the two organizations taken together.

- 4.20** An analysis of the results of a budget survey of 150 families showed an average monthly expenditure of ₹ 120 on food items with a standard deviation of ₹ 15. After the analysis was completed, it was noted that the figure recorded for one household was wrongly taken as ₹ 15 instead of ₹ 105. Determine the correct value of the average expenditure and its standard deviation.

- 4.21** The standard deviation of a distribution of 100 values was ₹ 2. If the sum of the squares of the actual values was ₹ 3,600, what was the mean of this distribution?

- 4.22** The hourly output of a new machine is four times that of the old machine. If the variance of the hourly output of the old machine in a period of n hours is 16, what is the variance of the hourly output of the new machine in the same period of n hours.

Coefficient of Variance

- 4.23** Two salesmen selling the same product show the following results over a long period of time:

	Salesman X	Salesman Y
Average Sales Volume per Month (₹)	30,000	35,000
Standard deviation	2,500	3,600

Which salesman seems to be more consistent in the volume of sales?

- 4.24** Suppose that samples of polythene bags from two manufacturers A and B are tested by a buyer for bursting pressure giving the following results:

<i>Bursting Pressure</i>	<i>Number of Bags</i>	
	<i>A</i>	<i>B</i>
5.0–9.9	2	9
10.0–14.9	9	11
15.0–19.9	29	18
20.0–24.9	54	32
25.0–29.9	11	27
30.0–34.9	5	13

- (a) Which set of bags has the highest bursting pressure?
 (b) Which has more uniform pressure? If prices are the same, which manufacturer's bags would be preferred by the buyer? Why?

- 4.25** The number of employees, average daily wages per employee, and the variance of daily wages per employee for two factories are given below:

	<i>Factory A</i>	<i>Factory B</i>
Number of employees :	50	100
Average daily wages (₹) :	120	85
Variance of daily wages (₹) :	9	16

- (a) In which factory is there greater variation in the distribution of daily wages per employee?
 (b) Suppose in factory B, the wages of an employee were wrongly noted as ₹ 120 instead of ₹ 100. What would be the correct variance for factory B?

- 4.26** The share prices of a company in Mumbai and Kolkata markets during the last 10 months are recorded below:

<i>Month</i>	<i>Mumbai</i>	<i>Kolkata</i>
January	105	108
February	120	117
March	115	120
April	118	130
May	130	100
June	127	125
July	109	125
August	110	120
September	104	110
October	112	135

Determine the arithmetic mean and standard deviation of prices of shares. In which market are the share prices more stable?

- 4.27** A person owns two petrol filling stations A and B. At station A, a representative sample of 200 consumers who purchase petrol was taken. The results were as follows:

<i>Number of Litres of Petrol Purchased</i>	<i>Number of Consumers</i>
0 and < 2	15
2 and < 4	40
4 and < 6	65
6 and < 8	40
8 and < 10	30
10 and over	10

A similar sample at station B users showed a mean of 4 litres with a standard deviation of 2.2 litres. At which station is the purchase of petrol relatively more variable?

Hints and Answers

- 4.9** $MAD = 1.239$; $\bar{x} = 63.89$
4.10 $\bar{x} = 10.68$; $MAD = 3.823$
4.11 $Med = 6.612$; $MAD = 2.252$
4.12 $\sigma^2 = 1$ and $\sigma = 1$
4.13 $\bar{x} = 14.013$ inches; $\sigma = 0.8706$ inches;
 $\bar{x} + 3\sigma = 14.884$ (largest size);
 $\bar{x} - 3\sigma = 13.142$ (smallest size)
4.14 $\bar{x} = ₹ 280.2$; $\sigma = ₹ 60.765$
4.15 Machine A: $\bar{x}_1 = 499.5$; $\sigma_1 = 7.14$;
 Machine B : $\bar{x}_2 = 500.5$; $\sigma_2 = 7.40$
4.16 $\bar{x} = 36.96$; $\sigma = 6.375$
4.17 $\sigma = 16.48$
4.18 $\bar{x}_{12} = 84$ minutes; $\sigma_{12} = 19.84$

4.19 $\bar{x}_{12} = ₹ 1306$; $\sigma_{12} = ₹ 53.14$

4.20 Corrected $\bar{x} = ₹ 120.6$ and

Corrected $\sigma = ₹ 12.4$

4.21 $\bar{x} = 5.66$

4.22 Variance (new machine) = 256 hours

4.23 Salesman X

4.24 Manufacture A: $\bar{x}_1 = 21$, $\sigma_1 = 4.875$ and C.V. = 23.32 per cent

Manufacturer B : $\bar{x}_2 = 21.81$, $\sigma_2 = 7.074$ and C.V. = 32.44 per cent; (a) Bags of manufacturer B have higher bursting pressure; (b) Bags of manufacturer A have more uniform pressure; (c) Bags of manufacturer A should be preferred by buyer as they have uniform pressure.

4.25 (a) CV(A) = 2.5;

CV(B) = 4.7. Variation in the distribution of daily wages per employee in factory B is more.

(b) Correct $\Sigma x = 100 \times 85 - 120 + 100$
= 8,480

Correct mean $\bar{x} = 8480/100 = 84.8$

Since $\sigma^2 = (\Sigma x^2/N) - (\bar{x})^2$

or $16 = (\Sigma x^2/100) - (85)^2$
= $(\Sigma x^2/100) - 7225$

or $\Sigma x^2 = 7,24,100$

Correct $\Sigma x^2 = 7,24,100 - (120)^2 + (100)^2$
= 7,19,700

Correct $\sigma^2 = (7,19,700/100) - (84.8)^2$
= 5.96

4.26 CV(Mumbai) = 7.24 per cent; CV(Kolkata) = 8.48 per cent. This shows more stability in Mumbai stock market.

4.27 CV(A) = 46.02 per cent; CV(B) = 55 per cent. The purchase of petrol is relatively more variable at station B.

Formulae Used

1. Range, R

Value of highest observation – Value of lowest observation = H – L

$$\text{Coefficient of range} = \frac{H - L}{H + L}$$

2. Interquartile range = $Q_3 - Q_1$

$$\text{Quartile deviation, QD} = \frac{Q_3 - Q_1}{2}$$

$$\text{Coefficient of QD} = \frac{Q_3 - Q_1}{Q_3 + Q_1}$$

3. Mean average deviation

For ungrouped data

(i) $\text{MAD} = \frac{\Sigma |x - \bar{x}|}{n}$ for sample

(ii) $\text{MAD} = \frac{\Sigma |x - \mu|}{N}$ for population

(iii) $\text{MAD} = \frac{\Sigma |x - \text{Me}|}{n}$ from median

For grouped data $\text{MAD} = \frac{\Sigma f |x - \bar{x}|}{\Sigma f}$

4. Coefficient of MAD = $\frac{\text{MAD}}{\bar{x} \text{ or Me}} \times 100$

5. Variance

Ungrouped data

$$\begin{aligned} \sigma^2 &= \frac{\Sigma (x - \bar{x})^2}{N} \\ &= \frac{\Sigma x^2}{N} - \left(\frac{\Sigma x}{N} \right)^2 \\ &= \frac{\Sigma d^2}{N} - \left(\frac{\Sigma d}{N} \right)^2 \end{aligned}$$

where $d = x - A$; A is any assumed A.M. value

Grouped data,

$$\sigma^2 = \left[\frac{\Sigma f d^2}{N} - \left(\frac{\Sigma f d}{N} \right)^2 \right] h$$

where $d = (m - A)/h$; h is the class interval and m is the mid-value of class intervals.

6. Standard deviation

Ungrouped data $\sigma = \sqrt{\sigma^2}$

Grouped data

$$\sigma = \sqrt{\frac{\sum f d^2}{N} - \left(\frac{\sum f d}{N}\right)^2} \times h$$

$$7. \text{ Coefficient of variation (CV)} = \frac{\sigma}{\bar{x}} \times 100$$

True or False Concepts Quiz

- Range is a measure of variation which gives us information about scatter of values around a measure of central tendency.
- When a distribution consists of different observations, s or σ are relatively large.
- The interquartile range is based upon only two values in the data set.
- Absolute measures of variation are used for comparing variability among observations in a data set.
- The semi-interquartile range is inappropriate to use with skewed distributions.
- Mean absolute deviation taken from median is least.
- The standard deviation is measured in the same unit as the observations in the data set.
- In a symmetrical distribution, semi-interquartile range is one-fourth of the range.
- The coefficient of variation is a relative measure of dispersion.
- The inter-quartile range measures the average range of the lower fourth of a distribution.
- For a symmetrical distribution, mean absolute deviation equals 4/5 of standard deviation.
- Variance indicates the average distance of any observation in the data set from the mean.
- Sample standard deviation provides an accurate estimate of the population standard deviation.
- Variance is the square of the standard deviation.
- Standard deviation can be calculated by taking deviation from any measure of central tendency.

Concepts Quiz Answers

1. F	2. T	3. T	4. F	5. T	6. T	7. T	8. T	9. T
10. F	11. T	12. T	13. F	14. T	15. F			

Review Self-practice Problems

- 4.28 A petrol filling station has recorded the following data for litres of petrol sold per automobile in a sample of 680 automobiles:

Petrol Sold (Litres)	Frequency
0–4	74
5–9	192
10–14	280
15–19	105
20–24	23
25–29	6

Compute the mean and standard deviation for the data.

- 4.29 A work-standards expert observes the amount of time (in minutes) required to prepare a sample of 10 business letters in the office with observations in ascending order: 5, 5, 5, 7, 9, 14, 15, 15, 16, 18.
- Determine the range and middle 70 percent range for the sample.
 - If the sample mean of the data is 10.9, then calculate the mean absolute deviation and variance.

- 4.30** ABCStereos, a wholesaler, was contemplating becoming the supplier to three retailers, but inventory shortages have forced him to select only one. ABC's credit manager is evaluating the credit record of these three retailers. Over the past 5 years these retailers' accounts receivable have been outstanding for the following average number of days. The credit manager feels that consistency, in addition to lowest average, is important. Based on relative dispersion, which retailer would make the best customer?

Lee : 62.20 61.80 63.40 63.00 61.70

Forest : 62.50 61.90 63.80 63.00 61.70

Davis : 62.00 61.90 63.00 63.90 61.50

- 4.31** A purchasing agent obtained samples of 60 watt bulbs from two companies. He had the samples tested in his own laboratory for length of life with the following results:

<i>Length of Life</i> (in hours)	<i>Samples from</i>	
	<i>Company A</i>	<i>Company B</i>
1700–1900	10	3
1900–2100	16	40
2100–2300	20	12
2300–2500	8	3
2500–2700	6	2

- Which company's bulbs do you think are better in terms of average life?
- If prices of both the companies are same, which company's bulbs would you buy and why?

- 4.32** The Chief Medical Officer of a hospital conducted a survey of the number of days 200 randomly chosen patients stayed in the hospital following an operation. The data are given below

Hospital stay (in days) :

1–3 4–6 7–9 10–12 13–15 16–18 19–21 22–24

Number of patients:

18 90 44 21 9 9 4 5

- Calculate the mean number of days patients stay in the hospital along with standard deviation of the same.
- How many patients are expected to stay between 0 and 17 days.

- 4.33** There are a number of possible measures of sales performance, including how consistent a sales person is, in meeting established sales goals. The following data represent the percentage of goal met by each of three sales persons over the last five years

Raman : 88 68 89 92 103

Sindhu : 76 88 90 86 79

Prasad : 104 88 118 88 123

Which salesman is most consistent. Suggest an alternative measure of consistency (if possible).

- 4.34** Gupta Machine Company has a contract with one of its customers to supply machined pump gears. One requirement is that the diameter of its gears be within specific limits. The following data is of diameters (in inches) of a sample of 20 gears:

4.01 4.00 4.02 4.03 4.00 3.98 3.99 3.99

4.01 4.02 3.99 3.98 3.97 4.00 4.02 4.01

4.02 4.00 4.01 3.99

What can Gupta say to his customers about the diameters of 95 percent of the gears they are receiving?

- 4.35** Public transportation and the automobiles are two options, an employee can use to get to work each day. Samples of time (in minutes) recorded for each option are shown below:

Public transportation:

28 29 32 37 33 25 29 32 41 34

Automobile:

29 31 33 32 34 30 31 32 35 33

- Compute the sample mean time to get to work for each option.
 - Compute the sample standard deviation for each option.
 - On the basis of your results from parts (a) and (b), which method of transportation should be preferred? Explain.
- 4.36** The mean and standard deviation of a set of 100 observations were worked out as 40 and 5, respectively, by a computer which, by mistake, took the value 50 in place of 40 for one observation. Find the correct mean and variance.

- 4.37** The number of employees, wages per employee and the variance of the wages per employee for two factories is given below:

	Factory A	Factory B
Number of employees	100	150
Average wage per employee per month (₹)	3200	2800
Variance of the wages per employee per month (₹)	625	729

- (a) In which factory is there greater variation in the distribution of wages per employee?
- (b) Suppose in factory B, the wages of an employee were wrongly noted as ₹ 3050 instead of ₹ 3650, what would be the correct variance for factory B?
- 4.38** Two factories A and B engaged in the same industry, the average weekly wages and standard deviations are as follows:

Factory	Average Weekly Wages (₹)	S.D. of Wages (₹)	No. of Wage Earners
A	460	50	100
B	490	40	80

- (a) Which factory, A or B, pays a higher amount as weekly wages?

Hints and Answers

- 4.28** $\bar{x} = 10.74$ litres per automobile, $\sigma = 5.00$ litres
- 4.29** (a) Range = H – L
 $= 18 - 5 = 15$ minutes
 Middle 70 per cent of R = $P_{85} - P_{15}$
 $= x_{(85/100) + (1/2)} - x_{(15/100) + (1/2)}$
 $= x_{(8.5 + 0.5)} - x_{(1.5 + 0.5)} = x_9 - x_2$
 $= 16 - 5 = 11$ minutes
- (b) $MAD = \sum |x - \bar{x}|$
 $= \frac{47}{10} = 4.7$ minutes
 $s^2 = \frac{\sum x^2 - n\bar{x}^2}{n - 1}$
 $= \frac{1,431 - 10(10.9)^2}{10 - 1}$
 $= 26.99$ minutes
- (b) Which factory shows greater variability in the distribution of wages?
- (c) What is the mean and standard deviation of all the workers in two factories taken together?
- 4.39** The mean of 5 observations is 4.4 and the variance is 8.24. If three of the five observations are 1, 2 and 6, find the other two.
- 4.40** The mean and standard deviation of normal distribution are 60 and 5, respectively. Find the inter-quartile range and the mean deviation of the distribution.
- 4.41** The mean and standard deviation of a set of 100 observations were found to be 40 and 5 respectively. But by mistake a value 50 was taken in place of 40 for one observation. Re-calculate the correct mean and standard deviation.
- 4.42** The mean and the standard deviation of a sample of 10 sizes were found to be 9.5 and 2.5, respectively. Later on, an additional observation became available. This was 15.0 and was included in the original sample. Find the mean and the standard deviation of 11 observations.
- 4.30** Lee : $\bar{x} = 62.42$, $\sigma = 0.7497$,
 $CV = (\sigma/\bar{x}) \times 100 = 1.20$ per cent
 Forest: $\bar{x} = 62.18$, $\sigma = 0.9257$,
 $CV = (\sigma/\bar{x}) \times 100 = 1.49$ per cent
 Davis : $\bar{x} = 62.46$, $\sigma = 0.9762$,
 $CV = (\sigma/\bar{x}) \times 100$
 $= 1.56$ per cent
 Based on CV, Lee would be the best customer.
- 4.31** For Company A:
 $\bar{x} = A + \frac{\sum fd}{N} \times h$
 $= 2200 - \frac{16}{60} \times 200$
 $= 2146.67$;

$$\sigma = \sqrt{\frac{\sum fd^2}{N} - \left(\frac{\sum fd}{N}\right)^2} \times h$$

$$= \sqrt{\frac{88}{60} - \left(\frac{-16}{60}\right)^2} \times 200 = 236.4$$

$$CV = (\sigma/\bar{x}) \times 100 = 11 \text{ per cent}$$

For Company B: $\bar{x} = 2070$; $\sigma = 158.8$

and $CV = 7.67$ per cent.

(a) Bulbs of Company A are better.

(b) $CV(B) < CV(A)$: Buy company B bulbs as their burning hours are more uniform.

$$4.32 \quad \bar{x} = \frac{\sum fm}{n} = \frac{1543}{200} = 7.715 \text{ days;}$$

$$\sigma^2 = \frac{\sum (x - \bar{x})^2}{n - 1} = \frac{4384.755}{199}$$

$$= 22.034$$

$$\sigma = \sqrt{22.034} = 4.69 \text{ days}$$

4.33	Sales Person	$\bar{x}$	σ	$CV = (\sigma/\bar{x}) \times 100$
	Raman	88	12.67	14.4%
	Sindhu	83.8	6.02	7.2%
	Prasad	104.2	16.35	15.7%

Sindhu is most consistent both in terms of σ and CV.

$$4.34 \text{ Diameter : } 3.97 \ 3.98 \ 3.99 \ 4.00 \ 4.01 \ 4.02 \ 4.03$$

Frequency: 1 2 4 4 4 4 1

$$\bar{x} = \frac{\sum x}{n} = 80.04/20 = 4.002 \text{ inches}$$

$$\sigma = \sqrt{\frac{\sum x^2 - n(\bar{x})^2}{n - 1}}$$

$$= \sqrt{\frac{320.325 - 20(4.002)^2}{19}} = 0.016 \text{ inches}$$

If distribution is bell-shaped, then 95 per cent of the gears will have diameters in the interval: $\bar{x} \pm 2\sigma = 4.002 \pm 2(0.016)$
 $= (3.970, 4.034) \text{ inches.}$

4.35 (a) Public : 32; Auto : 32 (b) Public : 4.64;
 Auto : 1.83 (c) Auto has less variability.

$$4.36 \text{ (i) } \bar{x} = \frac{\sum x}{n}$$

$$\text{or } \sum x = \bar{x} N = 40 \times 100 = 4,000$$

$$\text{Correct } \sum x = 4000 - 50 + 40 = 3990.$$

$$\text{Thus, Correct } \bar{x} = 3990 \div 100 = 39.9$$

$$\text{(ii) } \sigma^2 = \text{or } 25 = \frac{\sum x^2}{100} - (40)^2$$

$$\text{or } \sum x^2 = 1,62,500$$

$$\text{Correct } \sum x^2 = 1,62,500 - (50)^2 + (40)^2$$

$$= 1,62,500 - 2500 + 1600$$

$$= 1,61,600$$

$$\text{Correct } \sigma^2$$

$$= \frac{\text{Correct } \sum x^2}{N} - (\text{Correct } \bar{x})^2$$

$$= \frac{1,61,600}{100} - (39.9)^2 = 23.99$$

$$4.37 \text{ (a) } CV(A) = \frac{\sigma}{\bar{x}} \times 100 = \frac{\sqrt{625}}{3200} \times 100$$

$$= 0.781 ;$$

$$CV(B) = \frac{\sqrt{729}}{2800} \times 100 = 0.964$$

There is greater variation in the distribution of wages per employee in factory B.

$$\text{(b) } \bar{x} = \frac{\sum x}{N}$$

$$\text{or } \sum x = N \bar{x} = 150 \times 2800$$

$$= 4,20,000$$

$$\text{Correct } \sum x$$

$$= 4,20,000 - 3050 + 3650$$

$$= 4,20,600;$$

$$\text{Correct, } \bar{x} = \frac{4,20,600}{150} = 2,804$$

$$\text{Variance, } \sigma^2 = \frac{\sum x^2}{N} - (\bar{x})^2$$

$$\text{or } 729 = \frac{\sum x^2}{150} - (2800)^2$$

$$\text{or } \sum x^2 = 1,17,61,09,350$$

$$\text{Correct } \sum x^2 = 1,17,61,09,350 - (3050)^2 + (3650)^2 = 1,18,01,29,350$$

$$\text{Correct } \sigma^2$$

$$= \frac{\text{Correct } \sum x^2}{N} - (\text{Correct } \bar{x})^2$$

$$= \frac{1,18,01,29,350}{150} - (2804)^2 = 5113$$

$$4.38 \text{ (a) Total weekly wages: Factory A = } 460 \times 100 = ₹ 46,000; \text{ Factory B = } 490 \times 80 = ₹ 39,200.$$

Factory A pays a larger amount.

(b) CV (Factory A)

$$= \frac{\sigma}{\bar{x}} \times 100 = \frac{50}{460} \times 100 = 10.87 \text{ per cent};$$

CV (Factory B)

$$= \frac{40}{490} \times 100 = 8.16 \text{ per cent}$$

Since CV (A) > CV (B), Factory A shows greater variability in wages.

$$\begin{aligned} \text{(c) } \bar{x}_{12} &= \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2} \\ &= \frac{100 \times 460 + 80 \times 490}{100 + 80} \\ &= ₹ 473.33 \end{aligned}$$

$$\begin{aligned} \sigma_{12}^2 &= \frac{n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2)}{n_1 + n_2}; \\ &= \frac{100 \left\{ (50)^2 + (13.33)^2 \right\} + 80 \left\{ (40)^2 + (16.67)^2 \right\}}{100 + 80} \\ &= 2322.22 \end{aligned}$$

$$\begin{aligned} d_1 &= |\bar{x}_1 - \bar{x}_{12}| = |460 - 473.33| = 13.33 \\ d_2 &= |\bar{x}_2 - \bar{x}_{12}| = |490 - 473.33| = 16.67 \end{aligned}$$

$$\text{4.39 } \bar{x} = \frac{\Sigma x}{N}$$

$$\text{or } \Sigma x = N \bar{x} = 5 \times 4.4 = 22$$

Let two terms x_1 and x_2 are missing. Then,

$$x_1 + x_2 + 1 + 2 + 6 = 22$$

$$\text{or } x_1 + x_2 = 13$$

$$\text{Also } \sigma^2 = \frac{\Sigma x^2}{N} - (\bar{x})^2 \text{ or } 8.24$$

$$= \frac{\Sigma x^2}{5} - (4.4)^2$$

$$\text{or } \Sigma x^2 = 138$$

$$\begin{aligned} \therefore \Sigma x^2 &= x_1^2 + x_2^2 + (1)^2 + (2)^2 + (6)^2 \\ &= 138 \text{ or } x_1^2 + x_2^2 = 97 \end{aligned}$$

$$\text{Now } x_1^2 + x_2^2 = (x_1 + x_2)^2 - 2x_1x_2$$

$$\text{or } 97 = (13)^2 - 2x_1x_2 \text{ or } x_1x_2 = 36$$

$$\begin{aligned} (x_1 - x_2)^2 &= x_1^2 + x_2^2 - 2x_1x_2 \\ &= 97 - 2(36) \\ &= 25, \end{aligned}$$

$$\text{or } x_1 - x_2 = 5$$

Solving two equations $x_1 + x_2 = 13$ and $x_1 - x_2 = 5$, we have $x_1 = 9$ and $x_2 = 4$.

$$\text{4.40 } QD = \frac{2}{3} \sigma = \frac{2}{3} \times 5 = \frac{10}{3};$$

$$QD = \frac{Q_3 - Q_2}{2} = \frac{10}{3} \text{ or}$$

$$Q_3 - Q_2 = \frac{20}{3} = 6.67$$

Thus interquartile range is 6.67.

$$\text{4.41 } \text{Correct } \bar{x} = 39.9 \text{ and } \sigma = 4.9$$

$$\text{4.42 } \bar{x} = \frac{\Sigma x}{N} \text{ or } \Sigma x = N \bar{x} = 10 \times 9.5 = 95$$

Adding the 11th observation,

$$\text{We get } \Sigma x = 95 + 15 = 110.$$

$$\text{Then } \bar{x} = \frac{\Sigma x}{N} = 110 \div 11 = 10$$

$$\begin{aligned} \text{Also, } \sigma^2 &= \frac{\Sigma x^2}{N} - (\bar{x})^2 \text{ or } (2.5)^2 \\ &= \frac{\Sigma x^2}{10} - (9.5)^2 \end{aligned}$$

$$\text{or } \Sigma x^2 = 965$$

$$\text{Now value of } \Sigma x^2 = 965 + (15)^2 = 1190.$$

Then

$$\sigma^2 = \frac{\Sigma x^2}{N} - (\bar{x})^2 = \frac{1190}{11} - (10)^2 = 2.86$$

Glossary of Terms

Range: A measure of variability defined to be the difference between the largest and lowest values in the data set.

Interquartile range: A measure of variability defined to be the difference between the quartiles Q_3 and Q_1 .

Variance: A measure of variability based on the squared deviations of the observed values in the data set about the mean value.

Standard deviation: A measure of variability computed by taking the positive square root of the variance.

Coefficient of variation: A measure of relative variability computed by dividing the standard deviation by the mean, then multiplying by 100.

5

Skewness, Moments and Kurtosis

LEARNING OBJECTIVES

After studying this chapter, you should be able to

- know the complementary relationship of skewness with measures of central tendency and dispersion in describing a set of data.
- measure the degree to which individual values in the data set deviate from symmetry on both sides of the central value and the direction in which these are distributed.
- understand 'moments' as a convenient and unifying method for summarizing several descriptive statistical measures.

5.1 INTRODUCTION

In Chapter 4, we discussed measures of variation (or dispersion) to describe the spread of individual values in a data set around a central value. Such descriptive analysis of a frequency distribution remains incomplete until we measure the degree to which these individual values in the data set deviate from symmetry on both sides of the central value and the direction in which these are distributed. This analysis is important due to the fact that data sets may have the same mean and standard deviation but the frequency curves may differ in their shape. A frequency distribution of the set of values that is not 'symmetrical (normal)' is called *asymmetrical* or *skewed*. In a skewed distribution, extreme values in a data set move towards one side or tail of a distribution, thereby lengthening that tail. When extreme values move towards the upper or right tail, the distribution is positively skewed. When such values move towards the lower or left tail, the distribution is negatively skewed. Symmetrical and skewed distributions are shown in Fig. 5.1.

For a positively skewed distribution $A.M. > \text{Median} > \text{Mode}$, and for a negatively skewed distribution $A.M. < \text{Median} < \text{Mode}$. The relationship between these measures of central tendency is used to develop a **measure of skewness** called the *coefficient of skewness* to understand the degree to which these three measures differ.

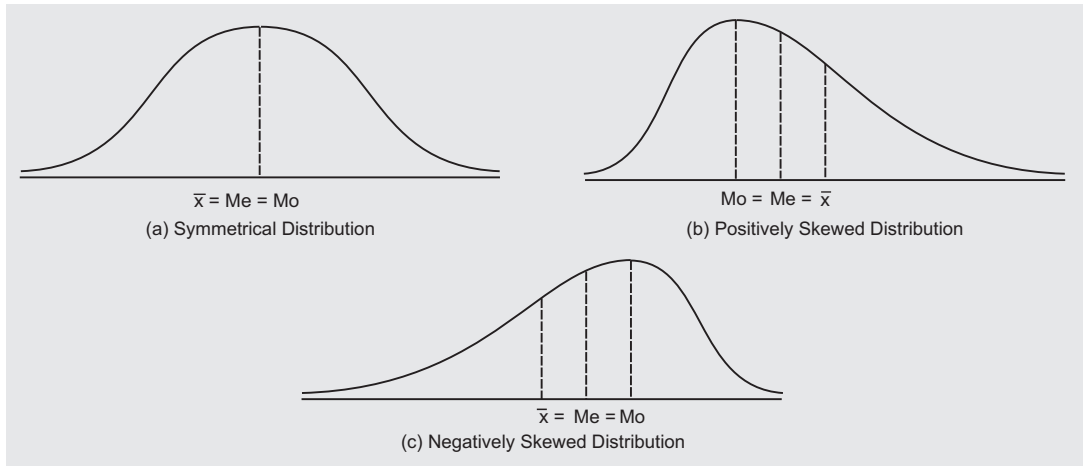


Figure 5.1 Comparison of Three Data Sets Differing in Shape

From the above discussion, two points of difference emerge between variation and skewness:

- (i) Variation indicates the amount of spread or dispersion of individual values in a data set around a central value, while skewness indicates the direction of dispersion, that is, away from symmetry.
- (ii) Variation is helpful in finding out the extent of variation among individual values in a data set, while skewness gives an understanding about the concentration of higher or lower values around the mean value.

5.2 MEASURES OF SKEWNESS

The degree of skewness in a distribution can be measured both in the *absolute* and *relative* sense. For an asymmetrical distribution, the distance between mean and mode may be used to measure the degree of skewness because the mean is equal to mode in a symmetrical distribution. Thus,

$$\begin{aligned}\text{Absolute } S_k &= \text{Mean} - \text{Mode} \\ &= Q_3 + Q_1 - 2 \text{ Median (if measured in terms of quartiles).}\end{aligned}$$

For a positively skewed distribution, $\text{Mean} > \text{Mode}$ and therefore S_k is a positive value, otherwise it is a negative value. This difference is taken to measure the degree of skewness because in an asymmetrical distribution, mean moves away from the mode. Larger the difference between mean and mode, whether positive or negative, more is the asymmetrical distribution or skewness. This difference, however, may not be desirable for the following reasons:

- (i) The difference between mean and mode is expressed in the same units as the distribution and therefore cannot be used for comparing skewness of two or more distributions having different units of measurement.
- (ii) The difference between mean and mode may be large in one distribution and small in another, although the shape of their frequency curves is the same.

In order to overcome these two shortcomings and to make valid comparisons between skewness of two or more distributions, the absolute difference has to be expressed in relation to the standard deviation—a measure of dispersion. Since we want to express any measure of skewness as a pure (relative) number, therefore this distance is expressed in terms of the unit of measurement in units of the standard deviation.

5.2.1 Relative Measures of Skewness

The following are three important relative measures of skewness.

Karl Pearson's Coefficient of Skewness

The measure suggested by Karl Pearson for measuring coefficient of skewness is given by

$$Sk_p = \frac{\text{Mean} - \text{Mode}}{\text{Standard deviation}} = \frac{\bar{x} - Mo}{\sigma} \quad (5-1)$$

where Sk_p = Karl Pearson's coefficient of skewness.

Since a mode does not always exist uniquely in a distribution, therefore it is convenient to define this measure using median. For a moderately skewed distribution, the following relationship holds:

$$\text{Mean} - \text{Mode} = 3(\text{Mean} - \text{Median}) \quad \text{or} \quad \text{Mode} = 3 \text{Median} - 2 \text{Mean}$$

When this value of mode is substituted in the Eqn. (5-1), we get

$$Sk_p = \frac{3(\bar{x} - \text{Med})}{\sigma} \quad (5-2)$$

Theoretically, the value of Sk_p varies between ± 3 . But for a moderately skewed distribution, value of $Sk_p = \pm 1$. Karl Pearson's method of determining coefficient of skewness is particularly useful in open-end distributions.

Bowley's Coefficients of Skewness

The method suggested by Prof. Bowley is based on the relative positions of the median and the quartiles in a distribution. If a distribution is symmetrical, then Q_1 and Q_3 would be at equal distances from the value of the median, that is,

$$\text{Median} - Q_1 = Q_3 - \text{Median}$$

$$\text{or} \quad Q_3 + Q_1 - 2 \text{Median} = 0 \quad \text{or} \quad \text{Median} = \frac{Q_3 + Q_1}{2}$$

This shows that the value of median is the mean value of Q_1 and Q_3 . Obviously in such a case, the absolute value of the coefficient of skewness will be zero.

When a distribution is asymmetrical, quartiles are not at equal distance from the median. The distribution is positively skewed, if $Q_1 - \text{Me} > Q_3 - \text{Me}$, otherwise negatively skewed.

The absolute measure of skewness is converted into a relative measure for comparing distributions expressed in different units of measurement. For this, absolute measure is divided by the inter-quartile range. That is,

$$\text{Relative } Sk_b = \frac{Q_3 + Q_1 - 2 \text{Med}}{Q_3 - Q_1} = \frac{(Q_3 - \text{Med}) - (\text{Med} - Q_1)}{(Q_3 - \text{Med}) + (\text{Med} - Q_1)} \quad (5-3)$$

In a distribution, if $\text{Med} = Q_1$, then $\text{Sk}_b = \pm 1$, but if $\text{Med} = Q_3$ then $\text{Sk}_b = -1$. This shows that the value of Sk_b varies between ± 1 for moderately skewed distribution. This method of measuring skewness is quite useful in those cases where (i) mode is ill-defined and extreme observations are present in the data and (ii) the distribution has open-ended classes. These two advantages of Bowley's coefficient of skewness indicate that it is not affected by extreme observations in the data set.

Remark: The values of Sk_b obtained by Karl Pearson's and Bowley's methods cannot be compared. On certain occasions it is possible that one of them gives a positive value while the other gives a negative value.

Kelly's Coefficient of Skewness

The relative measure of skewness suggested by Prof. Kelly is based on percentiles and deciles:

$$\text{Sk}_k = \frac{P_{10} + P_{90} - 2P_{50}}{P_{90} - P_{10}} \quad \text{or} \quad \frac{D_1 + D_9 - 2D_5}{D_9 - D_1} \quad (5-4)$$

This method is an extension of Bowley's method in the sense that Bowley's method is based on the middle 50 percent of the observations while this method is based on the observations between the 10th and 90th percentiles (or first and ninth deciles).

Example 5.1: Data of rejected items during a production process is as follows:

No of rejects	: 21-25	26-30	31-35	36-40	41-45	46-50	51-55
(per operator)							
No. of operators	: 5	15	28	42	15	12	3

Calculate the mean, standard deviation and coefficient of skewness and comment on the results.

Solution: The calculations for mean, mode and standard deviation are shown in Table 5.1

Table 5.1 Calculations for Mean, Mode and Standard Deviation

Class	Mid-value (<i>m</i>)	Frequency (<i>f</i>)	$d = \frac{m - A}{h} = \frac{m - 38}{5}$	<i>fd</i>	<i>fd</i> ²
21-25	23	5	-3	-15	45
26-30	28	15	-2	-30	60
31-35	33	28 $\leftarrow f_{m-1}$	-1	-28	28
36-40	38	42 $\leftarrow f_m$	0	0	0
41-45	43	15 $\leftarrow f_{m+1}$	1	15	15
46-50	48	12	2	24	48
51-55	53	3	3	9	27
		<u>N = 120</u>		<u>-25</u>	<u>223</u>

Let assumed mean is $A = 38$. Then,

$$\bar{x} = A + \frac{\sum fd}{N} \times h = 38 - \frac{25}{120} \times 5 = 36.96 \text{ rejects per operator}$$

$$\begin{aligned}\sigma &= \sqrt{\frac{\sum fd^2}{N} - \left(\frac{\sum fd}{N}\right)^2} \times h \\ &= \sqrt{\frac{223}{120} - \left(\frac{-25}{120}\right)^2} \times 5 = 6.736 \text{ rejects per operator}\end{aligned}$$

By inspection, mode lies in the class 36–40. Thus,

$$\begin{aligned}\text{Mo} &= l + \frac{f_m - f_{m-1}}{2f_m - f_{m-1} - f_{m+1}} \times h \\ &= 36 + \frac{42 - 28}{2 \times 42 - 28 - 15} \times 5 = 36 + \frac{14}{41} \times 5 = 37.71 \\ \text{Sk}_p &= \frac{\text{Mean} - \text{Mode}}{\text{Standard deviation}} = \frac{\bar{x} - \text{Mo}}{\sigma} \\ &= \frac{36.96 - 37.71}{6.73} = \frac{-0.75}{6.736} = -0.111.\end{aligned}$$

Since the coefficient of skewness, $S_k = -0.111$, the distribution is skewed to the left (negatively skewed). Thus, the concentration of the rejects per operator is more on the lower values of the distribution to the extent of 10.9 per cent.

Example 5.2: The following is the information about the settlement of an industrial dispute in a factory. Comment on the gains and losses from the point of view of workers and that of management:

	<i>Before</i>	<i>After</i>
No. of Workers	3000	2900
Mean wages (₹)	2200	2300
Median wages (₹)	2500	2400
Standard deviation	300	260

Solution: The comments on gains and losses from both worker's and management's point of view are as follows:

Total Wages Bill

$$\begin{array}{cc} \text{Before} & \text{After} \\ 3000 \times 2200 = 66,00,000 & 2900 \times 2300 = 66,70,000 \end{array}$$

The total wage bill has increased after the settlement of dispute, workers retained after the settlement are 50 workers less than the previous number.

After the settlement of dispute, the workers as a group are better off in terms of monetary gain. If the workers' efficiency remain same, then it is against the interest of management. But if the workers feel motivated, resulting in increased efficiency, then management can achieve higher productivity. This would be an indirect gain to management also.

Since workers retained after the settlement of dispute are less than the number employed before, it is against the interest of the workers.

Median Wages: The median wage after the settlement of dispute has come down from ₹ 2500 to ₹ 2400. This indicates that before the settlement, 50 per cent of the workers were getting wages above ₹ 2500 but after the settlement, they will be getting only ₹ 2400. It has certainly gone against the interest of the workers.

Uniformity in the Wage Structure: The extent of relative uniformity in the wage structure before and after the settlement can be determined by comparing the coefficient of variation as follows:

	Before	After
Coefficient of Variation (CV)	$\frac{300}{2200} \times 100 = 13.63$	$\frac{260}{2300} \times 100 = 11.30$

Since CV has decreased after the settlement from 13.63 to 11.30, the distribution of wages is more uniform after the settlement, that is, there is now comparatively less disparity in the wages received by the workers. Such a position is good for both the workers and the management in maintaining a cordial work environment.

Pattern of the Wage Structure: The nature and pattern of the wage structure before and after the settlement can be determined by comparing the coefficients of skewness.

	Before	After
Coefficient of skewness, Sk_p	$\frac{3(2200 - 2500)}{300} = -3$	$\frac{3(2300 - 2400)}{260} = -1.15$

Since coefficient of skewness is negative and has increased after the settlement, therefore it suggests that number of workers getting low wages has increased and that of workers getting high wages has decreased after the settlement.

Example 5.3: From the following data on age of employees, calculate the coefficient of skewness and comment on the result

Age below (years)	:	25	30	35	40	45	50	55
Number of employees	:	8	20	40	65	80	92	100

Solution: The data are given in a cumulative frequency distribution form. So, to calculate the coefficient of skewness, convert this data into a simple frequency distribution as shown in Table 5.2.

Table 5.2 Calculations for Coefficient of Skewness

Age (years)	Mid-value (m)	Number of Employees (f)	$d = \frac{(m - A)/h}{(m - 37.5)}$	fd	fd ²
20–25	22.5	8	–3	–24	72
25–30	27.5	12	–2	–24	48
30–35	32.5	20 $\leftarrow f_{m-1}$	–1	–20	20
35–40	37.5	25 $\leftarrow f_m$	0	0	0
40–45	42.5	15 $\leftarrow f_{m+1}$	1	15	15
45–50	47.5	12	2	24	48
50–55	52.5	8	3	24	72
		N = 100		–5	275

$$\text{Mean, } \bar{x} = A + \frac{\sum fd}{N} \times h = 37.5 - \frac{5}{100} \times 5 = 37.25$$

Mode value lies in the class interval 35–40. Thus,

$$\begin{aligned} \text{Mo} &= l + \frac{f_m - f_{m-1}}{2f_m - f_{m-1} - f_{m+1}} \times h \\ &= 35 + \frac{25 - 20}{2 \times 25 - 20 - 15} \times 5 = 35 + \frac{5}{15} \times 5 = 36.67 \end{aligned}$$

$$\begin{aligned} \text{Standard deviation, } \sigma &= \sqrt{\frac{\sum fd^2}{N} - \left(\frac{\sum fd}{N}\right)^2} \times h \\ &= \sqrt{\frac{275}{100} - \left(\frac{-5}{100}\right)^2} \times 5 = \sqrt{2.75 - 0.0025} \times 5 = 8.29 \end{aligned}$$

Karl Pearson's Coefficient of skewness:

$$\text{Sk}_p = \frac{\text{Mean} - \text{Mode}}{\sigma} = \frac{37.25 - 36.67}{8.29} = \frac{0.58}{8.29} = 0.07$$

The positive value of Sk_p indicates that the distribution is slightly positively skewed.

Example 5.4: (a) The sum of 50 observations is 500, its sum of squares is 6000 and median 12. Find the coefficient of variation and coefficient of skewness.

(b) For a moderately skewed distribution, the arithmetic mean is 100 and coefficient of variation 35, and Pearson's coefficient of skewness is 0.2. Find the mode and the median.

Solution: (a) Given that $N = 50$, $\sum x = 500$, $\sum x^2 = 6000$ and Median, $\text{Me} = 12$. Then,

$$\text{Mean, } \bar{x} = \frac{\sum x}{N} = \frac{500}{50} = 10$$

$$\text{Standard deviation, } \sigma = \sqrt{\frac{\sum x^2}{N} - (\bar{x})^2} = \sqrt{\frac{6000}{50} - (10)^2} = \sqrt{120 - 100} = 4.472$$

$$\text{Coefficient of Variation, CV} = \frac{\sigma}{\bar{x}} \times 100 = \frac{4.472}{10} \times 100 = 44.7 \text{ per cent}$$

$$\text{Mode} = 3 \text{ Median} - 2 \text{ Mean} = 3 \times 12 - 2 \times 10 = 16$$

$$\text{Coefficient of skewness, } \text{Sk}_p = \frac{\bar{x} - \text{Mo}}{\sigma} = \frac{10 - 16}{4.472} = -1.341$$

(b) Given that $\bar{x} = 100$, $\text{CV} = 35$ and $\text{Sk}_p = 0.2$.

$$\text{CV} = \frac{\sigma}{\bar{x}} \times 100 \quad \text{or} \quad 35 = \frac{\sigma}{100} \times 100 \quad \text{or} \quad \sigma = 35$$

$$\text{Also } \text{Sk}_p = \frac{\bar{x} - \text{Mo}}{\sigma} \quad \text{or} \quad 0.2 = \frac{100 - \text{Mo}}{35}, \text{ i.e. Mo} = 93$$

$$\text{Mode} = 3 \text{ Median} - 2 \text{ Mean} \quad \text{or} \quad 93 = 3 \text{ Med} - 2 \times 100 \quad \text{or} \quad \text{Med} = 97.7$$

Hence, Mode is 93 and Median is 97.7.

Example 5.5: The daily expenditure of 100 families is given below :

Daily Expenditure	: 0–20	20–40	40–60	60–80	80–100
No. of Families	: 13	?	27	?	16

If the mode of the distribution is 44, calculate the Karl Pearson coefficient of skewness.

Solution: Calculations for missing frequencies are shown below:

Daily Expenditure	Mid-value (m)	No. of Families (f)
0–20	10	13
20–40	30	x
40–60	50	27
60–80	70	y
80–100	90	16

Since mode is 44, modal class is 40–60. Thus,

$$\text{Mode, } M_0 = l + \frac{f_m - f_{m-1}}{2f_m - f_{m-1} - f_{m+1}}$$

$$\therefore 44 = 40 + \frac{27 - x}{54 - x - y} \times 20$$

$$\frac{44 - 40}{20} = \frac{27 - x}{54 - x - y}$$

$$\frac{1}{5} = \frac{27 - x}{54 - x - y}$$

$$\text{or } 54 - x - y = 135 - 5x$$

$$4x - y = 81$$

$$\text{Also } 13 + x + 27 + y + 16 = 100$$

$$\text{or } x + y = 44$$

Solving these two equations, we get $x = 25$, $y = 19$. Now the calculations for mean and standard deviation are as follows:

Class Interval	Mid-value m	f	$d = (m - 50)/20$	fd	fd^2
0–20	10	13	– 2	– 26	52
20–40	30	25	– 1	– 25	25
40–60	50	27	0	0	0
60–80	70	19	1	19	19
80–100	90	16	2	32	64
		100		0	160

$$\text{Mean, } \bar{x} = A + \frac{\Sigma fd}{N} \times h = 50 + \frac{0}{100} \times 20 = 50$$

$$\begin{aligned} \text{S.D., } \sigma &= \sqrt{\frac{\Sigma fd^2}{N} - \left(\frac{\Sigma fd}{N}\right)^2} \times h = \sqrt{\frac{160}{100} - \left(\frac{0}{100}\right)^2} \times 20 \\ &= \sqrt{1.6} \times 20 = 1.26 \times 20 = 25.20 \end{aligned}$$

$$\text{Hence, } Sk_p = \frac{\bar{x} - \text{Mode}}{\sigma} = \frac{50 - 44}{25.20} = 0.238$$

Example 5.6: Pearson's coefficient of skewness for a data distribution is 0.5 and coefficient of variation is 40 per cent. Its mode is 80. Find the mean and median of the distribution.

[Delhi Univ., B.Com.(Hons), 2004]

Solution: Given, $Sk_p = 0.5$, $CV = 40$ per cent and Mode = 80. Thus

$$CV = \frac{\sigma}{\bar{x}} \times 100 \quad \text{or} \quad 40 = \frac{\sigma}{\bar{x}} \times 100 \quad \text{or} \quad \sigma = 0.4 \bar{x}$$

$$\text{Also} \quad Sk_p = \frac{\text{Mean} - \text{Mode}}{\text{S.D.}} \quad \text{or} \quad 0.5 = \frac{\bar{x} - 80}{0.4 \bar{x}}$$

$$0.5 \times 0.4 \bar{x} = \bar{x} - 80, \text{ i.e. } \bar{x} = 100$$

Now

$$\text{Mode} = 3 \text{ Median} - 2 \text{ Mean}$$

$$80 = 3 \text{ Median} - 2(100) \quad \text{or} \quad \text{Med} = 280/3 = 93.33.$$

Example 5.7: The data on the profits (₹ in lakh) earned by 60 companies is as follows:

Profits	:	Below 10	10–20	20–30	30–40	40–50	50 and above
No. of Companies	:	5	12	20	16	5	2

- (a) Obtain the limits of profits of the central 50 per cent companies.
 (b) Calculate Bowley's coefficient of skewness.

Solution: (a) Calculations for different quartiles are shown in Table 5.3.

Table 5.3 Computation of Quartiles

Profits (₹ in lakh)	Frequency (f)	Cumulative Frequency (cf)
Below 10	5	5
10–20	12	17 ← Q_1 Class
20–30	20	37
30–40	16	53 ← Q_3 Class
40–50	5	58
50 and above	2	60
	<u>60</u>	

Q_1 = size of $(N/4)$ th observation = $(60/4)$ th = 15th observation. Thus, Q_1 lies in the class 10–20, and

$$Q_1 = l + \left\{ \frac{(N/4) - cf}{f} \right\} \times h$$

$$= 10 + \left\{ \frac{15 - 5}{12} \right\} \times 10 = 10 + 8.33 = 18.33 \text{ lakh}$$

Q_3 = size of $(3N/4)$ th observation = 45th observation. Thus, Q_3 lies in the class 30–40, and

$$Q_3 = l + \left\{ \frac{(3N/4) - cf}{f} \right\} \times h$$

$$= 30 + \left\{ \frac{45 - 37}{16} \right\} \times 10 = 30 + 5 = 35 \text{ lakh}$$

Hence, the profit of central 50 percent companies lies between ₹ 10.83 lakh and ₹ 35 lakh

$$\text{Coefficient of quartile deviation, Q.D.} = \frac{Q_3 - Q_1}{Q_3 + Q_1} = \frac{35 - 18.33}{35 + 18.33} = 0.313$$

(b) Median = size of $(N/2)$ th observation = 30th observation. Thus, median lies in the class 20–30 and

$$\text{Me} = l + \left\{ \frac{(N/2) - cf}{f} \right\} \times i = 20 + \left\{ \frac{30 - 17}{20} \right\} \times 10 = 20 + 6.5 = 26.5 \text{ lakh}$$

$$\text{Coefficient of skewness, Sk}_b = \frac{Q_3 + Q_1 - 2\text{Med}}{Q_3 - Q_1} = \frac{35 + 18.33 - 2(26.5)}{35 - 18.33} = 0.02$$

The positive value of Sk_b indicates that the distribution is positively skewed and therefore, there is a concentration of larger values on the right side of the distribution.

Example 5.8: Apply an appropriate measure of skewness to describe the following frequency distribution.

Age (yrs)	Number of Employees	Age (yrs)	Number of Employees
Below 20	13	35–40	112
20–25	29	40–45	94
25–30	46	45–50	45
30–35	60	50 and above	21

Solution: Since given frequency distribution is an open-ended distribution, Bowley's method of calculating skewness should be more appropriate. Calculations are shown in Table 5.4.

Table 5.4 Calculations for Bowley's Coefficient of Skewness

Age (yrs)	Number of Employees (f)	Cumulative Frequency (cf)
Below 20	13	13
20–25	29	42
25–30	46	88
30–35	60	148 $\leftarrow Q_1$ class
35–40	112	260
40–45	94	354 $\leftarrow Q_3$ class
45–50	45	399
50 and above	21	420
	<u>N = 420</u>	

Q_1 = size of $(N/4)$ th observation = $(420/4) = 105$ th observation. Thus, Q_1 lies in the class 30–35, and

$$Q_1 = l + \frac{(N/4) - cf}{f} \times h = 30 + \frac{105 - 88}{60} \times 5 = 30 + 1.42 = 31.42 \text{ yrs.}$$

Q_3 = size of $(3N/4)$ th observation = $(3 \times 420/4) = 315$ th observation. Thus, Q_3 lies in the class 40–45, and

$$Q_3 = l + \frac{(3N/4) - cf}{f} \times h = 40 + \frac{315 - 260}{94} \times 5 = 40 + 2.93 = 42.93 \text{ yrs}$$

Median = size of $(N/2)$ th = $(420/2) = 210$ th observation. Thus, median lies in the class 35–40, and

$$\text{Me} = l + \frac{(N/2) - cf}{f} \times h = 35 + \frac{210 - 148}{112} \times 5 = 35 + 2.77 = 37.77 \text{ yrs}$$

$$\begin{aligned} \text{Coefficient of skewness, } Sk_b &= \frac{Q_3 + Q_1 - 2\text{Med}}{Q_3 - Q_1} = \frac{42.93 + 31.42 - 2 \times 37.77}{42.93 - 31.42} \\ &= -\frac{1.19}{11.51} = -0.103 \end{aligned}$$

The negative value of Sk_b indicates that the distribution is negatively skewed.

Example 5.9: In a frequency distribution, coefficient of skewness based on quartiles is 0.6. If the sum of upper and lower quartile is 100 and median is 38, find the values of lower and upper quartiles. [Delhi Univ., B.Com.(Hons.), 1993]

Solution: Given $Sk_b = 0.6$; $Q_3 + Q_1 = 100$ and $\text{Med} = 38$

$$\begin{aligned} \text{We have } Sk_b &= \frac{Q_3 + Q_1 - 2\text{Med}}{Q_3 - Q_1} \quad \text{or} \quad 0.6 = \frac{100 - 2(38)}{Q_3 - Q_1} \\ Q_3 - Q_1 &= \frac{100 - 76}{0.6} = 40 \end{aligned}$$

Since $Q_3 + Q_1 = 100$, and $Q_3 - Q_1 = 40$, solving these equations we get $Q_3 = 70$ and $Q_1 = 30$

Example 5.10: Pearson's coefficient of skewness is 0.4, coefficient of variation is 30 per cent and the mode = 88. Find mean. [Delhi Univ., B.Com.(Hons.), 1997]

Solution: Given $Sk_p = 0.4$, $CV = \sigma/\bar{x} = 0.30$ and $\text{Mode} = 88$

$$\text{We have } Sk_p = \frac{\bar{x} - M_o}{\sigma} = \frac{\bar{x} - M_o}{CV \times \bar{x}} = \frac{1 - (M_o/\bar{x})}{CV}$$

$$\text{or } 0.40 = \frac{1 - (88/\bar{x})}{0.30} \quad \text{or} \quad 1 - \left(\frac{88}{\bar{x}}\right) = 0.12, \text{ i.e. } \bar{x} = 100$$

Example 5.11: The mean, mode and Q.D. of a distribution are 42, 36 and 15, respectively. If its Bowley's coefficient of skewness is $1/3$, find the values of the two quartiles. [Delhi Univ., B.Com.(Hons.), 2005]

Solution: Given $\bar{x} = 42$, $\text{Mode} = 36$, $\text{Q.D.} = 15$ and $Sk_b = 1/3$

$$\text{Since } \text{Q.D.} = \frac{Q_3 - Q_1}{2} = 15 \quad \text{or} \quad Q_3 - Q_1 = 30 \quad (\text{i})$$

$$\begin{aligned} \text{Also } \text{Mode} &= 3 \text{ Med} - 2 \text{ Mean} \\ 36 &= 3 \text{ Med} - 2(42), \text{ i.e. } \text{Med} = 40 \end{aligned}$$

$$\text{Now } Sk_b = \frac{Q_3 + Q_1 - 2 \text{Med}}{Q_3 - Q_1}$$

$$\text{or } \frac{1}{3} = \frac{Q_3 + Q_1 - 2 \times 40}{30}, \text{ i.e. } Q_3 - Q_1 = 90 \quad (\text{ii})$$

Since $Q_3 + Q_1 = 90$ and $Q_3 - Q_1 = 30$, solving these equations we get $Q_3 = 60$ and $Q_1 = 30$.

Conceptual Questions 5A

1. Explain the meaning of skewness using sketches of frequency curves. State the different measures of skewness that are commonly used. How does skewness differ from dispersion?
2. Measures of central tendency, dispersion and skewness are complementary to each other in describing a frequency distribution. Elucidate.
3. Distinguish between Karl Pearson's and Bowley's measure of skewness. Which one of these would you prefer and why?
4. Explain briefly the different methods of measuring skewness.
5. Distinguish between variation and skewness and point out the various methods of measuring skewness.
6. Briefly mention the tests which can be applied to determine the presence of skewness.
7. Explain the term 'skewness'. What purpose does a measure of skewness serve? Comment on some of the well-known measures of skewness.

[Kumaon Univ., MBA, 2000]

[Delhi Univ., MBA, 2000]

Self-practice Problems 5A

- 5.1** The following data relate to the profits (₹ in '000) of 1,000 companies:

Profits : 100–120 120–140 140–160 160–180
180–200 200–220 220–240

No. of 17 53 199 194
companies : 327 208 2

Calculate the coefficient of skewness and comment on its value.

- 5.2** A survey was conducted by a manufacturing company to find out the maximum price at which people would be willing to buy its product. The following table gives the stated price (in ₹) by 100 persons:

Price : 2.80–2.90 2.90–3.00 3.00–3.10
3.10–3.20 3.20–3.30

No. of 11 29 18
persons: 27 15

Calculate the coefficient of skewness and interpret its value.

- 5.3** Calculate coefficient of variation and Karl Pearson's coefficient of skewness from the data given below:

Marks (less than) : 20 40 60 80 100
No. of students : 18 40 70 90 100

- 5.4** The following table gives the length of the life (in hours) of 400 TV picture tubes:

Length of Life (in hours)	No. of Picture Tubes	Length of Life (in hours)	No. of Picture Tubes
4000–4199	12	5000–5199	55
4200–4399	30	5200–5399	36
4400–4599	65	5400–5599	25
4600–4799	78	5600–5799	9
4800–4999	90		

Compute the mean, standard deviation and coefficient of skewness.

- 5.5** Calculate Karl Pearson's coefficient of skewness from the following data:

Profit (₹ in lakh) : Below 20 40 60 80 100
No. of companies : 8 20 50 64 70

- 5.6** From the following information, calculate Karl Pearson's coefficient of skewness.

Measure	Place A	Place B
Mean	256.5	240.8
Median	201.0	201.6
S.D.	215.0	181.0

- 5.7 From the following data, calculate Karl Pearson's coefficient of skewness:

Marks (more than) :	0	10	20	30	40	50	60	70	80
No. of students :	150	140	100	80	80	70	30	14	0

- 5.8 Calculate Bowley's coefficient of skewness from the following data

Sales (₹ in lakh) :	Below 50	60	70	80	90
No. of companies :	8	20	40	65	80

- 5.9 The following table gives the distribution of weekly wages of 500 workers in a factory:

Weekly Wages (₹)	No. of Workers
Below 200	10
200–250	25
250–300	145
300–350	220
350–400	70
400 and above	30

- (a) Obtain the limits of income of the central 50 percent of the observed workers.

- (b) Calculate Bowley's coefficient of skewness.

- 5.10 Find Bowley's coefficient of skewness for the following frequency distribution:

No. of children per family :	0	1	2	3	4
No. of families :	7	10	16	25	18

- 5.11 In a frequency distribution, the coefficient of skewness based on quartiles is 0.6. If the sum of the upper and the lower quartiles is 100 and the median is 38, find the value of the upper quartile.

- 5.12 Calculate Bowley's measure of skewness from the following data:

Payment of Commission	No. of Salesmen	Payment of Commission	No. of Salesmen
100–120	4	200–220	80
120–140	10	220–240	32
140–160	16	240–260	23
160–180	29	260–280	17
180–200	52	280–300	7

Hints and Answers

$$\begin{aligned}
 5.1 \quad \bar{x} &= A + \frac{\sum fd}{N} \times h \\
 &= 170 + \frac{393}{1000} \times 20 = 177.86 \\
 Mo &= \frac{f_m - f_{m-1}}{2f_m - f_{m-1} - f_{m+1}} \times h \\
 &= 180 + \frac{233}{252} \times 20 = 190.55 \\
 \sigma &= h \sqrt{\frac{\sum fd^2}{N} - \left(\frac{\sum fd}{N}\right)^2} \\
 &= 20 \sqrt{\frac{1741}{1000} - \left(\frac{393}{1000}\right)^2} = 25.2 \\
 Sk_p &= \frac{\bar{x} - Mo}{\sigma} = \frac{177.86 - 190.55}{25.2} \\
 &= -0.5035
 \end{aligned}$$

$$\begin{aligned}
 5.2 \quad \bar{x} &= A + \frac{\sum fd}{N} \times h \\
 &= 3.05 + \frac{6}{100} \times 0.1 = 3.056 \\
 Mo &= l + \frac{f_m - f_{m-1}}{2f_m - f_{m-1} - f_{m+1}} \times h \\
 &= 2.9 + \frac{29 - 11}{2 \times 29 - 11 - 18} \times 0.1 = 2.962 \\
 \sigma &= h \sqrt{\frac{\sum fd^2}{N} - \left(\frac{\sum fd}{N}\right)^2} \\
 &= 0.1 \sqrt{\frac{160}{100} - \left(\frac{6}{100}\right)^2} = 0.1264 \\
 Sk_p &= \frac{\bar{x} - Mo}{\sigma} = \frac{3.056 - 2.962}{0.1264} = 0.744
 \end{aligned}$$

$$\begin{aligned}
 5.3 \quad \bar{x} &= A + \frac{\sum fd}{N} \times h \\
 &= 50 - \frac{18}{100} \times 20 = 46.4 \\
 \sigma &= h \sqrt{\frac{\sum fd^2}{N} - \left(\frac{\sum fd}{N}\right)^2} \\
 &= 20 \sqrt{\frac{154}{100} - \left(\frac{-18}{100}\right)^2} = 24.56 \\
 \text{C.V.} &= \frac{\sigma}{\bar{x}} \times 100 \\
 &= \frac{24.56}{46.4} \times 100 = 52.93 \\
 \text{Mo} &= l + \frac{f_m - f_{m-1}}{2f_m - f_{m-1} - f_{m+1}} \times h = 39.52; \\
 \text{Sk}_p &= 0.280
 \end{aligned}$$

$$\begin{aligned}
 5.4 \quad \bar{x} &= A + \frac{\sum fd}{N} \times h \\
 &= 4899.5 - \frac{108}{400} \times 200 = 4845.5 \\
 \sigma &= h \sqrt{\frac{\sum fd^2}{N} - \left(\frac{\sum fd}{N}\right)^2} \\
 &= 200 \sqrt{\frac{1368}{400} - \left(\frac{-108}{400}\right)^2} = 365.9
 \end{aligned}$$

Mo = Mode lies in the class 4800–4999; but real class interval is 4799.5–4999.5

$$\begin{aligned}
 &= l + \frac{f_m - f_{m-1}}{2f_m - f_{m-1} - f_{m+1}} \times h \\
 &= 4799.5 + \frac{12}{180 - 78 - 55} \times 200 \\
 &= 4850.56
 \end{aligned}$$

$$\text{Sk}_p = -0.014$$

$$\begin{aligned}
 5.5 \quad \bar{x} &= A + \frac{\sum fd}{N} \times h \\
 &= 50 - \frac{2}{70} \times 20 = 49.43 \\
 \text{Mo} &= l + \frac{f_m - f_{m-1}}{2f_m - f_{m-1} - f_{m+1}} \times h \\
 &= 40 + \frac{14}{12 + 14} \times 20 = 50.76
 \end{aligned}$$

$$\begin{aligned}
 \sigma &= h \sqrt{\frac{\sum fd^2}{N} - \left(\frac{\sum fd}{N}\right)^2} \\
 &= 20 \sqrt{\frac{80}{70} - \left(\frac{-2}{70}\right)^2} = 21.64; \\
 \text{Sk}_p &= -0.061
 \end{aligned}$$

$$5.6 \quad a \cong \bar{x} \text{ and } z \cong \text{Mo};$$

$$\text{Place A : Mode} = 3\text{Med} - 2\bar{x} = 90;$$

$$\text{Sk}_p = \frac{266.5 - 90}{215} = 0.823$$

$$\text{Place B : Mode} = 3\text{Med} - 2\bar{x} = 123.2;$$

$$\text{Sk}_p = \frac{240.8 - 123.2}{181} = 0.649$$

5.7

Marks	No. of Students
0–10	10
10–20	40
20–30	20
30–40	0
40–50	10
50–60	40
60–70	16
70–80	14

$$\bar{x} = 39.27; \quad \sigma = 22.81;$$

$$\begin{aligned}
 \text{Sk}_p &= \frac{3(\bar{x} - \text{Med})}{\sigma} = \frac{3(39.27 - 45)}{22.81} \\
 &= -0.754.
 \end{aligned}$$

5.8 Q_1 lies in the class 50–60; $Q_1 = 60$; Q_3 lies in the class 70–80; $Q_3 = 78$

Median (= Q_2) lies in the class 60–70; Med = 70

Bowley's coeff. of

$$\text{Sk}_b = \frac{Q_3 + Q_1 - 2\text{Med}}{Q_3 - Q_1} = -0.111$$

5.9 $Q_1 = (80/4) = 20$ th observation lies in the class 250–300; $Q_1 = 281.03$; $Q_3 = (3 \times 80)/4 = 60$ th observation lies in the class 300–350; $Q_3 = 344.32$

Median lies in the class 300–350,

Me = 315.9

Bowley's coeff. of $\text{Sk}_b = -0.111$ (negatively skewed distribution)

$$5.10 \quad Q_1 = \text{size of } \left(\frac{n+1}{4} \right) \text{th observation}$$

$$= 24\text{th observation} = 2$$

$$Q_3 = \text{size of } \frac{3(n+1)}{4} \text{th observation}$$

$$= 72\text{th observation} = 4$$

$$\text{Me} = \text{size of } \left(\frac{n+1}{2} \right) \text{th observation}$$

$$= 48\text{th observation};$$

$$\text{Sk}_b = \frac{4 + 2 - 2(3)}{4 - 2} = 0$$

$$5.11 \quad \text{Given } \text{Sk}_b = 0.6; Q_1 + Q_3 = 100; \text{Me} = 38;$$

$$Q_3 = ?$$

$$\text{Sk}_b = \frac{Q_3 + Q_1 - 2\text{Me}}{Q_3 - Q_1} \text{ or}$$

$$0.6 = \frac{100 - 2(38)}{Q_3 - (100 - Q_3)}, \text{ i.e. } Q_3 = 70$$

$$5.12 \quad Q_1 = (n/4)\text{th observation} = 67.5\text{th observation lies in class } 180\text{--}200;$$

$$Q_1 = 183.26$$

$$Q_3 = \left(\frac{3n}{4} \right) \text{th observation} = 202.5\text{th observation lies in class } 220\text{--}240;$$

$$Q_3 = 227.187$$

$$\text{Me} = (n/2)\text{th observation} = 135\text{th observation lies in class } 200\text{--}220;$$

$$\text{Me} = 206$$

$$\text{Sk}_b = \frac{Q_3 + Q_1 - 2\text{Me}}{Q_3 - Q_1} = -0.035$$

5.3 MOMENTS

According to R. A. Fisher, 'A quantity of data which by its mere bulk may be incapable of entering the mind is to be replaced by relatively few quantities which shall adequately represent the whole, or which, in other words, shall contain as much as possible, ideally the whole, of the relevant information contained in the original data.'

Among these 'relatively few quantities' are those which are known as **moments**. Two of them, the *mean* and the *variance*, have already been discussed and two other higher moments are in common use. The higher moments are basically used to describe the characteristic of any population rather than samples.

The measures of central tendency, variability and skewness which have been discussed to describe a frequency distribution may be classified into two groups:

(i) Percentile system

(ii) Moment system

The *percentile system* includes measures like median, quartiles, deciles, percentiles and so on. The value of these measures represents a given proportion of frequency distribution.

The *moment system* includes measures like mean, average deviation, standard deviation and so on. The value of these measures is obtained by taking the deviation of individual observations from a given origin. The term moment used in statistics is analogous to the term used in physics, where (i) size of class intervals represents the 'force' and (ii) deviation of mid-value of each class from an observation represents the distance.

While calculating moments, if deviations are taken from the actual mean, then such moments are denoted by the Greek letter μ (mu). On the other hand, if deviations are taken from some assumed mean (or arbitrary value other than zero), then moments are denoted by Greek letter ν (nu) or μ' .

5.3.1 Moments About Mean

Let $x_1, x_2, \dots, x_n$ be the n observations in a data set with mean $\bar{x}$. Then the r th moment about the actual mean of a variable both for ungrouped and grouped data is given by

For ungrouped data: $\mu_r = \frac{1}{n} \sum (x - \bar{x})^r$; $r = 1, 2, 3, 4$

For grouped data: $\mu_r = \frac{1}{n} \sum f(x - \bar{x})^r$; $r = 1, 2, 3, 4$; $N = \sum f_i$

For different values of $r = 1, 2, 3, 4$, different moments can be obtained as shown below:

Ungrouped data: $\mu_1 = \frac{1}{n} \sum (x - \bar{x}) = 0$; $\mu_2 = \frac{1}{n} \sum (x - \bar{x})^2 = \sigma^2$ (variance)

$$\mu_3 = \frac{1}{n} \sum (x - \bar{x})^3; \quad \mu_4 = \frac{1}{n} \sum (x - \bar{x})^4$$

For grouped data: $\mu_1 = \frac{1}{n} \sum f(x - \bar{x})$; $\mu_2 = \frac{1}{n} \sum f(x - \bar{x})^2$

$$\mu_3 = \frac{1}{n} \sum f(x - \bar{x})^3; \quad \mu_4 = \frac{1}{n} \sum f(x - \bar{x})^4$$

- (i) The *first moment*, μ_1 about origin gives the *mean* and is a measure of central tendency, i.e.

$$\mu_1 = \frac{1}{n} \sum (x - 0) = \frac{1}{n} \sum x \leftarrow \text{A.M. } (\bar{x})$$

- (ii) The *second moment*, μ_2 about the mean is known as *variance* and is a measure of dispersion.

$$\mu_2 = \frac{1}{n} \sum (x - \bar{x})^2 \leftarrow \text{Variance } (\sigma^2)$$

- (iii) The *third moment*, μ_3 about the mean indicates the symmetry or asymmetry of the distribution; its value is zero for symmetrical distribution.

$$\mu_3 = \frac{1}{n} \sum (x - \bar{x})^3 \leftarrow \text{Skewness}$$

- (iv) The *fourth moment*, μ_4 about the mean is a measure of Kurtosis (or flatness) of the frequency curve.

$$\mu_4 = \frac{1}{n} \sum (x - \bar{x})^4 \leftarrow \text{Kurtosis}$$

5.3.2 Moments About Arbitrary Point

When actual mean is in fractions, moments are first calculated about an assumed mean, say A, and then are converted about the actual mean as shown below:

For ungrouped data: $\mu'_r = \frac{1}{n} \sum (x - A)^r$; $r = 1, 2, 3, 4$

For grouped data: $\mu'_r = \frac{1}{n} \sum f(x - A)^r$; $r = 1, 2, 3, 4$

$$= \frac{1}{n} \sum f d^2 \times h^2, \text{ where } d = \frac{x - A}{h}$$

For $r = 1$, we have $\mu'_1 = \frac{1}{n} \sum (x - A) = \frac{1}{n} \sum x - A = \bar{x} - A$

5.3.3 Moments About Zero or Origin

The moments about zero or origin (for ungrouped data) are obtained as follows:

$$v_r = \frac{1}{n} \sum x^r; \quad r = 1, 2, 3, 4$$

The relationship among moments about zero and other moments is as follows:

$$\begin{aligned} v_1 &= A + \mu'_1, & v_2 &= \mu_2 + (v_1)^2 \\ v_2 &= \mu_3 + 3v_1v_2 - 2v_1^3, & v_4 &= \mu_4 + 4v_1v_3 - 6v_1^2v_2 + 3v_1^4 \end{aligned}$$

5.3.4 Relationship Between Moments About Mean and Moments About an Arbitrary Point

$$\begin{aligned} \mu_r &= \frac{1}{n} \sum (x - \bar{x})^r = \frac{1}{n} \sum \{x - A - (\bar{x} - A)\}^r = \frac{1}{n} \sum (x - A - \mu'_1)^r; \quad \mu'_1 = \bar{x} - A \\ &= \frac{1}{n} \left[\sum (x - A)^r - {}^r C_1 \mu'_1 \sum (x - A)^{r-1} + {}^r C_2 (\mu'_1)^2 \sum (x - A)^{r-2} + \dots + (-1)^r (\mu'_1)^r \right] \\ &= \mu'_r - {}^r C_1 \mu'_1 \mu'_{r-1} + {}^r C_2 (\mu'_1)^2 \mu'_{r-2} + \dots + (-1)^r (\mu'_1)^r \end{aligned} \quad (5-5)$$

From Eqn. (5-5) for various values of r , we have

$$\begin{aligned} \mu_1 &= \mu'_1; \quad r = 1 \\ \mu_2 &= \mu'_2 - (\mu'_1)^2; \quad r = 2 \\ \mu_3 &= \mu'_3 - 3\mu'_1\mu'_2 + 2(\mu'_1)^3; \quad r = 3 \\ \mu_4 &= \mu'_4 - 4\mu'_1\mu'_3 + 6\mu'_2(\mu'_1)^2 - 3(\mu'_1)^4; \quad r = 4 \end{aligned}$$

5.3.5 Moments in Standard Units

When expressed in standard units, moments about the mean of the population are usually denoted by the Greek letter α (alpha). Thus,

$$\begin{aligned} \alpha_r &= \frac{1}{n} \sum f z^r = \frac{1}{n} \sum f \left\{ \frac{x - \mu}{\sigma} \right\}^r \quad \text{by definition of } z = \frac{x - \mu}{\sigma} \\ &= \frac{1}{\sigma^r} \frac{1}{n} \sum f (x - \mu)^r = \frac{\mu_r}{\sigma^r} \end{aligned}$$

Hence, $\alpha_1 = 0$, $\alpha_2 = 1$, $\alpha_3 = \frac{\mu_3}{\sigma^3}$ and $\alpha_4 = \frac{\mu_4}{\sigma^4}$ for $r = 1, 2, 3$, and 4 , respectively.

In the notations used by Karl Pearson, we have

$$\begin{aligned} \text{(i) } \beta_1 \text{ (Beta one)} &= \alpha_3^2 = \frac{\mu_3^2}{\mu_2^3} \\ \text{(ii) } \beta_2 \text{ (Beta two)} &= \alpha_4 = \frac{\mu_4}{\mu_2^2} \end{aligned}$$

In the notations used by R.A. Fisher, we have

$$\begin{aligned} \text{(i) } \gamma_1 \text{ (Gamma one)} &= \alpha_3 = \sqrt{\beta_1} = \frac{\mu_3}{\mu_2^{3/2}} = \frac{\mu_3}{\sigma^3} \\ \text{(ii) } \gamma_2 \text{ (Gamma two)} &= \alpha_4 - 3 \text{ or } \beta_2 - 3 = \frac{\mu_4}{\mu_2^2} - 3 = \frac{\mu_4 - 3\mu_2^2}{\mu_2^2} \end{aligned}$$

All these coefficients are pure numbers and are independent of whatever unit the variable x may be expressed in. The values of α_3 and α_4 depend on the shape of the frequency curve and, therefore, can be used to distinguish between different shapes. Thus, α_3 or β_1 is a *measure of asymmetry about the mean or skewness* and $\beta_1 = 0$ for a *symmetrical distribution*. A curve with $\alpha_3 > 0$ is said have positive skewness and one with $\alpha_3 < 0$, negative skewness. For most distributions, α_3 lies between -3 and 3 .

The coefficient of skewness in terms of moments is given by

$$Sk = \frac{(\beta_2 + 3)\sqrt{\beta_1}}{2(5\beta_2 - 6\beta_1 - 9)}$$

If $\beta_1 = 0$ or $\beta_2 = -3$, then skewness is zero. But $\beta_2 = \frac{\mu_4}{\mu_2^2}$ cannot be zero and hence the only condition for skewness to be zero is $\beta_1 = 0$ and the coefficient has no sign.

5.3.6 Sheppard's Corrections For Moments

While calculating higher moments of grouped frequency distributions, it is assumed that frequencies are concentrated at the mid-values of class-intervals. However, it causes certain errors while calculating moments. W. E. Sheppard proved that, if

- (i) the frequency curve of the distribution is continuous,
- (ii) the frequency tapers off to zero at both ends, and
- (iii) the member of classes are not too large,

then the above assumption that frequencies are concentrated at the mid-value of the class intervals is corrected using Sheppard's corrections as follows:

$$\begin{aligned}\mu_2 (\text{corrected}) &= \mu_2 - \frac{h^2}{12} \\ \mu_4 (\text{corrected}) &= \mu_4 - \frac{1}{2}h^2\mu_2 + \frac{7}{240}h^4\end{aligned}$$

where h is the width of the class interval, μ_1 and μ_3 need no correction.

Example 5.12: The first four moments of a distribution about the origin are 1, 4, 10 and 46 respectively. Obtain the various characteristics of the distribution on the basis of the information given. Comment upon the nature of the distribution.

Solution: In the usual notations, we have

$$A = 0, \mu'_1 = 1, \mu'_2 = 4, \mu'_3 = 10 \text{ and } \mu'_4 = 46$$

$$\bar{x} = \text{first moment about origin} = \mu'_1 = 1$$

$$\text{Variance } (\sigma^2) = \mu_2 = \mu'_2 - (\mu'_1)^2 = 4 - 1 = 3$$

$$\text{S.D. } (\sigma) = \sqrt{\sigma^2} = \sqrt{3} = 1.732$$

$$\mu_3 = \mu'_3 - 3\mu'_2\mu'_1 + 2(\mu'_1)^3 = 10 - 3(4)(1) + 2(1)^3 = 0.$$

Karl Pearson's coefficient of skewness

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3} \quad \text{or} \quad \gamma_1 = \frac{\sqrt{\beta_1}}{\frac{\mu_3}{\mu_2^{3/2}}} = \frac{\mu_3}{\sigma^3} = 0$$

Substituting values in the above formula, we get $\gamma_1 = 0$ ($\beta_1 = 0$). This shows that the given distribution is symmetrical, and hence Mean = Median = Mode for the given distribution.

Example 5.13: The first three moments of a distribution about the value 1 of the variable are 2, 25 and 80. Find the mean, standard deviation and the moment-measure of skewness.

Solution: From the data of the problem, we have

$$\mu'_1 = 2, \mu'_2 = 25, \mu'_3 = 80 \text{ and } A = 1.$$

The moments about the arbitrary point $A = 1$ are calculated as follows:

$$\text{Mean, } \bar{x} = \mu'_1 + A = 2 + 1 = 3$$

$$\text{Variance, } \mu_2 = \mu'_2 - (\mu'_1)^2 = 25 - (2)^2 = 21$$

$$\mu_3 = \mu'_3 - 3\mu'_1\mu'_2 + 2(\mu'_1)^3 = 80 - 3(2)(25) + 2(2)^3 = -54$$

$$\text{Standard deviation, } \sigma = \sqrt{\mu_2} = \sqrt{21} = 4.58$$

$$\text{Moment-measure of skewness, } \gamma_1 = \sqrt{\beta_1} = \frac{\mu_3}{\mu_2^{3/2}} = \frac{-54}{(21)^{3/2}} = \frac{-54}{96.234} = -0.561$$

Example 5.14: Following is the data on daily earnings (in ₹) of employees in a company:

Earnings	:	50–70	70–90	90–110	110–130	130–150	150–170	170–190
No. of workers	:	4	8	12	20	6	7	3

Calculate the first four moments about the point 120. Convert the results into moments about the mean. Compute the value of γ_1 and γ_2 and comment on the result.

[Delhi Univ., MBA, 1990, 2002]

Solution: Calculations for first four moments are shown in Table 5.5:

Table 5.5 Computation of First Four Moments

Class	Mid-value (<i>m</i>)	Frequency (<i>f</i>)	$d = \frac{m - 120}{20}$	fd	fd^2	fd^3	fd^4
50–70	60	4	–3	–12	36	–108	324
70–90	80	8	–2	–16	32	–64	128
90–110	100	12	–1	–12	12	–12	12
110–130	120	20	0	0	0	0	0
130–150	140	6	1	6	6	6	6
150–170	160	7	2	14	28	56	112
170–190	180	3	3	9	27	81	243
		<u>60</u>		<u>–11</u>	<u>141</u>	<u>–41</u>	<u>825</u>

The moments about some arbitrary origin or point ($A = 120$) are given by

$$\begin{aligned} \mu'_r &= \frac{1}{n} \sum f(x - A)^r \text{ (for grouped data)} \\ &= \frac{1}{n} (\sum fd^r) h^r; \quad d = \frac{m - A}{h} \text{ or } m - A = hd \end{aligned}$$

For $A = 120$ and $x = m$, we get

$$\begin{aligned}\mu'_1 &= \frac{1}{n} \sum fd \times h = \frac{1}{60} (-11) \times 20 = -3.66 \\ \mu'_2 &= \frac{1}{n} \sum fd^2 \times h^2 = \frac{1}{60} (141) \times (20)^2 = 940 \\ \mu'_3 &= \frac{1}{n} \sum fd^3 \times h^3 = \frac{1}{60} (-41) (20)^3 = -5,466.66 \\ \mu'_4 &= \frac{1}{n} \sum fd^4 \times h^4 = \frac{1}{60} (825) (20)^4 = 22,00,000\end{aligned}$$

The moments about actual mean ($\mu'_2 = 940$) are given by

$$\begin{aligned}\mu_1 &= 0 \\ \mu_2 &= \mu'_2 - (\mu'_1)^2 = 940 - (-3.66)^2 = 926.55 \\ \mu_3 &= \mu'_3 - 3\mu'_2\mu'_1 + 2(\mu'_1)^3 = -5,466.66 - 3(940)(-3.66) + 2(-3.66)^3 \\ &= -5,466.66 + 10,340.094 - 98.59 = 4,774.83 \\ \mu_4 &= \mu'_4 - 4\mu'_3\mu'_1 + 6\mu'_2(\mu'_1)^2 - 3(\mu'_1)^4 \\ &= 22,00,000 - 4(-5,466.66)(-3.66) + 6(940)(-3.66)^2 - 3(-3.66)^4 \\ &= 22,00,000 - 80,178.50 + 7,582.03 - 542.27 = 21,95,107.20\end{aligned}$$

Since μ_3 is positive, therefore the given distribution is positively skewed. The relative measure of skewness is given by

$$\gamma_1 = \frac{\mu_3}{\mu_2^{3/2}} = \frac{\mu_3}{\mu_2 \sqrt{\mu_2}} = \frac{4774.83}{926.55 \sqrt{926.55}} = 0.169$$

Thus, $\beta_1 = \gamma_1^2 = 0.0285$. This implies that distribution is positively skewed to the right.

$$\begin{aligned}\beta_2 &= \frac{\mu_4}{\mu_2^2} = \frac{21,95,107.20}{(926.56)^2} = 2.56 \\ \gamma_2 &= \beta_2 - 3 = 2.56 - 3 = -0.44\end{aligned}$$

Since γ_2 is negative, the distribution is platykurtic.

5.4 KURTOSIS

The measure of kurtosis describes the degree of concentration of frequencies (observations) in a given distribution. That is, whether the observed values are concentrated more around the mode (a peaked curve) or away from the mode towards both tails of the frequency curve.

The word '**kurtosis**' comes from a Greek word meaning 'humped'. In statistics, it refers to the degree of flatness or peakedness in the region about the mode of a frequency curve. A few definitions of kurtosis are as follows:

- *The degree of kurtosis of a distribution is measured relative to the peakedness of a normal curve.*
— Simpson and Kafka
- *A measure of kurtosis indicates the degree to which a curve of a frequency distribution is peaked or flat-topped.*
— Croxten and Cowden
- *Kurtosis refers to the degree of peakedness of hump of the distribution.*
— C. H. Meyers

Two or more distributions may have identical average, variation and skewness, but they may show different degrees of concentration of values of observations around the mode, and hence may show different degrees of peakedness of the hump of the distributions as shown in Fig. 5.2.

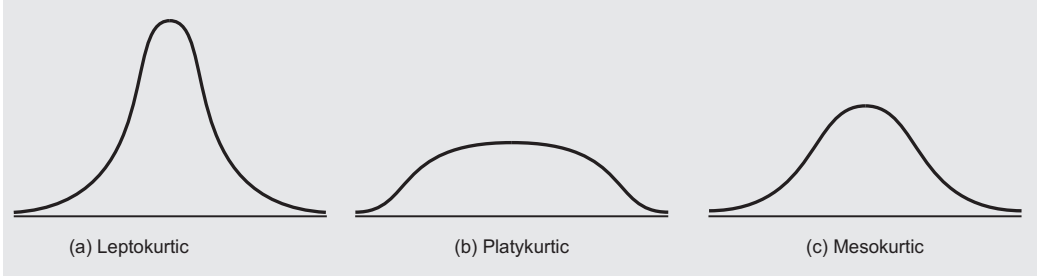


Figure 5.2 Shape of Three Different Curves Introduced by Karl Pearson

5.4.1 Measures of Kurtosis

The fourth standardized moment α_4 (or β_2) is a measure of *flatness* or *peakedness* of a single *humped* distribution (also called *Kurtosis*). For a normal distribution $\alpha_4 = \beta_2 = 3$ so that $\gamma_2 = 0$ and hence any distribution having $\beta_2 > 3$ will be peaked more sharply than the normal curve known as **leptokurtic** (narrow) while if $\beta_2 < 3$, the distribution is termed as **platykurtic** (broad).

The value of β_2 is helpful in selecting an appropriate measure of central tendency and variation to describe a frequency distribution. For example, if $\beta_2 = 3$, mean is preferred; if $\beta_2 > 3$ (leptokurtic distribution), median is preferred; while for $\beta_2 < 3$ (platykurtic distribution), quartile range is suitable.

Remark: W. S. Gosset explained different shapes of frequency curves as Platykurtic curves, like the platypus, are squat with short tails; leptokurtic curves are high with long tails like the Kangaroos noted for leaping.

Example 5.15: The first four moments of a distribution about the value 5 of the variable are 2, 20, 40 and 50. Show that the mean is 7. Also find the other moments, β_1 and β_2 , and comment upon the nature of the distribution.

Solution: From the data of the problem, we have

$$\mu'_1 = 2, \mu'_2 = 20, \mu'_3 = 40, \mu'_4 = 40 \text{ and } A = 5$$

Now the moments about the arbitrary point 5 are calculated as follows:

$$\text{Mean, } \bar{x} = \mu'_1 + A = 2 + 5 = 7$$

$$\text{Variance, } \mu_2 = \mu'_2 - (\mu'_1)^2 = 20 - (2)^2 = 16$$

$$\mu_3 = \mu'_3 - 3\mu'_1\mu'_2 + 2(\mu'_1)^3 = 40 - 3(2)(20) + 2(2)^3 = -64$$

$$\mu_4 = \mu'_4 - 4\mu'_1\mu'_3 + 6\mu'_2(\mu'_1)^2 - 3(\mu'_1)^4$$

$$= 50 - 4(2)(40) + 6(20)(2)^2 - 3(2)^4 = 162$$

The two constants, β_1 and β_2 , calculated from central moments are as follows:

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3} = \frac{(-64)^2}{(16)^3} = \frac{4096}{4096} = 1$$

$$\beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{162}{(16)^2} = \frac{162}{256} = 0.63$$

$$\gamma_1 = \frac{\mu_3}{\mu_2^{3/2}} = \frac{-64}{(16)^{3/2}} = -1 (< 0), \text{ distribution is negatively skewed.}$$

$$\gamma_2 = \beta_2 - 3 = 0.63 - 3 = -2.37 (< 0), \text{ distribution is platykurtic.}$$

Example 5.16: Find the standard deviation and kurtosis of the following set of data pertaining to kilowatt hours (kwh) of electricity consumed by 100 persons in a city.

Consumption

(in kwh)	:	0–10	10–20	20–30	30–40	40–50
Number of users	:	10	20	40	20	10

Solution: The calculations for standard deviation and kurtosis are shown in Table 5.5.

Table 5.5 Calculations of Standard Deviation and Kurtosis

Consumption (in kwh)	Number of Users (f)	Mid-Value (m)	$d = (m-A)/10$ $= (m-25)/10$	fd	fd ²
0–10	10	5	–2	–20	40
10–20	20	15	–1	–20	20
20–30	40	(25) ← A	0	0	0
30–40	20	35	1	20	20
40–50	10	45	2	20	40
	<u>100</u>			<u>0</u>	<u>120</u>

$$\bar{x} = A + \frac{\sum fd}{N} \times h = 25 + \frac{0}{100} \times 10 = 25$$

Since $\bar{x} = 25$ is an integer value, therefore we may calculate moments about the actual mean

$$\mu_r = \frac{1}{n} \sum f(x - \bar{x})^r = \frac{1}{n} \sum f(m - \bar{x})^r$$

Let $d = \frac{m - \bar{x}}{h}$ or $(m - \bar{x}) = hd$. Therefore,

$$\mu_r = h^r \frac{1}{n} \sum fd^r ; \quad h = \text{width of class intervals}$$

The calculations for moments are shown in Table 5.6.

Table 5.6 Calculations for Moments

Mid-value (m)	Frequency (f)	$d = \frac{m-25}{10}$	fd	fd ²	fd ³	fd ⁴
5	10	–2	–20	40	–80	160
15	20	–1	–20	20	–20	20
(25) ← A	40	0	0	0	0	0
35	20	1	20	20	20	20
45	10	2	20	40	80	160
	<u>100</u>		<u>0</u>	<u>120</u>	<u>0</u>	<u>360</u>

Moments about the origin $A = 25$ are

$$\mu_1 = h \frac{1}{N} \sum fd = 10 \times \frac{1}{100} \times 0 = 0$$

$$\mu_2 = h^2 \frac{1}{N} \sum fd^2 = (10)^2 \times \frac{1}{100} \times 120 = 120$$

$$\mu_3 = h^3 \frac{1}{N} \sum fd^3 = (10)^3 \times \frac{1}{100} \times 0 = 0$$

$$\mu_4 = h^4 \frac{1}{N} \sum fd^4 = (10)^4 \times \frac{1}{100} \times 360 = 36,000$$

$$\text{S.D. } (\sigma) = \sqrt{\mu_2} = \sqrt{120} = 10.95$$

Karl Pearson's measure of kurtosis is given by

$$\beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{36,000}{(120)^2} = 2.5$$

and therefore $\gamma_2 = \beta_2 - 3 = 2.5 - 3 = -0.50$

Since $\beta_2 < 3$ (or $\gamma_2 < 0$), distribution curve is platykurtic.

Example 5.17: Calculate the first four moments about the mean from the following data. Also calculate the value of β_1 and β_2 :

Marks	:	0–10	10–20	20–30	30–40	40–50	50–60	60–70
No. of Students	:	5	12	18	40	15	7	3

[Delhi Univ., B.Com.(Hons.), 1997]

Solution: Calculations for first four moments are shown below:

Marks (X)	Mid- value (m)	No. of Students (f)	$d = X - 35$	$d' = d/10$	fd'	fd'^2	fd'^3	fd'^4
0–10	5	5	–30	–3	–15	45	–135	405
10–20	15	12	–20	–2	–24	48	–96	192
20–30	25	18	–10	–1	–18	18	–18	18
30–40	35	40	0	0	0	0	0	0
40–50	45	15	10	1	15	15	15	15
50–60	55	7	20	2	14	28	56	112
60–70	65	3	30	3	9	27	81	243
		100			–19	181	–97	985

$$\mu'_1 = \text{First moment about arbitrary point } 35 = h \frac{\sum fd'}{\sum f} = \frac{-19}{100} \times 10 = -1.9$$

$$\mu'_2 = \text{Second moment about arbitrary point } 35 = h^2 \frac{\sum fd'^2}{\sum f} = \frac{181}{100} \times 100 = 181$$

$$\mu'_3 = h^3 \frac{\sum fd'^3}{\sum f} = (10)^3 \frac{-97}{100} = -970$$

$$\mu'_4 = h^4 \frac{\sum fd'^4}{\sum f} = (10)^4 \frac{985}{100} = 98,500$$

The moments about the mean value are calculated as follows:

$$\mu_1 = 0$$

$$\mu_2 = \mu'_2 - \mu'^2_1 = 181 - (-1.9)^2 = 177.39$$

$$\begin{aligned}\mu_3 &= \mu'_3 - 3\mu'_2 \mu'_1 + 2(\mu'_1)^3 \\ &= -970 - 3 \times 181 \times (-1.9) + 2(-1.9)^3 = 47.982\end{aligned}$$

$$\begin{aligned}\mu_4 &= \mu'_4 - 4\mu'_1 \mu'_3 + 6\mu'^2_2 (\mu'_1)^2 - 3(\mu'_1)^4 \\ &= 98,500 - 4(-970)(-1.9) + 6(181)(-1.9)^2 - 3(-1.9)^4 \\ &= 98,500 - 7,372 + 3,920.46 - 39.096 = 95,009.364\end{aligned}$$

$$\beta_1 = \frac{\mu_3}{\mu_2^{3/2}} = \frac{(47.982)^2}{(177.39)^3} = 0.0203$$

$$\beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{95,009.364}{(177.39)^2} = 3.019$$

Example 5.18: From the following data, calculate moments about assumed mean 25 and convert them into central moments :

$x :$	0-10	10-20	20-30	30-40
$f :$	1	3	4	2

[Delhi Univ., B.Com.(Hons.), 2000]

Solution: Calculations for moments about assumed mean, $A = 25$, are shown below:

x	Mid - value(m)	Frequency (f)	$d = x - 25$	d^2	d^3	fd	fd^2	fd^3	fd^4
0-10	5	1	-20	400	-8000	-20	400	-8000	1,60,000
10-20	15	3	-10	100	-1000	-30	300	-3000	30,000
20-30	25	4	0	0	0	0	0	0	0
30-40	35	2	10	100	1,000	20	200	2000	20,000
		10				-30	900	-9000	2,10,000

Let $A = 25$. Then $d = x - 25$. Thus,

$$\mu'_1 = \text{First moment about } 25 = \frac{\Sigma fd}{\Sigma f} = \frac{-30}{10} = -3$$

$$\mu'_2 = \text{Second moment about } 25 = \frac{\Sigma fd^2}{\Sigma f} = \frac{900}{10} = 90$$

$$\mu'_3 = \text{Third moment about } 25 = \frac{\Sigma fd^3}{\Sigma f} = \frac{-9000}{10} = -900$$

$$\mu'_4 = \text{Fourth moment about } 25 = \frac{\Sigma fd^4}{\Sigma f} = \frac{2,10,000}{10} = 21,000$$

$$\mu_1 = 0$$

$$\mu_2 = \mu'_2 - \mu'^2_1 = 90 - 9 = 81$$

$$\begin{aligned}\mu_3 &= \mu'_3 - 3\mu'_2 \mu'_1 + 2(\mu'_1)^3 \\ &= -900 - 3 \times 90 \times (-3) + 2 \times (-3)^3 = -900 + 810 - 54 = -144\end{aligned}$$

$$\begin{aligned}
 \mu_4 &= \mu'_4 - 4\mu'_3 \mu'_1 + 6\mu'_2 (\mu'_1)^2 - 3(\mu'_1)^4 \\
 &= 21,000 - 4(-900)(-3) + 6 \times 90 (-3)^2 - 3(-3)^4 \\
 &= 21,000 - 10,800 + 4860 - 243 = 14,817.
 \end{aligned}$$

Example 5.19: The first four moments of a distribution about value 2 are 1, 2.5, 5.5 and 16, respectively. Calculate the four moments about mean and comment on the nature of the distribution.

[Delhi Univ., B.Com.(Hons.), 2002]

Solution: The moments about the arbitrary value 2 are : $\mu'_1 = 1$, $\mu'_2 = 2.5$, $\mu'_3 = 5.5$ and $\mu'_4 = 16$

The calculations for moments about mean from the given values are shown below:

$$\begin{aligned}
 \mu_1 &= 0 \\
 \mu_2 &= \mu'_2 - (\mu'_1)^2 = 2.5 - (1)^2 = 1.5 \\
 \mu_3 &= \mu'_3 - 3\mu'_1 \mu'_2 + 2(\mu'_1)^3 = 5.5 - 3(1)(2.5) + 2(1)^3 = 5.5 - 7.5 + 2 = 0 \\
 \mu_4 &= \mu'_4 - 4\mu'_1 \mu'_3 + 6(\mu'_1)^2 \mu'_2 - 3(\mu'_1)^4 = 16 - 4(1)(5.5) + 6(2.5)(1)^2 - 3(1)^4 \\
 &= 16 - 22 + 15 - 3 = 6
 \end{aligned}$$

The values of β_1 and β_2 to comment upon the nature of distribution are

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3} = \frac{0}{(1.5)^3} = 0$$

Since $\beta_1 = 0$, the distribution is symmetrical. Also

$$\beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{6}{(1.5)^2} = \frac{6}{2.25} = 2.6$$

Since $\beta_2 < 3$, the distribution curve is platykurtic.

Example 5.20: Find the four moments about mean from the following data. Also decide whether it is a platykurtic distribution.

Central size of item (x)	: 1	2	3	4	5
Frequency (f)	: 2	3	5	4	1

[Delhi Univ., B.Com.(Hons.) 1996]

Solution: The calculations for first four moments about the value 3 are shown below:

x	f	d = x - 3	fd	fd ²	fd ³	fd ⁴
1	2	-2	-4	8	-16	32
2	3	-1	-3	3	-3	3
3	5	0	0	0	0	0
4	4	1	4	4	4	4
5	1	2	2	4	8	16
	15		-1	19	-7	55

Let $A = 3$. Then $d = x - 3$. Thus,

$$\begin{aligned}
 \mu'_1 &= \frac{\Sigma fd}{N} = \frac{-1}{15} = -0.07; \quad \mu'_2 = \frac{\Sigma fd^2}{N} = \frac{19}{15} = 1.27 \\
 \mu'_3 &= \frac{\Sigma fd^3}{N} = \frac{-7}{15} = -0.47; \quad \mu'_4 = \frac{\Sigma fd^4}{N} = \frac{55}{15} = 3.67
 \end{aligned}$$

Central moments or moments about mean are given by

$$\begin{aligned}\mu_1 &= 0, \\ \mu_2 &= \mu'_2 - (\mu'_1)^2 = 1.27 - (-0.07)^2 = 1.27 - 0.0049 = 1.2651 \\ \mu_3 &= \mu'_3 - 3\mu'_2 \mu'_1 + 2(\mu'_1)^3 \\ &= -0.47 - 3(1.2651)(-0.07) + 2(-0.07)^3 = -0.205015 \\ \mu_4 &= \mu'_4 - 4\mu'_3 \mu'_1 + 6\mu'_2 (\mu'_1)^2 - 3(\mu'_1)^4 \\ &= 3.67 - 4(-0.21)(-0.07) + 6(1.27)(-0.07)^2 - 3(-0.07)^4 = 3.648466 \\ \beta_2 &= \frac{\mu_4}{\mu_2^2} = \frac{3.6}{(1.2651)^2} = \frac{3.6}{1.6} = 2.25\end{aligned}$$

Since $\beta_2 < 3$, therefore distribution is platykurtic.

Example 5.21: If $\beta_1 = 1$ and $\beta_2 = 4$ and variance = 9, find the values of μ_3 and μ_4 . Comment upon the nature of the distribution. [Delhi Univ., B.Com.(Hons.), 1995]

Solution: Given $\beta_1 = 1$, $\beta_2 = 4$ and $\sigma^2 = 9$. We know that

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3} \quad \text{or} \quad 1 = \frac{\mu_3^2}{(9)^3} \quad (\text{since } \mu_2 = \sigma^2 = 9)$$

$$\mu_3^2 = 729, \text{ i.e. } \mu_3 = \sqrt{729} = 27$$

Also
$$\beta_2 = \frac{\mu_4}{\mu_2^2} \quad \text{or} \quad 4 = \frac{\mu_4}{(9)^2}, \text{ i.e. } \mu_4 = 324.$$

Example 5.22: In a certain distribution, the first four moments about 5 are 2, 20, 40 and 50. Calculate β_1 and β_2 . State whether the distribution is leptokurtic or platykurtic.

[Delhi Univ., B.Com.(Hons.), 2006]

Solution: Given, $A = 5$, $\mu'_1 = 2$, $\mu'_2 = 20$, $\mu'_3 = 40$ and $\mu'_4 = 50$. Values of central moments are calculated as follows:

$$\begin{aligned}\mu_1 &= 0 \\ \mu_2 &= \mu'_2 - (\mu'_1)^2 = 20 - (2)^2 = 16 \\ \mu_3 &= \mu'_3 - 3\mu'_2 \mu'_1 + 2(\mu'_1)^3 = 40 - 3(20)(2) + 2(2)^3 \\ &= 40 - 120 + 16 = -64 \\ \mu_4 &= \mu'_4 - 4\mu'_1 \mu'_3 + 6\mu'_2 (\mu'_1)^2 - 3(\mu'_1)^4 = 50 - 4(2)(40) + 6(20)(2)^2 - 3(2)^4 \\ &= 50 - 320 + 480 - 48 = 162 \\ \beta_1 &= \frac{\mu_3^2}{\mu_2^3} = \frac{(-64)^2}{(16)^3} = 1 \\ \beta_2 &= \frac{\mu_4}{\mu_2^2} = \frac{162}{(16)^2} = 0.633\end{aligned}$$

Since $\beta_2 < 3$, distribution is platykurtic.

Example 5.23: For a distribution, the mean is 10, standard deviation is 4, $\sqrt{\beta_1} = 1$ and $\beta_2 = 4$. Obtain the first four moments about the origin, i.e., zero. [Delhi Univ., B.Com.(Hons.), 1992]

Solution: Given $\mu_2 = \sigma^2 = (4)^2 = 16$. Also $\sqrt{\beta_1} = 1$ or $\beta_1 = 1$. Thus,

$$\beta_2 = \frac{\mu_4}{(\mu_2)^2} \quad \text{or} \quad 4 = \frac{\mu_4}{(16)^2}, \text{ i.e. } \mu_4 = 4 \times (16)^2 = 4 \times 256 = 1024$$

$$\text{Also} \quad \beta_1 = \frac{(\mu_3)^2}{(\mu_2)^3} \quad \text{or} \quad 1 = \frac{(\mu_3)^2}{(16)^3}, \text{ i.e. } \mu_3^2 = (16)^3 = 4096 \quad \text{or} \quad \mu_3 = \sqrt{4096} = 64$$

Now moments about zero (i.e. original):

$$v_1 = \bar{x} = 10$$

$$v_2 = \mu_2 + (v_1)^2 = 16 + (10)^2 = 116$$

$$v_3 = \mu_3 + 3v_2v_1 - 2(v_1)^3 = 64 + 3(116)(10) - 2(10)^3 \\ = 64 + 3480 - 2000 = 1544$$

$$v_4 = \mu_4 + 4v_3v_1 - 6v_2(v_1)^2 + 3(v_1)^4 \\ = 1024 + 4(1544)(10) - 6(116)(10)^2 + 3(10)^4 \\ = 1024 + 61760 - 69600 + 30000 = 23184$$

Example 5.24: Given the following information: $N = 100$, $\Sigma(x - 98) = 50$, $\Sigma(x - 98)^2 = 1970$; $\Sigma(x - 98)^3 = 2948$, $\Sigma(x - 98)^4 = 86752$. Do you think that the distribution is platykurtic?

[Delhi Univ., B.Com.(Hons.), 1993]

Solution: The most important measure of Kurtosis is the value of coefficient β_2 .

The greater the value of β_2 , more peaked the distribution will be. Let $d = \Sigma(x - 98)$. Then

$$\beta'_1 = \frac{\Sigma d}{n} = \frac{50}{100} = 0.5; \beta'_2 = \frac{\Sigma d^2}{n} = \frac{1970}{100} = 19.70$$

$$\beta'_3 = \frac{\Sigma d^3}{n} = \frac{2948}{100} = 29.48; \beta'_4 = \frac{\Sigma d^4}{n} = \frac{86752}{100} = 867.52$$

$$\mu_2 = \beta'_2 - (\beta'_1)^2 = 19.70 - (0.5)^2 = 19.70 - 0.25 = 19.45$$

$$\mu_3 = \beta'_3 - 3\beta'_1\beta'_2 + 2\beta'^3_1 = 29.48 - 3(0.5)(19.70) + 2(0.5)^3 = 0.18$$

$$\mu_4 = \beta'_4 - 4\beta'_1\beta'_3 + 6\beta'^2_2(\beta'_1)^2 - 3\beta'^4_1 \\ = 867.52 - 4(0.5)(29.48) + 6(19.70)(0.5)^2 - 3(0.5)^4 = 837.92$$

$$\beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{837.9225}{378.3025} = 2.215$$

Since the value of β_2 is less than 3, the distribution is platykurtic.

Example 5.25: The first four moments from mean of a distribution are: 0, 2.5, 0.7 and 18.75. The mean value is 11. Comment on the skewness and kurtosis of the distribution.

Solution: The measure of Skewness, β_1 , is given by

$$\text{and} \quad \beta_1 = \frac{\mu_3}{\mu_2^{\frac{3}{2}}} = \frac{0.7^2}{2.5^3} = \frac{0.49}{15.62} = 0.031 \text{ (Given)}$$

Since $\beta_1 \neq 0$, distribution is not perfectly symmetrical, it is skewed. Further β_2 is the measure of Kurtosis

$$\beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{18.75}{(2.5)^2} = \frac{18.75}{6.25} = 3$$

Since $\beta_2 = 3$, distribution is mesokurtic.

Example 5.26: The first four moments of the distribution about $A = 4$ are 1, 4, 10, 45. Obtain the various characteristics of the distribution on the basis of information given. Comment on the nature of distribution. [Delhi Univ., B.Com.(Hons.), 2003]

Solution: Given, $A = 4$, $\mu'_1 = 1$, $\mu'_2 = 4$, $\mu'_3 = 10$ and $\mu'_4 = 45$. Calculating central moments to obtain the various characteristics of the distribution as follows:

$$\begin{aligned}\mu_1 &= 0 \\ \mu_2 &= \mu'_2 - (\mu'_1)^2 = 4 - (1)^2 = 4 - 1 = 3 \\ \mu_3 &= \mu'_3 - 3\mu'_2 \mu'_1 + 2(\mu'_1)^3 = 10 - 3(4)(1) + 2(1)^3 = 0 \\ \mu_4 &= \mu'_4 - 4\mu'_3 \mu'_1 + 6\mu'_2 \mu_1^2 - 3(\mu'_1)^4 \\ &= 45 - 4(10)(1) + 6(4)(1)^2 - 3(1)^4 = 26\end{aligned}$$

Since $\mu_2 = \sigma^2 = 3$, therefore $\sigma = \sqrt{3} = 1.732$. Also $v_1 = A + \mu'_1 = 4 + 1 = 5$, therefore, Mean = 5 and coefficient of Skewness based on moments: $\beta_1 = \frac{\mu_3^2}{\mu_2^3} = \frac{0^2}{(3)^3} = 0$. This implies that the distribution is symmetrical. The measure of kurtosis, $\beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{26}{(3)^2} = \frac{26}{9} = 2.89 (< 3)$, distribution is platykurtik.

Example 5.27: Examine whether the following results of a piece of computation for obtaining the second central moments are consistent or not : $N = 120$, $\Sigma fx = -125$, $\Sigma fx^2 = 128$

[Delhi Univ., B.Com.(Hons.), 2005]

Solution: Second central moment, $\mu_2 (= \sigma^2) = \frac{\Sigma fx^2}{N} - \left(\frac{\Sigma fx}{N}\right)^2 = \frac{128}{120} - \left(\frac{-125}{120}\right)^2$
 $= 1.066 - 1.085 = -0.019$, which is negative

Since μ_2 being variance can never be negative, information is not consistent.

Example 5.28: For a distribution, mean = 10, variance = 16, $\gamma_1 = 1$ and $\beta_2 = 4$. Obtain the first four moments about origin, i.e. zero. Comment upon the nature of distribution.

[Delhi Univ., B.Com.(Hons.), 2005]

Solution: Given, Mean $\bar{x} = 10$, Variance, $\sigma^2 = 16$, $\gamma_1 = 1$, $\beta_2 = 4$. Thus

First moment about zero, $v_1 (= \bar{x}) = 10$

Var $\sigma^2 = \mu_2 =$ second moment about mean = 16

Second moment about zero, $v_2 = \mu_2 + v_1^2 = 16 + (10)^2 = 116$

Now $\gamma_1 = \sqrt{\beta_1}$ or $\gamma_1^2 = \beta_1$, i.e. $\beta_1 = 1$

We know that $\beta_1 = \frac{\mu_3^2}{\mu_2^3}$ or $1 = \frac{\mu_3^2}{(16)^3}$, i.e. $\mu_3 = 64$.

Third moment about zero, $v_3 = \mu_3 + 3v_2v_1 - 2v_1^3$
 $= 64 + 3(116)(10) - 2(10)^3 = 64 + 3480 - 2000 = 1544$

We are given that $\beta_2 = 4$, so $\beta_2 = \frac{\mu_4}{\mu_2^2} = 4$ or $\mu_4 = 4\mu_2^2 = 4(16)^2 = 1024$

Fourth Moment about zero, $v_4 = \mu_4 + 4v_3v_1 - 6v_2v_1^2 + 3v_1^4$
 $= 1024 + 4(1544)(10) - 6(116)(10)^2 + 3(10)^4$
 $= 1024 + 61760 - 69600 + 30000 = 23184$.

Conceptual Questions 5B

8. What do you understand by the terms skewness and kurtosis? Point out their role in analysing a frequency distribution.
9. Averages, dispersion, skewness and kurtosis are complementary to one another in understanding a frequency distribution? Elucidate.
10. Explain how the measure of skewness and kurtosis can be used in describing a frequency distribution.
11. Explain the terms 'skewness' and 'kurtosis' used in connection with the frequency distribution of a continuous variable. Give the different measures of skewness (any two of the measures to be given) and kurtosis.
12. What do you mean by 'kurtosis' in statistics? Explain one of the methods of measuring it.

Self-practice Problems 5B

- 5.13 Explain whether the following results of a piece of computation for obtaining the second central moment are consistent or not

$$n = 120, \Sigma fx = -125, \Sigma fx^2 = 128.$$

- 5.14 The first four central moments are 0, 4, 8 and 144. Examine the skewness and kurtosis.

- 5.15 Compute the first four moments about the mean from the following data:

Mid-value of variate : 5 10 15 20 25 30 35

Frequency : 8 15 20 32 23 17 5

Comment upon the nature of the distribution.

- 5.16 A record was kept over a period of 6 months by a sales manager to determine the average number of calls made per day by his six salesmen. The results are shown below:

Salesmen : A B C D E F

Average number of calls per day : 8 10 12 15 7 5

- (a) Compute a measure of skewness. Is the distribution symmetrical?
 - (b) Compute a measure of kurtosis. What does this measure mean?
- 5.17 Find the second, third and fourth central moments of the frequency distribution

given below. Hence find the measure of skewness and a measure of kurtosis of the following distribution:

Class limits	Frequency
100–104.9	7
105–109.9	13
110–114.9	25
115–119.9	25
120–124.9	30

- 5.18 Find the first four moments about the mean for the following distribution:

Class Interval : 60–62 63–65 66–68

69–71 72–74

Frequency : 5 18 42
27 8

- 5.19 Find the variance, skewness and kurtosis of the following frequency distribution by the method of moments:

Class interval : 0–10 10–20 20–30 30–40

Frequency : 1 4 3 2

- 5.20 Find the kurtosis for the following distribution

Class interval : 0–10 10–20 20–30 30–40

Frequency : 1 3 4 2

Comment on the nature of the distribution.

Hints and Answers

$$5.13 \quad \mu_2 = \sum fx^2 / N - (\sum fx / N)^2 \\ = 128/120 - (-125/120)^2 = -0.0146$$

since σ^2 cannot be negative, therefore the data is inconsistent.

$$5.14 \quad \beta_1 = \mu_3^2 / \mu_2^3 = 2/(4)^3 = 1;$$

$$\gamma_1 = + \sqrt{\beta_1} = 1$$

$$\beta_2 = \mu_4 / \mu_2^2 = 144/(4)^2 = 9;$$

$$\gamma_2 = \beta_2 - 3 = 6$$

$$5.16 \quad \beta_1 = 0.11; \quad \beta_2 = 1.97$$

$$5.17 \quad \mu_2 = 54; \quad \mu_3 = 100.5, \mu_4 = 7827;$$

$$\gamma_1 = + \sqrt{\beta_1} = 0.2533;$$

$$\gamma_2 = \beta_2 - 3 = -0.3158$$

$$5.18 \quad \mu_1 = 0, \quad \mu_2 = 8.527, \quad \mu_3 = -2.693, \\ \mu_4 = 199.375$$

$$5.19 \quad \sigma^2 = \mu_2 = 84, \quad \gamma_1 = + \sqrt{\beta_1} = 0.0935; \\ \beta_2 = 2.102$$

$$5.20 \quad \mu_2 = 81, \quad \mu_2 = 14817; \\ \beta_2 = \mu_4 / \mu_2^2 = 2.26$$

Formulae Used

1. Absolute measure of skewness

$$Sk = \bar{x} - \text{Mode or } Q_3 + Q_1 - 2 \text{ Med}$$

2. Coefficient of skewness

Karl Pearson's

$$Sk_p = \frac{\bar{x} - Mo}{\sigma} \quad \text{or} \quad \frac{3(\bar{x} - \text{Med})}{\sigma}$$

$$\text{Bowley's, } Sk_b = \frac{Q_3 + Q_1 - 2 \text{ Med}}{Q_3 - Q_1}$$

$$\text{Kelly's, } Sk_k = \frac{P_{90} + P_{10} - 2P_{50}}{P_{90} - P_{10}}$$

$$\text{or} \quad \frac{D_9 + D_1 - 2D_5}{D_9 - D_1}$$

$$5. \text{ Kurtosis } \gamma_1 = \sqrt{\beta_1} = \frac{\mu_3}{\mu_2^{3/2}}$$

$$\gamma_2 = \beta_2 - 3 = \frac{\mu_4}{\mu_2^2} - 3$$

6. For a normal curve, $\beta_2 = 3$ or $\gamma_2 = 0$; for a leptokurtic curve, $\beta_2 > 3$ or $\gamma_2 > 0$ and for a platykurtic curve, $\beta_2 < 3$ or $\gamma_2 < 0$.

Review Self-practice Problems

- 5.21 Find the coefficient of skewness from the following information:

Difference of two quartiles = 8;

Mode = 1;

Sum of two quartiles = 22; Mean = 8.

[Delhi Univ., B.Com. (Hons.), 1997]

- 5.22 From the data given below calculate the coefficient of variation:

Karl Pearson's coefficient of skewness = 0.42

Arithmetic mean = 86

Median = 80

[Osmania Univ., B.Com., 1998]

- 5.23 In a frequency distribution, the coefficient of skewness based on quartiles is 0.6. If the sum of upper and lower quartiles is 100 and the median is 38, find the value of the upper quartile.

- 5.24 The daily expenditure (in ₹) of 100 families is given below

Daily expenditure :

0-20 20-40 40-60 60-80 80-100

Number of families :

13 f_2 27 f_4 16

If mode of the distribution is 44, calculate Karl Pearson's coefficient of skewness.

- 5.25** Pearson's coefficient of skewness for a distribution is 0.4 and coefficient of variance is 30 per cent. Its mode is 88. Find the mean and median.
- 5.26** The following table gives the distribution of monthly wages of 500 workers in a factory:

Monthly Wages (₹ in hundred)	Number of Workers	Monthly Wages (₹ in hundred)	Number of Workers
15–20	10	30–35	220
20–25	25	35–40	70
25–30	145	40–45	30

Compute Karl Pearson's and Bowley's coefficient of skewness. Interpret your answer.

Hints and Answers

- 5.21** Mode = 3 Median – 2 Mean
or $11 = 3 \text{ Med} - 2 \times 8$ or $\text{Med} = 9$
 $Q_3 + Q_1 = 22$ and $Q_3 - Q_1 = 8$, i.e.
 $Q_3 = 15, Q_1 = 7$
Coefficient of skewness =
$$\frac{Q_3 + Q_1 - 2\text{Med}}{Q_3 - Q_1} = \frac{15 + 7 - 2(9)}{8} = 0.5$$
- 5.22** Mode = 3 Median – 2 Mean
 $= 3(80) - 2(86) = 68$
Coefficient of skewness = $\frac{\bar{x} - \text{Mode}}{\sigma}$
or $0.42 = \frac{86 - 68}{\sigma}$ or $\sigma = 42.86$
Coefficient of variation (CV)
 $= \frac{\sigma}{\bar{x}} \times 100$
 $= \frac{42.86}{68} \times 100 = 49.84$ per cent.
- 5.23** Given $\text{Sk} = 0.6, Q_1 + Q_3 = 100,$
 $\text{Med} = 38$
 $\text{Sk}_b = \frac{Q_3 + Q_1 - 2\text{Med}}{Q_3 - Q_1}$
or $0.6 = \frac{100 - 2 \times 38}{Q_3 - Q_1}$
 $0.6 = \frac{100 - 76}{Q_3 - (100 - Q_3)}$ or $Q_3 = 70$
- 5.24** Let the frequency for the class 20–40 be f_2 . Then frequency for the class 60–80 will be
 $f_4 = 100 - (13 + f_2 + 27 + 16) = 44 - f_2$

Expenditure	Number of Families (f)	Cumulative Frequency (cf)
0–20	13	13
20–40	f_2	$13 + f_2$
40–60	27	$40 + f_2$
60–80	$44 - f_2$	84
80–100	16	100

$$\begin{aligned}\text{Mode} &= l + \frac{f_m - f_{m-1}}{2f_m - f_{m-1} - f_{m+1}} \times h \\ &= 40 + \frac{27 - f_2}{54 - f_2 - 44} \times 20\end{aligned}$$

$$\text{or } f_2 = 25$$

Thus frequency for the class 20–40 is 25
and for the class 60–80 is $44 - 25 = 19$

$$\begin{aligned}\text{Apply the formula, } \text{Sk}_p &= \frac{\bar{x} - \text{Mo}}{\sigma} \\ &= \frac{50 - 44}{25.3} = 0.237\end{aligned}$$

- 5.25** Given $\text{Sk}_p = 0.4, \text{CV} = 0.30,$
Mode = 88
 $\text{Sk}_p = \frac{\bar{x} - \text{Mo}}{\sigma}$
 $= \frac{1 - (\text{Mo}/\bar{x})}{(\sigma/\bar{x})} = \frac{1 - (88/\bar{x})}{0.30};$
 $\text{CV} = \sigma/\bar{x}$ or $0.30 = \sigma/\bar{x}$
 $\frac{88}{\bar{x}} = 1 - 0.4 \times 0.3 = 0.88$ or $\bar{x} = 100$
Also, Mode = 3 Med – 2 $\bar{x}$
or $88 = 3 \text{ Med} - 2(100)$
or $\text{Med} = 96$

Glossary of Terms

Measure of skewness: The statistical technique to indicate the direction and extent of skewness in the distribution of numerical values in the data set.

Moments: Represent a convenient and unifying method for summarizing certain descriptive statistical measures.

Kurtosis: The degree of flatness or peakedness in the region around the mode of a frequency curve.

Leptokurtic: A frequency curve that is more peaked than the normal curve.

Platykurtic: A frequency curve that is flat-topped than the normal curve.

Mesokurtic: A frequency curve that is a normal (symmetrical) curve.

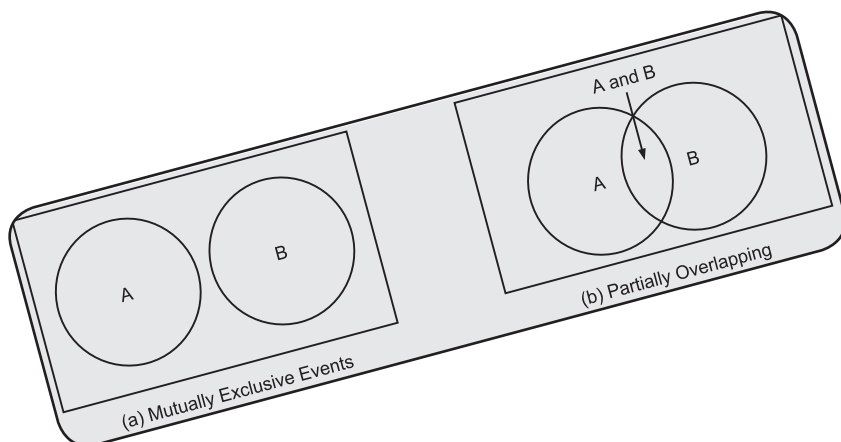
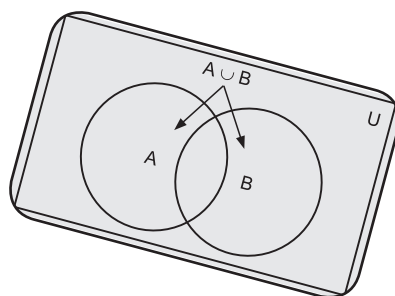
UNIT II

PROBABILITY, PROBABILITY DISTRIBUTIONS AND DECISION THEORY

Chapter 6: Fundamentals of Probability

Chapter 7: Probability Distributions

Chapter 8: Statistical Decision Theory



This page is intentionally left blank.

6

Fundamentals of Probability

LEARNING OBJECTIVES

After studying this chapter, you should be able to

- help yourself to understand the amount of uncertainty that is involved before making important decisions.
- understand fundamentals of probability and various probability rules that help you to measure uncertainty involving uncertainty.
- perform several analyses with respect to business decision involving uncertainty.

6.1 INTRODUCTION

Decision makers always face some degree of risk while selecting a particular course of action or strategy to solve a decision problem involving uncertainty. It is because each strategy can lead to a number of different possible outcomes (or results). Thus, it is necessary for the decision makers to enhance their capability of grasping the probabilistic situation so as to gain a deeper understanding of the decision problem and base their decisions on rational considerations. The knowledge of the concepts of probability, probability distributions and various related statistical techniques is therefore needed.

6.2 CONCEPTS OF PROBABILITY

In order to obtain a deeper understanding of probability, it is necessary to use certain terms and definitions more precisely.

Random Experiment

Random experiment (also called *act*, *trial*, *operation* or *process*) is an activity that leads to the occurrence of one and only one of several possible outcomes which is not likely to be known until

its completion, that is, the outcome is not perfectly predictable. This process has the properties that

- (i) all possible outcomes can be specified in advance.
- (ii) it can be repeated.
- (iii) the same outcome may not occur on various repetitions so that the actual outcome is not known in advance.

The variation among experimental outcomes caused by the effects of uncontrolled factors is called *random variation*. It is assumed that these effects vary randomly and unpredictably from one repetition of an experiment to the next.

The outcome (observation or measurement) generated by an experiment may or may not produce a numerical value. Few examples of experiments are as follows:

- (i) Measuring blood pressure of a group of individuals.
- (ii) Checking an automobile's petrol mileage.
- (iii) Tossing a coin and observing the face that appears.
- (iv) Testing a product to determine whether it is defective or an acceptable product.
- (v) Measuring daily rainfall, and so on.

In all such cases, there is uncertainty surrounding the outcome until an outcome is observed. For example, if we toss a coin, the outcome will not be known with certainty until either the head or the tail is observed. The number of outcomes may be finite or infinite depending on the nature of the experiment. For example, in the experiment of tossing a coin, the outcomes are finite and represented by the head and tail, whereas in the experiment of measuring the time between successive failures of an electronic device, the outcomes are infinite and represented by the time of failure.

The outcome of an experiment may be expressed in numerical or non-numerical value. For example,

- (i) counting the number of arrivals at a service window (numerical outcome)
- (ii) payment made by cash, cheque, or credit card (non-numerical outcome).

Although an individual outcome associated with a random experiment cannot be predicted exactly, the frequency of occurrence of such an outcome can be noted in a large number of repetitions and thus becomes the basis for resolving problems dealing with uncertainty.

Each experiment may result in one or more outcomes, which are called **events** and denoted by capital letters.

Sample Space

The set of all possible distinct outcomes (events) for a random experiment is called the sample space (or *event space*) provided.

- (i) no two or more of these outcomes can occur simultaneously
- (ii) exactly one of the outcomes must occur, whenever the experiment is performed

Sample space is denoted by the capital letter S.

Illustrations

1. Consider the experiment of recording a person's blood type. The four possible outcomes are the following simple events:

E_1 : Blood type A E_2 : Blood type B

E_3 : Blood type AB E_4 : Blood type O

The sample space is $S = \{E_1, E_2, E_3, E_4\}$.

Some experiments can be generated in stages and sample space can be displayed in a *tree diagram*. Each successive level of branching on the tree corresponds to a step required to generate the final outcome as shown in Fig. 6.1. The sample events in the tree diagram form the sample space $S = \{E_1, E_2, E_3, \dots, E_8\}$.

2. Consider the experiment of tossing two coins. The four possible outcomes are the following sample events

$E_1 = HH$ $E_2 = HT$ $E_3 = TH$ $E_4 = TT$

The sample space is $S = \{E_1, E_2, E_3, E_4\}$.

Event Types

A single possible outcome (or result) of an experiment is called a simple (or elementary) event. An **event** is the set (or collection) of one or more simple events of an experiment in the sample space and having a specific common characteristic. For example, for the above-defined sample space S , the collection (H, T), (T, H) is the event containing simple event as: H or T. Other examples of events are:

- More than 5 customers at a service facility in one hour
- Telephone calls lasting no more than 10 minutes
- 75 per cent marks or better in an examination
- Sales volume of a retail store more than ₹ 2,000 on a given day

Mutually Exclusive Events If two or more events cannot occur simultaneously in a single trial of an experiment, then such events are called mutually exclusive events or disjoint events. In other words, two events are mutually exclusive if the occurrence of one of them prevents or rules out the occurrence of the other. For example, the numbers 2 and 3 cannot occur simultaneously on the roll of a dice.

Symbolically, a set of events $\{A_1, A_2, \dots, A_n\}$ is mutually exclusive if . This means the intersection of two events is a null set ($\emptyset$); it is impossible to observe an event that is common in both A_i and A_j .

Collectively Exhaustive Events A list of events is said to be collectively exhaustive when all possible events that can occur from an experiment includes every possible outcome. That is, two or more events are said to be collectively exhaustive if one of the events must occur. Symbolically, a set of events $\{A_1, A_2, \dots, A_n\}$ is collectively exhaustive if the union of these events is identical with the sample space S . That is,

$$S = \{A_1 \cup A_2 \cup \dots \cup A_n\}$$

For example, being a male and female are mutually exclusive and collectively exhaustive events. Similarly, the number 7 cannot come upon the uppermost face during the experiment of rolling a dice because the number of faces uppermost has the sample space $S = \{1, 2, 3, 4, 5, 6\}$.

Independent and Dependent Events Two events are said to be *independent* if information about one tells nothing about the occurrence of the other. In other words, outcome of one event does not affect, and is not affected by, the other event. The outcomes of successive tosses of a coin are independent of its preceding toss. Increase in the population (in per cent) per year in India is independent of increase in wheat production (in per cent) per year in the USA.

However, two or more events are said to be dependent if information about one tells something about the other. That is, dependence between characteristics implies that a relationship exists, and therefore, knowledge of one characteristic is useful in assessing the occurrence of the other. For example, drawing of a card (say a queen) from a pack of playing cards without replacement reduces the chances of drawing a queen in the subsequent draws.

Compound Events When two or more events occur in connection with each other, then their simultaneous occurrence is called a compound event. These event may be (i) independent or (ii) dependent.

Equally Likely Events Two or more events are said to be equally likely if each has an equal chance to occur. That is, one of them cannot be expected to occur in preference to the other. For example, each number may be expected to occur on the uppermost face of a rolling die the same number of times in the long run.

Complementary Events If E is any subset of the sample space, then its complement denoted by (read as E-bar) contains all the elements of the sample space that are not part of E. If S denotes the sample space then

$$\bar{E} = S - E = \{\text{All sample elements not in } E\}$$

For example, if E represents companies with sales less than or equal to ₹ 25 lakh, written as $E = \{x : x \leq 25\}$, then this set is a complement of the set, $\bar{E} = \{x : x > 25\}$. Obviously such events must be mutually exclusive and collective exhaustive.

6.3 DEFINITION OF PROBABILITY

A general definition of probability states that **probability** is a numerical measure (between 0 and 1 inclusively) of the likelihood or chance of occurrence of an uncertain event. However, it does not tell us how to compute the probability. In this section, we shall discuss different conceptual approaches of calculating the probability of an event.

Classical Approach

This approach of defining the probability is based on the assumption that all the possible outcomes (finite in number) of an experiment are mutually exclusive and equally likely. It states that, during a random experiment, if there are 'a' possible outcomes where the favourable event A occurs and 'b' possible outcomes where the event A does not occur, and all these possible outcomes are mutually exclusive, exhaustive and equiprobable, then the probability that event A will occur is defined as

$$P(A) = \frac{a}{a+b} = \frac{\text{Number of favourable outcomes}}{\text{Total number of possible outcomes}} = \frac{c(A)}{c(S)}$$

For example, if a fair die is rolled, then on any trial, each event (face or number) is equally likely to occur since there are six equally likely exhaustive events, each will occur $1/6$ of the time, and therefore the probability of any one event occurring is $1/6$. Similarly for the process of selecting a card at random, each event or card is mutually exclusive, exhaustive and equiprobable. The probability of selecting any one card on a trial is equal to $1/52$, since there are 52 cards. Hence, in general, for a random experiment with n mutually exclusive, exhaustive and equiprobable events, the probability of any of the events is equal to $1/n$.

Since the probability of occurrence of an event is based on prior knowledge of the process involved, therefore this approach is often called *a priori classical probability approach*. This means we do not have to perform random experiments to find the probability of occurrence of an event. This also implies that no experimental data are required for computation of probability. Since the assumption of equally likely simple events can rarely be verified with certainty, therefore this approach is not used often other than in games of chance.

The assumption that all possible outcomes are equally likely may lead to a wrong calculation of probability in case some outcomes are more or less frequent in occurrence. For example, if we classify two children in a family according to their sex, then the possible outcomes in terms of number of boys in the family are 0, 1, 2. Thus, according to the **classical approach**, the probability for each of the outcomes should be $1/3$. However, it has been calculated that the probabilities are approximately $1/4$, $1/2$ and $1/4$ for 0, 1 and 2 boys, respectively. Similarly, we cannot apply this approach to find the probability of a defective unit being produced by a stable manufacturing process as there are only two possible outcomes, defective or non-defective.

Relative Frequency Approach

In situations where the outcomes of a random experiment are not all equally likely or when it is not known whether outcomes are equally likely, application of the classical approach is not desirable to quantify the possible occurrence of a random event. For example, it is not possible to state in advance, without repetitive trials of the experiment, the probabilities in cases like (i) whether a number greater than 3 will appear when die is rolled or (ii) if a lot of 100 items will include 10 defective items.

This approach of computing probability is based on the assumption that a random experiment can be repeated a large number of times under identical conditions where trials are independent to each other. While conducting a random experiment, we may or may not observe the desired event. But as the experiment is repeated many times that event may occur some proportion of time. Thus, the approach calculates *the proportion of the time (i.e. the relative frequency) with which the event occurs over an infinite number of repetitions of the experiment under identical conditions*. Since no experiment can be repeated in infinite number of times, therefore, a probability can never be exactly determined. However, we can approximate the probability of an event by recording the relative frequency with which the event has occurred over a finite number of repetitions of the experiment under identical conditions. For example, if a die is tossed n times and s denotes the number of times the event A (i.e., number 4, 5 or 6) occurs, then the ratio $P(A) = c(s)/n$ gives the proportion of times the event A occurs in n trials, and are also called relative frequencies of the event in n trials. Although our estimate about $P(A)$ may change after every trial, yet we will find that the proportion $c(s)/n$ tends to cluster around a unique central value as the number of trials n becomes even larger. This unique central value (also called probability of event A) is defined as

$$P(A) = \lim_{n \rightarrow \infty} \left\{ \frac{c(s)}{n} \right\}$$

where $c(s)$ represents the number of times that an event s occurs in n trials of an experiment.

Since the probability of an event is determined objectively by repetitive empirical observations of experimental outcomes, it is also known as *empirical probability*. Few situations to which this approach can be applied are follows:

- (i) Buying lottery tickets regularly and observing how often you win.
- (ii) Commuting to work daily and observing whether or not a certain traffic signal is red when you cross it.
- (iii) Observing births and noting how often the baby is a female
- (iv) Surveying many adults and determining what proportion smokes.

Subjective Approach

The **subjective approach** of calculating probability is always based on the degree of beliefs, convictions and experience concerning the likelihood of occurrence of a random event. It is thus a way to quantify an individual's beliefs, assessment and judgment about a random phenomenon. Probability assigned for the occurrence of an event may be based on just guess or on having some idea about the relative frequency of past occurrences of the event. This approach must be used when either sufficient data are not available or sources of information giving different results are not known.

6.4 FUNDAMENTAL RULES OF PROBABILITY

No matter which approach is used to define probability, the following fundamental rules must be satisfied. Let S be the sample space of an experiment that is partitioned into mutually exclusive and exhaustive events $A_1, A_2, \dots, A_n$ which may be elementary or compound. The probability of any event A in S is governed by the following rules:

- (i) Each probability should fall between 0 and 1, i.e. $0 \leq P(A_i) \leq 1$, for all i , where $P(A_i)$ is read as 'probability of event A_i '. In other words, the probability of an event is restricted to the range *zero to one* inclusive, where zero represents an impossible event and one represents a certain event.
For example, probability of the number seven occurring, on rolling a dice, $P(7) = 0$, because this number is an impossible event for this experiment.
- (ii) $P(S) = P(A_1) + P(A_2) + \dots + P(A_n) = 1$, where $P(S)$ is read as 'probability of the certain event'. This rule states that the sum of probabilities of all simple events constituting the sample space is equal to one. This also implies that if a random experiment is conducted, one of its outcomes in its sample space is certain to occur.
Similarly, the probability of an impossible event or an empty set is zero. That is $P(\Phi) = 0$.
- (iii) If events A_1 and A_2 are two elements in S and if occurrence of A_1 implies that A_2 occurs, that is, if A_1 is a subset of A_2 , then the probability of A_1 is less than or equal to the probability of A_2 . That is, $P(A_1) \leq P(A_2)$.
- (iv) $P(\bar{A}) = 1 - P(A)$, that is, the probability of an event that does not occur is equal to one minus the probability of the event that does occur (the probability rule for complementary events).

Glossary of Probability Terms

If A and B are two events, then

$A \cup B$ = an event which represents the occurrence of either A or B or both.

$A \cap B$ = an event which represents the simultaneous occurrence of A and B.

$\bar{A}$ = complement of event A and represents non-occurrence of A.

$\bar{A} \cap \bar{B}$ = both A and B do not occur.

$\bar{A} \cap B$ = event A does not occur but event B occurs.

$A \cap \bar{B}$ = event A occurs but event B does not occur.

$(A \cap \bar{B}) \cup (\bar{A} \cap B)$ = exactly one of the two events A and B occurs.

6.5 COUNTING RULES FOR DETERMINING THE NUMBER OF OUTCOMES

In order to assign probabilities to experimental outcomes, it is first necessary to identify and then count them. Following are three important rules for counting the experimental outcomes.

Multistep Experiments

The counting rule for multistep experiments helps us to determine the number of experimental outcomes without listing them. The rule is defined as:

If an experiment is performed in k stages with n_1 ways to accomplish the first stage, n_2 ways to accomplish the second stage ... and n_k ways to accomplish the kth stage, then the number of ways to accomplish the experiment is $n_1 n_2 \dots n_k$.

Illustrations

1. Tossing of two coins can be thought of as a two-step experiment in which each coin can land in one of two ways: head (H) and tail (T). Since the experiment involves two steps, forming the pair of faces (H or T), the total number of simple events in S will be

$$S = \{(H, H), (H, T), (T, H), (T, T)\}$$

The elements of S indicate that there are $2 \times 2 = 4$ possible outcomes.

When the number of alternative events in each of the several trials is same, that is, $n_1 = n_2 = \dots = n_k$, then the multi step method gives $n_1 n_2 \dots n_k = n^k$.

For example, if 4 coins are tossed simultaneously, then the number of experimental outcomes will be $2 \times 2 \times 2 \times 2 = 2^4 = 16$.

2. Suppose a person can take three routes from city A to city B, four from city B to city C and three from city C to city D. Then the possible routes for reaching from city A to D, while he must travel from A to B to C to D are : $(A \text{ to } B) \times (B \text{ to } C) \times (C \text{ to } D) = 3 \times 4 \times 3 = 36$ ways.

Combinations

Sometimes the ordering or arrangement of objects is not important, but only the objects that are chosen. For example, (i) you may not care in what order the books are placed on the shelf,

but only which books you are able to shelve. (ii) When a five-person committee is chosen from a group of 10 students, the order of choice is not important because all 5 students will be equal the members of committee.

This counting rule for combinations allows us to select r (say) number of outcomes from a collection of n distinct outcomes without caring in what order they are arranged. This rule is denoted by

$$C(n, r) = {}^nC_r = \frac{n!}{r!(n-r)!}$$

where $n! = n(n-1)(n-2) \dots 3 \times 2 \times 1$ and $0! = 1$.

The notation $!$ means *factorial*, for example, $4! = 4 \times 3 \times 2 \times 1 = 24$.

Important Results

1. ${}^nC_r = {}^nC_{n-r}$ and ${}^nC_n = 1$.
2. If n objects consist of all n_1 of one type, all n_2 of another type, and so on upto n_k of the k th type, then the total number of selections that can be made of 1, 2, 3 upto n objects is $(n_1 + 1)(n_2 + 1) \dots (n_k + 1) - 1$.
3. The total number of selections from n objects all different is $2^n - 1$.

Permutations

This rule of counting involves ordering or permutations. This rule helps us to compute the number of ways in which n distinct objects can be arranged, taking r of them at a time.

The total number of permutations of n objects taken r at a time is given by

$$P(n, r) = {}^nP_r = \frac{n!}{(n-r)!}$$

By permuting each combination of r objects among themselves, we shall obtain all possible permutations of n objects, r at a time. Each combination gives rise to $r!$ permutations, so that $r! C(n, r) = P(n, r) = n!/(n-r)!$.

Example 6.1: Of 10 electric bulbs, three are defective but it is not known which are defective. In how many ways can three bulbs be selected? How many of these selections will include at least one defective bulb?

Solution: Three bulbs out of 10 bulbs can be selected in ${}^{10}C_3 = 120$ ways. The number of selections which include exactly one defective bulb will be ${}^7C_2 \times {}^3C_1 = 63$.

Similarly, the number of selections which include exactly two and three defective bulbs will be ${}^7C_1 \times {}^3C_2 = 21$ and ${}^3C_3 = 1$, respectively. Thus, the total number of selections including at least one defective bulb is $63 + 21 + 1 = 85$.

Example 6.2: A bag contains 6 red and 8 green balls.

- (a) If one ball is drawn at random, then what is the probability of the ball being green?
- (b) If two balls are drawn at random, then what is the probability that one is red and the other green?

Solution: (a) Since the bag contains 6 red and 8 green balls, therefore it contains $6 + 8 = 14$ equally likely outcomes, that is, $S = \{r, g\}$. But one ball out of 14 balls can be drawn in ways, that is,

$${}^{14}C_1 = \frac{14!}{1!(14-1)!} = 14 \text{ ways}$$

Let A be the event of drawing a green ball. Then, out of these 8 green balls, one green ball can be drawn in ways:

$${}^8C_1 = \frac{8!}{1!(8-1)!} = 8$$

Hence,
$$P(A) = \frac{c(A)}{c(S)} = \frac{8}{14}$$

(b) All exhaustive number of cases, $c(S) = {}^{14}C_2 = \frac{14!}{2!(14-2)!} = 91$.

Also, out of 6 red balls, one red ball can be drawn in 6C_1 ways and out of 8 green balls, one green ball can be drawn in 8C_1 ways. Thus, the total number of favourable cases is

$$c(B) = {}^6C_1 \times {}^8C_1 = 6 \times 8 = 48.$$

Thus,
$$P(B) = \frac{c(B)}{c(S)} = \frac{48}{91}.$$

Example 6.3: A bag contains 2 white balls and 3 black balls. Four persons A, B, C and D in the order named, each draw one ball and do not replace it. The first to draw a white ball receives ₹20. Determine their expectation. [Delhi Univ., B.A. Eco. (Hons.), 1995]

Solution: The probability of drawing a white ball from the bag in the order of their name is defined as

$$P(A \text{ wins}) = P(A \text{ draw white ball in first draw}) = 2/5 = 0.4$$

$$P(B \text{ wins}) = P(A \text{ fails and } B \text{ draw white ball})$$

$$= \left(1 - \frac{2}{5}\right) \left(\frac{2}{4}\right) = \frac{3}{5} \times \frac{2}{4} = 0.3$$

$$P(C \text{ wins}) = P(A \text{ and } B \text{ both fails in first two draws and } C \text{ gets the white ball})$$

$$= \left(1 - \frac{2}{5}\right) \times \left(1 - \frac{2}{4}\right) \times \frac{2}{3} = 0.2$$

$$P(D \text{ wins}) = P(A, B, C \text{ all fails in first three draws and } D \text{ gets the white ball})$$

$$= \left(1 - \frac{2}{5}\right) \times \left(1 - \frac{2}{4}\right) \times \left(1 - \frac{2}{3}\right) \times \left(\frac{2}{2}\right) = 0.1$$

Expectations of A, B, C and D are:

$$A's \text{ Expectation} = 20 \times 0.4 = ₹ 8; \quad B's \text{ Expectation} = 20 \times 0.3 = ₹ 6$$

$$C's \text{ Expectation} = 20 \times 0.2 = ₹ 4; \quad D's \text{ Expectation} = 20 \times 0.1 = ₹ 2$$

Example 6.4: If n biscuits are distributed among N beggars, find the probability a particular beggar receives $r (< n)$ biscuits.

Solution: n biscuits can be distributed among N beggars in $(N)^n$ ways. Moreover, r biscuits can be given to any particular beggar in nC_r ways. The remaining $(n - r)$ biscuits will be distributed among the remaining $(N - 1)$ beggars in $(N - 1)^{n-r}$ ways. Thus, the total number of favourable ways are: ${}^nC_r(N - 1)^{n-r}$

Hence required probability is
$$\frac{{}^nC_r(N - 1)^{n-r}}{(N)^n}.$$

Example 6.5: Tickets are numbered from 1 to 100. They are well shuffled and a ticket is drawn at random. What is the probability that the drawn ticket has

- (a) an even number? (b) number 5 or a multiple of 5?
(c) a number which is greater than 75? (d) a number which is a square?

Solution: Since any of the 100 tickets can be drawn, therefore exhaustive number of cases are $c(S) = 100$.

- (a) Let A be the event of getting an even numbered tickets. Then, $c(A) = 50$, and hence

$$P(A) = 50/100 = 1/2$$

- (b) Let B be the event of getting a ticket bearing the number 5 or a multiple of 5, that is,

$$B = [5, 10, 15, 20, \dots, 95, 100]$$

which are 20 in number, $c(B) = 20$. Thus, $P(B) = 20/100 = 1/5$.

- (c) Let C be the event of getting a ticket bearing a number greater than 75, that is,

$$C = \{76, 77, \dots, 100\}$$

which are 25 in number, $c(C) = 25$. Thus, $P(C) = 25/100 = 1/4$.

- (d) Let D be the event of getting a ticket bearing a number which is a square, that is,

$$D = \{1, 4, 9, 16, 25, 36, 49, 64, 81, 100\}$$

which are 10 in number, $c(D) = 10$. Thus, $P(D) = 10/100 = 1/10$.

Conceptual Questions 6A

- (a) Discuss the different schools of thought on the interpretation of probability. How does each school define probability?

(b) Describe briefly the various schools of thought on probability. Discuss its importance in business decision making.

(c) Examine critically the different schools of thought on probability.
- Explain what you understand by the term probability. Discuss its importance in business decision making.
- (a) Give the classical and statistical definitions of probability and state the relationship, if any, between the two definitions.

(b) Critically examine the *a priori* definition of probability showing clearly the improvement which the empirical version of probability makes over it.
- Explain the meaning of each of the following terms:
 - Random phenomenon
 - Statistical experiment
 - Random event
 - Sample space
- What do you mean by probability? Explain the importance of probability.
- Distinguish between the two concepts in each of the following pairs:
 - Elementary event and compound events
 - Mutually exclusive events and overlapping events
 - Sample space and sample point

Self-practice Problems 6A

- 6.1** Three unbiased coins are tossed. What is the probability of obtaining:
 (a) all heads (b) two heads
 (c) one head (d) at least one head
 (e) at least two heads
 (f) all tails
- 6.2** A card is drawn from a well-shuffled deck of 52 cards. Find the probability of drawing a card which is neither a heart nor a king.
- 6.3** In a single throw of two dice, find the probability of getting (a) a total of 11, (b) a total of 8 or 11, and (c) same number on both the dice.
- 6.4** Five men in a company of 20 are graduates. If 3 men are picked out of the 20 at random, what is the probability that they are all graduates? What is the probability of at least one graduate?
- 6.5** A bag contains 25 balls numbered 1 through 25. Suppose an odd number is considered a 'success'. Two balls are drawn from the bag with replacement. Find the probability of getting
 (a) two successes
 (b) exactly one success
 (c) at least one success
 (d) no successes
- 6.6** A bag contains 5 white and 8 red balls. Two drawings of 3 balls are made such that (a) the balls are replaced before the second trial, and (b) the balls are not replaced before the second trial. Find the probability that the first drawing will give 3 white and the second, 3 red balls in each case.
- 6.7** Three groups of workers contain 3 men and one woman, 2 men and 2 women, and 1 man and 3 women respectively. One worker is selected at random from each group. What is the probability that the group selected consists of 1 man and 2 women?
- 6.8** What is the probability that a leap year, selected at random, will contain 53 Sundays?
- 6.9** A university has to select an examiner from a list of 50 persons, 20 of them women and 30 men, 10 of them knowing Hindi and 40 not, 15 of them being teachers and the remaining 35 not. What is the probability of the university selecting a Hindi-knowing woman teacher?

Hints and Answers

- 6.1** (a) $P(\text{all heads}) = 1/8$
 (b) $P(\text{two heads}) = 3/8$
 (c) $P(\text{one head}) = 3/8$
 (d) $P(\text{at least one head}) = 7/8$
 (e) $P(\text{at least two heads}) = 4/8 = 1/2$
 (f) $P(\text{all tails}) = 1/8$
- 6.2** $P(\text{neither a heart nor a king})$

$$= \frac{{}^{36}C_1}{{}^{52}C_1} = \frac{36}{52}$$
- 6.3** $c(S) = 36$; $P(\text{total of 11}) = 2/36$
 $P(\text{total of 9 or 11}) = 7/36$

$$\begin{aligned} \text{6.4 } P(\text{all graduate}) &= \frac{{}^5C_3 \times {}^{15}C_0}{{}^{20}C_3} \\ &= \frac{10 \times 1}{1140} = \frac{1}{114} \\ P(\text{no graduate}) &= \frac{{}^{15}C_3 \times {}^5C_0}{{}^{20}C_2} \\ &= \frac{455 \times 1}{1140} = \frac{91}{228} \\ P(\text{at least one graduate}) &= 1 - \frac{91}{228} \\ &= \frac{137}{228} \end{aligned}$$

- 6.5 (a) $P(\text{two successes}) = \frac{13}{25} \times \frac{13}{25} = \frac{169}{625}$
- (b) $P(\text{exactly one success})$
 $= \frac{13}{25} \times \frac{12}{25} + \frac{13}{25} \times \frac{12}{25} = \frac{312}{625}$
- (c) $P(\text{at least one success})$
 $= P(\text{exactly one success})$
 $+ P(\text{two successes})$
 $= \frac{312}{625} + \frac{169}{625} = \frac{481}{625}$
- (d) $P(\text{no successes}) = \frac{12}{25} \times \frac{12}{25} = \frac{144}{625}$

6.6 (a) *When balls are replaced:*

Total number of balls in the bag = 5 + 8
 $= 13$.

- 3 balls can be drawn from 13 in ${}^{13}C_3$ ways;
- 3 white balls can be drawn from 5 in 5C_3 ways;
- 3 red balls can be drawn from 8 in 8C_3 ways.
- The probability of 3 red balls in the second trial $= \frac{{}^5C_3}{{}^{13}C_3} = \frac{5}{143}$

- Probability of 3 red balls in the second trial $= \frac{{}^4C_2}{{}^{12}C_2} = \frac{28}{143}$

- The probability of the compound event

$$\frac{5}{143} \times \frac{28}{143} = \frac{140}{20449} = 0.007$$

(b) *When balls are not replaced:*

At the first trial, 3 white balls can be drawn in 5C_3 ways.

- The probability of drawing three white balls at the first trial $= \frac{{}^5C_3}{{}^{13}C_3} = \frac{5}{143}$
- When the white balls have been drawn and not replaced, the bag contains 2 white and 8 red balls. Therefore, at the second trial, 3 balls can be drawn from 10 in ${}^{10}C_3$ ways and 3 red balls can be drawn from 8 in 8C_3 ways.

- The probability of 3 red balls in the second trial

$$= \frac{{}^8C_3}{{}^{10}C_3} = \frac{7}{15}$$

- The probability of the compound event

$$= \frac{5}{142} \times \frac{7}{15} = \frac{7}{426} = 0.016.$$

6.7 There are three possibilities in this case:

- Man is selected from the 1st group and women from 2nd and 3rd groups; or
- Man is selected from the 2nd group and women from the 1st and 3rd groups; or
- Man is selected from the 3rd group and women from 1st and 2nd groups.

The probability of selecting a group of 1 man and 2 women is:

$$\left(\frac{3}{4} \times \frac{2}{4} \times \frac{3}{4} \right) + \left(\frac{2}{4} \times \frac{1}{4} \times \frac{3}{4} \right) + \left(\frac{1}{4} \times \frac{1}{4} \times \frac{2}{4} \right) = \frac{9}{32} + \frac{3}{32} + \frac{1}{32} = \frac{13}{32}.$$

6.8 A leap year consists of 366 days, therefore it contains 52 complete weeks and 2 extra days. These 2 days may make the following 7 combinations:

- Monday and Tuesday
- Tuesday and Wednesday
- Wednesday and Thursday
- Thursday and Friday
- Friday and Saturday
- Saturday and Sunday
- Sunday and Monday

Of these seven equally likely cases, only the last two are favourable. Hence the required probability is $2/7$.

6.9 Probability of selecting a woman = $20/50$;
 Probability of selecting a teacher = $15/50$
 Probability of selecting a Hindi-knowing candidate = $10/50$

Since the events are independent, the probability of the university selecting a Hindi-knowing woman teacher = $(20/50) \times (15/50) \times (10/50) = 3/125$.

6.6 RULES OF PROBABILITY AND ALGEBRA OF EVENTS

In probability, we use set theory notations to simplify the presentation of ideas. As discussed earlier in this chapter, the probability of the occurrence of an event A is expressed as:

$$P(A) = \text{probability of event A occurrence}$$

Such single probabilities are called **marginal (or unconditional) probabilities** because it is the probability of a single event occurring. In the coin tossing example, the marginal probability of a tail or head in a toss can be stated as $P(T)$ or $P(H)$.

6.6.1 Rules of Addition

The addition rules are helpful when we have two events and are interested in knowing the probability that at least one of the events occurs.

Mutually Exclusive Events The rule of addition for mutually exclusive (disjoint), exhaustive and equally likely events states that

If two events A and B are mutually exclusive, exhaustive, and equiprobable, then the probability of either event A or B or both occurring is equal to the sum of their individual probabilities.

This rule is expressed in the following formula

$$\begin{aligned} P(A \text{ or } B) &= P(A \cup B) = \frac{n(A \cup B)}{n(S)} = \frac{n(A) + n(B)}{n(S)} \\ &= \frac{n(A)}{n(S)} + \frac{n(B)}{n(S)} = P(A) + P(B) \end{aligned} \quad (6-1)$$

where $A \cup B$ (read as 'A union B') denotes the union of two events A and B, and it is the set of all sample points belonging to A or B or both. This rule can also be illustrated by the **Venn diagram** shown in Fig. 6.1. Here two circles contain all the sample points in events A and B. The overlap of the circles indicates that some sample points are contained in both A and B.

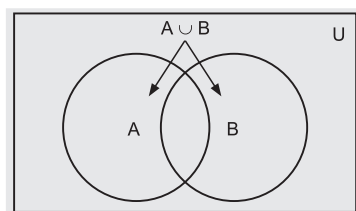


Figure 6.1 Union of Two Events

Illustration: Consider the pattern of arrival of customers at a service counter during the first hour it is open along with its probability:

No. of persons	:	0	1	2	3	4 or more
Probability	:	0.1	0.2	0.3	0.3	0.1

To understand the probability that either 2 or 3 persons will be there during the first hour, we have

$$P(2 \text{ or } 3) = P(2) + P(3) = 0.3 + 0.3 = 0.6$$

In particular, if there are n events in a sample space that are mutually exclusive, then the probability of the union of these events is given by

$$P(A_1 \cup A_2 \cup \dots \cup A_n) = P(A_1) + P(A_2) + \dots + P(A_n) \quad (6-2)$$

For example, if we are interested in knowing the probability that there will be two or more persons during the first hour, then using formula (6-2), we have

$$\begin{aligned} P(2 \text{ or more}) &= P(2, 3, 4 \text{ or more}) = P(2) + P(3) + P(4) \\ &= 0.3 + 0.3 + 0.1 = 0.7 \end{aligned}$$

Let A be any event and $\bar{A}$ be the *complement of A*. Obviously A and $\bar{A}$ are mutually exclusive and exhaustive events. Thus, either A occurs or it does not, is given by

$$P(A \text{ or } \bar{A}) = P(A) + P(\bar{A}) = P(A) + \{1 - P(A)\} = 1$$

$$\text{or} \quad P(A) = 1 - P(\bar{A}) \quad (6-3)$$

For example, if a dice is rolled, then the probability whether an odd number of spots occurs or does not.

Partially Overlapping (or Joint) Events If events A and B are not mutually exclusive, it is possible for both events to occur simultaneously? This means these events have some sample points in common. Such events are also called *joint* (or *overlapping*) *events*. The sample points in common (belong to both events) represent the joint event $A \cap B$ (read as: A intersection B). The addition rule in this case is stated as:

If two events A and B are not mutually exclusive, then the probability of either A or B or both occurring is equal to the sum of their individual probabilities minus the probability of A and B occurring together.

This rule is expressed in the following formula:

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

$$\text{or} \quad P(A \cup B) = P(A) + P(B) - P(A \cap B) \quad (6-4)$$

This addition rule can also be illustrated by the Venn diagram shown in Fig. 6.2.

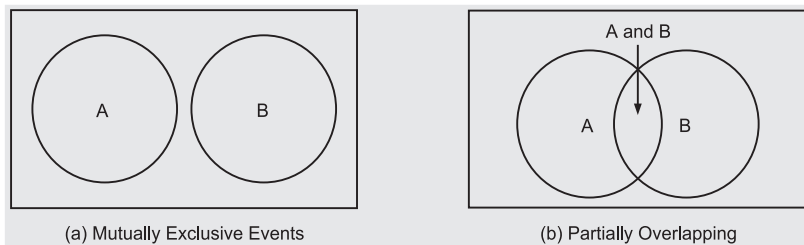


Figure 6.2 Partially Overlapping Events

Illustration: Suppose 70 per cent of all tourists who come to India will visit Agra while 60 per cent will visit Goa and 50 per cent of them will visit both Agra and Goa. The probability that a tourist will visit either Goa or Agra or both is obtained by applying formula (6-4) as follows:

$$\begin{aligned} P(\text{Agra or Goa}) &= P(\text{Agra}) + P(\text{Goa}) - P(\text{both Agra and Goa}) \\ &= 0.70 + 0.60 - 0.50 = 0.8 \end{aligned}$$

Consequently, the probability that a tourist will visit neither Agra nor Goa is calculated by

$$P(\text{neither Agra nor Goa}) = 1 - P(\text{Agra or Goa}) = 1 - 0.80 = 0.20$$

In particular, if there are three events that are not mutually exclusive, then the probability of the union of these events is given by

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C) \quad (6-5)$$

Remark: The rules of addition are applicable for calculating probability of events in case of simultaneous trials.

Example 6.6: What is the probability that a randomly chosen card from a deck of cards will either be a king or a heart.

Solution: Let event A and B be the king and heart in a deck of 52 cards, respectively. Then, it is given that

Card	Probability	Reason
King	$P(A) = 4/52$	4 kings in a deck of 52 cards
Heart	$P(B) = 13/52$	13 hearts in a deck of 52 cards
King of heart	$P(A \text{ and } B) = 1/52$	1 King of heart in a deck of 52 cards

Using the formula (6-4), we get

$$\begin{aligned} P(A \text{ or } B) &= P(A) + P(B) - P(A \text{ and } B) \\ &= \frac{4}{52} + \frac{13}{52} - \frac{1}{52} = \frac{16}{52} = 0.3077 \end{aligned}$$

Example 6.7: Of 1000 assembled components, 10 have a working defect and 20 have a structural defect. There is a good reason to assume that no component has both defects. What is the probability that randomly chosen component will have either type of defect?

Solution: Let the event A and B be the component which has working defect and has structural defect, respectively. Then it is given that

$$P(A) = 10/1000 = 0.01, P(B) = 20/1000 = 0.02 \text{ and } P(A \text{ and } B) = 0$$

The probability that a randomly chosen component will have either type of defect is given by

$$\begin{aligned} P(A \text{ or } B) &= P(A) + P(B) - P(A \text{ and } B) \\ &= 0.01 + 0.02 - 0.0 = 0.03 \end{aligned}$$

Example 6.8: A survey of 200 retail grocery shops revealed following monthly income pattern:

Monthly Income (₹)	Number of Shops
Under ₹ 20,000	102
20,000 to 30,000	61
30,000 and above	37

- What is the probability that a particular shop has monthly income under ₹ 20,000
- What is the probability that a shop selected at random has either an income between ₹ 20,000 and ₹ 30,000 or an income of ₹ 30,000 and more?

Solution: Let the events A, B and C represent the income under three categories, respectively.

(a) Probability that a particular shop has monthly income under ₹ 20,000 is $P(A) = 102/200 = 0.51$.

(b) Probability that shop selected at random has income between ₹ 20,000 and ₹ 30,000 or ₹30,000 and more is given by

$$P(A \text{ or } B) = P(A) + P(B) = \frac{61}{200} + \frac{37}{200} = 0.305 + 0.185 = 0.49$$

Example 6.9: From a sales force of 150 persons, one will be selected to attend a special sales meeting. If 52 of them are unmarried, 72 are college graduates, and $3/4$ of the 52 that are unmarried also college graduates. Find the probability that the sales person selected at random will be neither single nor a college graduate.

Solution: Let A and B be the events that a sales person selected is married and that he is a college graduate, respectively. Then, it is given that

$$P(A) = 52/150, P(B) = 72/150; P(A \text{ and } B) = (3/4) (52/150) = 39/150$$

The probability that a salesperson selected at random will neither be single nor a college graduate is

$$\begin{aligned} P(\bar{A} \cap \bar{B}) &= 1 - P(A \cup B) = 1 - \{P(A) + P(B) - P(A \cap B)\} \\ &= 1 - \left\{ \frac{52}{150} + \frac{72}{150} - \frac{39}{150} \right\} = \frac{13}{30} \end{aligned}$$

Example 6.10: The probability that X passed in maths is $2/3$. The probability that he passed in statistics is $4/9$. The probability that he passed in atleast one subject is $4/5$. What is the probability that he passed in both subjects ?
[GJ Univ., BBA, 2004]

Solution: Let A be the event that Mr. X passed in maths, i.e. $P(A) = 2/3$ and B be the event that Mr. X passed in statistics, i.e. $P(B) = 4/9$. Then $P(A \cup B) = 4/5$. The probability that he passed in at least one subject is given by

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\ \frac{4}{5} &= \frac{2}{3} + \frac{4}{9} - P(A \cap B) \\ P(A \cap B) &= \frac{2}{3} + \frac{4}{9} - \frac{4}{5} = \frac{30 + 20 - 36}{45} = \frac{14}{45} \end{aligned}$$

Example 6.11: The probability that a contractor will get a plumbing contract is $2/3$ and the probability that he will not get an electrical contract is $5/9$. If the probability of getting at least one contract is $4/5$, what is the probability that he will get both?

Solution: Let A and B denote the events that the contractor will get a plumbing and electrical contract, respectively. Given that

$$\begin{aligned} P(A) &= 2/3; P(B) = 1 - (5/9) = 4/9; P(A \cup B) = 4/5 \\ P(A \cap B) &= P(A) + P(B) - P(A \cup B) = \frac{2}{3} + \frac{4}{9} - \frac{4}{5} = \frac{14}{45} = 0.31 \end{aligned}$$

Thus, the probability that the contractor will get both the contracts is 0.31.

6.6.2 Rules of Multiplication

Statistically Independent Events When the occurrence of an event does not affect and is not affected by the probability of occurrence of any other event, the event is said to be a *statistically independent event*. There are three types of probabilities under statistical independence: *marginal*, *joint* and *conditional*.

- **Marginal Probability:** A marginal or unconditional probability is the simple probability of the occurrence of an event. For example, in a fair coin toss, the outcome of each toss is an event that is statistically independent of the outcomes of every other toss of the coin.
- **Joint Probability:** The probability of two or more independent events occurring together or in succession is called the **joint probability**. The joint probability of two or more independent events is equal to the product of their marginal probabilities. In particular, if A and B are independent events, the probability that both A and B will occur is given by

$$P(AB) = P(A \cap B) = P(A) \times P(B) \quad (6-6)$$

Suppose we toss a coin twice. The probability that in both the cases, the coin will turn up head is given by

$$P(H_1H_2) = P(H_1) \times P(H_2) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

The formula (6-6) is applied here because the probability of any outcome is not affected by any preceding outcome, in other words, outcomes are independent.

- **Conditional Probability:** It is the probability of a particular event occurring, given that another event has occurred. The **conditional probability** of event A, given that event B has already occurred, is written as: $P(A|B)$. Similarly, we may write $P(B|A)$. The vertical bar is read as 'given' and events appearing to the right of the bar are those that you know have occurred. Two events A and B are said to be independent if and only $P(A|B) = P(A)$ or $P(B|A) = P(B)$. Otherwise, events are said to be dependent.

Since, in the case of independent events the probability of occurrence of either of the events does not depend or affect the occurrence of the other, therefore in the coin tossing example, the probability of a head occurrence in the second toss, given that head resulted in the first toss, is still 0.5. That is, $P(H_2 | H_1) = 0.5 = P(H_2)$. It is because of the fact that the probabilities of heads and tails are the same for every toss and in no way influenced by whether it was a head or tail which occurred in the previous toss.

Statistically Dependent Events When the probability of an event is dependent upon or affected by the occurrence of any other event, the events are said to be **statistically dependent**. There are three types of probabilities under statistical dependence: *joint*, *conditional* and *marginal*.

- **Joint Probability:** If A and B are dependent events, then the joint probability as discussed under statistical dependence case is no longer equal to the product of their respective probabilities. That is, for dependent events

$$P(A \text{ and } B) = P(A \cap B) \neq P(A) \times P(B)$$

Accordingly, $P(A) \neq P(A | B)$ and $P(B) \neq P(B | A)$

The joint probability of events A and B occurring together or in succession under statistical dependences is given by

$$P(A \cap B) = P(A) \times P(B | A)$$

or $P(A \cap B) = P(B) \times P(A | B)$

- **Conditional Probability:** Under statistical dependence, the conditional probability of event B given that event A has already occurred, is given by

$$P(B | A) = \frac{P(A \cap B)}{P(A)}$$

Similarly, the conditional probability of A, given that event B has occurred, is

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

- **Marginal Probability:** The marginal probability of an event under statistical dependence is the same as the marginal probability of an event under statistical independence.

The marginal probability of events A and B can be written as

$$P(A) = P(A \cap B) + P(A \cap \bar{B})$$

$$\text{and} \quad P(B) = P(A \cap B) + P(\bar{A} \cap B)$$

Example 6.12: The odds against student X solving a Business Statistics problem are 8 to 6, and odds in favour of student Y solving the problem are 14 to 16.

- What is the chance that the problem will be solved if they both try independently of each other?
- What is the probability that none of them is able to solve the problem?

Solution: Let A = The event that the first student solves the problem

B = The event that the second student solves the problem

$$P(A) = \frac{6}{8 + 6} = \frac{6}{14} \quad \text{and} \quad P(B) = \frac{14}{14 + 16} = \frac{14}{30}$$

- Probability that the problem will be solved

$$\begin{aligned}
 &= P(\text{at least one of them solves the problem}) \\
 &= P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B) \\
 &= P(A) + P(B) - P(A) \times P(B) \quad (\text{because the events are independent}) \\
 &= \frac{6}{14} + \frac{14}{30} - \frac{6}{14} \times \frac{14}{30} = \frac{73}{105} = 0.695
 \end{aligned}$$

- Probability that neither A nor B solves the problem

$$\begin{aligned}
 P(\bar{A} \text{ and } \bar{B}) &= P(\bar{A}) \times P(\bar{B}) \\
 &= [1 - P(A)] \times [1 - P(B)] = \frac{8}{14} \times \frac{16}{30} = \frac{32}{105} = 0.305
 \end{aligned}$$

Example 6.13: The probability that a new marketing approach will be successful is 0.6. The probability that the expenditure for developing the approach can be kept within the original budget is 0.50. The probability that both of these objectives will be achieved is 0.30. What is the probability that at least one of these objectives will be achieved. For the two events described above, determine whether the events are independent or dependent.

Solution: Let

A = The event that the new marketing approach will be successful

B = The event that the expenditure for developing the approach can be kept within the original budget

Given that $P(A) = 0.60$, $P(B) = 0.50$ and $P(A \cap B) = 0.30$

Probability that both events A and B will be achieved is given by

$$\begin{aligned}
 P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\
 &= 0.60 + 0.50 - 0.30 = 0.80
 \end{aligned}$$

If events A and B are independent, then their joint probability is given by

$$P(A \cap B) = P(A) \times P(B) = 0.60 \times 0.50 = 0.30$$

Since this value is same as given in the problem, events are independent.

Example 6.14: A piece of equipment will function only when the three components A, B and C are working. The probability of A failing during one year is 0.15, that of B failing is 0.05, and that of C failing is 0.10. What is the probability that the equipment will fail before the end of the year?

Solution: Given that

$$P(A \text{ failing}) = 0.15; \quad P(A \text{ not failing}) = 1 - P(A) = 0.85$$

$$P(B \text{ failing}) = 0.05; \quad P(B \text{ not failing}) = 1 - P(B) = 0.95$$

$$P(C \text{ failing}) = 0.10; \quad P(C \text{ not failing}) = 1 - P(C) = 0.90$$

Since all the three events are independent, therefore the probability that the equipment will work is given by

$$P(\bar{A} \cap \bar{B} \cap \bar{C}) = P(\bar{A}) \times P(\bar{B}) \times P(\bar{C}) = 0.85 \times 0.95 \times 0.90 = 0.726$$

Probability that the equipment will fail before the end of the year is given by

$$\begin{aligned} P(A \cup B \cup C) &= 1 - P(\bar{A} \cap \bar{B} \cap \bar{C}) \\ &= 1 - P(\bar{A}) \times P(\bar{B}) \times P(\bar{C}) \\ &= 1 - \{0.85 \times 0.95 \times 0.90\} = 1 - 0.726 = 0.274 \end{aligned}$$

Example 6.15: A company has two plants to manufacture scooters. Plant I manufactures 80 per cent of the scooters and Plant II manufactures 20 per cent. In plant I, only 85 out of 100 scooters are considered to be of standard quality. In plant II, only 65 out of 100 scooters are considered to be of standard quality. What is the probability that a scooter selected at random came from plant I, if it is known that it is of standard quality?

Solution: Let A = The scooter purchased is of standard quality

B = The scooter is of standard quality and came from plant I

C = The scooter is of standard quality and came from plant II

D = The scooter came from plant I

The percentage of scooters manufactured in plant I that are of standard quality is 85 per cent of 80 per cent, that is, $0.85 \times (80/100) = 68$ per cent or $P(B) = 0.68$.

The percentage of scooters manufactured in plant II that are of standard quality is 65 per cent of 20 per cent, that is, $0.65 \times (20/100) = 13$ per cent or $P(C) = 0.13$.

The probability that a customer obtains a standard quality scooter from the company is $0.68 + 0.13 = 0.81$.

The probability that the scooters selected at random came from plant I, if it is known that it is of standard quality, is given by

$$P(D|A) = \frac{P(D \text{ and } A)}{P(A)} = \frac{0.68}{0.81} = 0.84$$

Example 6.16: A husband and wife appear in an interview for two vacancies in the same post. The probability of husband's selection is $1/7$ and that of wife's selection is $1/5$. What is the probability that

- (a) both of them will be selected,
- (b) only one of them will be selected, and
- (c) none of them will be selected.

Solution: Let A and B be the events of the husband's and wife's selection, respectively. Given that $P(A) = 1/7$ and $P(B) = 1/5$.

- (a) The probability that both of them will be selected is

$$P(A \text{ and } B) = P(A) \times P(B) = (1/7) \times (1/5) = 1/35 = 0.029$$

- (b) The probability that only one of them will be selected is

$$\begin{aligned} P[(A \text{ and } \bar{B}) \text{ or } (B \text{ and } \bar{A})] &= P(A \text{ and } \bar{B}) + P(B \text{ and } \bar{A}) \\ &= P(A) \times P(\bar{B}) + P(B) \times P(\bar{A}) \\ &= P(A) [1 - P(B)] + P(B) [1 - P(A)] \\ &= \frac{1}{7} \left(1 - \frac{1}{5}\right) + \frac{1}{5} \left(1 - \frac{1}{7}\right) = \left(\frac{1}{7} \times \frac{4}{5}\right) + \left(\frac{1}{5} \times \frac{6}{7}\right) \\ &= \frac{10}{35} = 0.286 \end{aligned}$$

- (c) The probability that none of them will be selected is

$$P(\bar{A}) \times P(\bar{B}) = (6/7) \times (4/5) = 24/35 = 0.686$$

Example 6.17: The odds that A speaks the truth is 3 : 2 and the odds that B speaks the truth is 5 : 3. In what percentage of cases are they likely to contradict each other on an identical point?

Solution: Let X and Y denote the events that A and B speak truth, respectively. Given that

$$P(X) = 3/5; P(\bar{X}) = 2/5; P(Y) = 5/8; P(\bar{Y}) = 3/8$$

The probability that A speaks the truth and B speaks a lie is $(3/5) (3/8) = 9/40$

The probability that B speaks the truth and A speaks a lie is $(5/8) (2/5) = 10/40$

So the compound probability is $\frac{9}{40} + \frac{10}{40} = \frac{19}{40}$

Hence, percentage of cases in which they contradict each other is $(19/40) \times 100 = 47.5$ per cent.

Example 6.18: A market survey was conducted in four cities to find out the preference for brand A soap. The responses are shown below:

	Delhi	Kolkata	Chennai	Mumbai
Yes	45	55	60	50
No	35	45	35	45
No opinion	5	5	5	5

- (a) What is the probability that a consumer selected at random, preferred brand A?
- (b) What is the probability that a consumer preferred brand A and was from Chennai?
- (c) What is the probability that a consumer preferred brand A, given that he was from Chennai?
- (d) Given that a consumer preferred brand A, what is the probability that he was from Mumbai?

Solution: The information from responses during market survey is as follows:

	<i>Delhi</i>	<i>Kolkata</i>	<i>Chennai</i>	<i>Mumbai</i>	<i>Total</i>
Yes	45	55	60	50	210
No	35	45	35	45	160
No opinion	5	5	5	5	20
Total	85	105	100	100	390

Let X denote the event that a consumer selected at random preferred brand A. Then

(a) The probability that a consumer selected at random preferred brand A is

$$P(X) = 210/390 = 0.5398$$

(b) The probability that a consumer preferred brand A and was from Chennai (C) is

$$P(X \cap C) = 60/390 = 0.1538$$

(c) The probability that a consumer preferred brand A, given that he was from Chennai, is

$$P(X|C) = \frac{P(A \cap C)}{P(C)} = \frac{60/390}{100/390} = \frac{0.153}{0.256} = 0.597$$

(d) The probability that the consumer belongs to Mumbai, given that he preferred brand A, is

$$P(M|X) = \frac{P(M \cap X)}{P(X)} = \frac{50/390}{210/390} = \frac{0.128}{0.538} = 0.238$$

Example 6.19: The probability that a trainee will remain with a company is 0.6. The probability that an employee earns more than ₹10,000 per month is 0.5. The probability that an employee who is a trainee remained with the company or who earns more than ₹10,000 per month is 0.7. What is the probability that an employee earns more than ₹10,000 per month given that he is a trainee who stayed with the company?

Solution: Let A and B be the events that a trainee who remained with the company and the event that an employee earns more than ₹ 10,000, respectively. Given that

$$P(A) = 0.6, P(B) = 0.5, \text{ and } P(A \text{ or } B) = P(A \cup B) = 0.7$$

The probability that an employee earns more than ₹ a 10,000, given that he is trainee who remained with the company, is given by

$$P(B|A) = \frac{P(A \text{ and } B)}{P(A)} = \frac{P(A \cap B)}{P(A)}$$

We know that

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

or

$$P(A \cap B) = P(A) + P(B) - P(A \cup B) = 0.6 + 0.5 - 0.7 = 0.4.$$

Hence, the required probability is

$$P(B|A) = \frac{(A \cap B)}{P(A)} = \frac{0.4}{0.6} = 0.667$$

Example 6.20: The personnel department of a company has records which show the following analysis of its 200 engineers.

<i>Age</i>	<i>Bachelor's Degree Only</i>	<i>Master's Degree</i>	<i>Total</i>
Under 30	90	10	100
30 to 40	20	30	50
Over 40	40	10	50
Total	150	50	200

If one engineer is selected at random from the company, find:

- The probability that he has only a bachelor's degree.
- The probability that he has a master's degree, given that he is over 40.
- The probability that he is under 30, given that he has only a bachelor's degree.

Solution: Let A, B, C and D denote the events that an engineer is under 30 years of age, 40 year of age, has bachelor's degree only, and has a master's degree, respectively. Therefore,

- The probability of an engineer who has only a bachelor's degree is $P(C) = 150/200 = 0.75$
- The probability of an engineer who has a master's degree, given that he is over 40 years, is

$$P(D|B) = \frac{P(D \cap B)}{P(B)} = \frac{10/200}{50/200} = \frac{10}{50} = 0.20$$

- The probability of an engineer who is under 30 years, given that he has only bachelor's degree, is

$$P(A|C) = \frac{P(A \cap C)}{P(C)} = \frac{90/200}{150/200} = \frac{90}{150} = 0.60$$

Self-practice Problems 6B

- Mr. X has 2 shares in a lottery in which there are 2 prizes and 5 blanks. Mr. Y has 1 share in a lottery in which there is 1 prize and 2 blanks. Show that the chance of Mr. X's success to that of Mr. Y's is 15 : 7.
- Explain whether or not each of the following claims could be correct:
 - A businessman claims that the probability that he will get contract A is 0.15 and that he will get contract B is 0.20. Furthermore, he claims that the probability of getting A or B is 0.50.
 - A market analyst claims that the probability of selling ten million rupees of plastic A or five million rupees of plastic B is 0.60. He also claims that the probability of selling ten million rupees of A and five million rupees of B is 0.45.
- The probability that an applicant for a Management Accountant's job has a postgraduate degree is 0.3, he has had some work experience as a chief Financial Accountant is 0.7, and that he has both is 0.2. Out of 300 applicants, approximately, what number would have either a postgraduate degree or some professional work experience?
- A can hit a target 3 times in 5 shots; B, 2 times in 5 shots; C, 3 times in 4 shots. They fire a volley. What is the probability that 2 shots hit?
- A problem in business statistics is given to five students, A, B, C, D and E. Their chances of solving are 1/2, 1/3, 1/4, 1/5 and 1/6, respectively. What is the probability that the problem will be solved?

- 6.15** A husband and wife appear in an interview for two vacancies for the same post. The probability of husband's selection is $1/7$ and that of wife's selection is $1/5$. What is the probability that
- only one of them will be selected?
 - both of them will be selected?
 - none of them will be selected?
- 6.16** There is 50-50 chance that a contractor's firm, A, will bid for the construction of a multi-storeyed building. Another firm, B, submits a bid and the probability is $3/5$ that it will get the job, provided that firm A does not submit a bid. If firm A submits a bid, the probability that firm B will get the job is only $2/3$. What is the probability that firm B will get the job?
- 6.17** Plant I of XYZ manufacturing organization employs 5 production and 3 maintenance foremen, plant II of same organization employs 4 production and 5 maintenance foremen. From any one of these plants, a single selection of two foremen is made. Find the probability that one of them would be a production and the other a maintenance foreman.
- 6.18** If a machine is correctly set up, it will produce 90 per cent acceptable items. If it is incorrectly setup, it will produce 40 per cent acceptable items. Past experience shows that 80 per cent of the setups are correctly done. If after a certain setup, the machine produces 2 acceptable items as the first 2 pieces, find the probability that the machine is correctly set up.
- [Delhi Univ., B.Com. (Hon.), 1998]*
- 6.19** An investment consultant predicts that the odds against the price of a certain stock going up during the next week are $2 : 1$ and odds in favour of the price remaining the same are $1 : 3$. What is the probability that the price of the stock will go down during the next week?
- 6.20** An article manufactured by a company consists of two parts A and B. In the process of manufacture of part A, 9 out of 100 are likely to be defective. Similarly, 5 out of 100 are likely to be defective in the manufacture of part B. Calculate the probability that the assembled part will not be defective.
- 6.21** A product is assembled from three components X, Y and Z, the probability of these components being defective is 0.01, 0.02 and 0.05, respectively. What is the probability that the assembled product will not be defective?
- 6.22** You note that your officer is happy in 60 per cent cases of your calls. You have also noticed that if he is happy, he accedes to your requests with a probability of 0.4, whereas if he is not happy, he accedes to your requests with a probability of 0.1. You call on him one day and he accedes to your request. What is the probability of his being happy?
- 6.23** In a colour preference experiment, eight toys are placed in a container. The toys are identical except for colour — two are red, and six are green. A child is asked to choose two toys at random. What is the probability that the child chooses the two red toys?

Hints and Answers

- 6.10** Considering Mr. X's chances of success.
- A = event that 1 share brings a prize and 1 share goes blank.
- B = event that both the shares bring prizes.
- C = event that X succeeds in getting atleast one prize = $A \cup B$.
- Since A and B are mutually exclusive, therefore

$$\begin{aligned} P(C) &= P(A \cup B) = P(A) + P(B) \\ &= \frac{{}^2C_1 \times {}^5C_1}{{}^7C_2} + \frac{{}^2C_2 \times {}^5C_0}{{}^7C_2} \end{aligned}$$

Similarly, if D denotes the event that Y succeeds in getting a prize, then we have

$$P(D) = \frac{{}^1C_1}{{}^3C_1} = \frac{1}{3}$$

X's chance of success: Y's chance of success

$$= \frac{15}{21} : \frac{1}{3} = 15 : 7.$$

6.11 (a) $P(A \cap B) = -0.15$

(b) $P(A) + P(B) = 1.05$

6.12 Let A = Applicant has P.G. degree;

B = Applicant has work experience;

Given, $P(A) = 0.3$, $P(B) = 0.7$, and

$P(A \cap B) = 0.2$. Therefore

$$300 \times P(A \cup B) = 300[P(A) + P(B) -$$

$$P(A \cap B)] = 240$$

6.13 The required event that two shots may hit the target, can happen in the following mutually exclusive cases:

(i) A and B hit and C fails to hit the target

(ii) A and C hit and B fails to hit the target

(iii) B and C hit and A fails to hit the target

Hence, the required probability that any two shots hit is given by, $P = P(i) + P(ii) + P(iii)$.

Let E_1 , E_2 , and E_3 be the event of hitting the target by A, B, and C respectively. Therefore,

$$P(i) = P(E_1 \cap E_2 \cap \bar{E}_3)$$

$$\begin{aligned} &= P(E_1) \times P(E_2) \times P(\bar{E}_3) \\ &= \left(\frac{3}{5}\right) \cdot \left(\frac{2}{5}\right) \cdot \left(1 - \frac{3}{4}\right) = \frac{6}{100} \end{aligned}$$

$$P(ii) = P(E_1 \cap \bar{E}_2 \cap E_3)$$

$$= \left(\frac{3}{5}\right) \left(1 - \frac{2}{5}\right) \left(\frac{3}{4}\right) = \frac{27}{100}$$

$$P(iii) = P(\bar{E}_1 \cap E_2 \cap E_3)$$

$$= \left(1 - \frac{3}{5}\right) \left(\frac{2}{5}\right) \left(\frac{3}{4}\right) = \frac{12}{100}$$

Since all the three events are mutually exclusive events, hence the required probability is given by

$$P(i) + P(ii) + P(iii)$$

$$= \frac{6}{100} + \frac{27}{100} + \frac{12}{100} = \frac{9}{20}$$

6.14 $P(\text{problem will be solved})$

$$= 1 - P(\text{problem is not solved})$$

$$= 1 - P(\text{all students fail to solve the problem})$$

$$= 1 - P(\bar{A} \cap \bar{B} \cap \bar{C} \cap \bar{D} \cap \bar{E})$$

$$= 1 - P(\bar{A}) P(\bar{B}) P(\bar{C}) P(\bar{D}) P(\bar{E})$$

$$= 1 - \left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{3}\right) \left(1 - \frac{1}{4}\right) \left(1 - \frac{1}{5}\right) \left(1 - \frac{1}{6}\right)$$

$$= 1 - \frac{1}{6} = \frac{5}{6}$$

6.15 $P(\text{only one of them will be selected})$

$$= P(H \cap \bar{W}) \cup (\bar{H} \cap W)$$

$$= P(H \cap \bar{W}) + P(\bar{H} \cap W)$$

$$= P(H) P(\bar{W}) + P(\bar{H}) P(W)$$

$$= \frac{1}{7} \left(1 - \frac{1}{5}\right) + \left(1 - \frac{1}{7}\right) \frac{1}{5} = \frac{2}{7}$$

(b) $P(\text{both of them will be selected})$

$$P(H \cap W) = P(H) P(W) = \frac{1}{35}$$

(c) $P(\text{none of them will be selected})$

$$P(\bar{H} \cap \bar{W}) = P(\bar{H}) P(\bar{W}) = \frac{24}{35}$$

6.16 $P(A) = 1/2$, $P(B | A) = 2/3$, and $P(B | \bar{A}) = 3/5$.

$$P(B) = P(A \cap B) + P(\bar{A} \cap B)$$

$$= P(A) P(B | A) + P(\bar{A}) P(B | \bar{A})$$

$$= \frac{1}{2} \times \frac{2}{3} + \frac{1}{2} \times \frac{3}{5} = \frac{19}{30}$$

6.17 Let E_1 and E_2 be the events that Plant I and II is selected respectively. Then, the probability of the event E that in a batch of 2, one is the production and the other is the maintenance man, is

$$P(E) = P(E_1 \cap E) + P(E_2 \cap E)$$

$$= P(E_1) P(E | E_1) + P(E_2) P(E | E_2)$$

$$= \frac{1}{2} \times \frac{{}^5C_1 \cdot {}^3C_1}{{}^8C_2} + \frac{1}{2} \times \frac{{}^4C_1 \cdot {}^5C_1}{{}^9C_2}$$

$$= \frac{1}{2} \times \frac{15}{28} + \frac{1}{2} \times \frac{5}{9} = \frac{275}{504}$$

6.18 Let A = event that the item is acceptable;

B_1 and B_2 = events that machine is correctly and incorrectly setup, respectively.

Given, $P(A|B_1) = 0.9$; $P(A|B_2) = 0.4$; $P(B_1) = 0.8$ and $P(B_2) = 0.2$. Then $P(B_1|A) = 0.9$.

- 6.19** $P(\text{price of a certain stock not going up}) = 2/3$
 $P(\text{price of a certain stock remaining same}) = 1/4$

The probability that the price of the stock will go down during the next week

$$= P(\text{price of the stock not going up and not remaining same})$$

$$= P(\text{price of the stock not going up} \times P(\text{price of the stock not remaining same}))$$

$$= \left(\frac{2}{3}\right) \times \left(1 - \frac{1}{4}\right) = \left(\frac{2}{3}\right) \times \left(\frac{3}{4}\right)$$

$$= \frac{1}{2} = 0.5$$

- 6.20** The assembled part will be defective if any of the parts is defective.

The probability of the assembled part being defective:

$$= P(\text{Any of the part is defective})$$

$$= P(A \cup B) = P(A) + P(B) - P(A) \times P(B)$$

$$= \frac{9}{100} + \frac{5}{100} - \left(\frac{9}{100}\right) \times \left(\frac{5}{100}\right) = 0.1355$$

The probability that assembled part is not defective

$$= 1 - 0.1355 = 0.8645.$$

- 6.21** Let A, B, and C denote the respective probabilities of components X, Y and Z being defective.

$$P(A) = 0.01, P(B) = 0.02, P(C) = 0.05$$

$$P(A \text{ or } B \text{ or } C)$$

$$= P(A) + P(B) + P(C) - P(AB) - P(BC) - P(AC) + P(ABC)$$

$$= 0.01 + 0.02 + 0.05 - 0.0002 - 0.0010 - 0.0005 + 0.00001$$

$$= 0.0783$$

Hence the probability that the assembled product will not be defective = $1 - 0.0783$ or 0.9217.

- 6.22** The probability that the officer is happy and accedes to requests = $0.6 \times 0.4 = 0.24$.

The probability that the officer is unhappy and accedes to requests = $0.4 \times 0.1 = 0.04$.

Total probability of acceding to requests = $0.24 + 0.04 = 0.28$.

The probability of his being happy if he accedes to requests = $0.24/0.28 = 0.875$.

- 6.23** Let R = Red toy is chosen and G = Green toy is chosen.

$$P(\text{Both toys are R})$$

$$= P(R \text{ on first choice} \cap R \text{ on second choice})$$

$$= P(R \text{ on first choice}) \times P(R \text{ on second choice} | R \text{ on first choice})$$

$$= (2/8) (1/7) = 1/28.$$

6.7 BAYES' THEOREM

The **Bayes' theorem** is useful in revising the original probability estimates of known outcomes as we gain additional information about these outcomes. The prior probabilities, when changed in the light of new information, are called *revised* or *posterior probabilities*.

Suppose $A_1, A_2, \dots, A_n$ represent n mutually exclusive and collectively exhaustive events with prior marginal probabilities $P(A_1), P(A_2), \dots, P(A_n)$. Let B be an arbitrary event with $P(B) \neq 0$ for which conditional probabilities $P(B|A_1), P(B|A_2), \dots, P(B|A_n)$ are also known. Given the information that outcome B has occurred, the revised (or posterior) probabilities $P(A_i|B)$ are determined with the help of Bayes' theorem using the formula:

$$P(A_i|B) = \frac{P(A_i \cap B)}{P(B)} \quad (6-8)$$

where the posterior probability of events A_i given event B is the conditional probability $P(A_i|B)$.

Since events $A_1, A_2, \dots, A_n$ are mutually exclusive and collectively exhaustive, the event B is bound to occur with either $A_1, A_2, \dots, A_n$. That is,

$$B = (A_1 \cap B) \cup (A_2 \cap B) \cup \dots \cup (A_n \cap B)$$

where the **posterior probability** of A_i given B is the conditional probability $P(A_i|B)$.

Since $(A_1 \cap B), (A_2 \cap B) \dots (A_n \cap B)$ are mutually exclusive, we get

$$\begin{aligned} P(B) &= P(A_1 \cap B) + P(A_2 \cap B) + \dots + P(A_n \cap B) = \sum_{i=1}^n P(A_i \cap B) \\ &= P(A_1) P(B|A_1) + P(A_2) P(B|A_2) + \dots + P(A_n) P(B|A_n) \\ &= \sum_{i=1}^n P(A_i) P(B|A_i) \end{aligned}$$

From formula (6-8) for a fixed i , we have

$$\begin{aligned} P(A_i | B) &= \frac{P(A_i \cap B)}{P(B)} = \frac{P(B|A_i) P(A_i)}{P(B)} \\ &= \frac{P(B|A_i) P(A_i)}{P(A_1) P(B|A_1) + P(A_2) P(B|A_2) + \dots + P(A_n) P(B|A_n)} \end{aligned}$$

Example 6.21: Suppose an item is manufactured by three machines X, Y and Z. All the three machines have equal capacity and are operated at the same rate. It is known that the percentages of defective items produced by X, Y and Z are 2, 7 and 12 per cent, respectively. All the items produced by X, Y and Z are put into one bin. From this bin, one item is drawn at random and is found to be defective. What is the probability that this item was produced on Y?

Solution: Let A be the defective item. We know the prior probability of defective items produced on X, Y and Z, that is, $P(X) = 1/3$; $P(Y) = 1/3$ and $P(Z) = 1/3$. We also know that

$$P(A|X) = 0.02, P(A|Y) = 0.07, P(A|Z) = 0.12$$

Now having known that the item drawn is defective, we want to know the probability that it was produced by Y. That is

$$\begin{aligned} P(Y|A) &= \frac{P(A|Y) P(Y)}{P(X) P(A|X) + P(Y) P(A|Y) + P(Z) P(A|Z)} \\ &= \frac{(0.07) (1/3)}{(1/3) (0.02) + (1/3) (0.07) + (1/3) (0.12)} = 0.33 \end{aligned}$$

Example 6.22: Assume that a factory has two machines. Past records show that machine 1 produces 30 per cent of the items of output and machine 2 produces 70 per cent of the items. Further, 5 per cent of the items produced by machine 1 were defective and only 1 per cent produced by machine 2 were defective. If a defective item is drawn at random, what is the probability that the defective item was produced by machine 1 or machine 2?

Solution: Let A_1 = Event of drawing an item produced by machine 1
 A_2 = Event of drawing an item produced by machine 2
 and D = Event of drawing a defective item produced either by machine 1 or machine 2

From the data in the problem, we know that

$$P(A_1) = 0.30, \quad P(A_2) = 0.70; \quad P(D | A_1) = 0.05, \quad P(D | A_2) = 0.1$$

The data of the problem can now be summarized as under:

Event	Prior Probability $P(A_i)$	Conditional Probability Event $P(D A_i)$	Joint Probability $P(A_i \text{ and } D)$	Posterior (revised) Probability $P(A_i/D) = \frac{P(A_i \text{ and } D)}{P(D)}$
(1)	(2)	(3)	(2) $\times$ (3)	
A_1	0.30	0.05	0.015	$0.015/0.022 = 0.682$
A_2	0.70	0.01	0.007	$0.007/0.022 = 0.318$

Here
$$P(D) = \sum_{i=1}^2 P(D | A_i) P(A_i) = 0.05 \times 0.30 + 0.01 \times 0.70 = 0.022$$

From the above table, the probability that the defective item was produced by machine 1 is 0.682 or 68.2 per cent and that by machine 2 is only 0.318 or 31.8 per cent. We may now say that the defective item is more likely drawn from the output produced by machine 1.

Example 6.23: A company uses a 'selling aptitude test' in the selection of salesmen. Past experience has shown that only 70 per cent of all persons applying for a sales position achieved a classification 'dissatisfactory' in actual selling, whereas the remainder were classified as 'satisfactory', 85 per cent had scored a passing grade in the aptitude test. Only 25 per cent of those classified dissatisfactory had passed the test on the basis of this information. What is the probability that a candidate would be a satisfactory salesman given that he passed the aptitude test?

Solution: Let A and B be the event representing 'unsatisfactory' classification as a salesman and 'passing the test', respectively. Now, the probability that a candidate would be 'satisfactory' salesman given that he passed the aptitude test is

$$P(A|B) = \frac{(0.70)(0.85)}{(0.70)(0.85) + (0.30)(0.25)} = \frac{0.595}{0.595 + 0.075} = 0.888$$

Assuming no change in the type of candidates applying for the selling positions, the probability that a random applicant would be satisfactory is 70 per cent. On the other hand, if the company only accepts an applicant if he passed the test, the probability increases to 88.8 per cent.

Example 6.24: In a bolt factory, machines A, B and C manufacture 25, 35 and 40 per cent of the total output, respectively. Of the total of their output, 5, 4 and 2 per cent are defective bolts. A bolt is drawn at random and is found to be defective. What is the probability that it was manufactured by machines A, B or C?

Solution: Let, A_i ($i = 1, 2, 3$) be the event of drawing a bolt produced by machine A, B and C, respectively. From the data we know that

$$P(A_1) = 0.25, \quad P(A_2) = 0.35 \text{ and } P(A_3) = 0.40$$

From the additional information, we know that

$$B = \text{the event of drawing a defective bolt}$$

Thus, $P(B|A_1) = 0.05$; $P(B|A_2) = 0.04$ and $P(B|A_3) = 0.02$

The table of posterior probabilities can be prepared as under:

<i>Event</i>	<i>Prior Probability</i> $P(A_i)$	<i>Conditional Probability</i> $P(B A_i)$	<i>Joint Probability</i>	<i>Posterior Probability</i>
(1)	(2)	(3)	(4) = (2) $\times$ (3)	(5) = (4)/P(B)
A_1	0.25	0.05	0.0125	$0.0125 \div 0.0345 = 0.362$
A_2	0.35	0.04	0.0140	$0.014 \div 0.0345 = 0.406$
A_3	0.40	0.02	0.0080	$0.008 \div 0.0345 = 0.232$
	1.00		0.0345	1.000

The above table shows the probability that the item was defective and produced by machine A is 0.362, machine B is 0.406 and machine C is 0.232.

Self-practice Problems 6C

- 6.24** In a post office, three clerks are assigned to process incoming mail. The first clerk, A, processes 40 per cent; the second clerk, B, processes 35 per cent; and the third clerk, C, processes 25 per cent of the mail. The first clerk has an error rate of 0.04, the second has an error rate of 0.06 and the third has an error rate of 0.03. A mail selected at random from a day's output is found to have an error. The postmaster wishes to know the probability that it was processed by clerk A or clerk B or clerk C.
- 6.25** A certain production process produces items 10 per cent of which defective. Each item is inspected before supplying to customers but 10 per cent of the time the inspector incorrectly classifies an item. Only items classified as good are supplied. If 820 items have been supplied in all, how many of them are expected to be defective?
- 6.26** A factory produces certain types of output by three machines. The respective daily production figures are: Machine A = 3000 units; Machine B = 2500 units; and Machine C = 4500 units. Past experience shows that 1 per cent of the output produced by machine A is defective. The corresponding fractions of defectives for the other two machines are 1.2 and 2 per cent respectively. An item is drawn at random from the day's production and is found to be defective. What is probability that it comes from the output of (a) Machine A; (b) Machine B; (c) Machine C?
- 6.27** In a bolt factory, machines A, B and C manufacture 25, 30 and 40 per cent of the total output, respectively. Of the total of their output 5, 4 and 2 per cent are defective bolts. A bolt is drawn at random from the lot and is found to be defective. What are the probabilities that it was manufactured by machines A, B, or C?
- 6.28** In a factory manufacturing pens, machines X, Y and Z manufacture 30, 30 and 40 per cent of the total production of pens, respectively. Of their output 4, 5 and 10 per cent of the pens are defective. If one pen is selected at random, and it is found to be defective, what is the probability that it is manufactured by machine Z?
- 6.29** A worker-operated machine produces a defective item with probability 0.01, if the worker follows the machine's operating instruction exactly, and with probability 0.03 if he does not. If the worker follows the instructions 90 per cent of the time, what proportion of all items produced by the machine will be defective?

6.30 Medical case histories indicate that different illnesses may produce identical symptoms. Suppose a particular set of symptoms, 'H' occurs only when one of three illnesses: A, B or C occurs, with probabilities

0.01, 0.005 and 0.02, respectively. The probability of developing the symptoms H given a illness A, B and C are 0.90, 0.95 and 0.75, respectively. Assuming that an ill person shows the symptoms H, what is the probability that a person has illness A?

Hints and Answers

6.24 Let A, B, and C = mail processed by first, second, and third clerk, respectively

E = mail containing error

Given $P(A) = 0.40$, $P(B) = 0.35$ and

$P(C) = 0.25$

$P(E | A) = 0.04$, $P(E | B) = 0.06$

and $P(E | C) = 0.03$

$$\therefore P(A | E) = \frac{P(A) P(E|A)}{P(E)}$$

$$= \frac{P(A) P(E|A)}{P(A) P(E|A) + P(B) P(E|B) + P(C) P(E|C)}$$

$$= \frac{0.40 \times 0.04}{0.40 (0.04) + 0.35 (0.06) + 0.25 (0.03)} = 0.36$$

$$\text{Similarly } P(B | E) = [P(B) P(E|B)] / P(E)$$

$$= 0.47$$

$$P(C | E) = [P(C) P(E|C)] / P(E)$$

$$= 0.17$$

6.25 $P(D)$ = Probability of defective item = 0.1;
 $P(\text{classified as good} | \text{defective}) = 0.1$

$$\therefore P(G) = \text{Probability of good item}$$

$$= 1 - P(D) = 1 - 0.1 = 0.9$$

$$P(\text{classified as good} | \text{good})$$

$$= 1 - P(\text{classified as good} | \text{defective})$$

$$= 1 - 0.1 = 0.9$$

$$\therefore P(\text{defective} | \text{classified as good})$$

$$= \frac{P(D) P(\text{classified as good} | \text{defective})}{[P(D) P(\text{classified as good} | D) + P(G) P(\text{classified as good} | G)]}$$

$$= \frac{0.1 \times 0.1}{0.1 \times 0.1 + 0.9 \times 0.9} = \frac{0.01}{0.82} = 0.012.$$

6.26 (a) 0.20 (b) 0.20 (c) 0.60

6.27 $P(A) = 0.37$, $P(B) = 0.40$, $P(C) = 0.23$

6.28 $P(Z) = 0.6639$

6.29 $P(A) = 0.012$

6.30 $P(A | H) = 0.3130$

6.8 EXPECTED VALUE AND VARIANCE OF A RANDOM VARIABLE

Expected Value The mean (also referred as **expected value**) of a random variable is the weighted average, where the possible values of random variable are weighted by the corresponding probabilities of occurrence. If x is a random variable with possible values $x_1, x_2, \dots, x_n$ occurring with probabilities $P(x_1), P(x_2), \dots, P(x_n)$, then the expected value of x denoted by $E(x)$ or μ is the sum of the values of the random variable weighted by the probability that the random variable takes on that value.

$$E(x) = \sum_{j=1}^n x_j P(x_j), \text{ provided } \sum_{j=1}^n P(x_j) = 1$$

If $E(x)$ is calculated in terms of rupees, then it is known as *expected monetary value* (EMV). For example, consider the price range of an item along with the probabilities as below:

Price, x	: 50	60	70	80
Probability, $P(x)$	: 0.2	0.5	0.2	0.1

Thus, the expected monetary value of the item is given by

$$EMV(x) = \sum_{j=1}^n x_j P(x_j) = 50 \times 0.2 + 60 \times 0.5 + 70 \times 0.2 + 80 \times 0.1 = ₹ 62.$$

Variance and Standard Deviation The expected value measures the *central tendency* of a probability distribution, while variance determines the *dispersion* or *variability* to which the possible random values differ among themselves.

The variance denoted by $\text{Var}(x)$ or σ^2 of a random variable x is the squared deviation of the individual values from their expected value or mean. That is

$$\begin{aligned}\text{Var}(x) &= E[(x - \mu)^2] = E[x^2 - 2x\mu + \mu^2] \\ &= E(x^2) - 2\mu E(x) + \mu^2 = E(x^2) - \mu^2\end{aligned}$$

$$\text{where } E(x^2) = \sum_{j=1}^n x_j^2 P(x_j) \text{ and } \mu E(x) = \sum_{j=1}^n x_j P(x_j)$$

The variance has the disadvantage of squaring the unit of measurement. Thus, if a random variable is measured in rupees, the variance will be measured in rupee squared. This shortcoming can be avoided by using *standard deviation* (σ_x) as a measure of dispersion so as to have the same unit of measurement. That is

$$\sigma_x = \sqrt{\text{Var}(x)} = \sqrt{\sum_{j=1}^n (x_j - \mu)^2 P(x_j)}$$

6.8.1 Properties of Expected Value and Variance

The following are the important properties of an expected value of a random variable:

1. The expected value of a constant c is constant. That is, $E(c) = c$, for every constant c .
2. The expected value of the product of a constant c and a random variable x is equal to constant c times, the expected value of the random variable. That is, $E(cx) = cE(x)$.
3. The expected value of a linear function of a random variable is same as the linear function of its expectation. That is, $E(a + bx) = a + bE(x)$.
4. The expected value of the product of two independent random variables is equal to the product of their individual expected values. That is, $E(xy) = E(x) E(y)$.
5. The expected value of the sum of the two independent random variables is equal to the sum of their individual expected values. That is, $E(x + y) = E(x) + E(y)$.
6. The variance of the product of a constant and a random variable X is equal to the constant squared times the variance of the random variable X . That is, $\text{Var}(cx) = c^2 \text{Var}(x)$.
7. The variance of the sum (or difference) of two independent random variables equals the sum of their individual variances. That is, $\text{Var}(x \pm y) = \text{Var}(x) \pm \text{Var}(y)$.

Example 6.25: From a bag containing 3 red balls and 2 white balls, a man is to draw two balls at random without replacement. He gains ₹ 20 for each red ball and ₹ 10 for each white one. What is the expectation of his draw?

Solution: Let x be the random variable denoting the number of red and white balls in a draw. Then x can take up the following values:

$$P(x = 2 \text{ red balls}) = \frac{{}^3C_2}{{}^5C_2} = \frac{3}{10}$$

$$P(x = 1 \text{ red and 1 white ball}) = \frac{{}^3C_1 \times {}^2C_1}{{}^5C_2} = \frac{3}{5}$$

$$P(x = 2 \text{ white balls}) = \frac{{}^2C_2}{{}^5C_2} = \frac{1}{10}$$

Thus, the probability distribution of x is:

Variable	:	2R	1R and 1W	2W
Gain, x	:	40	30	20
Probability, $P(x)$:		3/10	3/5	1/10

Hence, expected gain is $E(x) = 40(3/10) + 30(3/5) + 20(1/10) = ₹32$.

Example 6.26: In a cricket match played to benefit an ex-player, 10,000 tickets are to be sold at ₹500. The prize is a ₹12,000 fridge by lottery. If a person purchases two tickets, what is his expected gain?

Solution: The gain say x may take one of two values: he will either lose ₹1,000 (i.e. gain will be – ₹1,000) or win ₹(12,000–1,000) = ₹11,000, with probabilities 9,998/10,000 and 2/10,000, respectively. The probability distribution for the gain x is given below:

x (in ₹)	$P(x)$
–₹ 1000	9,998/10,000
₹ 11000	2/10,000

The expected gain will be

$$\begin{aligned}\mu &= E(x) = \sum x P(x) \\ &= -1000(9,998/10,000) + 11000(2/10,000) = -₹ 997.6\end{aligned}$$

The result implies that if the lottery were repeated an infinitely large number of times, average or expected loss will be ₹ 997.6.

Example 6.27: A market researcher at a major automobile company classified households by car ownership. The relative frequencies of households for each category of ownership are shown below:

Number of Cars per House hold	Relative Frequency
0	0.10
1	0.30
2	0.40
3	0.12
4	0.06
5	0.02

Calculate the expected value and standard deviation of the random variable and interpret the result.

Solution: The necessary calculations required to calculate expected and standard deviation of a random variable, say x are shown in Table 6.1.

Table 6.1 Calculations of Expected Value and Standard Deviation

<i>Number of Cars Per Households x</i>	<i>Relative Frequency, $P(x)$</i>	$x \times P(x)$	$x^2 \times P(x)$
0	0.10	0.00	0.00
1	0.30	0.30	0.30
2	0.40	0.80	1.60
3	0.12	0.36	1.08
4	0.06	0.24	0.96
5	0.02	0.10	0.50
		<u>1.80</u>	<u>4.44</u>

Expected value, $\mu = E(x) = \sum x P(x) = 1.80$. This value indicates that there are on an average 1.8 cars per household.

$$\text{Variance, } \sigma^2 = \sum x^2 P(x) - [E(x)]^2 = 4.44 - (1.80)^2 = 4.44 - 3.24 = 1.20$$

$$\text{Standard deviation } \sigma = \sqrt{1.20} = 1.095 \text{ cars.}$$

Conceptual Questions 6B

9. Define 'random variable'. How do you distinguish between discrete and continuous random variables? Illustrate your answer with suitable examples.
10. What do you understand by the expected value of a random variable?
11. What are the properties of expected value and variance of a random variable?

Self-practice Problems 6D

- 6.31 Anil company estimates the net profit on a new product it is launching to be ₹ 30,00,000 during the first year if it is 'successful' ₹ 10,00,000 if it is 'moderately successful'; and a loss of ₹ 10,00,000 if it is 'unsuccessful'. The firm assigns the following probabilities to its first year prospects for the product: Successful : 0.15, moderately successful: 0.25. What are the expected value and standard deviation of first year net profit for this product?
- 6.32 A box contains 12 items of which 3 are defective. A sample of 3 items is selected at random from this box. If x represents the number of defective items of 3 selected items, describe the random variable x completely and obtain its expectation.
- 6.33 50 per cent of all automobile accidents lead to property damage of ₹ 100, forty per cent lead to damage of ₹ 500, and ten per cent lead to total loss, that is, damage of ₹ 1800. If a car has a 5 per cent chance of being in an accident in a year, what is the expected value of the property damage due to that possible accident?
- 6.34 The probability that there is atleast one error in an account statement prepared by A is 0.2 and for B and C it is 0.25 and 0.4, respectively. A, B and C prepared 10, 16 and 20 statements, respectively. Find the expected number of correct statements in all.

- 6.35** A lottery sells 10,000 tickets at Re 1 per ticket, and the prize of ₹ 5000 will be given to the winner of the first draw. Suppose you have bought a ticket, how much should you expect to win?
- 6.36** The monthly demand for transistors is known to have the following probability distribution:

Demand (n)	:	1	2	3	4	5	6
Probability (P)	:	0.10	0.15	0.20	0.25	0.18	0.12

Determine the expected demand for transistors. Also obtain the variance. Suppose the cost (C) of producing ' n '

transistors is given by the relationship, $C = 10,000 + 500n$. Determine the expected cost.

- 6.37** A consignment of machine parts is offered to two firms, A and B, for ₹ 75,000. The following table shows the probabilities at which firms A and B will be able to sell the consignment at different prices.

Probability	Price (in ₹)	(at which the Consignment Can be Sold)			
		60,000	70,000	80,000	90,000
A	0.40	0.30	0.20	0.10	
B	0.10	0.20	0.50	0.20	

Which firm, A or B, will be more inclined towards this offer?

Hints and Answers

- 6.31** x : 3 1 - 1
 $P(x)$: 0.15 0.25 $1 - 0.15 - 0.25 = 0.60$
 $E(x) = ₹ 0.10$ million, $\text{Var}(x) = ₹ 2.19$ million, and $\sigma_x = ₹ 1.48$ million.
- 6.32** x : 0 1 2 3
 $P(x)$: 27/64 27/64 9/64 1/64;
 $E(x) = 0.75$
- 6.33** x : 100 500 1,800
 $P(x)$: 0.50 0.40 0.10
 $E(x) = ₹ 430$; $0.5 E(x) = ₹ 215$
- 6.34** $P(A) = 0.2$; $P(B) = 0.25$; $P(C) = 0.4$; and
 $P(\bar{A}) = 0.8$; $P(\bar{B}) = 0.75$; $P(\bar{C}) = 0.6$
 $E(x) = x_1 P(\bar{A}) + x_2 P(\bar{B}) + x_3 P(\bar{C}) = 32$
- 6.35** $P(\text{Win}) = \frac{9999}{10,000}$ and $\frac{1}{1000}$

$$E(x) = -1 \times \frac{9999}{10,000} + 4999 \times \frac{1}{1000} \\ = ₹ 3.9991$$

- 6.36** Expected demand for transistors,
 $E(n) = \sum np = 3.62$
 $E(C) = (10,000 + 500n)$
 $= 10,000 + 50 E(n) = ₹ 11,810.$

- 6.37** EMV (A)
 $= 6 \times 0.4 + 7 \times 0.3 + 8 \times 0.2 + 9 \times 0.1$
 $= ₹ 70,000.$
 EMV (B)
 $= 6 \times 0.1 + 7 \times 0.2 + 8 \times 0.5 + 9 \times 0.2$
 $= ₹ 78,000.$

Firm B will be more inclined towards the offer.

Formulae Used

- Counting methods for determining the number of outcomes
 - Multiplication method
 - $n_1 \times n_2 \times \dots \times n_k$
 - $n_1 \times n_2 \times \dots \times n_k = n^k$ when the event in each trial is the same
 - Number of Permutations ${}^nP_r = \frac{n!}{(n-r)!}$
 - Number of Combinations ${}^nC_r = \frac{n!}{r!(n-r)!}$
 - Classical or *a priori* approach of computing probability of an event A

$$P(A) = \frac{\text{Number of favourable cases for A}}{\text{All possible cases}}$$

$$= \frac{c(n)}{c(s)}$$
 - Relative frequency approach of computing probability of an event A in n trials of an experiment

$$P(A) = \lim_{n \rightarrow \infty} \frac{c(A)}{n}$$
 - Rule of addition of two events
 - When events A and B are mutually exclusive

$$P(A \text{ or } B) = P(A) + P(B)$$
 - When events A and B are not mutually exclusive

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$
 - Conditional probability
 - For statistically independent events

$$P(A|B) = P(A); P(B|A) = P(B)$$
 - For statistically dependent events

$$P(A|B) = \frac{P(A \text{ and } B)}{P(B)}$$

$$P(B|A) = \frac{P(A \text{ and } B)}{P(A)}$$
 - Rule of multiplication of two events
 - Joint probability of independent events

$$P(A \text{ and } B) = P(A) \times P(B)$$
 - Joint probability of dependent events

$$P(A \text{ and } B) = P(A|B) \times P(B)$$

$$P(A \text{ and } B) = P(B|A) \times P(A)$$
 - Rule of elimination
 - $P(B) = \sum P(A_i) P(B|A_i)$
 - $P(A) = \sum P(B_i) P(A|B_i)$
 - Baye's rule

$$P(A_i|B) = \frac{P(A \text{ and } B)}{P(B)}$$
 - Basic rules for assigning probabilities
 - The probability assigned to each experimental outcome

$$0 \leq P(A_i) \leq 1 \text{ for all } i$$
 - Sum of the probabilities for all the experimental outcomes

$$\sum P(A_i) = P(A_1) + P(A_2) + \dots + P(A_n) = 1$$
- Complement of an event, $P(A) = 1 - P(\bar{A})$

True or False Concepts Quiz

- The classical approach to probability theory requires that the total number of possible outcomes be known or calculated and that each of the outcomes be equally likely.
- For any two statistically independent events, $P(A \text{ or } B) = P(A) + P(B)$.
- The marginal probability of an event can be formed by all the possible joint probabilities which include the event as one of the events.
- The posterior probabilities help a decision maker to update his prior probabilities by using additional experimental data.
- The posterior probabilities are valid only when there are two elementary events and consistent outcomes.
- When someone says that the probability of occurrence of an event is 30 per cent, he is stating a classical probability.
- If $x = 4!/0!$, then x is not well defined.

8. Two events are mutually exclusive if their probabilities are less than one.
9. If events are mutually exclusive and collectively exhaustive, then posterior probabilities for these events can be equal to their prior probabilities.
10. *A priori* probability is estimated prior to receiving new information.
11. An event is one or more of the possible outcomes which result from conducting an experiment.
12. The condition of statistical independence arise when the occurrence of one event has no effect upon the probability of occurrence of any other event.
13. The collective exhaustive list of outcomes to an experiment contains every single outcome possible.
14. A marginal probability is also known as unconditional probability.
15. The relative frequency approach of assessing the probability of some event gives the greatest flexibility.

Concepts Quiz Answers

1. T	2. F	3. T	4. T	5. F	6. F	7. F	8. F	9. T
10. T	11. T	12. T	13. T	14. T	15. F			

Review Self-practice Problems

- 6.38 A human resource manager has found it useful to categorize engineering job applicants according to their degree in engineering and relevant work experience. Out of all applicants for the job 70 per cent have a degree with or without any work experience, and 60 per cent have work experience with or without the degree. Fifty per cent of the applicants have both the degree and relevant work experience.
 - (a) Determine the probability that a randomly selected job applicant has either the degree or relevant work experience.
 - (b) What is the probability that the applicant has neither the degree nor work experience?
- 6.39 A salesman is found to complete a sale with 10 per cent of potential customers contacted. If the salesman randomly selects two potential customers and calls on them, then (a) what is the probability that both the calls will result in sales? and (b) what is the probability that the two calls will result in exactly one sale?
- 6.40 Suppose 80 per cent of the material received from a vendor is of exceptional quality, while only 50 per cent of the material received from vendor B is of exceptional quality. However, the manufacturing capacity of vendor A is limited, and for this reason only 40 per cent of the material purchased comes from vendor A. The other 60 per cent comes from vendor B. An incoming shipment of material is inspected and it is found to be of exceptional quality. What is the probability that it came from vendor A.
- 6.41 The municipal corporation routinely conducts two independent inspections of each restaurant, with the restaurant passing only if both inspectors pass it. Inspector A is very experienced, and hence, passes only 2 per cent of restaurants that actually do have rules violations. Inspector B is less experienced and passes 7 per cent restaurants with violations. What is the probability that:
 - (a) A reports favourable, given that B has found a violation?

- (b) B reports favourable with a violation, given that inspector A passes it?
- (c) A restaurant with a violation is cleared by the corporation.

6.42 If a hurricane forms in the Indian Ocean, there is a 76 per cent chance that it will strike the western coast of India. From data gathered over the past 50 years, it has been determined that the probability of a hurricane's occurring in this area in any given year is 0.85. What is the probability that a hurricane will occur in the eastern Indian Ocean and strike India this year?

6.43 A departmental store has been the target of many shoplifters during the past month, but owing to increased security precautions, 250 shoplifters have been caught. Each shoplifter's sex is noted, also noted is whether he/she was a first time or repeat offender. The data are summarized in the table given below:

<i>Sex</i>	<i>First Time Offender</i>	<i>Repeat Offender</i>
Male	60	70
Female	44	76

Assuming that an apprehended shoplifter is chosen at random, find:

- (a) The probability that the shoplifter is male.
 - (b) The probability that the shoplifter is a first-time offender, given that the shoplifter is male.
 - (c) The probability that the shoplifter is female, given that the shoplifter is a repeat offender.
 - (d) The probability that the shoplifter is female, given that the shoplifter is a first-time offender.
- 6.44** The Deputy Commissioner of Police is trying to decide whether to schedule additional patrol units in two sensitive areas, A and B, in his district. He knows that on any given day during the past year, the probabilities of major crimes and minor crimes being committed in area A were 0.478 and 0.602, respectively, and that the corresponding probabilities in area B were 0.350 and 0.523. Assume that major and minor crimes occur independently of

each other and likewise that crimes in the two areas are independent of each other.

- (a) What is the probability that no crime of either type will be committed in the area A on a given day?
- (b) What is the probability that a crime of either type will be committed in the area B on a given day?
- (c) What is the probability that no crime of either type will be committed in either areas on a given day?

6.45 The result of conducting an examination in two papers, A and B, for 20 candidates were recorded as under: 8 passed in paper A, 7 passed in paper B, 8 failed in both papers. If out of these candidates one is selected at random, find the probability that the candidate (a) passed in both A and B, (b) failed only in A and (c) failed in A or B.

6.46 It is known from past experience that a football team will play 40 per cent of its matches on artificial turf this season. It is also known that a football player's chances of incurring a knee injury are 50 per cent higher if he is playing an artificial turf instead of grass. Further, if a player's probability of knee injury on artificial turf is 0.42, what is the probability that (a) a randomly selected player incurs a knee injury and (b) a randomly selected player with a knee injury, incurred the injury playing on grass?

6.47 In a locality of 5000 people, 1200 are above 30 years of age and 3000 are females. Out of 1200 who were above 30 years of age, 200 are females. A person is chosen at random and you are told that the person is female. What is the probability that she is above 30 years of age?

6.48 Suppose 5 men out of 100 and 25 women out of 1000 are colour blind. A colour blind person is chosen at random. What is the probability of his being male (assuming that male and females are equal in proportion).

6.49 Police Head Quarter classified crime by age (in years) of the criminal and whether the crime is violent or non-violent. A total of 150 crimes were reported in the last month as shown in the table below:

Type of Crime	Age (in years)			Total
	Under 20	20–40	Over 40	
Violent	27	41	14	82
Non-violent	12	34	22	68
	39	75	36	150

- (a) What is the probability of selecting a case to analyze and finding the crime was committed by someone less than 40 years old.
- (b) What is the probability of selecting a case that involved a violent crime or an offender less than 20 years old?
- (c) If two crimes are selected for review, then what is the probability that both are violent crimes?

6.50 With each purchase of a large pizza at a Pizza shop, the customer receives a coupon that can be scratched to see if a prize will be awarded. The odds of winning a free soft drink are 1 in 10, and the odds of winning a free large pizza are 1 in 50. You plan to eat lunch tomorrow at the shop. What is the probability.

- (a) That you will win either a large pizza or a soft drink?
- (b) That you will not win a prize?
- (c) That you will not win a prize on three consecutive visits to the Pizza shop?
- (d) That you will win at least one prize on one of your next three visits to the Pizza shop?

6.51 The boxes of men's shirts were received from the factor. Box 1 contained 25 sport shirts and 15 dress shirts. Box 2 contained 30 sport shirts and 10 dress shirts. One of the boxes was selected at random, and a shirt was chosen at random from that box to be inspected. The shirt was a sport shirt. Given this information, what is the probability that the sport shirt came from box 1?

6.52 There are four people being considered for the position of chief executive officer of an Enterprises. Three of the applications are over 60 years of age. Two are female, of which only one is over 60. All four applications are either over 60 years of age or female. What is the probability that a candidate is over 60 and female?

Hints and Answers

6.38 D = degree holders; W = with work experience

$$\begin{aligned} \text{(a)} \quad P(D \cup W) &= P(D \text{ or } W) \\ &= P(D) + P(W) - P(D \cap W) \\ &= 0.70 + 0.60 - 0.50 = 0.80 \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad P(\bar{D} \cap \bar{W}) &= 1.00 - P(D \cup W) \\ &= 1.00 - 0.80 = 0.20 \end{aligned}$$

6.39 S_1, S_2 = calls resulted in sales on both the customers, respectively

$$\begin{aligned} \text{(a)} \quad P(S_1 \text{ and } S_2) &= P(S_1 \cap S_2) \\ &= P(S_1) P(S_2) \\ &= 0.10 \times 0.10 = 0.01 \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad P(S_1 \cup S_2) &= P(S_2) P(\bar{S}_1) + P(S_1) P(\bar{S}_2) \\ &= 0.10 \times 0.90 + 0.90 \times 0.10 \\ &= 0.18 \end{aligned}$$

6.40 A = material supplied by vendor A
E = material is of exceptional quality.

$$\begin{aligned} P(A|E) &= \frac{P(A \cap E)}{P(E)} \\ &= \frac{P(A) P(E|A)}{P(A) P(E|A) + P(B) P(E|B)} \\ &= \frac{0.40 \times 0.80}{0.40 \times 0.80 + 0.60 \times 0.50} \\ &= \frac{0.32}{0.62} = 0.516 \end{aligned}$$

6.41 (a) $P(A | \bar{B}) = P(A) = 0.02$

(b) $P(B | \bar{A}) = P(B) = 0.07$

(c) $P(A \cap B) = P(A) P(B)$
 $= 0.02 \times 0.07 = 0.0014$

6.42 Let H = hurricane forming over Indian Ocean

W = hurricane hits western coast of India

$$P(H \cap W) = P(H) P(W | H) \\ = 0.76 \times 0.85 = 0.646$$

6.43 M = shoplifter is male

W = shoplifter is female

F = shoplifter is first time offender,

R = shoplifter is repeat offender

(a) $P(M) = (60 + 70) \div 250 = 0.520$

(b) $P(F|M) = P(F \cap M)/P(M) \\ = \frac{60}{250} \div \frac{130}{250} = 0.462$

(c) $P(W|R) = P(W \cap R)/P(R) \\ = \frac{76}{250} \div \frac{146}{250} = 0.521$

(d) $P(W|F) = P(W \cap F)/P(F) \\ = \frac{44}{250} \div \frac{104}{250} = 0.423$

6.44 M_1, M_2 = major crime in district A and B, respectively

m_1, m_2 = minor crime in district A and B, respectively

(a) $P(M_1 \cup m_1) \\ = P(M_1) + P(m_1) - P(M_1 \cap m_1) \\ = P(M_1) + P(m_1) - P(M_1) P(m_1) \\ = 0.478 + 0.602 - 0.478 \times 0.602 \\ = 0.792$

$$P(\bar{M}_1 \cap \bar{m}_1) = 1 - 0.792 = 0.208$$

(b) $P(M_2 \cup m_2) \\ = P(M_2) + P(m_2) - P(M_2) P(m_2) \\ = 0.350 + 0.523 - 0.350 \times 0.523 \\ = 0.690$

(c) $P(\text{crime in A}) = 0.792;$
 $P(\text{crime in B}) = 0.690$
 $P(\text{no crime in A and B}) \\ = 1 - P(\text{crime in at least A or B}) \\ = 1 - [P(A) + P(B) - P(A \text{ and B})] \\ = 1 - [P(A) + P(B) - P(A) P(B)] \\ = 0.064$

6.45 (a) $P(A \cap B) \\ = P(A) + P(B) - P(A \cup B) \\ = \frac{8}{20} + \frac{7}{20} - \left(1 - \frac{8}{20}\right) = \frac{3}{20}$

(b) $P(\bar{A} \cap B) = P(\bar{A}) \times P(B) \\ = \left(\frac{12}{20}\right) \left(\frac{7}{20}\right) = 0.21$

(c) $P(\bar{A} \cup \bar{B}) \\ = A(\bar{A}) + P(\bar{B}) - (\bar{A} \cap \bar{B}) \\ = \frac{12}{20} + \frac{13}{20} - \frac{8}{20} = \frac{17}{20}$

6.46 A = knee injury; B = playing on artificial turf

C = playing on grass

(a) $P(A \cap B) = P(B) P(A | B) \\ = 0.40 \times 0.42 = 0.168$

$$P(A \cap C) = P(C) P(A | C) \\ = 0.60 \times 0.28 = 0.168$$

$$\text{Thus } P(A) = P(A \cap B) + P(A \cap C) \\ = 0.168 + 0.168 = 0.336$$

(b) $P(C | A) = \frac{P(A \cap C)}{P(A)} \\ = \frac{P(C) P(A | C)}{P(C) P(A | C) + P(B) P(A | B)} \\ = \frac{0.168}{0.168 + 0.168} = \frac{1}{2}$

6.47 $P(A)$ = probability that a person chosen is above 30 years = $1200/5000 = 0.24$

$P(B)$ = probability that a person chosen is female = $3000/5000 = 0.60$

$P(A \text{ and } B) = P(A \cap B) \\ = \text{probability that a person chosen is above 30 years and a female} \\ = 200/5000 = 0.04$

$$\text{But } P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{0.04}{0.06} = 0.667$$

6.48 Let M, F and C denote male, female and colour blind persons. Then

$$P(M|C) = 5/100 = 1/20;$$

$$P(F|C) = 25/1000 = 1/40;$$

$$P(M) = P(F) = 1/2.$$

$$\text{Thus } P(C | M) = \frac{0.04}{0.06} \\ = \frac{P(M) P(M|C)}{P(M) P(M|C) + P(F) P(F|C)} \\ = \frac{(1/2)(1/20)}{(1/2)(1/20) + (1/2)(1/40)} = \frac{2}{3}$$

- 6.49** (a) $P(\text{crimes both violent and non-violent committed by a person less than 40 years old})$

$$= \frac{39}{150} + \frac{75}{150} = \frac{114}{150} = 0.76$$

- (b) $P(\text{crime of violent type or offender less than 20 years old})$

$$= \frac{82}{150} + \frac{39}{150} - \frac{27}{150} = \frac{94}{150} = 0.6267$$

- (c) $P(\text{both crimes are of violent nature})$

$$\frac{82}{150} \times \frac{81}{149} = \frac{6642}{22,350} = 0.2972$$

- 6.50** Let A and B represent the event of winning pizza and soft drink, respectively.

- (a) $P(A \text{ or } B) = P(A) P(\bar{B}) + P(\bar{A}) P(B)$

$$= \frac{1}{50} \times \frac{9}{10} + \frac{49}{50} \times \frac{1}{10} = \frac{9}{500} + \frac{49}{500}$$

$$= \frac{58}{500} = 0.116$$

- (b) $P(\text{no prize}) = [1 - P(A)] [1 - P(B)]$

$$= P(\bar{A}) P(\bar{B})$$

$$= \frac{49}{500} \times \frac{9}{10}$$

$$= \frac{441}{500} = 0.882$$

- (c) $P(\text{no prize on 3 visits})$

$$= [P(\text{no prize})]^3 = (0.882)^3 = 0.686$$

- (d) $P(\text{at least are prize}) = 1 - P(\text{no prize})$

$$= 1 - 0.686 = 0.314$$

- 6.51** Given $P(\text{sport shirt}) = 25/40$,

$$P(\text{dress shirt}) = 15/40 \text{ (in Box 1);}$$

$$P(\text{sport short}) = 30/40$$

$$P(\text{dress shirt}) = 10/40$$

$P(\text{shirt came from Box 1/shirt was sport-shirt})$

$$= \frac{P(\text{Box 1 and sport shirt})}{P(\text{sport shirt})}$$

$$= \frac{P(\text{sport shirt/Box 1}) P(\text{Box 1})}{P(\text{Box 1}) P(\text{sport/Box 1}) + P(\text{Box 2}) P(\text{sport shirt/Box 2})}$$

$$= \frac{(25/40) (1/2)}{(25/40) (1/2) + (30/40) (1/2)}$$

$$= \frac{0.625 \times 0.50}{0.625 \times 0.50 + 0.75 \times 0.50}$$

$$= \frac{0.3125}{0.6875} = 0.4545$$

- 6.52** $P(F \text{ and over 60 years}) = P(F) \times P(\text{over 60 years})$

$$= \frac{2}{4} \times \frac{1}{2} = 0.25$$

7

Probability Distributions

LEARNING OBJECTIVES

After studying this chapter, you should be able to

- define the terms random variable and probability distribution.
- distinguish between discrete and continuous probability distributions.
- describe the characteristics and compute probabilities using both discrete and continuous probability distributions.
- compute expected value and variance of a random variable.
- apply the concepts of probability distributions to real-life problems.

7.1 INTRODUCTION

In any probabilistic situation, each strategy (course of action) may lead to a number of different possible outcomes. For example, a product whose sale is estimated around 100 units may be equal to 100, less or more. Here the sale (i.e., an outcome) of the product is measured in real numbers but the volume of the sales is uncertain. The volume of sale which is an uncertain quantity and whose definite value is determined by chance is termed as *random (chance or stochastic) variable*. A listing of all the possible outcomes of a random variable with each outcome's associated probability of occurrence is called *probability distribution*. The numerical value of a random variable depends upon the outcome of an experiment and may be different for different trials of the same experiment. The set of all such values so obtained is called the *range space* of the random variable.

Illustration: If a coin is tossed twice, then the sample space of events, for this random experiment is

$$S = \{H H, T H, H T, T T\}$$

In this case, if the decision maker is interested to know the probability distribution for the number of heads on two tosses of the coin, then a random variable (x) may be defined as:

$$x = \text{number of H's occurred}$$

The values of x will depend on chance and may take the values: $H H = 2$, $H T = 1$, $T H = 1$, $T T = 0$. Thus, the range space of x is $\{0, 1, 2\}$.

When a random variable x is defined, a value is given to each simple event in the sample space. The probability of any particular value of x can then be found by adding the probabilities

for all the simple events that have that value of x . For example, the probabilities of occurrence of 'heads' can be associated with each of the random variable values. Supposing $P(x = r)$ represents the probability of the random variable taking the value r (here r represents the number of heads occurred). Then probabilities of occurrence of different number of heads are computed as:

Number of Heads (x)	Probability of Outcome $P(x)$
0	$P(x = 0) = P(TT) = P(T) \times P(T) = 0.5 \times 0.5 = 0.25$
1	$P(x = 1) = P(HT) + P(TH) = P(H) \times P(T) + P(T) \times P(H)$ $= 0.5 \times 0.5 + 0.5 \times 0.5 = 0.25 + 0.25 = 0.50$
2	$P(x = 2) = P(HH) = P(H) \times P(H) = 0.5 \times 0.5 = 0.25$

7.2 DISCRETE PROBABILITY DISTRIBUTIONS

Binomial Probability Distribution

Binomial probability distribution is a widely used probability distribution for a discrete random variable. This distribution describes discrete data resulting from an experiment called a *Bernoulli process* (named after Jacob Bernoulli, 1654–1705, the first of the Bernoulli family of Swiss mathematicians). For each trial of an experiment, *there are only two possible complementary (mutually exclusive) outcomes* such as, defective or good, head or tail, zero or one, boy or girl. In such cases, the outcome of interest is referred to as a 'success' and the other as a 'failure'. The term 'binomial' literally means two names.

Bernoulli process: *It is a process wherein an experiment is performed repeatedly, yielding either a success or a failure in each trial and where there is absolutely no pattern in the occurrence of successes and failures. That is, the occurrence of a success or a failure in a particular trial does not affect, and is not affected by, the outcomes in any previous or subsequent trials. The trials are independent.*

Conditions for Binomial Experiment The Bernoulli process involving a series of independent trials is based on certain conditions as under:

- There are only two mutually exclusive and collective exhaustive outcomes of the random variable and one of them is referred to as a *success* and the other as a *failure*.
- The random experiment is performed under the same conditions for a fixed and finite (also discrete) number of times, say n . Each observation of the random variable in a random experiment is called a *trial*. Each trial generates either a *success* denoted by p or a *failure* denoted by q .
- The outcome (i.e., success or failure) of any trial is not affected by the outcome of any other trial.
- All the observations are assumed to be independent of each of each other. This means that the probability of outcomes remains constant throughout the process. Thus, the probability of a success, denoted by p , remains constant from trial to trial. The probability of a failure is $q = 1 - p$.

To understand the Bernoulli process, consider the coin tossing problem where 3 coins are tossed. Suppose we are interested to know the probability of two heads. The possible sequence

of outcomes involving two heads can be obtained in the following three ways: HHT, HTH and THH.

The probability of each of the above sequences can be found by using the multiplication rule for independent events. Let the probability of a head be p and the probability of tail be q . The probability of each sequence can be written as:

$$ppq \quad pqp \quad qpp$$

Each of these probabilities can be written as p^2q , they are all equal.

Since three sequences correspond to the same event '2 heads', therefore the probability of 2 heads in 3 tosses is obtained by using the addition rule of probabilities for mutually exclusive events. Since the probability of each sequence is same, we can multiply p^2q (probability of one sequence) by 3 (number of possible sequences or orderings of 2 heads). Hence

$$P(2 \text{ heads}) = 3p^2q = {}^3C_2 p^2q$$

Here it may be noted that the possible sequences equals the binomial coefficient ${}^3C_2 = 3$. This coefficient represents the number of ways that three symbols, of which two are alike (i.e., 2H and one T), can be ordered (or arranged). In general, the binomial coefficient nC_r represents the number of ways that n symbols, of which r are alike, can be ordered.

Since events H and T are equally likely and mutually exclusive, therefore $p = 0.5$ and $q = 0.5$ for a toss of the coin. Thus, the probability of 2 heads in 3 tosses is

$$P(x = 2 \text{ heads}) = {}^3C_2 (0.5)^2 (0.5) = 3 (0.25) (0.5) = 0.375$$

Binomial Probability Function In general, for a binomial random variable, x the probability of success (occurrence of desired outcome) r number of times in n independent trials, regardless of their order of occurrence is given by the formula:

$$P(x = r \text{ successes}) = {}^nC_r p^r q^{n-r} = \frac{n!}{r!(n-r)!} p^r q^{n-r}, r = 0, 1, 2, \dots, n \quad (7-1)$$

where n = number of trials (specified in advance) or sample size

p = probability of success

$q = (1 - p)$, probability of failure

x = discrete binomial random variable

r = number of successes in n trials

In formula (7-1), the term $p^r q^{n-r}$ represents the probability of one sequence where r number of events (called successes) occur in n trials in a particular sequence, while the term nC_r represents the number of possible sequences (combinations) of r successes that are possible out of n trials.

Characteristics of the Binomial Distribution The expression (7-1) is known as **binomial distribution** with parameters n and p . Different values of n and p identify different binomial distributions which lead to different probabilities of r -values. The *mean* and *standard deviation* of a binomial distribution are computed in a shortcut manner as follows:

Mean, $\mu = np$,

Standard deviation, $\sigma = \sqrt{npq}$

Knowing the values of first two central moments $\mu_0 = 1$ and $\mu_1 = 1$, other central moments are given by

Second moment, $\mu_2 = npq$

Third moment, $\mu_3 = npq(q-p)$

Fourth moment, $\mu_4 = 3n^2p^2q^2 + npq(1-6pq)$

so that $\gamma_1 = \frac{\mu_3}{\mu_2^{3/2}} = \frac{q-p}{\sqrt{npq}}$, where $\beta_1 = \frac{n^2p^2q^2(q-p)^2}{n^3p^3q^3}$

and $\gamma_2 = \beta_2 - 3 = \frac{\mu_4}{\mu_2^2} - 3 = \frac{1-6pq}{npq}$, where $\beta_2 = \frac{3n^2p^2q^2 + npq(1-6pq)}{n^2p^2q^2}$

For a binomial distribution, *variance* < *mean*. This distribution is unimodal when np is a whole number, and $\text{mean} = \text{mode} = np$.

Example 7.1: In a binomial distribution with 6 independent trials, the probability of 3 and 4 success is found to be 0.2457 and 0.0819, respectively. Find the parameters p and q of the binomial distribution.

Solution: Given, $n = 6$, $P(3) = 0.2457$, $P(4) = 0.0819$. Applying the formula of binomial distribution, we get

$$P(x = r) = {}^nC_r p^r q^{n-r}; \quad r = 0, 1, \dots, 6$$

$$\text{For } r = 3, P(x=3) = {}^6C_3 p^3 q^3 \text{ or } 0.2457 = 20 p^3 q^3$$

$$\text{For } r = 4, P(x=4) = {}^6C_4 p^4 q^2 \text{ or } 0.0819 = 15 p^4 q^2$$

$$\text{Divide } P(x = 4) \text{ by } P(x = 3), \text{ we get, } \frac{0.0819}{0.2457} = \frac{15p^4q^2}{20p^3q^3} \text{ or } \frac{1}{3} = \frac{3p}{4q}$$

$$p = \frac{4}{9}q = \frac{4}{9}(1-p) \quad (\text{because } q = 1-p)$$

$$p + \frac{4}{9}p = \frac{4}{9}$$

$$13p = 4, \text{ i.e. } p = \frac{4}{13} \text{ and } q = 1-p = 1 - \frac{4}{13} = \frac{9}{13}.$$

Example 7.2: A salesman makes a sale on the average to 40 per cent of the customers he contacts. If 4 customers are contacted today, what is the probability that he makes sales to exactly two?

Solution: Let x be the random variable showing the number of sales. Given that

$$p = \text{probability of making sale} = 0.4, \text{ and } q = 1-p = 0.6$$

Apply binomial distribution formula:

$$P(x = r) = {}^nC_r p^r q^{n-r} = {}^4C_r (0.4)^r (0.6)^{4-r}$$

$$\text{when } r = 2, P(x = 2) = {}^4C_2 (0.4)^2 (0.6)^2 = 6(0.16)(0.36) = 0.3456$$

Hence, probability of making sales to exactly two customers is 0.3456.

Example 7.3: A brokerage survey reports that 30 per cent of individual investors have used a discount broker, i.e. one which does not charge the full commission. In a random sample of 9 individuals, what is the probability that

- exactly two of the sampled individuals have used a discount broker?
- not more than three have used a discount broker.
- at least three of them have used a discount broker.

Solution: The probability that individual investors have used a discount broker is $p = 0.30$, and therefore $q = 1-p = 0.70$

(a) Probability that exactly 2 of the 9 individual have used a discount broker is given by

$$\begin{aligned} P(x = 2) &= {}^9C_2 (0.30)^2 (0.70)^7 = \frac{9!}{(9-2)! 2!} (0.30)^2 (0.70)^7 \\ &= \frac{9 \times 8}{2} \times 0.09 \times 0.082 = 0.2657 \end{aligned}$$

(b) Probability that out of 9 randomly selected individuals not more than three have used a discount broker is given by

$$\begin{aligned} P(x \leq 3) &= P(x=0) + P(x=1) + P(x=2) + P(x=3) \\ &= {}^9C_0 (0.30)^0 (0.70)^9 + {}^9C_1 (0.30) (0.70)^8 + {}^9C_2 (0.30)^2 (0.70)^7 \\ &\quad + {}^9C_3 (0.30)^3 (0.70)^6 \\ &= 0.040 + 9 \times 0.30 \times 0.058 + 36 \times 0.09 \times 0.082 + 84 \times 0.027 \times 0.118 \\ &= 0.040 + 0.157 + 0.266 + 0.268 = 0.731 \end{aligned}$$

(c) Probability that out of 9 randomly selected individuals, at least three have used a discount broker is given by

$$\begin{aligned} P(x \geq 3) &= 1 - P(x < 3) = 1 - [P(x=0) + P(x=1) + P(x=2)] \\ &= 1 - [0.040 + 0.157 + 0.266] = 0.537 \end{aligned}$$

Example 7.4: Mr Gupta applies for a personal loan of ₹1,50,000 from a nationalized bank to repair his house. The loan offer informed him that over the years, bank has received about 2920 loan applications per year and that the probability of approval was, on average, above 0.85

(a) Mr Gupta wants to know the average and standard deviation of the number of loans approved per year.

(b) Suppose bank actually received 2654 loan applications per year with an approval probability of 0.82. What are the mean and standard deviation now?

Solution: (a) Assuming that approvals are independent from loan to loan, and that all loans have the same 0.85 probability of approval. Then

$$\text{Mean, } \mu = np = 2920 \times 0.85 = 2482$$

$$\text{Standard deviation, } \sigma = \sqrt{npq} = \sqrt{2920 \times 0.85 \times 0.15} = 19.295$$

(b) Mean, $\mu = np = 2654 \times 0.82 = 2176.28$

$$\text{Standard deviation, } \sigma = \sqrt{npq} = \sqrt{2654 \times 0.82 \times 0.18} = 19.792$$

Example 7.5: Suppose 10 per cent of new scooters will require warranty service within the first month of its sale. A scooter manufacturing company sells 1000 scooters in a month,

(a) Find the mean and standard deviation of scooters that require warranty service.

(b) Calculate the moment coefficient of skewness and kurtosis of the distribution.

Solution: Given that $p = 0.10$, $q = 1 - p = 0.90$ and $n = 1000$

(a) Mean, $\mu = np = 1000 \times 0.10 = 100$ scooters

$$\text{Standard deviation, } \sigma = \sqrt{npq} = \sqrt{1000 \times 0.10 \times 0.90} = 10 \text{ scooters (approx.)}$$

(b) Moment coefficient of skewness

$$\gamma_1 = \frac{\sqrt{\beta_1}}{\sqrt{npq}} = \frac{q - p}{\sqrt{npq}} = \frac{0.90 - 0.10}{9.48} = 0.084$$

Since γ_1 is more than zero, the distribution is positively skewed.

Moment coefficient of kurtosis, $\gamma_2 = \beta_2 - 3$

$$= \frac{1 - 6pq}{npq} = \frac{1 - 6(0.10)(0.90)}{90} = \frac{0.46}{90} = 0.0051$$

Since γ_2 is positive, the distribution is platykurtic.

Example 7.6: The incidence of occupational disease in an industry is such that the workers have 20 per cent chance of suffering from it. What is the probability that out of six workers 4 or more will come in contact of the disease?

Solution: The probability of a worker suffering from the disease is $p = 20/100 = 1/5$. Therefore, $q = 1 - p = 1 - (1/5) = 4/5$.

The probability of 4 or more, that is, 4, 5, or 6 coming in contact of the disease is given by

$$\begin{aligned} P(x \geq 4) &= P(x = 4) + P(x = 5) + P(x = 6) \\ &= {}^6C_4 \left(\frac{1}{5}\right)^4 \left(\frac{4}{5}\right)^2 + {}^6C_5 \left(\frac{1}{5}\right)^5 \left(\frac{4}{5}\right) + {}^6C_6 \left(\frac{1}{5}\right)^6 \\ &= \frac{15 \times 16}{15625} + \frac{6 \times 4}{15625} + \frac{1}{15625} = \frac{1}{15625} (240 + 24 + 1) \\ &= \frac{265}{15625} = 0.01696 \end{aligned}$$

Hence the probability that out of 6 workers 4 or more will come in contact of the disease is 0.01696.

Example 7.7: A multiple choice test contains 8 questions with 3 answers to each question (of which only one is correct). A student answers each question by rolling a balanced dice and checking the first answer if he gets 1 or 2, the second answer if he gets 3 or 4, and the third answer if he gets 5 or 6. To get a distinction, the student must secure at least 75 per cent correct answers. If there is no negative marking, what is the probability that the student secures a distinction?

Solution: Probability of a correct answer, p is one in three so that $p = 1/3$ and probability of wrong answer $q = 2/3$.

The required probability of securing a distinction (i.e., of getting the correct answer of at least 6 of the 8 questions) is given by

$$\begin{aligned} P(x \geq 6) &= P(x = 6) + P(x = 7) + P(x = 8) \\ &= {}^8C_6 \left(\frac{1}{3}\right)^6 \left(\frac{2}{3}\right)^2 + {}^8C_7 \left(\frac{1}{3}\right)^7 \left(\frac{2}{3}\right) + {}^8C_8 \left(\frac{1}{3}\right)^8 \\ &= \left(\frac{1}{3}\right)^6 \left[{}^8C_6 \left(\frac{2}{3}\right)^2 + {}^8C_7 \left(\frac{1}{3}\right) \left(\frac{2}{3}\right) + {}^8C_8 \left(\frac{1}{3}\right)^2 \right] \\ &= \frac{1}{729} \left(28 \times \frac{4}{9} + 8 \times \frac{2}{9} + \frac{1}{9} \right) = \frac{1}{729} (12.44 + 1.778 + 0.11) \\ &= 0.0197 \end{aligned}$$

Conceptual Questions 7A

- (a) Define binomial distribution stating its parameters, mean and standard deviation, and give two examples where such a distribution is ideally suited.
(b) Define binomial distribution. Point out its chief characteristics and uses.
- For a binomial distribution, is it true that the mean is the most likely value? Explain.
- Demonstrate that the binomial coefficient nC_r equals ${}^nC_{n-r}$ and illustrate this with a specific numerical example.
- What assumptions must be met for a binomial distribution to be applied to a real life situation?
- What is meant by the term parameter of a probability distribution? Relate the concept to the binomial distribution?
- What is a binomial coefficient and illustrate this with a specific numerical example.

Self-practice Problems 7A

- The normal rate of infection of a certain disease in animals is known to be 25 per cent. In an experiment with 6 animals injected with a new vaccine it was observed that none of the animals caught the infection. Calculate the probability of the observed result.
- Out of 320 families with 5 children each, what percentage would be expected to have (i) 2 boys and 3 girls, (ii) at least one boy? Assume equal probability for boys and girls.
- The incidence of a certain disease is such that on an average 20 per cent of workers suffer from it. If 10 workers are selected at random, find the probability that (i) exactly 2 workers suffer from the disease, (ii) not more than 2 workers suffer from the disease. Calculate the probability upto fourth decimal place.
- The mean of a binomial distribution is 40 and standard deviation 6. Calculate n , p and q .
- A student obtained answers with mean $\mu = 2.4$ and variance $\sigma^2 = 3.2$ for a certain problem given to him using binomial distribution. Comment on the result.
- The probability that an evening college student will graduate is 0.4. Determine the probability that out of 5 students (i) none, (ii) one, and (iii) at least one will graduate.
- The normal rate of infection of a certain disease in animals is known to be 25 per cent. In an experiment with 6 animals injected with a new vaccine it was observed that none of the animals caught infection. Calculate the probability of the observed result.
- Is there any inconsistency in the statement that the mean of a binomial distribution is 20 and its standard deviation is 4? If no inconsistency is found what shall be the values of p , q and n .
- Find the probability that in a family of 5 children there will be (i) at least one boy (ii) at least one boy and one girl (Assume that the probability of a female birth is 0.5).

Hints and Answers

- 7.1** Let P denote infection of the disease. Then $p = 25/100 = 1/4$ and $q = 3/4$.

$$P(x = 0) = {}^6C_0 \left(\frac{1}{4}\right)^0 \left(\frac{3}{4}\right)^6 = \frac{729}{4096}$$

- 7.2** (i) Given $p = q = 1/2$

$$\begin{aligned} P(\text{boy} = 2) &= {}^5C_2 p^2 q^3 \\ &= {}^5C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^3 \\ &= \frac{5}{16} \text{ or } 31.25 \text{ per cent} \end{aligned}$$

$$\begin{aligned} \text{(ii) } P(\text{boy} \geq 1) &= 1 - {}^5C_0 q^5 = 1 - \frac{1}{32} \\ &= \frac{31}{32} \text{ or } 97 \text{ per cent.} \end{aligned}$$

- 7.3** Probability that a worker suffers from a disease, $p = 1/5$ and $q = 4/5$.

$$\begin{aligned} P(x = r) &= {}^nC_r q^{n-r} p^r \\ &= {}^{10}C_r \left(\frac{1}{5}\right)^r \left(\frac{4}{5}\right)^{10-r} \\ &= {}^{10}C_r \frac{4^{10-r}}{5^{10}}; r = 0, 1, 2, \dots, 10 \end{aligned}$$

$$\text{(i) } P(x = 2) = {}^{10}C_2 \frac{4^{10-2}}{5^{10}} = 0.302$$

$$\begin{aligned} \text{(ii) } P(x = 0) + P(x = 1) + P(x = 2) \\ &= \frac{1}{5^{10}} ({}^{10}C_0 4^{10} + {}^{10}C_1 4^9 + {}^{10}C_2 4^8) \\ &= 0.678. \end{aligned}$$

- 7.4** Given $\mu = np = 40$ and $\sigma = \sqrt{npq} = 6$. Squaring σ , we get $npq = 36$ or $40q = 36$ or $q = 0.9$. Then $p = 1 - q = 0.1$. Since $np = 40$ or $n = 40/p = 40/0.1 = 400$.

- 7.5** Given $\sigma^2 = npq = 3.2$ and $\mu = np = 2.4$. Then $2.4q = 3.2$ or $q = 3.2/2.4 = 1.33$ (inconsistent result)

- 7.6** Given $p = 0.4$ and $q = 0.6$

$$\begin{aligned} \text{(i) } P(x = \text{no graduate}) &= {}^5C_0 (0.4)^0 (0.6)^5 \\ &= 1 \times 1 \times 0.0778 = 0 \\ \text{(ii) } P(x = 1) &= {}^5C_1 (0.4)^1 (0.6)^4 = 0.2592 \\ \text{(iii) } P(x \geq 1) &= 1 - P(x = 0) = 1 - 0.0778 \\ &= 0.9222 \end{aligned}$$

- 7.7** Probability of infection of disease = $25/100 = 0.25$; $q = 1 - p = 0.75$.

The first term in the expansion of $(q + p)^n$ is $\left(\frac{3}{4} + \frac{1}{4}\right)^6$ is ${}^6C_0 \left(\frac{3}{4}\right)^6 = 0.178$, which is also the required probability.

- 7.8** Given $\mu = np = 20$ and $\sigma = \sqrt{npq} = 4$ or $npq = 16$ or $20q = 16$ or $q = 16/20 = 0.80$ and then $p = 1 - q = 0.20$.

Hence $npq = 16$ gives $n = 16/pq = 16/(0.20 \times 0.80) = 100$.

- 7.9** Since $p = q = 0.5$, therefore,

$$\begin{aligned} \text{(i) } P(\text{boy} = 0) &= {}^5C_0 (0.5)^0 (0.5)^5 \\ &= 0.031 \end{aligned}$$

$$P(\text{at least one boy}) = 1 - 0.031 = 0.969$$

$$\begin{aligned} \text{(ii) } P(\text{at least 1B and 1G}) \\ &= {}^5C_1 (0.5)^1 (0.5)^4 \\ &\quad + {}^5C_2 (0.5)^2 (0.5)^3 + {}^5C_3 (0.5)^3 (0.5)^2 \\ &\quad + {}^5C_4 (0.5)^4 (0.5) \\ &= \frac{5}{32} + \frac{10}{32} + \frac{10}{32} + \frac{5}{32} = \frac{30}{32} \end{aligned}$$

Poisson Probability Distribution

Poisson distribution is named after the French mathematician S. Poisson (1781–1840). The Poisson process measures the number of occurrences of a particular outcome of a discrete random variable in a *predetermined time interval, space, or volume*, for which an *average number* of occurrences of the outcome is known or can be determined. In the Poisson process, the random variable values need counting. Such a count might be (i) number of telephone calls per hour coming into the switchboard, (ii) number of fatal traffic accidents per week in a city/state, (iii) number of patients arriving at a health centre every hour, (iv) number of organisms per unit volume of some fluid, (v) number of cars waiting for service in a workshop, (vi) number of flaws per unit length of some wire,

and so on. The Poisson probability distribution provides a simple, easy-to compute and accurate approximation to a binomial distribution when the probability of success, p is very small and n is large, so that $\mu = np$ is small, preferably $np > 7$. It is often called the 'law of improbable' events meaning that the probability, p , of a particular event's happening is very small. As mentioned above **Poisson distribution** occurs in business situations in which there are a few successes against a large number of failures or vice-versa (i.e. few successes in an interval) and has single independent events that are mutually exclusive. Because of this, the probability of success, p is very small in relation to the number of trials n , so we consider only the probability of success.

Conditions for Poisson Process The use of Poisson distribution to compute the probability of the occurrence of an outcome during a specific time period is based on the following conditions:

- (i) The outcomes within any interval occur randomly and independently of one another.
- (ii) The probability of one occurrence in a small time interval is proportional to the length of the interval and independent of the specific time interval.
- (iii) The probability of more than one occurrence in a small time interval is negligible when compared to the probability of just one occurrence in the same time interval.
- (iv) The average number of occurrences is constant for all time intervals of the same size.

Poisson Probability Function If the probability, p , of occurrence of an outcome of interest (i.e., success) in each trial is very small, but the number of independent trials n is sufficiently large, then the average number of times that an event occurs in a certain period of time or space, $\lambda = np$ is also small. Under these conditions, the binomial probability function

$$P(x = r) = {}^nC_r p^r q^{n-r} = \frac{n(n-1)(n-2)\dots(n-r+1)}{r!} p^r q^{n-r}$$

$$= \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \dots \left(1 - \frac{r-1}{n}\right) \frac{\lambda^r}{r!} \left(1 - \frac{\lambda}{n}\right)^{n-r}; np = \lambda \text{ or } p = \lambda/n$$

tends to for a fixed r . Thus, the Poisson probability distribution which approximates the binomial distribution is defined by the following probability function:

$$P(x = r) = \frac{\lambda^r e^{-\lambda}}{r!}, r = 0, 1, 2, \dots \quad (7-3)$$

where $e = 2.7183$.

Characteristics of Poisson Distribution Since Poisson probability distribution is specified by a process rate λ and the time period t , its mean and variance are identical and expressed in terms of the parameters: n and p as shown below:

- The *arithmetic mean* $\mu = E(x)$ of Poisson distribution is given by

$$\begin{aligned} \mu &= \sum x P(x) = \sum x \frac{e^{-\lambda} \lambda^x}{x!}, x = 1, 2, 3, \dots \text{ and } x P(x) = 0 \text{ for } x = 0 \\ &= \lambda e^{-\lambda} + \lambda^2 e^{-\lambda} + \frac{\lambda^3 e^{-\lambda}}{2!} + \dots + \frac{\lambda^r e^{-\lambda}}{(x-1)!} + \dots \\ &= \lambda e^{-\lambda} \left[1 + \lambda + \frac{\lambda^2}{2!} + \dots + \frac{\lambda^{r-1}}{(x-1)!} + \dots \right] = \lambda e^{-\lambda} e^{\lambda} = \lambda \end{aligned}$$

Thus, the mean of the distribution is $\mu = \lambda = np$.

- The variance σ^2 of Poisson distribution is given by

$$\begin{aligned}
 \sigma^2 &= E(x^2) - [E(x)]^2 = E(x^2) - \lambda^2 \\
 &= \sum x^2 \frac{e^{-\lambda} \lambda^x}{x!} - \lambda^2 = e^{-\lambda} \sum \frac{x(x-1) + x}{x!} \lambda^x - \lambda^2 \\
 &= \lambda^2 e^{-\lambda} \sum \frac{\lambda^{x-2}}{(x-2)!} + \lambda e^{-\lambda} \sum \frac{\lambda^{x-1}}{(x-1)!} - \lambda^2 \\
 &= \lambda^2 e^{-\lambda} e^{\lambda} + \lambda e^{-\lambda} e^{\lambda} - \lambda^2 = \lambda^2 + \lambda - \lambda^2 = \lambda
 \end{aligned}$$

Thus, the variance of the distribution is $\sigma^2 = \lambda = np$.

It is very rare for more than one event to occur during a short interval of time. The shorter the duration of interval, the occurrence of two or more events also becomes rare. The probability that exactly one event will occur in such an interval is approximately λ times its duration.

The typical application of Poisson distribution is for analysing queuing (or waiting line) problems in which arriving customers during an interval of time arrive independently and the number of arrivals depends on the length of the time interval. While applying Poisson distribution, if we consider a time period of different length, the distribution of number of events remains Poisson with the mean proportional to the length of the time period.

Example 7.8: What probability model is appropriate to describe a situation where 100 misprints are distributed randomly throughout the 100 pages of a book? For this model, what is the probability that a page observed at random will contain at least three misprints?

Solution: Since 100 misprints are distributed randomly throughout the 100 pages of a book, therefore on an average there is only one mistake on a page. This means the probability of there being a misprint, $p = 1/100$, is very small and the number of words, n , in 100 pages are vary large. Hence, Poisson distribution is best suited in this case.

Average number of misprints in one page, $\lambda = np = 100(1/100) = 1$. Therefore $e^{-\lambda} = e^{-1} = 0.3679$.

Probability of at least three misprints in a page is

$$\begin{aligned}
 P(x \geq 3) &= 1 - P(x < 3) = 1 - \{P(x = 0) + P(x = 1) + P(x = 2)\} \\
 &= 1 - [e^{-\lambda} + \lambda e^{-\lambda} + \frac{1}{2!} \lambda^2 e^{-\lambda}] \\
 &= 1 - \left\{ e^{-1} + e^{-1} + \frac{e^{-1}}{2!} \right\} = 1 - 2.5 e^{-1} = 1 - 2.5 (0.3679) \\
 &= 0.0802
 \end{aligned}$$

Example 7.9: A new automated production process has had an average of 1.5 breakdowns per day. Because of the cost associated with a breakdown, management is concerned about the possibility of having three or more breakdowns during a day. Assume that breakdowns occur randomly, that the probability of a breakdown is the same for any two time intervals of equal length, and that breakdowns in one period are independent of breakdowns in other periods. What is the probability of having three or more breakdowns during a day?

Solution: Given that $\lambda = np = 1.5$ breakdowns per day. Thus, probability of having three or more breakdowns during a day is given by

$$\begin{aligned}
 P(x \geq 3) &= 1 - P(x < 3) = 1 - [P(x=0) + P(x=1) + P(x=2)] \\
 &= 1 - \left[\frac{\lambda^0 e^{-\lambda}}{0!} + \frac{\lambda e^{-\lambda}}{1!} + \frac{\lambda^2 e^{-\lambda}}{2!} \right] \\
 &= 1 - e^{-\lambda} \left[1 + \lambda + \frac{1}{2} \lambda^2 \right] = 1 - 0.2231 \left[1 + 1.5 + \frac{1}{2} (1.5)^2 \right] \\
 &= 1 - 0.2231 (3.625) = 1 - 0.8088 = 0.1912
 \end{aligned}$$

Example 7.10: Suppose a life insurance company insures the lives of 5000 persons aged 42. If studies show the probability that any 42-years-old person will die in a given year to be 0.001, find the probability that the company will have to pay at least two claims during a given year.

Solution: Given that, $n = 5000$, $p = 0.001$, so $\lambda = np = 5000 \times 0.001 = 5$. Thus, the probability that the company will have to pay at least 2 claims during a given year is given by

$$\begin{aligned}
 P(x \geq 2) &= 1 - P(x < 2) = 1 - [P(x=0) + P(x=1)] \\
 &= 1 - [e^{-\lambda} + \lambda e^{-\lambda}] = 1 - [e^{-5} + 5e^{-5}] = 1 - 6e^{-5} \\
 &= 1 - 6 \times 0.0067 = 0.9598
 \end{aligned}$$

Example 7.11: A manufacturer who produces medicine bottles, finds that 0.1 per cent of the bottles are defective. The bottles are packed in boxes containing 500 bottles. A drug manufacturer buys 100 boxes from the producer of bottles. Using Poisson distribution, find how many boxes will contain:

- no defectives
- at least two defectives

Solution: Given that $p = 1$ per cent $= 0.001$, $n = 500$, $\lambda = np = 500 \times 0.001 = 0.5$

$$(a) P[x = 0] = e^{-\lambda} = e^{-0.5} = 0.6065$$

Therefore, the required number of boxes are : $0.6065 \times 100 = 61$ (approx.)

$$\begin{aligned}
 (b) P(x > 2) &= 1 - P(x \leq 1) = 1 - [P(x=0) + P(x=1)] \\
 &= 1 - [e^{-\lambda} + \lambda e^{-\lambda}] = 1 - [0.6065 + 0.5(0.6065)] \\
 &= 1 - 0.6065 (1.5) = 1 - 0.90975 = 0.09025.
 \end{aligned}$$

Therefore, the required number of boxes are $100 \times 0.09025 = 9$ (approx.)

Conceptual Questions 7B

- If x has a Poisson distribution with parameter λ , then show that $E(x)$ and $V(x) = \lambda$. Further, show that the Poisson distribution is a limiting form of the binomial distribution.
- What is Poisson distribution? Point out its role in business decision making. Under what conditions will it tend to become a binomial distribution?
- When can Poisson distribution be a reasonable approximation of the binomial?
- Discuss the distinctive features of Poisson distribution. When does a binomial distribution tend to become a Poisson distribution?
- Under what conditions is the Poisson probability distribution appropriate? How are its mean and variance calculated?

Self-practice Problems 7B

- 7.10** In a town, 10 accidents took place in a span of 50 days. Assuming that the number of accidents per day follows the Poisson distribution, find the probability that there will be three or more accidents in a day.
- 7.11** Find the probability that at most 5 defective bolts will be found in a box of 200 bolts if it is known that 2 per cent of such bolts are expected to be defective [you may take the distribution to be Poisson; $e^{-4} = 0.0183$].
- 7.12** On an average, one per cent in 400 items is defective. If the items are packed in boxes of 100, what is the probability that any given box of items will contain: (i) no defectives; (ii) less than two defectives; (iii) one or more defectives; and (iv) more than three defectives
- 7.13** It is given that 30 per cent of electric bulbs manufactured by a company are defective. Find the probability that a sample of 100 bulbs will contain (i) no defective, and (ii) exactly one defective.
- 7.14** A factory produces blades in packets of 10. The probability of a blade to be defective is 0.2 per cent. Find the number of packets having two defective blades in a consignment of 10,000 packets.
- 7.15** In a certain factory manufacturing razor blades, there is small chance $1/50$ for any blade to be defective. The blades are placed in packets, each containing 10 blades. Using an appropriate probability distribution, calculate the approximate number of packets containing not more than 2 defective blades in a consignment of 10,000 packets.
- 7.16** Suppose that a manufactured product has 2 defects per unit of product inspected. Using Poisson distribution, calculate the probabilities of finding a product without any defect, 3 defects, and 4 defects. (Given $e^{-2} = 0.135$)
- 7.17** A distributor received a shipment of 12 TV sets. Shortly after this shipment was received, the manufacturer informed that he had inadvertently shipped 3 defective sets. The distributor decided to test 4 sets randomly selected out of 12 sets received.
- (a) What is the probability that neither of the 4 sets tested was defective?
- (b) What is the mean and variance of defective sets.
- 7.18** Suppose a population contains 10 elements, 6 of which are defective. A sample of 3 elements is selected. What is the probability that exactly 2 are defective?
- 7.19** A transport company has a fleet of 15 trucks, used mainly to deliver fruits to wholesale market. Suppose 6 of the 15 trucks have brake problems. Five trucks were selected at random to be tested. What is the probability that 2 of those tested trucks have defective brakes?

Hints and Answers

- 7.10** The average number of accidents per day
 $= 10/50 = 0.2$
 $P(x \geq 3 \text{ accidents})$
 $= 1 - P(2 \text{ or less accidents})$
 $= 1 - [P(0) + P(1) + P(2)]$
 $= 1 - \left(e^{-2} + 0.2e^{-2} + \frac{e^{-2} \times 0.2 \times 0.2}{2} \right)$
- $= 1 - e^{-2} (1 + 0.2 + 0.02)$
 $= 1 - 0.8187 \times 1.22 \quad (\text{From table of } e^{-\lambda})$
 $= 1 - 0.999 = 0.001$
- 7.11** $p(\text{defective bolt}) = 2 \text{ per cent} = 0.02$. Given $n = 200$, so $\lambda = np = 200 \times 0.02 = 4$
- $P(0) = \frac{e^{-\lambda} \lambda^0}{0!} = e^{-4} = 0.0183$

$$P(x \leq 5)$$

$$= P(x = 0) + P(x = 1) + P(x = 2) +$$

$$P(x = 3) + P(x = 4) + P(x = 5)$$

$$= e^{-4} \left(1 + 4 + \frac{4^2}{2!} + \frac{4^3}{3!} + \frac{4^4}{4!} + \frac{4^5}{5!} \right)$$

$$= 0.0183 \times (643/15) = 0.7845.$$

$$\mathbf{7.13} \quad \lambda = np = 100 \times 0.30 = 3$$

$$P(x = 0) = e^{-\lambda} = e^{-3} = 0.05;$$

$$P(x = 1) = \lambda P(x = 0) = 3 \times 0.05 = 0.15$$

$$\mathbf{7.14} \quad \text{Given } n = 10, p = 0.002, \lambda = np = 10 \times 0.002 = 0.02.$$

$$\begin{aligned} P(x = 2) &= \frac{e^{-\lambda} \lambda^2}{2!} \\ &= \frac{e^{-0.02} (0.02)^2}{2!} \\ &= 0.000196 \end{aligned}$$

The required number of packets having two defective blades each in a consignment of 10,000 packets = $10,000 \times 0.000196 \cong 2$.

$$\mathbf{7.15} \quad \text{Given } N = 10,000, p = 1/50 \text{ and } n = 10.$$

Thus $\lambda = np = 0.20$ and

$$\begin{aligned} NP(x = 0) &= 10,000 e^{-\lambda} \\ &= 10,000 e^{-0.20} = 8187; \end{aligned}$$

$$\begin{aligned} P(x = 1) &= NP(x = 0) \times \lambda \\ &= 8187 \times 0.2 = 1637.4. \end{aligned}$$

$$\begin{aligned} NP(x = 2) &= NP(x = 1) \times \frac{\lambda}{2} \\ &= 1637.4 \times \frac{0.2}{2} \\ &= 163.74 \end{aligned}$$

The approximate number of packets containing not more than 2 defective blades in a consignment of 10,000 packets will be: $10,000 - (8187 + 1637.40 + 163.74) = 12$ approx.

$$\mathbf{7.16} \quad \text{Given average number of defects, } \lambda = 2.$$

$$P(x = 0) = e^{-\lambda} = e^{-2} = 0.135;$$

$$\begin{aligned} P(x = 1) &= P(x = 0) \times \lambda \\ &= 0.135 \times 2 = 0.27 \end{aligned}$$

$$\begin{aligned} P(x = 2) &= P(x = 1) \times \frac{\lambda}{2} \\ &= 0.27 \times \frac{2}{2} = 0.27 \end{aligned}$$

$$\begin{aligned} P(x = 3) &= P(x = 2) \times \frac{\lambda}{3} \\ &= 0.27 \times \frac{2}{3} = 0.18 \end{aligned}$$

$$\begin{aligned} P(x = 4) &= P(x = 3) \times \frac{\lambda}{4} \\ &= 0.18 \times \frac{2}{4} = 0.09 \end{aligned}$$

$$\mathbf{7.17} \quad \text{Given } N = 12, n = 4, m = 3 \text{ and } N - m = 9.$$

$$\begin{aligned} \text{(a) } P(x = 0) &= \frac{{}^3C_0 \times {}^9C_4}{{}^{12}C_4} = \frac{\frac{3!}{0!3!} \times \frac{9!}{4!5!}}{\frac{12!}{4!8!}} \\ &= \frac{1 \times 126}{495} = \frac{14}{55} \end{aligned}$$

$$\text{(b) Mean, } \mu = n \left(\frac{m}{N} \right) = 4 \left(\frac{3}{12} \right) = 1$$

$$\begin{aligned} \text{Variance, } \sigma^2 &= n \left(\frac{m}{N} \right) \left(\frac{N-m}{N} \right) \left(\frac{N-n}{N-1} \right) \\ &= 4 \left(\frac{3}{12} \right) \left(\frac{9}{12} \right) \left(\frac{8}{11} \right) = 0.5455 \end{aligned}$$

$$\mathbf{7.18} \quad P(x = 2) = \frac{{}^6C_2 \times {}^4C_1}{{}^{10}C_3} = \frac{15 \times 4}{120} = 0.50$$

$$\mathbf{7.19} \quad P(x = 2) = \frac{{}^9C_3 \times {}^6C_2}{{}^{15}C_5} = \frac{84 \times 15}{3003} = 0.4196$$

7.3 CONTINUOUS PROBABILITY DISTRIBUTIONS

If a random variable is discrete, then it is possible to assign a specific probability to each of its value and get the probability distribution for it. The sum of all the probabilities associated with the different values of the random variable is 1. However, not all experiments result in random variables that are discrete. Continuous random variables such as height, time, weight, monetary values, length of life of a particular product, etc. can take large number of observable values corresponding to points on a line interval much like the infinite number of gains of sand on a beach. The sum of probability to each of these infinitely large values is no longer sum to 1.

Unlike discrete random variables, continuous random variables do not have probability distribution functions specifying the exact probabilities of their specified values. Instead, probability distribution is created by distributing one unit of probability along the real line, much like distributing a handful of sand along a line. The probability of measurements (e.g. gains of sand) piles up in certain places resulting into a probability distribution called *probability density function*. Such distribution is used to find probabilities that the random variable falls into a specified interval of values.

Certain characteristics of probability density function for the continuous random variable x are follows:

- (i) The area under a continuous probability distribution is equal to 1.
- (ii) The probability $P(a \leq x \leq b)$ that random variable x value will fall into a particular interval from a to b is equal to the area under the density curve between the points (values) a and b .

Nature seems to follow a predictable pattern for many kinds of measurements. Most numerical values of a random variable are spread around the center, and greater the distance a numerical value has from the center, the fewer numerical values have that specific value. A frequency distribution of values of random variable observed in nature which follows this pattern is approximately bell shaped. A special case of distribution of measurements is called a **normal curve (or distribution)**.

If a population of numerical values follows a normal curve and x is the randomly selected numerical value from the population, then x is said to be *normal random variable*, which has a normal probability distribution.

7.4.1 Normal Probability Distribution Function

The formula that generates normal probability distribution is as follows:

$$f(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{(-1/2)[(x-\mu)/\sigma]^2}, \quad -\infty < x < \infty \quad (7-6)$$

where π = constant 3.1416

e = constant 2.7183

μ = mean of the normal distribution

σ = standard of normal distribution

The $f(x)$ values represent the relative frequencies (height of the curve) within which values of random variable x occur. The graph of a normal probability distribution with mean μ and standard deviation σ is shown in Fig. 7.1. The distribution is symmetric about its mean μ that locates at the centre.

Since the total area under the normal probability distribution is equal to 1, the symmetry implies that the area on either side of μ is 50 per cent or 0.5. The *shape* of the distribution is determined by μ and σ values.

In symbols, if a random variable x follows normal probability distribution with mean μ and standard deviation σ , then it is also expressed as: $x \sim N(\mu, \sigma)$.

Standard Normal Probability Distribution To deal with problems where the normal probability distribution is applicable more simply, it is necessary that a random variable x is standardized by expressing its value as the number of standard deviations (σ) it lies to the left or right of its mean (μ). The *standardized normal random variable* z (also called *z-statistic*, *z-score* or *normal variate*) is defined as:

$$z = \frac{x - \mu}{\sigma} \quad (7-7)$$

or equivalently $x = \mu + z\sigma$

A z-score measures the number of standard deviations that a value of the random variable x fall from the mean. From formula (7.7), we may conclude that

- (i) When x is less than the mean (μ), the value of z is negative
- (ii) When x is more than the mean (μ), the value of z is positive
- (iii) When $x = \mu$, the value of $z = 0$.

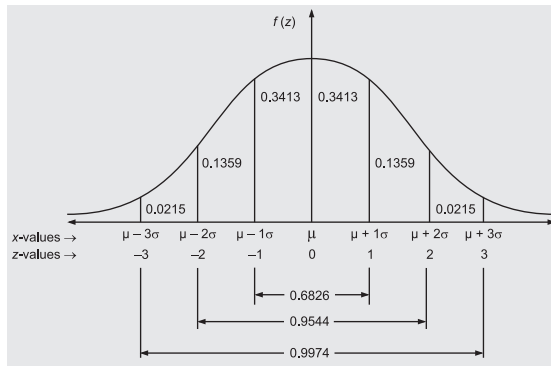


Figure 7.1 Standard Normal Distribution

Any normal probability distribution with a set of μ and σ value with random variable can be converted into a distribution called **standard normal probability distribution** z , as shown in Fig. 7.1, with mean $\mu_z = 0$ and standard deviation $\sigma_z = 1$ with the help of the formula (7-7).

A z -value measures the distance between a particular value of random variable x and the mean (μ) in units of the standard deviation (σ). With the value of z obtained by using the formula (7.7), we can find the area or probability of a random variable under the normal curve by referring to the standard distribution in Appendix. For example $z = \pm 2$ implies that the value of x is 2 standard deviations above or below the mean (μ).

Area Under the Normal Curve Since the range of normal distribution is infinite in both the directions away from μ , the *pdf* function $f(x)$ is never equal to zero. As x moves away from μ , $f(x)$ approaches x -axis but never actually touches it.

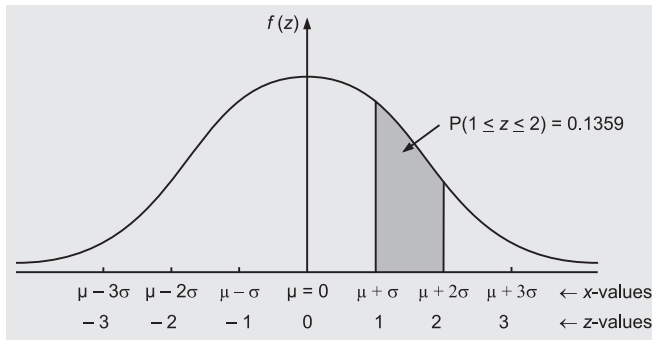


Figure 7.2 Diagram for Finding $P(1 < z < 2)$

The area under the standard normal distribution between the mean $z = 0$ and a specified positive value of z , say z_0 is the probability $P(0 \leq z \leq z_0)$ and can be read off directly from standard normal (z) tables. For example, area between $1 \leq z \leq 2$ is the proportion of the area under the curve which lies between the vertical lines erected at two points along the x -axis. A portion of the table is shown in Table 7.2. For example, as shown in Fig. 7.2, if x is σ away from μ , that is, the distance between x and μ is one standard deviation or $(x - \mu)/\sigma = 1$, then 34.134 per cent of the distribution lies between x and μ . Similarly, if x is at 2σ away from μ , that is, $(x - \mu)/\sigma = 2$, then the area will include 47.725 per cent of the distribution, and so on, as shown in Table 7.1.

Table 7.1: Area Under the Normal Curve

$z = \frac{x - \mu}{\sigma}$	Area Under Normal Curve Between x and μ
1.0	0.34134
2.0	0.47725
3.0	0.49875
4.0	0.49997

Since the normal distribution is symmetrical, Table 7.6 indicates that about 68.26 per cent of the normal distribution lies within the range $\mu - \sigma$ to $\mu + \sigma$. The other relationships derived from Table 7.1 are shown in Table 7.2 and Fig. 7.1.

Table 7.2: Percentage of the Area of the Normal Distribution Lying within the Given Range

Number of Standard Deviations from Mean	Approximate Percentage of Area under Normal Curve
$x \pm \sigma$	68.26
$x \pm 2\sigma$	95.45
$x \pm 3\sigma$	99.75

7.4.2 Approximation of Binomial and Poisson Distributions to Normal Distribution

The binomial distribution approaches a normal distribution with standardized variable, that is,

where
$$z = \frac{x - np}{\sqrt{npq}} \sim N(0, 1)$$

However, this approximation works well when both $np \geq 10$ and $npq \geq 10$

Similarly, Poisson distribution also approaches a normal distribution with standardized variable, that is,

$$z = \frac{x - \lambda}{\sqrt{\lambda}} \sim N(0, 1)$$

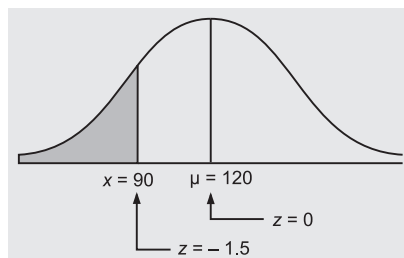
Example 7.12: 1000 light bulbs with a mean life of 120 days are installed in a new factory and their length of life is normally distributed with standard deviation of 20 days.

- (a) How many bulbs will expire in less than 90 days?
 (b) If it is decided to replace all the bulbs together, what interval should be allowed between replacements if not more than 10 per cent should expire before replacement?

Solution: (a) Given, $\mu = 120$, $\sigma = 20$ and $x = 90$.

$$\text{Then, } z = \frac{x - \mu}{\sigma} = \frac{90 - 120}{20} = -1.5$$

The area under the normal curve between $z = 0$ and -1.5 is 0.4332. Therefore area to the left of -1.5 is $0.5 - 0.4332 = 0.0668$. Thus, the expected number of bulbs to expire in less than 90 days will be $0.0668 \times 1000 = 67$ (approx.).



- (b) The value of z corresponding to an area 0.4 (0.5 - 0.10). Under the normal curve is 1.28. Therefore,

$$z = \frac{x - \mu}{\sigma} \text{ or } -1.28 = \frac{x - 120}{20} \text{ or}$$

$$x = 120 - 20(-1.28) = 94$$

Hence, the bulbs will have to be replaced after 94 days.

Example 7.13: The lifetimes of certain kinds of electronic devices have a mean of 300 hours and standard deviation of 25 hours. Assuming that the distribution of these lifetimes, which are measured to the nearest hour, can be approximated closely with a normal curve.

- (a) Find the probability that any one of these electronic devices will have a lifetime of more than 350 hours.
 (b) What percentage will have lifetimes of 300 hours or less?
 (c) What percentage will have lifetimes from 220 or 260 hours?

Solution: (a) Given, $\mu = 300$, $\sigma = 25$ and $x = 350$. Then,

$$z = \frac{x - \mu}{\sigma} = \frac{350 - 300}{25} = 2$$

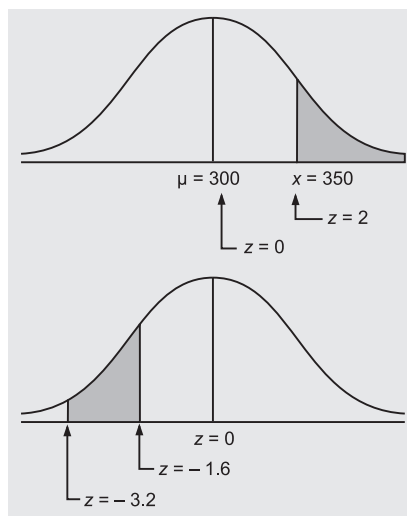
The area under the normal curve between $z = 0$ and $z = 2$ is 0.9772. Thus, the required probability is $1 - 0.9772 = 0.0228$.

$$(b) z = \frac{x - \mu}{\sigma} = \frac{300 - 300}{25} = 0$$

Therefore, the required percentage is $0.5000 \times 100 = 50$ per cent.

- (c) Given, $x_1 = 220$, $x_2 = 260$, $\mu = 300$ and $\sigma = 25$. Thus,

$$z_1 = \frac{220 - 300}{25} = -3.2 \text{ and } z_2 = \frac{260 - 300}{25} = -1.6$$



From the normal table, we have

$$P(z = -1.6) = 0.4452 \text{ and } P(z = -3.2) = 0.4903$$

Thus, the required probability is

$$P(z = -3.2) - P(z = -1.6) = 0.4903 - 0.4452 = 0.0541$$

Hence the required percentage = $0.0541 \times 100 = 5.41$ per cent.

Example 7.14: In an intelligence test given to 1500 candidates appearing in an examination carrying a maximum of 100 marks, it was found that marks were normally distributed with mean 39.5 and S.D. = 12.5. Determine the number of candidates who secured a first class for which a minimum of 60 marks is necessary.

Solution: Given Mean (μ) = 39.5, S.D. (σ) = 12.5 and $x = 60$. Then

$$\begin{aligned} P(x \geq 60) &= P\left(\frac{x - \mu}{\sigma} \geq \frac{60 - \mu}{\sigma}\right) = P\left(z \geq \frac{60 - 39.5}{12.5}\right) \\ &= P(z > 1.64) = 0.5 - P(z < 1.64) \\ &= 0.5 - 0.4495 = 0.0505 \end{aligned}$$

Students securing first class = $1500 \times 0.0505 = 76$ students (approx).

Example 7.15: The mean (μ) of a binomial distribution is 40 and standard deviation (σ) is 2. Find n , p , q .

Solution: Given mean ($\mu = np$) = 40 and S.D. ($\sigma = \sqrt{npq}$) = 2 or $npq = 4$. Then

$$\frac{npq}{np} = \frac{4}{40} = 0.1 \text{ or } q = 0.1 \text{ and } p = 1 - 0.1 = 0.9$$

$$\text{Mean } (np = 40) \text{ or } n \times 0.9 = 40, \text{ i.e. } n = 40/0.9 = 44.4$$

Example 7.16: Time taken by a crew of a company to construct a small bridge is a normal variate with mean 400 labour hours and standard deviation of 100 labour hours.

- What is the probability that the bridge gets constructed between 350 to 450 labour hours?
- If the company promises to construct the bridge in less than 450 labour hours and agrees to pay a penalty of ₹100 for each labour hour spent in excess of 450, what is the probability that the company pays a penalty of at least ₹2000? [Delhi Univ., B.A. Eco. (Hons.), 1998]

Solution: Let x be the time (in labour hours) to construct the bridge. Given mean, $\mu = 400$, and $\sigma = 100$.

- Probability that the bridge gets constructed between 350 and 450 labour hours

$$\begin{aligned} P(350 \leq x \leq 450) &= P\left(\frac{350 - \mu}{\sigma} \leq \frac{x - \mu}{\sigma} \leq \frac{450 - \mu}{\sigma}\right) \\ &= P\left(\frac{350 - 400}{100} \leq z \leq \frac{450 - 400}{100}\right) \\ &= P\left(\frac{-50}{100} \leq z \leq \frac{50}{100}\right) = P(-0.5 \leq z \leq 0.5) \\ &= 2P(0 \leq z \leq 0.5) = 2 \times 0.1915 = 0.3830. \end{aligned}$$

- The penalty for each labour hour in excess of 450 is ₹100. If the minimum penalty paid is ₹2000, then delay in completing the project is more than $2000/100 = 20$ hours.

Company will take a minimum $(450 + 20) = 470$ hrs to complete the bridge. Thus, the required probability is

$$\begin{aligned}
 P(x \geq 470) &= P\left(\frac{x - \mu}{\sigma} \geq \frac{470 - \mu}{\sigma}\right) = P\left(z \geq \frac{470 - 400}{100}\right) \\
 &= P\left(z \geq \frac{70}{100}\right) = P(z \geq 0.7) \\
 &= 0.5 - P(0 \leq z \leq 0.7) = 0.5 - 0.2580 = 0.2420.
 \end{aligned}$$

Example 7.17: The marks of the students in a certain examination are normally distributed with mean marks as 40 per cent and standard deviation marks at 20 per cent. On this basis, 60 per cent students failed. The result was moderated and 70 per cent student passed. Find the pass marks before and after moderation. [Delhi Unit., B.Com. (H.), 2002]

Solution: Let x be the per cent of marks before and after moderation to pass the examination. Given mean (μ) = 40 and SD(σ) = 20

Before Moderation: Let pass percentage of marks be x_1 per cent. Then 60 per cent students failed imply that 40 per cent students passed.

$$\begin{aligned}
 P(x \geq x_1) &= 0.40 \\
 P\left(z \geq \frac{x_1 - \mu}{\sigma}\right) &= 0.40 \\
 P(z \geq z_1) &= 0.40 \quad (\text{Because } z_1 \text{ is located to the right of } z = 0) \\
 0.5 - P(0 < z < z_1) &= 0.40, \text{ i.e. } P(0 < z < z_1) = 0.10
 \end{aligned}$$

$$\begin{aligned}
 \text{Now,} \quad z_1 &= \frac{x_1 - \mu}{\sigma} = 0.25 \quad (z = 0.25 \text{ from normal distribution table}) \\
 x_1 &= 0.25\sigma + \mu = 0.25 \times 20 + 40 = 45
 \end{aligned}$$

Hence, pass percentage is 45 per cent before moderation.

After Moderation: Let pass percentage of marks be x_2 per cent. Then

$$\begin{aligned}
 P(x \geq x_2) &= 0.70 \\
 P\left(Z \geq \frac{x_2 - \mu}{\sigma}\right) &= 0.70 \\
 \Rightarrow \quad P(z \geq -z_2) &= 0.70 \quad (\text{because } z_2 \text{ is located to left of } z = 0) \\
 \text{or} \quad P(z \geq -z_2) &= 0.5 + P(-z_2 < z < 0) \\
 0.70 &= 0.5 + P(0 \leq z \leq z_2), \text{ i.e. } P(0 \leq z \leq z_2) = 0.2
 \end{aligned}$$

From normal table, we get

$$\begin{aligned}
 z_2 &= 0.525 \quad \text{or} \quad \frac{x_2 - \mu}{\sigma} = -0.525 \\
 x_2 &= -0.525\sigma + \mu = -0.525(20) + 40 = 29.5
 \end{aligned}$$

Hence, pass percentage is 29.5 per cent after moderation.

Example 7.18: In a certain examination, the percentage of passes and distinctions were 46 and 9, respectively. Estimate the average marks obtained by the candidates, the minimum pass and distinction marks being 40 and 75, respectively (assume the distribution of marks to be normal).

Also determine what would have been the minimum qualifying marks for admission to a re-examination of the failed candidates, had it been desired that the best 25 per cent of them should be given another opportunity of being examined.

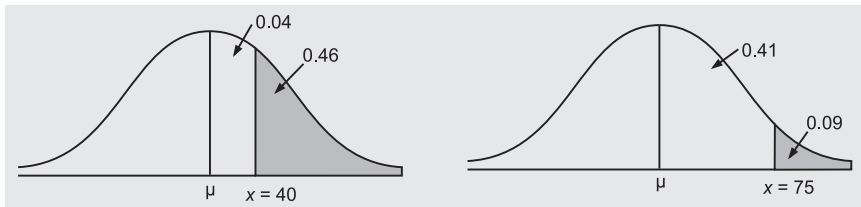
Solution: (a) Let μ be the mean and σ be the standard deviation of the normal distribution. The area to the right of the ordinate at $x = 40$ is 0.46 and hence the area between the mean and the ordinate at $x = 40$ is 0.04.

Now from the normal table, corresponding to 0.04, the standard normal variate, $z = 0.1$. Therefore, we have

$$\frac{40 - \mu}{\sigma} = 0.1 \text{ or } 40 - \mu = 0.1\sigma$$

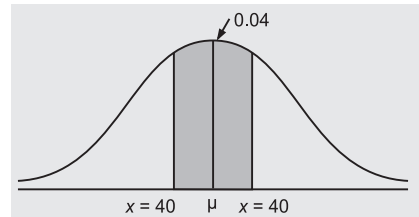
$$\text{Similarly, } \frac{75 - \mu}{\sigma} = 1.34 \text{ or } 75 - \mu = 1.34\sigma$$

Solving these equations, we get $\sigma = 28.23$ and $\mu = 37.18$ or 37.



(b) Let us assume that x_1 is the minimum qualifying marks for re-examination of the failed candidates.

The area to the right of $x = 40$ is 46 per cent. Thus the percentage of students failing = 54 and this is the area to the left of 40. We want that the best 25 per cent of these failed candidates should be given a chance to reappear. Suppose this area is equal to the shaded area in the diagram. This area is 25 per cent of 54 = 13.5 per cent = 0.1350.



The area between mean and ordinate at $x_1 = -(0.1350 - 0.04) = -0.0950$ (negative sign is included because the area lies to the left of the mean ordinates).

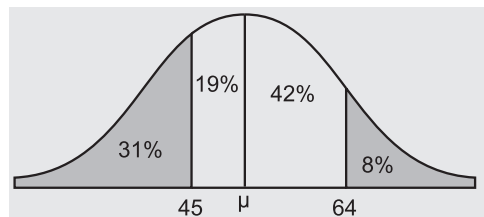
Corresponding to this area, the standard normal variate $z = -0.0378$. Thus, we write

$$\frac{x_1 - \mu}{\sigma} = -0.0378$$

$$\begin{aligned} x_1 &= \mu - 0.0378\sigma \\ &= 37.2 - (0.0378 \times 28.23) \\ &= 37.2 - 1.067 = 36.133 \text{ or } 36 \text{ (approx.)} \end{aligned}$$

Example 7.19: In a normal distribution, 31 per cent of the items are under 45 and 8 per cent are over 64. Find the mean and standard deviation of the distribution.

Solution: Since 31 per cent of the items are under 45, therefore the left of the ordinate at $x = 45$ is 0.31, and obviously the area to the right of the ordinate up to the mean is $(0.5 - 0.31) = 0.19$. The value of z corresponding to this area is 0.5. Hence,



$$z = \frac{45 - \mu}{\sigma} = -0.5 \text{ or } -\mu + 0.5\sigma = -45$$

As 8 per cent of the items are above 64, therefore area to the right of the ordinate at 64 is 0.08. Area to the left of the ordinate at $x = 64$ up to mean ordinate is $(0.5 - 0.08) = 0.42$ and the value of z corresponding to this area is 1.4. Hence,

$$z = \frac{64 - \mu}{\sigma} = 1.4 \text{ or } -\mu - 1.4\sigma = -64$$

From these two equations, we get $1.9\sigma = 19$ or $\sigma = 10$. Putting $\sigma = 10$ in the first equation, we get $\mu - 0.5 \times 10 = 45$ or $\mu = 50$. Thus, mean of the distribution is 50 and standard deviation 10.

If it is necessary to order in whole units, then the wholesale distributor should order 147 tonnes.

Example 7.20: Assume that the test scores from a college admissions test are normally distributed with a mean of 450 and a standard deviation of 100.

- What percentage of people taking the test score are between 400 and 500?
- Suppose someone received a score of 630. What percentage of the people taking the test score better? What percentage score worse?
- If a particular university will not admit any one scoring below 480, what percentage of the persons taking the test would be acceptable to the university?

Solution: (a) Given $\mu = 450$ and $\sigma = 100$. Let x be the test score. Then

$$z_1 = \frac{x - \mu}{\sigma} = \frac{500 - 450}{100} = 0.5$$

and

$$z_2 = \frac{x - \mu}{\sigma} = \frac{400 - 450}{100} = -0.5$$

The area under the normal curve between $z = 0$ and $z = 0.5$ is 0.1915

The required probability that the score falls between 400 and 500 is

$$P(400 \leq x \leq 500) = P(-0.5 \leq z \leq 0.5) = 0.1915 + 0.1915 = 0.3830$$

So the percentage of the people taking the test score between 400 and 500 is 38.30 per cent.

- (b) Given $x = 630$, $\mu = 450$ and $\sigma = 100$. Thus

$$z = \frac{x - \mu}{\sigma} = \frac{630 - 450}{100} = 1.8$$

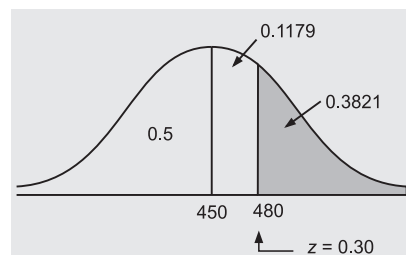
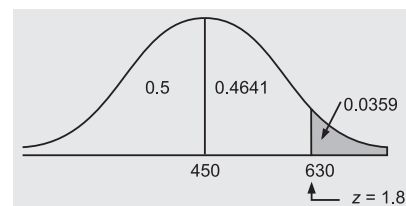
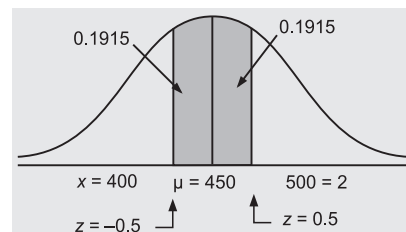
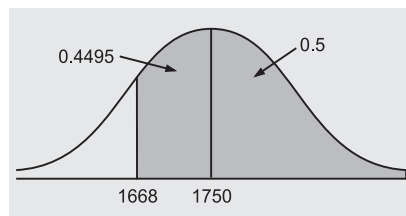
The area under the normal curve between $z = 0$ and $z = 1.8$ is 0.4641.

The probability that people taking the test score better is given by

$$\begin{aligned} P(x \geq 630) &= P(z \geq 1.8) \\ &= 0.5000 + 0.4641 = 0.9641 \end{aligned}$$

That is, 96.41 per cent people score better

The probability that people taking the test score worse is given by



$$P(x \leq 630) = P(z \leq 1.8) = 0.5000 - 0.4641 = 0.0359$$

That is, 3.59 per cent people score worse

(c) Given $x = 480$, $\mu = 450$ and $\sigma = 100$. Thus,

$$z = \frac{x - \mu}{\sigma} = \frac{480 - 450}{100} = 0.30$$

The area under the normal curve between $z=0$ and $z=0.30$ is 0.1179. So,

$$P(x \geq 480) = P(z \geq 0.30) = 0.5000 + 0.1179 = 0.6179$$

The percentage of people who score more than 480 and are acceptable to the university is 61.79 per cent.

Conceptual Questions 7C

12. State the conditions under which a binomial distribution tends to (i) Poisson distribution, (ii) normal distribution. Write down the probability functions of binomial and Poisson distributions.
13. Normal distribution is symmetric with a single peak. Does this mean that all symmetric distributions are normal? Explain.
14. When finding probabilities with a normal curve we always deal with intervals; the probability of a single value of x is defined equal to zero. Why is this so?
15. What are the parameters of normal distribution? What information is provided by these parameters?
16. What are the chief properties of normal distribution? Describe briefly the importance of normal distribution in statistical analysis.
17. Discuss the distinctive features of the binomial, Poisson, and normal distributions. When does a binomial distribution tend to become a normal distribution?

Self-practice Problems 7C

- 7.20** A cigarette company wants to promote the sales of X's cigarettes (brand) with a special advertising campaign. Fifty out of every thousand cigarettes are rolled up in gold foil and randomly mixed with the regular (special king sized, mentholated) cigarettes. The company offers to trade a new pack of cigarettes for each gold cigarette, a smoker finds in a pack of brand X. What is the probability that buyers of brand X will find $X = 0, 1, 2, 3, \dots$ gold cigarettes in a single pack of 10?

- 7.21** You are in charge of rationing in a State affected by food shortage. The following reports were received from investigators:

Daily calories of food available per adult during current period

Area	Mean	S.D.
A	2000	350
B	1750	100

The estimated daily requirement of an adult is taken as 2500 calories and the absolute minimum is 1000. Comment on the reported figures and determine which area in your opinion needs more urgent attention.

- 7.22** Assume the mean height of soldiers to be 68.22 inches with a variance of 10.8 inches. How many soldiers in a regiment of 1,000 would you expect to be over six feet tall?
- 7.23** In an intelligence test administered to 1000 students, the average score was 42 and standard deviation 24.

Find (a) the number of students exceeding a score of 50, (b) the number of students lying between 30 and 54 and (c) the value of the score exceeded by the top 100 students.

- 7.24** An aptitude test for selecting officers in a bank was conducted on 1000 candidates. The average score is 42 and the standard deviation of scores is 24. Assuming normal distribution for the scores, find:

- the number of candidates whose scores exceeds 58.
- the number of candidates whose scores lie between 30 and 66.

- 7.25** A workshop produces 2000 units of an item per day. The average weight of units

is 130 kg with a standard deviation of 10 kg. Assuming normal distribution, how many units are expected to weigh less than 142 kg?

- 7.26** Management of a company is considering adopting a bonus system to increase production. One suggestion is to pay a bonus on the highest 5 per cent of production based on past experience. Past records indicate that, on the average, 4000 units of a small assembly are produced during a week. The distribution of the weekly production is approximately normal with a standard deviation of 60 units. If the bonus is paid on the upper 5 per cent of production, the bonus will be paid on how many units or more?

Hints and Answers

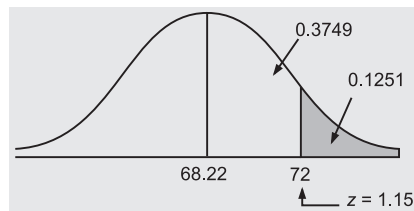
- 7.20** An experiment, with $n = 10$ trials, probability of finding a golden cigarette (a success) is $p = 50/100 = 0.05$. The expected number of golden cigarettes per pack is, $\lambda = np = 10(0.05)$.

Number of Golden Cigarettes per Pack	Probability
0	0.6065
1	0.3033
2	0.0758
3	0.0126
4	0.0016

- 7.21**
- | Area A | Area B |
|------------------------------|------------------------------|
| Mean $\pm 3\sigma$ | Mean $\pm 3\sigma$ |
| $= 2,000 \pm (3 \times 350)$ | $= 1,750 \pm (2 \times 100)$ |
| or 950 and 3,050 | 1,450 and 2,050 |
| calories | calories |

Since the estimated requirement is minimum of 1,000 calories, area A needs more urgent attention.

- 7.22** Assuming that the distribution of height is normal. Given that, $x = 72$ inches, $\mu = 68.22$, $\sigma = \sqrt{10.8} = 3.286$. Therefore,
- $$z = \frac{x - \mu}{\sigma} = \frac{72 - 68.22}{3.286} = 1.15$$



Area to the right of $z = 1.15$ from the normal table is $(0.5000 - 0.3749) = 0.1251$. Probability of getting soldiers above six feet is 0.1251 and their expected number is $0.1251 \times 1000 = 125$.

- 7.23** (a) Given $\mu = 42$, $x = 50$, $\sigma = 24$. Thus

$$z = \frac{x - \mu}{\sigma} = \frac{50 - 42}{24} = 0.333$$

Area to the right of $z = 0.333$ under the normal curve is $0.5 - 0.1304 = 0.3696$

Expected number of children exceeding a score of 50 are $0.3696 \times 1,000 = 370$.

- (b) Standard normal variate for score 30

$$z = \frac{x - \mu}{\sigma} = \frac{30 - 42}{24} = -0.5$$

Standard normal variate for score 54

$$z = \frac{x - \mu}{\sigma} = \frac{54 - 42}{24} = 0.5$$

Area between $z = 0$ to $z = 0.5 = 0.1915$

Area between $z = -0.5$ to $z = 0$ is 0.1915

Area between $z = -0.5$ to $z = 0.5$ is

$$0.1915 + 0.1915 = 0.3830$$

- 7.24** (a) Number of candidates whose score exceeds 58.

$$z = \frac{x - \mu}{\sigma} = \frac{60 - 42}{24} = 0.667$$

Area to the right of $z = 0.667$ under the normal curve is $(0.5 - 0.2476) = 0.2524$

Number of candidates whose score exceeds 60 is: $100 \times 0.2524 = 252.4$ or 252

(b) Number of candidates whose score lies between 30 and 66.

Standard normal variate corresponding to 30,

$$z = \frac{30 - 42}{24} = -0.5$$

Standard normal variate corresponding to 66,

$$z = \frac{66 - 42}{24} = 1$$

Area between $z = -0.5$ and $z = 1$, is

$$0.1915 + 0.3413 = 0.5328$$

Number of candidates whose score lies between 30 and 66 : $1000 \times 0.5328 = 532.8$ or 533

- 7.25** Given $N = 2000$, $\mu = 130$, $\sigma = 10$ and $x = 142$,

$$z = \frac{x - \mu}{\sigma} = \frac{142 - 130}{10} = 1.2 \approx 0.3849$$

$$P(x \leq 142) = 0.5 + 0.3849 = 0.8849$$

Expected number of units weighing less than 142 kg is $2000 \times 0.8849 = 1,770$ approx.

- 7.26** $z = \frac{x - \mu}{\sigma}$ or $1.65 = \frac{x - 4000}{60}$
 $x = 4,099$ units.

Formulae Used

1. Expected value of a random variable x

$$E(x) = \sum x P(x)$$

where x = value of the random variable

$P(x)$ = probability that the random variable will take on the value x .

2. Binomial probability distribution

- Probability of r success in n Bernoulli trials

$$P(x = r) = {}^nC_r p^r q^{n-r} = \frac{n!}{r!(n-r)!} p^r q^{n-r}$$

where p = probability of success

q = probability of failure, $q = 1 - p$

- Mean and standard deviation of binomial distribution

$$\text{Mean } \mu = np$$

$$\text{Standard deviation } \sigma = \sqrt{npq}$$

4. Poisson probability distribution

- Probability of getting exactly r occurrences of random event

$$P(x = r) = \frac{\lambda^r e^{-\lambda}}{r!}$$

where $\lambda = np$, mean number of occurrences per interval of time

$e = 2.71828$, a constant that represents the base of the natural logarithm system

- Mean and standard deviation of Poisson distribution

$$\lambda = np, \sigma = np$$

5. Normal distribution formula:

Number of standard deviations σ a value of random variable x is away from the mean μ of normal distribution:

$$z = \frac{x - \mu}{\sigma}$$

True or False Concepts Quiz

1. The expected value of a random variable describes the long range weighted average of its values.
2. The mean of the binomial distribution is greater than its variance.
3. A binomial distribution is positively skewed when $p > 0.5$.
4. The mean, median and mode always coincide in the normal distribution.
5. The expected value of a random variable is always a non-negative number.
6. The binomial distribution is symmetrical for any value of p (probability of success).
7. Poisson distribution generally describes arrivals at a service facility.
8. In a Bernoulli process, the probability of success must equal the probability of failure.
9. The symmetry of the normal distribution about its mean ensure that its tails extend indefinitely in both the positive and negative directions.
10. All normal distributions are defined by two measures – the mean and the standard deviation.
11. The expected value of a discrete random variable may be determined by taking an average of the values of the random variable.
12. Within 2σ limits from mean, the area under a normal curve is 95.45 per cent.
13. Any course of action that maximizes expected gain also minimizes expected loss.
14. The value of normal variate for some value of the random variable x lying in a normal distribution is the area between x and the mean μ of the distribution.
15. For a given binomial distribution with n fixed, if $p < 0.5$, then distribution will be skewed to the right.

Concepts Quiz Answers

1. T	2. T	3. T	4. T	5. F	6. F	7. T	8. F	9. T
10. F	11. F	12. T	13. F	14. T	15. T			

Review Self-practice Problems

- 7.27** The useful life of a certain brand of radial tyre has been found to follow a normal distribution with mean $\mu = 38,000$ km and standard deviations = 3000 km. If a dealer orders 500 tyres for sale, then
- (a) find the probability that a randomly chosen tyre will have a useful life of at least 35,000 km.
 - (b) find the approximate number of tyres that will last between 40,000 and 45,000 km.
 - (c) If an individual buys 2 tyres, then what is the probability that these tyres will last at least 38,000 km each?

- 7.28** The amount of time consumed by an individual at a bank ATM is found to be normally distributed with mean $\mu = 130$ seconds and standard deviation $\sigma = 45$ seconds.
- What is the probability that a randomly selected individual will consume less than 100 seconds at the ATM?
 - What is the probability that a randomly selected individual will spend between 2 to 3 minutes at the ATM?
 - Within what length of time do 20 per cent of individuals complete their job at the ATM?
 - What is the least amount of time required for individuals with top 5 per cent of required time?
- 7.29** An aptitude test for selecting officers in a bank was conducted on 1000 candidates. The average score is 42 and the standard deviation of scores is 24. Assuming normal distribution for the scores, find the
- number of candidates whose scores exceed 58.
 - number of candidates whose scores lie between 30 and 66.
- [Karnataka Univ., B.Com, 1995]*
- 7.30** The mean inside diameter of a sample of 500 washers produced by a machine is 5.02 mm and the standard deviation is 0.05 mm. The purpose for which these washers are intended allows a maximum tolerance in the diameter of 4.96 to 5.08 mm, otherwise the washers are considered defective. Determine the percentage of defective washers produced by the machine, assuming the diameters are normally distributed.
- 7.31** In a binomial distribution consisting of 5 independent trials, the probability of 1 and 2 successes are 0.4096 and 0.2048, respectively. Find the parameter p of the distribution.
- 7.32** The probability of a bomb hitting a target is 0.20. Two bombs are enough to destroy a bridge. If 6 bombs are aimed at the bridge, find the probability that the bridge will be destroyed.
- 7.33** In an Indian university, it has been found that 25 per cent of the students come from upper income families (U), 35 per cent from middle income families (M), and 40 per cent from lower income families (L). A sample of 10 students is taken at random. What is the probability that the sample will contain 5 students from U, 2 from M and 3 from L?
- 7.34** A firm uses a large fleet of delivery vehicles. Their record over a period of time (during which fleet size utilization may be assumed to have remained suitably constant) shows that the average number of vehicles unserviceable per day is 3. Estimate the probability on a given day when
- all vehicles will be serviceable.
 - more than 2 vehicles will be unserviceable.
- 7.35** The director, quality control of automobile company, while conducting spot checking of automatic transmission, removed ten transmissions from the pool of components and checked for manufacturing defects. In the past, only 2 per cent of the transmissions had such flaws. (Assume that flaws occur independently in different transmissions.)
- What is the probability that sample contains more than two transmissions with manufacturing flaws?
 - What is the probability that none of the selected transmission has any manufacturing flaw?
- 7.36** In the past 2 months, on an average, only 3 per cent of all cheques sent for clearance by a Group Housing Welfare Society (GHWS) have bounced. This month, the GHWS received 200 cheques. What is the probability that exactly ten of these cheques bounced?
- 7.37** Mr Tiwari is campaign manager for a candidate for Lok Sabha. General impression is that the candidate has the support of 40 per cent of registered voters. A random sample of 300 registered voters shows that 34 per cent would vote for the candidate. If 40 per cent of voters really are allied with the candidate, what is the probability that a sample of 300 voters would indicate 34 per cent or fewer on his side? Is it likely that the 40 per cent estimate is correct?

Hints and Answers

$$\begin{aligned} 7.27 \quad (a) \quad z &= \frac{x - \mu}{\sigma} \\ &= \frac{35,000 - 38,000}{3,000} = -1.00 \end{aligned}$$

$$\begin{aligned} P(x \geq 35,000) &= P(z \geq -1.00) \\ &= 0.500 + 0.3413 \\ &= 0.8413 \end{aligned}$$

$$\begin{aligned} (b) \quad z_1 &= \frac{x_1 - \mu}{\sigma} \\ &= \frac{40,000 - 38,000}{3,000} = 0.67 \end{aligned}$$

$$\begin{aligned} z_2 &= \frac{x_2 - \mu}{\sigma} \\ &= \frac{45,000 - 38,000}{3,000} = 2.33 \end{aligned}$$

$$\begin{aligned} P(40,000 \leq x \leq 45,000) \\ &= P(0.67 \leq z \leq 2.33) \\ &= 0.4901 - 0.2486 = 0.2415 \end{aligned}$$

$$(c) \quad P(x \geq 38,000) = P(z \geq 0) = 0.500$$

P(2 tyres each will last at least 38,000 km) = $(0.5000)^2 = 0.2500$ (based on the multiplication rule for the joint occurrence of independent events).

$$7.28 \quad (a) \quad z = \frac{x - \mu}{\sigma} = \frac{100 - 130}{45} = -0.67$$

$$\begin{aligned} P(x < 100) &= P(z < -0.67) \\ &= 0.5000 - 0.2486 \\ &= 0.2514 \end{aligned}$$

$$(b) \quad z_1 = \frac{120 - 130}{45} = -0.22;$$

$$z_2 = \frac{180 - 130}{45} = 1.11$$

$$\begin{aligned} P(120 \leq x \leq 180) &= P(-0.22 \leq z \leq 1.11) \\ &= 0.0871 + 0.3655 \\ &= 0.4526 \end{aligned}$$

$$\begin{aligned} (c) \quad x &= \mu + z\sigma = 130 + (-0.84) 45 \\ &= 92 \text{ sec} \end{aligned}$$

$$\begin{aligned} (d) \quad x &= \mu + z\sigma = 130 + (1.65) 45 \\ &= 204 \text{ sec} \end{aligned}$$

$$7.29 \quad (a) \quad z = \frac{x - \mu}{\sigma} = \frac{58 - 42}{24} = 0.67$$

$$\begin{aligned} P(x > 58) &= P(z > 0.67) \\ &= 0.5000 - 0.2476 = 0.2524 \end{aligned}$$

Expected number of candidates whose score exceeds 58 is $1000 (0.2524) = 2524$

$$(b) \quad z_1 = \frac{30 - 42}{24} = -0.50;$$

$$z_2 = \frac{66 - 42}{24} = 1.00$$

$$\begin{aligned} P(30 \leq x \leq 66) &= P(-0.50 \leq z \leq 1.00) \\ &= 0.1915 + 0.3413 \\ &= 0.5328 \end{aligned}$$

Expected number of candidates whose scores lie between 30 and 66 is $1000 (0.5328) = 5328$.

$$7.30 \quad z_1 = \frac{x - \mu}{\sigma} = \frac{4.96 - 5.02}{0.05} = -1.20;$$

$$z_2 = \frac{5.08 - 5.02}{0.05} = 1.20$$

$$\begin{aligned} P(4.96 \leq x \leq 5.08) &= P(-1.20 \leq z \leq 1.20) \\ &= 2P(0 \leq z \leq 1.20) \\ &= 2(0.3849) \\ &= 7698 \text{ per cent or } 76.98 \end{aligned}$$

per cent

Percentage of defective washers

$$= 100 - 76.98 = 23.02 \text{ per cent.}$$

$$\begin{aligned} 7.31 \quad \text{Given } n = 5; f(x = 1) &= {}^nC_1 p^1 q^{n-1} \\ &= {}^5C_1 p^1 q^4 = 5pq^4 \\ &= 0.4096 \end{aligned}$$

$$\begin{aligned} f(x = 2) &= {}^nC_2 p^2 q^{n-2} = {}^5C_2 p^2 q^3 \\ &= 10p^2 q^3 = 0.2048 \end{aligned}$$

$$\text{Thus, } \frac{f(x = 2)}{f(x = 1)} = \frac{10p^2 q^3}{5pq^4} = \frac{0.2048}{0.4096}$$

$$\text{or } \frac{2p}{q} = \frac{1}{2}$$

$$\text{or } 4p = q (= 1 - p), \text{ i.e. } p = 1/5$$

7.32 Given $p = 0.20$, $q = 0.80$ and $n = 6$. The bridge is destroyed if at least 2 of the bombs hit it. The required probability is

$$\begin{aligned} P(x \geq 2) &= P(x=1) + P(x=2) + \dots + P(x=6) \\ &= 1 - [P(x=0) + P(x=1)] \\ &= 1 - [{}^6C_0(0.80)^6 + {}^6C_1(0.20)(0.80)^5] \\ &= 1 - \frac{2048}{3125} = 0.345 \end{aligned}$$

7.33 Required probability

$$\begin{aligned} &= \frac{10!}{5!3!2!} (0.25)^2 (0.35)^2 (0.40)^3 \\ &= 1.2348 \end{aligned}$$

(Based on the rule of multinomial rule of probability)

7.34 (a) $P(x=0) = \frac{e^{-3}(3)^0}{0!} = 0.0497$

(b) $P(x > 2) = 1 - P(x \leq 2)$

$$\begin{aligned} &= 1 - [P(x=0) + P(x=1) + P(x=2)] \\ &= 1 - \left(e^{-3} + 3e^{-3} + \frac{9}{2}e^{-3} \right) \\ &= 1 - e^{-3} \left(1 + 3 + \frac{9}{2} \right) \\ &= 1 - \frac{17}{2} (0.0497) \\ &= 1 - 0.4224 = 0.5776 \end{aligned}$$

7.35 (a) Given, $p = 0.02$, $q = 0.98$, $n = 10$

$$\begin{aligned} P(x > 2 \text{ flaws}) &= 1 - [P(x=0) + P(x=1) + P(x=2)] \\ &= 1 - [{}^{10}C_0(0.98)^{10} + {}^{10}C_1(0.02)(0.98)^9 + {}^{10}C_2(0.02)^2(0.98)^8] \\ &= 1 - [0.8171 + 0.1667 + 0.0153] \\ &= 0.0009 \end{aligned}$$

(b) $P(x=0 \text{ flaw}) = {}^nC_0 p^0 q^n$

$$\begin{aligned} &= {}^{10}C_0 (0.02)^0 (0.98)^{10} \\ &= 10 (0.98)^{10} = 0.8171 \end{aligned}$$

7.36 Given $n = 200$, $p = 0.03$, $\lambda = np = 200(0.03) = 6$.

$$P(x=10) = \frac{e^{-\lambda} \lambda^r}{r!} = \frac{e^{-6} (6)^{10}}{10!} = 0.0413$$

7.37 Given $n = 300$, $p = 0.40$; $\mu = np = 300(0.40) = 120$;

$$\sigma = \sqrt{npq} = \sqrt{120(0.60)} = 8.48$$

$$P(x \leq 0.34 \times 300 = 102)$$

$$= P\left(z < \frac{x - \mu}{\sigma}\right) = P\left(z < \frac{102 - 120}{8.48}\right)$$

$$= P[z \leq -2.12] = 0.5000 - 0.4830$$

$$= 0.0170$$

Since the probability that the sample would indicate 34 per cent or less is very small, it is unlikely that the 40 per cent estimate is correct.

8

Statistical Decision Theory

LEARNING OBJECTIVES

After studying this chapter, you should be able to

- identify decision alternatives, the states of nature and payoff associated with every possible combination of decision alternatives and states of nature.
- revise subjective probability values included in the formulation of the decision problem.
- identify the one best decision alternative for decision situation of uncertainty and risk.
- develop managerial judgement to utilize the subjective as well as objective interpretation of probabilities in decision making.

8.1 INTRODUCTION

Making of a decision requires an enumeration of feasible and viable decision alternatives (courses of action or strategies), the projection of consequences associated with different alternatives, and a measure of effectiveness (of an objective) by which the most preferred decision alternative is identified. *Statistical decision theory* provides an analytical and systematic approach to the study of decision making wherein data concerning the occurrence of different outcomes (consequences) may be evaluated to enable the decision maker to identify the suitable decision alternative (or course of action).

Decision models that help decision makers to arrive at the best possible decisions are classified according to the *degree of certainty*. The scale of certainty can range from complete certainty to complete uncertainty. The region which falls between these two extreme points corresponds to decision making under risk (probabilistic problems).

Irrespective of the type of decision model, there are certain essential characteristics which are common to all as listed below.

Decision alternatives: There is a finite number of decision alternatives available with the decision-maker at each point in time when a decision is made. The number and type of such alternatives may depend on the previous decisions made and on what has happened subsequent to those

decisions. These alternatives are also called *courses of action* (*actions*, *acts*, or *strategies*) and are under the control of and known to the decision maker. These may be described numerically, such as stocking 100 units of a particular item, or non-numerically, such as conducting a market survey to know the likely demand of an item.

State of nature: A possible future condition (consequence or event) resulting from the choice of a decision alternative depends upon certain factors beyond the control of the decision maker. These factors are called states of nature (future). For example, if the decision is to carry an umbrella or not, the consequence (get wet or do not) depends on what action nature takes.

The states of nature are mutually exclusive and collectively exhaustive with respect to any decision problem. The states of nature may be described numerically, such as demand of 100 units of an item or non-numerically, such as an employees strike.

Payoff: A numerical value resulting from each possible combination of alternatives and states of nature is called payoff. The payoff values are always conditional values because of unknown states of nature.

A tabular arrangement of these conditional outcome (payoff) values is known as *payoff matrix*, as shown in Table 8.1.

Table 8.1: General Form of Payoff Matrix

<i>States of Nature</i>	<i>Courses of Action (Alternatives)</i>			
	S_1	S_2	$\dots$	S_n
N_1	p_{11}	p_{12}	$\dots$	p_{1n}
N_2	p_{21}	p_{22}	$\dots$	p_{2n}
$\vdots$	$\vdots$	$\vdots$		$\vdots$
N_m	p_{m1}	p_{m2}	$\dots$	p_{mn}

8.2 STEPS IN DECISION THEORY APPROACH

The decision making process involves the following steps:

1. Identifying and defining the problem.
2. Listing of all possible future events, called *states of nature*, which can occur in the context of the decision problem. Such events are not under the control of the decision maker because these are erratic in nature.
3. Identification of all the *courses of action* (alternatives or decision choices) which are available to the decision maker. The decision maker has control over these courses of action.
4. Expressing the payoffs (p_{ij}) resulting from each pair of course of action and state of nature. These payoffs are normally expressed as a monetary value.
5. Apply an appropriate mathematical decision theory model to select the best course of action from the given list on the basis of some criterion (measure of effectiveness) that results in the optimal (desired) payoff.

Example 8.1: A firm manufactures three types of products. The fixed and variable costs are given below:

	Fixed Cost (₹)	Variable Cost per Unit (₹)
Product A :	25,000	12
Product B :	35,000	9
Product C :	53,000	7

The likely demand (units) of the products is given below:

Poor demand :	3000
Moderate demand :	7000
High demand :	11,000

If the sale price of each type of product is ₹25, then prepare the payoff matrix.

Solution: Let D_1 , D_2 and D_3 be the poor, moderate and high demand, respectively. Then payoff is given by

$$\text{Payoff} = \text{Sales revenue} - \text{Cost}$$

The calculations for payoff (in thousand) for each pair of alternative demand (course of action) and the types of product (state of nature) are shown below:

$$\begin{aligned} D_1 A &= (3 \times 25) - 25 - (3 \times 12) = 14 \\ D_2 A &= (7 \times 25) - 25 - (7 \times 12) = 66 \\ D_1 B &= (3 \times 25) - 35 - (3 \times 9) = 13 \\ D_2 B &= (7 \times 25) - 35 - (7 \times 9) = 77 \\ D_1 C &= (3 \times 25) - 53 - (3 \times 7) = 1 \\ D_2 C &= (7 \times 25) - 53 - (7 \times 7) = 73 \\ D_3 A &= (11 \times 25) - 25 - (11 \times 12) = 118 \\ D_3 B &= (11 \times 25) - 35 - (11 \times 9) = 141 \\ D_3 C &= (11 \times 25) - 53 - (11 \times 7) = 145 \end{aligned}$$

The payoff values are shown in Table 8.2.

Table 8.2: Payoff Values (₹ in '000)

Product Type	Alternative Demand		
	D_1	D_2	D_3
A	14	66	118
B	13	77	141
C	1	73	145

8.3 TYPES OF DECISION-MAKING ENVIRONMENTS

Decisions are made based upon the data available about the occurrence of events as well as the decision situation (or environment). There are four types of decision making environments: *Certainty*, *uncertainty*, *risk* and *conflict*.

Decision Making Under Certainty In this case, the decision maker has complete knowledge (perfect information) of the consequence of every decision choice (course of action or alternative) with certainty. Obviously, he will select an alternative that yields the largest return (payoff) for the known future (state of nature). For example, the decision to purchase either N.S.C. (National Saving Certificate), Indira Vikas Patra, or deposit in N.S.S. (National Saving Scheme) is one in which it is reasonable to assume complete information about the future because there is no doubt that the Indian government will pay the interest when it is due and the principal at maturity.

In this decision model, only one possible state of nature (future) exists.

Decision Making Under Risk In this case, the decision maker has less than complete knowledge with certainty of the consequence of every decision choice (course of action). This means there is more than one state of nature (future) and for which he makes an assumption of the probability with which each state of nature will occur. For example, the probability of getting a head in the toss of a coin is 0.5.

Decision Making Under Uncertainty In this case, the decision maker is unable to specify the probabilities with which the various states of nature (future) will occur. Thus, decisions under uncertainty are taken with even less information than decisions under risk. For example, the probability that Mr X will be the prime minister of the country 15 years from now is not known.

8.4 DECISION-MAKING UNDER UNCERTAINTY

In the absence of knowledge about the probability of any state of nature (future) occurring, the decision maker must arrive at a decision only on the actual conditional payoff values, together with a policy (attitude). There are several different criteria of decision making in this situation. The criteria that we will discuss in this section include:

- (i) Maximax or Minimin
- (ii) Maximin or Minimax
- (iii) Equally Likely
- (iv) Criterion of Realism
- (v) Criterion of Regret

8.4.1 Criterion of Optimism (Maximum or Minimum)

In this criterion, the decision maker ensures that he should not miss the opportunity to achieve the largest possible profit (maximum) or lowest possible cost (minimum). Thus, he selects the alternative (decision choice or course of action) that represents the maximum of the maxima (or minimum of the minima) payoffs (consequences or outcomes). The working method is summarized as follows:

- (a) Locate the maximum (or minimum) payoff values corresponding to each alternative (or course of action), then
- (b) Select an alternative with the best anticipated payoff value (maximum for profit and minimum for cost).

Since in this criterion, the decision maker selects an alternative with the largest (or lowest) possible payoff value, it is also called an *optimistic decision criterion*.

8.4.2 Criterion of Pessimism (Maximum or Minimum)

In this criterion, the decision maker ensures that he would earn no less (or pay no more) than some specified amount. Thus, he selects the alternative that represents the maximum of the minima (or minimum of the maxima in case of loss) payoff in case of profits. The working method is summarized as follows:

- (a) Locate the minimum (or maximum in case of profit) payoff value in case of loss (or cost) data corresponding to each alternative, then
- (b) Select an alternative with the best anticipated payoff value (maximum for profit and minimum for loss or cost).

Since in this criterion, the decision maker is conservative about the future and always anticipates the worst possible outcome (minimum for profit and maximum for cost or loss), it is called a *pessimistic decision criterion*. This criterion is also known as *Wald's criterion*.

8.4.3 Equally Likely Decision (Laplace) Criterion

Since the probabilities of the states of nature are not known, it is assumed that all states of nature will occur with equal probability, that is, each state of nature is assigned an equal probability. As states of nature are mutually exclusive and collectively exhaustive, the probability of each of these must be $1/(\text{number of states of nature})$. The working method is summarized as follows:

- (a) Assign equal probability value to each state of nature by using the formula:

$$1 \div (\text{Number of states of nature})$$

- (b) Compute the expected (or average) payoff for each alternative (course of action) by adding all the payoffs and dividing by the number of possible states of nature or by applying the formula:

$$(\text{Probability of state of nature } j) \times (\text{Payoff value for the combination of alternative } i \text{ and state of nature } j)$$

- (c) Select the best expected payoff value (maximum for profit and minimum for cost).

This criterion is also known as the criterion of insufficient reason because, except in a few cases, some information of the likelihood of occurrence of the states of nature is available.

8.4.4 Criterion of Realism (Hurwicz Criterion)

This criterion suggests that a rational decision maker should be neither completely optimistic nor pessimistic and, therefore, must display a mixture of both. Hurwicz, who suggested this criterion, introduced the idea of a coefficient of optimism (denoted by α) to measure the decision maker's degree of optimism. This coefficient lies between 0 and 1, where 0 represents a complete pessimistic attitude about the future and 1 represents a complete optimistic attitude about the future. Thus, if α is the coefficient of optimism, then $(1 - \alpha)$ will represent the coefficient of pessimism.

The Hurwicz approach suggests that the decision maker must select an alternative that maximizes:

$$H(\text{Criterion of realism}) = \alpha(\text{Maximum in column}) + (1 - \alpha)(\text{Minimum in column})$$

The working method is summarized as follows:

- (a) Decide the coefficient of optimism α (alpha) and then the coefficient of pessimism $(1 - \alpha)$.

- (b) For each alternative, select the largest and smallest payoff values and multiply these with α and $(1 - \alpha)$ values, respectively. Then calculate the weighted average, H by using the above formula.

Select an alternative with the best anticipated weighted average payoff value.

8.4.5 Criterion of Regret

This criterion is also known as the opportunity loss decision criterion or minimax regret decision criterion because the decision maker feels regret after adopting a wrong course of action (or alternative) resulting in an opportunity loss of payoff. Thus, he always intends to minimize this regret. The working method is summarized as follows:

- (a) From the given payoff matrix, develop an opportunity loss (or regret) matrix as follows:
 - (i) Find the best payoff corresponding to each state of nature.
 - (ii) Subtract all other entries (payoff values) in that row from this value.
- (b) For each course of action (strategy or alternative) identify the worst or maximum regret value. Record this number in a new row.
- (c) Select the course of action (alternative) with the smallest anticipated opportunity loss value.

Example 8.2: A food products company is contemplating the introduction of a revolutionary new product with new packaging, or replace the existing product at a much higher price (S_1), or a moderate change in the composition of the existing product with a new packaging at a small increase in price (S_2), or a small change in the composition of the existing product except the word 'New' with a negligible increase in price (S_3). The three possible states of nature or events are: (i) high increase in sales (N_1), (ii) no change in sales (N_2), and (iii) decrease in sales (N_3). The marketing department of the company worked out the payoffs in terms of yearly net profits for each of the strategies of the three events (expected sales). This is represented in the following table:

Strategies	States of Nature		
	N_1	N_2	N_3
S_1	7,00,000	3,00,000	1,50,000
S_2	5,00,000	4,50,000	0
S_3	3,00,000	3,00,000	3,00,000

Which strategy should the concerned executive choose on the basis of

- (a) Maximin criterion?
- (b) Maximax criterion?
- (c) Minimax regret criterion?
- (d) Laplace criterion?

Solution: The payoff matrix is rewritten as follows:

- (a) *Maximin Criterion*

States of Nature	Strategies			
	S_1	S_2	S_3	
N_1	7,00,000	5,00,000	3,00,000	
N_2	3,00,000	4,50,000	3,00,000	
N_3	1,50,000	0,00,000	3,00,000	
Column minimum	1,50,000	0,00,000	3,00,000	← Maximin

The maximum of column minima is 3,00,000. Hence, the company should adopt strategy S_3 .

(b) *Maximax Criterion*

States of Nature	Strategies		
	S_1	S_2	S_3
N_1	7,00,000	5,00,000	3,00,000
N_2	3,00,000	4,50,000	3,00,000
N_3	1,50,000	0,00,000	3,00,000
Column maximum	7,00,000	5,00,000	3,00,000

↑ *Maximax*

The maximum of column maxima is 7,00,000. Hence, the company should adopt strategy S_1 .

(c) *Minimax Regret Criterion* Opportunity loss table is shown below:

States of Nature	Strategies		
	S_1	S_2	S_3
N_1	7,00,000 – 7,00,000 = 0	7,00,000 – 5,00,000 = 2,00,000	7,00,000 – 3,00,000 = 4,00,000
N_2	4,50,000 – 3,00,000 = 1,50,000	4,50,000 – 4,50,000 = 0	4,50,000 – 3,00,000 = 1,50,000
N_3	3,00,000 – 1,50,000 = 1,50,000	3,00,000 – 0 = 3,00,000	3,00,000 – 3,00,000 = 0
Column maximum	1,50,000	3,00,000	4,00,000

↑ *Minimax regret*

Hence the company should adopt the minimum opportunity loss strategy, S_1 .

(d) *Laplace Criterion* Since we do not know the probabilities of the states of nature, assume that they are equal. For this example, we would assume that each state of nature has a one-third probability of occurrence. Thus,

Strategy	Expected Return (₹)
S_1	$(7,00,000 + 3,00,000 + 1,50,000)/3 = \mathbf{3,83,333.33}$
S_2	$(5,00,000 + 4,50,000 + 0)/3 = 3,16,666.66$
S_3	$(3,00,000 + 3,00,000 + 3,00,000)/3 = 3,00,000$

Since the largest expected return is from strategy S_1 , the executive must select strategy S_1 .

Example 8.3: A manufacturer makes a product, of which the principal ingredient is a chemical, X. At the moment, the manufacturer spends ₹1000 per year on supply of X, but there is a possibility that the price may soon increase to four times its present figure because of a worldwide shortage of the chemical. There is another chemical Y, which the manufacturer could use in conjunction with a third chemical, Z, in order to give the same effect as chemical X. Chemicals Y and Z would together cost the manufacturer ₹3000 per year, but their prices are unlikely to rise. What action should the manufacturer take? Apply the maximin and minimax criteria for decision making and give two sets of solutions. If the coefficient of optimism is 0.4, find the course of action that minimizes the cost.

Solution: The data of the problem is summarized in the following table (negative figures in the table represent profit).

States of Nature	Courses of Action	
	S_1 (use Y and Z)	S_2 (use X)
N_1 (Price of X increases)	– 3000	– 4000
N_2 (Price of X does not increase)	– 3000	– 1000

(a) *Maximin Criterion*

States of Nature	Courses of Action	
	S_1	S_2
N_1	– 3000	– 4000
N_2	– 3000	– 1000
Column minimum	– 3000	– 4000

↑ *Maximin*

Maximum of column minima = – 3,000. Hence, the manufacturer should adopt action S_1 .

(b) *Minimax (or opportunity loss) Criterion*

States of Nature	Courses of Action	
	S_1	S_2
N_1	– 3000 – (– 3000) = 0	– 3000 – (– 4000) = 1000
N_2	– 1000 – (– 3000) = 2000	– 1000 – (– 1000) = 0
Maximum opportunity	– 2000	– 1000 ← <i>Minimax</i>

Hence, the manufacturer should adopt the minimum opportunity loss course of action S_2 .

(c) *Hurwicz Criterion:* Given the coefficient of optimism equal to 0.4, the coefficient of pessimism will be $1 - 0.4 = 0.6$. Then according to Hurwicz, a select course of action that optimizes (maximum for profit and minimum for cost) the payoff value

$$H = \alpha (\text{Best payoff}) + (1 - \alpha) (\text{Worst payoff})$$

$$= \alpha (\text{Maximum in column}) + (1 - \alpha) (\text{Minimum in column})$$

Course of Action	Best Payoff	Worst Payoff	H
S_1	– 3000	– 3000	– 3000
S_2	– 1000	– 4000	– 2800

Since course of action S_2 has the least cost (maximum profit) = $0.4(1000) + 0.6(4000) = 2800$, the manufacturer should adopt it.

Self-practice Problems 8A

- 8.1** The following matrix gives the payoff (in ₹) of different strategies (alternatives) S_1 , S_2 and S_3 against conditions (events) N_1 , N_2 , N_3 and N_4 .

Strategy	States of Nature			
	N_1	N_2	N_3	N_4
S_1	4000	– 100	6000	18,000
S_2	20,000	5000	400	0
S_3	20,000	15,000	– 2000	1000

Indicate the decision taken under the following approaches: (a) Pessimistic, (b) Optimistic, (c) Equal probability, (d) Regret and (e) Hurwicz criterion. The degree of optimism being 0.7.

- 8.2** In a toy manufacturing company, suppose the product acceptance probabilities are not known, but the following data is known:

Product	Anticipated First Year Profit (₹ in '000s)		
	Product Line		
Acceptance	Full	Partial	Minimal
Good	8	70	50
Fair	50	45	40
Poor	- 25	- 10	0

Determine the optimal decision under each of the following decision criteria and show how you arrived at it: (a) Maximax, (b) Maximin, (c) Equal likelihood and (d) Minimax regret?

- 8.3** The following is a payoff (in ₹) table for three strategies and two states of nature:

Strategy	State of Nature	
	N_1	N_2
S_1	40	60
S_2	10	- 20
S_3	- 40	150

Select a strategy using each of the following decision criteria: (a) Maximax, (b) Minimax regret, (c) Maximin and (d) Minimum risk, assuming equiprobable states.

- 8.4** Mr Sethi has ₹ 10,000 to invest in one of three options, A, B or C. The return on his investment depends on whether the economy experiences inflation, recession or no change at all. His possible returns under each economic condition are given below:

Strategy	State of Nature		
	Inflation	Recession	No Change
A	2000	1200	1500
B	3000	800	1000
C	2500	1000	1800

What would be his decision using the pessimistic criterion, optimistic criterion, equally likely criterion and regret criterion?

- 8.5** A manufacturer's representative has been offered a new product line. If he accepts the new line, he can handle it in one of the two ways. The best way according to the manufacturer would be to have a separate sales force to handle the new line exclusively. This would involve an initial investment of ₹1,00,000 in the office, equipment, and the hiring and training of the salesmen. On the other hand, if the new line could be handled by the existing sales force using the existing facilities, the initial investment would be only ₹30,000, principally for training of his present salesmen.

The new product sells for ₹250. The representative normally receives 20 per cent of the sale price on each unit sold of which 10 per cent is paid as commission to handle the new product. The manufacturer offers to pay 60 per cent of the sale price of each unit sold to the representative if the representative sets up a separate sales organization. Otherwise the normal 20 per cent will be paid. In either case, the salesman gets a 10 per cent commission. Based on the size of the territory and experience with other products, the representative estimates the probabilities of annual sales of the new product:

Sales (in units)	Probability
1000	0.10
2000	0.15
3000	0.40
4000	0.30
5000	0.05

- Set up a regret table.
- Find the expected regret of each course of action.
- Which course of action would have been best under the maximin criterion?

Hints and Answers

- 8.1 (a) S_2 , (b) S_2 or S_3 , (c) S_3 , (d) S_1 and (e) S_2 .
 8.2 (a) Full, (b) Minimal, (c) Full or partial and (d) Partial.
 8.3 (a) S_3 ; ₹150 (b) S_3 ; ₹80 (c) S_1 ; ₹40 and (d) S_3 ; ₹55.
 8.4 Choose A: ₹120, Choose B: ₹300, Choose C: ₹176.6 and Choose C: ₹50.
 8.5 Let S_1 = install new sales facilities
 S_2 = continue with existing sales facilities.

Therefore, payoff function corresponding to S_1 and S_2 would be

$$S_1 = -1,00,000 + 250 \times \frac{(30-10)}{100} \times \alpha \\ = -1,00,000 + 50\alpha$$

$$S_2 = -30,000 + 250 \times \frac{(20-10)}{100} \times \alpha \\ = -30,000 + 25\alpha$$

Equating the two, we get $-1,00,000 + 50\alpha = -30,000 + 25\alpha$ or $\alpha = 2800$.

8.5 DECISION MAKING UNDER RISK

Decision-making under risk is a probabilistic decision situation in which more than one state of nature exists and the decision maker has sufficient information to assign probability values to the likely occurrence of each of these states. Knowing the probability distribution of the states of nature, the best decision is to select that course of action which has the largest expected payoff value.

The most widely used criterion for evaluating various courses of action (alternatives) under risk is the *Expected Monetary Value (EMV)* or *Expected Utility*.

8.5.1 Expected Monetary Value (EMV)

The expected monetary value (EMV) for a given course of action is the weighted average payoff, which is the sum of the payoffs for each course of action multiplied by the probabilities associated with each state of nature. Mathematically, EMV is stated as follows:

$$\text{EMV (Course of action, } S_j) = \sum_{i=1}^m p_{ij} p_i$$

where m = number of possible states of nature

p_i = probability of occurrence of state of nature i

p_{ij} = payoff associated with state of nature N_i and course of action, S_j

Steps for Calculating EMV The various steps involved in the calculation of EMV are as follows:

- Construct a payoff matrix listing all possible courses of action and states of nature. Enter the conditional payoff values associated with each possible combination of course of action and state of nature along with the probabilities of the occurrence of each state of nature.
- Calculate the EMV for each course of action by multiplying the conditional payoffs by the associated probabilities and add these weighted values for each course of action.
- Select the course of action that yields the optimal EMV.

Example 8.4: Mr X often flies from town A to town B. He can use the airport bus which costs ₹25 but if he takes it, there is a 0.08 chance that he will miss the flight. The stay in a hotel costs

₹270 with a 0.96 chance of being on time for the flight. For ₹350 he can use a taxi which has a 99 per cent chance of being on time for the flight. If Mr X catches the plane on time, he will conclude a business transaction which will produce a profit of ₹10,000, otherwise he will lose it. Which mode of transport should Mr X use? Answer on the basis of the EMV criterion.

Solution: Computation of EMV of the various courses of action is shown in Table 8.3.

Table 8.3: Computation of EMV

States of Nature	Courses of Action								
	Bus			Stay in Hotel			Taxi		
	Cost	Prob.	Expected Value	Cost	Prob.	Expected Value	Cost	Prob.	Expected Value
Catches the flight	10,000 – 25 = 9975	0.92	9177	10,000 – 270 = 9730	0.96	9,340.80	10,000 – 350 = 9650	0.99	9553.50
Miss the flight	–25	0.08	–2.0	–270	0.04	–10.80	–350	0.01	–3.50
Expected monetary value (EMV)			9175			9330			9330

Comparing the EMV associated with each course of action indicates that course of action ‘Taxi’ is the logical alternative because it has the highest EMV.

Example 8.5: The manager of a flower shop promises his customers delivery within four hours on all orders. All flowers are purchased previous day and delivered to Parker by 8.00 a.m. the next morning. The daily demand for roses is as follows:

Dozens of roses	:	70	80	90	100
Probability	:	0.1	0.2	0.4	0.3

The manager purchases roses for ₹10 per dozen and sells them for ₹30. All unsold roses are donated to a local hospital. How many dozens of roses should Parker order each evening to maximize its profits? What is the optimum expected profit?

Solution: Since the number of roses (in dozens) purchased is under control of decision maker, purchase per day is considered as ‘course of action’ (decision choice) and the daily demand of the flowers is uncertain and only known with probability, therefore, it is considered as a ‘state of nature’ (event). From the data, it is clear that the flower shop must not purchase less than 70 or more than 100 dozen roses per day. Also, each dozen roses sold within a day yields a profit of $₹(30 - 10) = ₹20$, otherwise it is a loss of ₹10. Thus

$$\text{Marginal profit (MP)} = \text{Selling price} - \text{Cost} = 30 - 10 = ₹20$$

$$\text{Marginal loss (ML)} = \text{Loss on unsold roses} = ₹10$$

Using the information given in the problem, the various conditional profit (payoff) values for each combination of decision choice event are given by

$$\begin{aligned}\text{Conditional profit} &= \text{MP} \times \text{Roses sold} - \text{ML} \times \text{Roses not sold} \\ &= \begin{cases} 20D, & \text{if } D \geq S \\ 20D - 10(S - D) = 30D - 10S, & \text{if } D < S \end{cases}\end{aligned}$$

where D denotes the number of roses sold within a day and S the number of roses stocked.

The resulting conditional profit values and corresponding expected payoffs are computed in Table 8.4.

Table 8.4: Conditional Profit Value (Payoffs)

States of Nature (Demand per Day)	Probability	Conditional Profit (₹) due to Courses of Action (Purchase per Day)				Expected Payoff (₹) due to Courses of Action (Purchase per Day)			
		70	80	90	100	70	80	90	100
	(1)	(2)	(3)	(4)	(5)	(1) × (2)	(1) × (3)	(1) × (4)	(1) × (5)
70	0.1	140	130	120	110	14	13	12	11
80	0.2	140	160	150	140	28	32	30	28
90	0.4	140	160	180	170	56	64	72	68
100	0.3	140	160	180	200	42	48	54	60
Expected monetary value (EMV)						140	157	168	167

Since the highest EMV of ₹168 is corresponding to course of action 90, the flower shop should purchase ninety dozen roses everyday.

Example 8.6: A retailer purchases cherries every morning at ₹50 a case and sells them for ₹80 a case. Any case remaining unsold at the end of the day can be disposed of the next day at a salvage value of ₹20 per case (thereafter they have no value). Past sales have ranged from 15 to 18 cases per day. The following is the record of sales for the past 120 days:

Cases sold	:	15	16	17	18
Number of days	:	12	24	48	36

Find how many cases the retailer should purchase per day to maximize his profit.

Solution: Let N_i ($i = 1, 2, 3, 4$) be the possible states of nature (daily likely demand) and S_j ($j = 1, 2, 3, 4$) be all possible courses of action (number of cases of cherries to be purchased).

$$\text{Marginal profit (MP)} = \text{Selling price} - \text{Cost} = ₹(80 - 50) = ₹30$$

$$\text{Marginal Loss (ML)} = \text{Loss on unsold cases} = ₹(50 - 20) = ₹30$$

The conditional profit (payoff) values for each act-event combination are given by:

$$\begin{aligned}\text{Conditional Profit} &= \text{MP} \times \text{Cases sold} - \text{ML} \times \text{Cases unsold} \\ &= (80 - 50) (\text{Cases sold}) - (50 - 20) (\text{Cases unsold}) \\ &= 30S, & \text{if } S \geq N \\ &= (80 - 50) S - 30(N - S) = 60S - 30N & \text{if } S < N\end{aligned}$$

The resulting conditional profit values and corresponding expected payoffs are computed in Table 8.5.

Table 8.5: Conditional Profit Values (Payoffs)

States of Nature (Demand per Day)	Probability	Conditional Profit (₹) due to Courses of Action (Purchase per Day)				Expected Payoff (₹) due to Courses of Action (Purchase per Day)			
		15	16	17	18	15	16	17	18
	(1)	(2)	(3)	(4)	(5)	(1)×(2)	(1)×(3)	(1)×(4)	(1)×(5)
15	0.1	450	420	390	360	45	42	39	36
16	0.2	450	480	450	420	90	96	90	84
17	0.4	450	480	510	480	180	192	204	192
18	0.3	450	480	510	540	135	144	153	162
Expected monetary value (EMV)						450	474	486	474

Since the highest EMV of ₹486 is corresponding to course of action 17, the retailer must purchase 17 cases of cherries every morning.

Example 8.7: The probability of the demand for lorries for hiring on any day in a given district is as follows:

No. of lorries demanded	:	0	1	2	3	4
Probability	:	0.1	0.2	0.3	0.2	0.2

Lorries have a fixed cost of ₹90 each day to keep the daily hire charges (net of variable costs of running) ₹200. If the lorry-hire company owns 4 lorries, what is its daily expectation? If the company is about to go into business and currently has no lorries, how many lorries should it buy?

Solution: It is given that ₹90 is the fixed cost and ₹200 the variable cost. Now the payoff values with 4 lorries at the disposal of the decision maker are calculated as under:

No. of lorries

demand	:	0	1	2	3	4
Payoff	:	$0 - 90 \times 4$	$200 - 90 \times 4$	$400 - 90 \times 4$	$600 - 90 \times 4$	$800 - 90 \times 4$
(with 4 lorries)	=	- 360	- 160	40	240	440

Thus daily expectation is obtained by multiplying the payoff values with the given corresponding probabilities of demand:

$$\text{Daily Expectation} = (-360)(0.1) + (-160)(0.2) + (40)(0.3) + (240)(0.2) + (440)(0.2) = ₹80$$

The conditional payoffs and expected payoffs for each course of action are shown in Tables 8.6 and 8.7.

Table 8.6: Conditional Payoff Values

Number of Lorries	Probability	Conditional Payoff (₹) due to Decision to Purchase Lorries (Course of Action)				
		0	1	2	3	4
0	0.1	0	-90	-180	-270	-360
1	0.2	0	110	20	-70	-160
2	0.3	0	110	220	130	40
3	0.2	0	110	220	330	240
4	0.2	0	110	220	330	440

Table 8.7: Expected Payoffs and EMV

Number of Lorries	Probability	Conditional Payoff (₹) due to Decision to Purchase Lorries (Course of Action)				
		0	1	2	3	4
0	0.1	0	-9	-18	-27	-36
1	0.2	0	22	4	-14	-32
2	0.3	0	33	66	39	12
3	0.2	0	22	44	66	48
4	0.2	0	22	44	66	88
EMV		0	90	140	130	80

Since EMV of ₹140 for the course of action 2 is highest, the company should buy 2 lorries.

8.5.2 Expected Opportunity Loss (EOL)

An alternative approach to maximizing expected monetary value (EMV) is to minimize the **expected opportunity loss** (EOL) also called *expected value of regret*. The EOL is defined as the difference between the highest profit (or payoff) for a state of nature and the actual profit obtained for the particular course of action taken. In other words, EOL is the amount of payoff that is lost by not selecting the course of action that has the greatest payoff for the state of nature that actually occurs. The course of action due to which EOL is minimum is recommended.

Since EOL is an alternative decision criterion for decision making under risk, therefore, the results will always be the same as those obtained by EMV criterion discussed earlier. Thus, only one of the two methods should be applied to reach a decision. Mathematically, it is stated as follows:

$$\text{EOL (State of nature, } N_i) = \sum_{i=1}^m l_{ij} p_i$$

where l_{ij} = opportunity loss due to state of nature, N_i and course of action, S_j
 p_i = probability of occurrence of state of nature, N_i

Steps for Calculating EOL The various steps involved in the calculation of EOL are as follows:

- Prepare a conditional profit table for each course of action and state of nature combination along with the associated probabilities.
- For each state of nature, calculate the conditional opportunity loss (COL) values by subtracting each payoff from the maximum payoff for that outcome.
- Calculate EOL for each course of action by multiplying the probability of each state of nature with the COL value and then adding the values.
- Select a course of action for which the EOL value is minimum.

Example 8.8: A company manufactures goods for a market in which the technology of the product is changing rapidly. The research and development department has produced a new product which appears to have the potential for commercial exploitation. A further ₹60,000 is required for development testing.

The company has 100 customers and each customer might purchase at the most one unit of the product. Market research suggests that a selling price of ₹6000 for each unit with total variable costs of manufacturing and selling estimate are ₹2000 for each unit.

From previous experience, it has been possible to derive a probability distribution relating to the proportion of customers who will buy the product. It is as follows:

Proportion of customers	:	0.04	0.08	0.12	0.16	0.20
Probability	:	0.10	0.10	0.20	0.40	0.20

Determine the expected opportunity losses, given no other information than that stated above, and state whether or not the company should develop the product.

Solution: If p is the proportion of customers who purchase the new product, the conditional profit is: $(6000 - 2000) \times 100p - 60,000 = ₹(4,00,000p - 60,000)$

Let N_i ($i = 1, 2, \dots, 5$) be the possible states of nature, that is, proportion of the customers who will buy the new product and S_1 (develop the product) and S_2 (do not develop the product) be the two courses of action.

The conditional profit values (payoffs) for each pair of N_i 's and S_j 's are shown in Table 8.8.

Table 8.8: Conditional Profit Values (Payoffs)

State of Nature (Proportion of Customers, p)	Conditional Profit = ₹(4,00,000 p - 60,000)	
	Course of Action	
	S_1 (Develop)	S_2 (Do not Develop)
0.04	-44,000	0
0.08	-28,000	0
0.12	-12,000	0
0.16	4000	0
0.20	20,000	0

Opportunity loss values are shown in Table 8.9.

Table 8.9: Opportunity Loss Values

State of Nature	Probability	Conditional Profit (₹)		Opportunity Loss (₹)	
		S_1	S_2	S_1	S_2
0.04	0.1	-44,000	0	44,000	0
0.08	0.1	-28,000	0	28,000	0
0.12	0.2	-12,000	0	12,000	0
0.16	0.4	4000	0	0	4000
0.20	0.2	20,000	0	0	20,000

Using the given estimates of probabilities associated with each state of nature, the expected opportunity loss (EOL) for each course of action is given below:

$$\text{EOL}(S_1) = 0.1(44,000) + 0.1(28,000) + 0.2(12,000) + 0.4(0) + 0.2(0) = ₹9600$$

$$\text{EOL}(S_2) = 0.1(0) + 0.1(0) + 0.2(0) + 0.4(4000) + 0.2(20,000) = ₹5600$$

Since the company seeks to minimize the expected opportunity loss, the company should select course of action S_2 (do not develop the product) with minimum EOL.

8.5.3 Expected Value of Perfect Information (EVPI)

In decision making under risk, each state of nature is associated with the probability of its occurrence. However, if the decision maker can acquire *perfect* (complete and accurate) *information* about the occurrence of various states of nature, then he will be able to select a course of action that yields the desired payoff for whatever state of nature that actually occurs.

We have seen that the EMV or EOL criterion helps the decision maker to select a particular course of action that optimizes the expected payoff without any additional information. The *expected value of perfect information* (EVPI) represents the maximum amount of money, the decision maker has to pay to get this additional information about the occurrence of various states of nature before a decision has to be made. Mathematically it is stated as

$$\begin{aligned} \text{EVPI} &= \text{Expected profit (or value) with perfect information under certainty} \\ &\quad - \text{Expected profit without perfect information} \\ &= \sum_{i=1}^m p_{ij}(N_i)p_i - \text{EMV}^* \end{aligned}$$

where p_{ij} = best payoff for the state of nature, N_i

p_i = probability of the state of nature, N_i

EMV* = maximum expected monetary value

Example 8.9: A company needs to increase its production beyond its existing capacity. It has narrowed the alternatives to two approaches to do so: (a) expansion at a cost of ₹8 million, or (b) modernization at a cost of ₹5 million. Both approaches would require the same amount of time for implementation. Management believes that over the required payback period, demand will either be high or moderate. Since high demand is considered to be somewhat less likely than moderate demand, the probability of high demand has been set at 0.35. If the demand is high, expansion would gross an estimated additional ₹12 million but modernization only an additional ₹6 million, due to a lower maximum production capability. On the other hand, if the

demand is moderate, the comparable figures would be ₹7 million for expansion and ₹5 million for modernization.

- Calculate the conditional profit in relation to various action-and-outcome combinations and states of nature.
- If the company wishes to maximize its expected monetary value (EMV), should it modernize or expand?
- Calculate the EVPI.
- Construct the conditional opportunity loss table and also calculate EOL.

Solution: (a) Defining the states of nature: High demand or moderate demand (over which the company has no control) and courses of action (company's possible decisions): Expand or Modernize.

Since the probability that the demand is high is estimated at 0.35, the probability of moderate demand must be $(1 - 0.35) = 0.65$. The calculations for conditional profit values are shown in Table 8.10.

Table 8.10: Conditional Profit Table

State of Nature (Demand)	Conditional Profit (₹ in million) due to Course of Action	
	Expand (S_1)	Modernize (S_2)
High demand (N_1)	$12 - 8 = 4$	$6 - 5 = 1$
Moderate demand (N_2)	$7 - 8 = -1$	$5 - 5 = 0$

(b) The payoff table (Table 8.10) can be rewritten as follows along with the given probabilities of the states of nature.

Table 8.11: Payoff Table

State of Nature (Demand)	Probability	Conditional Profit (₹ in million) due to Course of Action	
		Expand	Modernize
High demand	0.35	4	1
Moderate demand	0.65	-1	0

The calculation of EMV for each course of action S_1 and S_2 is given below:

$$EMV(S_1) = 0.35(4) + 0.65(-1) = ₹0.75 \text{ million}$$

$$EMV(S_2) = 0.35(1) + 0.65(0) = ₹0.35 \text{ million}$$

Thus to maximize EMV, the company must choose the course of action S_1 (expand). The EMV of the optimal course of action is generally denoted by EMV^* . Therefore,

$$EMV^* = EMV(S_1) = ₹0.75 \text{ million}$$

(c) To calculate EVPI, we shall first calculate EPPI. For calculating EPPI, we choose the optimal course of action for each state of nature, multiply its conditional profit by the given probability to get weighted profit, and then sum these weights as shown in Table 8.12.

Table 8.12: Payoff Table

State of Nature (Demand)	Probability	Optimal Course of Action	Profit from Optimal Course of Action (₹ in million)	
			Conditional Profit	Weighted Profit
High demand	0.35	S_1	4	$4 \times 0.35 = 1.40$
Moderate demand	0.65	S_2	0	$0 \times 0.65 = 0$
Total				EPPI = 1.40

The optimal EMV* is ₹0.75 million corresponding to the course of action S_1 . Then

$$\begin{aligned}\text{EVPI} &= \text{EPPI} - \text{EMV}(S_1) \\ &= 1.40 - 0.75 = ₹0.65 \text{ million}\end{aligned}$$

In other words, if the company could get accurate information (or forecast) of demand (high or moderate), it should consider paying up to ₹0.65 million for such information.

The expected value of perfect information in business helps in getting an absolute upper bound on the amount that should be spent to get additional information on which to base a given decision.

(d) The opportunity loss values are shown in Table 8.13.

Table 8.13: Conditional Opportunity Loss Table

State of Nature (Demand)	Probability	Conditional Profit (₹ in million) due to Course of Action		Conditional Opportunity Loss (₹ in million) due to Course of Action	
		S_1	S_2	S_1	S_2
High demand (N_1)	0.35	4	1	0	3
Moderate demand (N_2)	0.65	-1	0	1	0

The conditional opportunity loss values may be explained as follows: If high demand (N_1) occurred, then the maximum profit of ₹4 million would be achieved by selecting course of action S_1 . Therefore, the selection of S_1 would result in zero opportunity loss, as it is the best decision that can be made if N_1 occurs. If course of action S_2 was chosen with a payoff of ₹1 million, then this would result in an opportunity loss of $4 - 1 = ₹3$ million. If moderate demand (N_2) occurred, then the best course of action would be S_2 with nil profit. Thus, opportunity loss would be associated with the selection of S_2 but if S_1 was selected, then the opportunity loss would be $0 - (-1) = ₹1$ million. That is, the company would have been ₹1 million worse off if it had chosen course of action S_2 .

Using the given estimates of probabilities associated with each state of nature, that is, $P(N_1) = 0.35$, and $P(N_2) = 0.65$, the expected opportunity losses for the two courses of action are:

$$\text{EOL}(S_1) = 0.35(0) + 0.65(1) = ₹0.65 \text{ million}$$

$$\text{EOL}(S_2) = 0.35(3) + 0.65(0) = ₹1.05 \text{ million}$$

Since the decision maker seeks to minimize the expected opportunity loss, he must select the course of action S_1 as it produces the smallest expected opportunity loss.

Example 8.10: A certain piece of equipment has to be purchased for a construction project at a remote location. This equipment contains an expensive part which is subject to random failure. Spares of this part can be purchased at the same time the equipment is purchased. Their unit cost is ₹1500 and they have no scrap value. If the part fails on the job and no spare is available, the part will have to be manufactured on a special order basis. If this is required, the total cost including down time of the equipment is estimated at ₹9000 for each such occurrence. Based on previous experience with similar parts, the following probability estimates of the number of failures expected over the duration of the project are provided as given below:

Failure :	0	1	2
Probability :	0.80	0.15	0.05

- Determine the optimal EMV* and the optimal number of spares to be purchased initially.
- Based on opportunity losses, determine the optimal course of action and optimal value of EOL.
- Determine the expected profit with perfect information and expected value of perfect information.

Solution: (a) Let N_1 (no failure), N_2 (one failure) and N_3 (two failures) be the possible states of nature (i.e., number of parts failures or number of spares required). Similarly, let S_1 (no spare purchased), S_2 (one spare purchased) and S_3 (two spares purchased) be the possible courses of action or strategies.

The conditional costs for each pair of course of action and state of nature combination are shown in Table 8.14.

Table 8.14: Conditional Cost

State of Nature (Spares required)	Course of Action (Number of spares purchased)	Purchase Cost (₹)	Emergency Cost (₹)	Total Conditional Cost (₹)
0	0	0	0	0
0	1	1500	0	1500
0	2	3000	0	3000
1	0	0	9000	9000
1	1	1500	0	1500
1	2	3000	0	3000
2	0	0	18,000	18,000
2	1	1500	9000	10,500
2	2	3000	0	3000

Using the conditional costs as given in Table 8.14 and the probabilities of the states of nature, the expected monetary value can be calculated for each of the three states of nature as shown in Table 8.15.

Table 8.15: Expected Monetary Value

State of Nature	Probability	Conditional cost (₹) due to Course of Action			Weighted Cost (₹) due to Course of Action		
		S_1	S_2	S_3	S_1	S_2	S_3
N_1	0.80	0	1500	3000	0.80(0) = 0	1200	2400
N_2	0.15	9000	1500	3000	0.15(9000) = 1350	225	450
N_3	0.05	18,000	10,500	3000	0.05(18,000) = 900	525	150
					EMV = 2250	1950	3000

Since the weighted cost—₹1950—is lowest due to course of action S_2 , it should be chosen. If the EMV is expressed in terms of profit, then

$$EMV^* = EMV(S_2) = -₹1950$$

Hence, the optimal number of spares to be purchased initially should be one.

(b) To determine the EOL, we must first find the COL. The calculations for conditional opportunity loss (COL) are shown in Table 8.16.

Table 8.16: Conditional Opportunity Loss (COL)

State of Nature	Conditional Cost (₹) due to Course of Action			Conditional Opportunity Loss (₹) due to Course of Action		
	S_1	S_2	S_3	S_1	S_2	S_3
N_1	0	1500	3000	0	1500	3000
N_2	9000	1500	3000	7500	0	1500
N_3	18,000	10,500	3000	15,000	7500	0

Since we are dealing with conditional costs rather than conditional profits, the lower value for each state of nature shall be considered for calculating opportunity losses.

The calculations for expected opportunity loss are shown in Table 8.17.

Table 8.17: Expected Opportunity Loss (EOL)

State of Nature	Probability	Conditional Cost (₹) due to Course of Action			Weighted Cost (₹) due to Course of Action		
		S_1	S_2	S_3	S_1	S_2	S_3
N_1	0.80	0	1500	3000	0.80(0) = 0	1200	2400
N_2	0.15	7500	0	1500	0.15 (7500) = 1125	0	225
N_3	0.05	15,000	7500	0	0.05 (15,000) = 750	375	0
					EMV = 1875	1575	2625

Since $EOL^* = EOL(S_2) = ₹ 1575$, adopt course of action S_2 and purchase one spare.

(c) The expected profit with perfect information (EPPI) can be determined by selecting the optimal course of action for each state of nature, multiplying its conditional values by the corresponding probability and then summing these products. The EPPI calculations are shown in Table 8.18.

Table 8.18: Computation of EPPI

States of Nature	Probability	Optimal Course of Action	Cost of Optimal Course of Action (₹)	
			Conditional Cost	Weighted Opportunity Loss
N_1	0.80	S_1	0	$0.80(0) = 0$
N_2	0.15	S_2	1500	$0.15 (1500) = 225$
N_2	0.05	S_3	3000	$0.05 (3000) = 150$
				<u>375</u>

Since expected profit with perfect information is ₹ 375, the expected value of perfect information is given by

$$EVPI = EPPI - EMV^* = -375 - (-1950) = ₹ 1575$$

Here it can be observed that $EVPI = EOL^* = ₹ 1575$.

Example 8.11: XYZ Company manufactures parts for passenger cars and sells them in lots of 10,000 parts each. The company has a policy of inspecting each lot before it is actually shipped to the retailer. Five inspection categories, established for quality control, represent the percentage of defective items contained in each lot. These are given in the following table. The daily inspection chart for the past 100 inspections shows the following rating or breakdown inspection.

The management is considering two possible courses of action:

(i) S_1 : Shut down the entire plant operations and thoroughly inspect each machine.

Rating	Proportion of Defective Items	Frequency
Excellent (A)	0.02	25
Good (B)	0.05	30
Acceptable (C)	0.10	20
Fair (D)	0.15	20
Poor (E)	0.20	<u>5</u>
		100

(ii) S_2 : Continue production as it now exists but offer the customer a refund for defective items that are found and subsequently returned.

The first alternative will cost ₹ 600 while the second alternative will cost the company ₹ 1 for each defective item that is returned. What is the optimum decision for the company? Find the EVPI.

Solution: Calculations of inspection and refund cost are shown in Table 8.19.

Table 8.19: Inspection and Refund Cost

Rating	Defect Rate	Probability	Cost		Opportunity Loss	
			Inspect	Refund	Inspect	Refund
A	0.02	0.25	600	200	400	0
B	0.05	0.30	600	500	100	0
C	0.10	0.20	600	1000	0	400
D	0.15	0.20	600	1500	0	900
E	0.20	0.05	600	2000	0	1400
		1.00	600*	800	170*	240

The cost of refund is calculated as follows:

$$\text{For lot A: } 10,000 \times 0.02 \times 1.00 = ₹ 200$$

The cost of refund for other lots is calculated in a similar manner.

The expected cost of refund is:

$$200 \times 0.25 + 500 \times 0.30 + \dots + 2000 \times 0.05 = ₹ 800$$

Now, the expected cost of inspection is:

$$600 \times 0.25 + 600 \times 0.30 + \dots + 600 \times 0.05 = ₹ 600$$

Since the cost of refund is more than the cost of inspection, the plant should be shut down for inspection. Also, $EVPI = EOL \text{ of inspection} = ₹ 170$

Example 8.12: A toy manufacturer is considering a project for manufacturing a dancing doll with three different movement designs. The doll will be sold at an average price of ₹ 10. The first movement design using 'gears and levels' will provide the lowest tooling at a set up cost of ₹ 1,00,000 and ₹ 5 per unit of variable cost. A second design with spring action will have a fixed cost of ₹ 1,60,000 and variable cost of ₹ 4 per unit. Yet another design with weights and pulleys will have a fixed cost of ₹ 3,00,000 and variable cost ₹ 3 per unit. One of the following demand events and its probabilities can occur for the doll:

	Demand (units)	Probability
Light demand	25,000	0.10
Moderate demand	1,00,000	0.70
Heavy demand	1,50,000	0.20

- Construct a payoff table for the above project.
- Which is the optimum design?
- How much can the decision maker afford to pay to obtain perfect information about the demand?

Solution: The calculations for EMV are shown in Table 8.20.

$$\begin{aligned} \text{Payoff} &= (\text{Demand} \times \text{Selling price}) - (\text{Fixed cost} + \text{Demand} \times \text{Variable cost}) \\ &= \text{Revenue} - \text{Total variable cost} - \text{Fixed cost} \end{aligned}$$

Table 8.20: EMV and Payoff Values

States of Nature (Demand)	Probability	Conditional Payoff (₹) due to Courses of Action (Choice of Movements)			Expected Payoff (₹) due to Courses of Action		
		Gears and Levels	Spring Action	Weights and Pulleys	Gears and Levels	Spring Action	Weights and Pulleys
Light	0.10	25,000	-10,000	-1,25,000	2500	-1000	-12,500
Moderate	0.70	4,00,000	4,40,000	4,00,000	2,80,000	3,08,000	2,80,000
Heavy	0.20	6,50,000	7,40,000	7,50,000	1,30,000	1,48,000	1,50,000
				EMV	4,12,500	4,55,000	4,17,500

Since EMV is largest for spring action, it must be selected.

Table 8.21: Expected Payoff with Perfect Information

States of Nature (Demand)	Probability	Courses of Action			Maximum Payoff	Maximum Payoff × Probability
		Gears and Levels	Spring Action	Weights and Pulleys		
Light	0.10	25,000	-10,000	-1,25,000	25,000	2500
Moderate	0.70	4,00,000	4,40,000	4,00,000	4,40,000	3,08,000
Heavy	0.20	6,50,000	7,40,000	7,50,000	7,50,000	1,50,000
						4,60,500

The maximum amount of money that the decision maker would be willing to pay to obtain perfect information regarding demand for the doll will be

$$\begin{aligned}
 \text{EVPI} &= \text{Expected payoff with perfect information} - \text{Expected payoff under uncertainty (EMV)} \\
 &= 4,60,500 - 4,55,000 = ₹ 5500
 \end{aligned}$$

Example 8.13: The demand pattern of cakes made in a bakery is as follows:

No. of cakes demanded	:	0	1	2	3	4	5
Probability	:	0.05	0.10	0.25	0.30	0.20	0.10

If the preparation cost is ₹ 3 per unit and selling price is ₹ 4 per unit, how many cakes should the baker bake to maximize profit?

Solution: Given that incremental cost (IC) is ₹ 3 per unit and incremental price (IP) ₹ 4 per unit, the minimum required probability of selling at least an additional unit of cake to justify the stocking of that unit is given by

$$p = \frac{\text{IC}}{\text{IC} + \text{IP}} = \frac{3}{3 + 4} = 0.75$$

This probability means that the bakery owner must have demand level k such that $P(\text{demand} \geq k) \geq p$.

The cumulative probabilities of greater than type are computed as shown in Table 8.22.

Table 8.22: Cumulative Probability

<i>Demand</i> (No. of cakes)	<i>Probability</i> $P(\text{Demand} = k)$	<i>Cumulative Probability</i> $P(\text{Demand} \geq k)$
0	0.05	1.00
1	0.10	0.95
2	0.25	0.85 ←
3	0.30	0.60
4	0.20	0.30
5	0.10	0.10

Since the highest value of k for which $P(\text{demand} \geq k)$ exceeds the critical ratio $p = 0.75$ is $k = 2$, the optimal decision is to prepare only two cakes.

8.6 DECISION TREE ANALYSIS

The decision making problems discussed so far are referred to as single stage decision problems, because the payoffs, states of nature, courses of action, and probabilities associated with the occurrence of states of nature are not subject to change.

However, situations may arise when a decision maker needs to revise his previous decisions on getting new information and make a sequence of other decisions. Thus, the problem becomes a multi-stage decision problem because the consequence of one decision affects future decisions. For example, in the process of marketing a new product, the first decision is often test marketing and the alternative courses of action might be either intensive testing or gradual testing. Given various possible consequences—good, fair or poor—the decision maker may be required to decide between redesigning the product, an aggressive advertising campaign, or complete withdrawal of product. Given that decision, there will be an outcome which will lead to another decision.

Decision tree analysis involves the construction of a diagram showing all the possible courses of action, states of nature, and the probabilities associated with the states of nature. The *decision diagram* looks very much like a drawing of a tree, therefore also called **decision tree**.

A decision tree consists of *nodes*, *branches*, *probability estimates* and *payoffs*. There are two types of nodes: *decision nodes* and *chance nodes*. A decision node is usually represented by a square and indicates places where a decision maker must make a decision. Each branch leading away from a decision node represents one of the several possible courses of action available to the decision maker. The chance node is represented by a circle and indicates a point at which the decision maker will discover the response to his decision, that is, different possible outcomes occurring from a chosen course of action.

Branches emanate from and connect various nodes and are either decisions or states of nature. There are two types of branches: *decision branches* and *chance branches*. Each branch leading away from a decision node represents a course of action or strategy that can be chosen at a decision point, whereas a branch leading away from a chance node represents the state of nature of a set of chance factors. Associated probabilities are indicated alongside of the respective chance branch. These probabilities are the likelihood that the chance outcome will assume the value assigned to the particular branch. Any branch that makes the end of the decision tree, that is, it is not followed by either another decision or chance node is called a *terminal branch*. A terminal branch can represent either a course of action or a chance outcome. The terminal points of a decision tree are said to be mutually exclusive points so that exactly one course of action will be chosen.

The *payoff* can be positive (i.e., revenue or sales) or negative (i.e., expenditure or cost) and they can be associated either with decisions or chance branches.

An illustration of a decision tree is shown in Fig. 8.1. It is possible for a decision tree to be deterministic or probabilistic and it can be further divided into single stage (a decision under condition of certainty) and a multistage (a sequence of decisions).

The optimal sequence of decisions in a tree is found by starting at the right-hand side and rolling backward. The aim of this operation is to maximize the return from the decision situation. At each node, an expected return is calculated (called the *position value*). If the node is a chance node, then the position value is calculated as the sum of the products of the probabilities or the branches emanating from the chance node and their respective position values. If the node is a decision node, then the expected return is calculated for each of its branches and the highest return is selected. The procedure continues until the initial node is reached. The position values for this node correspond to the maximum expected return obtainable from the decision sequence.

Remarks: *Decision Trees Versus Probability Trees:* Decision trees are basically an extension of probability trees. However, there are several basic differences:

- (a). The decision tree utilizes the concept of 'rollback' to solve a problem. This means starting at the right-hand terminus with the highest expected value of the tree and working back to the current or beginning decision point to determine the decision or decisions that should be made. Most decisions require trees with numerous branches and more than one decision point. It is the multiplicity of decision points that makes the rollback process necessary.
- (b). The probability tree is primarily concerned with calculating the correct probabilities, whereas the decision tree utilizes probability factors as a means of arriving at a final answer.
- (c). A most important feature of the decision tree, not found in probability trees, is that it takes time differences of future earnings into account. At any stage of the decision tree, it may be necessary to weigh the differences in immediate cost or revenue against differences in value at the next stage.

Example 8.14: You are given the following estimates concerning a research and development programme:

Decision D_i	Probability of Decision D_i Given Research R $P(D_i R)$	Outcome Number	Probability of Outcome x_i Given D_i $P(x_i D_i)$	Payoff Value of Outcome, x_i (₹ in '000)
Develop	0.5	1	0.6	600
		2	0.3	-100
		3	0.1	0
Do not develop	0.5	1	0.0	600
		2	0.0	-100
		3	1.0	0

Construct and evaluate the decision tree diagram for the above data. Show your workings for evaluation.

Solution: The decision tree of the given problem along with necessary calculations is shown in Fig. 8.2.

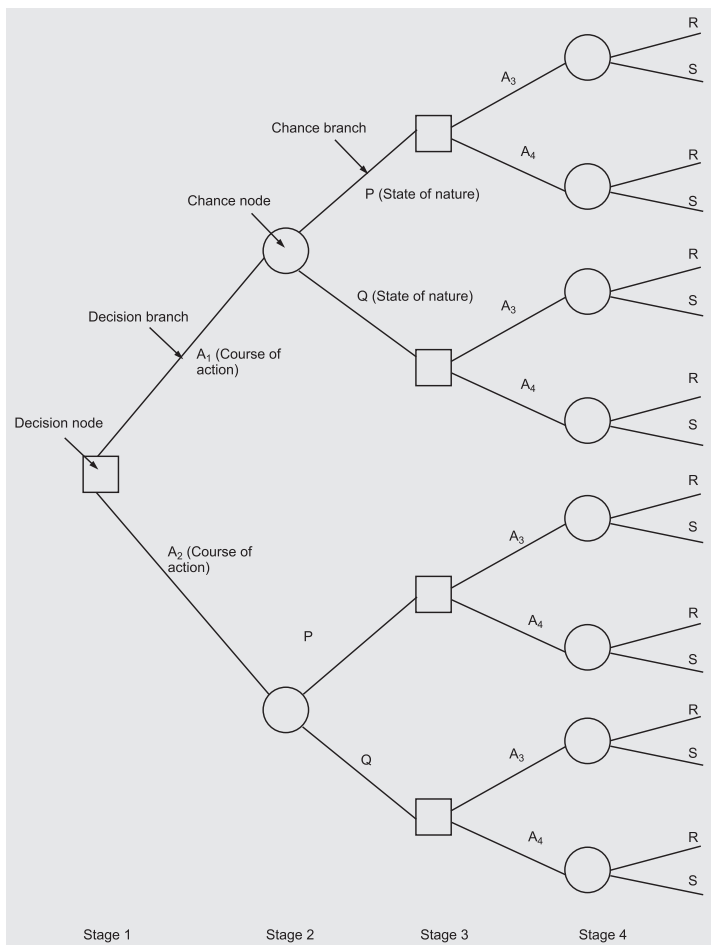


Figure 8.1 Decision Tree

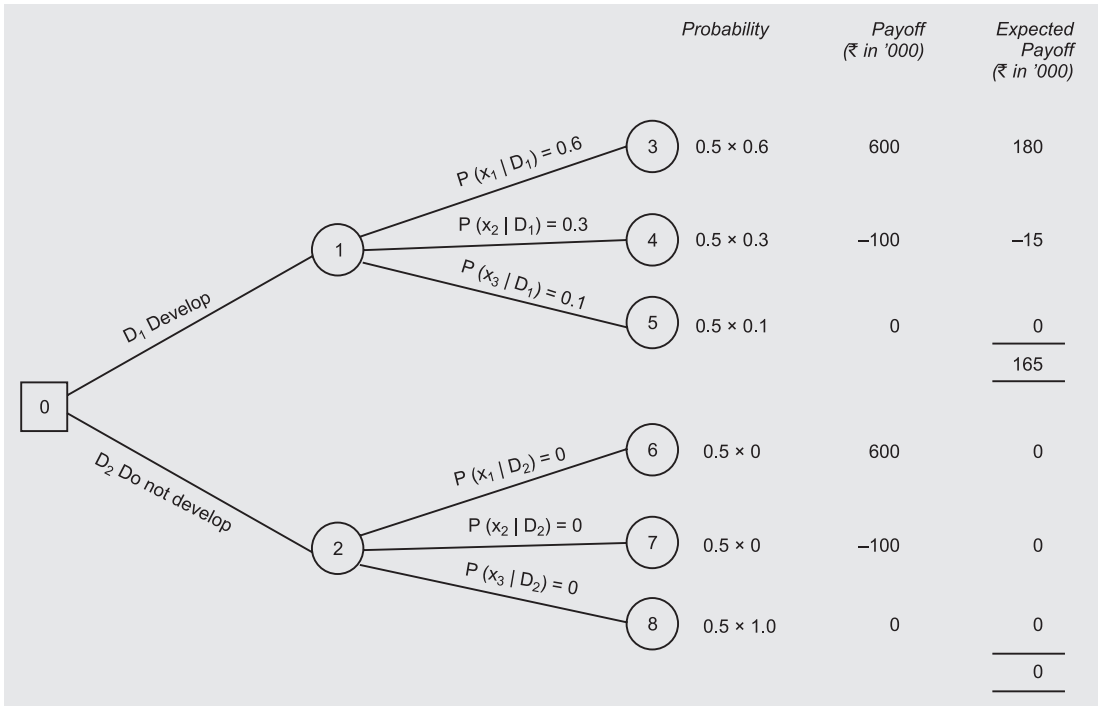


Figure 8.2 Decision Tree

Example 8.15: A glass factory specializing in crystal is developing a substantial backlog and the firm's management is considering three courses of action: Arrange for subcontracting (S_1), begin overtime production (S_2) and construct new facilities (S_3). The correct choice depends largely upon future demand which may be low, medium or high. By consensus, management ranks the respective probabilities as 0.10, 0.50 and 0.40. A cost analysis reveals the effect upon the profits that is shown in the table below:

Demand	Probability	Course of Action		
		S_1 (Subcontracting)	S_2 (Begin Overtime)	S_3 (Construct Facilities)
Low (L)	0.10	10	-20	-150
Medium (M)	0.50	50	60	20
High (H)	0.40	50	100	200

Show this decision situation in the form of a decision tree and indicate the most preferred decision and the corresponding expected value.

Solution: A decision tree which represents possible courses of action and states of nature is shown in Fig. 8.3. In order to analyse the tree, we start working backward from the end branches.

The most preferred decision at the decision node 0 is found by calculating the expected value of each decision branch and selecting the path (course of action) with high value.

Since node 3 has the highest EMV, therefore, the decision at node 0 will be to choose the course of action S_3 , that is, construct new facilities.

Example 8.16: A businessman has two independent investment portfolios A and B available to him, but he lacks the capital to undertake both of them simultaneously. He can choose A first and then stop, or if A is not successful, then take, B or vice versa. The probability of success of A is 0.6, while for B it is 0.4. Both investment schemes require an initial capital outlay of ₹ 10,000 and both return nothing if the venture is unsuccessful. Successful completion of A will return ₹ 20,000 (over cost) and successful completion of B will return ₹ 24,000 (over cost). Draw a decision tree and determine the best strategy.

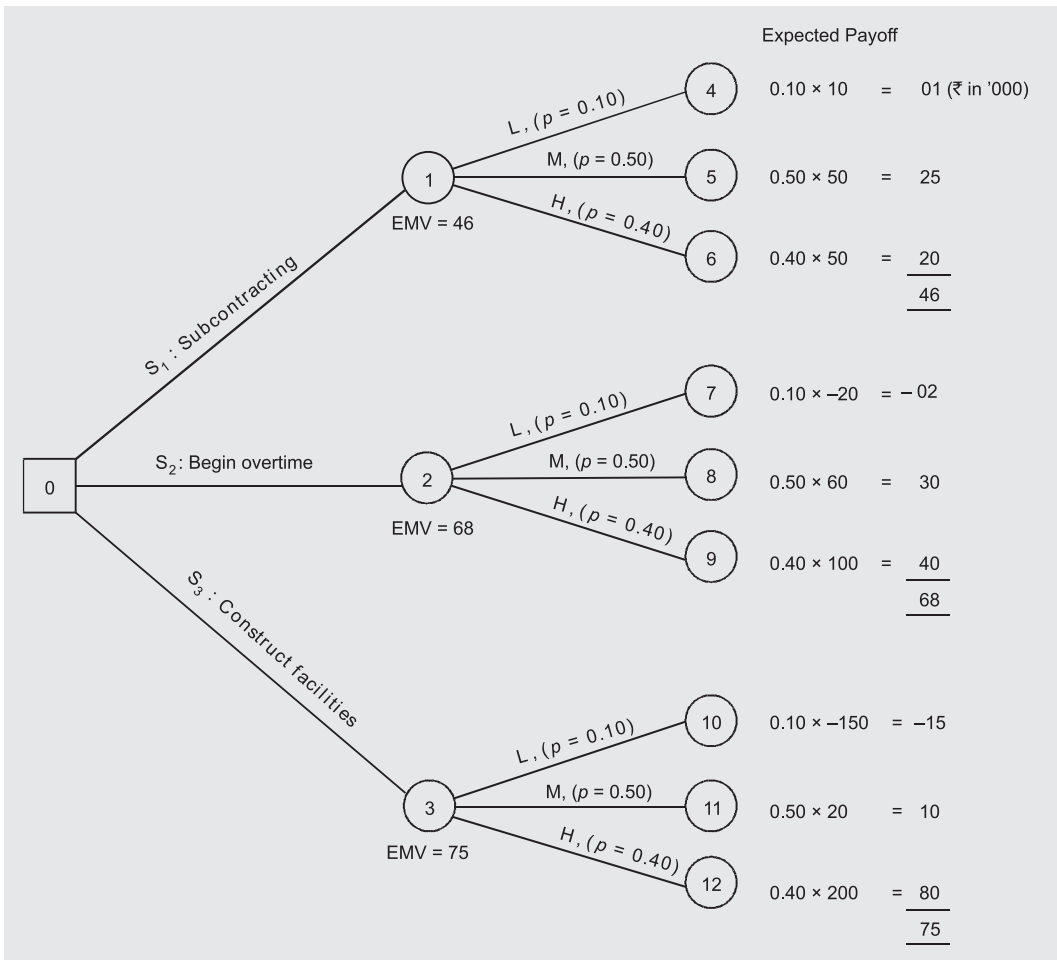


Figure 8.3 Decision Tree

Solution: The decision tree corresponding to the given information is depicted in Fig. 8.4 and Table 8.23.

Table 8.23: Evaluation of Decision and Chance Nodes

Decision Point	Outcome	Probability	Conditional Value (₹)	Expected Value (₹)
D ₃ (i) Accept A	Success	0.6	20,000	12000
	Failure	0.4	-10,000	-4000
				8000
	(ii) Stop	—	—	0
D ₂ (i) Accept B	Success	0.4	24,000	9600
	Failure	0.6	-10,000	-6000
				3600
	(ii) Stop	—	—	0
D ₁ (i) Accept A	Success	0.6	20,000 + 3600	14,160
	Failure	0.4	-10,000	-4000
				10,160
	(ii) Accept B	0.4	24,000 + 8000	12,800
	Failure	0.6	-10,000	-6000
				6800
	(iii) Do nothing	—	—	0

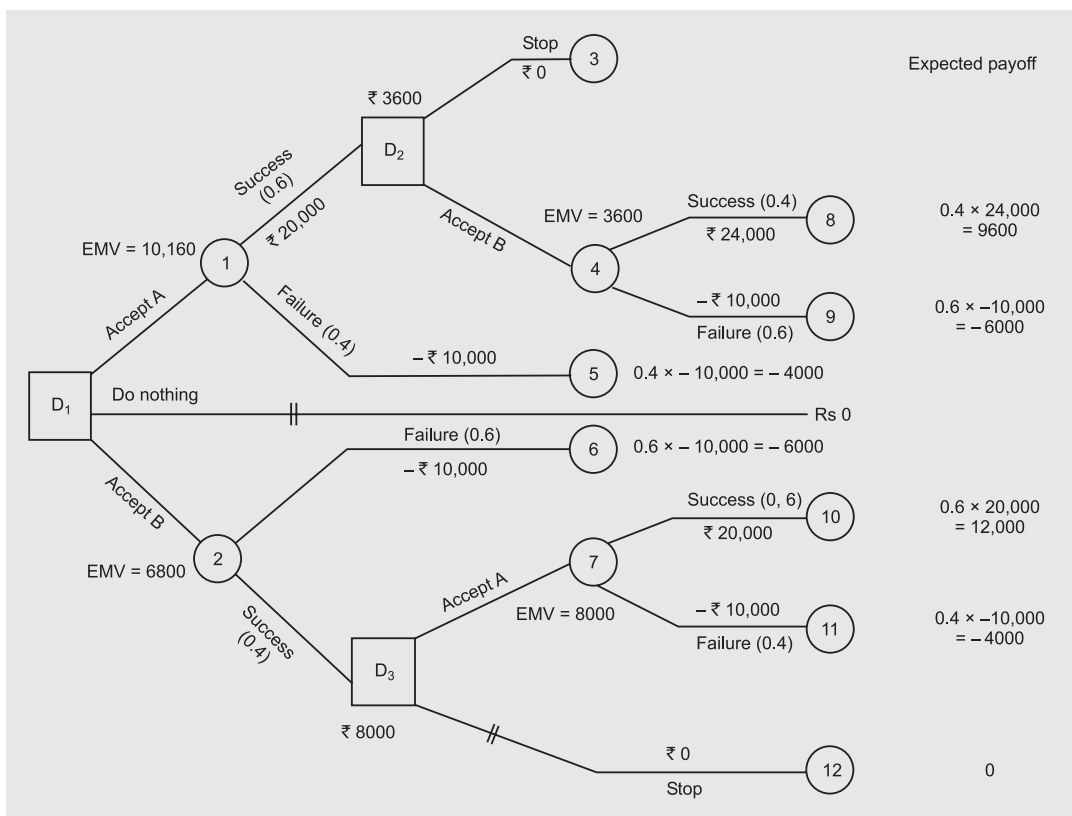


Figure 8.4 Decision Tree

Since the $EMV = ₹10,160$ at node 1 is the highest, therefore, the best strategy at node D_1 is to accept course of action A first and if A is successful, then accept B.

Example 8.17: The Oil India Corporation (OIC) is considering whether to go for an offshore oil drilling contract to be awarded in the Bombay High. If OIC bids, the value would be ₹600 million with a 65 per cent chance of gaining the contract. OIC may set up a new drilling operation or move their already existing operation, which has proved successful, to a new site. The probabilities of success and expected returns are as follows:

Outcome	New Drilling Operation		Existing Operation	
	Probability	Expected Revenue (₹ in million)	Probability	Expected Revenue (₹ in million)
Success	0.75	800	0.85	700
Failure	0.25	200	0.15	350

If the Corporation does not bid or lose the contract, they can use ₹ 600 million to modernize their operations. This would result in a return of either 5 per cent or 8 per cent on the sum invested with probabilities 0.45 and 0.55. (Assume that all costs and revenue have been discounted to present value.)

- Construct a decision tree for the problem showing clearly the courses of action.
- By applying an appropriate decision criterion recommend whether or not the Oil India Corporation should bid for the contract.

Solution: The decision tree based on the given information in the problem is depicted in Fig. 8.5 and Table 8.24.

Table 8.24: Evaluation of Decision and Chance Nodes

Decision Point	Outcome	Probability	Conditional Value (₹)	Expected Value (₹)
D_3 (i) Modernize	5% return	0.45	$600 \times 0.05 = 30$	13.5
	8% return	0.55	$600 \times 0.08 = 48$	26.4
				39.9
D_2 (i) Undertake new drilling operation	Success	0.75	800	600
	Failure	0.25	200	50
				650
	(ii) Move existing operation	Success	700	595
		Failure	350	52.5
				647.5
D_1 (i) Do not bid but modernize	5% return	0.45	$600 \times 0.05 = 30$	13.5
	8% return	0.55	$600 \times 0.08 = 48$	26.4
				39.9
	(ii) Bid	Success	650	422.50
		Failure	39.9	13.96
				436.46

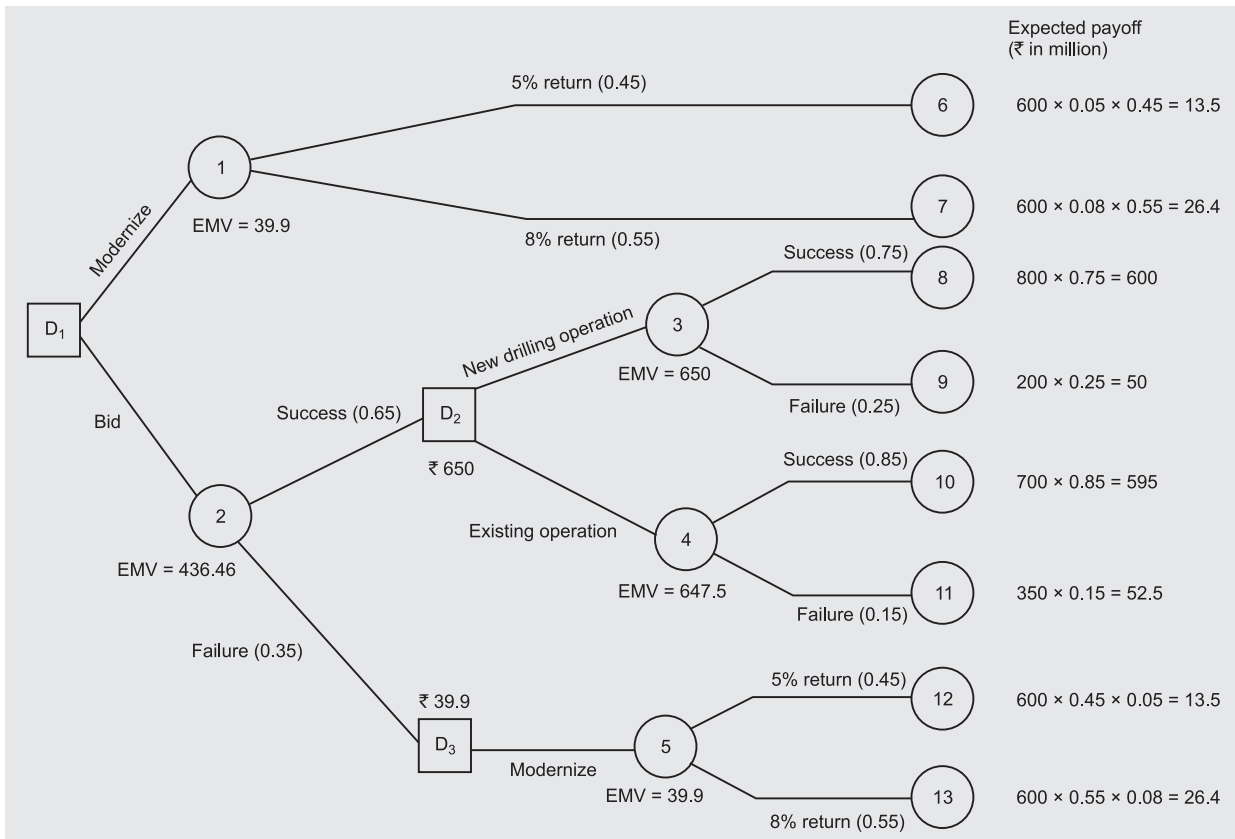


Figure 8.5 Decision Tree

Since the $EMV = ₹ 257.34$ at node 2 is highest, the best decision at node D_2 is to bid and, if successful, establish a new drilling operation.

Example 8.18: Matrix Company is planning to launch a new product which can be introduced initially in Western India or in the whole country. If the product is introduced only in Western India, the investment outlay will be ₹ 12 million. After two years, company can evaluate the project to determine whether it should cover the whole country. For such an expansion, it will have to incur an additional investment of ₹ 10 million. To introduce the product in the whole country right in the beginning would involve an outlay of ₹ 20 million. The product, in any case, will have a life of 5 years, after which the plant will have zero net value.

If the product is introduced only in Western India, demand would be high or low with the probabilities of 0.8 and 0.2, respectively, and annual cash inflow of ₹ 4 million and ₹ 2.5 million, respectively.

If the product is introduced in the whole country right in the beginning, the demand would be high or low with the probabilities of 0.6 and 0.4, respectively, and annual cash inflows of ₹ 8 million and ₹ 5 million, respectively.

Based on the observed demand in Western India, if the product is introduced in the entire country, the following probabilities would exist for high and low demand on an all India basis:

<i>Western India</i>	<i>Whole Country</i>	
	<i>High Demand</i>	<i>Low Demand</i>
High demand	0.90	0.10
Low demand	0.40	0.60

The hurdle rate applicable to this project is 12 per cent.

- Set up a decision tree for the investment situation.
- Advise Matrix Company on the investment policy it should follow. Support your advice with appropriate reasoning.

Solution: The decision tree based on the information given in the problem is depicted in Fig. 8.6 and Table 8.25.

Table 8.25: Evaluation of Decision and Chance Nodes

<i>Decision Point</i>	<i>Outcome</i>	<i>Probability</i>	<i>Conditional Value (₹ in million)</i>	<i>Expected Value (₹ in million)</i>
D ₃	(i) Expansion	High demand	8	7.2
		Low demand	5	0.5
				7.7 × 3 years = 23.1
				Less: Cost 10.0
				13.1
	(ii) Stop			0
				13.1
D ₂	(i) Expansion	High demand	8	3.2
		Low demand	5	3.0
				6.2 × 3 years = 18.6
				Less: Cost 10.0
				8.6
	(ii) No Expansion			0
				8.6
D ₁	(i) Introduction in whole country	High demand	8	4.8
		Low demand	5	2.0
				6.8 × 5 years = 34
				Less: Cost 20
				14
	(ii) Introduction in Western India	High demand	0.8	(4 + 13.1) = 13.68
		Low demand	0.2	(2.5 + 8.6) = 2.22
				15.90
	(iii) Do nothing			Less: Cost 12.00
				3.90

Since the EMV at node 2 is largest, make a decision to launch the product in the whole country.

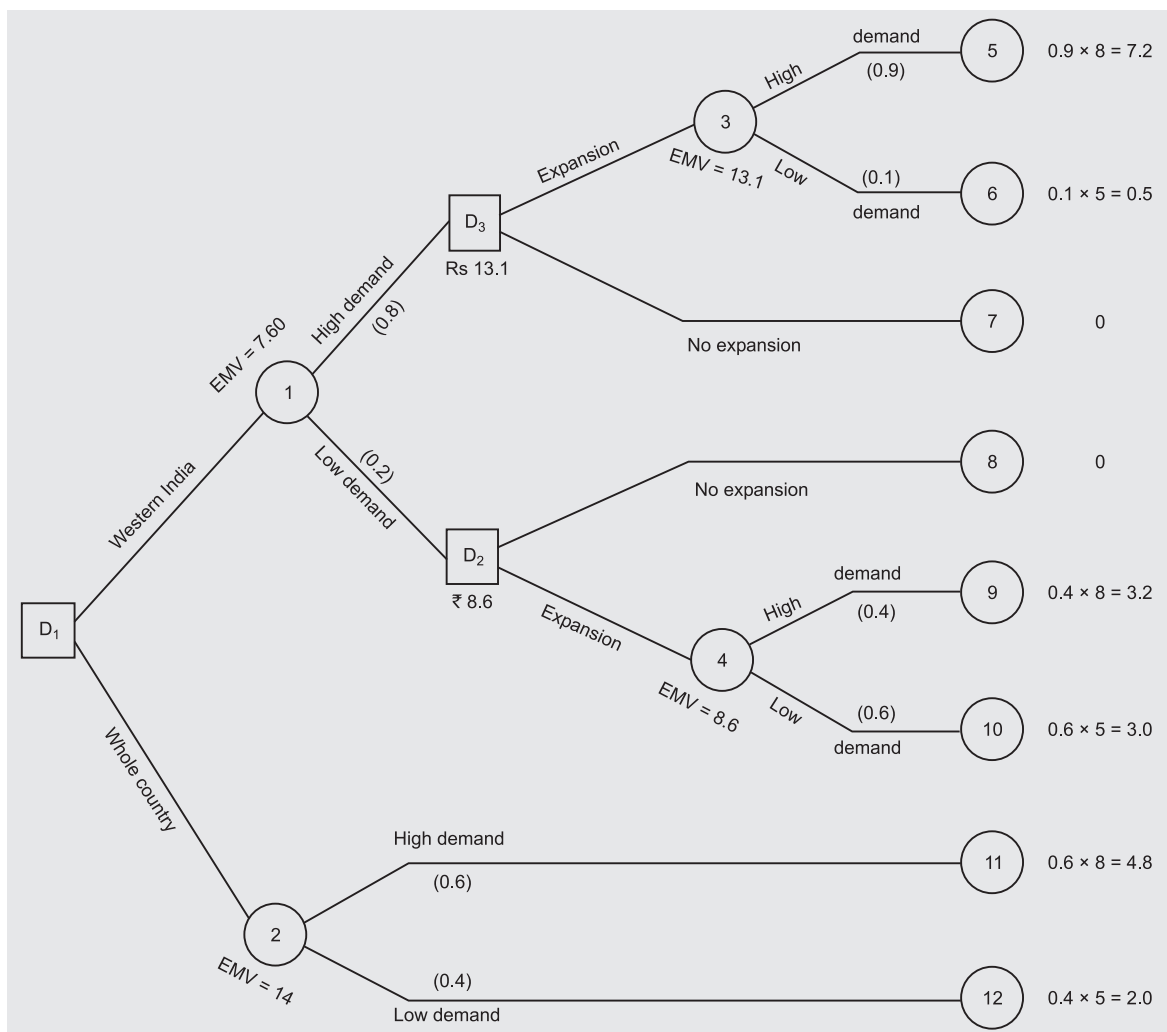


Figure 8.6 Decision Tree

Conceptual Questions 8A

1. Discuss the difference between decision making under certainty, uncertainty and risk.
2. What techniques are used to solve decision making problems under uncertainty? Which technique results in an optimistic decision? Which technique results in a pessimistic decision?
3. Explain the various quantitative methods which are useful for decision-making under uncertainty.
4. What is a scientific decision-making process? Discuss the role of the statistical method in such a process.
5. Give an example of a good decision you made that resulted in a bad outcome. Also give an example of a good decision you made that had a good outcome. Why was each decision good or bad?
6. Given the complete set of outcomes in a certain situation, how is the EMV determined for a specific course of action? Explain in your own words.
7. Explain the difference between expected opportunity loss and expected value of perfect information.
8. Indicate the difference between decision making under risk, and uncertainty in statistical decision theory.
9. Briefly explain 'expected value of perfect information' with examples.
10. Describe a business situation where a decision maker faces a decision under uncertainty and where a decision based on maximizing the expected monetary value cannot be made. How do you think the decision maker should make the required decision?

Self-practice Problems 8B

- 8.6** Raman Industries Ltd. has a new product which they expect has great potential. At the moment, they have two courses of action open to them: To test market (S_1) or to drop the product (S_2).

If they test it, it will cost ₹ 50,000 and the response could be positive or negative with probabilities 0.70 and 0.30, respectively. If it is positive, they could either market it with full effort or drop the product. If they market on a full scale, then the result might be low, medium, or high demand and the respective net payoffs would be—₹ 1,00,000, ₹ 1,00,000, or ₹ 5,00,000. These outcomes have probabilities of 0.25, 0.55, and 0.20, respectively. If the result of the test marketing is negative they may decide to drop the product. If, at any point, they drop the product, there is a net gain of ₹ 25,000 from the sale of scrap. All financial values have been discounted

to the present. Draw a decision tree for the problem and indicate the most preferred decision.

- 8.7** A manufacturing company has just developed a new product. On the basis of past experience, a product such as this will either be successful, with an expected gross return of ₹ 1,00,000, or unsuccessful, with an expected gross return of ₹ 20,000. Similar products manufactured by the company have a record of being successful about 50 per cent of the time. The production and marketing costs of the new product are expected to be ₹ 50,000.

The company is considering whether to market this new product or to drop it. Before making its decision, a test marketing effort can be conducted at a cost of ₹ 10,000. Based on past experience, test marketing results have been favourable

about 70 per cent of the time. Furthermore, products favourably tested have been successful 80 per cent of the time. However, when the test marketing result has been unfavourable, the product has been successful only 30 per cent of the time. What course of action should the company pursue?

- 8.8** The XYZ Company manufactures guaranteed tennis balls. At present time, approximately 10 per cent of the tennis balls are defective. A defective ball leaving the factory costs, the company ₹ 0.50 to honour its guarantee. Assume that all defective balls are returned. At a cost of ₹ 0.10 per ball, the company can conduct a test, which always correctly identifies both good and bad tennis balls.

(a) Draw a decision tree and determine the optimal course of action and its expected cost.

(b) At what testing cost should the company be indifferent to testing?

- 8.9** The Ore Mining Company is attempting to decide whether or not a certain piece of land should be purchased. The land cost is ₹ 3,00,000. If there are commercial ore deposits on the land, the estimated value of the property is ₹ 5,00,000. If no ore deposits exist, however, the property value is estimated at ₹ 2,00,000. Before purchasing the land, the property can be cored at a cost of ₹ 20,000. The coring will indicate if conditions are favourable or unfavourable for ore mining. If the coring report is favourable the probability of recoverable ore deposits on the land is 0.8, while if the coring report is unfavourable, the probability is only 0.2. Prior to obtaining any coring information, management estimates that the odds are 50–50 that ore is present on the land. Management has also received coring reports on pieces of land similar to the one in question and found that 60 per cent of the coring reports were favourable.

Construct a decision tree and determine whether the company should purchase

the land, decline to purchase it, or take a coring test before making its decision. Specify the optimal course of action and the EMV.

- 8.10** XYZ company, dealing with a newly invented telephone device, is faced with the problem of selecting one of the following courses of action available:

- manufacture the device itself; or
- be paid on a royalty basis by another manufacturer; or
- sell the rights for its invention for a lump sum.

The profit (₹ in '000s) which can be expected in each case and the probabilities associated with the level of sales are shown in the following table:

<i>Outcome</i>	<i>Prob</i>	<i>Manufacture Itself</i>	<i>Royalties</i>	<i>Sell all Rights</i>
High sales	0.1	75	35	15
Medium sales	0.3	25	20	15
Low sales	0.6	– 10	10	15

Represent the company's problem in the form of a decision tree. Redraw the decision tree further by introducing the following additional information:

- If it manufactures itself and sales are medium or high, then company has the opportunity of developing a new version of the telephone;
- From past experience, the company estimates that there is a 50 per cent chance of successful development,
- The cost of development is ₹ 15 and returns after deduction of development cost are ₹ 30 and ₹ 10 for high and medium sales, respectively.

- 8.11** An automobile owner faces the decision as to what deductible amount of comprehensive insurance coverage to select. Comprehensive coverage includes losses due to fire, vehicle theft, vandalism, and the forces of nature. The possible choices are zero deductible coverage for ₹ 60 per year or ₹ 50 deductible coverage for

₹ 45 per year. (The owner pays the first ₹ 50 of any loss of at least ₹ 500.) Considering accidents covered by the comprehensive portion of the policy, some of the owner's utility values are given below:

Amount (₹)	:	-95	-60	-50	-45	0
Utility	:	0.2	0.4	0.45	0.47	0.5

- Sketch the owner's utility curve. Is the owner a risk avoider, an EMV'er, or a risk taker?
- On the basis of the utility curve drawn in part (a), should the owner take the zero deductible or the ₹50 deductible comprehensive coverage?

Hints and Answers

- 8.6 The company should test market rather than drop the product. If the test market result is positive, the company should market the product otherwise drop it.
- 8.7 Test marketing should be done; EMV = ₹ 13,800.
- 8.8 (a) Do not test, ₹ 0.05; (b) ₹ 0.05
- 8.9 Test: If favourable—purchase; If not—do not purchase; ₹ 64,000

- 8.10 Royalties; IMV = ₹ 15.5

If p is the probability of high sale, then

- manufacture itself when $p > 0.24$
- pay on royalty basis when $0.07 < p < 0.24$
- sell all rights for lump sum when $p < 0.071$

- 8.11 (b) ₹ 50 deductible.

Formulae Used

1. Expected monetary value EMV (S_j)

$$= \sum_{i=1}^m p_{ij} p_i$$

p_i = probability of occurrence of state of nature i ($i = 1, 2, \dots, m$)

p_{ij} = payoff associated with state of nature, N_i and decision alternative, S_j

2. Expected opportunity loss EOL(N_i)

$$= \sum_{i=1}^m l_{ij} p_i$$

l_{ij} = opportunity loss due to state of nature N_i and decision alternative S_j

3. Expected value of perfect information

$$\text{EVPI} = \sum_{i=1}^m p_{ij}(N_i) - \text{Max EMV}$$

4. Expected profit with perfect information

$$\text{EPPI} = \text{EVPI} + \text{EMV}$$

True or False Concepts Quiz

- Classical methods of statistical inference are based on the use of sample data and the objective interpretation of probabilities.
- Techniques of statistical decision analysis have been developed with continuous as well as discrete variables.

3. The expected payoff criterion is based on the use of probability values as well as economic consequences.
4. A payoff table cannot include the probability value for each event.
5. Acts are referred to as the strategies available to the decision maker.
6. The expected payoff criterion identifies the best decision alternative for which the payoff is the highest.
7. The minimum EOL value is also the act that has the maximum expected payoff.
8. $EVPI = EP$ (with perfect information) + EP (under uncertainty).
9. $EVPI \neq EOL$ (best act under uncertainty).
10. Maximin criterion is reversed if the consequences are in terms of opportunity loss.

Concepts Quiz Answers

1. T	2. T	3. T	4. F	5. T	6. T	7. T	8. F	9. F
10. T								

Review Self-practice Problems

- 8.12** A producer of boats has estimated the following distribution of demand for a particular kind of boat:

No. demanded:	0	1	2	3	4	5	6
Probability	0.14	0.27	0.27	0.18	0.09	0.04	0.01

Each boat costs him ₹ 7000 and he sells them for ₹ 10,000 each. Boats left unsold at the end of the season must be disposed of for ₹ 6000 each. How many boats should be in stock so as to maximize his expected profit?

- 8.13** A small industry finds from past data that the cost of making an item is ₹ 25, the selling price is ₹ 30 if it is sold within a week, and it could be disposed of at ₹ 20 per item at the end of the week:

Weekly sales	<3	4	5	6	7	≥ 8
No. of weeks	0	10	20	40	30	0

Find the optimum number of items per week the industry should produce.

- 8.14** A firm makes pastries which it sells at ₹ 8 per dozen in special boxes containing one dozen each. The direct cost of pastries for the firm is ₹ 4.50 per dozen. At the end of the week, the stale pastries are sold off

at a lower price of ₹ 3.50 per dozen. The overhead expenses attributable to pastry production are ₹ 1.25 per dozen. Fresh pastries are sold in special boxes which cost 50 paise each and the stale pastries are sold wrapped in ordinary paper. The probability distribution of demand per week is as under:

Demand (in dozen):	0	1	2	3	4	5
Probability	0.01	0.14	0.2	0.5	0.1	0.05

Find the optimal production level of pastries per week.

- 8.15** The local football club wants your advice on the number of programmes that should be printed for each game. The cost of printing and production of programmes for each game, as quoted by the local printer, is ₹ 1000 plus 4 paise per copy. Advertising revenue which has been agreed upon for the season represents ₹ 800 for each game.

Programmes are sold for 15 paise each. A review of sales during the previous season indicates that the following pattern is expected to be repeated during the coming season of 50 games:

<i>Number of Programmes Sold</i>	<i>Number of Games</i>
10,000	5
20,000	20
30,000	15
40,000	10

Programmes not sold at the game are sold as waste paper to a paper manufacturer at one paisa per copy:

Assuming that the four options listed are the only possibilities:

- prepare a payoff table.
- determine the number of programmes that would provide the largest profit, if a constant number of programmes were to be printed for each game.
- calculate the profit which would arise from a perfect forecast of the number of programmes which would be sold at each game.

8.16 The probability distribution of monthly sales (units) of an item is as follows:

Monthly sales :	0	1	2	3	4	5	6
Probability :	0.01	0.06	0.25	0.30	0.22	0.10	0.06

The cost of carrying inventory (unsold during the month) is ₹ 30 per unit per month and the cost of unit shortage is ₹ 70. Determine the optimum stock to minimize expected cost.

8.17 A modern home appliances dealer finds that the cost of holding a mini cooking range in stock for a month is ₹ 200 (insurance, minor deterioration, interest on borrowed capital, etc.). Customers who cannot obtain a cooking range immediately tend to go to other dealers and he estimates that for every customer who cannot get immediate delivery, he loses an average of ₹ 500. The probabilities of a demand of 0, 1, 2, 3, 4, 5 mini cooking ranges in a month are 0.05, 0.10, 0.20, 0.30, 0.20 and 0.15, respectively. Determine the optimal stock level of cooking ranges. Also find the EVPI.

8.18 A TV dealer finds that the cost of holding a TV in stock for a week is ₹ 50. Customers who cannot obtain new TVs immediately

tend to go to other dealers and he estimates that for every customer who cannot get immediate delivery loses an average of ₹ 200. For one particular model of TV, the probabilities of a demand of 0, 1, 2, 3, 4 and 5 TVs in a week are 0.05, 0.10, 0.20, 0.30, 0.20, and 0.15, respectively.

- How many televisions per week should the dealer order? Assume there is no time lag between ordering and delivery.
- Compute the EVPI.
- The dealer is thinking of spending on a small market survey to obtain additional information regarding demand levels. How much should he be willing to spend on such a survey?

[Delhi Univ., MBA, 1988]

8.19 XYZ Co. Ltd wants to go in for a public share issue of ₹ 10 lakh (1 lakh shares of ₹ 10 each) as a part of its efforts to raise capital needed for its expansion programme. The company is optimistic that if the issue were made now, it would be fully taken up at a price of ₹ 30 per share.

However, the company is facing two crucial situations both of which may influence the share prices in the near future. These are:

- An impending wage dispute with assembly workers which could lead to a strike in the whole factory and could have an adverse effect on the share price.
- The possibility of substantial business in the export market which would increase the share price.

The four possible events and their expected effect on the company's share prices are envisaged as:

- E_1 : No strike, and no export business obtained—share price stays at ₹ 34.
 E_2 : Strike, and export business obtained—share price stays at ₹ 30.
 E_3 : No strike, and export business lost—share price hovers around ₹ 32.
 E_4 : Strike, and export business lost—share price drops to ₹ 16.

And the management has identified three possible strategies that the company could adopt, namely

S_1 : Issue 1,00,000 shares now.

S_2 : Issue 1,00,000 shares only after the outcome of (i) and (ii) above are known.

S_3 : Issue 50,000 shares now and 50,000 shares after the outcome (i) and (ii) are known.

You are required to:

- Draw up a payoff table for the company and determine the minimax regret solution. What alternative criteria might be used?
- Determine the optimum policy for the company using the criterion of maximizing expected payoff, given the estimate that the probability of a strike is 55 per cent and there is a 65 per cent chance of getting the export business, these probabilities being independent.
- Determine the expected value of perfect information for the company.

8.20 A company manufacturing large electrical equipment is anticipating the possibility of a total or a partial copper strike in the near future. It is attempting to decide whether to stockpile a large amount of copper, at an additional cost of ₹ 50,000; a small amount, costing an additional ₹ 20,000; or to stockpile no additional copper at all. The stockpiling costs, consisting of excess storage, holding and handling costs and so forth, are over and above the actual material costs.

If there is a partial strike, the company estimates that an additional cost of ₹ 50,000 for delayed orders will be incurred if there is no stockpile at all. If a total strike occurs, the cost of delayed orders is estimated at ₹ 1,00,000 if there is only a small stockpile and ₹ 2,00,000 with no stockpile. The company estimates the probability of a total strike as 0.1 and that of a partial strike as 0.3.

- Develop a conditional cost table showing the cost of all outcomes and course of action combinations.

- Determine the preferred course of action and its cost. What is the EPPI?
- Develop a conditional opportunity loss table.
- Without calculating the EOL, find the EVPI.

8.21 A well-known departmental store advertises female fashion garments from time to time in the Sunday press. Only one garment is advertised on each occasion. Experience of garments in the price range ₹ 30–40 leads the management to assess the statistical probability of demand at various levels after each advertisement as follows:

<i>Demand (No. of Garments)</i>	<i>After Advertising in One Newspaper</i>	<i>After Advertising in Two Newspapers</i>
30	0.10	0.00
40	0.25	0.15
50	0.40	0.35
60	0.25	0.40
70	0.00	0.10

The next garment to be advertised at ₹ 35 will cost ₹ 15 to make and can be disposed off to the trade, if unsold, for ₹ 10. The store's advertising agents will charge ₹ 37.50 for artwork and blockmaking for the advertisement and each newspaper will charge ₹ 50 for the insertion of the advertisement. You are required to:

- Calculate how many garments to the nearest 10 should be purchased by the store in order to maximize expected gross profit, after advertising in: (i) one newspaper (ii) two newspapers.
- Calculate the amount, if any, of the expected net profit after advertising: (i) in one newspaper (ii) two newspapers.

8.22 A car manufacturer uses a special control device in each car he produces. Two alternative methods can be used to detect and avoid a faulty device. Under the first method, each device is tested before it is installed. The cost is ₹ 2 per test.

Alternatively, the control device can be installed without being tested, and a faulty device can be detected and rendered after the car has been assembled, at a cost of ₹20 per faulty device.

Regardless of which method is used, faulty devices cannot be repaired and must be discarded.

A manufacturer purchases the control devices in batches of 10,000. Based on past experience, he estimates the proportion of defective components and the associated probability to be:

<i>Proportion of Faulty Devices</i>	<i>Probability</i>
0.08	0.20
0.12	0.70
0.16	0.10

Hints and Answers

8.12 Conditional profit value

$$\begin{aligned}
 &= MP \times \text{boats sold} - ML \times \text{boats unsold} \\
 &= (10000 - 7000) \text{ boats sold} \\
 &\quad - (7000 - 6000) \text{ boats unsold} \\
 &= \begin{cases} 300 D & ; D \geq S \\ 3000 D - 1000 (S - D) = 4000 D - 1000 S; & D < S \end{cases} \\
 &\quad \text{where } D \text{ is the number of boats sold and } S \\
 &\quad \text{is the number of boats produced} \\
 &\quad EMV^* = 4,080, \text{ stock 3 boats.}
 \end{aligned}$$

8.13 Conditional profit value

$$\begin{aligned}
 &= MP \times \text{item sold} - ML \times \text{item unsold} \\
 &= (30 - 25) \text{ item sold} - (25 - 20) \text{ item unsold} \\
 &= \begin{cases} (30 - 25) D & ; D \geq S \\ (30 - 25) D - (25 - 20) (S - D) = 10 D - 5 S; & D < S \end{cases} \\
 &\quad \text{where } D \text{ is the number of item sold; } S \text{ is} \\
 &\quad \text{the number of item produced.} \\
 &\quad EMV = ₹ 26, \text{ produce 6 items.}
 \end{aligned}$$

8.14 Conditional profit value

$$\begin{aligned}
 &= MP \times \text{boxes sold} - ML \times \text{boxes unsold} \\
 &= (8 - 4.50 - 1.25 - 0.50) \text{ boxes sold} \\
 &\quad - (4.50 + 1.125 - 5.50) \text{ boxes unsold} \\
 &= \begin{cases} 1.75 D & ; D \geq S \\ 1.75 D - 0.25 (S - D) = 2 D - 0.25 S; & D < S \end{cases}
 \end{aligned}$$

(a) Which inspection method should the manufacturer adopt?

(b) What is the expected value of perfect information (EVPI)?

8.23 Suppose a company has several independent investment opportunities each of which has an equal chance of gaining ₹ 1,00,000 or losing ₹ 60,000. What is the probability that the company will lose money on two such investments? On three such investments? On four such investments?

If a company has a number of independent investment opportunities, in each of which, the financial risk is relatively small compared to its overall asset position, why should the company try to maximize EMV, rather than expected utility?

EMV* = ₹ 4.28, produce 4 dozen pastries.

8.15 (a) Conditional profit

$$\begin{aligned}
 &= \text{Sales revenue} - \text{Cost} \\
 &= \{0.15 D + 800 + (P - D) \times 0.101\} \\
 &\quad - (1,000 + 0.04 P) \\
 &= \begin{cases} 0.14 D - 0.03 P - 20; & D < P \\ 0.11 D - 200 & ; D \geq P \end{cases}
 \end{aligned}$$

where P is the number of programmes printed and D is the demand for programmes.

(b) largest expected profit = ₹ 2260 accrues for P = 30,000 copies per game

(c) In case of perfect forecast, expected profit is:

$$900 \times 0.1 + 2000 \times 0.4 + 3100 \times 0.3 + 4200 \times 0.2 = ₹ 2660$$

$$EVPI = ₹ 2660 - ₹ 2260 = ₹ 400$$

$$\text{8.16 Cost function} = \begin{cases} 70 (D - S) & ; D \geq S \\ 200 (S - D); & D < S \end{cases}$$

where D = monthly demand (or sales); S = number of units purchased.

Since expected cost ₹ 46 is minimum for course of action 4, optimum stock to minimize the cost is 4 units per month.

$$8.17 \text{ Cost function} = \begin{cases} 500(D-S); & D \geq S \\ 200(S-D); & D < S \end{cases}$$

Since expected cost ₹ 315 is minimum for course of action 4, optimum stock to minimize the cost is 4 cooking ranges.

8.18 Cost function

$$= \begin{cases} 50S + 200(D-S); & D \geq S \\ 50D + 50(S-D); & D < S \end{cases}$$

where D = number of TV sets demanded;
 S = number of TV sets ordered (or stored)

Since expected cost ₹ 230 is minimum, dealer must order 4 TV sets per week.

$$\begin{aligned} \text{EVPI} &= \text{Minimum expected cost} \\ &\quad - \text{Expected cost under perfect information} = 230 - 147.5 \\ &= ₹ 82.5 \end{aligned}$$

8.19 (a) S_1 ; ₹ 4; Alternative: Maximum criterion; S_1 , ₹ 30.

(b) Adopt S_1 ; ₹ 30.075 lakh

$$\begin{aligned} \text{(c) EVPI} &= \text{EPPI} - \text{Optimal EMV} \\ &= 31.48 - 30.075 \\ &= ₹ 1.40 \text{ lakh.} \end{aligned}$$

8.20 (b) small stockpile; ₹ 30,000 – ₹ 11,000;
(d) ₹ 19,000

$$\begin{aligned} 8.21 \text{ Marginal profit (MP)} &= ₹ (35 - 15) \\ &= ₹ 20 \end{aligned}$$

$$\text{Marginal loss (ML)} = ₹ (15 - 10) = ₹ 5$$

$$\begin{aligned} \text{Conditional payoff} &= \text{MP} \times \text{garments sold} \\ &\quad - \text{ML} \times \text{garments unsold} \end{aligned}$$

$$(a) \text{ EMV } (S) = ₹ 600;$$

$$\text{EMV } (S_2) = ₹ 775;$$

$$\text{EMV } (S_3) = ₹ 887.5;$$

$$\text{EMV } (S_4) = ₹ 900;$$

$$\text{EMV } (S_5) = ₹ 850$$

Max. EMV is corresponding to course of action S_4 , i.e. purchase 60 garments.

$$(b) \text{ EMV } (S_2) = ₹ 800;$$

$$\text{EMV } (S_3) = ₹ 962.5;$$

$$\text{EMV } (S_4) = ₹ 1,037.5;$$

$$\text{EMV } (S_5) = ₹ 1,012.5.$$

Max. EMV is corresponding to course of action S_4 , i.e. purchase 60 garments.

8.22 (a) First alternative, ₹ 20,000

$$\begin{aligned} \text{(b) EVPI} &= \text{Expected cost under uncertainty} - \text{Expected cost with perfect information} \\ &= 20,000 - (16,000 \times 0.2 + 20,000 \times 0.7 + 20,000 \times 0.1) \\ &= 20,000 - 19,200 = ₹ 800 \text{ per batch.} \end{aligned}$$

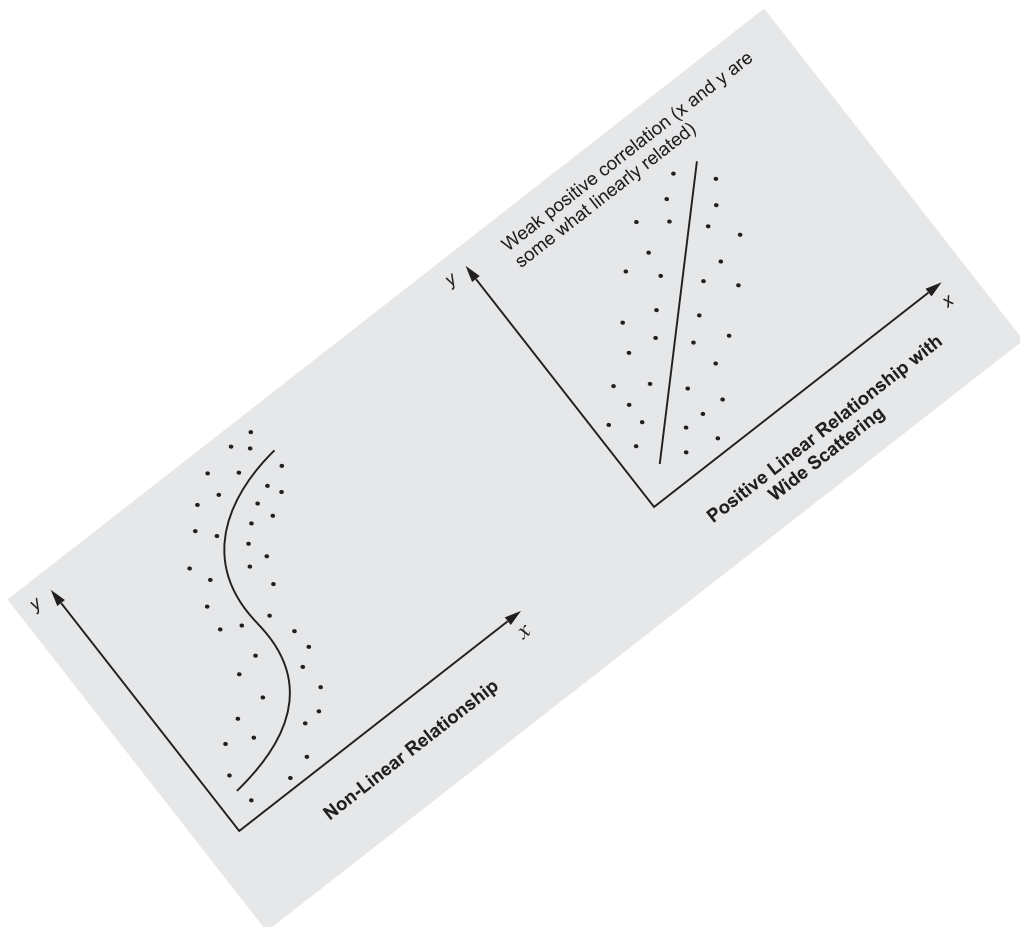
8.23 0.25; 0.50; 0.3125

UNIT III

SIMPLE CORRELATION AND REGRESSION ANALYSIS

Chapter 9: Correlation Analysis

Chapter 10: Regression Analysis



This page is intentionally left blank.

9

Correlation Analysis

LEARNING OBJECTIVES

After studying this chapter, you should be able to

- express quantitatively the degree and direction of the covariation or association between two variables.
- determine the validity and reliability of the covariation or association between two variables.
- provide a test of hypothesis to determine whether a linear relationship actually exists.

9.1 INTRODUCTION

The statistical methods, discussed so far, are used to analyse the data involving only one variable. Often an analysis of data concerning two or more quantitative variables is needed to look for any statistical relationship or association between them that can describe specific numerical features of the association. The knowledge of such a relationship is important to make inferences from the relationship between variables in a given situation. Few instances where the knowledge of an association or relationship between two variables would prove vital to make decision are:

- Family income and expenditure on luxury items.
- Yield of a crop and quantity of fertilizer used.
- Sales revenue and expenses incurred on advertising.
- Frequency of smoking and lung damage.
- Weight and height of individuals.

A statistical technique that is used to analyse the strength and direction of the relationship between two quantitative variables is called *correlation analysis*. A few definitions of correlation analysis are as follows:

- *An analysis of the relationship of two or more variables is usually called correlation.*

— A. M. Tuttle

- *When the relationship is of a quantitative nature, the appropriate statistical tool for discovering and measuring the relationship and expressing it in a brief formula is known as correlation.*

— Croxton and Cowden

The **coefficient of correlation** is a number that indicates the *strength (magnitude)* and *direction* of statistical relationship between two variables.

- The **strength** of the relationship is determined by the closeness of the points to a straight line when a pair of values of two variables are plotted on a graph. A straight line is used as the frame of reference for evaluating the relationship.
- The **direction** is determined by whether one variable generally increases or decreases when the other variable increases.

The importance of examining the statistical relationship between two or more variables can be divided into the following questions and accordingly requires the statistical methods to answer these questions:

- (i) Is there an association between two or more variables? If yes, what is the form and degree of that relationship?
- (ii) Is the relationship strong or significant enough to be useful to arrive at a desirable conclusion?
- (iii) Can the relationship be used for predictive purposes, that is, to predict the most likely value of a dependent variable corresponding to the given value of independent variable or variables?

9.2 SIGNIFICANCE OF MEASURING CORRELATION

The objective of any scientific research is to establish relationships between two or more variables (or observations) to arrive at some conclusion which is also near to reality. Finding such relationships is often an initial step for identifying causal relationships. Few advantages of measuring an association (or correlation) between two or more variables are as under:

1. Correlation analysis contributes to the understanding of economic behaviour, aids in locating the critically important variables on which others depend, may reveal to the economist the connections by which disturbances spread and suggest to him the paths through which stabilizing forces may become effective.
—W. A. Neiswanger
2. The effect of correlation is to reduce the range of uncertainty of our prediction. The prediction based on correlation analysis will be more reliable and near to reality.
— Tippet
3. In economic theory, we come across several types of variables which show some kind of relationship. For example, there exists a relationship between price, supply and quantity demanded; convenience, amenities and service standards are related to customer retention; yield of a crop related to quantity of fertilizer applied, type of soil, quality of seeds, rainfall and so on. Correlation analysis helps in quantifying precisely the degree of association and direction of such relationships.
4. Correlations are useful in the areas of healthcare such as determining the validity and reliability of clinical measures or in expressing how health problems are related to certain biological or environmental factors.

9.3 CORRELATION AND CAUSATION

There are at least three criteria for establishing a causal relationship; correlation is one of them. While drawing inferences from the value of correlation coefficient, we overlook the fact that it measures only the strength of a linear relationship and it does not necessarily imply a causal relationship. That is, there are several other explanations for finding a correlation.

The following factors should be examined to interpret the nature and extent of relationship between two or more variables:

1. **Chance coincidence:** A correlation coefficient may not reach any statistical significance, that is, it may represent a nonsense (spurious) or chance association. For example, (i) a positive correlation between growth in population and wheat production in the country has no statistical significance. Because each of the two events might have entirely different unrelated causes. (ii) While estimating the correlation in sales revenue and expenditure on advertisements over a period of time, the investigator must be certain that the outcome is not due to biased sampling or sampling error. That is, he needs to show that a correlation coefficient is statistically significant and not just due to random sampling error.
2. **Influence of third variable:** If the correlation coefficient does not establish any relationship, it can be used as a source for testing null and alternative hypotheses about a population. For example, it has been proved that smoking causes lung damage. However, given that there is often multiple reasons of health problems, the reason of stress cannot be ruled out. Similarly, there is a positive correlation between the yield of rice and tea because the crops are influenced by the amount of rainfall. But the yield of any one is not influenced by other.
3. **Mutual influence:** There may be a high degree of relationship between two variables but it is difficult to say as to which variable is influencing the other. For example, variables like price, supply and demand of a commodity are mutually correlated. According to the principle of economics, as the price of a commodity increases, its demand decreases, so price influences the demand level. But if demand of a commodity increases due to growth in population, then its price also increases. In this case, increased demand make an effect on the price. However, the amount of export of a commodity is influenced by an increase or a decrease in custom duties but the reverse is normally not true.

9.4 TYPES OF CORRELATIONS

There are three broad types of correlations:

1. Positive and negative
2. Linear and non-linear
3. Simple, partial and multiple

In this chapter, we will discuss simple linear positive or negative correlation analysis.

9.4.1 Positive and Negative Correlation

A positive (or direct) correlation refers to the same direction of change in the values of variables. In other words, if values of variables are varying (i.e., increasing or decreasing) in the same direction, then such correlation is referred to as **positive correlation**.

A **negative (or inverse) correlation** refers to the change in the values of variables in opposite direction.

The following examples illustrate the concept of positive and negative correlation.

Positive Correlation

Increasing	→ x	:	5	8	10	15	17
Increasing	→ y	:	10	12	16	18	20
Decreasing	→ x	:	17	15	10	8	5
Decreasing	→ y	:	20	18	16	12	10

Negative Correlation

Increasing $\rightarrow x$	:	5	8	10	15	17
Decreasing $\rightarrow y$	:	20	18	16	12	10
Decreasing $\rightarrow x$	:	17	15	12	10	6
Increasing $\rightarrow y$	:	2	7	9	13	14

It may be noted here that the change (increasing or decreasing) in values of both the variables is not proportional or fixed.

9.4.2 Linear and Non-linear Correlation

A linear correlation implies a constant change in one of the variable values with respect to a change in the corresponding values of another variable. In other words, a correlation is referred to as *linear correlation* when variations in the values of two variables have a constant ratio. The following example illustrates a linear correlation between two variables x and y .

x	:	10	20	30	40	50
y	:	40	60	80	100	120

When these pairs of values of x and y are plotted on a graph paper, the line joining these points would be a straight line.

A non-linear (or curvi linear) correlation implies an absolute change in one of the variable values with respect to changes in values of another variable. In other words, a correlation is referred to as a *non-linear correlation* when the amount of change in the values of one variable does not bear a constant ratio to the amount of change in the corresponding values of another variable. The following example illustrates a non-linear correlation between two variables x and y .

x	:	8	9	9	10	10	28	29	30
y	:	80	130	170	150	230	560	460	600

When these pair of values of x and y are plotted on a graph paper, the line joining these points would not be a straight line, rather it would be curvi linear.

9.4.3 Simple, Partial and Multiple Correlation

The distinction between simple, partial and multiple correlation is based upon the number of variables involved in the correlation analysis.

If only two variables are chosen to study correlation between them, then such a correlation is referred to as **simple correlation**. A study on the yield of a crop with respect to only amount of fertilizer, or sales revenue with respect to amount of money spent on advertisement, are a few examples of simple correlation.

In **partial correlation**, two variables are chosen to study the correlation between them, but the effect of other influencing variables is kept constant. For example, (i) yield of a crop is influenced by the amount of fertilizer applied, rainfall, quality of seed, type of soil and pesticides, (ii) sales revenue from a product is influenced by the level of advertising expenditure, quality of the product, price, competitors and distribution. In such cases, an attempt to measure the correlation between yield and seed quality, assuming that the average values of other factors exist, becomes a problem of partial correlation.

In **multiple correlation**, the relationship between more than three variables is considered

simultaneously for study. For example, employer–employee relationship in any organization may be examined with reference to training and development facilities; medical, housing and education to children facilities; salary structure and grievances handling system.

9.5 METHODS OF CORRELATION ANALYSIS

The correlation between two variables is represented by the letter r which takes on values between -1 and $+1$ only. Sometimes this measure is called the **Pearson product moment correlation coefficient**. The correlation coefficient is scale free and therefore its interpretation is independent of the units of measurement of two variables, say x and y .

In this chapter, the following methods of finding the correlation coefficient between any two variables x and y are discussed:

1. Scatter Diagram method
2. Karl Pearson's Coefficient of Correlation method
3. Spearman's Rank Correlation method
4. Method of Least squares

Figure 9.1 shows how the strength of the association between two variables is represented by the coefficient of correlation.

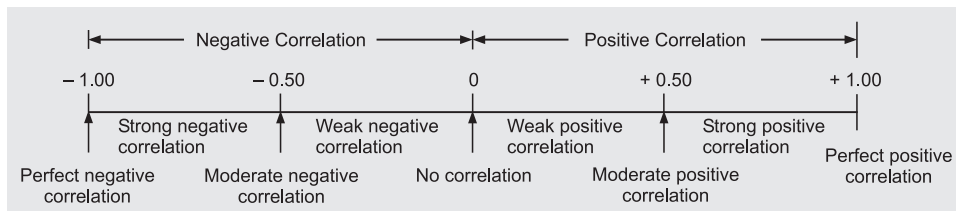


Figure 9.1 Interpretation of Correlation Coefficient

9.5.1 Scatter Diagram Method

The **scatter diagram** method is a quick at-a-glance method of determining of an apparent relationship between two variables, if any. A scatter diagram (or a graph) can be obtained on a graph paper by plotting observed (or known) pairs of values of variables x and y , taking the independent variable values on the x -axis and the dependent variable values on the y -axis.

It is common to try to draw a straight line through data points so that an equal number of points lie on either side of the line. The relationship between two variables x and y described by the data points is defined by this straight line.

In a scatter diagram, the horizontal and vertical axes are scaled in units corresponding to the variables x and y , respectively. Figure 9.2 shows examples of different types of relationships based on pairs of values of x and y in a sample data. The pattern of data points in the diagram indicates that the variables are related. If the variables are related, then the dotted line appearing in each diagram describes the relationship between the two variables.

The patterns depicted in Figures 9.2(a) and (b) represent linear relationships since the

patterns are described by straight lines. The pattern in Figure 9.2(a) shows a *positive* relationship since the value of y tends to increase as the value of x increases, whereas pattern in Figure 9.2(b) shows a *negative* relationship since the value of y tends to decrease as the value of x increases.

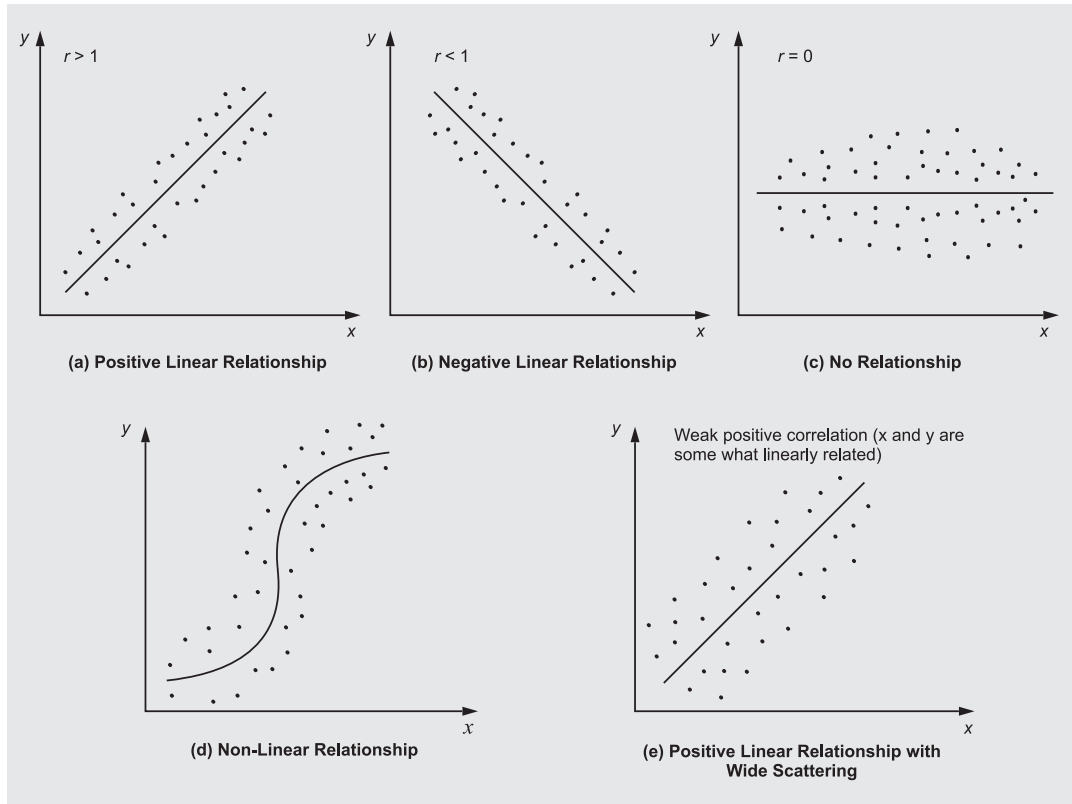


Figure 9.2 Typical Examples of Correlation Coefficient

The pattern depicted in Figure 9.2(c) illustrates very low or no relationship between the values of x and y , whereas Figure 9.2(d) represents a curvi linear relationship since it is described by a curve rather than a straight line. Figure 9.2(e) illustrates a positive linear relationship with a widely scattered pattern of points. The wider scattering indicates that there is a lower degree of association between the two variables x and y than there is in Figure 9.2(a).

Interpretation of Correlation Coefficients While interpreting correlation coefficient r , the following points should be taken into account:

- (i) A low value of r does not indicate that the variables are unrelated but indicates that the relationship is poorly described by a straight line. A non-linear relationship may also exist.
- (ii) A correlation does not imply a *cause-and-effect* relationship, it is merely an observed association.

Specific Features of the Correlation Coefficient Regardless of the type of correlation coefficient we use the following are the common among all of them:

- (i) The value of r depends on the slope of the line passing through the data points and the scattering of the pair of values of variables x and y about this line (for detail see Chapter 10).
- (ii) The sign of the correlation coefficient indicates the direction of the relationship. A positive correlation denoted by $+$ (positive sign) indicates that the two variables tend to increase (or decrease) together (a positive association) and a negative correlation by $-$ (minus sign) indicates that when one variable increases the other is likely to decrease (a negative association).
- (iii) The values of the correlation coefficient range from $+1$ to -1 regardless of the units of measurements of x and y .
- (iv) The value of $r = +1$ or -1 indicates that there is a perfect linear relationship between two variables, x and y . A perfect correlation implies that every observed pair of values of x and y falls on the straight line.
- (v) The magnitude of the correlation indicates the strength of the relationship, which is the overall closeness of the points to a straight line. The sign of the correlation does indicates about the strength of the linear relationship.
- (vi) Correlation coefficient is independent of the change of origin and scale of reference. In other words, its value remains unchanged when we subtract some constant from every given value of variables x and y (change of origin) and when we divide or multiply by some constant every given value of x and y (change of scale).
- (vii) Correlation coefficient is a pure number independent of the unit of measurement.
- (viii) The value of $r=0$ indicates that the straight line through the data is exactly horizontal, so that the value of variable x does not change the predicated value of variable y .
- (ix) The square of r , i.e., r^2 is referred to as *coefficient of determination*.

Further, from Figures 9.2(a)—(e) we conclude that the closer the value of r is to either $+1$ or -1 , the stronger is the association between x and y . Also, closer the value of r to 0 , the weaker the association between x and y appears to be.

Example 9.1: Given the following data:

Student	:	1	2	3	4	5	6	7	8	9	10
Management aptitude score	:	400	675	475	350	425	600	550	325	675	450
Grade point average	:	1.8	3.8	2.8	1.7	2.8	3.1	2.6	1.9	3.2	2.3

- (a) Draw this data on a graph paper.
- (b) Is there any correlation between per capita national income and per capita consumer expenditure? If yes, what is your opinion.

Solution: By taking an appropriate scale on the x and y axes, the pair of observations are plotted on a graph paper as shown in Figure 9.3. The scatter diagram in Figure 9.3 with straight line represents the relationship between x and y 'fitted' through it.

Interpretation: From the scatter diagram shown in Figure 9.3, it appears that there is a high degree of association between two variable values. It is because the data points are very close to a straight line passing through the points. This pattern of dotted points also indicates a high degree of linear positive correlation.

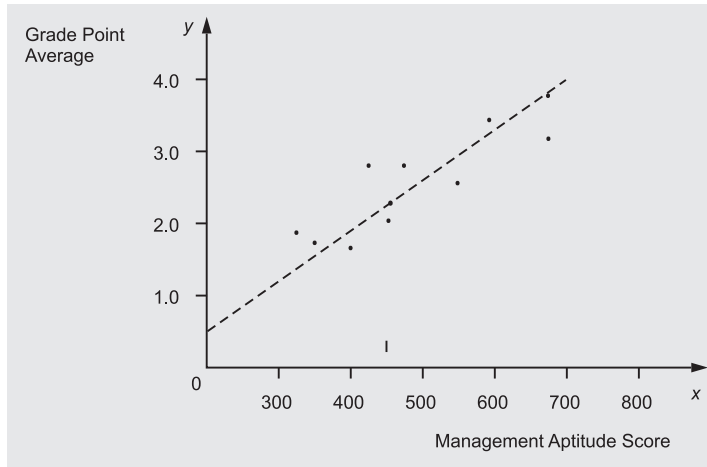


Figure 9.3 Scatter Diagram

9.5.2 Karl Pearson's Correlation Coefficient

Karl Pearson's correlation coefficient measures quantitatively the extent to which two variables x and y are correlated. For a set of n pairs of values of x and y , Pearson's correlation coefficient r is given by

$$r = \frac{\text{Covariance}(x, y)}{\sqrt{\text{var } x} \sqrt{\text{var } y}} = \frac{\text{Cov}(x, y)}{\sigma_x \sigma_y}$$

where $\text{Cov}(x, y) = \frac{1}{n} \sum (x - \bar{x})(y - \bar{y})$

$$\sigma_x = \sqrt{\frac{\sum (x - \bar{x})^2}{n}} \leftarrow \text{standard deviation of sample data on variable } x$$

$$\sigma_y = \sqrt{\frac{\sum (y - \bar{y})^2}{n}} \leftarrow \text{standard deviation of sample data on variable } y$$

Substituting mathematical formula for $\text{Cov}(x, y)$ and σ_x and σ_y , we have

$$r = \frac{\frac{1}{n} \sum (x - \bar{x})(y - \bar{y})}{\sqrt{\frac{\sum (x - \bar{x})^2}{n}} \sqrt{\frac{\sum (y - \bar{y})^2}{n}}} = \frac{n \sum xy - (\sum x)(\sum y)}{\sqrt{n \sum x^2 - (\sum x)^2} \sqrt{n \sum y^2 - (\sum y)^2}} \quad (9-1)$$

Step Deviation Method for Ungrouped Data When actual mean values $\bar{x}$ and $\bar{y}$ are in fraction, the calculation of Pearson's correlation coefficient can be simplified by taking deviations of x and y values from their assumed means A and B , respectively. That is, $d_x = x - A$ and $d_y = y - B$, where A and B are assumed means of x and y values. The formula (9-1) becomes

$$r = \frac{n \sum d_x d_y - (\sum d_x)(\sum d_y)}{\sqrt{n \sum d_x^2 - (\sum d_x)^2} \sqrt{n \sum d_y^2 - (\sum d_y)^2}} \quad (9-2)$$

Step Deviation Method for Grouped Data When data on x and y values are classified or grouped into a frequency distribution, the formula (9-2) is modified as

$$r = \frac{n \sum fd_x d_y - (\sum fd_x)(\sum fd_y)}{\sqrt{n \sum fd_x^2 - (\sum fd_x)^2} \sqrt{n \sum fd_y^2 - (\sum fd_y)^2}} \quad (9-3)$$

Assumptions of Using Pearson's Correlation Coefficient

- (i) Pearson's correlation coefficient is appropriate to calculate when both variables x and y are measured on an interval or a ratio scale.
- (ii) Both variables x and y are normally distributed, and that there is a linear relationship between these variables.
- (iii) The correlation coefficient is largely affected due to truncation of the range of values in one or both of the variables. This occurs when the distributions of both the variables greatly deviate from the normal shape.
- (iv) There is a cause and effect relationship between two variables that influences the distributions of both the variables. Otherwise correlation coefficient might either be extremely low or even zero.

Advantages and Disadvantages of Pearson's Correlation Coefficient The correlation coefficient is a numerical number between -1 and 1 that summarizes the magnitude as well as direction (positive or negative) of association between two variables. The limitations of Pearson's method are as follows:

- (i) The correlation coefficient always assumes a linear relationship between two variables, whether it is true or not.
- (ii) Great care must be exercised in interpreting the value of this coefficient as very often its value is misinterpreted.
- (iii) The value of the coefficient is unduly affected by the extreme values of two variable values.
- (iv) As compared with other methods the computational time required to calculate the value of r using Pearson's method is lengthy.

9.5.3 Probable Error and Standard Error of Coefficient of Correlation

The probable error (PE) of coefficient of correlation indicates extent to which its value depends on the condition of random sampling. If r is the calculated value of correlation coefficient in a sample of n pairs of observations, then the standard error SE_r of the correlation coefficient r is given by

$$SE_r = \frac{1 - r^2}{\sqrt{n}}$$

The probable error of the coefficient of correlation is calculated by the expression:

$$PE_r = 0.6745 SE_r = 0.6745 \frac{1 - r^2}{\sqrt{n}}$$

Thus with the help of PE_r , we can determine the range within which population coefficient of correlation is expected to fall using following formula: $\rho = r \pm PE_r$ where ρ (rho) represents population coefficient of correlation.

Remarks

1. If $r < PE_r$ then the value of r is not significant, that is, there is no relationship between two variables of interest.
2. If $r > 6PE_r$ then value of r is significant, that is, there exists a relationship between two variables.

Illustration: If $r = 0.8$ and $n = 25$, then PE_r is

$$PE_r = 0.6745 \frac{1 - (0.8)^2}{\sqrt{25}} = 0.6745 \frac{0.36}{5} = 0.048$$

Thus, the limits within which population correlation coefficient (ρ_r) should fall are

$$r \pm PE_r = 0.8 \pm 0.048 \quad \text{or} \quad 0.752 \leq \rho_r \leq 0.848$$

Example 9.2: If covariance of 10 pairs of items is 7, variance of x is 36, $\Sigma(y - \bar{y})^2 = 90$. Find out correlation coefficient, r .
[Delhi Univ., B.Com.(P), 2004]

Solution: Given, $\text{Cov}(x, y) = 7$, $n = 10$, $\sigma_x^2 = 36$, $\Sigma(y - \bar{y})^2 = 90$. Since $\text{Var}(x) (= \sigma_x^2) = 36$, i.e. Standard dev (σ_x) = 6.

Also
$$\sigma_y = \sqrt{\frac{\Sigma(y - \bar{y})^2}{n}} = \sqrt{\frac{90}{10}} = 3$$

Correlation coefficient,
$$r = \frac{\text{Cov}(x, y)}{\sigma_x \sigma_y} = \frac{7}{6 \times 3} = 0.39.$$

Example 9.3: Calculate Karl Pearson's coefficient of correlation from the following data. Interpret your result: $\sigma_x = 10$, $\sigma_y = 12$, $\bar{x} = 25$ and $\bar{y} = 35$. Summation of product of deviation from actual arithmetic means of two series is 24 and number of observations are 20.

[Delhi Univ., B.Com.(P), 2004]

Solution: Given $\sigma_x = 10$, $\sigma_y = 12$, $\bar{x} = 25$, $\bar{y} = 35$, $\Sigma(x - \bar{x})(y - \bar{y}) = 24$ and $n = 20$. Then

$$\text{Cov}(x, y) = \frac{\Sigma(x - \bar{x})(y - \bar{y})}{n} = \frac{24}{20} = 1.2$$

We know that,
$$r = \frac{\text{Cov}(x, y)}{\sigma_x \sigma_y} = \frac{1.2}{10 \times 12} = +0.01$$

Since magnitude of r is very small, correlation between x and y is negligible.

Example 9.4: If $r = 0.97$ and $n = 8$, find out the probable error of the coefficient of correlation and determine the limits for population correlation, r .

Solution: Given: $r = 0.97$, $n = 8$. Then

$$PE(r) = 0.6745 \frac{1 - r^2}{\sqrt{n}} = 0.6745 \frac{1 - (0.97)^2}{\sqrt{8}} = \frac{0.6745 \times 0.0591}{2.828} = 0.014$$

Limits of population correlation = $r \pm PE(r) = 0.97 \pm 0.014 = 0.956$ to 0.984

Example 9.5: Test the significance of correlation for $r = +0.4$ on (a) $n = 10$ and (b) $n = 100$.

Solution: Given, $n = 10$, $r = 0.4$. Then,

$$PE(r) = 0.6745 \frac{1-r^2}{\sqrt{n}} = 0.6745 \frac{1-(0.4)^2}{\sqrt{10}} = 0.18$$

Also, for test of significances $\frac{r}{P.E.(r)} = \frac{0.4}{0.18} = 2.22 < 6$

Since the ratio is less than 6, r is not significant.

When $r = 0.4$ and $n = 100$. Then,

$$PE(r) = 0.6745 \frac{1-r^2}{\sqrt{n}} = 0.6745 \frac{1-(0.4)^2}{\sqrt{100}} = 0.06$$

Also, for test of significance $\frac{r}{P.E.(r)} = \frac{0.4}{0.06} = 6.67 > 6$.

Since the ratio is more than 6, r is highly significant.

Example 9.6: Given the following information: $r = 0.8$, $\Sigma xy = 60$, $\sigma_y = 2.5$, $\Sigma x^2 = 90$. Find the number of items.

Solution: We know that

$$\sigma_y = \sqrt{\frac{\Sigma(y - \bar{y})^2}{n}} = \sqrt{\frac{\Sigma y^2}{n}}$$

or $2.5 = \sqrt{\frac{\Sigma y^2}{n}}$ or $6.25 = \frac{\Sigma y^2}{n}$, i.e. $\Sigma y^2 = 6.25n$

Given, $r = 0.8$, $\Sigma xy = 60$, $\Sigma y^2 = 6.25n$, $\Sigma x^2 = 90$. Apply the formula,

$$r = \frac{\Sigma xy}{\sqrt{\Sigma x^2 \times \Sigma y^2}}$$

$$0.8 = \frac{60}{\sqrt{90 \times 6.25n}}$$

$$\sqrt{90 \times 6.25n} = \frac{60}{0.8} = 75$$

Squaring on both sides, we get

$$90 \times 6.25n = 5625 \text{ or } n = \frac{5625}{90 \times 6.2} = \frac{5625}{558}, \text{ i.e. } n = 10$$

Example 9.7: The coefficient of correlation between two variables x and y is 0.4 and their covariance is 10. If variance of x series is 9, find variance of y .

Solution: Given, $r = 0.4$, $\text{Cov}(x, y) = 10$, and $\sigma_x^2 = 9$. To find variance, σ_y^2 of y , apply the formula:

$$r = \frac{\text{Cov}(x, y)}{\sigma_x \sigma_y}$$

$$0.4 = \frac{10}{3\sigma_y} \text{ or } \sigma_y = \frac{10}{3 \times 0.4}, \text{ i.e. } \sigma_y = 8.33$$

or Variance, $\sigma_y^2 = (8.33)^2 = 69.39$

9.5.4 The Coefficient of Determination

The squared value of the correlation coefficient denoted as r^2 is called **coefficient of determination**. It always has a value between 0 and 1. By squaring the correlation coefficient we retain information about the strength of the relationship but we lose information about the direction. *This measure represents the proportion (or percentage) of the total variability of the dependent variable, y that is accounted for or explained by the independent variable, x .* The proportion (or percentage) of variation in y that x can explain determines more precisely the extent or strength of association between two variables x and y .

- The coefficient of correlation r has been grossly overrated and is used entirely too much. Its square, coefficient of determination r^2 , is a much more useful measure of the linear covariation of two variables. The reader should develop the habit of squaring every correlation coefficient he finds cited or stated before coming to any conclusion about the extent of the linear relationship between two correlated variables. —Tuttle

Interpretation of Coefficient of Determination: Coefficient of determination is preferred for interpreting the strength of association between two variables because it is easier to interpret a percentage. Illustrates the meaning of the coefficient of determination:

- If $r^2 = 0$, then *no variation* in y can be *explained* by the variable x . It is shown in Figure 9.2(c) where x is of no value in predicting the value of y . There is *no association* between x and y .
- If $r^2 = 1$, then values of y are *completely explained* by x . There is *perfect association* between x and y .
- If $0 \leq r^2 \leq 1$, the degree of explained variation in y as a result of *variation in values of x* depends on the value of r^2 . Value of r^2 closer to 0 shows low proportion of variation in y explained by x . On the other hand, value of r^2 closer to 1 shows that variable x can predict the actual value of the variable y .

For example, let correlation between variable x (height) and variable y (weight) be $r = 0.70$. Now the coefficient of determination $r^2 = 0.49$ or 49 per cent implies that only 49 per cent of the variation in variable y (weight) can be accounted for in terms of variable x (height). The remaining 51 per cent of the variability may be due to other factors, say for instance, tendency to eat fatty foods.

It may be noted that even a relatively high correlation coefficient $r = 0.70$ accounts for less than 50 per cent of the variability. In this context, it is important to know that 'variability' refers to how values of variable y are scattered around its own mean value. That is, as in the above example, some people will be heavy, some average, some light. So we can account for 49 per cent of the total variability of weight (y) in terms of height (x) if $r = 0.70$. The greater the correlation coefficient, the greater the coefficient of determination, and the variability in dependent variable can be accounted for in terms of independent variable.

Example 9.8: The following table gives indices of industrial production and number of registered unemployed people (in lakh). Calculate the value of the correlation coefficient.

Year	:	1991	1992	1993	1994	1995	1996	1997	1998
Index of Production	:	100	102	104	107	105	112	103	99
Number Unemployed	:	15	12	13	11	12	12	19	26

Solution: Calculations of Karl Pearson's correlation coefficient are shown in the table below:

$$\bar{x} = \frac{\sum x}{n} = \frac{832}{8} = 104; \quad \bar{y} = \frac{\sum y}{n} = \frac{120}{8} = 15$$

Year	Production x	$dx = (x - \bar{x})$	d_x^2	Unemployed y	$d_y = (y - \bar{y})$	d_y^2	$d_x d_y$
1991	100	-4	16	15	0	0	0
1992	102	-2	4	12	-3	9	+6
1993	104	0	0	13	-2	4	0
1994	107	+3	9	11	-4	16	-12
1995	105	+1	1	12	-3	9	-3
1996	112	+8	64	12	-3	9	-24
1997	103	-1	1	19	+4	16	-4
1998	99	-5	25	26	+11	121	-55
	832	0	120	120	0	184	-92

Applying the formula,
$$r = \frac{n \sum d_x d_y - (\sum d_x)(\sum d_y)}{\sqrt{n \sum d_x^2 - (\sum d_x)^2} \sqrt{n \sum d_y^2 - (\sum d_y)^2}} = \frac{8 \times (-92)}{\sqrt{8 \times 120} \sqrt{8 \times 184}}$$

$$= \frac{-92}{10.954 \times 13.564} = \frac{-92}{148.580} = -0.619$$

Interpretation: Since coefficient of correlation, $r = -0.619$, is moderately negative, it indicates that there is a moderately large inverse correlation between the two variables. Hence we conclude that as the production index increases, the number of unemployed decreases and vice versa.

Example 9.9: Calculate coefficient of correlation between x and y

	X Series	Y Series
No. of items	15	15
Arithmetic Means	25	18
Sum of squares of deviations from arithmetic mean	136	138

Summation of product of deviations of x and y series from their respective means is 122.

[Delhi Univ., B.Com.(P) 1989, 2005, 2006]

Solution: Given, $n = 15$, $\bar{x} = 25$; $\bar{y} = 18$, $\Sigma(x - \bar{x})^2 = 136$; $\Sigma(y - \bar{y})^2 = 138$;
 $\Sigma(x - \bar{x})(y - \bar{y}) = 122$,

Let $X = x - \bar{x}$, $Y = y - \bar{y}$. Then $\Sigma X^2 = 136$, $\Sigma Y^2 = 138$, $\Sigma XY = 122$. Then

$$\text{Correlation coefficient, } r = \frac{\Sigma XY}{\sqrt{\Sigma X^2 \times \Sigma Y^2}} = \frac{122}{\sqrt{136 \times 138}} = \frac{122}{11.66 \times 11.75} = 0.89$$

Example 9.10: The total of the multiplication of deviation of x and y is 3044. Number of pairs of observations are 10, Total of deviations of x are -170, total of deviations of y are -20. Total of squares of deviations of $x = 8288$. Total of squares of deviations of $y = 2264$. Find out coefficient of correlation when the arbitrary means of x and y are 82 and 68, respectively.

[Delhi Univ., B.Com.(P), 2001]

Solution: Given $n = 10$, $A = 82$, $B = 68$, $\Sigma(x - A) = -170$, $\Sigma(y - B) = -20$, $\Sigma(x - A)^2 = 8288$
 $\Sigma(y - B)^2 = 2264$ and $\Sigma(x - A)(y - B) = 3044$,

Let $d_x = x - A$ and $d_y = y - B$. Then $\Sigma d_x = -170$, $\Sigma d_y = -20$, $\Sigma d_x^2 = 8288$, $\Sigma d_y^2 = 2264$,
 $\Sigma d_x d_y = 3044$

$$\begin{aligned}
 \text{Apply the formula, } r &= \frac{n \sum d_x d_y - \sum d_x \sum d_y}{\sqrt{n \sum d_x^2 - (\sum d_x)^2} \sqrt{n \sum d_y^2 - (\sum d_y)^2}} \\
 &= \frac{10(3044) - (-170)(-20)}{\sqrt{10(8288) - (-170)^2} \sqrt{10(2264) - (-20)^2}} \\
 &= \frac{30440 - 3400}{\sqrt{53980} \sqrt{22240}} = 0.78
 \end{aligned}$$

Example 9.11: The following table gives the distribution of items of production and also the relatively defective items among them according to size groups. Find the correlation coefficient between size and defect in quality.

Size-group	:	15–16	16–17	17–18	18–19	19–20	20–21
No. of items	:	200	270	340	360	400	300
No. of defective items	:	150	162	170	180	180	114

[Delhi Univ., B.Com., 1999]

Solution: Let group size be denoted by variable x and number of defective items by variable y . Calculations for Karl Pearson's correlation coefficient are shown below:

Size-Group	Mid-value (m)	$d_x = m - 17.5$	d_x^2	Per cent of Defective Items y	$d_y = y - 50$	d_y^2	$d_x d_y$
15–16	15.5	–2	4	75	25	625	–50
16–17	16.5	–1	1	60	10	100	–10
17–18	17.5	0	0	50	0	0	0
18–19	18.5	1	1	50	0	0	0
19–20	19.5	2	4	45	–5	25	–10
20–21	20.5	3	9	38	–12	144	–36
		<u>3</u>	<u>19</u>		<u>18</u>	<u>894</u>	<u>–106</u>

Substituting values in the formula of Karl Pearson's correlation coefficient r , we have

$$\begin{aligned}
 r &= \frac{n \sum d_x d_y - (\sum d_x)(\sum d_y)}{\sqrt{n \sum d_x^2 - (\sum d_x)^2} \sqrt{n \sum d_y^2 - (\sum d_y)^2}} \\
 &= \frac{6 \times (-106) - 3 \times 18}{\sqrt{6 \times 19 - (3)^2} \sqrt{6 \times 894 - (18)^2}} \\
 &= \frac{-636 - 54}{\sqrt{105} \sqrt{5040}} \\
 &= -\frac{690}{727.64} = -0.948
 \end{aligned}$$

Interpretation: Since value of r is negative, and is moderately close to -1 , the statistical association between x (size group) and y (per cent of defective items) is moderate and negative, we conclude that when size of group increases, the number of defective items decreases and vice versa.

Example 9.12: The following data relate to age of employees and the number of days they reported sick in a month.

Employees :	1	2	3	4	5	6	7	8	9	10
Age :	30	32	35	40	48	50	52	55	57	61
Sick days :	1	0	2	5	2	4	6	5	7	8

Calculate Karl Pearson's coefficient of correlation and interpret it. [Kashmir Univ., B.Com., 1997]

Solution: Let age and sick days be represented by variables x and y , respectively. Calculations for value of correlation coefficient are shown below:

Age (x)	$dx = x - \bar{x}$	d_x^2	Sick days (y)	$d_y = y - \bar{y}$	d_y^2	$d_x d_y$
30	-16	256	1	-3	9	48
32	-14	196	0	-4	16	56
35	-11	121	2	-2	4	22
40	-6	36	5	1	1	-6
48	2	4	2	-2	4	-4
50	4	16	4	0	0	0
52	6	36	6	2	4	12
55	9	81	5	1	1	9
57	11	121	7	3	9	33
61	15	225	8	4	16	60
460	0	1092	40	0	64	230

$$\bar{x} = \frac{\sum x}{n} = \frac{460}{10} = 46 \text{ and } \bar{y} = \frac{\sum y}{n} = \frac{40}{10} = 4$$

Substituting values in the formula of Karl Pearson's correlation coefficient r , we have

$$r = \frac{n \sum d_x d_y - (\sum d_x)(\sum d_y)}{\sqrt{n \sum d_x^2 - (\sum d_x)^2} \sqrt{n \sum d_y^2 - (\sum d_y)^2}} = \frac{10 \times 230}{\sqrt{10 \times 1092} \sqrt{10 \times 64}} = \frac{230}{264.363} = 0.870$$

Interpretation: Since value of r is positive, therefore age of employees and number of sick days are positively correlated to a high degree. Hence we conclude that as the age of an employee increases, he is likely to go on sick leave more often than others.

Example 9.13: Find the coefficient of correlation between price and sales from the following data.

Price (₹) :	103	98	85	92	90	84	88	90	94	95
Sales (units) :	500	610	700	630	670	800	800	570	700	680

[Delhi Univ., B.Com., 2006]

Solution: Let price and sales be represented by variables x and y , respectively. Calculations for value of correlation coefficient are shown below:

Price (x)	$d_x = x - 90$	d_x^2	Sales (y)	$d_y = y - 700$	d_y^2	$d_x d_y$
103	13	169	500	-200	40,000	-2,600
98	8	64	610	-90	8,100	-720
85	-5	25	700 ← B	0	0	0
92	2	4	630	-70	4,900	-140
90 ← A	0	0	670	-30	900	0
84	-6	36	800	100	10,000	-600
88	-2	4	800	100	10,000	-200
90	0	0	570	-130	16,900	0
94	4	16	700	0	0	0
95	5	25	680	-20	400	-100
	19	343		-340	91,200	-4360

Applying the formula of Karl Pearson's correlation coefficient, r , we have

$$\begin{aligned}
 r &= \frac{n \sum d_x d_y - \sum d_x \sum d_y}{\sqrt{n \sum d_x^2 - (\sum d_x)^2} \sqrt{n \sum d_y^2 - (\sum d_y)^2}} \\
 &= \frac{10(-4360) - 19(-340)}{\sqrt{10(343) - (19)^2} \sqrt{10(91,200) - (-340)^2}} \\
 &= \frac{-43,600 + 6460}{\sqrt{3430 - 361} \sqrt{9,12,000 - 1,15,600}} = \frac{-37,140}{49,430} = -0.751
 \end{aligned}$$

Example 9.14: The following table gives the frequency, according to the marks, obtained by 67 students in an intelligence test. Measure the degree of relationship between age and marks:

Test Marks	Age in years				Total
	18	19	20	21	
200-250	4	4	2	1	11
250-300	3	5	4	2	14
300-350	2	6	8	5	21
350-400	1	4	6	10	21
Total	10	19	20	18	67

[Allahabad Univ., B.Com., 1999]

Solution: Let age of students and marks obtained by them be represented by variables x and y , respectively. Calculations for correlation coefficient for this bivariate data are shown below:

Test Marks y m d _y	x d _x	Age in years				Total, f	fd _y	fd _y ²	fd _x d _y
		18 -1	19 0	20 1	21 2				
200–250 225 -1		4	0	-2	-2	11	-11	11	0
		4	4	2	1				
250–300 275 0		0	0	0	0	14	0	0	0
		3	5	4	2				
300–350 325 1		-2	0	8	10	21	21	21	16
		2	6	8	5				
350–400 375 2		-2	0	12	40	21	42	84	50
		1	4	6	10				
Total, f		10	19	20	18	n = 67	Σfd _y = 52	Σfd _y ² = 116	Σfd _x d _y = 66
fd _x		-10	0	20	36	Σfd _x = 46			
fd _x ²		10	0	20	72	Σfd _x ² = 102			
fd _x d _y		0	0	18	48	Σfd _x d _y = 66			

where $d_x = x - 19$, and $d_y = (m - 275)/50$

Substituting values in the formula of Karl Pearson's correlation coefficient, we have

$$\begin{aligned}
 r &= \frac{n \Sigma fd_x d_y - (\Sigma fd_x)(\Sigma fd_y)}{\sqrt{n \Sigma fd_x^2 - (\Sigma fd_x)^2} \sqrt{n \Sigma fd_y^2 - (\Sigma fd_y)^2}} \\
 &= \frac{67 \times 66 - 46 \times 52}{\sqrt{67 \times 102 - (46)^2} \sqrt{67 \times 116 - (52)^2}} \\
 &= \frac{4422 - 2392}{\sqrt{6834 - 2116} \sqrt{7772 - 2704}} \\
 &= \frac{2030}{\sqrt{4718} \sqrt{5068}} = \frac{2030}{68.688 \times 71.19} = 0.415
 \end{aligned}$$

Interpretation: Since the value of r is positive, therefore age of students and marks obtained in an intelligence test are positively correlated to the extent of 0.415. Hence, we conclude that as the age of students increases, score of marks in intelligence test also increases.

Example 9.15: Calculate the coefficient of correlation from the following bivariate frequency distribution:

Sales Revenue (₹ in lakh)	Advertising Expenditure (₹ in '000)				Total
	5-10	10-15	15-20	20-25	
75-125	4	1	—	—	5
125-175	7	6	2	1	16
175-225	1	3	4	2	10
225-275	1	1	3	4	9
Total	13	11	9	7	40

Solution: Let advertising expenditure and sales revenue be represented by variables x and y , respectively. The calculations for correlation coefficient are shown below:

Revenue y	Mid-value (m)	x $Mid-value(m)$ d_x	Advertising Expenditure				Total, f	fd_y	fd_y^2	$fd_x d_y$
			5-10	10-15	15-20	20-25				
			7.5	12.5	17.5	22.5				
			-1	0	1	2				
75-125	100	-2	$\begin{pmatrix} 8 \\ 4 \end{pmatrix}$	$\begin{pmatrix} 0 \\ 1 \end{pmatrix}$	$\begin{pmatrix} 0 \\ — \end{pmatrix}$	$\begin{pmatrix} 0 \\ — \end{pmatrix}$	5	-10	20	18
125-175	150	-1	$\begin{pmatrix} 7 \\ 7 \end{pmatrix}$	$\begin{pmatrix} 0 \\ 6 \end{pmatrix}$	$\begin{pmatrix} -2 \\ 2 \end{pmatrix}$	$\begin{pmatrix} -2 \\ 1 \end{pmatrix}$	16	-16	16	13
175-225	200	0	$\begin{pmatrix} 0 \\ 1 \end{pmatrix}$	$\begin{pmatrix} 0 \\ 3 \end{pmatrix}$	$\begin{pmatrix} 0 \\ 4 \end{pmatrix}$	$\begin{pmatrix} 0 \\ 2 \end{pmatrix}$	10	-0	0	10
225-275	250	1	$\begin{pmatrix} -1 \\ 1 \end{pmatrix}$	$\begin{pmatrix} 0 \\ 1 \end{pmatrix}$	$\begin{pmatrix} 3 \\ 3 \end{pmatrix}$	$\begin{pmatrix} 8 \\ 4 \end{pmatrix}$	9	-9	9	10
Total, f			13	11	9	7	$n = 40$	$\Sigma d_y = -17$	$\Sigma d_y^2 = 45$	$\Sigma fd_x d_y = 21$
fd_x			-13	0	9	14	$\Sigma fd_x = 10$			
fd_x^2			13	0	9	28	$\Sigma fd_x^2 = 50$			
$fd_x d_y$			14	0	1	6	$\Sigma fd_x d_y = 21$			

where $d_x = (m - 12.5)/5$ and $d_y = (m - 200)/50$

Substituting values in the formula of Karl Pearson's correlation coefficient, we have

$$\begin{aligned}
 r &= \frac{n \Sigma fd_x d_y - (\Sigma fd_x)(\Sigma fd_y)}{\sqrt{n \Sigma fd_x^2 - (\Sigma fd_x)^2} \sqrt{n \Sigma fd_y^2 - (\Sigma fd_y)^2}} = \frac{40 \times 21 - 10 \times -17}{\sqrt{40 \times 50 - (10)^2} \sqrt{40 \times 45 - (-17)^2}} \\
 &= \frac{840 + 170}{\sqrt{1900} \sqrt{1511}} = \frac{1010}{1694.373} = 0.596
 \end{aligned}$$

Interpretation: Since the value of r is positive, advertising expenditure and sales revenue are positively correlated to the extent of 0.596. Hence, we conclude that as expenditure on advertising increases, the sales revenue also increases.

Example 9.16: A computer while calculating the correlation coefficient between two variables x and y from 25 pairs of observations obtained the following results:

$$n = 25, \quad \Sigma x = 125, \quad \Sigma x^2 = 650 \quad \text{and} \quad \Sigma y = 100, \quad \Sigma y^2 = 460, \quad \Sigma xy = 508$$

It was, however, discovered at the time of checking that he had copied down two pairs of observations as

x	y	instead of	x	y
6	14		8	12
8	6		6	8

Obtain the correct value of correlation coefficient between x and y . [MD Univ., M.Com., 1998]

Solution: The corrected values for termed needed in the formula of Pearson's correlation coefficient are determined as follows:

$$\text{Correct } \Sigma x = 125 - (6 + 8 - 8 - 6) = 125$$

$$\text{Correct } \Sigma y = 100 - (14 + 6 - 12 - 8) = 100$$

$$\begin{aligned} \text{Correct } \Sigma x^2 &= 650 - \{(6)^2 + (8)^2 - (8)^2 - (6)^2\} \\ &= 650 - \{36 + 64 - 64 - 36\} = 650 \end{aligned}$$

$$\begin{aligned} \text{Correct } \Sigma y^2 &= 460 - \{(14)^2 + (6)^2 - (12)^2 - (8)^2\} \\ &= 460 - \{196 + 36 - 144 - 64\} = 436 \end{aligned}$$

$$\begin{aligned} \text{Correct } \Sigma xy &= 508 - \{(6 \times 14) + (8 \times 6) - (8 \times 12) - (6 \times 8)\} \\ &= 508 - \{84 + 48 - 96 - 48\} = 520 \end{aligned}$$

Applying the formula

$$\begin{aligned} r &= \frac{n \Sigma xy - (\Sigma x)(\Sigma y)}{\sqrt{n \Sigma x^2 - (\Sigma x)^2} \sqrt{n \Sigma y^2 - (\Sigma y)^2}} = \frac{25 \times 520 - 125 \times 100}{\sqrt{25 \times 650 - (125)^2} \sqrt{25 \times 436 - (100)^2}} \\ &= \frac{13,000 - 12,500}{\sqrt{625} \sqrt{900}} = \frac{500}{25 \times 30} = 0.667 \end{aligned}$$

Thus, the correct value of correlation coefficient between x and y is 0.667.

Example 9.17: In order to find the correlation coefficient between two variables x and y from 12 pairs of observations, the following calculations were made:

$$\Sigma x = 30, \quad \Sigma x^2 = 670, \quad \Sigma y = 5, \quad \Sigma y^2 = 285, \quad \Sigma xy = 344$$

On subsequent verification, it was discovered that the pair, $x=11$ and $y=4$, was copied wrongly, the correct values being $x = 10$ and $y = 14$. After making necessary correction, find the correlation coefficient.

Solution: The corrected value for the pair (x, y) required in the formula of correlation coefficient is calculated as follows:

$$\text{Correct } \Sigma x = 30 - 11 + 10 = 29$$

$$\text{Correct } \Sigma y = 5 - 4 + 14 = 15$$

$$\text{Correct } \Sigma x^2 = 670 - (11)^2 + (10)^2 = 670 - 121 + 100 = 649$$

$$\text{Correct } \Sigma y^2 = 285 - (4)^2 + (14)^2 = 285 - 16 + 196 = 465$$

$$\text{Correct } \Sigma xy = 344 - (11)(4) + (10)(14) = 344 - 44 + 140 = 440$$

$$\begin{aligned} \text{Applying the formula, } r &= \frac{n \Sigma xy - \Sigma x \Sigma y}{\sqrt{n \Sigma x^2 - (\Sigma x)^2} \sqrt{n \Sigma y^2 - (\Sigma y)^2}} = \frac{12(440) - (29)(15)}{\sqrt{12(649) - (29)^2} \sqrt{12(465) - (15)^2}} \\ &= \frac{4845}{83.35 \times 73.18} = \frac{4845}{6099.5} = 0.794 \end{aligned}$$

Thus, the correct value of correlation coefficient between x and y is 0.794.

Conceptual Questions 9A

- 9.1** Making use of the data summarized below, calculate the coefficient of correlation.

Case	x_1	x_2	Case	x_1	x_2
A	10	9	E	12	11
B	6	4	F	13	13
C	9	6	G	11	8
D	10	9	H	9	4

- 9.2** Find the correlation coefficient by Karl Pearson's method between x and y and interpret its value.

$x : 57 \ 42 \ 40 \ 33 \ 42 \ 45 \ 42 \ 44 \ 40 \ 56 \ 44 \ 43$

$y : 10 \ 60 \ 30 \ 41 \ 29 \ 27 \ 27 \ 19 \ 18 \ 19 \ 31 \ 29$

- 9.3** Calculate the coefficient of correlation between x and y from the following data and calculate the probable errors. Assume 69 and 112 as the mean value for x and y , respectively.

$x : 78 \ 89 \ 99 \ 60 \ 50 \ 79 \ 68 \ 61$

$y : 125 \ 137 \ 156 \ 112 \ 107 \ 136 \ 123 \ 108$

- 9.4** Calculate Karl Pearson's coefficient of correlation between age and playing habits from the data given below. Also calculate the probable error and comment on the value:

Age :	20	21	22	23	24	25
No. of students :	500	400	300	240	200	160
Regular players :	400	300	180	96	60	24

- 9.5** Find the coefficient of correlation between age and the sum assured (in '000 ₹) from the following table:

Age Group (years)	Sum Assured (in ₹)				
	10	20	30	40	50
20-30	4	6	3	7	1
30-40	2	8	15	7	1
40-50	3	9	12	6	2
50-60	8	4	2	—	—

- 9.6** Family income and its percentage spent on food in the case of one hundred families gave the following bivariate frequency distribution. Calculate the coefficient of correlation and interpret its value.

Food Ex- penditure (in per cent)	Monthly Family Income (₹)				
	2000- 3000	3000- 4000	4000- 5000	5000- 6000	6000- 7000
10-15	—	—	—	3	7
15-20	—	4	9	4	3
20-25	7	6	12	5	—
25-30	3	10	19	8	—

- 9.7** With the following data in 6 cities, calculate Pearson's coefficient of correlation between the density of population and death rate:

City	Area in Kilometres	Population (in '000)	No. of Deaths
A	150	30	300
B	180	90	1440
C	100	40	560
D	60	42	840
E	120	72	1224
F	80	24	312

[Sukhadia Univ., B.Com., 1998]

- 9.8** The coefficient of correlation between two variables x and y is 0.3. The covariance is 9. The variance of x is 16. Find the standard deviation of y series.

Hints and Answers

9.1 $\bar{x}_1 = 80/8 = 10$, $\bar{x}_2 = 64/8 = 8$;

$$r = \frac{43}{\sqrt{32} \sqrt{72}} = 0.896$$

9.2 $r = -0.554$

9.4 $r = 0.005$

9.3 $r = 0.997$

9.5 $r = -0.256$

9.6 $r = -0.438$

9.7 $r = 0.988$

9.8 Given $\sigma_x = \sqrt{16} = 4$;

$$r = \frac{\text{Cov}(x, y)}{9 \sigma_x \sigma_y}$$

or $0.3 = \frac{9}{4\sigma_y}$ or $\sigma_y = 7.5$.

9.5.5 Spearman's Rank Correlation Coefficient

This method of finding the correlation coefficient between two variables was developed by the British psychologist Charles Edward Spearman in 1904. This method is applied to measure the association between two variables when only *ordinal (or rank) data* are available. In other words, this method is applied in a situation in which quantitative measure of certain qualitative factors, such as judgement, brands personalities, TV programmes, leadership, colour, taste, cannot be fixed, but individual observations can be arranged in a definite order (also called rank). The ranking is decided by using a set of ordinal rank numbers with 1 for the individual observation ranked first either in terms of quantity or quality; and n for the individual observation ranked last in a group of n pairs of observations. Mathematically, Spearman's rank correlation coefficient is defined as:

$$R = 1 - \frac{6 \sum D^2}{n(n^2 - 1)}$$

where R = rank correlation coefficient

R_1 = rank of observations with respect to first variable

R_2 = rank of observations with respect to second variable

$D = R_1 - R_2$, difference in a pair of ranks

n = number of paired observations or individuals being ranked

The number '6' is placed in the formula as a scaling device, it ensures that the possible range of R is from -1 to 1 . While using this method we may come across three types of cases.

Advantages

- (i) This method is easy to understand and its application is simpler than Pearson's method.

- (ii) This method is useful for correlation analysis when variables are expressed in qualitative terms like beauty, intelligence, honesty and efficiency.
- (iii) This method is appropriate to measure the association between two variables if the data type is at least ordinal scaled (ranked).
- (iv) The sample data of values of two variables is converted into ranks either in ascending order or descending order for calculating degree of correlation between two variables.

Disadvantages

- (i) Values of both variables are assumed to be normally distributed and describing a linear relationship rather than non-linear relationship.
- (ii) A large computational time is required when number of pairs of values of two variables exceed 30.
- (iii) This method cannot be applied to measure the association between two variable grouped data.

Case I: When Ranks are Given

When observations in a data set are already arranged in a particular order (rank), take the differences in pairs of observations to determine d . Square these differences and obtain the total Σd^2 . Apply formula (9-4) to calculate correlation coefficient.

Example 9.18: The coefficient of rank correlation between debenture prices and share prices is found to be 0.143. If the sum of the squares of the differences in ranks is given to be 48, find the values of n .

Solution: The formula for Spearman's correlation coefficient is as follows:

$$R = 1 - \frac{6 \Sigma D^2}{n(n^2 - 1)}$$

Given, $R = 0.143$, and $\Sigma D^2 = 48$. Substituting values in the formula, we get

$$0.143 = 1 - \frac{6 \times 48}{n(n^2 - 1)} = 1 - \frac{288}{n^3 - n}$$

$$0.143 (n^3 - n) = (n^3 - n) - 288$$

$$0.857 n^3 - 0.857n - 288 = 0 \quad \text{or} \quad n^3 - n - 336 = 0$$

$$(n - 7) (n^2 + 7n + 48) = 0$$

This implies that either $n - 7 = 0$, that is $n = 7$ or $n^2 + 7n + 48 = 0$. But $n^2 + 7n + 48 = 0$ on simplification gives undesirable value of n because its discriminant $b^2 - 4ac$ is negative. Hence $n = 7$.

Example 9.19: When is rank correlation coefficient preferred to Karl Pearson's method? In a bivariate sample, the sum of squares of differences between the ranks of observed values of two variables is 231 and the correlation coefficient between them is -0.4 . Find the number of pairs.

[Delhi Univ., B.Com., 2006]

Solution: The association between two attributes such as Intelligence and Beauty, can be arranged in a serial order but the quantitative measurement of their value is difficult and cannot be calculated by the Pearson's correlation coefficient because definite values cannot be assigned to such attributes.

Given, $\Sigma D^2 = 231$, $R = -0.4$.

Rank correlation coefficient, $R = 1 - \frac{6\Sigma D^2}{n^3 - n}$, i.e. $-0.4 = 1 - \frac{6 \times 231}{n^3 - n}$

$$\text{or} \quad 1.4(n^3 - n) = 6 \times 231$$

$$n^3 - n = \frac{6 \times 231}{1.4} = 990$$

$$n^3 - n - 990 = 0$$

$$n^3 - n - 1000 + 10 = 0$$

$$n^3 - 10^3 - (n - 10) = 0$$

$$(n - 10)(n^2 - 10n - 99) = 0$$

That is, either $n - 10 = 0$, i.e. $n = 10$ or $n^2 - 10n - 99 = 0$ which gives imaginary value of n . Thus, $n = 10$.

Example 9.20: In a bivariate distribution, Spearman's coefficient of correlation is -0.25 . If the sum of the squares of differences of various ranks is 150. Find out the number of pairs of items.
[Delhi Univ., B.Com.(Pass), 2004]

Solution: Given, $R = -0.25$, $\Sigma D^2 = 150$.

Applying the formula:

$$R = 1 - \frac{6\Sigma D^2}{n^3 - n}$$

$$-0.25 = 1 - \frac{6 \times 150}{n^3 - n} \quad \text{or} \quad -1.25 = -\frac{900}{n^3 - n}$$

$$n^3 - n = \frac{900}{1.25} = 720 \quad \text{or} \quad n^3 - n - 720 = 0$$

$$n^3 - n - 729 + 9 = 0$$

$$(n - 9)(n^2 + 9n + 80) = 0$$

This implies that either $n - 9 = 0$ i.e. $n = 9$ or $n^2 + 9n + 80 = 0$. But $n^2 + 9n + 80 = 0$ gives imaginary values of n . Hence $n = 9$.

Example 9.21: The coefficient of rank correlation of the marks obtained by 10 students in Statistics and Accountancy was found to be 0.2. It was later discovered that the difference in ranks in the two subjects obtained by one of the students was wrongly taken as 9 instead of 7. Find the correct value of coefficient of rank correlation.

Solution: Given the incorrect rank correlation coefficient, $R = 0.2$.

Applying the formula:

$$R = 1 - \frac{6\Sigma D^2}{n(n^2 - 1)}$$

$$0.2 = 1 - \frac{6\Sigma D^2}{10(10^2 - 1)} = 1 - \frac{6\Sigma D^2}{10 \times 99}$$

$$\text{or} \quad -0.8 = -\frac{6\Sigma D^2}{990} \quad \text{or} \quad \Sigma D^2 = \frac{0.8 \times 990}{6} = 132$$

Corrected $\Sigma D^2 = 132 - (9)^2 + (7)^2 = 132 - 81 + 49 = 100$. Thus, correct value of

$$R = 1 - \frac{6(\text{Correct } \Sigma D^2)}{10(10^2 - 1)} = 1 - \frac{6 \times 100}{10 \times 99} = 1 - \frac{20}{33} = \frac{13}{33} = 0.39$$

Example 9.22: The ranks of 15 students in two subjects A and B are given below. The two numbers within brackets denote the ranks of a student in A and B subject, respectively.

(1, 10), (2, 7), (3, 2), (4, 6), (5, 4), (6, 8), (7, 3), (8, 1),
(9, 11), (10, 15), (11, 9), (12, 5), (13, 14), (14, 12), (15, 13)

Find Spearman's rank correlation coefficient.

Solution: Since ranks of students with respect to their performance in two subjects are given, calculations for rank correlation coefficient are shown below:

Rank in A R_1	Rank in B R_2	Difference $D = R_1 - R_2$	D^2
1	10	-9	81
2	7	-5	25
3	2	1	1
4	6	-2	4
5	4	1	1
6	8	-2	4
7	3	4	16
8	1	7	49
9	11	-2	4
10	15	-5	25
11	9	2	4
12	5	7	49
13	14	-1	1
14	12	2	4
15	13	2	4
			272

$$\begin{aligned} \text{Applying the formula, } R &= 1 - \frac{6 \sum D^2}{n^3 - n} = 1 - \frac{6 \times 272}{15\{(15)^2 - 1\}} \\ &= 1 - \frac{1632}{3360} = 1 - 0.4857 = 0.5143 \end{aligned}$$

The result shows a moderate degree positive correlation between performance of students in two subjects.

Example 9.23: An office has 12 clerks. The long-serving clerks feel that they should have a seniority increment based on length of service built into their salary structure. An assessment of their efficiency by their departmental manager and the Human Resource department produces a ranking of efficiency. This is shown below together with a ranking of their length of service.

Ranking according to length of service	:	1	2	3	4	5	6	7	8	9	10	11	12
Ranking according to efficiency	:	2	3	5	1	9	10	11	12	8	7	6	4

Do the data support the clerks' claim for seniority increment?

Solution: Since ranks are already given, calculations for rank correlation coefficient are shown below:

Rank According to Length of Service R_1	Rank According to Efficiency R_2	Difference $D = R_1 - R_2$	D^2
1	2	-1	1
2	3	-1	1
3	5	-2	4
4	1	3	9
5	9	-4	16
6	10	-4	16
7	11	-4	16
8	12	-4	16
9	8	1	1
10	7	3	9
11	6	5	25
12	4	8	64
			178

$$\begin{aligned}\text{Applying the formula, } R &= 1 - \frac{6 \sum D^2}{n(n^2 - 1)} = 1 - \frac{6 \times 178}{12(144 - 1)} \\ &= 1 - \frac{89}{143} = 1 - 0.622 = 0.378\end{aligned}$$

Since result shows a low degree positive correlation between length of service and efficiency, the claim of the clerks for a seniority increment based on length of service is not justified.

Example 9.24: Ten competitors in a beauty contest are ranked by three judges in the following order:

Judge 1	:	1	6	5	10	3	2	4	9	7	8
Judge 2	:	3	5	8	4	7	10	2	1	6	9
Judge 3	:	6	4	9	8	1	2	3	10	5	7

Use the rank correlation coefficient to determine which pair of judges has the nearest approach to common tastes in beauty.

Solution: The pair of judges who have the nearest approach to common choice in beauty can be obtained in ${}^3C_2 = 3$ ways as follows:

- (i) Judge 1 and judge 2
- (ii) Judge 2 and judge 3
- (iii) Judge 3 and judge 1

Calculations for comparing their ranking are shown below:

Judge 1 R_1	Judge 2 R_2	Judge 3 R_3	$D^2=(R_1 - R_2)^2$	$D^2=(R_2 - R_3)^2$	$D^2=(R_3 - R_1)^2$
1	3	6	4	9	25
6	5	4	1	1	4
5	8	9	9	1	16
10	4	8	36	16	4
3	7	1	16	36	4
2	10	2	64	64	0
4	2	3	4	1	1
9	1	10	64	81	1
7	6	5	1	1	4
8	9	7	1	4	1
			200	214	60

Applying the formula

$$R_{12} = 1 - \frac{6 \Sigma D^2}{n(n^2 - 1)} = 1 - \frac{6 \times 200}{10(100 - 1)} = 1 - \frac{1200}{990} = -0.212$$

$$R_{23} = 1 - \frac{6 \Sigma D^2}{n(n^2 - 1)} = 1 - \frac{6 \times 214}{10(100 - 1)} = 1 - \frac{1284}{990} = -0.297$$

$$R_{13} = 1 - \frac{6 \Sigma d^2}{n(n^2 - 1)} = 1 - \frac{6 \times 60}{10(100 - 1)} = 1 - \frac{360}{990} = 0.636$$

Since the correlation coefficient $R_{13} = 0.636$ is largest, the judges 1 and 3 have nearest approach to common choice for beauty.

Case 2: When Ranks are not Given

When pairs of observations in the data set are not ranked as in Case 1, the ranks are assigned by taking either the highest value or the lowest value as 1 for both the variable's values.

Example 9.25: Quotations of index numbers of security prices of a certain joint stock company are given below:

Year	Debenture Price	Share Price
1	97.8	73.2
2	99.2	85.8
3	98.8	78.9
4	98.3	75.8
5	98.4	77.2
6	96.7	87.2
7	97.1	83.8

Using the rank correlation method, determine the relationship between debenture prices and share prices. [Calicut Univ., B.Com., 1997]

Solution: Ranking from the lowest value for both the variables as shown below:

Debenture Price (x)	Rank	Share Price (y)	Rank	Difference $D = R_1 - R_2$	$D^2 = (R_1 - R_2)^2$
97.8	3	73.2	1	2	4
99.2	7	85.8	6	1	1
98.8	6	78.9	4	2	4
98.3	4	75.8	2	2	4
98.4	5	77.2	3	2	4
96.7	1	87.2	7	-6	36
97.1	2	83.8	5	-3	9
					= 62

$$\text{Applying the formula, } R = 1 - \frac{6 \sum D^2}{n^3 - n} = 1 - \frac{6 \times 62}{(7)^3 - 7} = 1 - \frac{372}{336} = 1 - 1.107 = -0.107$$

The result shows a low degree of negative correlation between the debenture prices and share prices of a certain joint stock company.

Example 9.26: An economist wanted to find out if there was any relationship between the unemployment rate in a country and its inflation rate. Data gathered from 7 countries for the year 2004 are given below:

Country	Unemployment Rate (per cent)	Inflation Rate (per cent)
A	4.0	3.2
B	8.5	8.2
C	5.5	9.4
D	0.8	5.1
E	7.3	10.1
F	5.8	7.8
G	2.1	4.7

Find the degree of linear association between a country's unemployment rate and its level of inflation.

Solution: Ranking from the lowest value for both the variables as shown below:

Unemployment Rate (x)	Rank R_1	Inflation Rate (y)	Rank R_2	Difference $D = R_1 - R_2$	$D^2 = (R_1 - R_2)^2$
4.0	3	3.2	1	2	4
8.5	7	8.2	5	2	4
5.5	4	9.4	6	-2	4
0.8	1	5.1	3	-2	4
7.3	6	10.1	7	-1	1
5.8	5	7.8	4	1	1
2.1	2	4.7	2	0	0
					18

Applying the formula,

$$R = 1 - \frac{6\sum D^2}{n^3 - n} = 1 - \frac{6 \times 18}{(7)^3 - 7} = 1 - \frac{108}{336} = 0.678$$

The result shows a moderately high degree of positive correlation between unemployment rate and inflation rate of seven countries.

Case 3: When Ranks are Equal

While ranking observations in the data set by taking either the highest value or lowest value as rank 1, we may come across a situation of more than one observations being of equal size. In such a case, the rank to be assigned to individual observations is an average of the ranks which these individual observations would have got had they differed from each other. For example, if two observations are ranked equal at third place, then the average rank of $(3 + 4)/2 = 3.5$ is assigned to these two observations. Similarly, if three observations are ranked equal at third place, then the average rank of $(3 + 4 + 5)/3 = 4$ is assigned to these three observations.

While equal ranks are assigned to a few observations in the data set, an adjustment is made in the Spearman rank correlation coefficient formula as given below:

$$R = 1 - \frac{6 \left\{ \sum D^2 + \frac{1}{12} (m_1^3 - m_1) + \frac{1}{12} (m_2^3 - m_2) + \dots \right\}}{n(n^2 - 1)}$$

where m_i ($i = 1, 2, 3, \dots$) stands for the number of times an observation is repeated in the data set for both variables.

Example 9.27: Find the rank correlation coefficient from the following marks awarded by the examiners in statistics:

Examiner	Marks Awarded by Examiner A	Marks Awarded by Examiner B	Marks Awarded by Examiner C
1	24	37	30
2	29	35	28
3	19	16	20
4	14	26	25
5	30	23	25
6	19	27	30
7	27	19	20
8	30	20	24
9	20	16	22
10	28	11	29
11	11	21	15

[Delhi Univ., B.Com., 2005]

Solution: Since ranks are not given, ranking from highest value for each examiner. Calculations for rank correlation coefficient are shown below:

Examiner	Marks Awarded by Examiner A	Rank	Marks Awarded by Examiner B	Rank	Marks Awarded by Examiner C	Rank	D_{AB}	D_{AC}	D_{BC}	D_{AB}^2	D_{BC}^2	D_{AC}^2
1	24	6	37	1	30	1.5	5	4.5	-0.5	25	20.25	0.25
2	29	3	35	2	28	4	1	-1	-2	1	1	4
3	19	8.5	16	9.5	20	9.5	-1	-1	0	1	1	0
4	14	10	26	4	25	5.5	6	4.5	-1.5	36	20.25	2.25
5	30	1.5	23	5	25	5.5	-3.5	-4	-0.5	12.25	16	0.25
6	19	8.5	27	3	30	1.5	5.5	7	1.5	30.25	49	2.25
7	27	5	19	8	20	9.5	-3	-4.5	-1.5	9	20.25	2.25
8	30	1.5	20	7	24	7	-5.5	-5.5	0	30.25	30.25	0
9	20	7	16	9.5	22	8	-2.5	-1	1.5	6.25	1	2.25
10	28	4	11	11	29	3	-7	1	8	49	1	64
11	11	11	21	6	15	11	5	0	-5	25	0	25
										225	160	102.5

Applying the formula:

$$\begin{aligned}
 R_{AB} &= 1 - \frac{6 \left[\Sigma D_{AB}^2 + \frac{1}{12}(m_1^3 - m_1) + \frac{1}{12}(m_2^3 - m_2) \right]}{n(n^2 - 1)} \\
 &= 1 - \frac{6 \left[225 + \frac{1}{12}(2^3 - 2) + \frac{1}{12}(2^3 - 2) \right]}{11 \times 120} = 1 - \frac{6 \times 226}{11 \times 120} = -\frac{1356}{1320} = -0.027 \\
 R_{AC} &= 1 - \frac{6 \left[160 + \frac{1}{12}(2^3 - 2) + \frac{1}{12}(2^3 - 2) + \frac{1}{12}(2^3 - 2) \right]}{11 \times 120} \\
 &= 1 - \frac{6 \times 161.5}{11 \times 120} = 0.266.
 \end{aligned}$$

Similarly $R_{BC} = 0.522$

Example 9.28: A financial analyst wanted to find out whether inventory turnover influences any company's earnings per share (in per cent). A random sample of 7 companies listed in a stock exchange were selected and the following data was recorded for each:

Company	Inventory Turnover (Number of Times)	Earnings per Share (per cent)
A	4	11
B	5	9
C	7	13
D	8	7
E	6	13
F	3	8
G	5	8

Find the strength of association between inventory turnover and earnings per share. Interpret this finding.

Solution: Ranking from lowest value for both the variables. Since there are tied ranks, the sum of the tied ranks is averaged and assigned to each of the tied observations as shown below.

<i>Inventory Turnover (x)</i>	<i>Rank R₁</i>	<i>Earnings Per Share (y)</i>	<i>Rank R₂</i>	<i>Difference D = R₁ - R₂</i>	<i>D² = (R₁ - R₂)²</i>
4	2	11	5	-3.0	9.00
5	3.5	9	4	-0.5	0.25
7	6	13	6.5	-0.5	0.25
8	7	7	1	6.0	36.00
6	5	13	6.5	-1.5	2.25
3	1	8	2.5	-1.5	2.25
5	3.5	8	2.5	1.0	1.00
					51.00

If may be noted that a value 5 of variable x is repeated twice ($m_1 = 2$) and values 8 and 13 of variable y is also repeated twice, so $m_2 = 2$ and $m_3 = 2$. Applying the formula:

$$\begin{aligned}
 R &= 1 - \frac{6 \left\{ \sum D^2 + \frac{1}{12} (m_1^3 - m_1) + \frac{1}{12} (m_2^3 - m_2) + \frac{1}{12} (m_3^3 - m_3) \right\}}{n(n^2 - 1)} \\
 &= 1 - \frac{6 \left\{ 51 + \frac{1}{12} (2^3 - 2) + \frac{1}{12} (2^3 - 2) + \frac{1}{12} (2^3 - 2) \right\}}{7(49 - 1)} \\
 &= 1 - \frac{6\{51 + 0.5 + 0.5 + 0.5\}}{336} = 1 - 0.9375 = 0.0625
 \end{aligned}$$

The result shows a very weak positive association between inventory turnover and earnings per share.

Example 9.29: Obtain the rank correlation coefficient between the variables x and y from the following pairs of observed values:

x :	50	55	65	50	55	60	50	65	70	75
y :	110	110	115	125	140	115	130	120	115	160

[Mangalore Univ., B.Com., 1997]

Solution: Ranking from lowest value for both the variables. Moreover, certain observations in both sets of data are repeated, the ranking is done in accordance with suitable average value as shown below.

Variable x	Rank R_1	Variable y	Rank R_2	Difference $D = R_1 - R_2$	$D^2 = (R_1 - R_2)^2$
50	2	110	1.5	0.5	0.25
55	4.5	110	1.5	3.0	9.00
65	7.5	115	4	3.5	12.25
50	2	125	7	-5.0	25.00
55	4.5	140	9	-4.5	20.25
60	6	115	4	2.0	4.00
50	2	130	8	-6.0	36.00
65	7.5	120	6	1.5	2.25
70	9	115	4	5.0	25.00
75	10	160	10	0.0	00.00
					134.00

It may be noted that for variable x , 50 is repeated thrice ($m_1 = 3$), 55 is repeated twice ($m_2 = 2$), and 65 is repeated twice ($m_3 = 2$). Also for variable y , 110 is repeated twice ($m_4 = 2$) and 115 thrice ($m_5 = 3$). Applying the formula:

$$\begin{aligned}
 R &= 1 - \frac{6 \left\{ \sum D^2 + \frac{1}{12} (m_1^3 - m_1) + \frac{1}{12} (m_2^3 - m_2) + \frac{1}{12} (m_3^3 - m_3) + \frac{1}{12} (m_4^3 - m_4) + \frac{1}{12} (m_5^3 - m_5) \right\}}{n(n^2 - 1)} \\
 &= 1 - \frac{6 \left\{ 134 + \frac{1}{12} (3^3 - 3) + \frac{1}{12} (2^3 - 2) + \frac{1}{12} (2^3 - 2) + \frac{1}{12} (2^3 - 2) + \frac{1}{12} (3^3 - 3) \right\}}{10(100 - 1)} \\
 &= 1 - \frac{6 [134 + 2 + 0.5 + 0.5 + 0.5 + 2]}{990} = 1 - \frac{6 \times 139.5}{990} = 1 - \frac{837}{990} \\
 &= 1 - 0.845 = 0.155
 \end{aligned}$$

The result shows a weak positive association between variables x and y .

Self-practice Problems 9A

- 9.9** The coefficient of rank correlation of the marks obtained by 10 students in statistics and accountancy was found to be 0.2. It was later discovered that the difference in ranks in two subjects obtained by one of the students was wrongly taken as 9 instead of 7. Find the correct coefficient of rank correlation. [Delhi Univ., B.Com., 1996]

- 9.10** The ranking of 10 students in accordance with their performance in two subjects A and B are as follows:

A :	6	5	3	10	2	4	9	7	8	1
B :	3	8	4	9	1	6	10	7	5	2

Calculate the rank correlation coefficient and comment on its value.

- 9.11** Calculate Spearman's coefficient of correlation between marks assigned to ten students by judges A and B in a certain competitive test as shown below:

Student	Marks by Judge A	Marks by Judge B
1	52	65
2	53	68
3	42	43
4	60	38
5	45	77
6	41	48
7	37	35
8	38	30
9	25	25
10	27	50

- 9.12** An examination of eight applicants for a clerical post was taken by a firm. From the marks obtained by the applicants in the accountancy and statistics papers, compute the rank correlation coefficient.

Applicant	:	A	B	C	D	E	F	G	H
Marks in accountancy	:	15	20	28	12	40	60	20	80
Marks in statistics	:	40	30	50	30	20	10	30	60

- 9.13** Seven methods of imparting business education were ranked by the MBA students of two universities as follows:

Method of Teaching	:	1	2	3	4	5	6	7
Rank by students of Univ. A	:	2	1	5	3	4	7	6
Rank by students of Univ. B	:	1	3	2	4	7	5	6

Calculate the rank correlation coefficient and comment on its value.

- 9.14** An investigator collected the following data with respect to the socio-economic status and severity of respiratory illness.

Patient	:	1	2	3	4	5	6	7	8
Socio-economic status (rank)	:	6	7	2	3	5	4	1	8
Severity of illness rank)	:	5	8	4	3	7	1	2	6

Calculate the rank correlation coefficient and comment on its value.

- 9.15** You are given the following data of marks obtained by 11 students in statistics in two

tests, one before and other after special coaching:

First Test (Before Coaching)	Second Test (After Coaching)
23	24
20	19
19	22
21	18
18	20
20	22
18	20
20	22
18	20
17	20
23	23
16	20
19	17

Do the marks indicate that the special coaching has benefited the students?

[Delhi Univ., M.Com., 1989]

- 9.16** Two departmental managers ranked a few trainees according to their perceived abilities. The ranking are given below:

Trainee	:	A	B	C	D	E	F	G	H	I	J
Manager A	:	1	9	6	2	5	8	7	3	10	4
Manager B	:	3	10	8	1	7	5	6	2	9	4

Calculate an appropriate correlation coefficient to measure the consistency in the ranking.

- 9.17** In an office some keyboard operators, who were already ranked on their speed, were also ranked on accuracy by their supervisor. The results were as follows:

Operator	:	A	B	C	D	E	F	G	H	I	J
Speed	:	1	2	3	4	5	6	7	8	9	10
Accuracy	:	7	9	3	4	1	6	8	2	10	5

Calculate the appropriate correlation coefficient between speed and accuracy.

- 9.18** The personnel department is interested in comparing the ratings of job applicants when measured by a variety of standard tests. The ratings of 9 applicants on interviews and standard psychological test are shown below:

<i>Applicant</i>	:	A	B	C	D	E	F	G	H	I
Interview	:	5	2	9	4	3	6	1	8	7
Standard test	:	8	1	7	5	3	4	2	9	6

Calculate Spearman's rank correlation coefficient and comment on its value.

Hints and Answers

9.9 Given $R = 0.2, n = 10$;

$$R = 1 - \frac{6 \Sigma D^2}{n(n^2 - 1)}$$

$$0.2 = 1 - \frac{6 \Sigma D^2}{10(100 - 1)} \text{ or } \Sigma D^2 = 100$$

$$\text{Correct value of } R = 1 - \frac{6 \times 100}{10 \times 99} = 0.394$$

9.10 $R = 1 - \frac{6 \Sigma D^2}{n(n^2 - 1)}$

$$= 1 - \frac{6 \times 36}{10(100 - 1)} = 0.782$$

9.11 $R = 1 - \frac{6 \Sigma D^2}{n(n^2 - 1)} = 1 - \frac{6 \times 76}{10(100 - 1)} = 0.539$

9.12

$$R = 1 - \frac{6 \left\{ \Sigma D^2 + \frac{1}{12} (m_1^3 - m_1) + \frac{1}{12} (m_2^3 - m_2) \right\}}{n(n^2 - 1)}$$

$$= 1 - \frac{6 \left\{ 81.5 + \frac{1}{12} (2^3 - 2) + \frac{1}{12} (3^3 - 3) \right\}}{8(64 - 1)} = 1$$

9.13 $R = 0.50$

9.14 $R = 0.477$

9.15 $R = 0.71$

9.16 $R = 0.842$

9.17 $R = 0.006$

9.18 $R = 0.817$

Conceptual Questions 9B

- What is the meaning of the coefficient of correlation?
- Explain the meaning and significance of the term correlation.
- What is meant by 'correlation'? Distinguish between positive, negative, and zero correlation.
- What are the numerical limits of r^2 and r ? What does it mean when r equals one? zero? minus one?
- What is correlation? Clearly explain its role with suitable illustration from simple business problems.
- What is the relationship between the coefficient of determination and the coefficient of correlation? How is the coefficient of determination interpreted?
- What is coefficient of rank correlation? Bring out its usefulness. How does this coefficient differ from the coefficient of correlation?
- What is Spearman's rank correlation coefficient? How does it differ from Karl Pearson's coefficient of correlation?
- (a) What is a scatter diagram? How do you interpret a scatter diagram?
(b) What is a scatter diagram? How does it help in studying the correlation between two variables, in respect of both its direction and degree?
- Define correlation coefficient ' r ' and give its limitations. What interpretation would you give if told that the correlation between the number of truck accidents per year and the age of the driver is $(-)$ 0.60 if only drivers with at least one accident are considered?

Formulae Used

1. Karl Pearson's correlation coefficient

$$r = \frac{\text{Covariance between } x \text{ and } y}{\sigma_x \sigma_y}$$

- Deviation from actual mean

$$r = \frac{\Sigma (x - \bar{x}) (y - \bar{y})}{\sqrt{\Sigma (x - \bar{x})^2} \sqrt{\Sigma (y - \bar{y})^2}}$$

- Deviation from assumed mean

$$r = \frac{n \Sigma d_x d_y - (\Sigma d_x) (\Sigma d_y)}{\sqrt{n \Sigma d_x^2 - (\Sigma d_x)^2} \sqrt{n \Sigma d_y^2 - (\Sigma d_y)^2}}$$

$$d_x = x - A, d_y = y - B$$

A, B = constants

- Bivariate frequency distribution

$$r = \frac{n \Sigma f d_x d_y - (\Sigma f d_x) (\Sigma f d_y)}{\sqrt{n \Sigma f d_x^2 - (\Sigma f d_x)^2} \sqrt{n \Sigma f d_y^2 - (\Sigma f d_y)^2}}$$

- Using actual values of x and y

$$r = \frac{n \Sigma x y - (\Sigma x) (\Sigma y)}{\sqrt{n \Sigma x^2 - (\Sigma x)^2} \sqrt{n \Sigma y^2 - (\Sigma y)^2}}$$

2. Standard error of correlation coefficient, r

$$SE_r = \frac{1 - r^2}{\sqrt{n}}$$

- Probable error of correlation coefficient, r

$$PE_r = 0.6745 \frac{1 - r^2}{\sqrt{n}}$$

3. Spearman's rank correlation coefficient

- Ranks are not equal

$$R = 1 - \frac{6 \Sigma D^2}{n(n^2 - 1)}$$

- Ranks are equal

$$R = 1 - \frac{6 \left[\Sigma d^2 + \frac{1}{12} (m_i^3 - m_i) \right]}{n(n^2 - 1)}$$

where, $i = 1, 2, \dots$

True or False Concepts Quiz

1. There are several types of correlation coefficients, the selection of which is determined by the level of scaling of the two variables.
2. When both variables are measured on an interval or ratio scale, Pearson's correlation coefficient is most appropriate.
3. To use Pearson's correlation coefficient, it is assumed that both variables are continuous and normally distributed.
4. When there is no linear association between two variables, the value of r will be close to zero.
5. A correlation coefficient $r = -1$ represents a very low linear correlation.
6. The coefficient of determination is the square of the correlation coefficient.
7. As the correlation coefficient approaches zero, the possible error in linear prediction increases.
8. The closer the correlation coefficient is to zero, the greater the predictive validity of a test.
9. If a correlation coefficient for reliability of a test is close to 1, then the test is unreliable.
10. Even a high correlation is not necessarily indicative of a causal relationship between two variables.
11. As the value of r increases, the proportion of variability of one variable y that can be accounted for another variable x decreases.
12. If the relationship between two variables is nonlinear, the value of the correlation coefficient must be negative.
13. Spearman's correlation coefficient is used where one or both variables are at least of interval scaling.

14. A scatter diagram is used to help to decide if the relationship between two variables is linear or curvilinear.
15. When calculating Spearman's correlation coefficient, Σd^2 is the sum of the square of the difference between the means.

Concepts Quiz Answers

1. T	2. T	3. T	4. T	5. F	6. T	7. T	8. F	9. F
10. T	11. F	12. F	13. F	14. F	15. F			

Review Self-practice Problems

- 9.19 The following are the monthly figures of the advertising expenditure and sales of a firm. It is generally found that advertising expenditure has its impact on sales generally after 2 months. Allowing for this time lag, calculate the coefficient of correlation.

Months	Advertising Expenditure	Sales	Months	Advertising Expenditure	Sales
Jan.	50	1200	July	140	2400
Feb.	60	1500	Aug.	160	2600
March	70	1600	Sep.	170	2800
April	90	2000	Oct.	190	2900
May	120	2200	Nov.	200	3100
June	150	2500	Dec.	250	3900

- 9.20 The coefficient of correlation between two variables x and y is 0.64. Their covariance is 16. The variance of x is 19. Find the standard deviation of y series.
- 9.21 Given $r = 0.8$, $\Sigma xy = 60$, $\sigma_y = 2.5$ and $\Sigma x^2 = 90$, find the number of observations, items. x and y are deviations from arithmetic mean. [Delhi Univ., B.Com., 1998]
- 9.22 Calculate the Karl Pearson's coefficient of correlation between age and playing habits from the data given below. Comment on the value.

Age	:	20	21	22	23	24	25
No. of students	:	500	400	300	240	200	160
Regular players	:	400	300	180	96	60	24

- 9.23 A survey regarding income and savings provided the following data:

Income (₹)	Saving (₹)			
	500	1000	1500	2000
40,000	8	4	—	—
6000	—	12	24	6
8000	—	9	7	2
10,000	—	—	10	5
12,000	—	—	9	4

Compute Karl Pearson's coefficient of correlation and interpret its value.

- 9.24 A company gives on-the-job training to its salesmen, followed by a test. It is considering whether it should terminate the services of any salesman who does not do well in the test. Following data give the test scores and sales (₹ in '000) made by nine salesmen during the last one year.
- Test scores: 14 19 24 21 26 22 15 20 19
Sales : 31 36 48 37 50 45 33 41 39
- Compute the coefficient of correlation between test scores and sales. Does it indicate that termination of the services of salesman with low test scores is justified?
- 9.25 Calculate the coefficient of correlation and its probable error from the following:

Subject	Per cent Marks in Final Year Exams	Per cent Marks in Sessions
Hindi	75	62
English	81	68
Physics	70	65
Chemistry	76	60
Maths	77	69
Statistics	81	72
Botany	84	76
Zoology	75	72

- 9.26** Following figures give the rainfall in inches for the year and the production (in '00 kg) for the Rabi crop and Kharif crops. Calculate Karl Pearson's coefficient of correlation, between rainfall and total production.

Rainfall	:	20	22	24	26	28	30	32
Rabi production	:	15	18	20	32	40	39	40
Kharif production	:	15	17	20	18	20	21	15

- 9.27** President of a consulting firm is interested in the relationship between environmental work factors and the employees' turnover rate. He defines environmental factors as those aspects of a job other than salary and benefits. He visited to similar plants and gave each plant a rating 1 to 25 on its environmental factors. He then obtained each plant's turnover rate (annual in percentage) examined the relationship.

Environmental rating	:	11	19	7	12	13	10	16	22	14	12
Turnover rate	:	6	4	8	3	7	8	3	2	5	6

Compute the correlation coefficient between turnover rate and environmental rating and test it.

- 9.28** Sixteen companies in a state have been ranked according to profit earned during a particular financial year, and the working capital for that year. Calculate the rank correlation coefficient.

Company	Rank (Profit)	Rank (Working capital)
A	1	13
B	2	16
C	3	14
D	4	15
E	5	10
F	6	12
G	7	4
H	8	11
I	9	5
J	10	9
K	11	8
L	12	3
M	13	1
N	14	6
O	15	7
P	16	2

- 9.29** Following are the percentage figures of expenditure incurred on clothing (₹ in '00) and entertainment (₹ in '00) by an average working class family in a period of 10 years.

Year	1989	90	91	92	93	94	95	96	97	98
Expenditure on clothing	: 24	27	31	32	20	25	33	30	28	22
Expenditure on entertainment	: 11	8	5	3	13	10	2	7	9	2

Compute Spearman's rank correlation coefficient and comment on the result.

Hints and Answers

9.19 $r = 0.918$

9.20 $r = \frac{\sum xy}{\sigma_x \sigma_y}$; $\sigma_x = \sqrt{9} = 3$;

$0.64 = 16 \frac{1}{9\sigma_y}$ or $\sigma_y = 1.2$

9.21 $r = \frac{\sum xy}{n\sigma_x\sigma_y}$ or $r^2 = \frac{(\sum xy)^2}{n^2\sigma_x^2\sigma_y^2}$;

$(0.8)^2 = \frac{(60)^2}{n^2(90/n) \times 6.25} = \frac{3600}{90n \times 6.25}$, i.e. $n = 10$

9.22 $r = -0.991$ **9.23** $r = 0.0522$

9.24 $r = 0.947$ **9.25** $r = 0.623$, $PE_r = 0.146$

9.26 $r = 0.917$ **9.27** $r = -0.801$

9.28 $R = -0.817$ **9.29** $R = -0.60$

10

Regression Analysis

LEARNING OBJECTIVES

After studying this chapter, you should be able to

- understand the role of descriptive statistics in summarization, description and interpretation of the data.
- understand the importance of summary measures to describe characteristics of a data set.
- use several numerical methods belonging to measures of central tendency to describe the characteristics of a data set.

10.1 INTRODUCTION

In Chapter 9, we introduced the concept of statistical relationship between two variables such as level of sales and amount of advertising; yield of a crop and the amount of fertilizer used and price of a product and its supply. The relationship between such variables indicates the degree and direction of their association, but fails to answer the following question:

- Is there any functional (or algebraic) relationship between two variables? If yes, can it be used to estimate the most likely value of one variable, given the value of other variable?

The statistical technique that expresses the relationship between two or more variables in the form of an equation to estimate the value of a variable based on the given value of another variable is called *regression analysis*. The variable whose value is estimated using the algebraic equation is called *dependent* (or *response*) *variable* and the variable whose value is used to estimate this value is called *independent* (*regressor* or *predictor*) *variable*. The linear algebraic equation used for expressing a dependent variable in terms of independent variable is called *linear regression equation*.

The basic differences between correlation and regression analysis are summarized as follows:

1. Developing an algebraic equation between two variables from sample data and predicting the value of one variable, given the value of the other variable is referred to as regression analysis, while measuring the strength (or degree) of the relationship between two variables is referred as correlation analysis. The sign of correlation coefficient indicates the nature (direct or inverse) of relationship between two variables, while the absolute value of correlation coefficient indicates the extent of relationship.
2. Correlation analysis determines an association between two variables x and y but not that they have a cause-and-effect relationship. Regression analysis, in contrast to correlation, determines the cause-and-effect relationship between x and y , that is, a change in the value

of independent variable x causes a corresponding change (effect) in the value of dependent variable y if all other factors that affect y remain unchanged.

3. In linear regression analysis, one variable is considered as dependent variable and other as independent variable, while in correlation analysis both variables are considered to be independent.
4. The coefficient of determination r^2 indicates the proportion of total variance in the dependent variable that is explained or accounted for by the variation in the independent variable. Since value of r^2 is determined from a sample, its value is subject to sampling error. Even if the value of r^2 is high, the assumption of a linear regression may be incorrect because it may represent a portion of the relationship that actually is in the form of a curve.

10.2 ADVANTAGES OF REGRESSION ANALYSIS

The following are some important advantages of regression analysis:

1. Regression analysis helps in developing a regression equation by which the value of a dependent variable can be estimated given a value of an independent variable.
2. Regression analysis helps to determine standard error of estimate to measure the variability or spread of values of a dependent variable with respect to the regression line. Smaller the variance and error of estimate, the closer the pair of values (x, y) fall about the regression line and better the line fits the data, that is, a good estimate can be made of the value of variable y . When all the points fall on the line, the standard error of estimate equals zero.
3. When the sample size is large ($df \geq 29$), the interval estimation for predicting the value of a dependent variable based on standard error of estimate is considered to be acceptable by changing the values of either x or y . The magnitude of r^2 remains the same regardless of the values of the two variables.

Remarks

1. The relationship between the dependent variable y and independent variable x exists and is linear. The average relationship between x and y can be described by a simple linear regression equation $y = a + bx + e$, where e is the deviation of a particular value of y from its expected value for a given value of independent variable x .
2. For every value of the independent variable x , there is an expected (or mean) value of the dependent variable y and these values are normally distributed. The mean of these normally distributed values fall on the line of regression.
3. The dependent variable y is a continuous random variable, whereas values of the independent variable x are fixed values and are not random.
4. The sampling error associated with the expected value of the dependent variable y is assumed to be an independent random variable distributed normally with mean zero and constant standard deviation. The errors are not related with each other in successive observations.
5. The standard deviation and variance of expected values of the dependent variable y about the regression line are constant for all values of the independent variable x within the range of the sample data.
6. The value of the dependent variable cannot be estimated for a value of an independent variable lying outside the range of values in the sample data.

10.3 PARAMETERS OF SIMPLE LINEAR REGRESSION MODEL

The fundamental aim of regression analysis is to determine a regression equation (line) that makes sense and fits the representative data such that the error of variance is as small as possible. This implies that the regression equation should adequately be used for prediction. J. R. Stockton stated that

- *The device used for estimating the values of one variable from the value of the other consists of a line through the points, drawn in such a manner as to represent the average relationship between the two variables. Such a line is called line of regression.*

The two variables x and y which are correlated can be expressed in terms of each other in the form of straight line equations called *regression equations*. Such lines should be able to provide the best fit of sample data to the population data. The algebraic expression of regression lines is written as

- The regression equation of y on x

$$y = a + bx$$

is used for estimating the value of y for given values of x .

- Regression equation of x on y

$$x = c + dy$$

is used for estimating the value of x for given values of y .

Remarks

1. When variables x and y are correlated perfectly (either positive or negative) these lines coincide, that is, we have only one line.
2. Higher the degree of correlation, nearer the two regression lines are to each other.
3. Lesser the degree of correlation, more the two regression lines are away from each other. That is, when $r = 0$, the two lines are at right angle to each other.
4. Two linear regression lines intersect each other at the point of the average value of variables x and y .

10.3.1 Regression Coefficients

To estimate values of population parameter β_0 and β_1 , under certain assumptions, the fitted or estimated regression equation representing the straight line regression model is written as:

$$\hat{y} = a + bx$$

where $\hat{y}$ = estimated average (mean) value of dependent variable y for a given value of independent variable x .

a or b_0 = y -intercept that represents average value of $\hat{y}$

b = slope of regression line that represents the expected change in the value of y for unit change in the value of x

To determine the value of $\hat{y}$ for a given value of x , this equation requires the determination of two unknown constants a (intercept) and b (also called regression coefficient). Once these constants are calculated, the regression line can be used to compute an estimated value of the dependent variable y for a given value of independent variable x .

The particular values of a and b define a specific linear relationship between x and y based on sample data. The coefficient ' a ' represents the *level of fitted line* (i.e., the distance of the line above or below the origin) when x equals zero, whereas coefficient ' b ' represents the *slope of the line* (a measure of the change in the estimated value of y for a one-unit change in x).

The regression coefficient ' b ' is also denoted as:

- b_{yx} (regression coefficient of y on x) in the regression line, $y = a + bx$
- b_{xy} (regression coefficient of x on y) in the regression line, $x = c + dy$

Properties of regression coefficients

1. The correlation coefficient is the geometric mean of two regression coefficients, that is, $r = \sqrt{b_{yx} \times b_{xy}}$.
2. If one regression coefficient is greater than one, then other regression coefficient must be less than one, because the value of correlation coefficient r cannot exceed one. However, both the regression coefficients may be less than one.
3. Both regression coefficients must have the same sign (either positive or negative). This property rules out the case of opposite sign of two regression coefficients.
4. The correlation coefficient will have the same sign (either positive or negative) as that of the two regression coefficients. For example, if $b_{yx} = -0.664$ and $b_{xy} = -0.234$, then $r = -\sqrt{0.664 \times 0.234} = -0.394$.
5. The arithmetic mean of regression coefficients b_{yx} and b_{xy} is more than or equal to the correlation coefficient r , that is, $(b_{yx} + b_{xy})/2 \geq r$. For example, if $b_{yx} = -0.664$ and $b_{xy} = -0.234$, then the arithmetic mean of these two values is $(-0.664 - 0.234)/2 = -0.449$, and this value is more than the value of $r = -0.394$.
6. Regression coefficients are independent of origin but not of scale.

10.4 METHODS TO DETERMINE REGRESSION COEFFICIENTS

Following are the methods to determine the parameters of a fitted regression equation:

10.4.1 Least-squares Normal Equations

Let $\hat{y} = a + bx$ be the least-squares line of y on x , where $\hat{y}$ is the estimated average value of dependent variable y . The line that minimizes the sum of squares of the deviations of the observed values of y from those predicted is the best-fitting line. Thus, the sum of residuals for any least-square line is minimum, where

$$L = \Sigma (y - \hat{y})^2 = \Sigma \{y - (a + bx)\}^2; \quad a, b = \text{constants}$$

Differentiating L with respect to a and b and equating to zero, we have

$$\frac{\partial L}{\partial a} = -2 \Sigma \{y - (a + bx)\} = 0$$

$$\frac{\partial L}{\partial b} = -2 \Sigma \{y - (a + bx)\}x = 0$$

Solving these two equations, we get the set of equations as:

$$\begin{aligned}\Sigma y &= na + b\Sigma x \\ \Sigma xy &= a\Sigma x + b\Sigma x^2\end{aligned}$$

where n is the total number of pairs of values of x and y in a sample data. These equations are called *normal equations* with respect to the regression line of y on x . After solving these equations for a and b , the values of a and b are substituted in the regression equation, $y = a + bx$.

Similarly if we have a least squares line $\hat{x} = c + dy$ of x on y , where $\hat{x}$ is the estimated mean value of dependent variable x , then the normal equations will be

$$\begin{aligned}\Sigma x &= nc + d\Sigma y \\ \Sigma xy &= n\Sigma y + d\Sigma y^2\end{aligned}$$

These equations are solved in the same manner as described above for constants c and d . The values of these constants are substituted to the regression equation $x = c + dy$.

Alternative method to calculate value of constants

Instead of using the algebraic method to calculate values of a and b , we may directly use the results of the solutions of these normal equation.

The gradient ' b ' (regression coefficient of y on x) and ' d ' (regression coefficient of x on y) are calculated as

$$b = \frac{S_{xy}}{S_{xx}}, \quad \text{where} \quad S_{xy} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = \sum_{i=1}^n x_i y_i - \frac{1}{n} \sum_{i=1}^n x_i \sum_{i=1}^n y_i$$

$$S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2 = \sum_{i=1}^n x_i^2 - \frac{1}{n} \left(\sum_{i=1}^n x_i \right)^2$$

$$\text{and } d = \frac{S_{yx}}{S_{yy}}, \quad \text{where} \quad S_{yy} = \sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{i=1}^n y_i^2 - \frac{1}{n} \left(\sum_{i=1}^n y_i \right)^2$$

Since the regression line passes through the point $(\bar{x}, \bar{y})$, the mean values of x and y and the regression equations can be used to find the value of constants a and c as follows:

$$a = \bar{y} - b\bar{x} \text{ for regression equation of } y \text{ on } x$$

$$c = \bar{x} - d\bar{y} \text{ for regression equation of } x \text{ on } y$$

The calculated values of a , b and c , d are substituted in the regression line $y = a + bx$ and $x = c + dy$, respectively, to determine the exact relationship.

Example 10.1: Use least-squares regression line to estimate the increase in sales revenue expected from an increase of 7.5 per cent in advertising expenditure.

Firm	Annual Percentage Increase in Advertising Expenditure	Annual Percentage Increase in Sales Revenue
A	1	1
B	3	2
C	4	2
D	6	4
E	8	6
F	9	8
G	11	8
H	14	9

Solution: Assume sales revenue (y) is dependent on advertising expenditure (x). Calculations for regression line using following normal equations are shown in Table 10.1

$$\Sigma y = na + b\Sigma x \text{ and } \Sigma xy = a\Sigma x + b\Sigma x^2$$

Table 10.1 Calculation for Normal Equations

Sales Revenue y	Advertising Expenditure, x	x^2	xy
1	1	1	1
2	3	9	6
2	4	16	8
4	6	36	24
6	8	64	48
8	9	81	72
8	11	121	88
9	14	196	126
40	56	524	373

Approach 1 (Normal Equations):

$$\Sigma y = na + b\Sigma x \text{ or } 40 = 8a + 56b$$

$$\Sigma xy = a\Sigma x + b\Sigma x^2 \text{ or } 373 = 56a + 524b$$

Solving these equations, we get $a = 0.072$ and $b = 0.704$. Substituting these values in the regression equation

$$y = a + bx = 0.072 + 0.704x$$

For $x = 7.5$ per cent or 0.075 increase in advertising expenditure, the estimated increase in sales revenue will be $y = 0.072 + 0.704 (0.075) = 0.1248$ or 12.48 per cent

Approach 2 (Short-cut method):

$$b = \frac{S_{xy}}{S_{xx}} = \frac{93}{132} = 0.704,$$

$$\text{where } S_{xy} = \Sigma xy - \frac{\Sigma x \Sigma y}{n} = 373 - \frac{40 \times 56}{8} = 93$$

$$S_{xx} = \Sigma x^2 - \frac{(\Sigma x)^2}{n} = 524 - \frac{(56)^2}{8} = 132$$

The intercept ' a ' on the y -axis is calculated as:

$$a = \bar{y} - b\bar{x} = \frac{40}{8} - 0.704 \times \frac{56}{8} = 5 - 0.704 \times 7 = 0.072$$

Substituting the values of $a = 0.072$ and $b = 0.704$ in the regression equation, we get

$$y = a + bx = 0.072 + 0.704x$$

For $x = 0.075$, we have $y = 0.072 + 0.704 (0.075) = 0.1248$ or 12.48 per cent.

Example 10.2: The owner of a small garment shop is hopeful that his sales are rising significantly week by week. Treating the sales for the previous six weeks as a typical example of this rising trend, he recorded them in ₹ 1000's and analysed the results.

Week :	1	2	3	4	5	6
Sales :	2.69	2.62	2.80	2.70	2.75	2.81

Fit a linear regression equation to suggest to him the weekly rate at which his sales are rising and use this equation to estimate expected sales for the 7th week.

Solution: Assume sales (y) is dependent on weeks (x). Then the normal equations for regression equation: $y = a + bx$ are written as:

$$\Sigma y = na + b\Sigma x \text{ and } \Sigma xy = a\Sigma x + b\Sigma x^2$$

Calculations for sales during various weeks are shown in Table 10.2.

Table 10.2 Calculations of Normal Equations

Week (x)	Sales (y)	x^2	xy
1	2.69	1	2.69
2	2.62	4	5.24
3	2.80	9	8.40
4	2.70	16	10.80
5	2.75	25	13.75
6	2.81	36	16.86
21	16.37	91	57.74

The gradient ' b ' is calculated as

$$b = \frac{S_{xy}}{S_{xx}} = \frac{0.445}{17.5} = 0.025; \quad S_{xy} = \Sigma xy - \frac{\Sigma x \Sigma y}{n} = 57.74 - \frac{21 \times 16.37}{6} = 0.445$$

$$S_{xx} = \Sigma x^2 - \frac{(\Sigma x)^2}{n} = 91 - \frac{(21)^2}{6} = 17.5$$

The intercept ' a ' on the y -axis is calculated as

$$\begin{aligned} a &= \bar{y} - b\bar{x} = \frac{16.37}{6} - 0.025 \times \frac{21}{6} \\ &= 2.728 - 0.025 \times 3.5 = 2.64 \end{aligned}$$

Substituting the values $a = 2.64$ and $b = 0.025$ in the regression equation, we have

$$\begin{aligned} y &= a + bx = 2.64 + 0.025x \\ &= 2.64 + 0.025(7) = 2.815, \text{ for } x = 7 \end{aligned}$$

Hence the expected sales during the 7th week is likely to be ₹ 2.815 (₹ in 1000's).

10.4.2 Deviations Method

Calculations to least-squares normal equations become lengthy when values of x and y are large. Thus, the following two methods may be used to reduce the computational time.

(a) **Deviations Taken from Actual Mean Values of x and y** If deviations of actual values of variables x and y are taken from their mean values $\bar{x}$ and $\bar{y}$, then the regression equations can be written as

- Regression equation of y on x
 $y - \bar{y} = b_{yx}(x - \bar{x})$

- Regression equation of x on y
 $x - \bar{x} = b_{xy}(y - \bar{y})$

where b_{yx} = regression coefficient of
y on x.

The value of b_{yx} can be calculated by using
the following formula

$$b_{yx} = \frac{\Sigma(x - \bar{x})(y - \bar{y})}{\Sigma(x - \bar{x})^2}$$

where b_{xy} = regression coefficient
of x on y.

The value of b_{xy} can be calculated by
using the formula

$$b_{xy} = \frac{\Sigma(x - \bar{x})(y - \bar{y})}{\Sigma(y - \bar{y})^2}$$

- (b) **Deviations Taken from Assumed Mean Values for x and y** If mean value of either x or y or both are in fractions, then we must prefer to take deviations of actual values of variables x and y from their assumed means.

- Regression equation of y on x
 $y - \bar{y} = b_{yx}(x - \bar{x})$

$$\text{where } b_{yx} = \frac{n \Sigma d_x d_y - (\Sigma d_x)(\Sigma d_y)}{n \Sigma d_x^2 - (\Sigma d_x)^2}$$

n = number of observations

$d_x = x - A$; A is assumed mean of x

$d_y = y - B$; B is assumed mean of y

- Regression equation of x on y
 $x - \bar{x} = b_{xy}(y - \bar{y})$

$$\text{where } b_{xy} = \frac{n \Sigma d_x d_y - (\Sigma d_x)(\Sigma d_y)}{n \Sigma d_y^2 - (\Sigma d_y)^2}$$

n = number of observations

$d_x = x - A$; A is assumed mean of x

$d_y = y - B$; B is assumed mean of y

- (c) **Regression Coefficients in Terms of Correlation Coefficient** If deviations are taken from actual mean values, then the values of regression coefficients can be alternatively calculated as follows:

$$b_{yx} = \frac{\Sigma(x - \bar{x})(y - \bar{y})}{\Sigma(x - \bar{x})^2}$$

$$= \frac{\text{Covariance}(x, y)}{\sigma_x^2} = r \frac{\sigma_y}{\sigma_x}$$

$$b_{xy} = \frac{\Sigma(x - \bar{x})(y - \bar{y})}{\Sigma(y - \bar{y})^2}$$

$$= \frac{\text{Covariance}(x, y)}{\sigma_y^2} = r \frac{\sigma_x}{\sigma_y}$$

Example 10.3: Compute the two regression coefficients using the value of actual mean value of X and Y from the data given below and then work out the value of r.

X	: 7	4	8	6	5
Y	: 6	5	9	8	2

[GJU (Hisar), BBA, 2004]

Solution: Calculations for two regression coefficients are given below:

X	Y	$x = X - \bar{X}$	$y = Y - \bar{Y}$	xy	x^2	y^2
7	6	+1	0	0	1	0
4	5	-2	-1	2	4	1
8	9	+2	+3	6	4	9
6	8	0	+2	0	0	4
5	2	-1	-4	4	1	16
30	30	$\Sigma x = 0$	$\Sigma y = 0$	$\Sigma xy = 12$	$\Sigma x^2 = 10$	$\Sigma y^2 = 30$

$$\bar{X} = \frac{\Sigma X}{n} = \frac{30}{5} = 6 \quad \text{and} \quad \bar{Y} = \frac{\Sigma Y}{n} = \frac{30}{5} = 6$$

$$\text{Regression coefficient of X on Y: } b_{xy} = \frac{\Sigma xy}{\Sigma y^2} = \frac{12}{30} = 0.4$$

$$\text{Regression coefficient of Y on X: } b_{yx} = \frac{\Sigma xy}{\Sigma x^2} = \frac{12}{10} = 1.2$$

Correlation coefficient, $r = \sqrt{b_{xy} \times b_{yx}} = \sqrt{1.2 \times 0.4} = \sqrt{0.48} = 0.69$.

Example 10.4: Use following data to find out the two lines of regression and compute the Karl Pearson's coefficient of correlation.

$\Sigma x = 250$, $\Sigma y = 300$, $\Sigma xy = 7,900$, $\Sigma x^2 = 6,500$, $\Sigma y^2 = 10,000$ and $n = 10$

Solution: Calculating $\bar{x}$, $\bar{y}$, b_{xy} and b_{yx} with the help of given data as follows:

$$\bar{x} = \frac{\Sigma x}{n} = \frac{250}{10} = 25; \quad \bar{y} = \frac{\Sigma y}{n} = \frac{300}{10} = 30$$

$$b_{xy} = \frac{n \Sigma xy - \Sigma x \Sigma y}{n \Sigma y^2 - (\Sigma y)^2} = \frac{10(7900) - (250)(300)}{10(10000) - (300)^2} = \frac{4000}{10,000} = 0.4$$

Let the regression line of x on y be

$$x - \bar{x} = b_{xy}(y - \bar{y}) \quad \text{or} \quad x - 25 = 0.4(y - 30)$$

$$\text{i.e.} \quad x = 0.4y - 12 + 25 = 0.4y + 13$$

$$\text{Also} \quad b_{yx} = \frac{n \Sigma xy - \Sigma x \Sigma y}{n \Sigma x^2 - (\Sigma x)^2} = \frac{10(7900) - (250)(300)}{10(6500) - (250)^2} = \frac{4000}{2500} = 1.6$$

Let the regression line of y on x be

$$y - \bar{y} = b_{yx}(x - \bar{x}) \quad \text{or} \quad y - 30 = 1.6(x - 25)$$

$$\text{i.e.} \quad y = 1.6x - 40 + 30 = 1.6x - 10$$

Hence, the coefficient of correlation is, $r = \sqrt{b_{xy} \times b_{yx}} = \sqrt{0.4 \times 1.6} = 0.8$.

Example 10.5: A departmental store gives training to its salesmen which is followed by a test. The store is considering whether it should terminate the service of any salesman who does not do well in the test. The following data gives the scores and sales made by nine salesmen during a certain period:

Test Scores	: 14	19	24	21	26	22	15	20	19
Sales ('00 ₹)	: 31	36	48	37	50	45	33	41	39

Calculate the correlation coefficient between test scores and the sales. Does it indicate that the termination of services of low test scores is justified? If the firm wants a minimum sales volume of ₹ 3000, what is the minimum test score that will ensure continuation of service? Also estimate the most probable sales volume of a sales making a score of 28.

[Delhi Univ., B.Com.(Hons.), 1998, 2002]

Solution: Let test scores be x and sales be y . Calculations required to determine correlation coefficient are shown below:

Test Scores (x)	$d_x = x - 20$	d_x^2	Sales (y)	$d_y = y - 40$	d_y^2	$d_x d_y$
14	-6	36	31	-9	81	54
19	-1	1	36	-4	16	4
24	4	16	48	8	64	32
21	1	1	37	-3	9	-3
26	6	36	50	10	100	60
22	2	4	45	5	25	10
15	-5	25	33	-7	49	35
20	0	0	41	1	1	0
19	-1	1	39	-1	1	1
180	0	120	360	0	346	193

$$\bar{x} = \frac{\Sigma x}{n} = \frac{180}{9} = 20 \quad \text{and} \quad \bar{y} = \frac{\Sigma y}{n} = \frac{360}{9} = 40$$

$$\begin{aligned} \text{Correlation coefficient, } r &= \frac{x \Sigma d_x d_y - (\Sigma d_x)(\Sigma d_y)}{\sqrt{x \Sigma d_x^2 - (\Sigma d_x)^2} \sqrt{x \Sigma d_y^2 - (\Sigma d_y)^2}} \\ &= \frac{193}{\sqrt{120 \times 346}} = \frac{193}{203.76} = 0.947 \end{aligned}$$

The value of r shows that there is a high degree of correlation between test scores (x) and sales (y). This implies that the persons having low test score will not be able to make good sales hence termination is justified.

Regression equation of x on y is

$$\begin{aligned} x - \bar{x} &= b_{xy} (y - \bar{y}); \quad \text{where } b_{xy} = \frac{n \Sigma d_x d_y - (\Sigma d_x)(\Sigma d_y)}{x \Sigma d_y^2 - (\Sigma d_y)^2} = \frac{193}{346} = 0.558 \\ x - 20 &= 0.557 (y - 40) \\ x &= 0.557 y - 22.28 + 20 = 0.557 y - 2.28 \end{aligned}$$

If sales is ₹ 3000, i.e. $y = 30$, then $x = 0.557 (30) - 2.28 = 14.43$. Hence, test score is 14.43 when sales volume is ₹ 3000.

Regression equation of y on x :

$$\begin{aligned} y - \bar{y} &= b_{yx} (x - \bar{x}); \quad \text{where } b_{yx} = \frac{n \Sigma d_x d_y}{n \Sigma d_x^2} = \frac{193}{120} = 1.6 \\ y - 40 &= 1.6 (x - 20) \\ y &= 1.6x - 32 + 40 = 1.6x + 8 \end{aligned}$$

If test score is $x = 28$, then $y = 1.6(28) + 8 = 52.8$. Thus, sales volume is $52.8 \times 100 = ₹ 5280$.

Example 10.6: The following data relate to the scores obtained by 9 salesmen of a company in an intelligence test and their weekly sales (₹ in 1000's)

Salesmen	:	A	B	C	D	E	F	G	H	I
Test scores	:	50	60	50	60	80	50	80	40	70
Weekly sales	:	30	60	40	50	60	30	70	50	60

- Obtain the regression equation of sales on intelligence test scores of the salesmen.
- If the intelligence test score of a salesman is 65, what would be his expected weekly sales.

[HP Univ., M.Com., 1996]

Solution: Assume weekly sales (y) as dependent variable and test scores (x) as independent variable. Calculations for the following regression equation are shown in Table 10.3.

$$y - \bar{y} = b_{yx} (x - \bar{x})$$

$$(a) \quad \bar{x} = \frac{\Sigma x}{n} = \frac{540}{9} = 60; \quad \bar{y} = \frac{\Sigma y}{n} = \frac{450}{9} = 50$$

Table 10.3 Calculation for Regression Equation

Weekly Sales, x	$dx = x - 60$	d_x^2	Test Score, y	$dy = y - 50$	d_y^2	$d_x d_y$
50	-10	100	30	-20	400	200
60	0	0	60	10	100	0
50	-10	100	40	-10	100	100
60	0	0	50	0	0	0
80	20	400	60	10	100	200
50	-10	100	30	-20	400	200
80	20	400	70	20	400	400
40	-20	400	50	0	0	0
70	10	100	60	10	100	100
540	0	1600	450	0	1600	1200

$$b_{yx} = \frac{n \sum d_x d_y - (\sum d_x)(\sum d_y)}{n \sum d_x^2 - (\sum d_x)^2} = \frac{1200}{1600} = 0.75$$

Substituting values in the regression equation, we have

$$y - 50 = 0.75 (x - 60) \text{ or } y = 5 + 0.75x$$

For test score $x = 65$ of salesman, we have $y = 5 + 0.75 (65) = 53.75$

Hence we conclude that the weekly sales is expected to be ₹ 53.75 (₹ in 1000's) for a test score of 65.

Example 10.7: A company is introducing a job evaluation scheme in which all jobs are graded by points for skill, responsibility, and so on. Monthly pay scales (₹ in 1000's) are then drawn up according to the number of points allocated and other factors such as experience and local conditions. To date the company has applied this scheme to 9 jobs.

Job	:	A	B	C	D	E	F	G	H	I
Points	:	5	25	7	19	10	12	15	28	16
Pay (₹)	:	3.0	5.0	3.25	6.5	5.5	5.6	6.0	7.2	6.1

- Find the least-squares regression line for linking pay scales to points.
- Estimate the monthly pay for a job graded by 20 points.

Solution: Assume monthly pay (y) as the dependent variable and job grade points (x) as the independent variable. Calculations for the following regression equation are shown in Table 10.4.

$$y - \bar{y} = b_{yx} (x - \bar{x})$$

$$(a) \quad \bar{x} = \frac{\sum x}{n} = \frac{137}{9} = 15.22 \quad \text{and} \quad \bar{y} = \frac{\sum y}{n} = \frac{48.15}{9} = 5.35$$

Table 10.4 Calculations for Regression Equation

Grade Points, x	$d_x = x - 15$	d_x^2	Pay Scale, y	$d_y = y - 5$	d_y^2	$d_x d_y$
5	-10	100	3.0	-2.0	4	20
25	10	100	5.0 ← B	0	0	0
7	-8	64	3.25	-1.75	3.06	14
19	4	16	6.5	1.50	2.25	6
10	-5	25	5.5	0.50	0.25	-2.5
12	-3	9	5.6	0.60	0.36	-1.8
15 ← A	0	0	6.0	1.00	1.00	0
28	13	169	7.2	2.2	4.84	28.6
16	1	1	6.1	1.1	1.21	1.1
137	2	484	48.15	3.15	16.97	65.40

Since mean values $\bar{x}$ and $\bar{y}$ are non-integer value, therefore deviations are taken from assumed mean as shown in Table 10.4.

$$b_{yx} = \frac{n \sum d_x d_y - (\sum d_x)(\sum d_y)}{n \sum d_x^2 - (\sum d_x)^2} = \frac{9 \times 65.40 - 2 \times 3.15}{9 \times 484 - (2)^2} = \frac{582.3}{4352} = 0.134$$

Substituting values in the regression equation, we have

$$y - \bar{y} = b_{yx}(x - \bar{x}) \text{ or } y - 5.35 = 0.134(x - 15.22) = 0.134x - 2.039$$

$$y = 0.134x + 3.311$$

(b) For job grade point $x=20$, the estimated average pay scale is given by

$$y = 3.311 + 0.134x = 3.311 + 0.134(20) = 5.991$$

Hence, likely monthly pay for a job with grade points 20 is ₹ 5991.

Example 10.8: Find two regression equations from the data given below:

x	: 57	58	59	59	60	61	62	64
y	: 77	78	75	78	82	82	79	81

[GJU (Hisar), BBA, 2005]

Solution: Calculations of two regression equations as shown below:

x	$d_x = x - 60$	y	$d_y = y - 79$	$d_x d_y$	dx^2	dy^2
57	-3	77	-2	6	9	4
58	-2	78	-1	2	4	1
59	-1	75	-4	4	1	16
59	-1	78	-1	1	1	1
60 ← A	0	82	3	0	0	9
61	1	82	3	3	1	9
62	2	79 ← B	0	0	4	0
64	4	81	2	8	16	4
480	0	632	0	24	36	44

$$\bar{x} = \frac{\Sigma x}{n} = \frac{480}{8} = 60 \quad \text{and} \quad \bar{y} = \frac{\Sigma y}{n} = \frac{632}{8} = 79$$

We know that

$$b_{xy} = \frac{n \Sigma d_x d_y - \Sigma d_x \Sigma d_y}{n \Sigma d_y^2 - (\Sigma d_y)^2} = \frac{8 \times 24}{8 \times 44} = 0.55$$

$$b_{yx} = \frac{n \Sigma d_x d_y - \Sigma d_x \Sigma d_y}{n \Sigma d_x^2 - (\Sigma d_x)^2} = \frac{8 \times 24}{8 \times 36} = 0.67$$

Regression equation of x on y : $x - \bar{x} = b_{xy}(y - \bar{y})$

$$x - 60 = 0.55(y - 79)$$

$$x = 0.55y - 43.45 + 60 = 0.55y + 16.5$$

Regression equation of y on x : $y - \bar{y} = b_{yx}(x - \bar{x})$

$$y - 79 = 0.67(x - 60)$$

$$y = 0.67x - 40.2 + 79 = 0.67x + 38.8.$$

Example 10.9: The following data give the age and blood pressure of 10 women:

Age	:	56	42	36	47	49	42	60	72	63	55
Blood pressure	:	147	125	118	128	145	140	155	160	149	150

- Find the correlation coefficient between age and blood pressure.
- Determine the least squares regression equation of blood pressure on age.
- Estimate the blood pressure of a woman whose age is 45 years.

Solution: Assume blood pressure (y) as the dependent variable and age (x) as the independent variable. Calculations for regression equation of blood pressure on age are shown in Table 10.5.

Table 10.5 Calculations for Regression Equation

Age, x	$d_x = x - 49$	d_x^2	Blood, y	$d_y = y - 145$	d_y^2	$d_x d_y$
56	7	49	147	2	4	14
42	-7	49	125	-20	400	140
36	-13	169	118	-27	729	351
47	-2	4	128	-17	289	34
(49) ← A	0	0	(145) ← B	0	0	0
42	-7	49	140	-5	25	35
60	11	121	155	10	100	110
72	23	529	160	15	225	345
63	14	196	149	4	16	56
55	6	36	150	5	25	30
522	32	1202	1417	-33	1813	1115

- Coefficient of correlation between age and blood pressure is given by

$$r = \frac{n \Sigma d_x d_y - \Sigma d_x \Sigma d_y}{\sqrt{n \Sigma d_x^2 - (\Sigma d_x)^2} \sqrt{n \Sigma d_y^2 - (\Sigma d_y)^2}}$$

$$\begin{aligned}
 &= \frac{10(1115) - (32)(-33)}{\sqrt{10(1202) - (32)^2} \sqrt{10(1813) - (-33)^2}} \\
 &= \frac{11150 + 1056}{\sqrt{12020 - 1024} \sqrt{18130 - 1089}} = \frac{12206}{13689} = 0.892
 \end{aligned}$$

We may conclude that there is a high degree of positive correlation between age and blood pressure.

(b) The regression equation of blood pressure on age is given by

$$\begin{aligned}
 y - \bar{y} &= b_{yx} (x - \bar{x}) \\
 \bar{x} &= \frac{\Sigma x}{n} = \frac{522}{10} = 52.2 \quad \text{and} \quad \bar{y} = \frac{\Sigma y}{n} = \frac{1417}{10} = 141.7
 \end{aligned}$$

and

$$b_{yx} = \frac{n \Sigma d_x d_y - \Sigma d_x \Sigma d_y}{n \Sigma d_x^2 - (\Sigma d_x)^2} = \frac{10(1115) - 32(-33)}{10(1202) - (32)^2} = \frac{12206}{10996} = 1.11$$

Substituting these values in the above equation, we have

$$y - 141.7 = 1.11 (x - 52.2) \text{ or } y = 83.758 + 1.11x$$

This is the required regression equation of y on x .

(c) For a woman whose age is 45, the estimated average blood pressure will be

$$y = 83.758 + 1.11(45) = 83.758 + 49.95 = 133.708$$

Hence, the likely blood pressure of a woman of 45 years is 134.

Example 10.10: The General Sales Manager of Kiran Enterprises—an enterprise dealing in the sale of readymade men's wear—is toying with the idea of increasing his sales to ₹ 80,000. On checking the records of sales during the last 10 years, it was found that the annual sale proceeds and advertisement expenditure were highly correlated to the extent of 0.8. It was further noted that the annual average sale has been ₹ 45,000 and annual average advertisement expenditure ₹ 30,000, with a variance of ₹ 1600 and ₹ 625 in advertisement expenditure respectively.

In view of the above, how much expenditure on advertisement would you suggest the General Sales Manager of the enterprise to incur to meet his target of sales?

Solution: Assume advertisement expenditure (y) as the dependent variable and sales (x) as the independent variable. Then the regression equation advertisement expenditure on sales is given by

$$(y - \bar{y}) = r \frac{\sigma_y}{\sigma_x} (x - \bar{x})$$

Given $r = 0.8$, $\sigma_x = 40$, $\sigma_y = 25$, $\bar{x} = 45,000$ and $\bar{y} = 30,000$. Substituting these values in the above equation, we have

$$\begin{aligned}
 (y - 30,000) &= 0.8 \frac{25}{40} (x - 45,000) = 0.5 (x - 45,000) \\
 y &= 0.5x - 22,500 + 30,000 = 7500 + 0.5x
 \end{aligned}$$

When a sales target is fixed at $x = 80,000$, the estimated amount likely to be spent on advertisement would be

$$y = 7500 + 0.5 \times 80,000 = 7500 + 40,000 = ₹ 47,500$$

Example 10.11: You are given the following information about advertising expenditure and sales:

	Advertisement (x) (₹ in lakh)	Sales (y) (₹ in lakh)
Arithmetic mean, $\bar{x}$	10	90
Standard deviation, σ	3	12

Correlation coefficient = 0.8

- Obtain the two regression equations.
- Find the likely sales when advertisement budget is ₹ 15 lakh.
- What should be the advertisement budget if the company wants to attain sales target of ₹ 120 lakh?

Solution: (a) Regression equation of x on y is given by

$$x - \bar{x} = r \frac{\sigma_x}{\sigma_y} (y - \bar{y})$$

Given $\bar{x} = 10$, $r = 0.8$, $\sigma_x = 3$, $\sigma_y = 12$ and $\bar{y} = 90$. Substituting these values in the above regression equation, we have

$$x - 10 = 0.8 \frac{3}{12} (y - 90) \text{ or } x = -8 + 0.2y$$

Regression equation of y on x is given by

$$(y - \bar{y}) = r \frac{\sigma_y}{\sigma_x} (x - \bar{x})$$

$$y - 90 = 0.8 \frac{12}{3} (x - 10) \text{ or } y = 58 + 3.2x$$

- Substituting $x = 15$ in regression equation of y on x . The likely average sales volume would be

$$y = 58 + 3.2(15) = 58 + 48 = 106$$

Thus, the likely sales for advertisement budget of ₹ 15 lakh is ₹ 106 lakh.

- Substituting $y = 120$ in the regression equation of x on y . The likely advertisement budget to attain desired sales target of ₹ 120 lakh would be

$$x = -8 + 0.2y = -8 + 0.2(120) = 16$$

Hence, the likely advertisement budget of ₹ 16 lakh should be sufficient to attain the sales target of ₹ 120 lakh.

Example 10.12: For 100 students of a class, the regression equation of marks in statistics (x) on marks in economics (y) is: $3y - 5x + 180 = 0$. If average marks in economics is 50 and variance of marks in statistics is $(4/9)$ of variance of marks in economics, find mean marks in statistics and the coefficient of correlation between them.

[Delhi Univ., B.Com. (Hons.), 2005]

Solution: Regression equation of x on y : $3y - 5x + 180 = 0$ or $x = \frac{3y}{5} + \frac{180}{5}$

Thus, regression coefficient of x on y , $b_{xy} = \frac{3}{5}$

Also it is given that $\bar{y} = 50$. To find $\bar{x}$, put $y = 50$ in the regression equation, we get

$$x = \frac{3y}{5} + \frac{180}{5} = \frac{3 \times 50}{5} + \frac{180}{5} = 6$$

It is given that, $\sigma_x^2 = \frac{4}{9}\sigma_y^2$, i.e. $\sigma_x = \frac{2}{3}\sigma_y$ or $\frac{\sigma_x}{\sigma_y} = \frac{2}{3}$

Since, $b_{xy} = \frac{3}{5}$ or $r \cdot \frac{\sigma_x}{\sigma_y} = \frac{3}{5}$

$$r\left(\frac{2}{3}\right) = \frac{3}{5} \text{ or } r = \frac{3}{5} \times \frac{3}{2} = \frac{9}{10} = 0.9$$

Example 10.13: Given that

Regression equation of y on x is $y = 20 + 0.4x$

Mean of $x = 30$

Correlation coefficient = 0.8

Find regression equation of x on y .

[Delhi Univ., B.Com., 2005]

Solution: Since regression equation of y on x : $y = 20 + 0.4x$, this implies that $b_{yx} = 0.4$

Also, $\bar{x} = 30$, put $x = 30$ in the regression equation of y on x , we get

$$y = 20 + 0.4(30) = 20 + 12 = 32$$

Thus $\bar{y} = 32$

We know that, $r^2 = b_{xy} \times b_{yx}$ or $(0.8)^2 = b_{xy}(0.4)$, i.e. $b_{xy} = \frac{0.64}{0.40} = 1.6$

Regression equation of x on y is:

$$x - \bar{x} = b_{xy}(y - \bar{y})$$

$$x - 30 = 1.6(y - 32) = 1.6y - 51.2 + 30 = 1.6y - 21.2.$$

Example 10.14: Given

	<i>X Series</i>	<i>Y Series</i>
Mean	18	100
Standard deviation	14	20

Coefficient of correlation = 0.8

Find the most probable value of Y when $X = 70$ and most probable value of X when $Y = 90$.

[Delhi Univ., B.Com. (Hons.), 2005]

Solution: Given: $\bar{X} = 18$, $\bar{Y} = 100$, $\sigma_X = 14$, $\sigma_Y = 20$, and $r = 0.8$

Regression equation of X on Y : $X - \bar{X} = \frac{r\sigma_X}{\sigma_Y}(Y - \bar{Y})$

$$X - 18 = \frac{(0.8)(14)}{20}(Y - 100) = 0.56(Y - 100)$$

$$X = 0.56Y - 56 + 18 = 0.56Y - 38$$

If $Y = 90$, then $X = 0.56(90) - 38 = 12.4$

Regression equation of Y on X : $Y - \bar{Y} = \frac{r\sigma_Y}{\sigma_X}(X - \bar{X})$

$$Y - 100 = \frac{(0.8)(20)}{14}(X - 18) = 1.14(X - 18)$$

$$Y = 1.14X - 20.52 + 100 = 1.14X + 79.48$$

If $X = 70$, then $Y = 1.14(70) + 79.48 = 159.28$.

Example 10.15: Given $x = 4y + 5$ and $y = kx + 4$ are the lines of regression of x on y and y on x respectively. If k is positive, prove that it cannot exceed $1/4$. If $k = 1/16$, find the means of two variables and correlation coefficient between them. [Delhi Univ., B.Com., 2006]

Solution: Regression equation of x on y : $x = 4y + 5$. This implies that $b_{xy} = 4$

Regression equation of y on x : $y = kx + 4$. This implies that $b_{yx} = k$

We know that, $r^2 = b_{yx} \times b_{xy} = k \times 4$. Now if $r < 1$, then $r^2 < 1$ and hence $4k < 1$ or $k < 1/4$

If $k = 1/16$, then $y = (1/16)x + 4$. Thus the regression coefficient of y on x becomes $b_{yx} = 1/16$. Then the correlation coefficient is

$$r = \sqrt{b_{yx} \times b_{xy}} = \sqrt{(1/16) \times 4} = \sqrt{1/4} = 0.5.$$

Solving two regression equations: $x = 4y + 5$ and $y = (1/16)x + 4$, for x and y , we get $\bar{x} = 28$ and $\bar{y} = 5.76$.

Example 10.16: In a partially destroyed laboratory record of an analysis of regression data, the following results only are legible:

Variance of $x = 9$

Regression equations : $8x - 10y + 66 = 0$ and $40x - 18y = 214$

Find on the basis of the above information:

- The mean values of x and y .
- Coefficient of correlation between x and y .
- Standard deviation of y .

Solution: (a) Since two regression lines always intersect at a point $(\bar{x}, \bar{y})$ representing mean values of the variables involved, solving given regression equations to get the mean values $\bar{x}$ and $\bar{y}$ as shown below:

$$\begin{aligned} 8x - 10y &= -66 \\ 40x - 18y &= 214 \end{aligned}$$

Multiplying the first equation by 5 and subtracting from the second, we have

$$32y = 544 \text{ or } y = 17, \text{ i.e. } \bar{y} = 17$$

Substituting the value of y in the first equation, we get

$$8x - 10(17) = -66 \text{ or } x = 13, \text{ that is, } \bar{x} = 13$$

(b) To find correlation coefficient r between x and y , we need to determine the regression coefficients b_{xy} and b_{yx} .

Rewriting the given regression equations in such a way that the coefficient of dependent variable is less than one at least in one equation.

$$8x - 10y = -66 \text{ or } 10y = 66 + 8x \text{ or } y = \frac{66}{10} + \frac{8}{10}x$$

That is, $b_{yx} = 8/10 = 0.80$

$$40x - 18y = 214 \text{ or } 40x = 214 + 18y \text{ or } x = \frac{214}{40} + \frac{18}{40}y$$

That is, $b_{xy} = 18/40 = 0.45$

Hence coefficient of correlation, r between x and y is given by

$$r = \sqrt{b_{xy} \times b_{yx}} = \sqrt{0.45 \times 0.80} = 0.60$$

(c) To determine the standard deviation of y , consider the formula:

$$b_{yx} = r \frac{\sigma_y}{\sigma_x} \text{ or } \sigma_y = \frac{b_{yx} \sigma_x}{r} = \frac{0.80 \times 3}{0.6} = 4$$

Example 10.17: There are two series of index numbers, P for price index and S for stock of a commodity. The mean and standard deviation of P are 100 and 8, and of S are 103 and 4, respectively. The correlation coefficient between the two series is 0.4. With these data, work out a linear equation to read off values of P for various values of S. Can the same equation be used to read off values of S for various values of P?

Solution: The regression equation to read off values of P for various values S is given by

$$P = a + bS \text{ or } (P - \bar{P}) = r \frac{\sigma_p}{\sigma_s} (S - \bar{S})$$

Given $\bar{P} = 100$, $\bar{S} = 103$, $\sigma_p = 8$, $\sigma_s = 4$, $r = 0.4$. Substituting these values in the above equation, we have

$$P - 100 = 0.4 \frac{8}{4} (S - 103) \text{ or } P = 17.6 + 0.8 S$$

This equation cannot be used to read off values of S for various values of P. Thus to read off values of S for various values of P, we use another regression equation of the form:

$$S = c + dP \text{ or } S - \bar{S} = r \frac{\sigma_s}{\sigma_p} (P - \bar{P})$$

Substituting given values in this equation, we have

$$S - 103 = 0.4 \frac{4}{8} (P - 100) \text{ or } S = 83 + 0.2P$$

Example 10.18: A panel of Judges A and B graded seven debators and independently awarded the following marks:

Debator	:	1	2	3	4	5	6	7
Marks of A	:	40	34	28	30	44	38	31
Marks of B	:	32	39	26	30	38	34	28

An eighth debator was awarded 36 marks by Judge A while Judge B was not present. If Judge B was also present, how many marks would you expect him to award to eighth debator assuming same degree of relationship exists in judgement?

Solution: Let marks of A be denoted by x and that of B by y . Also, let $A = 30$ and $B = 30$ be assumed as mean value of x -series and y -series, respectively. The calculations required for regression equations are shown below:

Marks of A(x)	$d_x = x - 30$	d_x^2	Marks of B(y)	$d_y = y - 30$	d_y^2	$d_x d_y$
40	10	100	32	2	4	20
34	4	16	39	9	81	36
28	-2	4	26	-4	16	8
30 ← A	0	0	30 ← B	0	0	0
44	14	196	38	8	64	112
38	8	64	34	4	16	32
31	1	1	28	-2	4	-2
245	35	381	227	17	185	206

$$\bar{x} = \frac{\Sigma x}{n} = \frac{245}{7} = 35 \text{ and } \bar{y} = \frac{\Sigma y}{n} = \frac{227}{7} = 32.43$$

Regression coefficient of y on x :

$$\begin{aligned} b_{yx} &= \frac{n \sum d_x d_y - \sum d_x \sum d_y}{n \sum d_x^2 - (\sum d_x)^2} = \frac{7(206) - (35)(17)}{7(381) - (35)^2} \\ &= \frac{1442 - 595}{2667 - 1225} = \frac{847}{1442} = 0.587 \end{aligned}$$

Regression equation of y on x is

$$\begin{aligned} y - \bar{y} &= b_{yx}(x - \bar{x}) \\ y - 32.43 &= 0.587(x - 35) \\ y &= 0.587x - 20.545 + 32.43 = 0.587x + 11.885 \end{aligned}$$

When Judge A awarded $x = 36$ marks to eighth debator, the marks awarded to eighth debator by Judge B are:

$$y = 0.587(36) + 11.885 = 33 \text{ marks}$$

Example 10.19: The lines of regression of a bivariate distribution are as follows: $5x - 145 = -10y$ and $14y - 208 = -8x$, variance of $x = 4$. Find out mean values of x and y and standard deviation of y . Also find correlation coefficient between x and y . [Delhi Univ., B.Com. (Hons.), 2001]

Solution: Let the regression equation of x on y be

$$\begin{aligned} 5x - 145 &= -10y \\ x &= \frac{-10y}{5} + \frac{145}{5} = -2y + 29 \end{aligned}$$

Hence, regression coefficient of x on y becomes $b_{xy} = -2$

Let the regression equation of y on x be

$$\begin{aligned} 14y - 208 &= -8x \\ y &= \frac{208}{14} - \frac{8x}{14} \end{aligned}$$

Hence regression coefficient of y on x becomes $b_{yx} = \frac{-8}{14}$

Now the coefficient of correlation r between x and y is given by

$$r = \sqrt{b_{yx} \times b_{xy}} = \sqrt{(-8/14) \times (-2)} = 1.07$$

Since value of r cannot exceed one, consider another set of regression equations:

Regression equation of x on y is

$$14y - 208 = -8x \text{ or } x = \frac{208}{8} - \frac{14}{8}y, \text{ i.e. } b_{xy} = \frac{-14}{8} = -\frac{7}{4}$$

Regression equation of y on x is

$$5x - 145 = -10y \text{ or } y = \frac{145}{10} - \frac{5}{10}x, \text{ i.e. } b_{yx} = \frac{-5}{10} = -\frac{1}{2}$$

$$\text{Now } r = \sqrt{b_{yx} \times b_{xy}} = \sqrt{-\frac{1}{2} \times \left(-\frac{7}{4}\right)} = \sqrt{0.875} = \pm 0.93$$

Since b_{xy} and b_{yx} both are negative, r should also be negative. Hence $r = -0.93$.

$$\text{We know that } b_{yx} = \frac{r\sigma_y}{\sigma_x} \text{ or } -\frac{1}{2} = \frac{-0.93\sigma_y}{2}, \text{ where } \sigma_x = 2$$

$$\text{or } \sigma_y = \frac{1}{0.93} = 1.075$$

Solving the two regression equations simultaneously to get $\bar{x}$ and $\bar{y}$.

$$5x - 145 = -10y \quad \text{or} \quad 5x + 10y = 145 \quad (\text{i})$$

$$14y - 208 = -8x \quad \text{or} \quad 8x + 14y = 208 \quad (\text{ii})$$

Multiply (i) by 8 and (ii) by 5 and subtracting, we get $10y = 120$ or $y = 12$

Substitute $y = 12$ in (i), we get $5x + 10(12) = 145$ or $x = 5$. Hence $\bar{x} = 5$ and $\bar{y} = 12$.

Example 10.20: The two regression lines obtained in a correlation analysis of 60 observations are

$$5x = 6y + 24 \quad \text{and} \quad 1000y = 768x - 3708$$

What is the correlation coefficient and what is its probable error? Show that the ratio of the coefficient of variability of x to that of y is $5/24$. What is the ratio of variances of x and y ?

Solution: Rewriting the regression equations:

$$5x = 6y + 24 \quad \text{or} \quad x = \frac{6}{5}y + \frac{24}{5}, \text{ i.e. } b_{xy} = \frac{6}{5}$$

$$1000y = 768x - 3708 \quad \text{or} \quad y = \frac{768}{1000}x - \frac{3708}{1000}, \text{ i.e. } b_{yx} = \frac{768}{1000}$$

We know that $b_{xy} = r \frac{\sigma_x}{\sigma_y} = \frac{6}{5}$ and $b_{yx} = r \frac{\sigma_y}{\sigma_x} = \frac{768}{1000}$, therefore

$$r^2 = b_{xy} \times b_{yx} = \frac{6}{5} \times \frac{768}{1000} = 0.9216$$

Hence, $r = \sqrt{0.9216} = 0.96$.

Since both b_{xy} and b_{yx} are positive, the correlation coefficient is positive and $r = 0.96$.

$$\text{Probable error of } r, PE_r = 0.6745 \frac{1 - r^2}{\sqrt{n}} = 0.6745 \frac{1 - (0.96)^2}{\sqrt{60}} = \frac{0.0529}{7.7459} = 0.0068$$

Solving the given regression equations for x and y , we get $\bar{x} = 6$ and $\bar{y} = 1$ because regression lines passed through the point $(\bar{x}, \bar{y})$.

$$\text{Since } b_{xy} = r \frac{\sigma_x}{\sigma_y} \quad \text{or} \quad \frac{6}{5} = 0.96 \frac{\sigma_x}{\sigma_y} \quad \text{i.e.} \quad \frac{\sigma_x}{\sigma_y} = \frac{6}{5 \times 0.96} = \frac{5}{4}$$

$$\text{Also the ratio of the coefficient of variability} = \frac{\sigma_x / \bar{x}}{\sigma_y / \bar{y}} = \frac{\bar{y}}{\bar{x}} \cdot \frac{\sigma_x}{\sigma_y} = \frac{1}{6} \times \frac{5}{4} = \frac{5}{24}.$$

Conceptual Questions 10A

10.1 The following calculations have been made for prices of twelve stocks (x) at the Calcutta Stock Exchange on a certain day along with the volume of sales in thousands of shares (y). From these calculations, find the regression equation of price of stocks

on the volume of sales of shares.

$$\Sigma x = 580, \quad \Sigma y = 370, \quad \Sigma xy = 11494,$$

$$\Sigma x^2 = 41658, \quad \Sigma y^2 = 17206.$$

10.2 A survey was conducted to study the relationship between expenditure (in ₹)

on accommodation (x) and expenditure on food and entertainment (y) and the following results were obtained:

	Mean	Standard Deviation
• Expenditure on accommodation	173	63.15
• Expenditure on food and entertainment	47.8	22.98

Coefficient of correlation $r = 0.57$

Write down the regression equation and estimate the expenditure on food and entertainment if the expenditure on accommodation is ₹ 200.

- 10.3** The following data give the experience of machine operators and their performance ratings given by the number of good parts turned out per 100 pieces:

Operator : 1 2 3 4 5 6 7 8
experience (x) : 16 12 18 4 3 10 5 12

Performance

ratings (y) : 87 88 89 68 78 80 75 83

Calculate the regression lines of performance ratings on experience and estimate the probable performance if an operator has 7 years experience.

- 10.4** A study of prices of a certain commodity at Delhi and Mumbai yield the following data:

	Delhi	Mumbai
• Average price per kilo (₹)	2.463	2.797
• Standard deviation	0.326	0.207
• Correlation coefficient between prices at Delhi and Mumbai $r = 0.774$		

Estimate from the above data the most likely price (a) at Delhi corresponding to the price of ₹ 2.334 per kilo at Mumbai (b) at Mumbai corresponding to the price of 3.052 per kilo at Delhi

- 10.5** The following table gives the aptitude test scores and productivity indices of 10 workers selected at random:

Aptitude

scores (x) : 60 62 65 70 72 48 53 73 65 82

Productivity

index (y) : 68 60 62 80 85 40 52 62 60 81

Calculate the two regression equations and estimate (a) the productivity index of a worker whose test score is 92, (b) the test score of a worker whose productivity index is 75.

- 10.6** A company wants to assess the impact of R&D expenditure (₹ in 1000s) on its annual profit (₹ in 1000's). The following table presents the information for the last eight years:

Year	R & D Expenditure	Annual Profit
1991	9	45
1992	7	42
1993	5	41
1994	10	60
1995	4	30
1996	5	34
1997	3	25
1998	2	20

Estimate the regression equation and predict the annual profit for the year 2002 for an allocated sum of ₹ 1,00,000 as R&D expenditure.

- 10.7** Obtain the two regression equations from the following bivariate frequency distribution:

Sales Revenue (₹ in lakh)	Advertising Expenditure 6(₹ in 1000)			
	5-15	15-25	25-35	35-45
75-125	3	4	4	8
125-175	8	6	5	7
175-225	2	2	3	4
225-275	3	3	2	2

Estimate (a) the sales corresponding to advertising expenditure of ₹ 50,000, (b) the advertising expenditure for a sales revenue of ₹ 300 lakh, (c) the coefficient of correlation.

- 10.8** The personnel manager of an electronic manufacturing company devises a manual test for job applicants to predict their production rating in the assembly department. In order to do this, he selects a random sample of 10 applicants. They are given the test and later assigned a

production rating. The results are as follows:

Worker	:	A	B	C	D	E	F	G	H	I	J
Test score	:	53	36	88	84	86	64	45	48	39	69
Production rating	:	45	43	89	79	84	66	49	48	43	76

Fit a linear least-squares regression equation of production rating on test score.

- 10.9** Find the regression equation showing the capacity utilization on production from the following data:

	Average	Standard Deviation
• Production (in lakh units) :	35.6	10.5
• Capacity utilization (in percentage) :	84.8	8.5
• Correlation coefficient $r = 0.62$		

Estimate the production when the capacity utilization is 70 per cent.

- 10.10** Suppose that you are interested in using past expenditure on R&D by a firm to predict current expenditures on R&D. You got the following data by taking a random sample of firms, where x is the amount spent on R&D (in lakh of rupees) 5 years ago and y is the amount spent on R&D (in lakh of rupees) in the current year:

x : 30 50 20 80 10 20 20 40

y : 50 80 30 110 20 20 40 50

- (a) Find the regression equation of y on x .
 (b) If a firm is chosen randomly and $x = 10$, can you use the regression to predict the value of y ? Discuss.

- 10.11** The following data relates to the scores obtained by a salesmen of a company in an intelligence test and their weekly sales (₹ in 1000's):

Salesman	:	A	B	C	D	E	F	G	H	I
intelligence										
Test score	:	50	60	50	60	80	50	80	40	70
Weekly sales	:	30	60	40	50	60	30	70	50	60

- (a) Obtain the regression equation of sales on intelligence test scores of the salesmen.

- (b) If the intelligence test score of a salesman is 65, what would be his expected weekly sales?

[HP Univ., M.Com., 1996]

- 10.12** Two random variables have the regression equations:

$$3x + 2y - 26 = 0 \text{ and } 6x + y - 31 = 0$$

- (a) Find the mean values of x and y and coefficient of correlation between x and y .
 (b) If the variance of x is 25, then find the standard deviation of y from the data.

- 10.13** For a given set of bivariate data, the following results were obtained

$$\bar{x} = 53.2, \bar{y} = 27.9$$

Regression coefficient of y on $x = -1.5$ and
 Regression coefficient of x and $y = -0.2$.

Find the most probable value of y when $x = 60$.

- 10.14** In trying to evaluate the effectiveness in its advertising campaign, a firm compiled the following information:

Calculate the regression equation of sales on advertising expenditure. Estimate the probable sales when advertisement expenditure is ₹ 60 thousand.

Year	Adv. expenditure (₹ in 1000's)	Sales (₹ in lakh)
1996	12	5.0
1997	15	5.6
1998	17	5.8
1999	23	7.0
2000	24	7.2
2001	38	8.8
2002	42	9.2
2003	48	9.5

Hints and Answers

10.1 $\bar{x} = \Sigma x/n = 580/12 = 48.33$

$\bar{y} = \Sigma y/n = 370/12 = 30.83$

$$b_{xy} = \frac{\Sigma xy - n\bar{x}\bar{y}}{\Sigma y^2 - n(\bar{y})^2}$$

$$= \frac{11494 - 12 \times 48.33 \times 30.83}{17206 - 12(30.83)^2} = -1.102$$

Regression equation of x on y :

$$x - \bar{x} = b_{xy} (y - \bar{y})$$

$$x - 48.33 = -1.102 (y - 30.83)$$

or $x = 82.305 - 1.102y$

10.2 Given $\bar{x} = 172$, $\bar{y} = 47.8$, $\sigma_x = 63.15$,
 $\sigma_y = 22.98$, and $r = 0.57$

Regression equation of food and entertainment (y) on accomodation (x) is given by

$$y - \bar{y} = r \frac{\sigma_y}{\sigma_x} (x - \bar{x})$$

$$y - 47.8 = 0.57 \frac{22.98}{63.15} (x - 173)$$

or $y = 11.917 + 0.207x$

For $x = 200$, we have

$$y = 11.917 + 0.207(200) = 53.317$$

10.3 Let the experience and performance rating be represented by x and y respectively.

$$\bar{x} = \Sigma x/n = 80/8 = 10;$$

$$\bar{y} = \Sigma y/n = 648/8 = 81$$

$$b_{yx} = \frac{n \Sigma d_x d_y - \Sigma d_x \Sigma d_y}{n \Sigma d_x^2 - (\Sigma d_x)^2} = \frac{247}{218} = 1.133;$$

where $d_x = x - \bar{x}$, $d_y = y - \bar{y}$

Regression equation of y on x

$$y - \bar{y} = b_{yx} (x - \bar{x})$$

or $y - 81 = 1.133 (x - 10)$

or $y = 69.67 + 1.133x$

When $x = 7$, $y = 69.67 + 1.133(7)$
 $= 77.60 \approx 78$

10.4 Let price at Mumbai and Delhi be represented by x and y , respectively

(a) Regression equation of y on x

$$y - \bar{y} = r \frac{\sigma_y}{\sigma_x} (x - \bar{x})$$

$$y - 2.463 = 0.774 \frac{0.326}{0.207} (x - 2.797)$$

For $x = ₹ 2.334$, the price at Delhi would be $y = ₹ 1.899$.

(b) Regression on equation of x on y

$$x - \bar{x} = r \frac{\sigma_x}{\sigma_y} (y - \bar{y})$$

$$\text{or } x - 2.791 = 0.774 \frac{0.207}{0.326} (y - 2.463)$$

For $y = ₹ 3.052$, the price at Mumbai would be $x = ₹ 3.086$.

10.5 Let aptitude score and productivity index be represented by x and y , respectively.

$$\bar{x} = \Sigma x/n = 650/10 = 65;$$

$$\bar{y} = \Sigma y/n = 650/10 = 65$$

$$b_{xy} = \frac{n \Sigma d_x d_y - (\Sigma d_x)(\Sigma d_y)}{n \Sigma d_y^2 - (\Sigma d_y)^2}$$

$$= \frac{1044}{1752} = 0.596$$

where $d_x = x - \bar{x}$; $d_y = y - \bar{y}$

(a) Regression equation of x on y

$$x - \bar{x} = b_{xy} (y - \bar{y})$$

or $x - 65 = 0.596 (y - 65)$

or $x = 26.26 + 0.596y$

When $y = 75$, $x = 26.26 + 0.596(75)$
 $= 70.96 \approx 71$

(b) $b_{yx} = \frac{n \Sigma d_x d_y - (\Sigma d_x)(\Sigma d_y)}{n \Sigma d_x^2 - (\Sigma d_x)^2}$

$$= \frac{1044}{894} = 1.168$$

$$y - \bar{y} = b_{yx} (x - \bar{x})$$

or $y - 65 = 1.168 (x - 65)$

or $y = -10.92 + 1.168x$

When $x = 92$, $y = -10.92 + 1.168(92) = 96.536 \approx 97$

- 10.6** Let R&D expenditure and annual profit be denoted by x and y , respectively

$$\bar{x} = \Sigma x/n = 45/8 = 5.625;$$

$$\bar{y} = \Sigma y/n = 297/8 = 37.125$$

$$\begin{aligned} b_{yx} &= \frac{n \Sigma d_x d_y - (\Sigma d_x)(\Sigma d_y)}{n \Sigma d_x^2 - (\Sigma d_x)^2} \\ &= \frac{8 \times 238 - (-3)(1)}{8 \times 57 - (-3)^2} = 4.266 \end{aligned}$$

where $d_x = x - 6$, $d_y = y - 37$

Regression equation of annual profit on R&D expenditure

$$\begin{aligned} y - \bar{y} &= b_{yx}(x - \bar{x}) \\ y - 37.125 &= 4.266(x - 5.625) \end{aligned}$$

or $y = 13.29 + 4.266x$

For $x = ₹ 1,00,000$ as R&D expenditure, we have from above equation $y = ₹ 4,26,613$ as annual profit.

- 10.7** Let sales revenue and advertising expenditure be denoted by x and y , respectively

$$\begin{aligned} \bar{x} &= A + \frac{\Sigma fd_x}{n} \times h \\ &= 150 + \frac{12}{66} \times 50 = 159.09 \end{aligned}$$

$$\begin{aligned} \bar{y} &= B + \frac{\Sigma fd_y}{n} \times k \\ &= 30 - \frac{26}{66} \times 10 = 26.06 \end{aligned}$$

$$\begin{aligned} b_{xy} &= \frac{n \Sigma fd_x d_y - (\Sigma fd_x)(\Sigma fd_y)}{n \Sigma fd_y^2 - (\Sigma fd_y)^2} \times \frac{h}{k} \\ &= \frac{66(-14) - 12(-26)}{66(100) - (-26)^2} \times \frac{50}{10} = -0.516 \end{aligned}$$

- (a) Regression equation of x on y

$$\begin{aligned} x - \bar{x} &= b_{xy}(y - \bar{y}) \\ x - 159.09 &= -0.516(y - 26.06) \end{aligned}$$

or $x = 172.537 - 0.516y$

For $y = 50$, $x = 146.737$

- (b) Regression equation of y on x

$$\begin{aligned} b_{yx} &= \frac{n \Sigma fd_x d_y - (\Sigma fd_x)(\Sigma fd_y)}{n \Sigma fd_x^2 - (\Sigma fd_x)^2} \times \frac{k}{h} \\ &= \frac{66(-14) - 12(-26)}{66(70) - (12)^2} \times \frac{10}{50} \\ &= -0.027. \end{aligned}$$

$$y - \bar{y} = b_{yx}(x - \bar{x})$$

$$y - 26.06 = -0.027(x - 159.09)$$

$$y = 30.355 - 0.027x$$

For $x = 300$, $y = 22.255$

$$\begin{aligned} \text{(c) } r &= \sqrt{b_{xy} \times b_{yx}} \\ &= -\sqrt{0.516 \times 0.027} = -0.1180 \end{aligned}$$

- 10.8** Let test score and production rating be denoted by x and y , respectively.

$$\bar{x} = \Sigma x/n = 612/10 = 61.2;$$

$$\bar{y} = \Sigma y/n = 622/10 = 62.2$$

$$\begin{aligned} b_{yx} &= \frac{n \Sigma d_x d_y - (\Sigma d_x)(\Sigma d_y)}{n \Sigma d_x^2 - (\Sigma d_x)^2} \\ &= \frac{10 \times 3213 - 2 \times 2}{10 \times 3554 - (2)^2} = 0.904 \end{aligned}$$

Regression equation of production rating (y) on test score (x) is given by

$$\begin{aligned} y - \bar{y} &= b_{yx}(x - \bar{x}) \\ y - 62.2 &= 0.904(x - 61.2) \\ y &= 6.876 + 0.904x \end{aligned}$$

- 10.9** Let production and capacity utilization be denoted by x and y , respectively.

(a) Regression equation of capacity utilization (y) on production (x)

$$\begin{aligned} y - \bar{y} &= r \frac{\sigma_y}{\sigma_x}(x - \bar{x}) \\ y - 84.8 &= 0.62 \frac{8.5}{10.5}(x - 35.6) \\ y &= 66.93 + 0.502x \end{aligned}$$

(b) Regression equation of production (x) on capacity utilization (y)

$$x - \bar{x} = r \frac{\sigma_x}{\sigma_y}(y - \bar{y})$$

$$x - 35.6 = 0.62 \frac{10.5}{8.5} (y - 84.8)$$

$$x = -29.3483 + 0.7659y$$

When $y = 70$, $x = -29.3483 + 0.7659(70)$
 $= 24.2647$

Hence the estimated production is 2,42,647 units when the capacity utilization is 70 per cent.

10.10 $\bar{x} = \Sigma x/n = 270/8 = 33.75$;

$$\bar{y} = \Sigma y/n = 400/8 = 50$$

$$b_{yx} = \frac{n \Sigma d_x d_y - (\Sigma d_x)(\Sigma d_y)}{n \Sigma d_x^2 - (\Sigma d_x)^2}$$

$$= \frac{8 \times 4800 - 6 \times 0}{8 \times 3592 - (6)^2} = 1.338;$$

where $d_x = x - 33$ and $d_y = y - 50$

Regression equation of y on x

$$y - \bar{y} = b_{yx}(x - \bar{x})$$

$$y - 50 = 1.338(x - 33.75)$$

$$y = 4.84 + 1.338x$$

For $x = 10$, $y = 18.22$

10.11 Let intelligence test score be denoted by x and weekly sales by y

$$\bar{x} = 540/9 = 60; \quad \bar{y} = 450/9 = 50,$$

$$b_{yx} = \frac{n \Sigma dx dy - (\Sigma dx)(\Sigma dy)}{n \Sigma d_x^2 - (\Sigma d_x)^2}$$

$$= \frac{9 \times 1200}{9 \times 1600} = 0.75$$

Regression equation of y on x :

$$y - \bar{y} = b_{yx}(x - \bar{x})$$

$$y - 50 = 0.75(x - 60)$$

$$y = 5 + 0.75x$$

For $x = 65$, $y = 5 + 0.75(65) = 53.75$

10.12 (a) Solving two regression lines:

$$3x + 2y = 6 \quad \text{and} \quad 6x + y = 31$$

we get mean values as $\bar{x} = 4$ and $\bar{y} = 7$

(b) Rewriting regression lines as follows:

$$3x + 2y = 26 \quad \text{or} \quad y = 13 - (3/2)x,$$

$$\text{So } b_{yx} = -3/2$$

$$6x + y = 31 \quad \text{or} \quad x = 31/6 - (1/6)y,$$

$$\text{So } b_{xy} = -1/6$$

Correlation coefficient,

$$r = \sqrt{b_{xy} \times b_{yx}} = -\sqrt{(3/2)(1/6)} = -0.5$$

Given, $\text{Var}(x) = 25$, so $\sigma_x = 5$. Calculate σ_y using the formula:

$$b_{yx} = r \frac{\sigma_y}{\sigma_x}$$

$$\text{or} \quad -\frac{3}{2} = 0.5 \frac{\sigma_y}{5} \quad \text{or} \quad \sigma_y = 15$$

10.13 The regression equation of y on x is stated as

$$y - \bar{y} = b_{yx}(x - \bar{x}) = r \cdot \frac{\sigma_y}{\sigma_x}(x - \bar{x})$$

Given, $\bar{x} = 53.20$; $\bar{y} = 27.90$, $b_{yx} = -1.5$
 $b_{xy} = -0.2$

$$\text{Thus } y - 27.90 = -1.5(x - 53.20)$$

$$\text{or} \quad y = 107.70 - 1.5x$$

For $x = 60$, we have y

$$= 107.70 - 1.5(60) = 17.7$$

$$\text{Also } r = \sqrt{b_{yx} \times b_{xy}}$$

$$= -\sqrt{1.5 \times 0.2} = -0.5477$$

10.14 Let advertising expenditure and sales be denoted by x and y respectively.

$$\bar{x} = \Sigma x/n = 217/8 = 27.125; \quad \bar{y} = \Sigma y/n = 58.1/8 = 7.26$$

$$b_{yx} = \frac{n \Sigma dx dy - (\Sigma dx)(\Sigma dy)}{n \Sigma d_x^2 - (\Sigma dx)^2}$$

$$= \frac{8(172.2) - (25)(2.1)}{8(1403) - (25)^2}$$

$$= \frac{1325.1}{10599} = 0.125$$

Thus regression equation of y on x is:

$$y - \bar{y} = b_{yx}(x - \bar{x})$$

$$\text{or} \quad y - 7.26 = 0.125(x - 27.125)$$

$$y = 3.87 + 0.125x$$

When $x = 60$, the estimated value of y is
 $y = 3.87 + 0.125(60) = 11.37$

Conceptual Questions

- (a) Explain the concept of regression and point out its usefulness in dealing with business problems.
(b) Distinguish between correlation and regression. Also point out the properties of regression coefficients.
- Explain the concept of regression and point out its importance in business forecasting.
- Under what conditions can there be one regression line? Explain.
- Why should a residual analysis always be done as part of the development of a regression model?
- What are the assumptions of simple linear regression analysis and how can they be evaluated?
- What is the meaning of the standard error of estimate?
- What is the interpretation of y-intercept and the slope in a regression model?
- What are regression lines? With the help of an example illustrate how they help in business decision making.
- Point out the role of regression analysis in business decision-making. What are the important properties of regression coefficients?
- (a) Distinguish between correlation and regression analysis.
(b) The coefficient of correlation and coefficient of determination are available as measures of association in correlation analysis. Describe the different uses of these two measures of association.
- What are regression coefficients? State some of the important properties of regression coefficients.
- What is regression? How is this concept useful to business forecasting?
- What is the difference between a prediction interval and a confidence interval in regression analysis?
- Explain what is required to establish evidence of a cause-and-effect relationship between y and x with regression analysis.
- What technique is used initially to identify the kind of regression model that may be appropriate.
- (a) What are regression lines? Why is it necessary to consider two lines of regression?
(b) In case the two regression lines are identical, prove that the correlation coefficient is either +1 or -1. If two variables are independent, show that the two regression lines cut at right angles.

Formulae Used

- Normal equations

$$\Sigma y = an + b\Sigma x$$

$$\Sigma xy = a\Sigma x + b\Sigma y^2$$
- Simple linear regression equation based on sample data $y = a + bx$
- Regression coefficient in sample regression equation $b = \hat{y}$

$$a = \bar{y} - b\bar{x}$$
- (a) Regression equation of y on x

$$y - \bar{y} = b_{yx} (x - \bar{x})$$
 (b) Regression equation of x on y

$$x - \bar{x} = b_{xy} (y - \bar{y})$$
- (a) $b_{xy} = r \frac{\sigma_x}{\sigma_y}$ and $b_{yx} = r \frac{\sigma_y}{\sigma_x}$
(b) $b_{xy} = \frac{n \Sigma d_x d_y - (\Sigma d_x)(\Sigma d_y)}{n \Sigma d_x^2 - (\Sigma d_x)^2}$

True or False Concepts Quiz

1. A statistical relationship between two variables does not indicate a perfect relationship.
2. A dependent variable in a regression equation is a continuous random variable.
3. The residual value is required to estimate the amount of variation in the dependent variable with respect to the fitted regression line.
4. Standard error of estimate is the conditional standard deviation of the dependent variable.
5. Standard error of estimate is a measure of scatter of the observations about the regression line.
6. If one of the regression coefficients is greater than one the other must also be greater than one.
7. The signs of the regression coefficients are always same.
8. Correlation coefficient is the geometric mean of regression coefficients.
9. If the sign of two regression coefficients is negative, then sign of the correlation coefficient is positive.
10. Correlation coefficient and regression coefficient are independent.
11. The point of intersection of two regression lines represents average value of two variables.
12. The two regression lines are at right angle when the correlation coefficient is zero.
13. When value of correlation coefficient is one, the two regression lines coincide.
14. The product of regression coefficients is always more than one.
15. The regression coefficients are independent of the change of origin but not of scale.

Concepts Quiz Answers

1. T	2. T	3. T	4. T	5. T	6. F	7. T	8. T	9. F
10. F	11. T	12. T	13. T	14. T	15. T			

Review Self-practice Problems

10.15 Given the following bivariate data:

x :	-1	5	3	2	1	1	7	3
y :	-6	1	0	0	1	2	1	5

- (a) Fit a regression line of y on x and predict y if $x = 10$.
- (b) Fit a regression line of x on y and predict x if $y = 2.5$.

10.16 Find the most likely production corresponding to a rainfall of 40 inches from the following data:

	<i>Rainfall</i> (in inches)	<i>Production</i> (in quintals)
Average	30	50
Standard deviation	5	10

Coefficient of correlation $r = 0.8$.

10.17 The coefficient of correlation between the ages of husbands and wives in a community was found to be $+0.8$, the average of husbands age was 25 years and that of wives age 22 years. Their standard deviations were 4 and 5 years, respectively. Find with the help of regression equations:

- (a) the expected age of husband when wife's age is 16 years.
 (b) the expected age of wife when husband's age is 33 years.

10.18 You are given below the following information about advertisement expenditure and sales:

	<i>Adv. Exp. (x)</i> (₹ in crore)	<i>Sales (y)</i> (₹ in crore)
Mean	20	120
Standard deviation	5	25

Correlation coefficient 0.8

- (a) Calculate the two regression equations.
 (b) Find the likely sales when advertisement expenditure is ₹ 25 crore.
 (c) What should be the advertisement budget if the company wants to attain sales target of ₹ 150 crore?

10.19 For 50 students of a class, the regression equation of marks in Statistics (x) on the marks in Accountancy (y) is $3y - 5x + 180 = 0$. The mean marks in Accountancy is 44 and the variance of marks in Statistics is $\frac{9}{16}$ th of the variance of marks in Accountancy. Find the mean marks in Statistics and the coefficient of correlation between marks in the two subjects.

10.20 The HRD manager of a company wants to find a measure which he can use to fix the monthly income of persons applying for a job in the production department. As an experimental project, he collected data on 7 persons from that department referring to years of service and their monthly income.

Years of service : 11 7 9 5 8 6 10

Income (₹ in 1000's) : 10 8 6 5 9 7 11

- (a) Find the regression equation of income on years of service.
 (b) What initial start would you recommend for a person applying for the job after having served in a similar capacity in another company for 13 years?
 (c) Do you think other factors are to be considered (in addition to the years of service) in fixing the income with

reference to the above problems? Explain.

10.21 The following table gives the age of cars of a certain make and their annual maintenance costs. Obtain the regression equation for costs related to age.

Age of cars (in years)	:	2	4	6	8
Maintenance costs (₹ in 100's)	:	10	20	25	30

10.22 An analyst in a certain company was studying the relationship between travel expenses in rupees (y) for 102 sales trips and the duration in days (x) of these trips. He has found that the relationship between y and x is linear. A summary of the data is given below:

$$\Sigma x = 510; \Sigma y = 7140; \Sigma x^2 = 4150; \Sigma xy = 54,900 \text{ and } \Sigma y^2 = 7,40,200$$

- (a) Estimate the two regression equations from the above data.
 (b) A given trip takes seven days. How much money should a salesman be allowed so that he will not run short of money?

10.23 The quantity of a raw material purchased by ABC Ltd. at specified prices during the post 12 months is given below:

Month	Price per kg (in ₹)	Quantity (in kg)	Month	Price per kg (in ₹)	Quantity (in kg)
Jan	96	250	July	112	220
Feb	110	200	Aug	112	220
March	100	250	Sept	108	200
April	90	280	Oct	116	210
May	86	300	Nov	86	300
June	92	300	Dec	92	250

- (a) Find the regression equations based on the above data.
 (b) Can you estimate the approximate quantity likely to be purchased if the price shoots up to ₹ 124 per kg?
 (c) Hence or otherwise obtain the coefficient of correlation between the price prevailing and the quantity demanded.

Hints and Answers

10.15 $\bar{x} = \Sigma x/n = 21/8 = 2.625$;

$\bar{y} = \Sigma y/n = 4/8 = 0.50$

$$b_{yx} = \frac{n \Sigma d_x d_y - (\Sigma d_x)(\Sigma d_y)}{n \Sigma d_x^2 - (\Sigma d_x)^2}$$

$$= \frac{8 \times 30 - (-3)(-12)}{8 \times 45 - (-1)^2} = 0.568;$$

$d_x = x - 3$; $d_y = y - 3$.

Regression equation of y on x :

$$y - \bar{y} = b_{yx}(x - \bar{x})$$

$$y - 0.5 = 0.568(x - 2.625)$$

$$y = -0.991 + 0.568x$$

(b) $b_{xy} = \frac{n \Sigma d_x d_y - (\Sigma d_x)(\Sigma d_y)}{n \Sigma d_y^2 - (\Sigma d_y)^2}$

$$= \frac{8 \times 30 - (-3)(-12)}{8 \times 84 - (-12)^2} = 0.386$$

Regression equation of x on y :

$$x - \bar{x} = b_{xy}(y - \bar{y})$$

$$x - 2.625 = 0.386(y - 0.5)$$

$$x = 2.432 + 0.386y$$

10.16 Let x = rainfall y = production by y .
The expected yield corresponding to a rainfall of 40 inches is given by regression equation of y on x .

Given $\bar{y} = 50$, $\sigma_y = 10$, $\bar{x} = 30$, $\sigma_x = 5$, $r = 0.8$

$$y - \bar{y} = r \frac{\sigma_y}{\sigma_x}(x - \bar{x});$$

$$y - 50 = 0.8 \frac{10}{5}(x - 30)$$

$$y = 2 + 1.6x$$

For $x = 40$, $y = 2 + 1.6(40) = 66$ quintals.

10.17 Let x = age of wife y = age of husband.

Given $\bar{x} = 25$, $\bar{y} = 22$, $\sigma_x = 4$, $\sigma_y = 5$, $r = 0.8$

(a) Regression equation of x on y :

$$x - \bar{x} = r \frac{\sigma_x}{\sigma_y}(y - \bar{y})$$

$$x - 25 = 0.8 \frac{4}{5}(y - 22)$$

$$x = 10.92 + 0.64y$$

When age of wife is $y = 16$; $x = 10.92 + 0.64(16) = 21$ approx. (husband's age)

(b) Left as an exercise

10.18 (a) Regression equation of x on y :

$$x - \bar{x} = b_{xy}(y - \bar{y}) = r \frac{\sigma_x}{\sigma_y}(y - \bar{y})$$

$$x - 20 = 0.8 \frac{5}{25}(y - 120)$$

$$x = 0.8 + 0.16y$$

Regression equation of y on x :

$$y - \bar{y} = r \frac{\sigma_y}{\sigma_x}(x - \bar{x})$$

$$y - 120 = 0.8 \frac{25}{5}(x - 20)$$

$$y = 40 + 4x$$

(b) When advertisement expenditure is of ₹ 25 crore, likely sales is

$$y = 40 + 4x = 40 + 4(25) = 140 \text{ crore.}$$

(c) For $y = 150$, $x = 0.8 + 0.16y$

$$= 0.8 + 0.16(150) = 24.8$$

10.19 Let x = marks in Statistics and y = marks in Accountancy.

Given, $3y - 5x + 180 = 0$

or $x = (3/5)y + (180/5)$

For $y = 44$, $x = (3/5) \times 44 + (180/5) = 62.4$

Regression coefficient of x on y , $b_{xy} = 3/5$

Coefficient of regression

$$b_{xy} = r \frac{\sigma_x}{\sigma_y} = \frac{\sqrt{9}}{\sqrt{16}} \text{ (given)}$$

or $\frac{3}{5} = r \frac{\sqrt{9}}{\sqrt{16}}$ or $\frac{3}{5} = \frac{3r}{4}$

Hence $3r = 2.4$ or $r = 0.8$

10.20 Let x = years of service and y = income.

(a) Regression equation of y on x :

$$b_{yx} = \frac{n \Sigma xy - (\Sigma x)(\Sigma y)}{n \Sigma x^2 - (\Sigma x)^2}$$

$$= \frac{7 \times 469 - 56 \times 56}{7 \times 476 - (56)^2} = 0.75$$

$$y - \bar{y} = b_{yx} (x - \bar{x})$$

$$y - 8 = 0.75 (x - 8)$$

$$y = 2 + 0.75x$$

- (b) When $x = 13$ years, the average income would be

$$y = 2 + 0.75x = 2 + 0.75(13)$$

$$= ₹ 11,750$$

- 10.21** Let x = age of cars and

y = maintainance costs.

The regression equation of y on x :

$$\bar{x} = \Sigma x/n = 20/4 = 5;$$

$$\bar{y} = \Sigma y/n = 85/4 = 21.25$$

$$\begin{aligned} \text{and } b_{yx} &= \frac{n \Sigma xy - \Sigma x \Sigma y}{n \Sigma x^2 - (\Sigma x)^2} \\ &= \frac{7 \times 490 - 20 \times 85}{7 \times 120 - (20)^2} = 3.93 \end{aligned}$$

$$y - \bar{y} = b_{yx} (x - \bar{x})$$

$$y - 21.25 = 3.93 (x - 5)$$

$$y = 1.6 + 3.93x$$

- 10.22** $\bar{x} = \Sigma x/n = 510/102 = 5;$

$$\bar{y} = \Sigma y/n = 7140/102 = 70$$

Regression coefficients:

$$\begin{aligned} b_{xy} &= \frac{n(\Sigma xy) - (\Sigma x)(\Sigma y)}{n \Sigma y^2 - (\Sigma y)^2} \\ &= \frac{102 \times 54900 - 510 \times 7140}{102 \times 740200 - (7140)^2} = 0.08 \end{aligned}$$

$$b_{yx} = \frac{n(\Sigma xy) - (\Sigma x)(\Sigma y)}{n \Sigma x^2 - (\Sigma x)^2}$$

$$= \frac{102 \times 54900 - 510 \times 7140}{102 \times 4150 - (510)^2} = 12$$

Regression lines:

$$x - \bar{x} = b_{xy} (y - \bar{y})$$

$$x - 5 = 0.08 (y - 70) \text{ or } x = 0.08y - 0.6$$

$$\text{and } y - \bar{y} = b_{yx} (x - \bar{x})$$

$$y - 70 = 12 (x - 5) \text{ or } y = 12x + 10$$

$$\text{When } x = 7, y = 12 \times 7 + 10 = 94$$

- 10.23** Let price be denoted by x and quantity by y

$$\bar{x} = \Sigma x/n = 1200/12 = 100 ;$$

$$\bar{y} = \Sigma y/n = 2980/12 = 248.33$$

- (a) Regression coefficients:

$$b_{xy} = \frac{n \Sigma d_x d_y - (\Sigma d_x)(\Sigma d_y)}{n \Sigma d_y^2 - (\Sigma d_y)^2} = -0.26$$

$$b_{yx} = \frac{n \Sigma d_x d_y - (\Sigma d_x)(\Sigma d_y)}{n \Sigma d_x^2 - (\Sigma d_x)^2} = -3.244$$

Regression lines:

$$x - \bar{x} = b_{xy} (y - \bar{y})$$

$$x - 100 = -0.26 (y - 248.33)$$

$$\text{or } x = -0.26y + 164.56$$

$$\text{and } y - \bar{y} = b_{yx} (x - \bar{x})$$

$$y - 248.33 = -3.244 (x - 100)$$

$$y = -3.244x + 572.73$$

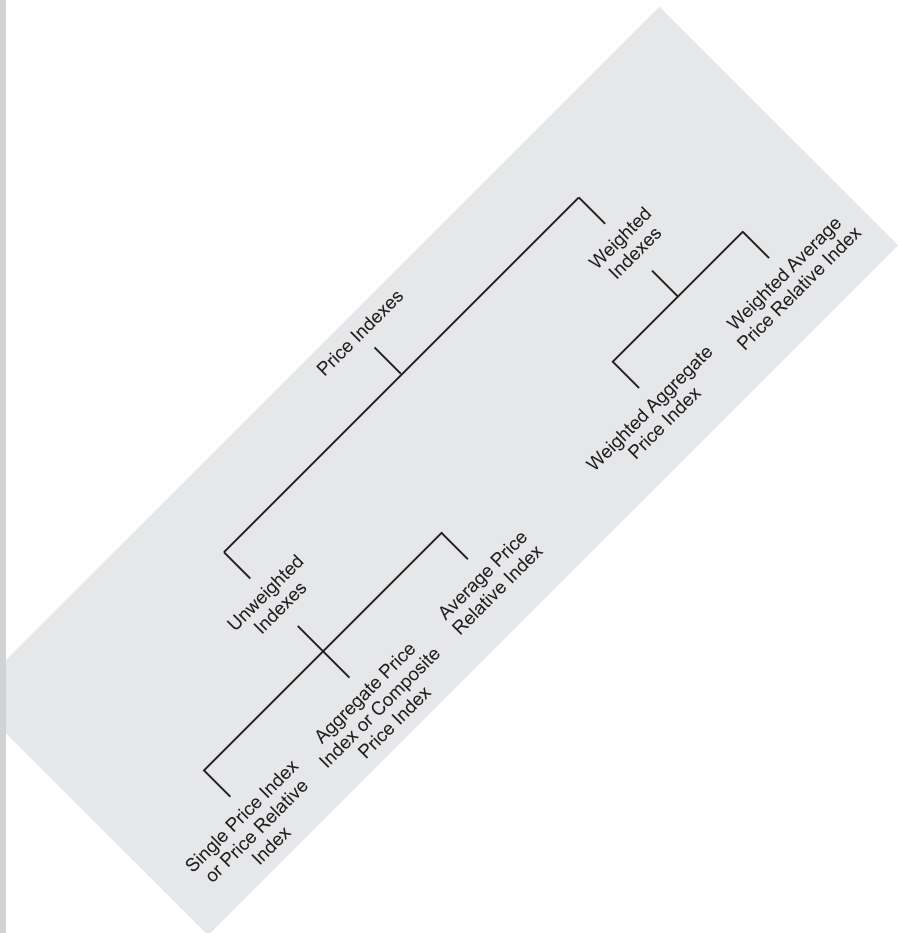
- (b) For $x = 124,$

$$\begin{aligned} y &= -3.244 \times 124 + 572.73 \\ &= 170.474 \end{aligned}$$

UNIT IV

INDEX NUMBERS

Chapter 11: Index Numbers



This page is intentionally left blank.

11

Index Numbers

LEARNING OBJECTIVES

After studying this chapter, you should be able to

- explain the purpose of index numbers.
- compute indexes to measure price changes and quantity changes over time.
- revise the base period of a series of index numbers.
- explain and derive link relatives.
- discuss the limitations of index number construction.

11.1 INTRODUCTION

We know that most values change and therefore may want to know how much change has taken place over a period of time. For example, we may want to know how much the prices of different items essential to a household have increased or decreased so that necessary adjustments can be made in the monthly budget. An organization may be concerned with the way in which prices paid for raw materials, annual income and profit, commodity prices, share prices, production volume, advertising budget and wage bills, have changed over a period of time. However, while prices of a few items may have increased, others may have decreased over a given period of time. Consequently in all such situations, an average measure needs to be defined to compare such differences from one time period to another. *Index numbers* are yardsticks for describing such difference.

An *index number* can be defined as a relative measure describing the average changes in any quantity over time. In other words, an index number measures the changing value of prices, quantities, or values over a period of time in relation to its value at some fixed point in time, called the *base period*. This resulting ratio of the current value to a base value is multiplied by 100 to express the index as a percentage. Since an index number is constructed as a ratio of a measure taken during one time period to that same measure taken during another time period (called base period), it has no unit and is always expressed as a percentage term as follows:

$$\text{Index number} = \frac{\text{Current period value}}{\text{Base period value}} \times 100$$

Indexes may be based at any convenient period, which is occasionally adjusted, and these are published at any convenient frequency. Examples of some indexes are:

Daily	Stock market prices
Monthly	Unemployment figures
Yearly	Gross National Product (GNP)

Index numbers were originally developed by economists for monitoring and comparing different groups of goods. For decision making in business, it is sometimes essential to understand and manipulate different published index series and to construct one's own index series.

11.2 INDEX NUMBER

Definition of index numbers can be classified into the following three broad categories:

1. A measure of change

- It is a numerical value characterizing the change in complex economic phenomena over a period of time or space. —Maslow
- An index number is a quantity which, by reference to a base period, shows by its variations the changes in the magnitude over a period of time. In general, index numbers are used to measure changes over time in magnitudes which are not capable of direct measurement. —John I. Raffin
- An index number is a statistical measure designed to show changes in variables or a group of related variables with respect to time, geographic location or other characteristics. —Speigel
- Index number is a single ratio (usually in percentages) which measures the combined (i.e., averaged change of several variables between two different times, places or situations. —A. M. Tuttle

2. A device to measure change

- Index numbers are devices measuring differences in the magnitude of a group of related variables. —Corxton and Cowden
- An index number is a device which shows by its variation the changes in a magnitude which is not capable of accurate measurement in itself or of direct valuation in practice. —Wheldom

3. A series representing the process of change

- Index numbers are series of numbers by which changes in the magnitude of a phenomenon are measured from time to time or place to place. —Horace Secris
- A series of index numbers reflects in its trend and fluctuations the movements of some quantity of which it is related. —B. L. Bowley
- An index number is a statistical measure of fluctuations in a variable arranged in the form of a series, and using a base period for making comparisons. —L. J. Kaplan

11.3 TYPES OF INDEX NUMBERS

Index numbers are broadly classified into three categories: (i) price indexes, (ii) quantity indexes, and (iii) value indexes. A brief description of each of these is as follows:

Price Indexes These indexes are of two categories:

- Single price index
- Composite prices index

The single price index measures the percentage change in the current price per unit of a product to its base period price. To facilitate comparisons with other years, the actual per unit price is converted into a price relative, which expresses the unit price in each period as a percentage of unit price in a base period. Price relatives are very helpful to understand and interpret changing economic and business conditions over time. Table 11.1 illustrates the calculations of price relatives.

Table 11.1 Calculation of Price Index (Base year = 1996)

Year	Total Wage Bill (₹ millions)	Ratio	Price Index or Percentage Relative
(1)	(2)	(3) = (2)/11.76	(4) = (3) × 100
2000	11.76	11.76/11.76 = 1.0	100.0
2001	12.23	12.23/11.76 = 1.039	103.9
2002	12.84	12.84/11.76 = 1.091	109.1
2003	13.35	13.35/11.76 = 1.135	113.5
2004	13.82	13.82/11.76 = 1.175	117.5

From Table 11.1, it is observed that the price relative of 113.5 in 2003 shows an increase of 13.5 per cent in wage bill compared to the base year 2000.

A composite price index measures the average price change for a basket of related items from a base period to the current period. For example, the wholesale price index reflects the general price level for a group of items (or a basket of items) taken as a whole.

The retail price index reflects the general changes in the retail prices of various items including food, housing and clothing. In India, the Bureau of Labour statistics publishes retail price index. The consumer price index, a special type of retail price index, is the primary measure of the cost of living in a country. The consumer price index is a weighted average price index with fixed weights. The weightage applied to each item in the basket of items is derived from the urban and rural families.

Quantity Index A quantity index measures the relative changes in quantity levels of a group (or basket) of items consumed or produced, such as agricultural and industrial production, imports and exports, between two time periods. The method of constructing quantity indexes is the same as that of price index except that the quantities are vary from period to period.

The two most common quantity indexes are the weighted relative of aggregates and the weighted average of quantity relative index.

Value Index A value index measures the relative changes in total monetary worth of an item, such as inventories, sales, or foreign trade, between the current and base periods. The value of an item is determined by multiplying its unit price by the quantity under consideration. The value index can also be used to measure differences in a given variable in different locations. For example, the comparative cost of living shows that in terms of cost of goods and services, it is cheaper to live in a small city than in metro cities.

Special Purpose Indexes A few index numbers such as industrial production, agricultural production and productivity can also be constructed separately depending on the nature and degree of relationship between groups and items.

- Index number, almost alone in the domain of social sciences, may truly be called an exact science, if it be permissible to designate as science, the theoretical foundations of a useful art.

—Irving Fisher.

11.4 CHARACTERISTICS AND USES OF INDEX NUMBERS

Based on the definitions and types of index numbers discussed earlier in this chapter, the following characteristics and uses of index number emerge.

11.4.1 Characteristics of Index Numbers

1. **Index numbers are specialized averages:** According to R. L. Corner—‘*An index number represents a special case of an average, generally weighted average, compiled from a sample of items judged to be representative of the whole*’.

‘Average’ is a single figure representing the characteristic of a data set. This figure can be used as a basis for comparing two or more data sets provided the unit of measurement of observations in all sets is the same. However, index numbers which are considered as a special case of average can be used for comparison of two or more data sets expressed in different units of measurement.

The consumer price index, for example, which represents a price comparison for a group of items—food, clothing, fuel and house rent, are expressed in different units. An average of prices of all these items expressed in different units is obtained by using the technique of price index number calculation.

2. **Index numbers measure the change in the level of phenomena in percentages:** Since index numbers are considered as a special case of an average, these are used to represent, in one single figure, the increase or decrease (expressed in terms of percentage) in the value of a variable. For example, a quantity index number of 110 for cars sold in a given year when compared with that of a base year would mean that cars sales in the given year were 10 per cent higher than in the base year (value of index number in base period is always equal to 100). Similarly, a quantity index number of 90 in a given year would indicate that the number of cars sold in the given period were 10 per cent less than in the base year.
3. **Index numbers measure changes in a variety of phenomena which cannot be measured directly:** According to Bowley—‘*Index numbers are used to measure the changes in some quantity which we cannot observe directly. . .*’

It is not possible, for example, to directly measure the changes in the import-export activities of a country. However, it is possible to study relative changes in import and export activities by studying the variations in factors such as raw materials available, technology, competitors, quality and other parameters which affect import and export, and are capable of direct measurement. Similarly, cost of living cannot be measured in quantitative terms directly, we can only study relative changes in it by studying the variations in certain other factors connected to it.

4. **Index numbers measure the effect of changes in relation to time or place:** Index numbers are used to compare changes which take place over periods of time, between locations, and in categories. For example, cost of living may be different at two different places at the same or cost of living in one city can be compared across two periods of time.

11.4.2 Uses of Index Numbers

According to G. Simpson and F. Kafka — ‘*Today Index numbers are one of the most widely used statistical tools. They are used to feel the pulse of the economy and they have come to be used as indicators of inflationary or deflationary tendencies*’. Other important uses of index number can be summarized as follows:

1. **Index numbers act as economic barometers:** A barometer is an instrument that is used to measure atmospheric pressure. Index numbers are used to feel the pressure of the economic and business behaviour, as well as to measure ups and downs in the general economic condition of a country. For example, the composite index number of indexes of prices, industrial output, foreign exchange reserves, and bank deposits, could act as an economic barometer.
2. **Index numbers help in policy formulation:** Many aspects of economic activity are related to price movements. The price indexes can be used as indicators of change in various segments of the economy. For example, by examining the price indexes of different segments of a firm's operations, the management can assess the impact of price changes and accordingly take some remedial and/or preventive actions.

In the same way, by examining the population index, the government can assess the need to formulate a policy for health, education and other utilities.

3. **Index numbers reveal trends and tendencies:** An index number is defined as a relative measure describing the average change in the level of a phenomenon between the current period and a base period. This property of the index number can be used to reflect typical patterns of change in the level of a phenomenon. For example, by examining the index number of industrial production, agricultural production, imports, exports, and wholesale and retail prices for the last 8–10 years, we can draw the trend of the phenomenon under study and also draw conclusions as to how much change has taken place due to the various factors.
4. **Index numbers help to measure purchasing power:** In general, the purchasing power is not associated with a particular individual; rather it is related to an entire class or group. Furthermore, it is not associated with the cost of a single item, because individuals purchase many different items in order to live. Consequently, earnings of a group of people or class must be adjusted with a price index that provides an overall view of the purchasing power for the group.

For example, suppose a person earns ₹ 1000 per month in 1990. If an item costs ₹ 100 in that year, the person could purchase $1000 \div 100 = 10$ units of the item with one month's earnings. But if in year 2000, the same person earns ₹ 2000 per month but the item cost is ₹ 250, then he could purchase $2000 \div 250 = 8$ units of the item. Hence, the effect of monthly earning relative to the particular item is less in year 2000 than in 1990 as a lesser number of units of the items can be purchased with current earnings. By dividing the item price in both the years, we can eliminate the effect of price and determine the real purchasing power for that item. For instance, in 1990, the purchasing power was $10 \div 1000 = 0.10$ or 10 paise which it was ₹ 0.125 or 12.5 paise in 2000.

5. **Index numbers help in deflating various values:** When real rupee value is computed, the base period is earlier than the given years for which this value is being determined. Thus, the adjustment of current rupee value to real terms is referred to as *deflating a value series* because prices typically increase over time.

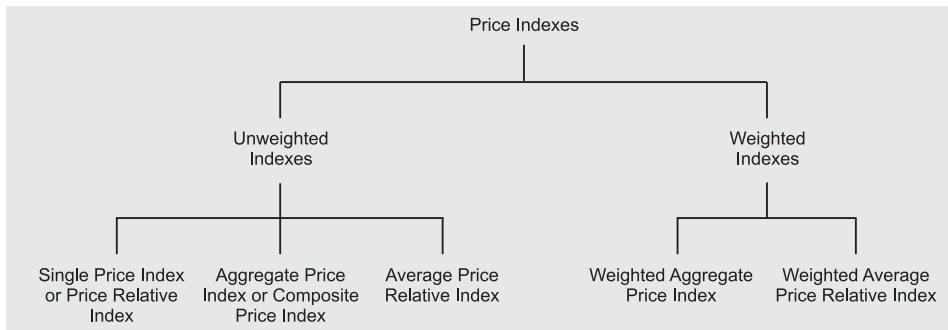
The price index number is helpful in deflating the national income to remove the effect of inflation over a long term, so that we may understand whether there is any change in the real income of the people or not. The retail price index is often used to compute real changes in earnings and expenditure as it compares the purchasing power of money at different points in time. It is generally accepted as a standard measure of inflation even though calculated from a restricted basket of goods.

Conceptual Questions 11A

1. Explain the significance of index numbers.
2. Explain the differences among the three principal types of indexes: price, quantity, and value.
3. How are index numbers constructed? What is their purpose?
4. What is an index number? Describe briefly its applications in business and industry.
5. What does an index number measure? Explain the nature and uses of index numbers.
6. Index numbers are economic barometers. Explain this statement and mention the limitations of index numbers (if any).
7. What are the basic characteristics of an index number?
8. Since value of the base year is always 100, it does not make any difference which period is selected as the base on which to construct an index. Comment.
9. What are the main uses of an index number?
10. What is meant by the term deflating a value series?

11.5 METHODS FOR CONSTRUCTION OF PRICE INDEXES

Various types of price indexes and their methods of construction can be classified into broad categories as shown in the chart below:



11.6 UNWEIGHTED PRICE INDEXES

The unweighted price indexes are further classified into three groups as shown above in the chart. The method of calculating each of these is discussed below.

11.6.1 Single Price Index

A single unweighted price index number measures the percentage change in price for a single item or a basket of items between any two time periods. Unweighted implies that all the values considered in calculating the index are of equal importance.

An unweighted single price index is calculated by dividing the price of an item in the given period by the price of the same item in the base period. To facilitate comparison with other years, the actual price of the item can be converted into a *price relative*, which expresses the unit price in each year (period) as a percentage of the unit price in a base year.

The general formula for calculating the single price index or price relative index is

$$\text{Single price index in period } n = \frac{p_n}{p_0} \times 100$$

where p_n = price per unit of an item in the n th year

p_0 = price per unit of an item in the base year

Example 11.1: The retail price of a typical commodity over a period of four years is given below:

Year	:	2000	2001	2002	2003
Price (₹)	:	24.60	25.35	26.00	26.50

- Find the price index based on 2000 prices.
- Find the percentage change in price between consecutive years (base year = 2000) and the percentage increase between consecutive years.

Solution: (a) For the prices of the commodity with base year 2000, the price relatives for one unit of the commodity in the years 2000 to 2003 are given in Table 11.2.

Table 11.2 Price Relatives

Year	Price (₹)	Price Relatives	Percentage Change
2000	24.60	100	—
2001	25.35	$\frac{25.35}{24.60} \times 100 = 103.04$	3.04
2002	26.00	$\frac{26}{24.60} \times 100 = 105.69$	2.65
2003	26.50	$\frac{26.50}{24.60} \times 100 = 107.72$	2.03

- The percentage change in price relative is divided by the index, it has come from and multiplied by 100 for finding percentage increase.

$$\text{For year 2001: } \frac{103.04 - 100}{100} \times 100 = 3.04 \text{ per cent}$$

$$\text{For year 2002: } \frac{105.69 - 103.04}{103.04} \times 100 = 2.57 \text{ per cent}$$

$$\text{For year 2003: } \frac{107.72 - 105.69}{105.69} \times 100 = 1.92 \text{ per cent}$$

11.6.2 Aggregate Price Index

An **aggregate index price** or *composite price index* measures the average price change for a basket of related items from the base period to the current period. For example, to measure the change in the cost of living over a period of time, we need the index that measures the change based on the price changes for a variety of commodities including food, housing, clothing, transportation, health care and so on. Since the number of commodities is large, therefore a sample of commodities should be selected for calculating the aggregate price index.

Irrespective of the units of measurement in which prices of several commodities are quoted, the steps of the method to calculate an aggregate price index are summarized as follows:

- (i) Add the unit prices of a group of commodities in the year of interest.
- (ii) Add the unit prices of a group of commodities in the base year.
- (iii) Divide the sum obtained in step (i) by the sum obtained in step (ii), and multiply the quotient by 100.

From the sample of commodities or items included in the calculation of index, we cannot expect a true reflection of price changes for all commodities. This calculation provides us with only a rough estimate of price change.

A formula of calculating an unweighted aggregate price index is defined as

$$\text{Aggregate price index } P_{01} = \frac{\sum p_1}{\sum p_0} \times 100 \quad (12-1)$$

where p_1 = unit price of a commodity in the current period of interest

p_0 = unit price for a commodity in the base period

Example 11.2: The following are two sets of retail prices of a typical family's shopping basket. The data pertain to retail prices during 2001 and 2002.

Commodity	Unit Price (₹)	
	2001	2002
Milk (1 litre)	18	20
Eggs (1 dozen)	15	18
Butter (1 kg)	120	150
Bread (500 gm)	9	11

Calculate the simple aggregate price index for 2002 using 2000 as the base year.

Solution: Calculations for aggregate price index are shown in Table 11.3.

Table 11.3 Calculation of Aggregate Price Index

Commodity	Unit Price (₹)	
	2000 (p_0)	2002 (p_1)
Milk (1 litre)	18	20
Eggs (1 dozen)	15	18
Butter (1 kg)	120	150
Bread (500 gm)	9	11
Total	162	199

The unweighted aggregate price index for expenses on a few food items in 2002 is given by

$$P_{01} = \frac{\sum p_1}{\sum p_0} \times 100 = \frac{199}{162} \times 100 = 122.84$$

The value $P_{01} = 122.84$ implies that the price of food items included in the price index has increased by 22.83 per cent over the period 2000 to 2002.

Limitations of an Unweighted Aggregate Price Index

1. The unweighted aggregate approach of calculating a composite price index is heavily influenced by the items with large per unit price. Consequently items with relatively low unit price are dominated by the high unit price items.

2. Equal weights are assigned to every commodity included in the index irrespective of the relative importance of the commodity in terms of the amount purchased by a typical consumer. In other words, it did not attach more weight or importance to the price change of a high-use commodity than it did to a low-use commodity. For example, a family may purchase 30 packets of 500 gm bread in a month while it is unusual to buy 30 kg butter every month. A substantial price change for slow-moving items like butter, ghee can distort an index.

Due to these limitations, the unweighted index is not widely used in statistical analyses. These limitations suggest the use of weighted index. There are two methods to calculate weighted index, and these will be discussed later in the chapter.

11.6.3 Average Price Relative Index

This index is an improvement over the aggregate price index because it is not affected by the unit in which prices are quoted. However, it also suffers from the problem of equal importance (weight) given to all the items or commodities included in the index.

Steps of the method to calculate **average price relative** index are summarized as follows:

- (i) Select a base year, and then divide the price of each commodity in the current year by the price in the base year, to obtain price relatives.
- (ii) Divide the sum of the price relatives of all commodities by the number of commodities used in the calculation of the index.
- (iii) Multiply the average value obtained in step (ii) by 100 to express it in percentage.

The formula for computing the index is as follows:

$$\text{Average price relative index } P_{01} = \frac{1}{n} \sum \left(\frac{p_1}{p_0} \right) 100 \quad (11-2)$$

where n = number of commodities included in the calculation of the index.

The average used in computing the index of price relatives could be arithmetic mean or geometric mean. When geometric mean is used for averaging the price relatives, the formula (11-2) becomes

$$\log P_{01} = \frac{1}{n} \sum \log \left\{ \left(\frac{p_1}{p_0} \right) 100 \right\} = \frac{1}{n} \sum \log P \text{ where } P = \left(\frac{p_1}{p_0} \right) 100$$

Then
$$P_{01} = \text{antilog} \left\{ \frac{1}{n} \sum \log P \right\}$$

Advantages

This index has the following advantages over the aggregate price index:

- (i) The value of this index is not affected by the units in which prices of commodities are quoted. The price relatives are pure numbers and therefore are independent of the original units in which they are quoted.
- (ii) Equal importance is given to each commodity and extreme commodities do not influence the index number.

Limitations

Despite the few advantages mentioned above, this index is not popular on account of the following limitations:

- (i) Since it is an unweighted index, therefore each price relative is given equal importance. However in actual practice a few price relatives are more important than others.
- (ii) Although arithmetic mean is often used to calculate the average of price relatives, it also has a few biases. The use of geometric mean is computationally difficult. Other

measures of central tendency such as median, mode and harmonic mean, are almost never used for calculating this index.

- (iii) Index of price relatives does not satisfy all criteria such as identity, time reversal, and circular properties, laid down for an ideal index. These criteria will be discussed later in the chapter.

Example 11.3: From the data given below, construct the index of price relatives for the year 2002 taking 2001 as base year using (a) arithmetic mean and (b) geometric mean.

<i>Expenses on</i>	:	<i>Food</i>	<i>Rent</i>	<i>Clothing</i>	<i>Education</i>	<i>Misc.</i>
Price (₹), 2001	:	1800	1000	700	400	700
Price (₹), 2002	:	2000	1200	900	500	1000

Solution: (a) Calculations of Index number using arithmetic mean (A.M.) is shown in Table 11.4.

Table 11.4 Calculation of Index Using A.M.

<i>Expenses on</i>	<i>Price in</i> <i>2001(p_0)</i>	<i>Price in</i> <i>2000(p_1)</i>	<i>Price Relatives</i> $\frac{p_1}{p_0} \times 100$
Food	1800	2000	111.11
Rent	1000	1200	120.00
Clothing	700	900	128.57
Education	400	500	125.00
Miscellaneous	700	1000	142.86
			627.54

$$\text{Average of price relative index } P_{01} = \frac{1}{n} \sum \left(\frac{p_1}{p_0} \right) 100 = \frac{1}{5} (627.54) = 125.508$$

Hence, we conclude that prices of items included in the calculation of index have increased by 25.508 per cent in 2002 as compared to the base year 2001.

(b) Index number using geometric mean (G.M.) is shown in Table 11.5.

$$\begin{aligned} \text{Average price relative index } P_{01} &= \text{antilog} \left\{ \frac{1}{n} \sum \log P \right\} = \text{antilog} \left\{ \frac{1}{5} (10.4856) \right\} \\ &= \text{antilog} (2.0971) = 125.00 \end{aligned}$$

Table 11.5: Calculations of Index Using G.M.

<i>Expenses on</i>	<i>Price in</i> <i>2001(p_0)</i>	<i>Price in</i> <i>2002(p_2)</i>	<i>Price Relatives</i> $P = \frac{p_1}{p_0} \times 100$	<i>log P</i>
Food	1800	2000	111.11	2.0457
Rent	1000	1200	120.00	2.0792
Clothing	700	900	128.57	2.1090
Education	400	500	125.00	2.0969
Miscellaneous	700	1000	142.86	2.1548
				10.4856

Self-practice Problems 11A

- 11.1** The following data concern monthly salaries for the different classes of employees within a small factory over a 3-year period.

Employee Class	Salary per Month		
	1998	1999	2000
A	2300	2500	2600
B	1900	2000	2300
C	1700	1700	1800
D	1000	1100	1300

Using 1998 as the base year, calculate the simple aggregate price index for the years 1999 and 2000.

- 11.2** The following data describe the average salaries (₹ in '000) for the employees in a company over ten consecutive years.

Year	:	1	2	3	4	5
Average salary :		10.9	11.4	12.0	12.7	13.6
Year	:	6	7	8	9	10
Average salary :		14.4	15.0	15.5	16.3	13.6

- Calculate an index for these average salaries using year 5 as the base year.
- Calculate the percentage points change between consecutive years.

- 11.3** A State Govt. had compiled the information shown below regarding the price of the three essential commodities: wheat, rice, and sugar. From the commodities listed, the corresponding price indicates the average price for that year. Using 1998 as the base year, express the price for the years 2000 to 2002, in terms of unweighted aggregate index.

Commodity	1998	1999	2000	2001	2002
Wheat	4	6	8	10	12
Rice	16	20	24	30	36
Sugar	8	10	16	20	24

- 11.4** Following are the prices of commodities in 2003 and 2004. Calculate a price index based on price relatives, using the geometric mean.

Year	Commodity					
	A	B	C	D	E	F
2003	45	60	20	50	85	120
2004	55	70	30	75	90	130

- 11.5** A textile worker in the city of Mumbai earns ₹ 3500 per month. The cost of living index for a particular month is given as 136. Using the following information, find out the amount of money he spent on house rent and clothing.

Group	Expenditure (₹)	Group Index
Food	1400	180
Clothing	x	150
House rent	y	100
Food and lighting	560	110
Misc.	630	80

[Delhi Univ., B.Com., 1997]

- 11.6** In 1996, for working class people, wheat was selling at an average price of ₹ 160 per 10 kg, cloth at ₹ 40 per metre, house rent ₹ 10,000 per house, and other items at ₹ 100 per unit. By 1997 the cost of wheat rose by ₹ 40 per 10 kg, house rent by ₹ 1500 per house, and other items doubled in price. The working class cost of living index for the year 1997 (with 1996 as base) was 160. By how much did the cloth price rise during the period 1996–97?

- 11.7** From the following data, calculate an index number using family budget method for the year 1996 with 1995 as the base year.

Commodity	Quantity (in units)		Price (in ₹) per unit	
	in 1995		1995	1996
A	110		8.00	12.00
B	25		6.00	7.50
C	10		5.00	5.25
D	20		48.00	60.00
E	25		15.00	16.50
F	30		9.00	27.00

[Karnataka Univ., B.Com., 1997]

- 11.8** The following table gives the annual income of a teacher and the general index of price during 1990–97. Prepare the index number to show the change in the real income of the teacher and comment on price increase:

<i>Year</i>	<i>Income</i>	<i>Index</i>
1990	4000	100
1991	4400	130
1992	4800	160
1993	5200	220
1994	5600	270
1995	6000	330
1996	6400	400
1997	6800	490

[HP Univ., B.Com., 1997]

Hints and Answers

- 11.1** Simple aggregate price index

$$P_{0,89} = \frac{7300}{6900} \times 100$$

$$= 105.8 \text{ for the year 1999}$$

$$P_{0,90} = \frac{8000}{6900} \times 100$$

$$= 115.9 \text{ for the year 2000}$$

- 11.2** (a)

Year	:	1	2	3	4	5
Index number	:	80.1	83.8	88.2	93.4	100
Year	:	6	7	8	9	10
Index number	:	105.9	110.3	114.0	119.9	129.4

For example, index for

$$\text{Year 1: } 10.9 \div 13.6)100 = 80.1$$

$$\text{Year 2: } (11.4 \div 13.6)100 = 83.8$$

(b)	<i>Year</i>	<i>Index Number</i>	<i>Percentage Point Change</i>
	1	80.1	—
	2	83.8	3.7
	3	88.2	4.4
	4	93.4	5.2
	5	100.0	6.6
	6	105.9	5.9
	7	110.3	4.4
	8	114.0	3.7
	9	119.9	5.9
	10	129.4	9.5

- 11.3** Aggregate price

1998	1999	2000	2001	2002
100	133.33	137.78	125	120

- 11.4**

<i>Commodity</i>	$P = \frac{P_1}{P_0} \times 100$	<i>Log P</i>
A	122.22	2.0872
B	116.67	2.0669
C	150.00	2.1761
D	150.00	2.1761
E	105.88	2.0248
F	108.33	2.0348

$$P_{01} = \text{antilog} \left\{ \frac{1}{n} \log P \right\}$$

$$= \text{antilog} \left\{ \frac{1}{6} (12.5659) \right\}$$

$$= \text{antilog} (2.0948) = 124.4$$

- 11.5** Let expenditure on clothing be x and on house rent be y . Then as per conditions given, we have

$$3500 = 1400 + x + y + 560 + 630$$

$$\text{or } x + y = 910 \quad (\text{i})$$

Multiplying expenditure with group index and equating it to 136, we get

$$136 = \frac{(1400 \times 180) + (x \times 150) + (y \times 100) + (560 \times 110) + (630 \times 80)}{3500}$$

$$136 = \frac{2,52,000 + 150x + 100y + 61,600 + 50,400}{3500}$$

$$4,76,000 = 2,52,000 + 150x + 100y + 61,600 + 50,400$$

$$150x + 100y = 1,12,000 \quad (\text{ii})$$

Multiplying Eq. (i) by 150 and subtracting it from (ii), we get

$$50y = 24,500 \text{ or } y = ₹ 490 \text{ (house rent)}$$

Substituting the value of y in Eq. (i):

$$x + 490 = 910 \text{ or } x = ₹ 420 \text{ (clothing)}$$

11.6 Let the rise in price of cloth be x .

Commodity	Price	Index	Price 1997	Index
Wheat	160	100	200	$\frac{200}{160} \times 100$ = 125
Cloth	40	100	x	$\frac{x}{40} \times 100$ = $2.5x$
House rent	10,000	100	11,500	$\frac{11,500}{10,000} \times 100$ = 115
Miscellaneous	100	100	200	$\frac{200}{100} \times 100$ = 200
Total				$440 + 2.5x$

The index for 1997 as given is 160. Therefore, the sum of the index numbers of the four commodities would be $160 \times 4 = 640$. Thus, $440 + 2.5x = 640$ or $x = 80$. Hence the rise in the price of cloth was ₹ 40 ($80 - 40$) per metre.

11.7

Commodity	Quantity (Q)	p_0	p_1	$P = \frac{p_1}{p_0} \times 100$	PQ
A	110	8	12.00	150	16,500
B	25	6	7.50	125	3125
C	10	5	5.25	105	1050
D	20	48	60.00	125	2500
E	25	15	16.50	110	2750
F	30	9	27.00	300	9000
Total	210				34,925

$$\begin{aligned} \text{Index number} &= \frac{\Sigma PQ}{\Sigma Q} \\ &= \frac{34,925}{210} = 166.31 \end{aligned}$$

11.8

Year	Income (₹)	Index	Real Income (₹)	Real Income Index
1990	4000	100	$\frac{4000}{100} \times 100 = 4000.00$	100.00
1991	4400	130	$\frac{4400}{130} \times 100 = 3384.62$	84.62
1992	4800	160	$\frac{4800}{160} \times 100 = 3000.00$	75.00
1993	5200	220	$\frac{5200}{220} \times 100 = 2363.64$	59.09
1994	5600	270	$\frac{5600}{270} \times 100 = 2074.07$	51.85
1995	6000	330	$\frac{6000}{330} \times 100 = 1818.18$	45.45
1996	6400	400	$\frac{6400}{400} \times 100 = 1600.00$	40.00
1997	6800	490	$\frac{6800}{490} \times 100 = 1387.76$	34.69

11.7 WEIGHTED PRICE INDEXES

While constructing weighted price indexes, rational weights are assigned to all items or commodities in an explicit manner. Such weights indicate the relative importance of items or commodities included in the calculation of an index. The weights used are of two types, *quantity weights* and *value weights*. There are two price indexes that are commonly in use

1. Weighted aggregate price index
2. Weighted average of price relative index

11.7.1 Weighted Aggregate Price Index

In a weighted aggregate price index, each item in the basket of items chosen for calculation of the index is assigned a weight according to its importance. In most cases, the quantity of usage is the best measure of importance. Hence, we should obtain a measure of the quantity of usage for the various items in the group. This explicit weighting allows us to gather more information than just the change in price over a period of time as well as improve the accuracy of the general price level estimate.

Weight is assigned to each item in the basket in various ways and the weighted aggregates are also used in different ways to calculate an index. A few methods (or approaches) to determine weights (value) to be assigned to each item in the basket are as follows:

- Laspeyre's method
- Paasche's method
- Dorbish and Bowley's method
- Fisher's ideal method
- Marshall-Edgeworth's method
- Walsch's method
- Kelly's method

Laspeyre's Weighting Method

This method suggests to treat quantities as constant at *base period* level and are used for weighting price of each item or commodities both in base period and current period. Since this index number depends upon the same base price and quantity, therefore one can directly compare the index of one period with another. The formula for calculating *Laspeyre's price index* named after the statistician Laspeyre's is given by

$$\text{Laspeyre's price index} = \frac{\sum p_1 q_0}{\sum p_0 q_0} \times 100$$

where p_1 = prices in the current period

p_0 = prices in the base period

q_0 = quantities consumed in the base period

Advantages

The main advantage of this method is that it uses only one quantity measure based on the base period and therefore we need not keep record of quantity consumed in each period. Moreover, having used the same base period quantity, we can compare the index of one period with another directly.

Disadvantages

We know that the consumption of commodities decreases with relatively large increases in price and vice versa. Since in this index, the fixed quantity weights are determined from the base period usage, it does not adjust such changes in consumption and therefore tends to result in a bias in the value of the composite price index.

Example 11.4: Compute the cost of living index number using Laspeyre's method from the following information:

<i>Commodity</i>	<i>Unit Consumption in Base Period</i>	<i>Price in Base Period</i>	<i>Price in Current Period</i>
Wheat	200	1.0	1.2
Rice	50	3.0	3.5
Pulses	50	4.0	5.0
Ghee	20	20.0	30.0
Sugar	40	2.5	5.0
Oil	50	10.0	15.0
Fuel	60	2.0	2.5
Clothing	40	15.0	18.0

Solution: Calculation of cost of living index by Laspeyre's method is shown in Table 11.6.

Table 11.6 Laspeyre's Method

<i>Commodity</i>	<i>Base Period Quantity (q_0)</i>	<i>Base Period Price (p_0)</i>	<i>Current Price (p_1)</i>	p_1q_0	p_0q_0
Wheat	200	1.0	1.2	240	200
Rice	50	3.0	3.5	175	150
Pulses	50	4.0	5.0	250	200
Ghee	20	20.0	30.0	600	400
Sugar	40	2.5	5.0	200	100
Oil	50	10.0	15.0	750	500
Fuel	60	2.0	2.5	150	120
Clothing	40	15.0	18.0	720	600
Total	510			3085	2270

$$\text{Cost of living index} = \frac{\sum p_1 q_0}{\sum p_0 q_0} \times 100 = \frac{3085}{2270} \times 100 = 135.9$$

Paasche's Weighting Method

In the Paasche's method, the price of each item or commodity is weighted by the quantity in the current period instead of the base year as used in Laspeyre's method. Paasche's formula for calculating the index is given by

$$\text{Paasche price index} = \frac{\sum p_1 q_1}{\sum p_0 q_1}$$

where p_1 = prices in current year

p_0 = prices in base year

q_1 = quantities in current year

Advantages

The Paasche's method combines the effects of changes in price and quantity consumption patterns during the current year. It provides a better estimate of changes in the economy than Laspeyre's method. If the prices or quantities of all commodities or items change in the same ratio, then the values of the Laspeyre's and Paasche's indexes will be same.

Disadvantages

This method requires knowledge of the quantities consumed of all commodities in each period. Getting the data on the quantities for each period is either expensive or time-consuming. Moreover, each year the index number for the previous year requires recomputation to reflect the effect of the new quantity weights. Thus, it is difficult to compare indexes of different periods when calculated by the Paasche's method.

Example 11.5: For the following data, calculate the price index number of 1999 with 1998 as the base year using: (a) Laspeyre's method and (b) Paasche's method.

Commodity	1998		1999	
	Price	Quantity	Price	Quantity
A	20	8	40	6
B	50	10	60	5
C	40	15	50	15
D	20	20	20	25

Solution: Table 11.7 presents the information necessary for both Laspeyre's and Paasche's methods.

Table 11.7 Calculation of Laspeyre's and Paasche's and Paasche's Indexes

Commodity	Base Period, 1998		Current year, 1999		$p_1 q_0$	$p_0 q_0$	$p_1 q_1$	$p_0 q_1$
	Price (p_0)	Quantity (q_0)	Price (p_1)	Quantity (q_1)				
A	20	8	40	6	320	160	240	120
B	50	10	60	5	600	500	300	250
C	40	15	50	15	750	600	750	600
D	20	20	20	25	400	400	500	500
					2070	1660	1790	1470

$$\text{Laspeyre's price index} = \frac{\sum p_1 q_0}{\sum p_0 q_0} \times 100 = \frac{2070}{1660} \times 100 = 124.7$$

$$\text{Paasche's price index} = \frac{\sum p_1 q_1}{\sum p_0 q_1} \times 100 = \frac{1790}{1470} \times 100 = 121.77$$

The Paasche's price index shows a price level increase of 21.77 per cent, while Laspeyre's index shows a price level increase of 24.7 per cent. Hence, we may conclude that Paasche's index shows a trend towards less expensive commodities.

Dorbish and Bowley's Method

This method (or approach) is the simple *arithmetic mean* of the Laspeyre's and Paasche's indexes. This index takes into account the influence of quantity weights of both base period and current period. The formula for calculating the index using Dorbish and Bowley method is given by

$$\text{Dorbish and Bowley's price index} = \frac{1}{2} \left\{ \frac{\sum p_1 q_0}{\sum p_0 q_0} + \frac{\sum p_1 q_1}{\sum p_0 q_1} \right\} 100$$

Fisher's Ideal Method

This method (or approach) is the *geometric mean* of the Laspeyre's and Paasche's indexes and the formula is given by

$$\text{Fisher's ideal price index} = \sqrt{\frac{\sum p_1 q_0}{\sum p_0 q_0} \times \frac{\sum p_1 q_1}{\sum p_0 q_1}} \times 100$$

Advantages

Fisher's method is also called ideal method due to following reasons:

- (i) The formula is based on geometric mean which is considered to be the best average for constructing index numbers.
- (ii) The formula takes into account both base year and current year quantities as weights. Thus it avoids the bias associated with the Laspeyre's and Paasche's indexes.
- (iii) This method satisfies essential tests required for a index, that is, time reversal test and factor reversal test.

Disadvantages

The calculation of index using this method requires more computation time. Although the index number is theoretically better than others discussed previously, it is not fit for common use because it requires current quantity weights every time an index is calculated.

Example 11.6: Compute index number from the following data using Fisher's ideal index formula.

Commodity	1999		2000	
	Price	Quantity	Price	Quantity
A	12	10	15	12
B	15	7	20	5
C	24	5	20	9
D	5	16	5	14

Solution: Table 11.8 presents the information necessary for Fisher's method to calculate the index.

Table 11.8 Calculations of Fisher Ideal Index

Commodity	Base Year, 1999		Current Year, 2000		$p_1 q_0$	$p_0 q_0$	$p_1 q_1$	$p_0 q_1$
	(q_0)	(p_0)	(q_1)	(p_1)				
A	12	10	15	12	144	120	180	150
B	15	7	20	5	75	105	100	140
C	24	5	20	9	216	120	180	100
D	5	16	5	14	70	80	70	80
					505	425	530	470

$$\begin{aligned} \text{Fisher's ideal price index} &= \sqrt{\frac{\sum p_1 q_0}{\sum p_0 q_0} \times \frac{\sum p_1 q_1}{\sum p_0 q_1}} \times 100 = \sqrt{\frac{505}{425} \times \frac{530}{470}} \times 100 \\ &= \sqrt{1.3399} \times 100 = 1.1576 \times 100 = 115.76 \end{aligned}$$

Hence, we conclude that the price level has increased by 15.76 per cent in the year 2000.

Example 11.7: Calculate from the following data, the Fisher's ideal index number for the year 2000:

Commodity	1999		2000	
	Price (₹)	Expenditure on Quantity Consumed (₹)	Price (₹)	Expenditure on Quantity Consumed (₹)
A	8	200	65	1950
B	20	1400	30	1650
C	5	80	20	900
D	10	360	15	300
E	27	2160	10	600

Solution: Table 11.9 presents the information necessary for Fisher's method to calculate the index.

Table 11.9 Calculations of Fisher's Ideal Index

Commodity	Base Year, 1999		Current Year, 2000		$p_1 q_0$	$p_0 q_0$	$p_1 q_1$	$p_0 q_1$
	(p_0)	(q_0)	(p_1)	(q_1)				
A	8	$200/8 = 25$	65	$1950/65 = 30$	1625	200	1950	240
B	20	$1400/20 = 70$	30	$1650/30 = 55$	2100	1400	1650	1100
C	5	$80/5 = 16$	20	$900/20 = 45$	320	80	900	225
D	10	$360/10 = 36$	15	$300/15 = 20$	540	360	300	200
E	27	$2160/27 = 80$	10	$600/10 = 60$	800	2160	600	1620
					5385	4200	5400	3385

$$\begin{aligned} \text{Fisher's ideal price index} &= \sqrt{\frac{\sum p_1 q_0}{\sum p_0 q_0} \times \frac{\sum p_1 q_1}{\sum p_0 q_1}} \times 100 = \sqrt{\frac{5385}{4200} \times \frac{5400}{3385}} \times 100 \\ &= 1.430 \times 100 = 143. \end{aligned}$$

Hence, we conclude that the price level has increased by 43 per cent in the year 2000.

Example 11.8: Compute index numbers from the following data using (i) Laspeyre's, (ii) Paasche's and (iii) Fisher's Ideal formulae:

Commodity	Base Year		Current Year	
	Quantity	Price	Quantity	Price
A	8	4	10	9
B	7	3	8	4
C	6	4	7	8
D	5	2	5	4

[Delhi Univ., B.Com. (P), 1990]

Solution: The following table presents information necessary for calculating required index numbers.

Commodity	Base Year		Current Year		$p_1 q_0$	$p_0 q_0$	$p_1 q_1$	$p_0 q_1$
	Price (p_0)	Qty. (q_0)	Price (p_1)	Qty. (q_1)				
A	4	8	9	10	72	32	90	40
B	3	7	4	8	28	21	32	24
C	4	6	8	7	48	24	56	28
D	2	5	4	5	20	10	20	10
					168	87	198	102

$$\text{Laspeyre's index} = \frac{\sum p_1 q_0}{\sum p_0 q_0} \times 100 = \frac{168}{87} \times 100 = 193.10.$$

$$\text{Paasche's index} = \frac{\sum p_1 q_1}{\sum p_0 q_1} \times 100 = \frac{198}{102} \times 100 = 194.12.$$

$$\text{Fisher's index} = \sqrt{\frac{\sum p_1 q_0}{\sum p_0 q_0} \times \frac{\sum p_1 q_1}{\sum p_0 q_1}} \times 100 = \sqrt{\frac{168}{87} \times \frac{198}{102}} \times 100 = 193.61.$$

Example 11.9: Given that $\sum p_1 q_1 = 250$, $\sum p_0 q_0 = 150$; Paasche's index number = 150 and Dorbish and Bowley's index number = 145. Find out Fisher's ideal index number and Marshall Edgeworth index number.

[Delhi Univ., B.Com. (Hons.), 1992, 2005]

Solution: Given, $\sum p_1 q_1 = 250$, $\sum p_0 q_0 = 150$

$$\text{Paasche's index number} = \frac{\sum p_1 q_1}{\sum p_0 q_1} \times 100 \quad \text{or} \quad 150 = \frac{250}{\sum p_0 q_1} \times 100$$

$$\text{or} \quad 150 \sum p_0 q_1 = 25,000, \text{ i.e. } \sum p_0 q_1 = \frac{25,000}{150} = \frac{500}{3} = 167 \text{ (approx)}$$

$$\text{Dorbish-Bowley's index number} = \frac{1}{2} \left(\frac{\sum p_1 q_0}{\sum p_0 q_0} + \frac{\sum p_1 q_1}{\sum p_0 q_1} \right) \times 100$$

$$\text{or} \quad 145 = \left(\frac{\sum p_1 q_0}{150} + \frac{250}{167} \right) 50 \quad \text{or} \quad \frac{145}{50} = \frac{\sum p_1 q_0}{150} + \frac{250}{167}$$

$$\text{or} \quad \frac{145}{50} - \frac{250}{167} = \frac{\sum p_1 q_0}{150} \quad \text{or} \quad 1.40 = \frac{\sum p_1 q_0}{150}$$

$$\text{or} \quad \sum p_1 q_0 = 1.40 \times 150 = 210$$

$$\begin{aligned} \text{Fisher's index number} &= \sqrt{\frac{\sum p_1 q_0}{\sum p_0 q_0} \times \frac{\sum p_1 q_1}{\sum p_0 q_1}} \times 100 = \sqrt{\frac{210}{150} \times \frac{250}{167}} \times 100 \\ &= 100 \times \sqrt{1.40 \times 1.50} = 100 \sqrt{2.10} = 100 \times 1.449 = 144.9 \end{aligned}$$

Example 11.10: Give that $\sum p_1 q_1 = 250$, $\sum p_0 q_0 = 150$, Paasche's index number = 150 and Dorbish and Bowley's index number = 145. Find Fisher's Ideal index number and $\sum p_1 q_0$.

[Delhi Univ., B.Com. (Hons.), 2005]

$$\text{Solution: Paasche's Index number} = \frac{\sum p_1 q_1}{\sum p_0 q_1} \times 100 \quad \text{or} \quad 150 = \frac{250}{\sum p_0 q_1} \times 100$$

$$\text{or} \quad \Sigma p_0 q_1 = \frac{250}{150} \times 100 = \frac{500}{3} = 167 \text{ (approx.)}$$

$$\text{Dorbish and Bowley's index number} = \frac{L + P}{2} \text{ or } 145 = \frac{L + 150}{2}, \text{ i.e. } L = 140$$

$$\text{But } L = \frac{\Sigma p_1 q_0}{\Sigma p_0 q_0} \times 100, \text{ or } 140 = \frac{\Sigma p_1 q_0}{150} \times 100$$

$$\text{or} \quad \Sigma p_1 q_0 = \frac{140 \times 150}{100} = 210$$

$$\text{Fisher's ideal index} = \sqrt{L \times P} = \sqrt{140 \times 150} = 144.91$$

Marshall-Edgeworth Method

In this method, the sum of base year and current year quantities are considered as the weight to calculate the index. The formula for constructing the index is

$$\text{Marshall-Edgeworth price index} = \frac{\Sigma (q_0 + q_1) p_1}{\Sigma (q_0 + q_1) p_0} \times 100 = \frac{\Sigma q_0 p_1 + \Sigma q_1 p_1}{\Sigma q_0 p_0 + \Sigma q_1 p_0} \times 100$$

where notations have their usual meaning.

The disadvantage with this formula is the same as that of Paasche index and Fisher's ideal index in the sense that it also needs current quantity weights every time an index is constructed.

Walsch's Method

In this method, the quantity weight used is the geometric mean of the base and current year quantities. The formula for constructing the index is

$$\text{Walsch's price index} = \frac{\Sigma p_1 \sqrt{q_0 q_1}}{\Sigma p_0 \sqrt{q_0 q_1}} \times 100$$

Although this index satisfies the time reversal test, it needs current quantity weight every time an index is constructed.

Kelly's Method

The method suggested by T.L. Kelly for the construction of index number is

$$\text{Kelly's price index} = \frac{\Sigma p_1 q}{\Sigma p_0 q} \times 100$$

where q = fixed weight.

This method is also called the *fixed weight aggregate method* because instead of using base period or current period quantities as weights, it uses weights from a representative period. The representative weights are referred to as *fixed weight*. The fixed weights and the base period prices do not have to come from the same period.

Advantages

An important advantage of this index is that it does not need yearly changes in the weights. One can select a different period for fixed weight other than base period. This can improve the accuracy of the index. Moreover, the base period can also be changed without changing the fixed weight. The weights should be appropriate and should indicate the relative importance of various commodities. This weight may be kept fixed until new data are available to revise the index.

Disadvantages: One disadvantage with this index is that it does not take into account the weight either of the base year or of the current year.

Example 11.11: It is stated that the Marshall-Edgeworth index number is a good approximation of the ideal index number. Verify this statement using the following data:

Commodity	2002		2003	
	Price	Quantity	Price	Quantity
A	2	74	3	82
B	5	125	4	140
C	7	40	6	33

Solution: Table 11.10 presents the information necessary to calculate Fisher and Marshall-Edgeworth indexes.

Table 11.10 Calculations of Fisher's Ideal and Marshall-Edgeworth's Index

Commodity	Base Year, 2002		Current Year, 2003		$p_1 q_0$	$p_0 q_0$	$p_1 q_1$	$p_0 q_1$
	(p_0)	(q_0)	(p_1)	(q_1)				
A	2	74	3	82	222	148	246	164
B	5	125	4	140	500	625	560	700
C	7	40	6	33	240	280	198	231
					962	1053	1004	1095

$$\begin{aligned}\text{Fisher ideal price index} &= \sqrt{\frac{\sum p_1 q_0}{\sum p_0 q_0} \times \frac{\sum p_1 q_1}{\sum p_0 q_1}} \times 100 = \sqrt{\frac{962}{1053} \times \frac{1004}{1095}} \times 100 \\ &= \sqrt{0.838} \times 100 = 0.9154 \times 100 = 91.54\end{aligned}$$

$$\begin{aligned}\text{Marshall-Edgeworth price index} &= \frac{\sum p_1 (q_0 + q_1)}{\sum p_0 (q_0 + q_1)} \times 100 = \frac{\sum p_1 q_0 + \sum p_1 q_1}{\sum p_0 q_0 + \sum p_0 q_1} \times 100 \\ &= \frac{962 + 1004}{1053 + 1095} \times 100 = 0.9152 \times 100 = 91.52\end{aligned}$$

Hence, we conclude that Fisher's method and Marshall-Edgeworth method provide almost the same value of the index.

Example 11.12: Compute Laspeyre's, Paasche's, Fisher's, and Marshall-Edgeworth's index numbers from the following data:

Item	1998		1999	
	Price	Quantity	Price	Quantity
A	5	25	6	30
B	3	8	4	10
C	2	10	3	8
D	10	4	3	5

Solution: Table 11.11 presents the information necessary to calculate several indexes.

Table 11.11 Calculations of Indexes

Item	Base Year, 1998		Current Year, 1999		$p_1 q_0$	$p_0 q_0$	$p_1 q_1$	$p_0 q_1$
	(p_0)	(q_0)	(p_1)	(q_1)				
A	5	25	6	30	150	125	180	150
B	3	8	4	10	32	24	40	30
C	2	10	3	8	30	20	24	16
D	10	4	3	5	12	40	15	50
					224	209	259	246

$$\text{Laspeyre's price index} = \frac{\sum p_1 q_0}{\sum p_0 q_0} \times 100 = \frac{224}{209} \times 100 = 107.18$$

$$\text{Paasche's price index} = \frac{\sum p_1 q_1}{\sum p_0 q_1} \times 100 = \frac{259}{246} \times 100 = 105.28$$

$$\text{Fisher's ideal price index} = \sqrt{L \times P} = \sqrt{107.17 \times 105.28} = 106.22$$

$$\begin{aligned} \text{Marshall-Edgeworth's price index} &= \frac{\sum p_1 (q_0 + q_1)}{\sum p_0 (q_0 + q_1)} \times 100 = \frac{\sum p_1 q_0 + \sum p_1 q_1}{\sum p_0 q_0 + \sum p_0 q_1} \times 100 \\ &= \frac{224 + 259}{209 + 246} \times 100 = 110.55 \end{aligned}$$

11.7.2 Weighted Average of Price Relative Index

Unlike the unweighted average of price relative, the weighted average of price relative is determined by using the quantity consumed in the base period for weighting the items or commodities. The value (in rupees) of each item or commodity included in the calculation of composite index is determined by multiplying the price of each item by its quantity consumed.

The formula for constructing the weight average of price relatives index using base values is

$$\begin{aligned} \text{Weighted average of price relative index, } P_{01} &= \frac{\sum \{(p_1/p_0) \times 100\} (p_0 q_0)}{\sum p_0 q_0} = \frac{\sum PV}{\sum V} \\ &= \frac{\sum p_1 q_0}{\sum p_0 q_0} \times 100 \end{aligned}$$

where $V (= p_0 q_0)$ = base period value

$P (= p_1/p_0 \times 100)$ = price relative

This formula is equivalent to Laspeyre's method for any given problem.

If we wish to compute a weighted average of price relative using $V = p_0 q_1$, then the above formula becomes

$$P_{01} = \frac{\sum \{(p_1/p_0) \times 100\} (p_0 q_1)}{\sum p_0 q_1} = \frac{\sum p_1 q_1}{\sum p_0 q_1} \times 100$$

This formula is equivalent to Paasche's method for any given problem.

If instead of using weighted arithmetic average we wish to use weighted geometric mean, then the above formula becomes

$$P_{01} = \frac{\sum V \times \log P}{\sum V}; \quad P = \frac{p_1}{p_0} \times 100 \text{ and } V = p_0 q_0$$

Advantages

1. Different index numbers constructed using average price relative with same base can be combined to form a new index.
2. Weighted average of price relative method is suitable to construct an index by selecting one item from each of the many subgroups of items. In such a case, the values of each subgroup may be used as weights.

Example 11.13: Compute price index by using weighted average of price relative method based on the following data.

Items	Quantity (kg) (q_0)	Price per kg (₹)	
		p_0	p_1
A	20	3.0	4.0
B	40	1.5	1.6
C	10	1.0	1.5

Solution: The following table presents the necessary information to calculate the weighted average price relative index.

Items	p_0	q_0	p_1	$V = p_0 q_0$	Price Relatives $P = (p_1/p_0) \times 100$	Weighted Average Relatives PV
A	3	20	4	60	$\frac{4}{3} \times 100 = 133.33$	8000
B	1.5	40	1.6	60	$\frac{1.6}{1.5} \times 100 = 106.67$	6400
C	1	10	1.5	10	$\frac{1.5}{1} \times 100 = 150$	1500
				130		15,900

$$\text{Weighted average of price relative index, } P_{01} = \frac{\Sigma PV}{\Sigma V} = \frac{15,900}{130} = 122.31.$$

Example 11.14: A large manufacturer purchases an identical component from three different suppliers that differ in unit price and quantity supplied. The relevant data for 2000 and 2001 are given below:

Supplier	Quantity Index in (2000)	Unit Price (₹)	
		2000	2001
A	20	18	20
B	40	12	14
C	10	15	16

Construct a weighted average price relative index using (a) arithmetic mean and (b) geometric mean.

Solution: Table 11.12 presents the information necessary to calculate the weight average price relative index.

Table 11.12 Calculations of Weighted Average of Price Relatives

Supplier	Prices in		Quantity in	Percentage	Base Value	Weighted Percentage
	2000	2001	2000	Price Relative	$V = p_0 q_0$	Relative PV
	p_0	p_1	q_0	$P = \frac{p_1}{p_0} \times 100$		
A	18	20	20	$(20/18) \times 100 = 111.11$	360	39,999.60
B	12	14	40	$(14/12) \times 100 = 116.67$	480	56,001.60
C	15	16	10	$(16/15) \times 100 = 106.67$	150	16,000.50
					990	1,12,001.70

(a) Weighted average of price relative index

$$P_{01} = \frac{[\Sigma(p_1 / p_0) \times 100] p_0 q_0}{\Sigma p_0 q_0} = \frac{1,12,001.70}{990} = 113.13$$

The value of P_{01} implies that there has been 13.13 per cent increase in price from year 2000 to 2001.

(b)

Table 11.13 Calculations of Weighted Geometric Mean of Price Relatives

Supplier	Prices in		Quantity in	Base Value	Percentage	$\log P$	$V \log P$
	2000	2001	2000	Value	Price Relative		
	p_0	p_1	q_0	$V = p_0 q_0$	$P = \frac{p_1}{p_0} \times 100$		
A	18	20	20	360	111.11	2.046	736.56
B	12	14	40	480	116.67	2.067	992.16
C	15	16	10	150	106.67	2.028	304.20
				990			2032.92

Weighted geometric mean of price relatives

$$P_{01} = \text{antilog} \left(\frac{\Sigma V \log P}{\Sigma V} \right) = \text{antilog} \left(\frac{2032.92}{990} \right)$$

$$= \text{antilog} (2.0535) = 113.11$$

11.8 QUANTITY OR VOLUME INDEXES

A quantity index measures the percentage change in consumption, production or distribution level of either an individual item or a basket of items from one time period to another. When constructing quantity indexes, it is necessary to *hold price levels constant over time* to isolate the effect of quantity (consumption level) changes only. For example, agricultural production is measured using a quantity index because it eliminates effects of fluctuating prices. Any of the methods, such as the relative method (both simple and weighted) or the aggregative method,

which take into account weights to construct price indexes can also be used to calculate quantity indexes. The weights in quantity index numbers are prices. Therefore, quantity indexes can be easily derived from a price indexes by interchanging the p 's and q 's.

Seven quantity indexes analogous to the seven price indexes already discussed in the previous section can be constructed as given below:

$$\text{Laspeyre's quantity index, } Q_L = \frac{\sum V_0 (q_1 / q_0)}{\sum V_0} \times 100 = \frac{\sum q_1 p_0}{\sum q_0 p_0} \times 100$$

where $V_0 = p_0 q_0$, values of base year consumption at base year prices

$$\text{Similarly, Paasche's quantity index, } Q_P = \frac{\sum V_1 (q_1 / q_0)}{\sum V_1} \times 100 = \frac{\sum q_1 p_1}{\sum q_0 p_1} \times 100$$

where $V_1 = p_1 q_1$ values of base year consumption at current year prices

$$\text{Fisher's quantity index, } Q_F = \sqrt{Q_L \times Q_P} = \sqrt{\frac{\sum q_1 p_0}{\sum q_0 p_1} \times \frac{\sum q_1 p_1}{\sum q_0 p_1}} \times 100$$

The formula for computing a weighted average of quantity relative index is also the same as used to compute a price index. The formula for this type of quantity index is

$$\text{Weighted average of quantity relative index} = \frac{\sum \left(\frac{q_1}{q_0} \times 100 \right) (q_0 p_0)}{\sum q_0 p_0}$$

where q_1 = quantities for the current period

q_0 = quantities for the base period

Example 11.15: From the following data, construct quantity index number by (i) Fisher's method and (ii) Marshall-Edgeworth's method:

Commodities	Base Year		Current Year	
	Price (₹)	Quantity (₹)	Price (₹)	Quantity (₹)
A	25	40	2,000	50
B	22	18	1,200	30
C	54	16	1,320	44
D	20	40	1,350	45
E	18	30	630	15

[Delhi Univ., B.Com.(Hons.), 1990]

Solution: The following table presents necessary information to calculate required index numbers.

Commodity	Base Year		Current Year		$q_1 p_0$	$q_0 p_0$	$q_1 p_1$	$q_0 p_1$
	p_0	q_0	p_1^*	q_1				
A	25	40	40	50	1250	1000	2000	1600
B	22	18	40	30	660	396	1200	720
C	54	16	30	44	2376	864	1320	480
D	20	40	30	45	900	800	1350	1200
E	18	30	42	15	270	540	630	1260
					5456	3600	6500	5260

* Current year prices have been calculated by dividing the expenditure by quantity.

$$\begin{aligned}\text{Fisher's quantity index number, } Q_{01} &= \sqrt{\frac{\sum q_1 p_0}{\sum q_0 p_0} \times \frac{\sum q_1 p_1}{\sum q_0 p_1}} \times 100 \\ &= \sqrt{\frac{5456}{3600} \times \frac{6500}{5260}} \times 100 = \sqrt{\frac{35,464}{18,936}} \times 100 = 187.28\end{aligned}$$

$$\begin{aligned}\text{Marshall Edgeworth quantity index, } Q_{01} &= \frac{\sum q_1 (p_1 + p_0)}{\sum q_0 (p_1 + p_0)} \times 100 \\ &= \frac{\sum q_1 p_1 + \sum q_1 p_0}{\sum q_0 p_1 + \sum q_0 p_0} \times 100 = \frac{6500 + 5456}{5260 + 3600} \times 100 = 134.94.\end{aligned}$$

Example 11.16: Obtain Laspeyre's price index number and Paasche's quantity index number from the following data:

Item	Price (₹ per Unit)		Quantity (Units)	
	Base Year	Current Year	Base Year	Current Year
1	2	5	20	15
2	4	8	4	5
3	1	2	10	12
4	5	10	5	6

[Mangalore Univ., B.Com., 1997]

Solution: Table 11.14 presents the information necessary to calculate Laspeyre's price index and Paasche's quantity indexes.

Table 11.14 Calculations on Laspeyre's Price Index and Paasche's Quantity Index

Item	Price		Quantity		$p_1 q_0$	$p_0 q_0$	$q_1 p_1$	$p_0 q_1$
	p_0	p_1	q_0	q_1				
1	2	5	20	15	100	40	75	30
2	4	8	4	5	32	16	40	20
3	1	2	10	12	20	10	24	12
4	5	10	5	6	50	25	60	30
					202	91	199	92

$$\text{Laspeyre's price index} = \frac{\sum p_1 q_0}{\sum p_0 q_0} \times 100 = \frac{202}{91} \times 100 = 221.98$$

$$\text{Paasche's quantity index} = \frac{\sum q_1 p_1}{\sum q_0 p_1} \times 100 = \frac{199}{92} \times 100 = 216.30$$

Example 11.17: Compute the quantity index by using Fisher's formula from the data given below:

Commodity	2002		2003	
	Price (₹/Unit)	Total Value	Price (₹/Unit)	Total Value
A	5	50	4	48
B	8	48	7	49
C	6	18	5	20

Solution: The base year quantity q_0 and current year quantity q_1 for individual commodity can be calculated as shown in Table 11.15.

$$q_0 \text{ (for 2002)} = \frac{\text{Total value}}{\text{Price}} = \frac{50}{5} = 10; \quad \frac{48}{4} = 6 \text{ and } \frac{18}{6} = 3$$

$$q_1 \text{ (for 2003)} = \frac{\text{Total value}}{\text{Price}} = \frac{48}{4} = 12; \quad \frac{49}{7} = 7 \text{ and } \frac{20}{5} = 4$$

Table 11.15 Calculations for Fisher's Quantity Index

Commodity	Price, 2002		Quantity, 2003		$q_1 p_0$	$q_0 p_0$	$q_1 p_1$	$q_0 p_1$
	p_0	q_0	p_1	q_1				
A	5	10	4	12	60	50	48	40
B	8	6	7	7	56	48	49	42
C	6	3	5	4	24	18	20	15
					140	116	117	97

Substituting values in the formula, we get

$$\begin{aligned} \text{Fisher's quantity index} &= \sqrt{\frac{\sum q_1 p_0}{\sum q_0 p_0} \times \frac{\sum q_1 p_1}{\sum q_0 p_1}} \times 100 \\ &= \sqrt{\frac{140}{116} \times \frac{117}{97}} \times 100 = 145.57 \end{aligned}$$

Example 11.18: Calculate the weighted average of quantity relative index from the following data:

Commodity	Quantity (Units)		Price (₹/Unit)
	2000	2002	2000
A	10	12	100
B	15	20	75
C	8	10	80
D	20	25	60
E	50	60	500

Solution: Table 11.16 presents the information necessary to calculate the weighted average of quantity relative index.

Table 11.16 Calculations of a Weighted Average of Quantity Relatives Index

Commodity	Quantity (units)		Price (₹/unit)	Percentage	Base	Weighted
	2000	2002	in 2000	Relatives	Value	Relatives
	q_0	q_1	p_0	$(q_1/q_0) \times 100$	$q_0 p_0$	$[(q_1/q_0) \times 100] \times q_0 p_0$
A	10	12	100	$(12/10) \times 100 = 120$	1000	1,20,000.00
B	15	20	75	$(20/15) \times 100 = 133.33$	1125	1,49,996.25
C	8	10	80	$(10/8) \times 100 = 125$	640	80,000.00
D	20	25	60	$(25/20) \times 100 = 125$	1200	1,50,000.00
E	50	60	500	$(60/50) \times 100 = 120$	25,000	30,00,000.00
					28,965	34,99,996.25

$$\begin{aligned}\text{Weighted average of quantity relatives index} &= \frac{\sum \{(q_1 / q_0) \times 100\} (q_0 p_0)}{\sum q_0 p_0} \\ &= \frac{34,99,996.25}{28,965} = 120.835\end{aligned}$$

11.9 VALUE INDEXES

A value index number measures the percentage change in the total value of either an individual item or a basket of items from one time period to another. The value of an item or commodity is obtained by multiplying its price and quantity. Since value is determined both by price and quantity, a value index measures the combined effects of price and quantity changes. A simple value ratio is equal to the value of the current year divided by the value of the base year. If this ratio is multiplied by 100, we get the value index as

$$\text{Value index, } V = \frac{\sum p_1 q_1}{\sum p_0 q_0} \times 100$$

If the values are given directly, then the value index number is given by

$$\text{Value index, } V = \frac{\sum V_1}{\sum V_0}$$

where V_0 = values at the base year or period

V_1 = values at the current year or period

Such indexes are not weighted because they take into account both the price and quantity. These indexes are, however, not very popular because the situation revealed by price and quantities are not fully revealed by the values. A value index does not distinguish between the effects of its components, namely price and quantity.

Self-practice Problems 11B

11.9 The following table contains information from the raw material purchase records of a small factory for the year 2002 and 2003:

Commodity	2002		2003	
	Price	Total	Price	Total
	₹/Unit)	Value	(₹/Unit)	Value
A	5	50	6	72
B	7	84	10	80
C	10	80	12	96
D	4	20	5	30
E	8	56	8	64

Calculate Fisher's ideal index number.

11.10 The subgroup indexes of the consumer price index number for urban non-

manual employees of an industrial centre for a particular year (with base 1990 = 100) were:

Food	200
Clothing	130
Fuel and Lighting	120
Rent	150
Miscellaneous	140

The weights are 60, 8, 7, 10 and 15 respectively. It is proposed to fix dearness allowance in such a way as to compensate fully the rise in the prices of food and house rent. What should be the dearness allowance, expressed as a percentage of wage?

- 11.11** The owner of a small shop selling food items collected the following information regarding the price and quantity sold of a particular item.

Item	Average Price (₹/Unit)		Quantity Sold (Units)	
	2002	2003	2002	2003
A	1	2	10	5
B	1	x	5	2

If the ratio between Laspeyre's (L) and Paasche's (P) Index number is: L:P = 28:27, then find the value of x.

- 11.12** An increase of 50 per cent in the cost of a certain consumable product raises the cost of living of a certain family by 5 per cent. What percentage of its cost of living was due to buying that product before the change in the price?

- 11.13** Calculate Fisher's ideal index from the data given below:

Commodity	Base Year, 2000		Current Year, 2001	
	Price	Value	Price	Value
A	10	30	12	48
B	15	60	15	75
C	5	50	8	96
D	2	10	3	15

[HP Univ., M.Com., 1995]

- 11.14** Using the data given below, calculate the price index number for the year 1998 by (i) Laspeyre's formula, (ii) Paasche's formula and (iii) Fisher's formula considering 1989 as the base year.

Commodity	Price (₹/Unit)		Quantity (in 1000 kg)	
	1989	1998	1989	1998
Rice	9.3	4.5	100	90
Wheat	6.4	3.7	11	10
Pulses	5.1	2.7	5	3

- 11.15** It is stated that the Marshall-Edgeworth's index is a good approximation of the ideal

index number. Verify using the following data:

Commodity	1996		2000	
	Price	Quantity	Price	Quantity
A	2	74	3	82
B	5	125	4	140
C	7	40	6	33

- 11.16** In preparation for an appropriations hearing, the DCP of a city zone has collected the following information:

Type of Crime	2000	2001	Weight
Robberies	13	8	6
Car thefts	15	22	5
Cycle thefts	249	185	4
Pocket picking	328	259	1
Theft by servants	497	448	2

Calculate the index of crime for 2001, using 2000 as the base period.

- 11.17** Using Paasche's formula compute the quantity index for the year 1993 with 1985 as base year.

Commodity	Quantity (in Units)		Value (in ₹)	
	1985	1993	1985	1993
A	100	150	500	900
B	80	100	320	500
C	60	72	150	360
D	30	33	360	297

- 11.18** Calculate a weighted average of relative quantity index using 1995 as base period.

Commodity	Quantity (in 1000 kg)		Price (₹/kg)
	1995	1999	1995
Wheat	29	24	3.80
Corn	3	2.5	2.91
Soyabeans	12	14	6.50

- 11.19** Divide the values by price and obtain quantity figures and then calculate Fisher's ideal price index.

Hints and Answers

- 11.10** Let the income of the consumer be ₹ 100. He spent ₹ 60 on food and ₹ 10 on house rent in 1990. The index of food is 200 and the house rent ₹ 150 for the particular year for which the data are given. In order to maintain the same consumption standards regarding two items, he will have to spend ₹ 120 on food and ₹ 15 on house rent. Further the weights of other items are constant; in order to maintain the same standard he will have to spend $120+8+7+15+5 = ₹ 155$. Hence the dearness allowance should be 55 per cent.

11.11

Item	p_0	q_0	p_1	q_1	p_1q_0	p_0q_0	p_1q_1	p_0q_1
A	1	10	2	5	20	10	10	5
B	1	5	x	2	$5x$	5	$2x$	2
					$20 + 5x$	15	$10 + 2x$	7

$$\text{Laspeyre's index} = \frac{\sum p_1q_0}{\sum p_0q_0} = \frac{20 + 5x}{15}$$

$$\text{Paasche's index} = \frac{\sum p_1q_1}{\sum p_0q_1} = \frac{10 + 2x}{7}$$

$$\text{Given } \frac{(20 + 5x)/15}{(10 + 2x)/7} = \frac{28}{27}$$

$$\text{or } \frac{20 + 5x}{15} \times \frac{7}{10 + 2x} = \frac{28}{27} \quad \text{or } x = 4$$

- 11.12** Let the cost of the article before the increase be x . After increase it will be $150x \div 100 = 1.5x$. The rise $1.5x - x = 0.5x$ is equivalent to an increase of 5 per cent in the cost of living. The increases in the cost of living was $1.05y - y = 0.05y$.

Hence $0.5x = 0.05y$ or $x = 0.05y/0.5 = 0.1y = 10$ per cent of y . Thus the expenditure on that item was 10 per cent of the cost of living.

11.13

Commodity	p_0	q_0	p_1	q_1	p_1q_0	p_0q_0	p_1q_1	p_0q_1
A	10	3	12	4	36	30	48	40
B	15	4	15	5	60	60	75	75
C	5	10	8	12	80	50	96	60
D	2	5	3	5	15	10	15	10
					191	150	234	185

$$P_{01} = \sqrt{\frac{\sum p_1q_0}{\sum p_0q_0} \times \frac{\sum p_1q_1}{\sum p_0q_1}} \times 100$$

$$= \sqrt{\frac{191}{150} \times \frac{234}{185}} \times 100 = 126.9$$

11.14

Commodity	Price (₹)		Quantity (in 1000 kg)					
	1989	1998	1989	1998				
	p_0	p_1	q_0	q_1	p_0q_0	p_1q_0	p_0q_1	p_1q_1
Rice	9.3	4.5	100	90	930.0	450.0	837.0	405.0
Wheat	6.4	3.7	11	10	70.4	40.7	64.0	37.0
Pulses	5.1	2.7	5	3	25.5	13.5	15.3	8.1
					1025.9	504.2	916.3	450.1

$$\text{Laspeyre's index} = \frac{\sum p_1q_0}{\sum p_0q_0} \times 100$$

$$= \frac{504.2}{1025.9} \times 100 = 49.15$$

$$\text{Paasche's index} = \frac{\sum p_1q_1}{\sum p_0q_1} \times 100$$

$$= \frac{450.1}{916.3} \times 100 = 49.12$$

$$\text{Fisher's index} = \sqrt{\frac{\sum p_1q_0}{\sum p_0q_0} \times \frac{\sum p_1q_1}{\sum p_0q_1}} \times 100$$

$$= \sqrt{49.15 \times 49.12} = 49.135$$

11.15

Commo- dity	1996		2000					
	$\bar{p}_0$	q_0	$\bar{p}_1$	q_1	p_0q_0	p_0q_1	p_1q_0	p_1q_1
A	2	74	3	82	148	164	222	246
B	5	125	4	140	625	700	500	560
C	7	40	6	33	280	231	240	198
					1053	1095	962	1004

Marshall-Edgeworth index

$$= \frac{\sum p_1q_0 + \sum p_1q_1}{\sum p_0q_0 + \sum p_0q_1} \times 100$$

$$= \frac{962 + 1004}{1053 + 1095} \times 100 = 91.53$$

Fisher's Ideal index

$$= \sqrt{\frac{\sum p_1q_0}{\sum p_0q_0} \times \frac{\sum p_1q_1}{\sum p_0q_1}} \times 100 = 83.77$$

11.16

Type of Crime	2000	2001	Weight, W	Crime Relative R	RW
Robberies	13	8	6	$(8/13) \times 100$ = 61.54	369.24
Car thefts	15	22	5	$(22/15) \times 100$ = 146.67	733.35
Cycle thefts	249	185	4	$(185/249) \times 100$ = 74.30	297.20
Pocket picking	328	259	1	$(259/328) \times 100$ = 78.96	78.96
Thefts by servants	497	448	2	$(448/497) \times 100$ = 90.14	180.28
			18		1659.03

$$\text{Crime index} = \frac{\Sigma RW}{\Sigma W} = \frac{1659.03}{18} = 92.17$$

11.17

Commodity	Quantity		Price		
	1985	1993	1993		
	q_0	q_1	p_1	$q_1 p_1$	$q_0 p_1$
A	100	150	$900/150=6$	900	600
B	80	100	$500/100=5$	500	400
C	60	72	$360/72=5$	360	300
D	30	33	$297/33=9$	297	270
				2057	1570

$$\begin{aligned} \text{Paasche's quantity index} &= \frac{\Sigma q_1 p_1}{\Sigma q_0 p_1} \times 100 \\ &= \frac{2057}{1570} \times 100 \\ &= 131.02 \end{aligned}$$

11.18

Commodity	Quantity 1995 q_1	Price 1999 p_1	Per- cent $\frac{p_1}{p_0} \times 100$	Base Relatives $\frac{q_1}{q_0} \times 100$	Weighted Value $q_0 p_0$	Relatives (6)= (4)×(5)
	(1)	(2)	(3)	(4)	(5)	
Wheat	29	24	3.80	83	110.20	9,146.60
Corn	3	2.5	2.91	83	8.73	724.59
Soyabeans	12	14	6.50	117	78.00	9126.00
					196.93	18,997.19

Weighted average of relative quantity index

$$\begin{aligned} &= \frac{\Sigma \{(q_1 / q_0) \times 100\} (q_0 p_0)}{\Sigma q_0 p_0} \\ &= \frac{18,997.19}{196.93} = 96. \end{aligned}$$

11.19

Commodity	p_0	q_0	p_1	q_1	$p_1 q_0$	$p_0 q_0$	$q_1 p_1$	$q_0 p_1$
A	5	10	6	12	60	50	72	60
B	7	12	10	8	120	84	80	56
C	10	8	12	8	96	80	96	80
D	4	5	5	6	25	20	30	24
E	8	7	8	8	56	56	64	64
					357	290	342	284

Fisher's ideal price index:

$$\begin{aligned} &\sqrt{\frac{\Sigma p_1 q_0}{\Sigma p_0 q_0} \times \frac{\Sigma p_1 q_1}{\Sigma p_0 q_1}} \times 100 \\ &= \sqrt{\frac{357}{290} \times \frac{342}{284}} \times 100 = 121.65 \end{aligned}$$

11.10 TESTS OF ADEQUACY OF INDEXES

So far we have discussed several methods to construct unweighted and weighted index numbers. However, the problem still remains of selecting an appropriate method for the construction of an index number in a given situation. The following tests have been suggested to select the adequacy of an index number:

- Time reversal test
- Factor reversal test
- Circular test

11.10.1 Time Reversal Test

The time reversal test is used to test whether a given method will work both backwards and forwards with respect to time. The test is that the formula for calculating the index number should be such that it will give the same ratio between one point of comparison and another no matter which of the two is taken as base. In other words, a price or quantity index for a given period with respect to the preceding period is equal to the reciprocal of the price or quantity index when periods are interchanged. For example, if P_{01} is a price index in the current year '1' with base of preceding year '0' and P_{10} is a price index for the base year '0' based on the current year '1', then the following relation should be satisfied:

$$P_{01} = \frac{1}{P_{10}} \text{ or } P_{01} \times P_{10} = 1 \text{ and } Q_{01} \times Q_{10} = 1$$

This test is not satisfied by the Laspeyre's Index and the Paasche's Index. The methods which satisfy the time reversal test are:

- Fisher's ideal index method
- Simple geometric mean of price relatives
- Aggregates with fixed weights (Kelly's formula)
- Marshall-Edgeworth's method
- Weighted geometric mean of price relatives when fixed weights are used
- Walsch's formula

For example, let us see how Fisher's ideal index formula satisfies the time reversal test.

$$P_{01} = \sqrt{\frac{\sum p_1 q_0}{\sum p_0 q_0} \times \frac{\sum p_1 q_1}{\sum p_0 q_1}}$$

For calculating P_{10} , the time is interchanged so that p_0 becomes p_1 and p_1 becomes p_0 . Similarly q_0 becomes q_1 and q_1 becomes q_0 , we get

$$P_{10} = \sqrt{\frac{\sum p_0 q_1}{\sum p_1 q_1} \times \frac{\sum p_0 q_0}{\sum p_1 q_0}}$$

$$P_{01} \times P_{10} = \sqrt{\frac{\sum p_1 q_0}{\sum p_0 q_0} \times \frac{\sum p_1 q_1}{\sum p_0 q_1} \times \frac{\sum p_0 q_1}{\sum p_1 q_1} \times \frac{\sum p_0 q_0}{\sum p_1 q_0}} = 1$$

Since $P_{01} \times P_{10} = 1$, Fisher's ideal index satisfies the test.

11.10.2 Factor Reversal Test

According to Fisher—'Just as each formula should permit the interchange of two items without giving inconsistent result so it ought to permit interchanging the prices and quantities without giving inconsistent result, i.e. the two results multiplied together should give the true value ratio.' In other words, the change in the price when multiplied by the change in quantity should represent the total change in value. Thus, if the price of a commodity has doubled during a certain period and in this period the quantity has trebled, then the total change in the value should be six times the former level. That is, if p_1 and p_0 represent the prices and q_1 and q_0 , the quantities in the current and the base periods, respectively, then the price index for period '1' with base year '0' and the quantity index for period '1' with base year '0' is given by

$$P_{01} \times Q_{01} = \frac{\sum p_1 q_1}{\sum p_0 q_0}$$

The factor reversal test is satisfied only by the Fisher's ideal price index as shown below:

$$P_{01} = \sqrt{\frac{\sum p_1 q_0}{\sum p_0 q_0} \times \frac{\sum p_1 q_1}{\sum p_0 q_1}}$$

Changing p to q and q to p , we get the quantity index:

$$Q_{01} = \sqrt{\frac{\sum q_1 p_0}{\sum q_0 p_0} \times \frac{\sum q_1 p_1}{\sum q_0 p_1}}$$

Multiplying P_{01} and Q_{01} , we get

$$\sqrt{\frac{\sum p_1 q_0}{\sum p_0 q_0} \times \frac{\sum p_1 q_1}{\sum p_0 q_1} \times \frac{\sum q_1 p_0}{\sum q_0 p_0} \times \frac{\sum q_1 p_1}{\sum q_0 p_1}} = \sqrt{\frac{\sum p_1 q_1}{\sum p_0 q_0} \times \frac{\sum q_1 p_1}{\sum q_0 p_0}} = \frac{\sum p_1 q_1}{\sum p_0 q_0}$$

This result implies that the Fisher's index formula could be used for constructing both price and quantity indexes.

11.10.3 Circular Test

This test measure the change in price over a period of years when the shifting of base is desirable in a circular fashion. For example, if an index is constructed for the year 2000 with the base of 1999 and another index for 1999 with the base of 1998, then it should be possible for us to directly get an index for the year 2000 with the base of 1998. If the index calculated directly does not give an inconsistent value, the circular test is said to be satisfied. If P_{ab} is price index for period 'b' with base period 'a'; P_{bc} is the price index for period 'c' with base period 'b' and P_{ca} is the price index for period 'a' with base period 'c', then an index is said to satisfy the circular test provided: $P_{ab} \times P_{bc} \times P_{ca} = 1$

Example 11.19: Calculate Fisher's ideal index from the following data and prove that it satisfy both Time Reversal and Factor Reversal test.

Items	Base Year		Current Year	
	Price	Quantity	Price	Quantity
A	6	50	10	60
B	2	100	2	120
C	4	60	6	60

[Delhi Univ., B.Com.(Pass), 2005]

Solution: The following table presents the necessary information for constructing the Fisher's ideal index number:

Items	Base Year		Current Year		$p_0 q_0$	$p_0 q_1$	$p_1 q_0$	$p_1 q_1$
	p_0	q_0	p_1	q_1				
A	6	50	10	60	300	360	500	600
B	2	100	2	120	200	240	200	240
C	4	60	6	60	240	240	360	360
					740	840	1060	1200

$$\text{Fisher's index, } P_{01} = \sqrt{\frac{\sum p_1 q_0}{\sum p_0 q_0} \times \frac{\sum p_1 q_1}{\sum p_0 q_1}} \times 100 = \sqrt{\frac{1060 \times 1200}{740 \times 840}} \times 100 = 143$$

Time Reversal Test, i.e. $P_{01} \times P_{10} = 1$

$$P_{01} = \sqrt{\frac{\sum p_1 q_0}{\sum p_0 q_0} \times \frac{\sum p_1 q_1}{\sum p_0 q_1}} = \sqrt{\frac{1060}{740} \times \frac{1200}{840}} \quad \text{and} \quad P_{10} = \sqrt{\frac{\sum p_0 q_1}{\sum p_1 q_1} \times \frac{\sum p_0 q_0}{\sum p_1 q_0}} = \sqrt{\frac{840}{1200} \times \frac{740}{1060}}$$

$$P_{01} \times P_{10} = \sqrt{\frac{1060}{740} \times \frac{1200}{840} \times \frac{840}{1200} \times \frac{740}{1060}} = 1$$

Hence, the time reversal test is satisfied.

Factor Reversal Test, i.e. $P_{01} \times Q_{01} = \frac{\sum p_1 q_1}{\sum p_0 q_0}$

$$P_{01} \times Q_{01} = \sqrt{\frac{\sum p_1 q_0}{\sum p_0 q_0} \times \frac{\sum p_1 q_1}{\sum p_0 q_1} \times \frac{\sum q_1 p_0}{\sum q_0 p_0} \times \frac{\sum q_1 p_1}{\sum q_0 p_1}} = \sqrt{\frac{1060}{740} \times \frac{1200}{840} \times \frac{840}{740} \times \frac{1200}{1060}}$$

$$= \sqrt{\frac{(1200)^2}{(740)^2}} = \frac{1200}{740} = \frac{\sum p_1 q_1}{\sum p_0 q_0}$$

Hence, factor reversal test is satisfied.

Example 11.20: Construct Fisher's price index using following data and show how it satisfies the time and factor reversal tests.

Commodity	2002		2003	
	Quantity	Price	Quantity	Price
A	20	12	30	14
B	13	14	15	20
C	12	10	20	15
D	8	6	10	4
E	5	8	5	6

Solution: Table 11.17 presents all the necessary information for constructing the Fisher's ideal index number.

Table 11.17 Calculation of Fisher's Ideal Index

Commodity	2002		2003		$p_1 q_0$	$p_0 q_0$	$q_1 p_1$	$p_0 q_1$
	q_0	p_0	q_1	p_1				
A	20	12	30	14	280	240	420	360
B	13	14	15	20	260	182	300	210
C	12	10	20	15	180	120	300	200
D	8	6	10	4	32	48	40	60
E	5	8	5	6	30	40	30	40
					782	630	1090	870

Fisher's ideal price index

$$P_{01} = \sqrt{\frac{\sum p_1 q_0}{\sum p_0 q_0} \times \frac{\sum p_1 q_1}{\sum p_0 q_1}} = \sqrt{\frac{782}{630} \times \frac{1090}{870}} = 1.2471$$

Time Reversal Test: This test is satisfied when $P_{01} \times P_{10} = 1$

$$P_{10} = \sqrt{\frac{\sum p_0 q_1}{\sum p_1 q_1} \times \frac{\sum p_0 q_0}{\sum p_1 q_0}} = \sqrt{\frac{870}{1090} \times \frac{630}{782}} = 0.8019$$

$$P_{01} \times P_{10} = \sqrt{\frac{782}{630} \times \frac{1090}{870} \times \frac{870}{1090} \times \frac{630}{782}} = \sqrt{1.2471 \times 0.8019} = 1$$

Hence, time reversal test is satisfied.

Factor Reversal Test: This test is satisfied when $P_{01} \times Q_{01} = \frac{\sum p_1 q_1}{\sum p_0 q_0}$

$$P_{01} = \sqrt{\frac{\sum p_1 q_0}{\sum p_0 q_0} \times \frac{\sum p_1 q_1}{\sum p_0 q_1}} \text{ and } Q_{01} = \sqrt{\frac{\sum q_1 p_0}{\sum q_0 p_0} \times \frac{\sum q_1 p_1}{\sum q_0 p_1}}$$

Thus, $P_{01} \times Q_{01} = \sqrt{\frac{782}{630} \times \frac{1090}{870} \times \frac{870}{630} \times \frac{1090}{782}} = \frac{1090}{630}$ which is the value of $\frac{\sum p_1 q_1}{\sum p_0 q_0}$

Hence, factor reversal test is also satisfied.

Example 11.21: Calculate Fisher's ideal index from the data given below and show that it satisfies the time reversal test.

Commodity	2000		2001	
	Price	Quantity	Price	Quantity
A	10	49	12	50
B	12	25	15	20
C	18	10	20	12
D	20	5	40	2

Solution: Table 11.18 presents information necessary for Fisher's method to calculate the index.

Table 11.18 Calculation of Fisher's Ideal Index

Commodity	2000		2001		$p_1 q_0$	$p_0 q_0$	$p_1 q_1$	$p_0 q_1$
	p_0	q_0	p_1	q_1				
A	10	49	12	50	588	490	600	500
B	12	25	15	20	375	300	300	240
C	18	10	20	12	200	180	240	216
D	20	5	40	2	200	100	80	40
					1363	1070	1220	996

$$\text{Fisher's index } P_{01} = \sqrt{\frac{\sum p_1 q_0}{\sum p_0 q_0} \times \frac{\sum p_1 q_1}{\sum p_0 q_1}} \times 100 = \sqrt{\frac{1363}{1070} \times \frac{1220}{996}} \times 100 = 124.9$$

Time Reversal Test: This test is satisfied when $P_{01} \times P_{10} = 1$

$$P_{10} = \sqrt{\frac{\sum p_0 q_1}{\sum p_1 q_1} \times \frac{\sum p_0 q_0}{\sum p_1 q_0}}$$

Thus,

$$P_{01} \times P_{10} = \sqrt{\frac{\sum p_1 q_0}{\sum p_0 q_0} \times \frac{\sum p_1 q_1}{\sum p_0 q_1} \times \frac{\sum p_0 q_1}{\sum p_1 q_1} \times \frac{\sum p_0 q_0}{\sum p_1 q_0}}$$

$$= \sqrt{\frac{1363}{1070} \times \frac{1220}{996} \times \frac{996}{1220} \times \frac{1070}{1363}} = \sqrt{1} = 1$$

Hence, the time reversal test is satisfied.

Example 11.22: With the help of the following data, show that the index number calculated on the basis of arithmetic mean is not reversible while the index number calculated on the basis of geometric mean is reversible. Make comparison between arithmetic mean and geometric mean:

Commodity	Price in 1998	Price in 1999
A	40	60
B	50	80
C	20	40
D	20	10

[Delhi Univ., B.Com. (Hons.), 2003]

Solution: Calculations for price relatives using 1998 and 1999 as base year are shown in the table below:

Commodity	Price in 1998 (p_0)	Price in 1999 (p_1)	Price Relative (p_1/p_0) $\times 100$ Using 1998 as Base	Price Relative (p_0/p_1) $\times 100$ Using 1999 as Base
A	40	60	150	66.67
B	50	80	160	62.50
C	20	40	200	50.00
D	20	10	50	200.00
			560	379.17

$$\text{A.M. of price relatives (with 1998 as base)} = \frac{560}{4} = 140 (= P_{01})$$

$$\text{A.M. of price relatives (with 1999 as base)} = \frac{379.17}{4} = 94.79 (= P_{10})$$

Time Reversal Test is said to be satisfied if $P_{01} \times P_{10} = 1$ (ignoring the factor 100). In the given situation the A.M. of price relatives $P_{01} \times P_{10} = 1.4 \times 0.9479 \neq 1$

$$\begin{aligned} \text{G.M. of price relatives (ignoring the factor 100) with 1998 as base is} \\ = (1.5 \times 1.6 \times 2 \times 0.5)^{1/4} = 1.2446 (= P_{01}) \end{aligned}$$

$$\begin{aligned} \text{G.M. of price relatives (ignoring the factor 100) with 1999 as base is} \\ = (0.6667 \times 0.6250 \times 0.5 \times 2)^{1/4} = 0.8034 (= P_{10}) \end{aligned}$$

In case of G.M., the relation $P_{01} \times P_{10} = 1$. Further it may be noted that the index number using A.M. is higher than the index number using G.M.

Example 11.23: If the ratio between Laspeyers and Paasche's index number is 28 : 27. Find missing figure in the following table.

Commodities	Base Year		Current Year	
	Price	Quantity	Price	Quantity
X	1	10	2	5
Y	1	5	–	2

Solution: Calculations required to find the missing figure are shown below:

Items	Base Year		Current Year		$p_0 q_0$	$p_0 q_1$	$p_1 q_0$	$p_1 q_1$
	p_0	q_0	p_1	q_1				
X	1	10	2	5	10	5	20	10
Y	1	5	x	2	5	2	$5x$	$2x$

$$\text{Laspeyres index, } L = \frac{\sum p_1 q_0}{\sum p_0 q_0} \times 100 = \frac{20 + 5x}{15} \times 100$$

$$\text{Paasche's index, } P = \frac{\sum p_1 q_1}{\sum p_0 q_1} \times 100 = \frac{10 + 2x}{7} \times 100$$

Since $\frac{L}{P} = \frac{28}{27}$, therefore

$$\frac{\frac{20 + 5x}{15} \times 100}{\frac{10 + 2x}{7} \times 100} = \frac{20 + 5x}{15} \times \frac{7}{10 + 2x} = \frac{28}{27}$$

$$\frac{4 + x}{3} \times \frac{7}{2(5 + x)} = \frac{28}{27} \quad \text{or} \quad \frac{4 + x}{5 + x} = \frac{8}{9}$$

$$9(4 + x) = 8(5 + x)$$

$$36 + 9x = 40 + 8x \quad \text{or} \quad x = 40 - 36 = 4$$

Hence, the missing figure is 4.

Example 11.24: Calculate Fisher's price index for the following data and prove that it satisfies both Time Reversal and Factor Reversal test:

Items	Quantity (kg)		Price (₹)	
	2001	2002	2001	2002
Wheat	8	10	20	30
Sugar	6	9	14	18
Tea	2	5	15	20

[Delhi Univ., B.Com. (Pass), 2004]

Solution: Calculations required to calculate Fisher's price index are shown below:

Commodity	Quantity (kg)		Price (₹ per kg)		$p_1 q_0$	$p_0 q_0$	$p_1 q_1$	$p_0 q_1$
	2001 (q_0)	2002 (q_1)	2001 (p_0)	2002 (p_1)				
Wheat	8	10	20	30	240	160	300	200
Sugar	6	9	14	18	108	84	162	126
Tea	2	5	15	20	40	30	100	75
					388	274	562	401

$$P_{01} = \sqrt{\frac{\Sigma p_1 q_0}{\Sigma p_0 q_0} \times \frac{\Sigma p_1 q_1}{\Sigma p_0 q_1}} \times 100 = \sqrt{\frac{388}{274} \times \frac{562}{401}} \times 100 = \sqrt{1.985} \times 100 = 140.88 \text{ (App.)}$$

Time Reversal Test: $P_{01} \times P_{10} = 1$

$$P_{01} \times P_{10} = \sqrt{\frac{\Sigma p_1 q_0}{\Sigma p_0 q_0} \times \frac{\Sigma p_1 q_1}{\Sigma p_0 q_1}} \times \sqrt{\frac{\Sigma p_0 q_1}{\Sigma p_1 q_1} \times \frac{\Sigma p_0 q_0}{\Sigma p_1 q_0}} = \sqrt{\frac{388}{274} \times \frac{562}{401} \times \frac{401}{562} \times \frac{274}{388}} = \sqrt{1} = 1$$

Hence, Fisher's index satisfies time reversal test.

$$\text{Factor Reversal Test: } P_{01} \times Q_{01} = \frac{\Sigma p_1 q_1}{\Sigma p_0 q_0} = V_{01}$$

$$\begin{aligned} P_{01} \times Q_{01} &= \sqrt{\frac{\Sigma p_1 q_0}{\Sigma p_0 q_0} \times \frac{\Sigma p_1 q_1}{\Sigma p_0 q_1}} \times \sqrt{\frac{\Sigma q_1 p_0}{\Sigma q_0 p_0} \times \frac{\Sigma q_1 p_1}{\Sigma q_0 p_1}} = \sqrt{\frac{388}{274} \times \frac{562}{401}} \times \sqrt{\frac{401}{274} \times \frac{562}{388}} \\ &= \sqrt{\frac{562}{274} \times \frac{562}{274}} = \frac{562}{274} = \frac{\Sigma p_1 q_1}{\Sigma p_0 q_0} = V_{01} \end{aligned}$$

11.11 CHAIN INDEXES

The various formulae discussed so far assumed that the base period is any fixed period. The base period is the immediately preceding year of the current year. Moreover, the index of a given period on a given fixed base was not affected by changes in the relevant values of any other year. But in the chain base method, the data of each period is related with that of the immediately preceding period and not with any fixed period. This means that for the index of 2000, the base would be 1999 and for the index of 1999, the base would be 1998 and similarly of the index of 1998 the base would be 1997. Such index numbers are very useful in comparing current period data with the preceding period's data. Fixed base index in such a case does not give an appropriate comparison, because all prices are based on the fixed base period which may be far away for the current period and the preceding period.

For constructing an index by the chain base method, a series of indexes are computed for each period with preceding period as the base. These indexes are known as *link index* or *link relatives*. The steps of calculating link relatives are summarized below:

- Express the data of a particular period as a percentage of the preceding period's data. This is called the *link relative*.
- These link relatives can be chained together. This is done by multiplying the link relative of the current year by the *chain index* of the previous year and dividing the product by 100. Thus,

$$\text{Chain index for current year} = \frac{\text{Link relative of current year} \times \text{Chain index of previous year}}{100}$$

The chain index is useful for long-term comparison whereas link relatives are used for a comparison with the immediately preceding period. The fixed base indexes compiled from the original data and the chain indexes compiled from link relatives give the same value of index provided there is only one commodity whose indexes are being constructed.

Remarks Chain relatives differ from fixed base relatives in computation. Chain relatives are computed from link relatives whereas fixed base relatives are computed directly from original data.

$$\text{Link relative} = \frac{\text{Price relative for the current period}}{\text{Price relative for the previous period}} \times 100$$

$$\text{Price relative} = \frac{\text{Current period's link relative} \times \text{Previous period's price relatives}}{100}$$

Multiplying the link relatives $P_{01}, P_{12}, P_{23}, \dots, P_{(n-1)n}$ successively is known as the chaining process that gives link relatives with a common base:

$$\begin{aligned} P_{01} &= \text{First link} \\ P_{02} &= P_{01} \times P_{12} \\ P_{03} &= (P_{01} \times P_{12}) \times P_{23} = P_{02} \times P_{23} \\ &\vdots \\ &\vdots \\ P_{0n} &= P_{0(n-1)} \times P_{(n-1)n} \end{aligned}$$

Advantages

The following are a few advantages of chain base index:

- (i) The chain base indexes enable us to make comparisons with the *previous and not any distant past period*. Thus, these index are very useful in the analysis of business data.
- (ii) The chain base method permits us to introduce new commodities and delete the existing ones which are obsolete without any recalculation of the entire series.
- (iii) The index numbers calculated by the chain base method are relatively free from cyclical and seasonal variations.

Disadvantage

The main disadvantage of the chain base index is that it is not useful for long-term comparisons of chained percentages in a time series. The process of chaining link relatives is computationally difficult.

Conversion of chain base index (CBI) to fixed base index (FBI)

$$\text{Current period FBI} = \frac{\text{Current period CBI} \times \text{Previous period FBI}}{100}$$

Example 11.25: Construct an index by the chain base method based on the following data of the wholesale prices of a certain commodity.

<i>Year</i>	<i>Price</i>	<i>Year</i>	<i>Price</i>
1994	37	2000	48
1995	39	2001	49
1996	43	2002	54
1997	48	2003	56
1998	48	2004	57
1999	52		

Solution: Computation of the chain base index number is shown in Table 11.19.

Table 11.19 Chain Base Indexes

<i>Year</i>	<i>Price</i>	<i>Link Relative</i>	<i>Chain Base Index Numbers</i> (Base Year 1985 = 100)
1994	37	100.00	100
1995	39	$(39/37) \times 100 = 105.41$	$(105.41/100) \times 100 = 105.41$
1996	43	$(43/39) \times 100 = 110.26$	$(110.26/100) \times 105.41 = 116.23$
1997	48	$(48/43) \times 100 = 111.63$	$(111.63/100) \times 116.23 = 129.75$
1998	48	$(48/48) \times 100 = 100.00$	$(100/100) \times 129.75 = 129.75$
1999	52	$(52/48) \times 100 = 108.33$	$(108.33/100) \times 129.75 = 140.56$
2000	48	$(48/52) \times 100 = 92.30$	$(92.30/100) \times 140.56 = 129.74$
2001	49	$(49/48) \times 100 = 102.08$	$(102.08/100) \times 129.74 = 132.43$
2002	54	$(54/49) \times 100 = 110.20$	$(110.20/100) \times 132.43 = 145.95$
2003	56	$(56/54) \times 100 = 103.70$	$(103.70/100) \times 145.95 = 151.35$
2004	57	$(57/56) \times 100 = 101.79$	$(110.79/100) \times 151.35 = 154.06$

Example 11.26: Prepare fixed base index numbers from the chain base index numbers given below:

Year	:	1996	1997	1998	1999	2000	2001
Chain Index	:	94	104	104	93	103	102

Solution: Computation of fixed base indexes is shown in Table 11.20 using the following formula:

$$\text{Fixed base index (FBI)} = \frac{\text{Current period CBI} \times \text{Previous period FBI}}{100}$$

Table 11.20 Fixed Base Index Numbers

<i>Year</i>	<i>Chain Index</i>	<i>Fixed Base Index</i>
1996	94	94
1997	104	$\frac{104 \times 94}{100} = 97.76$
1998	104	$\frac{104 \times 97.76}{100} = 101.67$
1999	93	$\frac{93 \times 101.67}{100} = 94.55$
2000	103	$\frac{103 \times 94.55}{100} = 97.39$
2001	102	$\frac{102 \times 97.39}{100} = 99.34$

Example 11.27: Calculate the chain base index number and fixed base index number from the following data:

Commodity	1998	1999	2000	2001	2002
A	4	6	8	10	12
B	16	20	24	30	36
C	8	10	16	20	24

Solution: Computation of the chain index number and fixed base index number is shown in Tables 11.21 and 11.22.

Table 11.21 Chain Base Index Number

Commodity	Link Relatives Based on Preceding Year				
	1998	1999	2000	2001	2002
A	100	$\frac{6}{4} \times 100 = 150$	$\frac{8}{6} \times 100 = 133.33$	$\frac{10}{8} \times 100 = 125$	$\frac{12}{10} \times 100 = 120$
B	100	$\frac{20}{16} \times 100 = 125$	$\frac{24}{20} \times 100 = 120$	$\frac{30}{24} \times 100 = 125$	$\frac{36}{30} \times 100 = 120$
C	100	$\frac{10}{8} \times 100 = 125$	$\frac{16}{10} \times 100 = 160$	$\frac{20}{16} \times 100 = 125$	$\frac{24}{20} \times 100 = 120$
Total link relatives	300	400	413.33	375	360
Average link relatives	100	133.33	137.78	125	120
Chain base index	100	$\frac{133.33 \times 100}{100} = 133.33$	$\frac{137.78 \times 133.33}{100} = 183.70$	$\frac{125 \times 183.70}{100} = 229.63$	$\frac{120 \times 229.63}{100} = 275.56$

Table 11.22 Fixed Base Index Number

Commodity	Price Relatives (Base 1998 = 100)				
	1998	1999	2000	2001	2002
A	100	$\frac{6}{4} \times 100 = 150$	$\frac{8}{4} \times 100 = 200$	$\frac{10}{4} \times 100 = 250$	$\frac{12}{4} \times 100 = 300$
B	100	$\frac{20}{16} \times 100 = 125$	$\frac{24}{16} \times 100 = 150$	$\frac{30}{16} \times 100 = 187.5$	$\frac{36}{16} \times 100 = 225$
C	100	$\frac{10}{8} \times 100 = 125$	$\frac{16}{8} \times 100 = 200$	$\frac{20}{8} \times 100 = 250$	$\frac{24}{8} \times 100 = 300$
Total	300	400	550	687.5	825
Average (Fixed base index number)	100	133.33	183.33	229.17	275

Example 11.28: Shift the base from 2004 to 2006 in the data given below:

<i>Year</i>	<i>Index (2004 = 100)</i>
2001	87.27
2002	90.91
2003	95.40
2004	100.00
2005	104.00
2006	106.00
2007	112.00

Solution: Calculations required to shift the base from 2004 to 2006 are shown below:

<i>Year</i>	<i>Index (2004 = 100)</i>	<i>Index (2006 = 100)</i>
2001	87.27	$\frac{87.27}{106} \times 100 = 82.33$
2002	90.91	$\frac{90.91}{106} \times 100 = 85.76$
2003	95.40	$\frac{95.4}{106} \times 100 = 90.00$
2004	100.00	$\frac{100}{106} \times 100 = 94.34$
2005	104.00	$\frac{104}{106} \times 100 = 98.11$
2006	106.00	$\frac{106}{106} \times 100 = 100$
2007	112.00	$\frac{112}{106} \times 100 = 105.66$

Example 11.29: Given below are two sets of indices. For the purpose of continuity of records, you are required to construct a combined series with the year 1993 as the base:

<i>Year</i>	<i>First Set Price Relative</i>	<i>Second set Link Relatives</i>
1990	100	–
1991	120	–
1992	125	–
1993	150	–
1994	–	110
1995	–	120
1996	–	95
1997	–	105

Solution: Price relative = $\frac{\text{Link relative} \times \text{Previous year price relative}}{100}$

$$\text{For 1994 : Price Relative} = \frac{110 \times 150}{100} = 165$$

$$\text{For 1995 : Price Relative} = \frac{120 \times 165}{100} = 198 \text{ and so on.}$$

Year	Price Relative With 1990 = 100	Price Relative With 1993 as Base
1989	87.27	
1990	100	$100 \times \frac{100}{150} = 66.66$
1991	120	$120 \times \frac{100}{150} = 80$
1992	125	$125 \times \frac{100}{150} = 83.33$
1993	150	$150 \times \frac{100}{150} = 100$
1994	$\frac{150 \times 110}{100} = 165$	$165 \times \frac{100}{150} = 110$
1995	$\frac{165 \times 120}{100} = 198$	$198 \times \frac{100}{150} = 132$
1996	$\frac{198 \times 95}{100} = 188.1$	$188.1 \times \frac{100}{150} = 125.4$
1997	$188.1 \times \frac{10^5}{100} = 197.5$	$197.5 \times \frac{100}{150} = 131.66$

Example 11.30: Prepare a spliced series of index numbers with 2003 as base from the following series:

Year	1998	1999	2000	2001	2002	2003	2004
Index A	100	120	135	—	—	—	—
Index B	—	—	100	115	125	145	—
Index C	—	—	—	—	—	100	110

[Delhi Univ., BCom (Hons.), 2005]

Solution: The spliced series of index numbers with 2003 as base is shown below:

Year	Index A	Index B	Splicing Index A to B with Base 2000	Index C	Splicing All Index Number with Base 2003
1998	100	—	$\frac{100}{135} \times 100 = 74.07$	—	$\frac{100}{145} \times 74.07 = 51.082$
1999	120	—	$\frac{100}{135} \times 120 = 88.89$	—	$\frac{100}{145} \times 88.89 = 61.30$
2000	135	100	—	100	$\frac{100}{145} \times 100 = 68.97$
2001	—	115	—	115	$\frac{100}{145} \times 115 = 79.31$
2002	—	125	—	125	$\frac{100}{145} \times 125 = 86.21$
2003	—	145	—	145	100
2004	—	—	—	110	110

Example 11.31: A price index number series was started in 1997 as base. By 2001, it rose by 15 per cent. The link relative for year 2002 was 95. In this year, a new series was started. This new series rose by 25 per cent in next year. During year 2004, the price level was 5 per cent higher than 2003. However in 2005, they were 6 per cent higher than 2004. Splice the two series and calculate the index number for various years by shifting base to 2003.

Solution: Calculations required for splicing of indices are shown below:

Year	Index A (1997 = 100)	Index B (2002 = 100)	Splicing A to B (2002 = 100)
1997	100	—	$\frac{100}{109.25} \times 100 = 91.53$
2001	$100 + \left(\frac{15}{100} \times 100 \right) = 115$	—	$\frac{100}{109.25} \times 115 = 105.26$
2002	$\frac{95 \times 115}{100} = 109.25^*$	100	100
2003	—	$100 + \frac{25}{100} \times 100 = 125$	125
2004	—	$125 + \frac{5}{100} \times 125 = 131.25$	131.25
2005	—	$131.25 + \frac{6}{100} \times 131.25 = 139.125$	139.125

* Price for year 2002 = $\frac{\text{Link relative for 2002} \times \text{Previous year price}}{100}$

Shifting Base to 2003

Year	Index (2002 = 100)	Index (2003 = 100)
1997	91.53	$\frac{91.53}{125} \times 100 = 73.224$
2001	105.26	$\frac{105.26}{125} \times 100 = 84.21$
2002	100	$\frac{100}{125} \times 100 = 80$
2003	125	100
2004	131.25	$\frac{131.25}{125} \times 100 = 105$
2005	139.125	$\frac{139.125}{125} \times 100 = 111.3$

Example 11.32: The following table gives the annual income of a person and the general price index number for the period 1988 to 1996. Prepare index number to show the changes in the real income of the person.

Year	Annual Income (₹)	Price Index Number
1988	36,000	100
1989	42,000	120
1990	50,000	145
1991	55,000	160
1992	60,000	250
1993	64,000	320
1994	68,000	450
1995	72,000	530
1996	75,000	600

[Delhi Univ., B.Com. (H), 2000]

Solution: Calculations required to calculate real income index number are shown below:

Year	Annual Income (₹)	Price Index Number	Real Wage = $\frac{\text{Annual income}}{\text{Price index}} \times 100$	Real Wage Index 1988 = 100
1988	36,000	100	$\frac{36,000}{100} \times 100 = 36,000$	100
1989	42,000	120	$\frac{42,000}{120} \times 100 = 35,000$	$\frac{35,000}{36,000} \times 100 = 97.22$
1990	50,000	145	$\frac{50,000}{145} \times 100 = 34482.7$	$\frac{34,482.7}{36,000} \times 100 = 95.79$

(Continued)

Year	Annual Income (₹)	Price Index Number	Real Wage = $\frac{\text{Annual income}}{\text{Price index}} \times 100$	Real Wage Index 1988 = 100
1991	55,000	160	$\frac{55,000}{160} \times 100 = 34,375$	$\frac{34,375}{36,000} \times 100 = 95.49$
1992	60,000	250	$\frac{60,000}{250} \times 100 = 24,000$	$\frac{24,000}{36,000} \times 100 = 66.67$
1993	64,000	320	$\frac{64,000}{320} \times 100 = 20,000$	$\frac{20,000}{36,000} \times 100 = 55.56$
1994	68,000	450	$\frac{68,000}{450} \times 100 = 15,111.1$	$\frac{15,111.1}{36,000} \times 100 = 41.98$
1995	72,000	530	$\frac{72,000}{530} \times 100 = 13,584.9$	$\frac{13,584.9}{36,000} \times 100 = 37.74$
1996	75,000	600	$\frac{75,000}{600} \times 100 = 12,500$	$\frac{12,500}{36,000} \times 100 = 34.72$

Self-practice Problems 11C

- 11.20** Calculate Fisher's Ideal index from the data given below and show that it satisfies the time reversal and factor reversal tests.

Commodity	Base Year		Current Year	
	Quantity	Price	Quantity	Price
A	12	10	15	12
B	15	7	20	5
C	24	5	20	9
D	5	16	5	14

- 11.21** Splice the following two index number series, continuing series A forward and series B backward.

Year	: 1998	1999	2000	2001	2002	2003
Series A :	100	120	150	—	—	—
Series B :	—	—	100	110	120	150

- 11.22** Calculate the chain base index number chained to 1994 from the average price of following three commodities:

Commodity	1999	2000	2001	2002	2003
Wheat	4	6	8	10	12
Rice	16	20	24	30	36
Sugar	8	10	16	20	24

- 11.23** The following table gives the annual income of a clerk and the general index number of price during 1994–98. Prepare the index number to show the changes in the real income of the teacher.

Year	Income (₹)	Price Index No.	Year	Income (₹)	Price Index No.
1994	36,000	100	1999	64,000	290
1995	42,000	104	2000	68,000	300
1996	50,000	115	2001	72,000	320
1997	55,000	160	2002	75,000	330
1998	60,000	280	—	—	—

- 11.24** From the following average price of groups of commodities given in rupees per unit, find the chain base index number with 1994 as the base year:

Group	1994	1995	1996	1997	1998
I	2	3	4	5	6
II	8	10	12	15	18
III	4	5	18	10	12

[Agra Univ., B.Com., 1998]

- 11.25** Given the following data:

Year	Weekly Take-home Pay (Wages)	Consumer Price Index
1998	109.50	112.8
1999	112.20	118.2
2000	116.40	127.4
2001	125.08	138.2
2002	135.40	143.5
2003	138.10	149.8

- What was the real average weekly wage for each year?
 - In which year did the employees have the greatest buying power?
 - What percentage increase in the weekly wages for the year 2003 is required (if any) to provide the same buying power that the employees enjoyed in the year in which they had the highest real wages?
- 11.26** Using the following data construct Fisher's ideal index and show that it satisfies the factor reversal and time reversal tests:

Commodity	Price (in ₹/Unit)		Number of Units	
	Base Year	Current Year	Base Year	Current Year
A	6	8	10	12
B	10	10	5	8
C	5	7	8	10
D	15	20	12	15
E	20	25	15	10

- 11.27** From the data given below, calculate Fisher's ideal index and show that it satisfies the time reversal and factor reversal test:

Commodity	1998		1999	
	Price	Quantity	Price	Quantity
A	12	20	14	30
B	14	13	20	15
C	10	12	15	20
D	6	8	4	10
E	8	5	6	5

- 11.28** Calculate the index number by using Paasche's method, and Fisher's method.

Commodity	p_1	q_1	p_0	q_0
A	5	14	3	8
B	8	18	6	25
C	3	25	1	40
D	15	36	12	48
E	9	14	7	18
F	7	13	5	19

Hints and Answers

11.20

Commodity	q_0	p_0	q_1	p_1	p_1q_0	p_0q_0	p_1q_1	p_0q_1
A	12	10	15	12	144	120	180	150
B	15	7	20	5	75	105	100	140
C	24	5	20	9	216	120	180	100
D	5	16	5	14	70	80	70	80
					505	425	530	470

$$P_{01} = \sqrt{\frac{\sum p_1q_0}{\sum p_0q_0} \times \frac{\sum p_1q_1}{\sum p_0q_1}} = \sqrt{\frac{505}{425} \times \frac{530}{470}} = 1.1576$$

$$P_{10} = \sqrt{\frac{\sum p_0q_1}{\sum p_1q_1} \times \frac{\sum p_0q_0}{\sum p_1q_0}} = \sqrt{\frac{470}{530} \times \frac{425}{505}} = 0.8639$$

Time Reversal Test

$$P_{01} \times P_{10} = \sqrt{\frac{505}{425} \times \frac{530}{670} \times \frac{470}{530} \times \frac{425}{505}} = \sqrt{1} = 1$$

Factor Reversal Test

$$P_{01} \times Q_{01} = \frac{\sum p_1q_1}{\sum p_0q_0};$$

$$Q_{01} = \sqrt{\frac{\sum q_1p_0}{\sum q_0p_0} \times \frac{\sum q_1p_1}{\sum q_0p_1}}$$

$$= \sqrt{\frac{470}{425} \times \frac{530}{505}} = 1.0773$$

$$P_{01} \times Q_{01} = \sqrt{\frac{505}{425} \times \frac{530}{470} \times \frac{470}{425} \times \frac{530}{505}}$$

$$= \sqrt{\frac{530}{425} \times \frac{530}{425}}$$

$$= \frac{530}{425} \text{ which is equal to } \frac{\sum p_1q_1}{\sum p_0q_0}$$

11.21

Year	Series A	Series B	Series B Spliced to A	Series A Spliced to B
1998	100	—	—	$\left(\frac{100}{150}\right) \times 100$ = 66.66
1999	120	—	—	$\left(\frac{100}{150}\right) \times 120$ = 80.00

Year	Series A	Series B	Series B Spliced to A	Series A Spliced to B
2000	150	100	$\left(\frac{150}{100}\right) \times 100$ = 150	$\left(\frac{100}{150}\right) \times 150$ = 100.00
2001	—	110	$\left(\frac{150}{100}\right) \times 110 = 165$	
2002	—	120	$\left(\frac{150}{100}\right) \times 120 = 180$	
2003	—	150	$\left(\frac{150}{100}\right) \times 150 = 225$	

11.22

Commodity	Relatives Based on the Preceding Year				
	1999	2000	2001	2002	2003
Wheat	100	150	133.33	125	120
Rice	100	125	120.00	125	120
Sugar	100	125	160.00	125	120
Total	300	400	413.33	375	360
Average of link relatives	100	133.33	137.78	125	120
Chain index (1999 = 100)	100	$\frac{133.33 \times 100}{100} = 133.33$	$\frac{137.78 \times 133.33}{100} = 183.70$	$\frac{125 \times 183.70}{100} = 229.63$	$\frac{120 \times 229.63}{100} = 275.56$

11.23

Year	Income (₹)	Price Index	Real Income	Real Income Index
1994	360	100	$(360/100) \times 100$ = 360.00	100.00
1995	420	104	$(420/104) \times 100$ = 403.85	112.18
1996	500	115	$(500/115) \times 100$ = 434.78	120.77
1997	550	160	$(550/160) \times 100$ = 343.75	95.49
1998	600	280	$(600/280) \times 100$ = 214.29	59.52
1999	640	290	$(640/290) \times 100$ = 220.69	61.30
2000	680	300	$(680/300) \times 100$ = 226.67	62.96
2001	720	320	$(720/320) \times 100$ = 225.00	62.52
2002	750	330	$(750/330) \times 100$ = 227.27	63.13

11.24

Group	1994		1995		1996		1997		1998	
	Price	Link Relative	Price	Link Relative	Price	Link Relative	Price	Link Relative	Price	Link Relative
I	2	100	3	150	4	133.3	5	125	6	120
II	8	100	10	125	12	120.0	15	125	18	120
III	4	100	5	125	18	160.0	10	125	12	120
Total		300		400		413.3		375		360
Average of link relatives		100		133.33		137.77		125		120
Chain index		100		$\frac{133.33}{100} \times 100$		$\frac{137.77}{100} \times 133.33$		$\frac{125}{100} \times 183.69$		$\frac{120}{100} \times 229.61$
(1994 = 100)				= 133.33		= 183.69		= 229.61		= 275.53

11.25 (a) Average weekly wage can be obtained by using the following formula:

$$\text{Real wage} = \frac{\text{Money wage}}{\text{Price index}} \times 100$$

Year	Weekly Take-home Pay (₹)	Consumer Price Index	Real wages
1998	109.50	112.8	$\frac{109.5}{112.8} \times 100$ = 97.07
1999	112.20	118.2	$\frac{112.2}{118.2} \times 100$ = 94.92
2000	116.40	127.4	$\frac{116.4}{127.4} \times 100$ = 91.37
2001	125.08	138.2	$\frac{125.08}{138.2} \times 100$ = 90.51
2002	135.40	143.5	$\frac{135.4}{143.5} \times 100$ = 94.36
2003	138.10	149.8	$\frac{138.10}{149.8} \times 100$ = 92.19

(b) Since real wage was maximum in the year 1998, the employees had the greatest buying power in that year.

(c) The percentage increase in the weekly wages for the year 2003 required to provide the same buying power that the employees had in 1998:

$$\text{Absolute difference} = 97.07 - 92.19 = 4.88.$$

11.26

Comm	p_0	p_1	q_0	q_1	p_1q_0	p_0q_0	p_1q_1	p_0q_1
A	6	8	10	12	80	60	96	72
B	10	10	5	8	50	50	80	80
C	5	7	8	10	56	40	70	60
D	15	20	12	15	240	180	300	225
E	20	25	15	10	375	300	250	200
					801	630	796	637

$$P_{01} = \sqrt{\frac{\sum p_1q_0}{\sum p_0q_0} \times \frac{\sum p_1q_1}{\sum p_0q_1}};$$

$$P_{10} = \sqrt{\frac{\sum p_0q_1}{\sum p_1q_1} \times \frac{\sum p_0q_0}{\sum p_1q_0}}$$

$$\text{Time Reversal Test: } P_{01} \times P_{10} = 1$$

$$P_{01} \times P_{10} = \sqrt{\frac{801}{630} \times \frac{796}{637} \times \frac{637}{796} \times \frac{630}{801}} = \sqrt{1} = 1$$

$$\text{Factor Reversal Test: } P_{01} \times Q_{01} = \frac{\sum p_1q_1}{\sum p_0q_0} = \frac{796}{630}$$

$$Q_{01} = \sqrt{\frac{\sum q_1p_0}{\sum q_0p_0} \times \frac{\sum q_1p_1}{\sum q_0p_1}}$$

$$= \sqrt{\frac{637}{630} \times \frac{796}{801}}$$

$$P_{01} \times Q_{01}$$

$$= \sqrt{\frac{801}{630} \times \frac{796}{637} \times \frac{637}{630} \times \frac{796}{801}} = \frac{796}{630}$$

$$\text{which is equal to } \frac{\sum p_1q_1}{\sum p_0q_0}$$

11.27

Comm	1998		1999		p_1q_0	p_0q_0	p_1q_1	p_0q_1
	p_0	q_0	p_1	q_1				
A	12	20	14	30	280	240	420	360
B	14	13	20	15	260	182	300	210
C	10	12	15	20	180	120	300	200
D	6	8	4	10	32	48	40	60
E	8	5	6	5	30	40	30	40
					782	630	1090	870

Fisher's Ideal Index:

$$P_{01} = \sqrt{\frac{\sum p_1q_0}{\sum p_0q_0} \times \frac{\sum p_1q_1}{\sum p_0q_1}} \times 100$$

$$= \sqrt{\frac{782}{630} \times \frac{1090}{870}} \times 100 = 124.7$$

Time Reversal Test:

$$P_{01} \times P_{10} = \sqrt{\frac{\sum p_1q_0}{\sum p_0q_0} \times \frac{\sum p_1q_1}{\sum p_0q_1} \times \frac{\sum p_0q_1}{\sum p_1q_1} \times \frac{\sum p_0q_0}{\sum p_1q_0}}$$

$$= \sqrt{\frac{782}{630} \times \frac{1090}{870} \times \frac{870}{1090} \times \frac{630}{782}} = 1$$

11.28

Comm	p_1	q_1	p_0	q_0	p_1q_1	p_0q_1	p_1q_0	p_0q_0
A	5	14	3	8	70	42	40	24
B	8	18	6	25	144	108	200	150
C	3	25	1	40	75	25	120	40
D	15	36	12	48	540	432	720	576
E	9	14	7	18	126	98	162	126
F	7	13	5	19	91	65	133	95
					1046	770	1375	1011

(i) Paasche's Index,

$$P_{01} = \frac{\sum p_1q_1}{\sum p_0q_1} \times 100 = \frac{1046}{770} \times 100 = 135.84$$

(ii) Fisher's Index,

$$P_{01} = \sqrt{\frac{\sum p_1q_0}{\sum p_0q_0} \times \frac{\sum p_1q_1}{\sum p_0q_1}} \times 100$$

$$= \sqrt{\frac{1375}{1011} \times \frac{1046}{770}} \times 100 = 135.92$$

11.13 CONSUMER PRICE INDEXES

The consumer price index, also known as the *cost of living index* or *retail price index*, is constructed to measure the amount of money which consumers of a particular class have to pay to get a basket of goods and services at a particular point of time in comparison to what they paid for the same in the base year.

The need for constructing consumer price indexes arises because the general indexes do not highlight the effects of rise or fall in prices of various commodities consumed by different classes of people on their cost of living. Moreover, different classes of people consume different types of commodities and even the same type of commodities are not consumed in the same proportion by different classes of people. To study the effect of rise or fall in prices of different types of commodities, the Cost of Living Index (CLI) are constructed separately for different classes of people.

The problem in constructing consumer price indexes arises because variations in prices of commodities have to be studied from the point of view of consumers living in different regions or places. Since retail prices in different places differ and the pattern of consumption is also not identical at different places, therefore people living in different regions pay different prices to purchase various commodities. Moreover, the relative importance of various commodities to all people is not identical. Therefore, we cannot construct one CLI for the whole country.

11.13.1 Uses of Consumer Price Index (CPI) Number

The importance of the CPI can be seen from the following:

- (i) The CPI are used to formulate economic policy, escalate income payments and measure real earnings.
- (ii) The CPI are used to measure purchasing power of the consumer in rupees. The purchasing power of the rupee is the value of a rupee in a given year as compared to a base year. The formula for calculating the purchasing power of the rupee is

$$\text{Purchasing power} = \frac{1}{\text{Consumer price index}} \times 100$$

- (iii) When a time series is concerned with such rupee values as retail sales amounts or wage rates, the price index is most frequently used to achieve deflation of such time series. The process of deflating can be expressed in the form of a formula as

$$\text{Real wage} = \frac{\text{Money value}}{\text{Consumer price index}} \times 100$$

- (iv) The CPI is used in wage negotiations and wage contracts. Automatic adjustment of wages or the dearness allowance component of the wages is done on the basis of the consumer price index.

11.13.2 Construction of a Consumer Price Index

The CPI is a weighted aggregate price index with fixed weights. The need for weighting arises because the relative importance of various commodities or items for different classes of people is not the same. The percentage of expenditure on different commodities by an average family constitutes the individual weights assigned to the corresponding price relatives and the percentage expenditure on five well-accepted groups of commodities namely: (i) food, (ii) clothing, (iii) fuel and lighting, (iv) house rent and (v) miscellaneous.

The weight applied to each commodity in the market basket is derived from a usage survey of families throughout the country. The consumer price index or cost of living index numbers are constructed by the following two methods:

Aggregate expenditure method or weighted aggregate method

This method is similar to the Laspeyre's method of constructing a weighted index. To apply this method, the quantities of various commodities consumed by a particular class of people are assigned weights on the basis of quantities consumed in the base year. Mathematically, it is stated as

$$\text{Consumer price index} = \frac{\text{Total expenditure in current period}}{\text{Total expenditure in base period}} \times 100 = \frac{\sum p_1 q_0}{\sum p_0 q_0} \times 100$$

where p_1 and p_0 = prices in the current period and base period, respectively
 q_0 = quantities consumed in the base period

Family budget method or method of weighted average of price relatives

To apply this method, the family budget of a large number of people, for whom the index is meant, are carefully studied. Then the aggregate expenditure of an average family on various

commodities is estimated. These values constitute the weights. Mathematically, consumer price index is stated as

$$\text{Consumer price index} = \frac{\sum PV}{\sum V} \times 100$$

when P = price relatives, $p_1/p_0 \times 100$

V = Value weight, p_0q_0

Example 11.33: Owing to change in prices, the consumer price index of the working class in a certain area rose in a month by one quarter of what it was prior to 225. The index of food became 252 from 198, that of clothing from 185 to 205, of fuel and lighting from 175 to 195, and that of miscellaneous from 138 to 212. The index of rent, however, remained unchanged at 150. It was known that the weight of clothing, rent and fuel, and lighting were the same. Find out the exact weight of all the groups.

Solution: Suppose the weights of items included in the group are as follows:

- Food x
- Fuel and Lighting z
- Rent z
- Miscellaneous y
- Clothing z

Therefore, the weighted index in the beginning of the month would be as follows:

	<i>Index</i> <i>I</i>	<i>Weight</i> <i>W</i>	<i>IW</i>
Food	198	x	$198x$
Clothing	185	z	$185z$
Fuel and Lighting	175	z	$175z$
Rent	150	z	$150z$
Miscellaneous	138	y	$138y$
		$x + y + 3z$	$198x + 138y + 510z$

$$\text{Index number} = \frac{198x + 138y + 510z}{x + y + 3z}$$

Similarly the weighted index at the end of the month would be as follows:

	<i>I</i>	<i>W</i>	<i>IW</i>
Food	252	x	$252x$
Clothing	205	z	$205z$
Fuel and Lighting	195	z	$195z$
Rent	150	z	$150z$
Miscellaneous	212	y	$212y$
		$x + y + 3z$	$252x + 212y + 550z$

$$\text{Index number} = \frac{252x + 212y + 550z}{x + y + 3z}$$

The weighted index at the end of the month was 225 (given). This index is a rise from the first index by one quarter. Therefore, the index at the beginning was $(4/5)$ th of $225 = 180$.

Hence the weighted index at the beginning of the month was

$$180 = \frac{198x + 138y + 510z}{x + y + 3z}$$

$$180x + 180y + 540z = 198x + 138y + 510z$$

$$18x - 42y - 30z = 0 \quad \text{(i)}$$

Similarly the weighted index at the end of month was

$$225 = \frac{252x + 212y + 550z}{x + y + 3z}$$

$$225x + 225y + 675z = 252x + 212y + 550z$$

$$27x - 13y - 125z = 0 \quad \text{(ii)}$$

Let the total weight be equal to 100. Hence

$$x + y + 3z = 100 \quad \text{(iii)}$$

Multiplying Eq. (iii) by 18 and subtracting from (i), we get

$$-60y - 84z = -1800 \text{ or } 60y + 84z = 1800 \quad \text{(iv)}$$

Multiplying (iii) by 27, and subtracting from Eq. (ii), we get

$$-40y - 206z = -2700 \text{ or } 40y + 206z = 2700$$

Multiplying Eq. (iv) by 20, and Eq. (v) by 30 and subtracting, we get

$$-4500z = -45000 \text{ or } z = 10$$

Substituting the value of z in Eq. (iv), we have

$$60y + (84 \times 10) = 1800 \text{ or } y = 10$$

Substituting the value of y and z in Eq. (iii), we have

$$x + 16 + (3 \times 10) = 100 \text{ or } x = 54$$

Thus, the exact weights are:

• Food	54	• Clothing	10
• Fuel and Lighting	10	• Rent	10
• Miscellaneous	16		

Example 11.34: Incomplete information obtained from a partially destroyed records on cost of living analysis is given below:

Group	Group Index	Percent of Total Expenditure
Food	268	60
Clothing	280	Not available
Housing	210	20
Fuel and Electricity	240	5
Miscellaneous	260	Not available

The cost of living index with per cent of total expenditure as weight was found to be 255.8. Estimate the missing weights. [Delhi Univ., B.Com.(Hons.), 2005]

Solution: Let the weights for clothing be x_1 and for miscellaneous be x_2 . Then

$$60 + x_1 + 20 + 5 + x_2 = 100$$

$$x_1 + x_2 = 15 \quad \dots(i)$$

$$255.8 = \frac{268 \times 60 + 280x_1 + 210 \times 20 + 240 \times 5 + 260x_2}{100}$$

$$25580 = 16080 + 280x_1 + 4200 + 1200 + 260x_2 = 21480 + 280x_1 + 260x_2$$

$$4100 = 260(x_1 + x_2) + 20x_1$$

$$= 260 \times 15 + 20x_1 = 3900 + 20x_1$$

(Since, $x_1 + x_2 = 15$)

$$200 = 20x_1 \text{ or } x_1 = 10$$

$$x_2 = 15 - 10 = 5 \text{ and, then } x_1 = 10, x_2 = 5.$$

Example 11.35: Calculate the index number using (a) Aggregate expenditure method, and (b) Family budget method for the year 2000 with 1995 as the base year from the following data:

Commodity	Quantity (in Units)	Price (in ₹/Unit)	Price (in ₹/Unit)
	1990	1990	2000
A	100	8.00	12.00
B	25	6.00	7.50
C	10	5.00	5.25
D	20	48.00	52.00
E	25	15.00	16.50
F	30	9.00	27.00

Solution: Calculations of cost of living index are shown in Tables 11.23 and 11.24.

Table 11.23 Index Number by Aggregative Expenditure Method

Commodity	Price (₹ per unit) in		Quantity (in units)		
	1990	2000	in 1990		
	p_o	p_1	q_o	$p_o q_o$	$p_1 q_o$
A	8.00	12.00	100	800.00	1200.00
B	6.00	7.50	25	150.00	187.50
C	5.00	5.25	10	50.00	52.50
D	48.00	52.00	20	960.00	1040.00
E	15.00	16.50	25	375.00	412.50
F	9.00	27.00	30	270.00	810.00
				2605.00	3702.50

Table 11.24 Index Number by Family Budget Method

Commodity	Price (₹ per unit) in		Quantity (in units)	Price Relatives	Weights	PW
	1990	2000				
	p_0	p_1	q_0	$P (P_1/p_0) \times 100$	$W = p_0 q_0$	
A	8.00	12.00	100	150.00	800	1,20,000.00
B	6.00	7.50	25	125.00	150	18,750.00
C	5.00	5.25	10	105.00	50	5,250.00
D	48.00	52.00	20	108.33	960	1,03,996.80
E	15.00	16.50	25	110.00	375	41,250.00
F	9.00	27.00	30	300.00	270	81,000.00
					2605	3,70,246.80

$$\text{Cost of living index} = \frac{\sum p_1 q_0}{\sum p_0 q_0} \times 100 = \frac{3702.5}{2605} \times 100 = 142.13$$

$$\text{Cost of living index} = \frac{\sum PW}{\sum W} = \frac{3,70,246.8}{2605} = 142.129$$

The small difference observed between the index by the Aggregative Method (142.13) and the index by the Family Budget Method (142.123) is due to the approximation in the value of price relatives (= 108.33) in commodity D.

Example 11.36: The monthly income of a person is ₹ 10,500. It is given that cost of living index for a particular month is 136. Find out the money spent by that person on food and on clothing.

Item	Expenditure (₹)	Index
Food	—	180
Rent	1470	100
Clothing	—	150
Fuel and Power	1680	110
Misc	1890	80

[Delhi Univ., B Com (Hons.), 2001]

Solution: Let amount spent on food be x and on clothing be y

Item	Expenditure E	Index I	$I \times E$
Food	x	180	$180x$
Rent	1470	100	1,47,000
Clothing	y	150	$150y$
Fuel and Power	1680	110	1,84,800
Miscellaneous	1890	80	1,51,200
	10,500		$180x + 150y + 4,83,000$

$$\text{Consumer Price Index (CPI)} = \frac{\sum I \times E}{\sum E}$$

$$136 = \frac{180x + 150y + 4,83,000}{10,500}$$

$$136 (10,500) = 180x + 150y + 4,83,000$$

$$180x + 150y = 14,28,000 - 4,83,000 = 9,45,000 \quad (i)$$

$$\text{Given } 10,500 \text{ (Expenditure)} = x + 1470 + y + 1680 + 1890$$

$$x + y = 10500 - 5040 = 5460 \quad (ii)$$

Multiplying (ii) by 180 and subtracting from (i), we get

$$180x + 150y = 9,45,000$$

$$180x + 180y = 9,82,800$$

$$-30y = -37,800 \text{ or } y = 1,260$$

Putting $y = 1,260$ in (ii), we get $x = 4,200$.

Example 11.37: The consumer price index in a particular town and the weights according to different groups of items were as follows:

Food 55, Fuel 15, Clothing 10, Rent 12 and Miscellaneous 8. In October 1999, the dearness allowance was fixed by a mill of that town at 182 per cent of worker's wages which fully compensated for the rise in the prices of food and rent but did not compensate for anything else. Another mill of the same town paid dearness allowance of 46.5 per cent which compensated for the rise in fuel and miscellaneous groups. It is known that the rise in food is double the rise in fuel and the rise in miscellaneous group is double the rise in rent. Find the rise in food, fuel, rent and miscellaneous groups. [Delhi Univ., B.Com.(Hons.), 2002]

Solution: Let rise in fuel be x and rise in rent be y . Then, rise in food will be $2x$ and rise in miscellaneous group will be $2y$.

First mill compensated fully for rise in food and rent but did not compensate for anything else. Dearness allowance was fixed at 182 per cent, i.e. ₹ 282 paid against ₹ 100.

Index after rise for first mill is 282.

	Index (I)	Weight (W)	$W \times I$
Food	$2x$	55	$110x$
Fuel	100	15	1500
Clothing	100	10	1000
Rent	y	12	$12y$
Miscellaneous	100	8	800
		100	$3300 + 110x + 12y$

$$\text{Index} = \frac{\Sigma W \times I}{\Sigma W}$$

$$282 = \frac{3300 + 110x + 12y}{100}$$

$$28,200 - 3300 = 110x + 12y$$

$$110x + 12y = 24900$$

Second mill paid dearness allowance at the rate of 46.5 per cent.

(i)

	Index (I)	Weight (W)	$W \times I$
Food	100	55	5500
Fuel	x	15	$15x$
Clothing	100	10	1000
Rent	y	12	$12y$
Miscellaneous	$2y$	8	$16y$
Total		100	$7700 + 15x + 16y$

$$\text{Index} = \frac{\Sigma W \times I}{\Sigma W}$$

$$146.5 = \frac{7700 + 15x + 16y}{100}$$

$$14650 - 7700 = 15x + 16y$$

$$15x + 16y = 6950$$

(ii)

By multiplying (i) by 4 and (ii) by 3 and subtracting (ii) from (i), we get

$$440x + 48y = 99600$$

$$45x + 48y = 20850$$

$$395x = 78750, \text{ i.e. } x = 199.367$$

Substituting $x = 199.367$ in (ii), we get $16y = 6950 - 2990 = 360$ or $y = 247.5$

(a) Hence rise in fuel shall be 199.37 and rise in food shall be 398.74

(b) Rise in rent shall be 247.5 and rise in Miscellaneous groups shall be 495.

Conceptual Questions 11B

11. (a) Discuss the various problems faced in the construction of index numbers.
(b) Explain the problem faced in the construction of cost of living index.
12. Discuss the importance and use of weights in the construction of general price index numbers.
13. What is Fisher's Ideal index? Why is it called ideal? Show that it satisfies both the time reversal test as well as the factor reversal test.
14. Laspeyre's price index generally shows an upward trend in the price changes while Paasche's method shows a downward trend on them. Elucidate the statement.
15. Explain the Time Reversal Test and Factor Reversal Test with the help of suitable examples.
16. Distinguish between deflating and splicing of index numbers.
17. What is the cost of living index number? Is it the same as the consumer price index number?
18. What is the chain base method of construction of index numbers and how does it differ from the fixed base method?
19. It is said that index numbers are a specialized type of averages. How far do you agree with this statement? Explain briefly the Time Reversal and Factor Reversal Tests.
20. What are the Factor Reversal and Circular tests of consistency in the selection of an appropriate index formula? Verify whether Fisher's Ideal Index satisfies such tests.
21. What is the major difference between a weighted aggregate index and a weighted average of relatives index?
22. What are the tests to be satisfied by a good index number? Examine how far they are met by Fisher's Ideal index number.
23. What are the tests prescribed for a good index number? Describe the index number which satisfies these tests.

Formulae Used

1. Price relatives in period n , $P_{0n} = \frac{\Sigma W \times I}{\Sigma W}$

Quantity relative in period n , Q_{0n}

$$= \frac{3300 + 110x + 12y}{100}$$

Value relative in period n , $V_{0n} = \frac{\Sigma W \times I}{\Sigma W}$

2. Unweighted aggregate price index in period n

$$P_{0n} = \frac{7700 + 15x + 16y}{100}$$

Simple average of price relative

$$P_{0n} = \frac{p_n}{p_0} \times 100$$

Simple G.M. of price relative

$$P_{0n} = \text{antilog} \left[\frac{1}{n} \Sigma \left(\frac{p_n}{p_0} \right) \times 100 \right]$$

Simple aggregate quantity index

$$Q_{0n} = \frac{\Sigma p_n}{\Sigma p_0} \times 100$$

3. Weighted aggregate price indexes

(a) Weighted aggregate method in period n

$$P_{0n} = \frac{\Sigma p_n q}{\Sigma p_0 q} \times 100$$

Laspeyre's index, $I_p(L) = \frac{\Sigma p_n q_0}{\Sigma p_0 q_0} \times 100$

Paasche's index, $I_p(P) = \frac{\Sigma p_n q_n}{\Sigma p_0 q_n} \times 100$

Marshall-Edgeworth's index

$$I_p(M-E) = \frac{\Sigma p_n (q_0 + q_n)}{\Sigma p_0 (q_0 + q_n)} \times 100$$

Dorbish and Bowley's index

$$I_p(D-B) = \frac{1}{2} (L + P) \times 100$$

Fisher's ideal index, $= \sqrt{L \times P} \times 100$

(b) Weighted average of price relatives in period n

$$P_{0n} = \frac{\Sigma \left(\frac{p_n}{p_0} \times 100 \right) W}{\Sigma W}$$

Weighted average of price relatives

$$P_{0n} = \frac{\Sigma \left(\frac{p_1}{p_0} \times 100 \right) (p_0 q_0)}{\Sigma p_0 q_0}$$

(base year value as weights)

Weighted average of price relatives

$$P_{0n} = \frac{\Sigma \left(\frac{p_1}{p_0} \times 100 \right) (p_1 q_1)}{\Sigma p_1 q_1}$$

(current year value as weights)

4. Quantity indexes

(a) Unweighted quantity index in period n

$$Q_{0n} = \frac{\Sigma q_n}{\Sigma q_0} \times 100$$

Simple average of quantity relative

$$Q_{0n} = \frac{1}{n} \Sigma \left(\frac{q_n}{q_0} \times 100 \right)$$

(b) Weighted quantity index in period n

$$Q_{0n} = \frac{\Sigma q_n W}{\Sigma q_0 W} \times 100$$

5. Tests for adequacy or consistency

Time reversal test: $P_{0n} \times P_{n0} = 1$

Factor reversal test: $P_{0n} \times Q_{0n} = \frac{\Sigma p_n q_n}{\Sigma p_0 q_0}$

Circular test: $P_{01} \times P_{12} \times P_{23} \times \dots \times P_{(n-1)n} \times P_{n0} = 1$

6. Link relative

$$= \frac{\text{Current period price}}{\text{Price of the preceding period}} \times 100$$

Chain index

$$= \frac{\text{Current period's link relative} \times \text{Preceding period's chain index}}{100}$$

True or False Concepts Quiz

1. Like all statistical tools, index numbers must be used with great caution.
2. For constructing index numbers, the best method on theoretical grounds is not the best method from practical point of view.
3. The Fisher Ideal Index number is a compromise between two well known indexes—not a right compromise, economically, for the statistician.
4. The real problem while constructing a index number is whether he shall leave weighting to chance or seek to rationalize it.
5. Like relatives are based on the idea that one series can be converted into another because time reversibility holds.
6. Index numbers are the signs and guide-posts along the business highway that indicate to the businessman how he should drive or manage.
7. Index numbers measure change in magnitude of a group of distinct but related variables.
8. Prices should be for the same unit of quantity in index numbers.
9. Quantity relatives are used to measure changes in the volume of consumption.
10. Weighting of index number makes them more representative.
11. Cost of living index numbers are based on retail prices of items of consumption.
12. Splicing means constructing one continuous series from two index series on the basis of a common base.
13. Chain indexes give the same result as do fixed base index numbers.
14. Weighted average of relatives and weighted aggregative methods render the same result.
15. Paasche's formula is a weighted aggregate index with quantity weights in the base year.

Concepts Quiz Answers

1. T	2. F	3. F	4. F	5. T	6. T	7. T	8. F	9. T
10. T	11. T	12. T	13. T	14. T	15. F			

Review Self-practice Problems

- 11.29** Construct an index number for each year from the following average annual price of cotton with 1989 as the base year.

Year	Price (₹)	Year	Price (₹)
1989	75	1994	70
1990	50	1995	69
1991	65	1996	75
1992	60	1997	84
1993	72	1998	80

- 11.30** The price quotations of four different commodities for 1996 and 1997 are given

below. Calculate the index number for 1997 with 1996 as base by using (i) the simple average of price relatives and (ii) the weighted average of price relatives.

Comm	Unit	Weight (₹ 1000)	Price (in ₹ per unit)	
			1996	1997
A	Kg	5	2.00	4.50
B	Quintal	7	2.50	3.20
C	Dozen	6	3.00	4.50
D	Kg	2	1.00	1.80

- 11.31** From the chain base index numbers given below, find the fixed base index numbers.

Year	:	1996	1997	1998	1999	2000
Chain base index :		80	110	120	90	140

- 11.32** The following are the group index numbers and the group weights of an average working class family's budget. Construct the cost of living number.

Group	Index Number	Weight
Food	330	50
Clothing	208	10
Fuel and lighting	200	12
House rent	162	12
Miscellaneous	180	16

- 11.33** In 1998, for working class people, wheat was selling at an average price of ₹ 120 per 20 kg. Cloth ₹ 20 per metre, house rent ₹ 300 per house and other items ₹ 100 per unit. By 1998 cost of wheat rose by ₹ 180 per 20 kg, rent by ₹ 450 Per house and other items doubled in price. The working class cost of living index for the year 1998 with 1988 as base was 160. By how much did the price of cloth rise during the period?

- 11.34** Calculate the cost of living index from the following data:

Items	Quantity Consumed per Year in the Given Year	Price (in ₹ per Unit) Base Year	Given Year
Rice (qtl)	2.50×12	12	25
Pulses (kg)	3×12	0.4	0.6
Oil (litre)	2×12	1.5	2.2
Clothing (metres)	6×12	0.75	1.0
Housing (per month)	—	20	30
Miscellaneous (per month)	—	10	15

- 11.35** Compute the Consumer Price Index number from the following:

Group	Base Year Price (₹)	Current Year Price (₹)	Weight Per cent
Food	400	550	35
Rent	250	300	25
Clothing	500	600	15
Fuel	200	350	20
Entertainment	150	225	5

- 11.36** In calculating a certain cost of living index number, the following weights were used: Food 15, Clothing 3, Rent 4, Fuel and Light 2 and Miscellaneous 1. Calculate the index for the period when the average percentage increases in prices of items in the various groups over the base period were 32, 54, 47, 78 and 58, respectively.

Suppose a business executive was earning ₹ 2050 in the base period, what should be his salary in the current period if his standard of living is to remain the same?

- 11.37** Construct the cost of living index number from the following data:

Group	Weights	Group Index
Food	47	247
Fuel and Lighting	7	293
Clothing	8	289
House rent	13	100
Miscellaneous	14	236

- 11.38** During a certain period, the cost of living index goes up from 110 to 200 and the salary of a worker is also raised from ₹ 3250 to ₹ 5000. Does the worker really gain, and if so, by how much in real terms?

- 11.39** An enquiry into the budgets of middle class families in a certain city gave the following information:

Expenses	Food 35%	Fuel 10%	Clothing 20%	Rent 15%	Miscellaneous 20%
Prices (₹)					
1990	: 150	25	75	30	40
Prices (₹)					
1991	: 145	23	65	30	45

What is the Cost of Living Index number of 1991 as compared with that of 1990?

Hints and Answers

11.29

Year	Price	Index Number (Base 1989)
1989	75	100
1990	50	$\frac{50}{75} \times 100 = 66.67$
1991	65	$\frac{65}{75} \times 100 = 86.67$
1992	60	$\frac{60}{75} \times 100 = 80.00$
1993	72	$\frac{72}{75} \times 100 = 96.00$
1994	70	$\frac{70}{75} \times 100 = 93.33$
1995	69	$\frac{69}{75} \times 100 = 92.00$
1996	75	$\frac{75}{75} \times 100 = 100.00$
1997	84	$\frac{84}{75} \times 100 = 112.00$
1998	80	$\frac{80}{75} \times 100 = 106.67$

11.30

Comm	Unit	Weight	Price	$P = \frac{p_1}{p_0} \times 100$			
		(₹, 1000)	1996	1997			
		W	p_0	p_1	P	PW	
A	Kg	5	2.00	4.50	225	1,125	
B	Quint	7	2.50	3.20	128	896	
C	Dozen	6	3.00	4.50	150	900	
D	Kg	2	1.00	1.80	180	360	
		20			683	3,281	

(i) Simple average of price relatives

$$P_{01} = \frac{1}{n} \sum \left(\frac{p_1}{p_0} \times 100 \right) = \frac{683}{4} = 170.75$$

(ii) Weighted average of price relatives method:

$$P_{01} = \frac{\sum PW}{\sum W} = \frac{3281}{20} = 164.05$$

11.31 The formula for converting a Chain Base Index (CBI) number to a Fixed Base Index (FBI) Number is:

Current years FBI

$$= \frac{\text{Current years CBI} \times \text{Previous years FBI}}{100}$$

Conversion of CBI to FBI

Year	Chain Base Index	Fixed Base Index
1996	80	80
1997	110	$\frac{110 \times 80}{100} = 88$
1998	120	$\frac{120 \times 88}{100} = 105.60$
1999	90	$\frac{90 \times 105.6}{100} = 95.04$
2000	140	$\frac{140 \times 95.04}{100} = 133.06$

11.32

Groups	Index No. P	Weights W	PW
Food	330	50	16,500
Clothing	208	10	2080
Fuel and Lighting	200	12	2400
House Rent	162	12	1944
Miscellaneous	180	16	2880
		100	25,804

$$\text{Cost of living index} = \frac{\sum PW}{\sum W} = \frac{25,804}{100} = 258.04$$

11.33

Commodity	Price 1998	Index No. 1998	Price 1998	Index No. 1998
Wheat	120	100	180	$\frac{180}{120} \times 100 = 150$
Cloth	20	100	x	$\frac{x}{20} \times 100 = 5x$
House rent	300	100	450	$\frac{450}{300} \times 100 = 150$
Miscell	100	100	200	$\frac{200}{100} \times 100 = 200$
				$500 + 5x$

The index for 1998 is 160. Thus the sum of the index numbers of the four commodities would be $160 \times 4 = 640$. Hence, $500 + 5x = 640$ or $x = 28$. The rise in the price of cloth was ₹ 8 per metre.

11.34

Items	Quantity				
	Consumed	q_1	p_0	p_1	$p_0 q_1$
Rice (qtl)	2.50	12	12.00	25.0	750.0
Pulses (kg)	3	12	0.40	0.6	21.6
Oil (litres)	2	12	1.50	2.2	52.8
Clothing (mt)	6	12	0.75	1.0	72.0
Housing (per month)	—		20	30	360.0
Miscellaneous (per month)	—		10	15	180.0
					1436.4
					824.4

$$\begin{aligned} \text{Cost of living index} &= \frac{\sum p_1 q_1}{\sum p_0 q_1} \times 100 \\ &= \frac{1436.4}{824.4} \times 100 = 174.24 \end{aligned}$$

11.35

Group	p_0	p_1	$P = \frac{p_1}{p_0} \times 100$	Weight W	PW
Food	400	550	137.5	35	4812.5
Rent	250	300	120.0	25	3000.0
Clothing	500	600	120.0	15	1800.0
Fuel	200	350	175.0	20	3500.0
Entertainment	150	225	150.0	5	750.0
Total				100	13,862.5

$$\begin{aligned} \text{Consumer price index} &= \frac{\sum PW}{\sum W} = \frac{13,862.5}{100} \\ &= 138.63 \end{aligned}$$

11.36

Group	Average Price Per cent	Group Index Increase in P	Price Weight W	PW
Food	32	132	15	1980
Clothing	54	154	3	462
Rent	47	147	4	588
Fuel and light	78	178	2	356
Miscellaneous	58	158	1	158
Total			25	3544

$$\text{Cost of living index} = \frac{\sum PW}{\sum W} = \frac{3544}{25} = 141.76$$

For maintaining the same standard, the business executive should get = ₹ 2906.08.

11.37

Group	Weights (W)	Group Index (P)	PW
Food	47	247	11,609
Fuel and lighting	7	293	2051
Clothing	8	289	2312
House rent	13	100	1300
Miscellaneous	14	236	3304
Total	89		20,576

$$\text{Cost of living} = \frac{\sum PW}{\sum W} = \frac{20,576}{89} = 231.19$$

11.38 Real wage of ₹ 3250

$$\begin{aligned} &= \frac{\text{Actual wage}}{\text{Cost of living index}} \times 100 \\ &= \frac{3250}{110} \times 100 = ₹ 2954.54 \end{aligned}$$

$$\text{Real wage of Rs. 5000} = \frac{5000}{200} \times 100$$

$$= ₹ 2500 \text{ which is less than } ₹ 2954.54$$

Since the real wage of ₹ 5000 is less than that of ₹ 3250, the worker does not really gain, real wage decrease by ₹ (2954.54 – 2500) = ₹ 454.54.

11.38

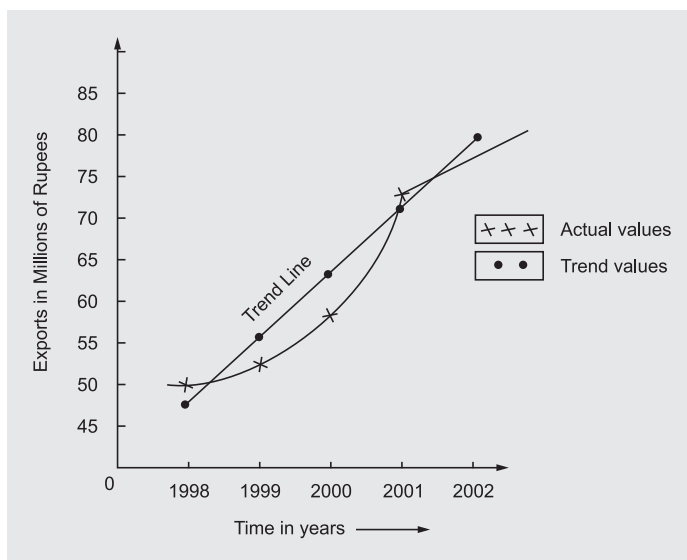
Expenses on	1990 p_0	1991 p_1	$P = \frac{p_1}{p_0} \times 100$	Weight W	PW
Food	150	145	96.67	35	3383.33
Fuel	25	23	92.00	10	920.00
Clothing	75	65	86.67	20	1733.33
Rent	30	30	100.00	15	1500.00
Miscell	40	45	112.50	20	2250.00
Total				100	9786.66

Cost of living index for 1991

$$= \frac{\sum PW}{\sum W} = \frac{9786.66}{100} = 97.87.$$

TIME SERIES ANALYSIS

Chapter 12: Time Series Analysis



This page is intentionally left blank.

12

Time Series Analysis

LEARNING OBJECTIVES

After studying this chapter, you should be able to

- understand the pattern of the historical data and then extrapolate the pattern into the future.
- understand the different approaches to forecasting that can be applied in business.
- gain a general understanding of time series forecasting techniques.
- learn how to decompose time series data into their various components and to forecast by using decomposition techniques.

12.1 INTRODUCTION

Forecasting takes the historical data to assess the future consequences of existing decisions and to evaluate the consequences of decisions (actions or strategies). For example, inventory is ordered without certainty of future sales; new equipment is purchased despite uncertainty about the demand for products; investments are made without knowing profits in future; alternative staff mix is made without knowing the increase in the level of service that can be provided, and so on.

Forecasting is essential to make reliable and accurate estimates of what will happen in the future, in the face of uncertainty. A flow chart of forecasts and the decision making process is shown in Fig. 12.1. In general, the decisions are influenced by the chosen strategy with regard to an organization's future priorities and activities. Once decisions are taken, the consequences are measured in terms of expectation to achieve the desired products/services levels.

Decisions also get influenced by the additional information obtained from the forecasting method used. Such information and the perceived accuracy of the forecasts may also affect the strategy formulation of an organization. Thus an organization needs to establish a monitoring system to compare planned performance with the actual. Divergence, if any, and no matter what the cause of such divergence between the planned and actual performance, should be fed back into the forecasting process, to generate new forecasts. A few objectives of forecasting are as follows:

- (i) The creation of plans of action because it is not possible to evolve a system of business control without an acceptable system of forecasting.
- (ii) Monitoring of the continuing progress of action plans based on forecasts.
- (iii) The forecast provides a warning system of the critical factors to be monitored regularly because they might drastically affect the performance of the plan.

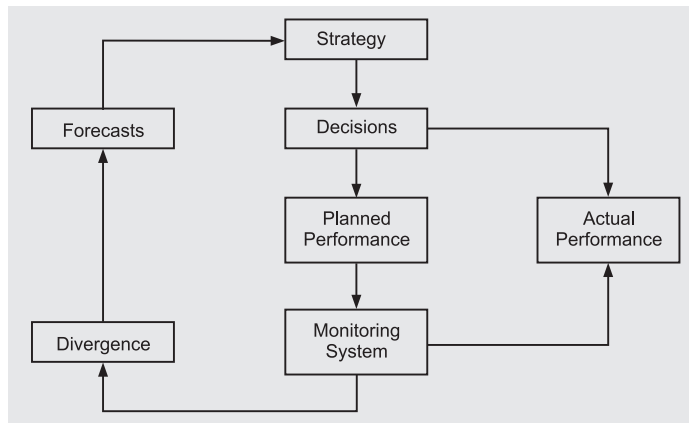


Figure 12.1 Decision-making Process and Forecasts

12.2 FORECASTING METHODS

Forecasting methods may be classified as either quantitative or qualitative (opinion or judgment). Figure 12.2 provides an overview of the types of forecasting methods.

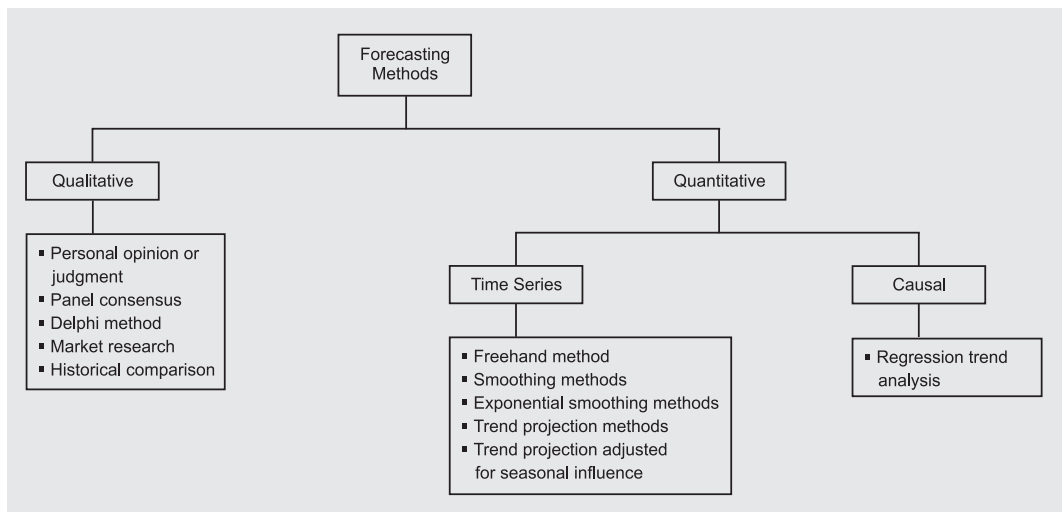


Figure 12.2 Forecasting Methods

12.2.1 Quantitative Forecasting Methods

These methods can be used when

- (i) past information about the variable being forecast is available.

- (ii) information can be quantified.
- (iii) a reasonable assumption is that the pattern of the past will continue into the future.

The quantitative methods of forecasting are further classified into two categories:

Time Series Forecasting Methods A time series is a set of measurements of a variable that are ordered through time. The time variable does not fluctuate arbitrarily. It moves uniformly always in the same direction, from past to future, Thus we can exercise some freedom of choice as to the times at which observations can be made. The time series data are gathered on a given variable characteristic over a period of time at regular intervals.

The time series forecasting methods attempt to account for changes over a period of time at regular intervals by examining patterns, cycles or trends to predict the outcome for a future time period.

Causal Forecasting Methods These methods are based on the assumptions that the variable value which we intend to forecast has a cause-effect relationship with one or more other variables. A linear regression analysis which depends upon the causal relationship or interaction of two or more variables is called causal forecasting method (Fig. 12.2).

12.2.2 Qualitative Forecasting Methods

These methods consist of collecting the opinions and judgments of individuals who are expected to have the best knowledge of current activities or future plans of the organization. For example, knowledge of demand trend and customer plans are often known to marketing executives or product managers. Through regular contact with customers, the marketing and sales personnel are presumably familiar with individual customers or retail market segment. Management usually maintains broader market information on trends by product line, geographic area, customer groups, and so on.

Qualitative forecasting methods have the advantage that they can incorporate subjective experience as inputs along with objectives data. It is the human brain that permits assimilation of all types of information and the ultimate issuance of a prediction.

Since each human being has different knowledge, experience, and perspective of reality, intuitive forecasts are likely to differ from one individual to another. Furthermore, the less they are based upon fact and quantified data, the less they lend themselves to analysis and resolution of differences of opinion. The quantification of data gives them a more precise meaning than words which are inexact and are capable of being misunderstood. Also, if the forecasts prove to be inaccurate there is an objective basis for improvement the next time around.

12.3 TIME SERIES ANALYSIS

A time series is a set of numerical values of some variable obtained at regular period over time. The series is usually tabulated or graphed in a manner that readily conveys the behaviour of the variable under study. Figure 12.3 presents the export of cement (in tonnes) by a cement company between 1999 and 2009. The graph suggests that the series is time dependent. The management of the company is interested in determining how the series is dependent on time and in developing a means of predicting future levels with some degree of reliability. The nature of the time dependence is often analysed by decomposing the time series into its components.

Year	Export (tonnes)
1999	2
2000	3
2001	6
2002	10
2003	8
2004	7
2005	12
2006	14
2007	14
2008	18
2009	19

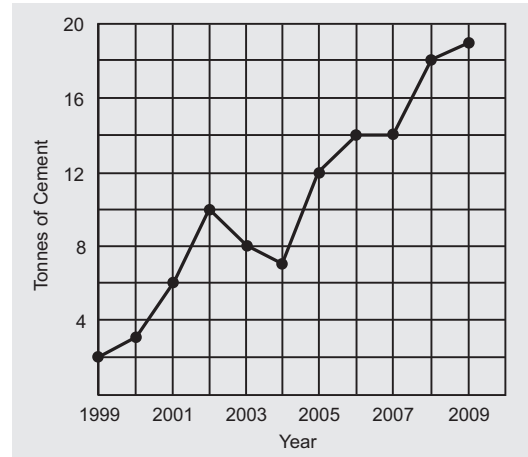


Figure 12.3 Export of Cement

12.3.1 Objectives of Time Series Analysis

- The assumption underlying time series analysis is that the future will look like the past, that is, the various factors which have already influenced the patterns of change in the value of the variable under study will continue to do so in more or less the same manner in the future. In other words, some underlying pattern exists in historical data. Thus one of the objective of time series analysis is to identify the pattern and isolate the influencing factors (or effects) for prediction purposes as well as for future planning and control.
- The review and evaluation of progress made on the basis of a plan are done on the basis of time series data. For example, the progress of our Five-Year Plans is judged by the annual growth rates in the Gross National Product (GNP). Similarly the evaluation of our policy of controlling inflation and price rise is done by the study of various price indices which are based on the analysis of time series.

12.3.2 Time Series Patterns

We assume that time series data consist of an underlying pattern accompanied by random fluctuations. This may be expressed in the following form:

$$\begin{array}{lcl} \text{Actual value of the} & = & \text{Mean value of the} & + & \text{Random deviation from mean value} \\ \text{variable at time } t & & \text{variable at time } t & & \text{of the variable at time } t \end{array}$$

$$\hat{y} = \text{Pattern} + e$$

where $\hat{y}$ is the forecast variable at period t ; pattern is the mean value of the forecast variable at period t and represents the underlying pattern, and e is the random fluctuation from the pattern that occurs of the forecast variable at period t .

12.3.3 Components of a Time Series

The **time series** data contain four components: *trend*, *cyclicity*, *seasonality* and *irregularity*. Not all time series have all these components. Figure 12.4 shows the effects of these time series components over a period of time.

Trend Sometimes a time series displays a steady tendency of either upward or downward movement in the average (or mean) value of the forecast variable y over time. Such a tendency is called a trend. When observations are plotted against time, a straight line describes the increase or decrease in the time series over a period of time.

Cycles An upward and downward movement in the variable value about the trend time over a time period are called cycles. A business cycle may vary in length, usually more than a year but less than 5 to 7 years. The movement is through four phases: from *peak* (prosperity) to *contradiction* (recession) to *trough* (depression) to *expansion* (recovery or growth) as shown in Figure 12.4.

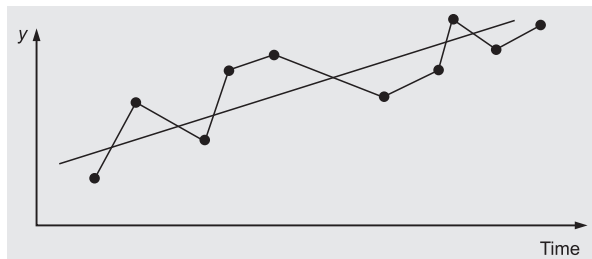


Figure 12.4 Time series Effects

Seasonal It is a special case of a cycle component of time series in which fluctuations are repeated usually within a year (e.g. daily, weekly, monthly, quarterly) with a high degree of regularity. For example, average sales for a retail store may increase greatly during festival seasons.

Irregular Irregular variations are rapid charges or *bleeps* in the data caused by short-term unanticipated and non-recurring factors. Irregular fluctuations can happen as often as day to day.

12.4 TIME SERIES DECOMPOSITION MODELS

The analysis of time series consists of two major steps:

- (i) Identifying the various factors or influences which produce the variations in the time series
- (ii) Isolating, analysing and measuring the effect of these factors independently, by holding other things constant.

The purpose of decomposition models is to break a time series into its components: Trend (T), Cyclical (C), Seasonality (S), and Irregularity (I). Decomposition of time series aims to isolate influence of each of the four components on the actual series so as to provide a basis for forecasting. There are many models by which a time series can be analysed; two models commonly used for decomposition of a time series are discussed below.

12.4.1 Multiplicative Model

The actual values of a time series, represented by Y can be found by multiplying four components at a particular time period. The effect of four components on the time series is interdependent. The multiplicative time series model is defined as:

$$Y = T \times C \times S \times I \leftarrow \text{Multiplicative model}$$

The multiplicative model is appropriate in situations where the effect of C , S and I is measured in relative sense and is not in absolute sense. The geometric mean of C , S and I is assumed to be less than one. For example, let the actual sales for period of 20 months be $Y_{20} = 423.36$. Further let this value be broken down into its components as trend component (mean sales) 400; effect of current cycle (0.90) which decreases sales by 10 per cent seasonality of the series (1.20) that increases sales by 20 per cent. Thus, besides the random fluctuation, the expected value of sales, for this period is $400 \times 0.90 \times 1.20 = 432$. If the random factor decreases sales by 2 per cent in this period, then the actual sales value will be $432 \times 0.98 = 423.36$.

12.4.2 Additive Model

In this model, it is assumed that the effect of various components can be estimated by adding the various components of a time series. It is stated as

$$Y = T + C + S + I \leftarrow \text{Additive model}$$

Here C , S and I are absolute quantities and can have positive or negative values. It is assumed that these four components are independent of each other. However in real-life time series data, this assumption does not hold good.

Conceptual Questions 12A

- Briefly describe the steps that are used to develop a forecasting system.
- What is forecasting? Discuss in brief the various theories and methods of business forecasting.
- For what purpose, do we apply time series analysis to data collected over a period of time?
- How can one benefit from determining past patterns?
- What is the difference between a causal model and a time series model?
- What is a judgmental forecasting model, and when is it appropriate?
- Explain clearly the different components into which a time series may be analysed. Explain any method for isolating trend values in a time series.
- Explain what you understand by time series. Why is time series considered to be an effective tool of forecasting?
- Explain briefly the additive and multiplicative models of time series. Which of these models is more popular in practice and why?
- Identify the four principal components of a time series and explain the kind of change, over time, to which each applies.
- What is the advantage of reducing a time series into its four components?
- Despite great limitations of statistical forecasting, forecasting techniques are invaluable to the economist, the businessman and the government. Explain.
 - Why are forecasts important to organizations?
 - Explain the difference between the terms: seasonal variation and cyclical variation.
 - Give reasons why the seasonal component in the time series is not constant? Give examples where you believe the seasonality may change.
- Identify the classical components of a time series and indicate how each is accounted for in forecasting.

12.5 QUANTITATIVE FORECASTING METHODS

The quantitative forecasting methods fall into two general categories:

- Time series methods
- Causal methods

The *time series methods* are concerned with taking some observed historical pattern for some variable and projecting this pattern into the future using a mathematical formula. These methods do not attempt to suggest why the variable under study will take some future value. This limitation of the time series approach is taken care by the application of a causal method.

The *causal method* tries to identify factors which influence the variable in some way or cause it to vary in some predictable manner. The two causal methods, regression analysis and correlation analysis have already been discussed previously.

A few time series methods such as *freehand curves* and *moving averages* simply describe the given data values, while other methods such as *semi-average* and *least squares* help to identify a trend equation to describe the given data values.

12.5.1 Freehand (or Graphical) Method

A freehand curve drawn smoothly through the data values is often an easy and, perhaps, adequate representation of the data. From Fig. 12.3, it appears that a straight line connecting the 1994 and 2004 exports volumes is a fairly good representation of the given data.

The forecast can be obtained simply by extending the trend line. A trend line fitted by the freehand method should confirm to the following conditions:

- (i) The trend line should be smooth—a straight line or mix of long gradual curves.
- (ii) The sum of the vertical deviations of the observations above the trend line should equal the sum of the vertical deviations of the observations below the trend line.
- (iii) The sum of squares of the vertical deviations of the observations from the trend line should be as small as possible.
- (iv) The trend line should bisect the cycles so that area above the trend line should be equal to the area below the trend line, not only for the entire series but as much as possible for each full cycle.

Example 12.1: Fit a trend line to the following data by using the freehand method.

Year	: 1997	1998	1999	2000	2001	2002	2003	2004
Sales turnover	: 80	90	92	83	94	99	92	104.
(₹ in lakh)								

Solution: Figure 12.5 presents the freehand graph of sales turnover (₹ in lakh) from 1997 to 2004. Forecast can be obtained simply by extending the trend line

Limitations of freehand method

- (i) This method is highly subjective because the trend line depends on personal judgment and therefore what happens to be a good-fit for one individual may not be so for another.
- (ii) The trend line drawn cannot have much value if it is used as a basis for predictions.
- (iii) It is very time consuming to construct a freehand trend if a careful and conscientious job is to be done.

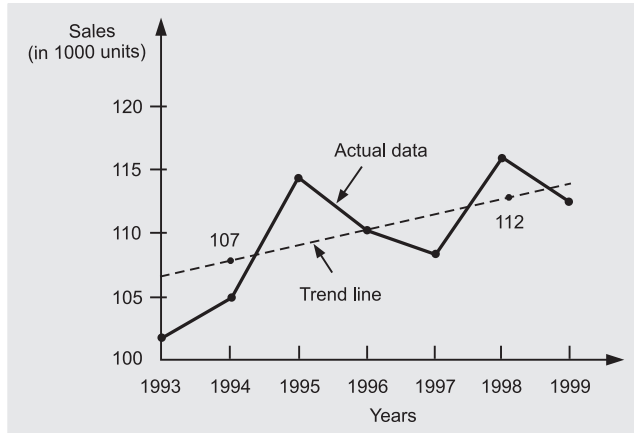


Figure 12.5 Graph of Sales Turnover

12.5.2 Smoothing Methods

The objective of smoothing methods is to smoothen out the random variations due to irregular components of the time series and thereby provide us with an overall impression of the pattern of movement in the data over time. In this section, we shall discuss three smoothing methods:

- (i) Moving averages
- (ii) Weighted moving averages
- (iii) Semi-averages

The data requirements for the techniques to be discussed in this section are minimal and these techniques are easy to use and understand.

Moving Averages

If we attempt to observe the movement of some variable values over a period of time and try to project this movement into the future, then it is essential to smooth out first the irregular pattern in the historical values of the variable, and later use this as the basis for a future projection. This can be done by using the technique of **moving averages**.

This method is a subjective method and depends on the length of the period chosen for calculating moving averages. To remove the effect of cyclical variations, the period chosen should be an integer value that corresponds to or is a multiple of the estimated average length of a cycle in the series.

The moving averages which serve as an estimate of the next period's value of a variable given a period of length n is expressed as

$$\text{Moving average, } MA_{t+1} = \frac{\Sigma\{D_t + D_{t-1} + D_{t-2} + \dots + D_{t-n+1}\}}{n}$$

where t = current time period

D = actual data which is exchanged each period

n = length of time period

In this method, the term 'moving' is used because it is obtained by summing and averaging the values from a given number of periods, each time deleting the oldest value and adding a new value.

Advantage

The major *advantage* of a moving average is the opportunity it provides to focus on the long-term trend (and cyclical) movements in a time series without the obscuring effect of short-term 'noise' influences.

Limitations

The *limitation* of this method is that it is highly subjective and dependent on the length of period chosen for constructing the averages. Moving averages have the following three limitations:

- (i) As the size of n (the number of periods averaged) increases, it smoothes the variations better, but it also makes the method less sensitive to real changes in the data.
- (ii) It is difficult to choose the optimal length of time for which to compute the moving average. Moving averages cannot be found for the first and last $k/2$ periods in a k -period moving average.
- (iii) Moving averages cannot pick-up trends very well. Since these are averages, it will always stay within past levels and will not predict a change to either a higher or lower level.
- (iv) It causes a loss of information (data values) at either end of the original time series.
- (v) Moving averages do not usually adjust for such time series effects as trend, cycle or seasonality.

Example 12.2: Shown is production volume (in '000 tonnes) for a product. Use these data to compute a three-year moving average for all available years. Also determine the trend and short-term error.

Year	Production (in '000 tonnes)	Year	Production (in '000 tonnes)
1995	21	2000	22
1996	22	2001	25
1997	23	2002	26
1998	25	2003	27
1999	24	2004	26

Solution: The first average is computed for the first 3 years as follows:

$$\text{Moving average (year 1-3)} = \frac{21 + 22 + 23}{3} = 22$$

The first three-year moving average can be used to forecast the production volume in fourth year, 1998. Because 25,000 tonnes production was made in 1998, the error of the forecast is $\text{Error}_{1998} = 25,000 - 22,000 = 3000$ tonnes.

Similarly, the moving average calculation for the next 3 years is

$$\text{Moving average (year 2-4)} = \frac{22 + 23 + 25}{3} = 23.33$$

A complete summary of three-year moving average calculations is given in Table 12.1.

Odd and Even Number of Years When the chosen period of length n is an odd number, the moving average period is centred on i (middle period in the consecutive sequence of n periods). For instance with $n = 5$, $\text{MA}_3(5)$ is centred on the third year, $\text{MA}_4(5)$ is centred on the fourth year..., and $\text{MA}_9(5)$ is centred on the ninth year.

No moving average can be obtained for the first $(n - 1)/2$ years or the last $(n - 1)/2$ year of the series. Thus for a 5-year moving average, we cannot make computations for the just two years or the last two years of the series.

Table 12.1 Calculation of Trend and Short-term Fluctuations

Year	Production y	3-Year Moving Total	3-Yearly Moving Average (Trend values) $\hat{y}$	Forecast Error ($y - \hat{y}$)
1995	21	—	—	—
1996	22	→ (21 + 22 + 23) = 66	66/3 = 22.00	0
1997	23	→ (22 + 23 + 25) = 70	70/3 = 23.33	-0.33
1998	25	→ (23 + 25 + 24) = 72	72/3 = 24.00	1.00
1999	24	71	23.67	0.33
2000	22	71	23.67	-1.67
2001	25	73	24.33	0.67
2002	26	78	26.00	0
2003	27	→ (26 + 27 + 26) = 79	79/3 = 26.33	0.67
2004	26	—	—	—

When the chosen period of length n is an even numbers, equal parts can easily be formed and an average of each part is obtained. For example, if $n = 4$, then the first moving average M_3 (placed at period 3) is an average of the first four data values, and the second moving average M_4 (placed at period 4) is the average of data values 2 through 5. The average of M_3 and M_4 is placed at period 3 because it is an average of data values for period 1 through 5.

Example 12.3: Assume a four-year cycle and calculate the trend by the method of moving average from the following data relating to the production of tea in India:

Year	Production (million lbs)	Year	Production (million lbs)
1987	464	1992	540
1988	515	1993	557
1989	518	1994	571
1990	467	1995	586
1991	502	1996	612

Solution: The first four-year moving average is:

$$MA_3(4) = \frac{464 + 515 + 518 + 467}{4} = \frac{1964}{4} = 491.00$$

This moving average is centred on the middle value, that is, the third year of the series. Similarly,

$$MA_4(4) = \frac{515 + 518 + 467 + 502}{4} = \frac{2002}{4} = 500.50$$

This moving average is centred on the fourth year of the series.

Table 12.2 presents the data along with the computations of four-year moving averages.

<i>Year</i>	<i>Production (mn lbs)</i>	<i>four-Yearly Moving Totals</i>	<i>four-Yearly Moving Average</i>	<i>four-Yearly Moving Average Centred</i>
1987	464	—	—	—
1988	515	—	—	—
		→ 1964	491.00	
1989	518			→ 495.75
		→ 2002	500.50	
1990	467			→ 503.63
		→ 2027	506.75	
1991	502			→ 509.63
		2066	512.50	
1992	540			527.50
		2170	542.50	
1993	557			553.00
		2254	563.50	
1994	571			572.50
		2326	581.50	
1995	586	—	—	—
1996	612	—	—	—

In moving averages, each observation is given equal importance (weight). However, it may be desired to place more weight (importance) on certain periods of time than on others. So *a moving average in which some time periods are weighted differently than others is called a weighted moving average*. In such a case, different values may be assigned to compute a weighted average of the most recent n values. Choice of weights is somewhat arbitrary because there is no set formula to determine them. In most cases, the most recent observation receives the most weightage, and the weight decreases for older data values.

$$\text{Weighted moving average} = \frac{\sum (\text{Weight for period } n) (\text{Data value in period } n)}{\sum \text{Weights}}$$

<i>Weight Applied</i>	<i>Month</i>
3	Last month
2	Two months ago
1	Three months ago
6	

Months :	1	2	3	4	5	6	7	8	9	10	11	12
Actual sales : (in units)	10	12	13	16	19	23	26	30	28	18	16	14

Solution: The results of three-month weighted average are shown in Table 12.3

$$\begin{aligned}\bar{x}_{\text{weighted}} &= 3M_{t-1} + 2M_{t-2} + 1M_{t-3} \\ &= \frac{1}{6} [3 \times \text{Sales last month} + 2 \times \text{Sales two months ago} + 1 \times \text{Sales three months ago}]\end{aligned}$$

Table 12.3 Weighted Moving Average

Month	Actual Sales	Three-month Weighted Moving Average
1	10	—
2	12	—
3	13	—
4	16	$\frac{1}{6} [(3 \times 13) + (2 \times 12) + (1 \times 10)] = \frac{73}{6}$
5	19	$\frac{1}{6} [(3 \times 16) + (2 \times 13) + (1 \times 12)] = \frac{43}{3}$
6	23	$\frac{1}{6} [(3 \times 19) + (2 \times 16) + (1 \times 13)] = 17$
7	26	$\frac{1}{6} [(3 \times 23) + (2 \times 19) + (1 \times 16)] = \frac{123}{6}$
8	30	$\frac{1}{6} [(3 \times 26) + (2 \times 23) + (1 \times 19)] = \frac{143}{6}$
9	28	$\frac{1}{6} [(3 \times 30) + (2 \times 26) + (1 \times 23)] = \frac{55}{2}$
10	18	$\frac{1}{6} [(3 \times 28) + (2 \times 30) + (1 \times 26)] = \frac{85}{3}$
11	16	$\frac{1}{6} [(3 \times 18) + (2 \times 28) + (1 \times 30)] = \frac{70}{3}$
12	14	$\frac{1}{6} [(3 \times 16) + (2 \times 18) + (1 \times 28)] = \frac{56}{3}$

Example 12.5: A food processor uses a moving average to forecast next month's demand. Past actual demand (in units) is shown below:

Month :	43	44	45	46	47	48	49	50	51
Actual demand :	105	106	110	110	114	121	130	128	137

- Compute a simple five-month moving average to forecast demand for month 52.
- Compute a weighted three-month moving average where the weights are highest for the latest months and descend in order of 3, 2, 1.

Solution: Calculations for five-month moving average are shown in Table 12.4.

- Five-month average demand for month 52 is

$$\frac{\sum x}{\text{Number of periods}} = \frac{114 + 121 + 130 + 128 + 137}{5} = 126 \text{ units}$$

Table 12.4 Five-month Moving Average

<i>Month</i>	<i>Actual Demand</i>	<i>five-month Moving Total</i>	<i>five-month Moving Average</i>
43	105	—	—
44	106	—	—
45	110	545	109.00
46	110	561	112.2
47	114	585	117.0
48	121	603	120.6
49	130	630	126.0
50	128	—	—
51	137	—	—

(b) Weighted three-month average as per weights is as follows:

$$\bar{x}_{\text{weighted}} = \frac{\sum \text{Weight} \times \text{Data value}}{\sum \text{weight}}$$

where Month Weight $\times$ Value = Total

$$\begin{array}{rcl} 51 & 3 & \times 137 = 411 \\ 50 & 2 & \times 128 = 256 \\ 49 & 1 & \times 130 = 130 \\ \hline & 6 & \quad \quad \quad 797 \end{array}$$

$$\bar{x}_{\text{weighted}} = \frac{797}{6} = 133 \text{ units.}$$

Semi-Average Method

The semi-average method permits us to estimate the slope and intercept of the trend line quite easily if a linear function will adequately describe the data. The procedure is simply to divide the data into two parts and compute their respective arithmetic means. These two points are plotted corresponding to their midpoint of the class interval covered by the respective part and then these points are joined by a straight line, which is the required trend line. The arithmetic mean of the first part is the intercept value, and the slope is determined by the ratio of the difference in the arithmetic mean of the number of years between them, that is, the change per unit time. The resultant is a time series of the form: $\hat{y} = a + bx$. The $\hat{y}$ is the calculated trend value, and a and b are the intercept and slope values, respectively. The equation should always be stated completely with reference to the year where $x = 0$ and a description of the units of x and y .

The semi-average method of developing a trend equation is relatively easy to commute and may be satisfactory if the trend is linear. If the data deviate much from linearity, the forecast will be biased and less reliable.

Example 12.6: Fit a trend line to the following data by the method of semi-average and forecast the sales for the year 2002.

Year	Sales of Firm (thousand units)	Year	Sales of Firm (thousand units)
1993	102	1997	108
1994	105	1998	116
1995	114	1999	112
1996	110		

Solution: Since number of years are odd in number, therefore divide the data into equal parts (A and B) of 3 years ignoring the middle year (1996). The average of part A and B is

$$\bar{y}_A = \frac{102 + 105 + 114}{3} = \frac{321}{3} = 107 \text{ units}$$

$$\bar{y}_B = \frac{108 + 116 + 112}{3} = \frac{336}{3} = 112 \text{ units}$$

Part A is centred upon 1994 and part B on 1998. Plot points 107 and 112 against their middle years, 1994 and 1998. By joining these points, we obtain the required trend line as shown in Fig. 12.6. The line can be extended and be used for prediction.

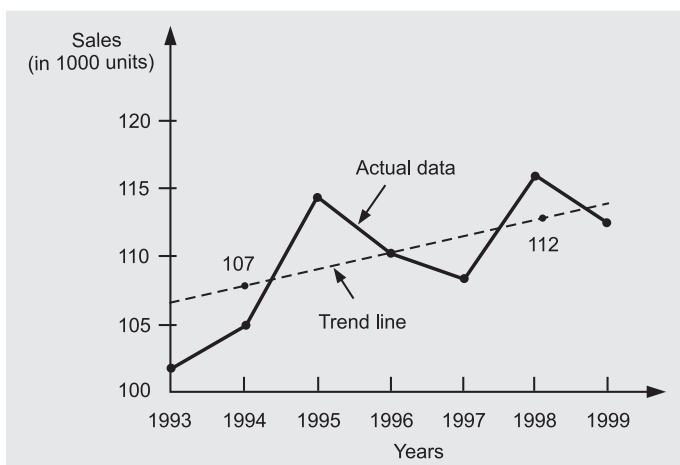


Figure 12.6 Trend Line by the Method of Semi-average

To calculate the time series $\hat{y} = a + bx$, we need

$$\begin{aligned} \text{Slope} = b &= \frac{\Delta y}{\Delta x} = \frac{\text{change in sales}}{\text{change in year}} \\ &= \frac{112 - 107}{1998 - 1994} = \frac{5}{4} = 1.25 \end{aligned}$$

$$\text{Intercept} = a = 107 \text{ units at } 1994$$

Thus, the trend line is $\hat{y} = 107 + 1.25x$

Since 2002 is eight-year distant from the origin (1994), therefore we have

$$\hat{y} = 107 + 1.25(8) = 117$$

Example 12.7: In the study of sales, a company obtained the following trend equation: $y_c = 16 + 2x$ (Origin 1995, x unit = 1 year, y = total number of units sold).

The company has the physical facilities to provide only 30 units in a year and it believes that atleast for the next decade trend will continue as before. Find

- What is the average annual increase in the number of units sold?
- By which year the company's expected sales have equalled to its present capacity?
- Estimate the sales for the year 1998. [Delhi Univ., B.Com.(Hons.), 2003]

Solution: (a) Trend equation is $y_c = 16 + 2x$. Since slope of this line is $b = 2$, therefore average annual increase is 2 units.

- Since the company's present capacity is 30 units, substituting $y = 30$ in the trend equation, we get $30 = 16 + 2x$ or $x = 7$. Thus, in seven years, the company's expected sales have equalled the present capacity. Since 1995 is taken as origin, therefore required year would be $1995 + 7 = 2002$ year.

- Since 1995 is origin, therefore, for estimating sales of 1998, putting $x = 1998 - 1995 = 3$ in the trend equation, we get $y = 16 + 2(3) = 22$ units.

Example 12.8: Trend equation for yearly sales (₹ in '000) for a commodity is: $y = 81.6 + 28.8x$ (unit of $x = 1$ year, origin is July 16, 1991). Adjust the trend equation to find the monthly trend values with January 1992 as origin and find the trend values for March 1992.

[Delhi Univ., B.Com.(Hons.), 2003]

Solution: Annual trend equation is $y = 81.6 + 28.8x$. Therefore, monthly trend equation is

$$y = \frac{81.6}{12} + \frac{28.8}{12 \times 12}x$$

Here x unit = one month and origin = July 16, 1991

Since the required origin is January 1992 i.e., Jan. 16, 1992, the trend equation will be obtained by increasing x by 6 months because, Jan. 16, 1992–July 16, 1992 = 6 months

$$y = 6.8 + 0.2(x + 6) = 0.2x + 8$$

To find trend value for March 1992, we will put $x = 2$ in trend equation as March 16, 1992–Jan. 16, 1992 = 2 months

$$y = 0.2(2) + 8 = 8.4 \text{ (₹ in thousand)}$$

Example 12.9: Give below is the quarterly trend equation for sales (₹ in thousand) of a commodity:

$$y_c = 130 + 1.8x$$

[Origin: first quarter of 2002; x unit = 1 quarter, y unit = average quarterly sales (₹ in thousand)]

Convert the above equation to annual trend equation and estimate the sales for the year 2006. [Delhi Univ., B.Com.(Hons.), 2005]

Solution: Quarterly trend equation is: $y_c = 130 + 1.8x$; origin as first quarter of 2002, i.e. February 15, 2002. To convert it into annual trend, shift origin to June 30, 2002 (middle of year 2002). That is shift x by June 30, 2002–February 15, 2002 = 4.5 months or 1.5 quarters

Thus, the trend equation with June 30, 2002 as origin becomes

$$y_c = 130 + 1.8(x + 1.5) = 132.7 + 1.8x$$

The annual trend equation then is

$$y_c = 132.7 \times 4 + (1.8)(16x) = 530.8 + 28.8x$$

Putting $x = 4$ to get expected sales for 2006: $y_c = 530.8 + 28.8(4) = 530.8 + 115.2 = ₹ 646$.

Self-practice Problems 12A

- 12.1** The owner of a small company manufactures a product. Since he started the company, the number of units of the product he has sold is represented by the following time series:

Year :	1995	1996	1997	1998	1999	2000	2001
Units sold	100	120	95	105	108	102	112

Find the trend line that describes the trend by using the method of semi-averages.

- 12.2** Fit a trend line to the following data by the freehand method:

Year	Production of Steel (million tonnes)	Year	Production of Steel (million tonnes)
1995	20	2000	25
1996	22	2001	23
1997	24	2002	26
1998	21	2003	25
1999	23		

- 12.3** A State Government is studying the number of traffic fatalities in the state resulting from drunken driving for each of the last 12 months:

Month	Accidents
1	280
2	300
3	280
4	280
5	270
6	240
7	230
8	230
9	220
10	200
11	210
12	200

Find the trend line that describes the trend by using the method of semi-averages.

- 12.4** Calculate the three-month moving averages from the following data:

Jan.	Feb.	March	April	May	June
57	65	63	72	69	78
July	Aug.	Sept.	Oct.	Nov.	Dec.
82	81	90	92	95	97

[Osmania Univ., B.Com., 1996]

- 12.5** Gross revenue data (₹ in million) for a Travel Agency for a 11-year period is as follows:

Year	Revenue
1995	3
1996	6
1997	10
1998	8
1999	7
2000	12
2001	14
2002	14
2003	18
2004	19

Calculate a three-year moving average for the revenue earned.

- 12.6** The owner of small manufacturing company has been concerned about the increase in manufacturing costs over the past 10 years. The following data provide a time series of the cost per unit for the company's leading product over the past 10 years.

Year	Cost per Unit	Year	Cost per Unit
1995	332	2000	405
1996	317	2001	410
1997	357	2002	427
1998	392	2003	405
1999	402	2004	438

Calculate a five-year moving average for the unit cost of the product.

- 12.7** The following data provide a time series of the number of Commercial and Industrial units failures during the period 1989–2004.

<i>Year</i>	<i>No. of Failures</i>	<i>Year</i>	<i>No. of Failures</i>
1989	23	1997	9
1990	26	1998	13
1991	28	1999	11
1992	32	2000	14
1993	20	2001	12
1994	12	2002	9
1995	12	2003	3
1996	10	2004	1

Calculate a five-year and seven-year moving average for the number of units failure.

- 12.8** Estimate the trend values using the data given by taking a four-year moving average :

<i>Year</i>	<i>Value</i>	<i>Year</i>	<i>Value</i>
1990	12	1997	100
1991	25	1998	82
1992	39	1999	65
1993	54	2000	49
1994	70	2001	34
1995	87	2002	20
1996	105	2003	7

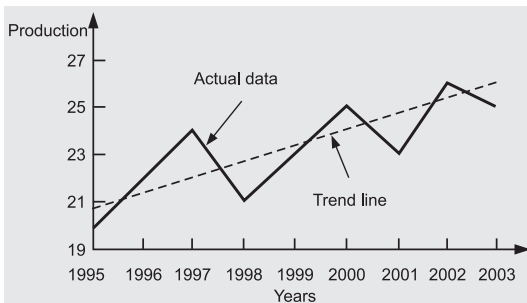
Hints and Answers

12.1

<i>Year (y)</i>	<i>Units Sold (x)</i>	
1995	10	} $315/3 = 105.00 = a$
1996	12	
1997	9	
1998	105	
1999	10	} $322/3 = 107.33 = b$
2000	10	
2001	11	

Trend line $y = 105 + 107.33x$.

12.2



12.3

<i>Month</i>	<i>Accidents</i>
1	280
2	300
3	280
4	280
5	270
6	240
7	230
8	230
9	220
10	200
11	210
12	200

Average of first 6 months,

$$a = 1650/6 = 275$$

Average of last 6 months, $b = 1290/6 = 215$

Trend line $y = 275 + 215x$.

12.4

Month	Values	three-month Total	three-month Moving Average
Jan.	57	—	—
Feb.	65	185	$185/3 = 61.67$
March	63	200	$200/3 = 66.67$
April	72	204	$204/3 = 68.00$
May	69	219	73.00
June	78	229	76.33
July	82	241	80.33
Aug.	81	253	84.33
Sept.	90	263	87.67
Oct.	92	277	92.33
Nov.	95	284	94.67
Dec.	97	—	—

12.5

Year	Revenue	three-year Moving Total	three-year Moving Average
1995	3	—	—
1996	6	19	$19/3 = 6.33$
1997	10	24	$24/3 = 8.00$
1998	8	25	$25/3 = 8.33$
1999	7	27	9
2000	12	33	11
2001	14	40	13.33
2002	14	46	15.33
2003	18	51	17.00
2004	19	—	—

12.6

Year	Per Unit Cost	five-year Moving Total	five-year Moving Average
1995	332	—	—
1996	317	—	—
1997	357	1800	$1800/5 = 360.0$
1998	392	1873	$1873/5 = 374.6$
1999	402	1966	$1966/5 = 393.2$
2000	405	2036	407.2
2001	410	2049	409.8
2002	427	2085	417.0
2003	405	—	—
2004	438	—	—

12.7

Year	Number of Failures	five- year Moving Total	five- year Moving Average	seven- year Moving Total	seven- year Moving Average
1989	23	—	—	—	—
1990	26	—	—	—	—
1991	28	129	25.8	—	—
1992	32	118	23.6	153	21.9
1993	20	104	20.8	140	20.0
1994	12	86	17.2	123	17.6
1995	12	63	12.6	108	15.4
1996	10	56	11.2	87	12.4
1997	9	55	11.0	81	11.6
1998	13	57	11.4	81	11.6
1999	11	59	11.8	78	11.1
2000	14	59	11.8	71	10.1
2001	12	49	9.8	63	9.0
2002	9	39	7.8	—	—
2003	3	—	—	—	—
2004	1	—	—	—	—

12.8

Year	Value	4-year Moving Total	Moving Average
1990	12	—	—
1991	25	—	—
1992	39	130	$130/4 = 32.5$
1993	54	188	$188/4 = 47.0$
1994	70	250	$250/4 = 62.5$
1995	87	316	79.0
			$(32.5 + 47)/2 = 39.75$
			$(47 + 62.5)/2 = 54.75$
			70.75
			84.75

<i>Year</i>	<i>Value</i>	<i>4-year Moving Total</i>	<i>4-year Moving Average</i>	<i>4-year Moving Total</i>	<i>Moving Moving Average</i>	<i>Year</i>	<i>Value</i>	<i>4-year Moving Total</i>	<i>4-year Moving Average</i>	<i>4-year Moving Total</i>	<i>Moving Moving Average</i>
	<i>Centred</i>					<i>Average</i>	<i>Centred</i>				
		362	90.5					230	57.5		
1996	105				92.00	2000	49				49.75
		374	93.5					168	42.0		
1997	100				90.75	2001	34				34.75
		352	88.0					110	27.5		
1998	82				81.00	2002	20	—	—		—
		296	74.0			2003	7	—	—		—
1999	65				65.75						

12.6 TREND PROJECTION METHODS

A *trend* is the long-run general direction (upward, downward or constant) of a business climate over a period of several year. It is best represented by a straight line.

The trend projection method fits a trend line to a time series data and then projects medium-to-long-range forecasts. Several possible trend fits can be explored (such as exponential and quadratic) depending upon movement of time series data. In this section, we will discuss linear, quadratic and exponential trend models. Since seasonal effects can compound trend analysis, it is assumed that no seasonal effects occur in the data or are removed before establishing the trend.

Reasons to study trend: A few reasons to study trends are as follows:

- The study of trend helps to describe the long-run general direction (upward, downward and constant) of a business climate over a period of several years.
- The study allows us to use trends as an aid in making intermediate and long-range forecasting projections in the future.
- The study of trends help to isolate and then eliminate its influencing effects on the time series model.

12.6.1 Linear Trend Model

The *method of least squares* from regression analysis is used to find the *trend line of best fit* to a time series data. The regression trend line ($\hat{y}$) is defined by the following equation:

$$\hat{y} = a + bx$$

where $\hat{y}$ = predicted value of the dependent variable

a = y-axis intercept

b = slope of the regression line (or the rate of change in y for a given change in x)

x = independent variable (which is *time* in this case)

The trend line of best fit has the properties that (i) the summation of all vertical deviations about it is zero, that is, $\Sigma (y - \hat{y}) = 0$, (ii) the summation of all vertical deviations squared is a

minimum, that is, $\Sigma (y - \hat{y})$ is least and (iii) the line goes through the mean values of variables x and y . For linear equations, it is found by the simultaneous solution for a and b of the two normal equations:

$$\Sigma y = na + b\Sigma x \quad \text{and} \quad \Sigma xy = a\Sigma x + b\Sigma x^2$$

where the data can be coded so that $\Sigma x = 0$, two terms in these equations drop out and we have

$$\Sigma y = na \quad \text{and} \quad \Sigma xy = b\Sigma x^2$$

Coding is easily done with time series data. For coding the data, we choose the centre of the time period as $x = 0$ and have an equal number of plus and minus periods on each side of the trend line which sum to zero.

Alternately, we can also find the values of constants a and b for any regression line as

$$b = \frac{\Sigma xy - n\bar{x}\bar{y}}{\Sigma x^2 - n(\bar{x})^2} \quad \text{and} \quad a = \bar{y} - b\bar{x}$$

Example 12.10: Below are given the figures of production (in thousand quintals) of a sugar factory:

Year	:	1995	1996	1997	1998	1999	2000	2001
Production	:	80	90	92	83	94	99	92

- Fit a straight line trend to these figures.
- Plot these figures on a graph and show the trend line.
- Estimate the production in 2004.

[Bangalore Univ., B.Com., 1998]

Solution: (a) Using normal equations and the sugar production data, we can compute constants a and b as shown in Table 12.5:

Table 12.5 Calculation for Least-squares Equation

Year	Time Period (x)	Production (y)	x^2	xy	Trend Values $\hat{y}$
1995	1	80	1	80	84
1996	2	90	4	180	86
1997	3	92	9	276	88
1998	4	83	16	332	90
1999	5	94	25	470	92
2000	6	99	36	594	94
2001	7	92	49	644	96
	<u>28</u>	<u>630</u>	<u>140</u>	<u>2576</u>	

$$\bar{x} = \frac{\Sigma x}{n} = \frac{28}{7} = 4, \quad \bar{y} = \frac{\Sigma y}{n} = \frac{630}{7} = 90$$

$$b = \frac{\Sigma xy - n\bar{x}\bar{y}}{\Sigma x^2 - n(\bar{x})^2} = \frac{2576 - 7(4)(90)}{140 - 7(4)^2} = \frac{56}{28} = 2$$

$$a = \bar{y} - b\bar{x} = 90 - 2(4) = 82$$

Therefore, linear trend component for the production of sugar is:

$$\hat{y} = a + bx = 82 + 2x$$

The slope $b = 2$ indicates that over the past 7 years, the production of sugar had an average growth of about 2 thousand quintals per year.

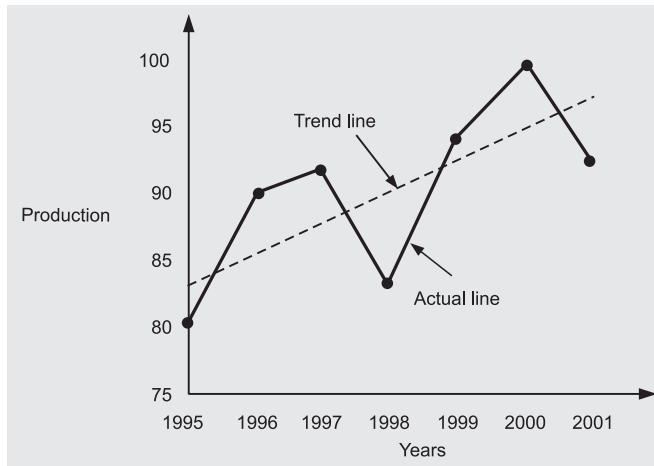


Figure 12.7 Linear Trend for Production of Sugar

- (b) Plotting points on the graph paper, we get an actual graph representing production of sugar over the past 7 years. Join the point $a = 82$ and $b = 2$ (corresponds to 1996) on the graph, we get a trend line as shown in Fig. 12.7.
- (c) The production of sugar for year 2004 will be

$$\hat{y} = 82 + 2(10) = 102 \text{ thousand quintals}$$

Example 12.11: The following table relates to the tourist arrivals (in millions) during 1994 to 2000 in India:

Year	: 1994	1995	1996	1997	1998	1999	2000
Tourists arrivals :	18	20	23	25	24	28	30

Fit a straight line trend by the method of least squares and estimate the number of tourists that would arrive in the year 2004.

Solution: Using normal equations and the tourists arrival data, we can compute constants a and b as shown in Table 12.6:

Table 12.6 Calculations for Least-squares Equation

<i>Year</i>	<i>Time Scale (x)</i>	<i>Tourist Arrivals (y)</i>	<i>xy</i>	<i>x²</i>
1994	-3	18	-54	9
1995	-2	20	-40	4
1996	-1	23	-23	1
1997	0	25	0	0
1998	1	24	24	1
1999	2	28	56	4
2000	3	30	90	9
	<u>0</u>	<u>168</u>	<u>53</u>	<u>28</u>

$$\hat{x} = \frac{\Sigma x}{n} = 0, \bar{y} = \frac{\Sigma y}{n} = \frac{168}{7} = 24$$

$$b = \frac{\Sigma xy - n\bar{x}\bar{y}}{\Sigma x^2 - n(\bar{x})^2} = \frac{53}{28} = 1.893;$$

$$a = \bar{y} - b\bar{x} = 24 - 1.893(0) = 24$$

Therefore, the linear trend component for arrival of tourists is

$$\hat{y} = a + bx = 24 + 1.893x$$

The estimated number of tourists that would arrive in the year 2004 are

$$\hat{y} = 24 + 1.893(7) = 37.251 \text{ million (measured from 1997 = origin)}$$

Example 12.12: From the following data, calculate trend by method of least squares:

Year	: 1970	1971	1972	1973	1974	1975	1976
Profit ('000 ₹)	: 300	700	600	800	900	700	1000

[Delhi Univ., B.Com. (P) 1985]

Solution: Using normal equation, the calculations required to determine trend are shown below:

<i>Year (t)</i>	<i>Profit (₹ 000) y</i>	<i>x = t - 1973</i>	<i>x²</i>	<i>xy</i>
1970	300	-3	9	-900
1971	700	-2	4	-1400
1972	600	-1	1	-600
1973	800	0	0	0
1974	900	1	1	900
1975	700	2	4	1400
1976	1000	3	9	3000
	<u>5000</u>	<u>0</u>	<u>28</u>	<u>2400</u>

Equation of straight line trend, $y_c = a + bx$

$$\text{where } a = \frac{\Sigma y}{n} = \frac{5000}{7} = 714.2 \text{ and } b = \frac{\Sigma xy}{\Sigma x^2} = \frac{2400}{28} = 85.71$$

Hence the trend line becomes

$$\begin{aligned}
 y_c &= a + bx = 714.2 + 85.71x \\
 y_{1970} &= 714.2 + 85.71(-3) = 457.15; Y_{1971} = 714.2 + 85.71(-2) = 542.86 \\
 y_{1972} &= 714.2 + 85.71(-1) = 628.57; Y_{1973} = 714.2 + 85.71(0) = 714.28 \\
 y_{1974} &= 714.2 + 85.71(1) = 799.99; Y_{1975} = 714.2 + 85.71(2) = 885.70 \\
 y_{1976} &= 714.2 + 85.71(3) = 971.41
 \end{aligned}$$

Example 12.13: Calculate trend values by method of least squares from the data given below and also estimate the sales for year 1991:

Year	: 1986	1987	1988	1989	1990	
Sales (in crores) :	12	18	20	23	27	[Delhi Univ., B.Com.(P), 1992]

Solution: Using normal equations, calculations required to estimate the sales for the year 1991 are shown below:

Year (t)	$x = t - 1988$	y (sales)	x^2	xy
1986	-2	12	4	-24
1987	-1	18	1	-18
1988	0	20	0	0
1989	1	23	1	23
1990	2	27	4	54
Total	0	100	10	35

Let the straight line trend be: $y_c = a + bx$, where $a = \frac{\Sigma y}{n} = \frac{100}{5} = 20$ and $b = \frac{\Sigma xy}{\Sigma x^2} = \frac{35}{10} = 3.5$

Hence, $y_c = 20 + 3.5x$. Putting $x = 3$, in the trend line to estimate sales for year 1991 as follows:

$$Y_{1991} = 20 + 3.5(3) = ₹ 30.5 \text{ crore.}$$

Example 12.14: Fit a straight line trend to the following data by least squares method after summing the given quarterly data to yearly data. Also tabulate short-term fluctuations.

Export of Cotton Textile (Million ₹)

Year	Quarter I	Quarter II	Quarter III	Quarter IV
1998	10	13	14	12
1999	12	14	15	13
2000	13	15	18	14
2001	15	18	21	18
2002	15	22	23	20

Plot the trend values and actual values and draw the trend line. [Delhi Univ., B.Com.(Hons.), 1989]

Solution: Let the year 2000 be origin. Also x represents time in years and y represents exports in millions of rupees.

Convert the quarterly data into yearly data as follows:

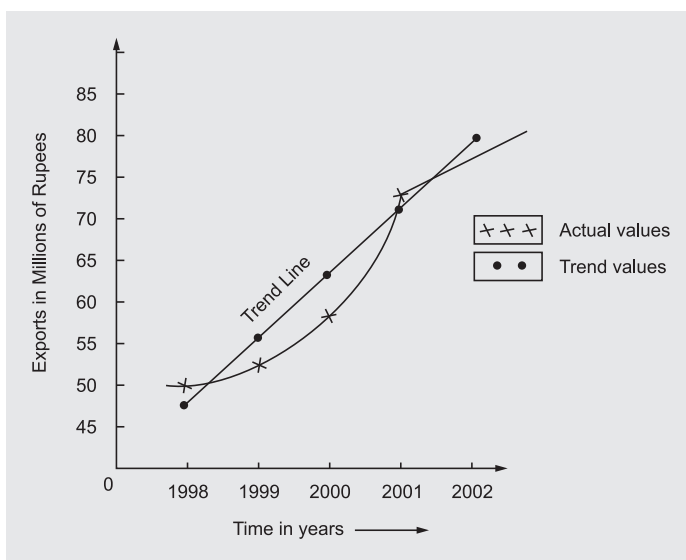
Year	Quarter I	Quarter II	Quarter III	Quarter IV	Yearly total
1998	10	13	14	12	49
1999	12	14	15	13	54
2000	13	15	18	14	60
2001	15	18	21	18	72
2002	15	22	23	20	80

Calculations required to fit a straight-line trend are shown below:

Year	x ($t - 2000$)	x^2	y	xy	$y_c = 63 + 8x$	Short term Fluctuations $y - y_c$
1998	-2	4	49	-98	$63 + 8(-2) = 47$	$49 - 47 = 2$
1999	-1	1	54	-54	$63 + 8(-1) = 55$	$54 - 55 = -1$
2000	0	0	60	0	$63 + 8(0) = 63$	$60 - 63 = -3$
2001	1	1	72	72	$63 + 8(1) = 71$	$72 - 71 = 1$
2002	2	4	80	160	$63 + 8(2) = 79$	$80 - 79 = 1$
	<u>0</u>	<u>10</u>	<u>315</u>	<u>80</u>		

Let the straight line trend be $y = a + bx$, where $a = \frac{\Sigma y}{n} = \frac{315}{5} = 63$ and $b = \frac{\Sigma xy}{\Sigma x^2} = \frac{80}{10} = 8$
Hence, $y = a + bx = 63 + 8x$

Plotting trend values on a graph, the trend line so obtained is shown below:



12.6.2 Exponential Trend Model

When the given values of dependent variable y form approximately a geometric progression while the corresponding independent variable x values form an arithmetic progression, the relationship between variables x and y is given by an exponential function, and the best-fitting curve is said to describe the *exponential trend*. Data from the fields of biology, banking, and economics frequently exhibit such a trend. For example, growth of bacteria, money accumulating at compound interest, sales or earnings over a short period, and so on, follow exponential growth.

The characteristic property of this law is that the rate of growth, that is, the rate of change of y with respect to x is proportional to the values of the function. The following function has this property.

$$y = a b^x, a > 0$$

The letter b is a fixed constant, usually either 10 or e , where a is a constant to be determined from the data.

To assume that the law of growth will continue is usually unwarranted, so only short-range predictions can be made with any considerable degree of reliability.

If we take logarithms (with base 10) of both sides of the above equation, we obtain

$$\log y = \log a + (c \log b) x$$

For $b = 10$, $\log b = 1$, but for $b = e$, $\log b = 0.4343$ (approx.). In either case, this equation is of the form

$$y' = c + dx \quad (i)$$

where $y' = \log y$, $c = \log a$ and $d = c \log b$.

Equation (i) represents a straight line. A method of fitting an exponential trend line to a set of observed values of y is to fit a straight trend line to the logarithms of the y -values.

In order to find out the values of constants a and b in the exponential function, the two normal equations to be solved are

$$\begin{aligned} \Sigma \log y &= n \log a + \log b \Sigma x \\ \Sigma x \log y &= \log a \Sigma x + \log b \Sigma x^2 \end{aligned}$$

When the data is coded so that $\Sigma x = 0$, the two normal equations become

$$\begin{aligned} \Sigma \log y &= n \log a \quad \text{or} \quad \log a = \frac{1}{n} \Sigma \log y \\ \text{and} \quad \Sigma x \log y &= \log b \Sigma x^2 \quad \text{or} \quad \log b = \frac{\Sigma x \log y}{\Sigma x^2} \end{aligned}$$

Coding is easily done with time series data by simply designating the center of the time period as $x = 0$, and have equal number of plus and minus period on each side which sum to zero.

Example 12.15: The sales (₹ in million) of a company for the years 1995 to 1999 are

Year	: 1997	1998	1999	2000	2001
Sales	: 1.6	4.5	13.8	40.2	125.0

Find the exponential trend for the given data and estimate the sales for 2004.

Solution: The computational time can be reduced by coding the data. For this consider $u = x - 3$. The necessary computations are shown in Table 12.7.

Table 12.7 Calculation for Least-squares Equation

Year	Time Period x	$u = x - 3$	u^2	Sales y	$\log y$	$u \log y$
1997	1	-2	4	1.60	0.2041	-0.4082
1998	2	-1	1	4.50	0.6532	-0.6532
1999	3	0	0	13.80	1.1390	0
2000	4	1	1	40.20	1.6042	1.6042
2001	5	2	4	125.00	2.0969	4.1938
			10		5.6983	4.7366

$$\log a = \frac{1}{n} \Sigma \log y = \frac{1}{5} (5.6983) = 1.1397$$

$$\log b = \frac{\Sigma u \log y}{\Sigma u^2} = \frac{4.7366}{10} = 0.4737$$

Therefore, $\log y = \log a + (x + 3) \log b = 1.1397 + 0.4737x$

For sales during 2004, $x = 3$, and we obtain

$$\log y = 1.1397 + 0.4737(3) = 2.5608$$

or

$$y = \text{antilog}(2.5608) = 363.80$$

Example 12.16: Fit an exponential trend to the following data:

Year	: 2001	2002	2003	2004	2005	2006	2007
Sales (in lakhs of Rs.)	: 32	47	65	92	132	190	275

Solution: Calculations to fit an exponential trend to the given data are shown below:

Year	x	Sales (y)	$\log y$	x^2	$x \log y$
2001	-3	32	1.5051	9	-4.5153
2002	-2	47	1.6721	4	-3.3442
2003	-1	65	1.8129	1	1.8129
2004	0	92	1.9638	0	0
2005	1	132	2.1206	1	2.1206
2006	2	190	2.2788	4	4.5576
2007	3	275	2.4393	9	7.3179
	0	833	13.7926	28	4.3237

Let year 2004 be the origin and the exponential trend equation be $y = ab^x$. Then normal equations are

$$\Sigma \log y = n \log a + \log b \Sigma x \quad \text{or} \quad 13.7926 = 7 \log a \quad \text{or} \quad \log a = 1.9704$$

$$\text{Also} \quad \Sigma x \log y = \log a \Sigma x + \log b \Sigma x^2$$

$$4.3237 = 0 + 28 \log b \quad \text{or} \quad \log b = 0.154$$

Then $\log y = 1.9704 + 0.154x$

For sales in 2008, $x = 4$. Thus, $\log y = 1.9704 + 0.154(4) = 2.5864$.

Hence $y = \text{Antilog}(2.5864) = 385.9$

12.6.3 Changing the Origin and Scale of Equations

When a moving average or trend value is calculated, it is assumed to be centred in the middle of the month (fifteenth day) or the year (July 1). Similarly, the forecast value is assumed to be centred in the middle of the future period. However, the reference point (origin) can be shifted, or the units of variables x and y are changed to monthly or quarterly values if desired. The procedure is as follows:

- (i) Shift the origin, simply by adding or subtracting the desired number of periods from independent variable x in the original forecasting equation.
- (ii) Change the time units from annual values to monthly values by dividing independent variable x by 12.
- (iii) Change the y units from annual to monthly values, the entire right-hand side of the equation must be divided by 12.

Example 12.17: The following forecasting equation has been derived by a least-squares method:

$$= 10.27 + 1.65x \quad (\text{Base year: 1997; } x = \text{years; } y = \text{tonnes/year})$$

Rewrite the equation by

- (a) shifting the origin to 2002.
- (b) expressing x units in months, retaining y in tonnes/year.
- (c) expressing x units in months and y in tonnes/month.

Solution: (a) Shifting of origin can be done by adding the desired number of period 5 (1997 to 2002) to x in the given equation. That is

$$\hat{y} = 10.27 + 1.65(x + 5) = 18.52 + 1.65x$$

where 2002 = 0, x = years, y = tonnes/year.

(b) Expressing x units in months

$$\hat{y} = 10.27 + \frac{1.65x}{12} = 10.27 + 0.14x$$

where July 1, 1997 = 0, x = months, y = tonnes/year.

(c) Expressing y in tonnes/month, retaining x in months

$$\hat{y} = \frac{1}{12}(10.27 + 0.14x) = 0.86 + 0.01x$$

where July 1, 1997 = 0, x = months, y = tonnes/month.

Remarks

1. If both x and y are to be expressed in months together, then divide constant ' a ' by 12 and constant ' b ' by 24. It is because data are sums of 12 months. Thus, monthly trend equation becomes

$$\text{Linear trend} : \hat{y} = \frac{a}{12} + \frac{b}{24}x$$

$$\text{Parabolic trend} : \hat{y} = \frac{a}{12} + \frac{b}{144}x + \frac{c}{1728}x^2$$

But if data are given as monthly averages per year, then value of 'a' remains unchanged, 'b' is divided by 12 and 'c' by 144.

2. The annual trend equation can be reduced to quarterly trend equation as

$$\hat{y} = \frac{a}{4} + \frac{b}{4 \times 12}x = \frac{a}{4} + \frac{b}{48}x$$

Self-practice Problems 12B

- 12.9** The general manager of a building materials production plant feels that the demand for plasterboard shipments may be related to the number of construction permits issued in the country during the previous quarter. The manager has collected the data shown in the table.

Construction Permits	Plasterboard Shipments
15	6
9	4
40	16
20	6
25	13
25	9
15	10
35	16

- (a) Use the normal equations to derive a regression forecasting equation.
 (b) Determine a point estimate for plasterboard shipments when the number of construction permits is 30.
- 12.10** A company that manufactures steel, observed the production of steel (in metric tonnes) represented by the time series:

Year : 1996 1997 1998 1999 2000 2001 2002
 Production

of steel: 60 72 75 65 80 85 95

- (a) Find the linear equation that describes the trend in the production of steel by the company.
 (b) Estimate the production of steel in 2003.

- 12.11** Fit a straight line trend by the method of least squares to the following data. Assuming that the same rate of change continues, what would be the predicted earning (₹ in lakh) for the year 2004?

Year : 1995 1996 1997 1998 1999 2000 2001 2002
 Earnings:

38 40 65 72 69 60 87 95

[Agra Univ., B.Com. 1996; MD Univ., B.Com., 1998]

- 12.12** The sales (₹ in lakh) of a company for the years 1990 to 1996 are given below:

Year : 1998 1999 2000 2001 2002 2003 2004
 Sales : 32 47 65 88 132 190 275

Find trend values by using the equation $y_c = a b^x$ and estimate the value for 2005.

[Delhi Univ., B.Com., 1996]

- 12.13** A company that specializes in the production of petrol filters has recorded the following production (in '000 units) over the last 7 years.

Years : 1995 96 97 98 99 00 01
 Production : 42 49 62 75 92 122 158

- (a) Develop a second-degree estimating equation that best describes these data.
 (b) Estimate the production in 2005.
- 12.14** In 1996, a firm began downsizing in order to reduce its costs. One of the results of these cost cutting measures has been a decline in the percentage of private industry jobs that are managerial. The following data show the percentage of females who are managers from 1996 to 2003.

Years : 1996 97 98 99 00 01 02 03
 Percentage : 6.7 5.3 4.3 6.1 5.6 7.9 5.8 6.1

- (a) Develop a linear trend line for this time series through 2001 only.
 (b) Use this trend to estimate the percentage of females who are managers in 2004.

12.15 A company develops, markets, manufactures, and sells integrated wide-area network

access products. The following are annual sales (₹ in million) data from 1998 to 2004.

Year : 1998 1999 2000 2001 2002 2003 2004
 Sales : 16 17 25 28 32 43 50

- (a) Develop the second-degree estimating equation that best describes these data.
 (b) Use the trend equation to forecast sales for 2005.

Hints and Answers

12.9 (a)

x	y	xy	x^2	y^2
15	6	90	225	36
9	4	36	81	16
40	16	640	1,600	256
20	6	120	400	36
25	13	325	625	169
25	9	225	625	81
15	10	150	225	100
35	16	560	1,225	256
184	80	2,146	5,006	950

$n = 8$ pairs of observations;

$$\bar{x} = 184/8 = 23; \bar{y} = 80/8 = 10$$

$$\Sigma y = na + b\Sigma x$$

or $80 = 8a + 184b$

$$\Sigma xy = a\Sigma x + b\Sigma x^2$$

or $2,146 = 184a + 5,006b$

After solving equations, we get $a = 0.915$ and $b = 0.395$. Therefore, the equation is

$$\hat{y} = 0.915 + 0.395x$$

- (b) For $x = 30$, we have $\hat{y} = 0.915 + 0.395(30) = 13$ shipments (approx.)

12.10 $a = \Sigma y/n = 532/7 = 76$;

$$b = \Sigma xy/\Sigma x^2 = 136/28 = 4.857$$

(a) Trend line $\hat{y} = a + bx = 76 + 4.857x$

(b) For 2003, $x = 4$, $\hat{y} = 76 + 4.857(4) = 95.428$ metric tonnes.

12.11 $a = \Sigma y/n = 526/8 = 65.75$;

$$b = \Sigma xy/\Sigma x^2 = 616/168 = 3.667$$

$$\text{Trend line : } \hat{y} = a + bx = 65.75 + 3.667x$$

For 2004, $x = 11$,

$$\hat{y} = 65.75 + 3.667(11) = ₹ 106.087 \text{ lakh.}$$

$$\mathbf{12.12} \quad \log a = \frac{1}{n} \Sigma \log y = \frac{1}{7} (13.7926) = 1.9704$$

$$\log b = \frac{\Sigma x \log y}{\Sigma x^2} = \frac{4.3237}{28} = 0.154$$

$$\begin{aligned} \text{Thus } \log y &= \log a + x \log b \\ &= 1.9704 + 0.154x \end{aligned}$$

For 2005, $x = 4$;

$$\log y = 1.9704 + 0.154(4) = 2.5864$$

$$y = \text{antilog}(2.5864)$$

$$= ₹ 385.9 \text{ lakh.}$$

12.13

Year	Period	Deviation from 1998 (x)	x^2	x^4	y	xy	x^2y
1995	1	-3	9	81	42	-126	378
1996	2	-2	4	16	49	-98	196
1997	3	-1	1	1	62	-62	62
1998	4	0	0	0	75	0	0
1999	5	1	1	1	92	+92	92
2000	6	2	4	16	122	+244	488
2001	7	3	9	81	158	+474	1422
		0	28	196	600	524	2638

- (a) Solving the equations

$$\Sigma y = na + c\Sigma x^2$$

or $600 = 7a + 28c$

$$\Sigma x^2 y = a \Sigma x^2 + c \Sigma x^4$$

$$\text{or } 2638 = 28a + 196c$$

$$\Sigma xy = b \Sigma x^2$$

$$\text{or } 524 = 28b$$

$$\text{We get } a = 80.05, b = 18.71 \text{ and}$$

$$c = -1.417$$

$$\text{Hence } \hat{y} = a + bx + cx^2$$

$$= 80.05 + 18.71x - 1.417x^2$$

$$(b) \text{ For 2005, } x = 8;$$

$$\hat{y} = 80.05 + 18.71(8) - 1.417(8)^2$$

$$= ₹ 139.042 \text{ thousand.}$$

12.14

Year	Time Period	Deviation from 2001	Percentage of Females	xy	x ²
		x	y		
1996	1	-5	6.7	-33.5	25
1997	2	-4	5.3	-21.2	16
1998	3	-3	4.3	-12.9	9
1999	4	-2	6.1	-12.2	4
2000	5	-1	5.6	-5.6	1
2001	6	0	7.9	0	0
2002	7	1	5.8	5.8	1
2003	8	2	6.1	12.2	4
		-12	47.8	-67.4	60

(a) Solving the equations

$$\Sigma y = na + b \Sigma x$$

$$\text{or } 47.8 = 8a - 12b$$

$$\Sigma xy = a \Sigma x + b \Sigma x^2$$

$$\text{or } -67.4 = -12a + 60b$$

$$\text{We get } a = 6.28 \text{ and } b = 0.102$$

$$\text{Hence } = a + bx = 6.128 + 0.102x$$

$$(b) \text{ For 2004, } x = 3; \hat{y} = 6.128 + 0.102(3)$$

$$= 6.434 \text{ per cent.}$$

12.15

Year	Time Period	Deviation from 2001 (x)	Sales y	xy	x ²	x ⁴	x ² y
1998	1	-3	16	-48	9	81	144
1999	2	-2	17	-34	4	16	68
2000	3	-1	25	-25	1	1	25
2001	4	0	28	0	0	0	0
2002	5	1	32	32	1	1	32
2003	6	2	43	86	4	16	172
2004	7	3	50	150	9	81	450
		0	211	161	28	196	891

(a) Solving the equations

$$\Sigma y = na + c \Sigma x^2$$

$$\text{or } 211 = 7a + 28c$$

$$\Sigma x^2 y = a \Sigma x^2 + c \Sigma x^4$$

$$\text{or } 891 = 28a + 196c$$

$$\Sigma xy = b \Sigma x^2$$

$$\text{or } 161 = 28b$$

$$\text{We get } a = 27.904, b = 5.75 \text{ and}$$

$$c = 0.559$$

$$\hat{y} = a + bx + cx^2$$

$$= 27.904 + 5.75x + 0.559x^2$$

$$\text{For 2005, } x = 4$$

$$\hat{y} = 27.904 + 5.75(4) + 0.559(4)^2$$

$$= 59.848$$

12.7 MEASUREMENT OF SEASONAL EFFECTS

As mentioned earlier that time series data consists of four components: trend, cyclical effects, seasonal effects and irregular fluctuations. In this section, we will discuss techniques for identifying seasonal effects in a time series data. Seasonal effect is defined as the repetitive and predictable pattern of data behaviour in a time series around the trend line during particular time intervals of the year. In order to measure (or detect) the seasonal effect, time period must be less than one year such as days, weeks, months or quarters.

Seasonal effects arises as the result of natural changes in the seasons during the year or may result due to habits, customs or festivals that occur at the same time year after year.

We have three main reasons to study seasonal effects:

- (i) The description of the seasonal effect provides a better understanding of the impact of this component upon a particular time series.
- (ii) Once the seasonal pattern that exists is established, seasonal effect can be eliminated from the time series in order to observe the effect of the other components, such as cyclical and irregular components. Elimination of seasonal effect from the series is referred to as *deseasonalizing* or *seasonal adjusting* of the data.
- (iii) Trend analysis may be adequate for long-range forecast, but for short-run predictions knowledge of seasonal effects on time series data is essential for projection of past pattern into the future.

Remarks

1. In an additive time series model, we can estimate the seasonal component as

$$S = Y - (T + C + I)$$

In the absence of C and I, we have $S = Y - T$. That is, the seasonal component is the difference between actual data values in series and the trend values.

2. One of the technique for isolating the effects of seasonality is decomposition. The process of decomposition begins by determining TC for each and dividing the time series data (TCSI) by TC. The resulting expression contains seasonal effects along with irregular fluctuations

$$\frac{T.C.S}{T.C} = S.I.$$

A method for eliminating irregular fluctuations can be applied, leaving only the seasonal effects as shown below:

$$\text{Seasonal effect} = \frac{T.S.C}{T.C.I} = \frac{Y}{T.C.I} \times 100\%$$

3. The process of eliminating the effects of seasonality from a time series data is referred to as *de-seasonalization* or *seasonal adjustment*. The data can be deseasonalized by dividing the actual values Y by final adjusted seasonal effects, and is expressed as

$$\frac{Y}{S} = \frac{T.S.C.I}{S} = T.C.I \times 100 \text{ per cent} \quad \leftarrow \text{Multiplicative}$$

$$Y - S = (T + S + C + I) - S = T + C + I \quad \leftarrow \text{Additive Model}$$

Each adjusted seasonal index measures the average magnitude of seasonal influence on the actual values of the time series for a given period within a year. By subtracting the base index of 100 (which represents the T and C components) from each seasonal index, the extent of the influence of seasonal force can be measured.

12.7.1 Seasonal Index

Seasonal effects are measured in terms of an index, called *seasonal index*, attached to each period of the time series within a year. Hence, if monthly data are considered, there are 12 separate seasonal indexes, one for each month. Similarly for quarterly data, there are four separate

indexes. A seasonal index is an average that indicates the percentage deviation of actual values of the time series from a base value which excludes the short-term seasonal influences. The base time series value represents the trend/cyclical influences only.

The following four methods are used to construct seasonal indexes to measure seasonal effects in the time series data:

- (i) Method of simple averages
- (ii) Ratio-to-trend method
- (iii) Ratio-to-moving average method
- (iv) Link relatives method

12.7.2 Method of Simple Averages

This method is also called **average percentage method** because this method expresses the data of each month or quarter as a percentage of the average of the year. The steps of the method are summarized below:

- (i) Average the unadjusted data by years and months (or quarters if quarterly data are given).
- (ii) Add the figures of each month and obtain the averages by dividing the monthly totals by the number of years. Let the averages for 12 months be denoted by $\bar{x}_1, \bar{x}_2, \dots, \bar{x}_{12}$.
- (iii) Obtain an average of monthly averages by dividing the total of monthly averages by 12. That is

$$\bar{\bar{x}} = \frac{\bar{x}_1 + \bar{x}_2 + \dots + \bar{x}_{12}}{12}$$

- (iv) Compute seasonal indexes for different months by expressing monthly averages as percentages of the grand average $\bar{\bar{x}}$ as follows:

$$\begin{aligned} \text{Seasonal index for month } i &= \frac{\text{Monthly average for month } i}{\text{Average of monthly averages}} \times 100 \\ &= \frac{\bar{x}_i}{\bar{\bar{x}}} \times 100 \quad (i = 1, 2, \dots, 12) \end{aligned}$$

It is important to note that the average of the indexes will always be 100, that is, sum of the indexes should be 1200 for 12 months, and sum should be 400 for 4 quarterly data. If the sum of these 12 months percentages is not 1200, then the monthly percentage so obtained are adjusted by multiplying these by a suitable factor $[1200 \div (\text{sum of the 12 values})]$.

Example 12.18: The seasonal indexes of the sale of readymade garments in a store are given below:

Quarter	Seasonal Index
January to March	98
April to June	90
July to September	82
October to December	130

If the total sales of garments in the first quarter is worth ₹ 1,00,000, determine how much worth of garments of this type should be kept in stock to meet the demand in each of the remaining quarters.

[Delhi Univ., B.Com., 1996]

Solution: Calculations of seasonal index for each quarter and estimated stock (in ₹) are shown in Table 12.8.

Table 12.8 Calculation of Estimated Stock

<i>Quarter</i>	<i>Seasonal Index (SI)</i>	<i>Estimated Stock (₹)</i>
Jan. —March	98	1,00,000.00
April —June	90	91,836.73*
July —Sept.	82	83,673.47
Oct. —Dec.	130	1,32,653.06

* These figures are calculated as follows:

$$\text{Seasonal index for second quarter} = \frac{\text{Figure for first quarter} \times \text{SI for second quarter}}{\text{SI for first quarter}}$$

$$\text{Seasonal index for third quarter} = \frac{\text{Figure for first quarter} \times \text{SI for third quarter}}{\text{SI for first quarter}}$$

Example 12.19: Use the method of monthly averages to determine the monthly indexes for the data of production of a commodity for the years 2002 to 2004.

<i>Month</i>	<i>2002</i>	<i>2003</i>	<i>2004</i>
January	15	23	25
February	16	22	25
March	18	28	35
April	18	27	36
May	23	31	36
June	23	28	30
July	20	22	30
August	28	28	34
September	29	32	38
October	33	37	47
November	33	34	41
December	38	44	53

Solution: Computation of seasonal index by average percentage method based on the data is shown in Table 12.9.

$$\text{Monthly Average : } 1080/12 = 90; 360/12 = 30; 1200/12 = 100$$

Table 12.9 Calculation of Seasonal Indexes

<i>Month</i>	<i>2002</i>	<i>2003</i>	<i>2004</i>	<i>Monthly Total for 3 Years</i>	<i>Monthly Averages for 3 Years</i>	<i>Percentage Average of Monthly Averages</i>
Jan.	15	23	25	63	21	70
Feb.	16	22	25	63	21	70
March	18	28	35	81	27	90
April	18	27	36	81	27	90
May	23	31	36	90	30	100
June	23	28	30	81	27	90
July	20	22	30	72	24	80
Aug.	28	28	34	90	30	100
Sept.	29	32	38	99	33	110
Oct.	33	37	47	117	39	130
Nov.	33	34	41	108	36	120
Dec.	38	44	53	135	45	150
				1080	360	1200

The average of monthly averages is obtained by dividing the total of monthly averages by 12. In column 7 each monthly average for 3 years have been expressed as a percentage of the averages. For example, the percentage for January is

$$\text{Monthly index for January} = (21/30) \times 100 = 70;$$

$$\text{February} = (21/30) \times 100 = 70$$

$$\text{March} = (27/30) \times 100 = 90, \text{ and so on}$$

Example 12.20: The data on prices (₹ in per kg) of a certain commodity during 2000 to 2004 are shown below:

<i>Quarter</i>	<i>Years</i>				
	<i>2000</i>	<i>2001</i>	<i>2002</i>	<i>2003</i>	<i>2004</i>
I	45	48	49	52	60
II	54	56	63	65	70
III	72	63	70	75	84
IV	60	56	65	72	66

Compute the seasonal indexes by the average percentage method and obtain the deseasonalized values.

Solution: Calculations for quarterly averages are shown in Table 12.10.

Table 12.10 Calculation Seasonal Indexes

Year	Quarters			
	I	II	III	IV
2000	45	54	72	60
2001	48	56	63	56
2002	49	63	70	65
2003	52	65	75	72
2004	60	70	84	66
Quarterly total	254	308	364	319
Quarterly average	50.8	61.6	72.8	63.8
Seasonal index	81.60	98.95	116.94	102.48

$$\text{Average of quarterly averages} = \frac{50.8 + 61.6 + 72.8 + 63.8}{4} = \frac{249}{4} = 62.25$$

$$\text{Thus, Seasonal index for quarter I} = \frac{50.8}{62.25} \times 100 = 81.61$$

$$\text{Seasonal index for quarter II} = \frac{61.6}{62.25} \times 100 = 98.96$$

$$\text{Seasonal index for quarter III} = \frac{72.8}{62.25} \times 100 = 116.95$$

$$\text{Seasonal index for quarter IV} = \frac{63.8}{62.25} \times 100 = 102.48$$

Deseasonalized Values Seasonal influences are removed from a time series data by dividing the actual y value for each quarter by its corresponding seasonal index:

$$\text{Deseasonalized value} = \frac{\text{Actual quarterly value}}{\text{Seasonal index of corresponding quarter}} \times 100$$

The deseasonalized y values which are measured in the same unit as the actual values, reflect the collective influence of *trend*, *cyclical* and *irregular* forces. The deseasonalized values are given in Table 12.11.

Limitations of the Method of Simple Averages This method is the simplest of all the methods for measuring seasonal variation. However, the limitation of this method is that it assumes that there is no trend component in the series, that is, $\text{CSI} = 0$ or trend is assumed to have little impact on the time series. This assumption is not always justified.

Table 12.11 Calculation for Least-squares Equation

Year	Quarters			
	I	II	III	IV
2000	55.14	54.57	61.57	58.54
2001	58.82	56.59	53.87	54.64
2002	60.00	63.66	59.85	63.42
2003	63.72	65.68	64.13	70.25
2004	73.52	70.74	71.83	64.40

12.7.3 Ratio-to-Trend Method

This method is also known as the *percentage trend method*. This method is an improvement over the method of simple averages. Because here it is assumed that seasonal variation for a given month is a constant fraction of trend. The ratio-to-trend method isolates the seasonal factor when the following ratios are computed:

$$\frac{T \cdot S \cdot C}{T} = \text{SCI}$$

The steps of the method are summarized as follows:

- (i) Compute the trend values by applying the least-squares method.
- (ii) Eliminate the trend value. In a multiplicative model, the trend is eliminated by dividing the original data values by the corresponding trend values and multiplying these ratios by 100. The values so obtained are free from trend.
- (iii) Arrange the percentage data values obtained in Step (ii) according to months or quarters as the case may be for the various years.
- (iv) Find the monthly (or quarterly) averages of figures arranged in Step (iii) with any one of the usual measures of central tendency—arithmetic mean, median.
- (v) Find the grand average of monthly averages found in Step (iv). If the grand average is 100, then the monthly averages represent seasonal indexes. Otherwise, an adjustment is made by multiplying each index by a suitable factor [$1200/(\text{sum of the 12 values})$] to get the final seasonal indexes.

Example 12.21: Quarterly sales data (₹ in million) in a super bazar are presented in the following table for a four-year period

Year	Quarters			
	I	II	III	IV
2000	60	80	72	68
2001	68	104	100	88
2002	80	116	108	96
2003	108	152	136	124
2004	160	184	172	164

Calculate the seasonal index for each of the four quarters using the ratio-to-trend method.

Solution: Calculations to obtain annual trend values from the given quarterly data using the method of least-squares are shown in Table 12.12.

Table 12.12 Calculation of Trend Values

Year	Yearly Total (1)	Yearly Average $y = (2)/4$	Deviation From Mid-Year x	x^2	xy	Trend Values $\hat{y}$
2000	280	70	-2	4	-140	64
2001	360	90	-1	1	-90	88
2002	400	100	0	0	0	0
2003	520	130	1	1	130	112
2004	680	170	2	4	340	160
		560		10	240	

Solving the following normal equations, we get

$$\begin{aligned}\Sigma y &= na + b\Sigma x & \text{or } 560 &= 5a & \text{or } a &= 112 \\ \Sigma xy &= a\Sigma x + b\Sigma x^2 & \text{or } 240 &= 10b & \text{or } b &= 24\end{aligned}$$

Thus, the yearly fitted trend line is $y = 112 + 24x$. The value of $b = 24$ indicates yearly increase in sales. Thus the quarterly increment will be $24/4 = 6$.

To calculate quarterly trend values, consider first the year 2000. The trend value for this year is 64. This is the value for the middle of the year 2000, that is, half of the 2nd quarter and half of the 3rd quarter. Since quarterly increment is 6, the trend value for the 2nd quarter of 2000 would be $64 - (6/2) = 61$ and for the 3rd quarter it would be $64 + (6/2) = 67$. The value for the 1st quarter of 2000 would be $61 - 6 = 55$ and for the 4th quarter it would be $67 + 6 = 73$. Similarly, trend values of the various quarters of other years can be calculated as shown in Table 12.13.

Table 12.13 Quarterly Trend Values

Year	Quarters			
	I	II	III	IV
2000	55	61	67	73
2001	79	85	91	97
2002	103	109	115	121
2003	127	133	139	145
2004	151	157	163	169

After getting the trend values, the given data values in the time series are expressed as percentages of the corresponding trend values in Table 12.13. Thus for the 1st quarter of 2000, this percentage would be $(60/55) \times 100 = 109.09$; for the 2nd quarter it would be $(80/61) \times 100 = 131.15$, and so on. Other values can be calculated in the same manner as shown in Table 12.14.

The total of average of seasonal indexes is 403.12 (>400). Thus, we apply the correction factor $(400/403.12) = 0.992$. Now each quarterly average is multiplied by 0.992 to get the adjusted seasonal index as shown in Table 12.14.

Table 12.14 Ratio-to-Trend Values

Year	Quarters			
	I	II	III	IV
2000	109.09	131.15	107.46	93.15
2001	86.08	122.35	109.89	90.72
2002	77.67	106.42	93.91	79.34
2003	85.04	114.29	97.84	85.52
2004	105.96	117.20	105.52	97.04
Total	463.84	591.41	514.62	445.77
Average	92.77	118.28	102.92	89.15
Adjusted				= 403.12
seasonal index	92.02	117.33	102.09	88.44

The seasonal index 92.02 in the first quarter means that on average sales trend to be depressed by the presence of seasonal forces to the extent of approx. $(100 - 92.02) = 7.98$ per cent. Alternatively, values of time series would be approx. $(7.98/92.02) \times 100 = 8.67$ per cent higher had seasonal influences not been present.

Example 12.22: The production of a commodity during 1993-98 is given below. Fit the second degree parabola to these data and estimate the production for the year 2000:

Year	: 1993	1994	1995	1996	1997	1998
Production	: 10	12	13	15	18	20
('000 tonnes)						[Delhi Univ., B.Com., 2002]

Solution: Second degree parabolic trend equation is given by

$$y_C = a + bx + cx^2$$

To find the values of constants a , b and c , the normal equations are

$$\Sigma y = na + b\Sigma x + c\Sigma x^2$$

$$\Sigma xy = a\Sigma x + b\Sigma x^2 + c\Sigma x^3$$

$$\Sigma x^2y = a\Sigma x^2 + b\Sigma x^3 + c\Sigma x^4$$

Calculations required to calculate values of constants considering 1995 as origin are shown below:

Year	y	x	x^2	x^3	x^4	xy	x^2y
1993	10	-2	4	-8	16	-20	40
1994	12	-1	1	-1	1	-12	12
1995	13	0	0	0	0	0	0
1996	15	1	1	1	1	15	15
1997	18	2	4	8	16	36	72
1998	20	3	9	27	81	60	180
	88	3	19	27	115	79	319

Putting values to the normal equations, we get

$$88 = 6a + 3b + 19c \quad (i)$$

$$79 = 3a + 19b + 27c \quad (ii)$$

$$319 = 19a + 27b + 115c \quad (iii)$$

Multiply equation (i) by 9 and subtracting from (iii), we get $35a + 56c = 473$ (iv)

Multiply eq. (i) by 19 and (ii) by 3 and subtract, we get $105a + 280c = 1435$ (v)

Multiply eq. (iv) by 3 and subtract (v) from it, we get $112c = 16$ or $c = 0.143$

Putting $c = 0.143$ in (iv) we get $35a + 56(0.143) = 473$ or $a = 13.285$

Multiply eq. (ii) by 19 and (iii) by 3 and subtract we get

$$280b + 168c = 544 \text{ or } 35b + 21c = 68 \quad (vi)$$

$$35b + 21(0.143) = 68 \text{ or } b = 1.858$$

Hence the parabolic equation becomes

$$y = 13.285 + 1.858x + 0.143x^2$$

Also for $x = 5$

$$y_{2000} = 13.285 + 1.858(5) + 0.143(5)^2 = 26.15$$

Example 12.23: The prices of a commodity during 2001-2006 are given below. Fit a parabola $Y = a + bx + cx^2$ to these data. Estimate the price for the year 2007.

Year	:	2001	2002	2003	2004	2005	2006
Price (₹)	:	100	107	128	144	181	192

[Delhi Univ., B.Com.(Hons.), 2006]

Solution: Normal equations are given by

$$\Sigma y = na + b\Sigma x + c\Sigma x^2$$

$$\Sigma xy = a\Sigma x + b\Sigma x^2 + c\Sigma x^3$$

$$\Sigma x^2y = a\Sigma x^2 + b\Sigma x^3 + c\Sigma x^4$$

Let the year 2004 be taken as origin. Then x unit = years, y unit = price in ₹. Calculations required to calculate values of constants a , b and c in normal equation is shown below.

Year	y	x	x^2	x^3	x^4	xy	x^2y
2001	100	-3	9	-27	81	-300	900
2002	107	-2	4	-8	16	-214	428
2003	128	-1	1	-1	1	-128	128
2004	144	0	0	0	0	0	0
2005	181	1	1	1	1	181	181
2006	192	2	4	8	16	384	768
	852	-3	19	-27	115	-77	2405

Putting values to normal equations, we get

$$852 = 6a - 3b + 19c \quad (i)$$

$$-77 = -3a + 19b - 27c \quad (ii)$$

$$2405 = 19a - 27b + 115c \quad (iii)$$

By simplify equations (i), (ii) and (iii), we get

$$a = 147.97, b = 21.44 \text{ and } c = 1.5.$$

Parabolic trend equation is given by

$$y = 147.97 + 21.44x + 1.5x^2$$

Put $x = 3$ for the year 2007, we get

$$y = 147.97 + 21.44(3) + 1.5(3)^2 = 225.79.$$

Example 12.24: (a) The trend equation for the yearly sales of a commodity with 1st July, 1991 as origin is $y_C = 96 + 28.8x + 4x^2$, where x unit = 1 year. Determine the monthly trend equation with Jan. 1992 as origin.

(b) Compute trend values for August 1991.

[Delhi Univ., B.Com.(Hons.), 2002]

Solution: Given trend equation $y_C = 96 + 28.8x + 4x^2$ (origin: 1st July, 1991, x unit = one year)

(a) To obtain monthly trend equation divide 96 (i.e. a) by 12, 28.8 (i.e. b) by (12×12) and 4 (i.e. c) by $(12 \times 12 \times 12)$

$$y_C = \frac{96}{12} + \frac{28.8}{12 \times 12}x + \frac{4}{12 \times 12 \times 12}x^2 = 8 + 0.2x + 0.0023x^2 \quad (i)$$

(origin: 1st July, 1991, x unit 1 month)

To change the origin from 1st July, 1991 to January, 1992, x shall be increased by 6.5. That is

$$\begin{aligned} y_C &= 8 + 0.2(x + 6.5) + 0.0023(x + 6.5)^2 \\ &= 8 + 0.2(x + 6.5) + 0.0023(x^2 + 13x + 42.25) \\ &= 8 + 0.2x + 1.3 + 0.0023x^2 + 0.03x + 0.097 \\ &= 9.397 + 0.23x + 0.0023x^2 \end{aligned} \quad (ii)$$

(b) To get trend value for August 1991, replace x by 1.5 in (i)

$$y_C = 8 + 0.2(1.5) + 0.0023 \times (1.5)^2 = 8.305.$$

12.7.4 Ratio-to-Moving Average Method

This method is also called the *percentage moving average method*. In this method, the original values in the time series data are expressed as percentages of moving averages instead of percentages of trend values in the ratio-to-trend method. The steps of the method are summarized as follows:

- (i) Find the centred 12 monthly (or 4 quarterly) moving averages of the original data values in the time series.
- (ii) Express each original data value of the time series as a percentage of the corresponding centred moving average values obtained in Step (i). In other words, in a multiplicative time series model, we get

$$\frac{\text{Original data values}}{\text{Trend values}} \times 100 = \frac{\text{TCSI}}{\text{TC}} \times 100 = (\text{SI}) \times 100 \text{ per cent}$$

This implies that the ratio-to-moving average represents the seasonal and irregular components.

- (iii) Arrange these percentages according to months or quarter of given years. Find the averages over all months or quarters of the given years.
- (iv) If the sum of these indexes is not 1200 (or 400 for quarterly figures), multiply them by a correction factor = $1200/(\text{sum of monthly indexes})$. Otherwise, the 12 monthly averages will be considered as seasonal indexes.

Example 12.25: Calculate the seasonal index by the ratio-to-moving method from the following data:

Year	Quarters			
	I	II	III	IV
2001	75	60	54	59
2002	86	65	63	80
2003	90	72	66	85
2004	100	78	72	93

Solution: Calculations for 4 quarterly moving averages and ratio-to-moving averages are shown in Tables 12.15 and 12.16.

Table 12.15 Calculation of Ratio-to-Moving Averages

Year	Quarter	Original Values	four-Quarter Moving Total	four-Quarter Moving Average	2 × four-Quarter Moving Average	Ratio-to-Moving Average (Per cent)
		$Y = TCSI$	Total	Average	Average TC	$\frac{Y}{TC} = (SI)100\%$
2001	1	75	—	—	—	—
	2	60	—	—	—	—
	3	54	248	507	63.375	54/63.375 = 85.21
	4	59	259	523	65.375	59/65.375 = 90.25
2002	1	86	264	537	67.125	128.12
	2	65	273	567	70.875	91.71
	3	63	294	592	74.000	85.14
	4	80	298	603	75.375	106.14
2003	1	90	305	613	76.625	117.46
	2	72	308	621	77.625	92.75
	3	66	313	636	79.500	83.02
	4	85	323	652	81.500	104.29
2004	1	100	329	664	83.000	120.48
	2	78	335	678	84.750	92.04
	3	72	343	—	—	—
	4	93	—	—	—	—

Table 12.16 Calculation of Seasonal Index

Year	Quarters			
	I	II	III	IV
2001	—	—	85.21	90.25
2002	128.12	91.71	85.14	106.14
2003	117.46	92.75	83.02	104.29
2004	120.48	92.04	—	—
Total	366.06	276.49	253.37	300.68
Seasonal average	91.52	69.13	63.34	75.17 = 299.16
Adjusted seasonal index	122.36	92.43	84.70	100.51 $\cong$ 400

The total of seasonal averages is 299.16. Therefore, the corresponding correction factor would be $400/299.16 = 1.337$. Each seasonal average is multiplied by the correction factor 1.337 to get the adjusted seasonal indexes shown in Table 12.17.

Advantages and Disadvantages of Ratio-to-Moving Average Method This is the most widely used method for measuring seasonal variations because it eliminates both trend and cyclical variations from the time series. However, if cyclical variations are not regular, then this method is not capable of eliminating them completely. Seasonal indexes calculated by this method will contain some effect of cyclical variations.

The only disadvantage of this method is that six data values at the beginning and the six data values at the end are not taken into consideration for calculation of seasonal indexes.

12.7.5 Link Relative Method

This method is also known as *Pearson's method*. The percentages obtained by this method are called *link relatives* as these link each month to the preceding one. The steps involved in this method are summarized below:

- (i) Convert the monthly (or quarterly) data into link relatives by using the following formula:

$$\text{Link relative for a particular month} = \frac{\text{Data value of current month}}{\text{Data value of preceding month}} \times 100$$

- (ii) Calculate the average of link relatives of each month using either median or arithmetic mean.
- (iii) Convert the link relatives (LR) into chain relatives (CR) by using the formula:

$$\text{CR for a particular month} = \frac{[\text{LR of current month (or quarter)} \times \text{CR of preceding month (or quarter)}]}{100}$$

The CR for the first month (or quarter) is assumed to be 100.

- (iv) Compute the new chain relative for January (first month) on the basis of December (last month) using the formula:

$$\text{New CR for January} = \frac{\text{CR of January} \times \text{CR of December}}{100}$$

The new CR is usually not equal to 100 and therefore needs to be multiplied with the monthly correction factor

$$d = \frac{1}{12} (\text{New CR for January} - 100)$$

If the figures are given quarterly, then the correction factor would be

$$d = \frac{1}{4} (\text{New CR of first quarter} - 100)$$

The corrected CR for other months can be calculated by using the formula:

$$\text{Corrected CR for } k\text{th month} = \text{Original CR of } k\text{th month} - (k - 1) d$$

where $k = 1, 2, 3, \dots, 12$

- (v) Find the mean of the corrected chain index. If it is 100, then the corrected chain indexes represent the seasonal variation indexes. Otherwise divide the corrected CR of each month (or quarter) by the mean value of corrected CR and then multiply by 100 to get the seasonal variation indexes.

Example 12.26: Apply the method of link relatives to the following data and calculate seasonal indexes.

Year	Quarters			
	I	II	III	IV
1999	68	62	61	63
2000	65	58	56	61
2001	68	63	63	67
2002	70	59	56	62
2003	60	55	51	58

Solution: Computations of link relatives (LR) are shown in Table 12.17 by using the following formula:

$$\text{Link relative of any quarter} = \frac{\text{Data value of current quarter}}{\text{Data value of preceeding quarter}} \times 100$$

Table 12.17 Computation of Link Relatives

Year	Quarters			
	I	II	III	IV
1999	—	91.18	98.39	103.28
2000	103.17	89.23	96.55	108.93
2001	111.48	92.65	100.00	106.35
2002	104.48	84.29	94.92	110.71
2003	96.77	91.67	92.73	113.73
Total of LR	415.91	449.02	482.59	543.00
Arithmetic mean of LR	103.98	89.80	96.52	108.60
Chain relatives (CR)	100	$\frac{89.80 \times 100}{100}$ = 89.80	$\frac{96.52 \times 89.80}{100}$ = 86.67	$\frac{108.60 \times 86.67}{100}$ = 94.12

The new chain relatives for the first quarter on the basis of last quarter is calculated as follows:

$$\text{New CR} = \frac{\text{LR of first quarter} \times \text{CR of previous quarter}}{100} = \frac{103.98 \times 94.12}{100} = 97.9$$

Since new CR is not equal to 100, therefore we need to apply quarterly correction factor as

$$d = \frac{1}{4} (\text{New CR of first quarter} - 100) = \frac{1}{4} (97.9 - 100) = -0.53$$

Thus the corrected (or adjusted) CR for other quarters is shown in Table 12.18. For this we use the formula:

$$\text{Corrected CR for } k\text{th quarter} = \text{Original CR of } k\text{th quarter} - (k - 1) d$$

where $k = 1, 2, 3, 4$.

Table 12.18 Calculation of Link Relatives

Quarter	I	II	III	IV
Corrected CR	100	$89.80 - (-0.53)$ $= 90.33$	$86.67 - 2(-0.53)$ $= 87.73$	$94.13 - 3(-0.53)$ $= 95.72$
Seasonal indexes	$\frac{100}{93.44} \times 100$ $= 107.02$	$\frac{90.33}{93.44} \times 100$ $= 96.67$	$\frac{87.73}{93.44} \times 100$ $= 93.89$	$\frac{95.71}{93.44} \times 100$ $= 102.43$

$$\text{Mean of corrected CR} = \frac{100 + 90.33 + 87.73 + 95.71}{4} = 93.44$$

$$\text{Seasonal variation index} = \frac{\text{Corrected C.R.}}{\text{Mean of corrected C.R.}} \times 100$$

Example 12.27: Apply the method of link relatives to the following data and calculate the seasonal index:

Year	Quarters			
	I	II	III	IV
2000	45	54	72	60
2001	48	56	63	56
2002	49	63	70	65
2003	52	65	75	72
2004	60	70	84	86

Solution : Computations of link relatives (LR) using the following formula are shown in Table 12.19.

$$\text{LR of any quarter} = \frac{\text{Data value of current quarter}}{\text{Data value of preceding quarter}} \times 100$$

Table 12.19 Computation of Link Relatives

<i>Year</i>	<i>Quarters</i>			
	<i>I</i>	<i>II</i>	<i>III</i>	<i>IV</i>
2000	—	120	133.33	83.33
2001	80.00	116.67	112.50	88.89
2002	87.50	128.57	111.11	92.86
2003	80.00	125.00	115.38	96.00
2004	83.33	116.67	120.00	102.38
Total of LR	330.83	606.91	592.32	463.46
Arithmetic mean of LR	83.30	121.38	118.46	92.70
Chain relatives	100	$\frac{121.38 \times 100}{100}$	$\frac{118.46 \times 121.38}{100}$	$\frac{92.70 \times 143.79}{100}$
(CR)		= 121.38	= 143.79	= 133.29

The new chain relatives for the first quarter on the basis of the preceding quarter is calculated as follows:

$$\begin{aligned} \text{New CR} &= \frac{\text{LR of first quarter} \times \text{CR of previous quarter}}{100} \\ &= \frac{83.30 \times 126.42}{100} = 105.30 \end{aligned}$$

Since the new CR is more than 100, therefore we need to apply a quarterly correction factor as

$$\begin{aligned} d &= \frac{1}{4} (\text{New CR of first quarter} - 100) \\ &= \frac{1}{4} (105.30 - 100) = 1.325 \end{aligned}$$

Thus, the corrected (or adjusted) CR for other quarters is shown in Table 12.20. For this, we use the formula

$$\text{Corrected CR for } k\text{th quarter} = \text{Original CR of } k\text{th quarter} - (k - 1) d$$

where $k = 1, 2, 3, 4$.

Table 12.20 Corrected CR

<i>Quarters</i>	<i>I</i>	<i>II</i>	<i>III</i>	<i>IV</i>
Corrected CR	100	$121.38 - 1.32$ = 120.06	$143.78 - 2(1.32)$ = 141.14	$126.42 - 3(1.32)$ = 122.46
Seasonal indexes	$\frac{100}{120.92} \times 100$ = 82.70	$\frac{120.06}{120.92} \times 100$ = 99.30	$\frac{141.14}{120.92} \times 100$ = 116.72	$\frac{122.46}{120.92} \times 100$ = 101.27

$$\text{Mean of corrected CR} = \frac{100 + 120.06 + 141.14 + 122.46}{4} = 120.92$$

$$\text{Seasonal variation index} = \frac{\text{Corrected C.R.}}{\text{Mean of corrected C.R.}} \times 100$$

Advantages and Disadvantages of Link Relative Method This method is much simpler than the ratio-to-trend or the ratio-to-moving average methods. In this method, the LR of the first quarter (or month) is not taken into consideration as compared to ratio-to-trend method, where 6 values each at the beginning and at the end periods (month) are lost.

This method eliminates the trend but it is possible only if there is a straight line (linear) trend in the time series—which is generally not formed in business and economic series.

12.8 MEASUREMENT OF CYCLICAL VARIATIONS—RESIDUAL METHOD

As mentioned earlier that a typical time series has four components: secular trend (T), seasonal variation (S), cyclical variation (C), and irregular variation (I). In a multiplicative time series model, these components are written as

$$y = TCSI$$

The deseasonalization data can be adjusted for trend analysis by dividing these by the corresponding trend and seasonal variation values. Thus we are left with only cyclical (C) and irregular (I) variations in the data set as shown below:

$$\frac{y}{T.S} = \frac{TCSI}{TS} = CI$$

The moving averages of an appropriate period may be used to eliminate or reduce the effect of irregular variations and thus left behind only the cyclical variations.

The procedure of identifying cyclical variation is known as the *residual method*. Recall that cyclical variations in time series tend to oscillate above and below the secular trend line for periods longer than one year. The steps of residual method are summarized as follows:

- (i) Obtain seasonal indexes and deseasonalized data.
- (ii) Obtain trend values and expressed seasonalized data as percentages of the trend values.
- (iii) Divide the original data (y) by the corresponding trend values (T) in the time series to get SCI. Further divide SCI by S to get CI.
- (iv) Smooth out irregular variations by using moving averages of an appropriate period but of short duration, leaving only the cyclical variation.

12.9 MEASUREMENT OF IRREGULAR VARIATIONS

Since irregular variations are random in nature, no particular procedure can be followed to isolate and identify these variations. However, the residual method can be extended one step further by dividing CI by the cyclical component (C) to identify the irregular component (I).

Alternately, trend (T), seasonal (S), and cyclical (C) components of the given time series are estimated then the residual is taken as the irregular variation. Thus, in the case of multiplicative time-series model, we have

$$\frac{Y}{TCS} = \frac{TCSI}{TCS} = I$$

where S and C are in fractional form and not in percentages.

Conceptual Questions 12B

15. (a) Under what circumstances can a trend equation be used to forecast a value in a series in the future? Explain.
- (b) What are the advantages and disadvantages of trend analysis? When would you use this method of forecasting?
16. What effect does seasonal variability have on a time series? What is the basis for this variability for an economic time series?
17. What is measured by a moving average? Why are 4-quarter and 12-month moving averages used to develop a seasonal index?
18. Briefly describe the moving average and least squares methods of measuring trend in time series.
19. Explain the simple average method of calculating indexes in the context of time series analysis.
20. Distinguish between ratio-to-trend and ratio-to-moving average as methods of measuring seasonal variations. Which is better and why?
21. Distinguish between trend, seasonal variations, and cyclical variations in a time series. How can trend be isolated from variations?

Self-practice Problems 12C

- 12.16** Apply the method of link relatives to the following data and calculate seasonal indexes.

Quarter	1999	2000	2001	2002	2003
I	6.0	5.4	6.8	7.2	6.6
II	6.5	7.9	6.5	5.8	7.3
III	7.8	8.4	9.3	7.5	8.0
IV	8.7	7.3	6.4	8.5	7.1

- 12.17** A company estimates its sales for a particular year to be ₹ 24,00,000. The seasonal indexes for sales are as follows:

Month	Seasonal Index	Month	Seasonal Index
January	75	July	102
February	80	August	104
March	98	September	100
April	128	October	102
May	137	November	82
June	119	December	73

Using this information, calculate estimates of monthly sales of the company. (Assume that there is no trend).

- 12.18** Calculate the seasonal index from the following data using the average method:

Year	Quarter			
	I	II	III	IV
2000	72	68	80	70
2001	76	70	82	74
2002	74	66	84	80
2003	76	74	84	78
2004	78	74	86	82

[Kerala Univ., B.Com., 1996]

- 12.19** Calculate seasonal index numbers from the following data:

Year	Quarter			
	I	II	III	IV
1998	108	130	107	93
1999	86	120	110	91
2000	92	118	104	88
2001	78	100	94	78
2002	82	110	98	86
2003	106	118	105	98

- 12.20** On the basis of quarterly sales (₹ in lakh) of a certain commodity for the years 2003–2004, the following calculations were made:

Trend : $y = 20 + 0.5t$ with origin at first quarter of 2003

where t = time unit (one quarter)

y = quarterly sales (₹ in lakh)

Seasonal variations:

Quarter	:	1	2	3	4
Seasonal index	:	80	90	120	110

Estimate the quarterly sale for the year 2003 using multiplicative model.

Hints and Answers

12.16

	<i>Year</i>	<i>Quarters</i>			
		<i>I</i>	<i>II</i>	<i>III</i>	<i>IV</i>
	1999	—	108.3	120.0	111.5
	2000	62.1	146.3	106.3	86.9
	2001	93.2	95.6	143.1	68.8
	2002	112.5	80.6	129.3	113.3
	2003	77.6	110.6	109.6	88.8
Arithmetic average	$\frac{345.4}{4} = 86.35$	$\frac{541.4}{5} = 108.28$	$\frac{608.3}{5} = 121.66$	$\frac{469.3}{5} = 93.86$	
Chain relatives	100	$\frac{100 \times 108.28}{100} = 108.28$	$\frac{121.66 \times 108.28}{100} = 131.73$	$\frac{93.86 \times 131.73}{100} = 123.64$	
Corrected chain relatives	100	$108 - 1.675 = 106.325$	$131.73 - 3.35 = 128.38$	$123.64 - 5.025 = 118.615$	
Seasonal indexes	$\frac{100 \times 100}{113.4} = 88.18$	$\frac{106.325}{113.4} \times 100 = 93.76$	$\frac{128.38}{113.4} \times 100 = 113.21$	$\frac{118.615}{113.4} \times 100 = 104.60$	

12.17 Seasonal indexes are usually expressed as percentages. The total of all the seasonal indexes is 1200.

Seasonal effect = Seasonal index + 100

The yearly sales being ₹ 24,00,000, the estimated monthly sales for a specified month:

Estimated sales

$$\begin{aligned}
 &= \frac{\text{Annual sales}}{12} \times \text{Seasonal effect} \\
 &= \frac{24,00,000}{12} \times \text{Seasonal effect} \\
 &= 2,00,000 \times \text{Seasonal effect}
 \end{aligned}$$

<i>Month</i> (1)	<i>Seasonal Index</i> (2)	<i>Seasonal Effect</i> (3) = (2) ÷ 100	<i>Estimated Sales</i> (4) = (3) × 2,00,000
January	75	0.75	1,50,000
February	80	0.80	1,60,000
March	98	0.98	1,96,000
April	128	1.28	2,56,000
May	137	1.37	2,74,000
June	119	1.19	2,38,000
July	102	1.02	2,04,000
August	104	1.04	2,08,000
September	100	1.00	2,00,000
October	102	1.02	2,04,000
November	82	0.82	1,64,000
December	73	0.73	1,46,000
	1200	12.00	24,00,000

12.18

Year	Quarters			
	I	II	III	IV
2000	72	68	80	70
2001	76	70	82	74
2002	74	66	84	80
2003	76	74	84	78
2004	78	74	86	82
Total	376	352	416	384
Average	75.2	70.4	83.2	76.8
Seasonal index	98.43	92.15	108.9	100.52

$$\text{Grand average} = \frac{75.2 + 70.4 + 83.2 + 76.8}{4}$$

$$= \frac{305.6}{4} = 76.4$$

Seasonal index for quarter

$$k = \frac{\text{Average of quarter } k}{\text{Grand average}} \times 100$$

12.19

Year	Quarters			
	I	II	III	IV
1998	108	130	107	93
1999	86	120	110	91
2000	92	118	104	88
2001	78	100	94	78
2002	82	110	98	86
2003	106	118	105	98

Total	552	696	618	534
Average	92	116	103	89
Seasonal Index	$\frac{92}{100} \times 100$	$\frac{116}{100} \times 100$	$\frac{103}{100} \times 100$	$\frac{89}{100} \times 100$
	= 92	= 116	= 103	= 89

Sales in different quarters:

I: ₹ 20,000; II: $20,000 \times 1.16 = ₹ 23,200$

III: $20,000 \times 1.03 = ₹ 20,600$;

IV: $20,000 \times 0.89 = ₹ 17,800$

12.20

Quarter of 2003	Time Unit	Trend (T) Values $y = 20 + 0.5t$	Seasonal Estimated Effect or Sales Index (S) TS (₹ in lakh)	
1	4	$20 + 0.5 \times 4 = 22.0$	0.80	17.60
2	5	$20 + 0.5 \times 5 = 22.5$	0.90	20.25
3	6	$20 + 0.5 \times 6 = 23.0$	1.20	27.60
4	7	$20 + 0.5 \times 7 = 23.5$	1.10	25.85

Formulae Used

1. Secular trend line

- Linear trend model

$$y = a + bx$$

$$\text{where } a = \bar{y} - b\bar{x}; b = \frac{\sum xy - n\bar{x}\bar{y}}{\sum x^2 - n(\bar{x})^2}$$

- Exponential trend model

$$y = ab^x;$$

$$\log a = \frac{1}{n} \sum \log y; \log b = \frac{\sum x \log y}{\sum x^2}$$

- Parabolic trend model

$$y = a + bx + cx^2$$

$$\bullet \text{ where } a = \frac{\sum y - c \sum x^2}{n}; b = \frac{\sum xy}{\sum x^2}$$

$$c = \frac{n \sum x^2 y - \sum x^2 \sum y}{n \sum x^4 - (\sum x^2)^2}$$

2. Moving average

$$MA_{t+1} = \frac{\sum \{D_t + D_{t-1} + \dots + D_{t-n+1}\}}{n}$$

where t = current time period

D = actual data value

n = length of time period

Review Self-practice Problems

- 12.21** A sugar mill is committed to accepting beets from local producers and has experienced the following supply pattern (in thousands of tons/year and rounded).

Year	Tonnes	Year	Tonnes
1990	100	1995	400
1991	100	1996	400
1992	200	1997	600
1993	600	1998	800
1994	500	1999	800

The operations manager would like to project a trend to determine what facility additions will be required by 2004

- Sketch a freehand curve and extend it to 2004. What would be your 2004 forecast based upon the curve?
 - Compute a three-year moving average and plot it as a dotted line on your graph.
- 12.22** Use the data of Problem 12.21 and the normal equations to develop a least squares line of best fit. Omit the year 1990.
- State the equation when the origin is 1995.
 - Use your equation to estimate the trend value for 2004.
- 12.23** A forecasting equation is of the form:

$$\hat{y}_c = 720 + 144x$$

(2003 = 0, x unit = 1 year, y = annual sales)

- Forecast the annual sales rate for 2003 and also for one year later.
 - Change the time (x) scale to months and forecast the annual sales rate at July 1, 2003, and also at one year later.
 - Change the sales (y) scale to monthly and forecast the monthly sales rate at July 1, 2003, and also at one year later.
- 12.24** Data collected on the monthly demand for an item were as shown below:

January	100
February	90
March	80
April	150
May	240
June	320
July	300
August	280
September	220

- What conclusion can you draw with respect to the length of moving average versus smoothing effect?
- Assume that the 12-month moving average centred on July was 231. What is the value of the ratio-to-moving average that would be used in computing a seasonal index?

- 12.25** Consider the following time series data:

Week :	1	2	3	4	5	6
Value :	8	13	15	17	16	9

- Develop a 3-week moving average for this time series. What is the forecast for week 7?
- Use $\alpha = 0.2$ to compute the exponential smoothing values for the time series. What is the forecast for week 7?

- 12.26** Below are given the figures of production (in million tonnes) of a cement factory:

Year :	1990	1992	1993	1994	1995	1996	1999
Production :	77	88	94	85	91	98	90

- Fit a straight line trend by the 'least squares method' and tabulate the trend values.
- Eliminate the trend. What components of the time series are thus left over?
- What is the monthly increase in the production of cement?

- 12.27** The sale of commodity in tonnes varied from January, 2000 to December, 2000 in the following manner:

280	300	280	280	270	240
230	230	220	200	210	200

Fit a trend line by the method of semi-averages.

- 12.28** Fit a parabolic curve of the second degree to the data given below and estimate the value for 2002 and comment on it.

Year : 1996 1997 1998 1999 2000

Sales

(₹ in '000) : 10 12 13 10 8

- 12.29** Given below are the figures of production of a sugar (in 1000 quintals) factory:

Year : 1991 1992 1993 1994 1995 1996 1997

Production: 40 45 46 42 47 49 46

Fit a straight line trend by the method of least squares and estimate the value for 2001.

Hints and Answers

- 12.21** (a) Forecasts is around 1200 (thousand) tonnes

(b) Averages are: 133, 300, 433, 500, 433, 466, 600 and 733.

- 12.22** (a) $\hat{y} = 489 + 75x$ (1995 = 0, x = years, y = tonnes in thousand)

(b) 11,64,000 tonnes

- 12.23** (a) 720 units when $x = 0$, 864 units when $x = 1$.

(b) $\hat{y} = 720 + 12x$ (July 1, 2003 = 0; x unit = 1 month; y = annual sales rates in units)
720 units per year; 864 units per year.

(c) $\hat{y} = 60 + x$ (July 1, 2003 = 0, x unit = 1 month; y = monthly sales rates in units)
60 units per month; 72 units per month.

- 12.24** (a) Longer average yield more smoothing; (b) 1.3

- 12.25** (a)

Week (1)	Values (2)	Forecast (3)	Forecast Error (4) = (2) - (3)	Squared Forecast Error
1	8	—	—	—
2	13	—	—	—
3	15	—	—	—
4	17	12	5	25
5	16	15	1	1
6	9	16	-7	49

Forecast for week 7 is $(17 + 16 + 9)/3 = 14$.

(b)

Week (t)	Values y_t	Forecast F_t	Forecast Error $y_t - F_t$	Squared Error $(y_t - F_t)^2$
1	8	—	—	—
2	13	8.00	5.00	25.00
3	15	9.00	6.00	36.00
4	17	10.20	6.80	46.24
5	16	11.56	4.44	19.71
6	9	12.45	-3.45	11.90
				138.85

Forecast for week 7 is $0.2(9) + (1 - 0.2)(12.45) = 11.76$.

- 12.26** (a)

Year	Production (in m. tonnes)	Deviation From 1994	Trend Values		
	y	x	xy	x^2	
1990	77	-4	-308	16	83.299
1992	88	-2	-176	4	86.051
1993	94	-1	-94	1	87.427
1994	85	0	0	0	88.803
1995	91	1	91	1	90.179
1996	98	2	196	4	91.555
1999	90	5	450	25	95.683
		623	1	159	51

Solving the normal equations

$$\begin{aligned}\Sigma y &= na + b\Sigma x & 623 &= 7a + b \\ \Sigma xy &= a\Sigma x + b\Sigma x^2 & 159 &= a + 51b\end{aligned}$$

we get $a = 88.803$ and $b = 1.376x$. Thus,

$$\hat{y} = a + bx = 88.803 + 1.376x$$

Substituting $x = -4, -2, -1, 0, 1, 2, 5$ to get trend values as shown above in the table.

- (b) After eliminating the trend, we are left with S, C, and I components of time series.
- (c) Monthly increase in the production of cement is given by $b/12 = 1.376/12 = 0.115$.

12.27

Month (in tonnes)	Sales
January	280
February	300
March	280
April	280
May	270
June	240
July	230
August	230
September	220
October	200

Total = 1650 of first six months;
Average = $\frac{1650}{6} = 275$
Total = 1390 of last six months;
Average = $\frac{1390}{6} = 231.67$

Plot 275 and 231.67 in the middle of March-April 2000 and that of September and October 2000. By joining these two points we get a trend line which describes the given data.

12.28

Year	Sales	Period				
	y	x	xy	x^2	x^2y	x^4
1996	10	-2	-20	4	40	16
1997	12	-1	-12	1	12	1
1998	13	0	0	0	0	0
1999	10	1	10	1	10	1
2000	8	2	16	4	32	16
	53	0	-6	10	94	34

Parabolic trend line : $y = a + bx + cx^2$

$$\begin{aligned}a &= \frac{\Sigma y - c\Sigma x^2}{n} = \frac{53 - 0.857 \times 10}{5} = 8.886 \\ b &= \frac{\Sigma xy}{\Sigma x^2} = \frac{-6}{10} = -0.6 \\ c &= \frac{n\Sigma x^2y - \Sigma x^2\Sigma y}{n\Sigma x^4 - (\Sigma x^2)^2} = \frac{5(94) - 10(53)}{5(34) - (10)^2} \\ &= -0.857 \\ y &= 8.886 - 0.6x - 0.857x^2\end{aligned}$$

For 2002, $x = 4$;

$$\begin{aligned}y &= 8.886 - 0.6(4) - 0.857(4)^2 \\ &= -7.226\end{aligned}$$

12.29

Year	Production (‘000 qtls)	Deviations from 1994		
	y	x	xy	x^2
1991	40	-3	-120	9
1992	45	-2	-90	4
1993	46	-1	-46	1
1994	42	0	0	0
1995	47	1	47	1
1996	49	2	98	4
1997	46	3	138	9
	315	0	27	28

$$\hat{y} = a + bx; a = \Sigma y/n = 315/7 = 45$$

$$b = \frac{\Sigma xy}{\Sigma x^2} = \frac{27}{28} = 0.964$$

$$\hat{y} = 45 + 0.964x$$

$$y_{2001} = 45 + 0.964(7)$$

$$= 45 + 6.748 = 51.748$$

BUSINESS STATISTICS
B.Com. (Hons.) First Year
Paper IV—2008

Time : 3 hours

Maximum Marks : 75

Note: The maximum marks printed on the Question Paper are applicable for the candidates registered with the School of Open Learning for B.Com. (Hons.). These marks will, however, be scaled down proportionately in respect of the students of regular colleges, at the time of posting of awards for compilation of result.

Attempt **all** the questions.

Q. 1. (a) The following information relating to works in an industrial town is given:

<i>Item of Consumption</i>	<i>Consumer Price Index 2005 (2000 = 100)</i>	<i>Proportion of Expenditure on Item</i>
Food	132	60%
Clothing	154	12%
Fuel and lighting	147	16%
Housing	178	8%
Miscellaneous	158	4%

Average wage per month in 2000 is ₹ 2,000. What should be dearness allowance expressed as per cent of wages? What should be the average wage per worker per month in 2005 in that town so that the standard of living of worker does not fall below the 2000 level? 10

Solution: Price index in 2005

$$= \frac{132 \times 60 + 154 \times 12 + 147 \times 16 + 178 \times 8 + 158 \times 4}{100}$$

$$= \frac{7920 + 1848 + 2352 + 1424 + 632}{100} = \frac{14176}{100} = 141.76$$

If the pay in Year 2000 is ₹ 2,000, then the pay in Year 2005 will be

$$\frac{2000 \times 141.76}{100} = ₹ 2,835.20$$

Dearness allowance = ₹ 2,835.20 – ₹ 2,000 = ₹ 835.20

Or

Two factories manufacture the same machine parts. Each part is classified as having either 0, 1, 2 or 3 manufacturing defects. The joint probability for this is given below:

<i>Manufacturer</i>	<i>Number of Defects</i>			
	0	1	2	3
X	0.1250	0.0625	0.1875	0.1250
Y	0.0625	0.0625	0.1250	0.2500

- (i) A part is observed to have no defect. What is the probability that it was produced by X manufacturer?
- (ii) A part is known to have been produced by manufacturer X. What is the probability that the part has no defects?
- (iii) A part is known to have two or more defects. What is the probability that it was manufactured by X?
- (iv) A part is known to have one or more defects. What is the probability that it was manufactured by Y?

10

Solution:

Manufacturer	Number of Defects				Total
	0	1	2	3	
X	0.1250	0.0625	0.1875	0.1250	0.5
Y	0.0625	0.0625	0.1250	0.2500	0.5
Total	0.1875	0.1250	0.3125	0.3750	1.0

$$(i) \text{ No defect, Produced by X} = \frac{0.1250}{0.1875} = 0.67$$

$$(ii) \text{ Produced by X, No defect} = \frac{0.1250}{0.5} = 0.25$$

$$(iii) \text{ Two or more defects, Produced by X} = \frac{0.1875 + 0.1250}{0.3125 + 0.3750} = \frac{0.3125}{0.6875} = 0.4545$$

$$(iv) \text{ One or more defects, Produced by Y} = \frac{0.0625 + 0.1250 + 0.2500}{0.1250 + 0.3125 + 0.3750} = \frac{0.4375}{0.8125} = 0.5385$$

(b) Distinguish between any two:**(i) Ratio-to-trend and ratio-to-moving average method****(ii) Dispersion and Skewness****(iii) Laspeyre's and Paasche's index number**

5

Answer: (i) Ratio-to-trend: This method calculates seasonal index. The various steps involved in the calculation of seasonal index in this method are as follows:

- Trend values are obtained by the method of least squares.
- Original data of month to month is divided by trend values and the ratios so obtained are multiplied by 100.
- Values so obtained are free of trend and then these values have to be free only from irregular and cyclical movements.
- Now figures for the various months are taken as an average using any one of the measures of Central Tendency (usually median is preferred because median is an average which is not affected by very high and low values)
- The seasonal index for each month is now expressed as a percentage of month's average.

Remark: The calculations can be made on quarterly (periodical) basis as required.

Ratio-to-moving averages: This method is mostly used for measuring seasonal index. Steps are as under:

- Remove seasonal variations by taking 12 months (centred) moving average.
- A four-quarter (centred) moving average is taken if a quarterly data is given instead of monthly data.
- Original data for each month is expressed as a percentage of the centred 12 months by moving averages.
- Original data for each quarter (if quarterly figures are given) is expressed as a percentage of the centred four-quarterly moving average.
- Each monthly item or quarterly item of the original data is divided by the corresponding 12 monthly moving average or four quarterly moving averages.
- These percentages for a given month may also be averaged to cancel out the irregular fluctuations.

The ratio to moving average method for measuring seasonal variation is regarded as the most satisfactory method because the index so obtained does not fluctuate so much as the index based on straight line trends. Moving average used for eliminating cyclical variation is a better approach than the trend method approach.

(ii) Distinguish between Dispersion and skewness

(iii) Distinction between Laspeyre's and Paasche's Index Number

- Laspeyre's index number uses base year quantities as weights whereas Paasche's index number uses current year quantities as weights.
- Laspeyre's index number suffers from upward bias whereas Paasche's index number suffers from downward bias.

Q.2. (a) The mean, mode and Q.D. of a distribution are 42, 36 and 15, respectively. If its Bowley's coefficient of skewness is $1/3$, find the values of two quartiles. Also state the empirical relationship between mean, mode and median. 9

Solution: Given $Q.D. = \frac{Q_3 - Q_1}{2} = 15$ or $Q_3 - Q_1 = 30$; Mean = 42, and Mode = 36 ... (i)

Empirical relationship between Mean, Median and Mode is

$$\text{Mean} - \text{Mode} = 3 (\text{Mean} - \text{Median})$$

$$\text{or } 42 - 36 = 3 (42 - \text{Median})$$

$$\text{or } 2 = 42 - \text{Median, i.e. Median } (=Q_2) = 40$$

Bowley's coefficient of skewness

$$\frac{1}{3} = \left[\frac{Q_3 + Q_1 - 2Q_2}{Q_3 - Q_1} \right] = \frac{Q_3 + Q_1 - 2 \times 40}{30}$$

$$\text{or } 10 = Q_3 + Q_1 - 80, \text{ i.e. } Q_3 + Q_1 = 90 \quad \dots (ii)$$

Solving Eqn. (i) and (ii), we get $Q_2 = 60$, and $Q_1 = 30$

Or

Why do we have two regression lines? Can there be only one regression line? In the estimation of regression equation of two variables X and Y , the following results were obtained:

$$\Sigma x = 900, \Sigma y = 700, n = 10, \Sigma x^2 = 6360, \Sigma y^2 = 2860 \text{ and } \Sigma xy = 3900.$$

Obtain two regression equations.

9

Solution: Regression equations have been derived under different assumptions and therefore the regression relation is irreversible, hence one equation is not sufficient to find the values of the two variables. So two regression equations are required, but if the correlation between the two variables X and Y is perfect, i.e., perfect positive or perfect negative, one equation is sufficient.

Given that: $\Sigma x = 900$, $\Sigma y = 700$, $n = 10$, $\Sigma x^2 = 6360$, $\Sigma y^2 = 2860$ and $\Sigma xy = 3900$.

$$b_{yx} = \frac{\frac{\Sigma xy}{n} - \frac{\Sigma x}{n} \frac{\Sigma y}{n}}{\frac{\Sigma x^2}{n} - \left(\frac{\Sigma x}{n}\right)^2} = \frac{\frac{3,900}{10} - \left(\frac{900}{10}\right)\left(\frac{700}{10}\right)}{\left(\frac{6360}{10}\right) - \left(\frac{900}{10}\right)^2} = \frac{390 - 90 \times 70}{636 - 8100}$$

Since denominator that represents variance of x can never be negative, therefore the data as given in the question is wrong:

Considering $\Sigma x = 90$ and $\Sigma y = 70$. Then the regression coefficients are calculated as

$$b_{yx} = \frac{\frac{\Sigma xy}{n} - \left(\frac{\Sigma x}{n}\right)\left(\frac{\Sigma y}{n}\right)}{\frac{\Sigma x^2}{n} - \left(\frac{\Sigma x}{n}\right)^2} = \frac{390 - 9 \times 7}{636 - 81} = \frac{327}{555} = 0.589$$

$$b_{xy} = \frac{\frac{\Sigma xy}{n} - \left(\frac{\Sigma x}{n}\right)\left(\frac{\Sigma y}{n}\right)}{\frac{\Sigma y^2}{n} - \left(\frac{\Sigma y}{n}\right)^2} = \frac{390 - 9 \times 7}{286 - 49} = \frac{327}{237} = 1.3797$$

The two regression lines are

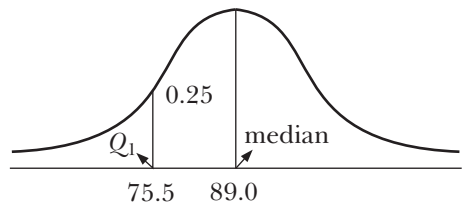
$$\begin{aligned} y - \bar{y} &= b_{yx}(x - \bar{x}) & \text{and} & & x - \bar{x} &= b_{xy}(y - \bar{y}) \\ \text{or } y - 7 &= 0.589(x - 9) & \text{and} & & x - 9 &= 1.3797(y - 7) \\ y &= 0.589x + 1.699 & & & x &= 1.3797y + -0.6579 \end{aligned}$$

- (b) The median and Q_1 of a normal distribution are 89.0 and 75.5 respectively. Calculate the standard deviation. What is the relationship between Binomial, Normal and Poisson distribution? 6

Solution: Given, Median = 89.0 and $Q_1 = 75.5$

$$\begin{aligned} \text{At } Q_1: \quad Z &= \frac{75.5 - 89.0}{\sigma} \\ 0.675 &= \frac{13.5}{\sigma}, \text{ i.e. } \sigma = \frac{13.5}{0.675} = 20 \end{aligned}$$

[Area under the standard normal curve between $Z = 0$ and 0.675 is 0.25]



Normal distribution is regarded as a limiting case of Poisson distribution because parameter $\lambda (=np) \rightarrow \infty$, where λ is the mean = variance. Also normal distribution is a limiting case of binomial probability distribution when:

- (i) The number of trials is indefinitely large, i.e., $n \rightarrow \infty$.
- (ii) Neither the probability of success nor the probability of failure in one trial is very small, i.e. both $np \geq 5$ and $n(1-p) \geq 5$.

Or

Explain briefly statistical decision theory. What are the ingredients of a decision problem? 6

Solution: Decision theory provides a framework which enables us to justify as to how and why a decision maker has selected a particular strategy out of the various strategies (or courses of action.)

Main ingredients of decision problems are:

- (i) **Acts** (Alternatives or Courses of action)
- (ii) **States of Nature:** The various possible outcomes of any course of action. Selection of any course of action is under the control of the decision maker but State of Nature is a matter of chance and is not under the control of any decision maker.
- (iii) **Payoff table:** It is the matrix of outcomes due to the various combination of state of nature and courses of action.
- (iv) **Opportunity Loss or Regret:** It is defined as the difference between the pay off for action i and the pay off for the action that would have the largest pay off under the state of nature j .

Q.3. (a) From the given ratios of observed trend values (per cent), calculate seasonal indices. If sales for 2001 are expected to be ₹ 2,000 lakh, what are the likely sales for individual quarters?

Years	Quarters			
	I	II	III	IV
1997	80	95	80	110
1998	101	104	90	110
1999	100	95	90	100
2000	115	110	100	120

(b) Out of additive and multiplicative models in time series analysis, which is better and why? 10

Solution:

Years	Quarters			
	I	II	III	IV
1997	80	95	80	110
1998	101	104	90	110
1999	100	95	90	100
2000	115	110	100	120
Total	396	404	360	440
Average	99	101	90	110
Seasonal Indices	99	101	90	110

Since the average of the averages = $\frac{99 + 101 + 90 + 110}{4} = 100$, Seasonal Index is the same as the Average Index

$$\text{Deseasonalised sales for first quarter} = 2000 \times \frac{100}{99} = ₹ 2020.20 \text{ lakh}$$

$$\text{Second quarter} = \frac{2020.20 \times 101}{100} = ₹ 2040.402$$

$$\text{Third quarter} = \frac{2020.2 \times 90}{100} = ₹ 1818.18$$

$$\text{Fourth quarter} = \frac{2020.2 \times 110}{100} = ₹ 2222.22$$

(b) Multiplicative model is always preferred in Business and Economics because it assumes that the four components of the time series are due to different causes but they are not necessarily independent and they can affect each other. In a time series model S , C and I are not viewed as absolute amounts but as relative variations. Except for trend component T , the other components S , C and I are expressed as rates (or indices) fluctuating above or below 'one' such that the Geometric means of all the $S = S_t$ values in year, $C = C_t$ values in a cycle or $I = I_t$ values in long-term period are unity:

$$Y = TSCI$$

$$\log Y = \log T + \log S + \log C + \log I$$

which is an additive model fitted to the logarithms of the given time series values.

Or

Fit a second degree parabola to the following data:

10

X	:	1	2	3	4	5
Y	:	1090	1220	1390	1625	1915

Solution: Calculations for Parabolic Trend are as follows:

X	Y	$U = X - 3$	$V = \frac{Y - 1450}{5}$	U^2	U^3	U^4	UV	U^2V
1	1090	-2	-72	4	-8	16	144	-288
2	1220	-1	-46	1	-1	1	46	-46
3	1390	0	-12	0	0	0	0	0
4	1625	1	35	1	1	1	35	35
5	1915	2	93	4	8	16	186	372
Total		0	-2	10	0	34	411	73

Let the parabola of best fit of V on U be

$$V = a + bU + cU^2 \quad (i)$$

where $U = X - 3$ and
$$V = \frac{Y - 1450}{5} \quad (ii)$$

Then the normal equations for estimating a , b and c are:

$$\Sigma V = na + b\Sigma U + c\Sigma U^2$$

$$\Sigma UV = a\Sigma U + b\Sigma U^2 + c\Sigma U^3$$

$$\Sigma U^2V = a\Sigma U^2 + b\Sigma U^3 + c\Sigma U^4$$

Substituting values of U and V , we get

$$-2 = 5a + 10c \quad (iii)$$

$$411 = 10b \quad (vi)$$

$$73 = 10a + 34c \quad (v)$$

From Eq (iv), we get, $411 = 10b$, i.e. $b = \frac{411}{10} = 41.1$ (vi)

Multiply Eq. (iii) by -2 and add with Eq. (v), we get $73 + 4 = 34c - 20c = 14c$ i.e. $c = 77/14 = 5.5$

Substituting the value of c in Eq. (iii), we get

$$50 = -2 - 10(5.5) = -57, \text{ i.e. } a = \frac{-57}{5} = -11.4$$

Substituting the values of a , b and c in Eq. (i), the parabola of best fit of V on U becomes

$$V = -11.4 + 41.1U + 5.5U^2 \quad (vi)$$

where U and V are given by (ii).

Substituting the values of U and V from (ii) in (vi), the second degree parabola of best fit of Y on X becomes:

$$\begin{aligned} \frac{Y - 1450}{5} &= -11.4 + 41.1(X - 3) + 5.5(X - 3)^2 \\ &= -11.4 + 41.1X - 123.3 + 5.5(X^2 - 6X + 9) \\ &= 5.5X^2 + (41.1 - 33)X + (-11.4 - 123.3 + 49.5) \\ &= 5.5X^2 + 8.1X - 85.2 \\ Y - 1450 &= 27.5X^2 + 40.5X - 426 \\ Y &= 27.5X^2 + 40.5X + 1024 \end{aligned}$$

(b) For a mesokurtic distribution, the first moment about 7 is 23 and second moment about origin is 1000. Find coefficient of variation and fourth moment about mean. 5

Solution: Given, $\frac{\Sigma(x - 7)}{n} = 23$ or $\frac{\Sigma x}{n} - 7 = 23$ or $\frac{\Sigma x}{n} = 30$ (= Mean) and $\frac{\Sigma x^2}{n} = 1000$

$$\text{Since, SD}(\sigma) = \sqrt{\frac{\Sigma x^2}{n} - \left(\frac{\Sigma x}{n}\right)^2} = \sqrt{1000 - (30)^2} = \sqrt{1000 - 900} = \sqrt{100} = 10 \text{ therefore,}$$

$$\text{Coefficient of variation} = \frac{\text{SD}}{\text{Mean}} \times 100 = \frac{10}{30} \times 100 = \frac{100}{3} \text{ per cent}$$

For a mesokurtic distribution

$$\frac{\mu_4}{(\mu_2)^2} = 3 \text{ or } \frac{\mu_4}{(10)^2} = 3, \text{ i.e. } \mu_4 = 300$$

Or

Consider the following distributions:

	<i>A</i>	<i>B</i>
$\bar{X}$	100	90
Median	90	80
σ	10	10

(i) Distribution A has the same degree of variation as distribution B.

(ii) Both the distributions have the same degree of skewness.

True/False? Comment giving reasons.

5

Solution: (i) C.V. for $A = \frac{\text{SD}}{\text{Mean}} \times 100 = \frac{10}{100} \times 100 = 10\%$

$$\text{C.V. for } B = \frac{\text{SD}}{\text{Mean}} \times 100 = \frac{10}{90} \times 100 = \frac{100}{9} \%$$

Thus, both do not have the same degree of variation

(ii) Karl Pearson's measure of skewness for

$$\text{Distribution } A = \frac{3(\text{Mean} - \text{Median})}{\sigma} = \frac{3(100 - 90)}{10} = 3$$

$$\text{Similarly, Distribution } B = \frac{3(90 - 80)}{10} = 3$$

Thus both have the same degree of skewness.

Q.4. (a) A food products company is contemplating the introduction of a revolutionary new product with new packaging to replace the existing product at much higher price (S_1) or a moderate change in the composition of existing product with a new packaging at a small increase in price (S_2) or a small change in the composition of the existing except the word 'new' with negligible increase in price (S_3). The three possible states of nature are: (i) high increase in sales (N_1), (ii) No change in sales (N_2) and (iii) decrease in sales (N_3). The marketing department of a company worked out payoffs in terms of yearly net profits for each of these strategies of these events.

State of Nature → Strategies ↓	Payoff in ₹		
	N_1	N_2	N_3
S_1	7,00,000	3,00,000	1,50,000
S_2	5,00,000	4,50,000	0
S_3	3,00,000	3,00,000	3,00,000

Which strategy should the executive concerned choose on the basis of:

(i) Maximin criterion

(ii) Maximax criterion

(iii) Minimax criterion

(iv) Laplace criterion?

10

Solution: (i) Maximin criterion. Under this criterion, a strategy is selected which maximises the minimum payoffs.

Strategy	Minimum payoffs
S_1	1,50,000
S_2	0
S_3	3,00,000

The executive should choose strategy S_3 .

(ii) Maximax criterion. In this criterion, a strategy is selected which gives the maximum payoffs.

Strategy	Maximum payoffs
S_1	7,00,000
S_2	5,00,000
S_3	3,00,000

The executive should choose strategy S_1 .

(iii) **Minimax regret criterion.** Necessary steps to adopt this criterion are:

- Determine the opportunity loss for each strategy by subtracting from maximum payoff of a state of nature, the pay-offs for that state of nature for all the strategies.
- Determine the maximum of opportunity loss for each strategy.
- Select the strategy which minimises the maximum of the loss.

Opportunity Loss Table

Strategy	N_1	N_2	N_3
S_1	$7,00,000 - 7,00,000$ $= 0$	$4,50,000 - 3,00,000$ $= 1,50,000$	$3,00,000 - 1,50,000$ $= 1,50,000$
S_2	$7,00,000 - 5,00,000$ $= 2,00,000$	$4,50,000 - 4,50,000$ $= 0$	$3,00,000 - 0$ $= 3,00,000$
S_3	$7,00,000 - 3,00,000$ $= 4,00,000$	$4,50,000 - 3,00,000$ $= 1,50,000$	$3,00,000 - 3,00,000$ $= 0$

Since minimum opportunity loss is for strategy S_1 , executive should choose strategy S_1 to minimise the maximum of the losses.

(iv) **Laplace criterion.** Necessary steps to adopt this criterion are:

- Assigning equal opportunity to each state of nature.
- Calculating the expected monetary value EMV.
- Selecting that strategy for which EMV is maximum.

S_1	$\frac{1}{3}(7,00,000 + 3,00,000 + 1,50,000) = \frac{11,50,000}{3} = 3,83,333.33$
S_2	$\frac{1}{3}(5,00,000 + 4,50,000 + 0) = \frac{9,50,000}{3} = 3,16,666.66$
S_3	$\frac{1}{3}(3,00,000 + 3,00,000 + 3,00,000) = \frac{9,00,000}{3} = 3,00,000$

Since the EMV is highest for strategy S_1 , hence the executive should select strategy S_1 .

Or

In 500 small scale industrial units, the return on investment ranged from 0 to 30per cent, no unit sustaining any loss. 5per cent of industrial units had returns exceeding 0per cent but not exceeding 5per cent. 15per cent of units had returns exceeding 5per cent but not exceeding 10per cent. Median and upper quartile rate of return was 15per cent and 20per cent respectively. The uppermost layer of returns exceeding 25per cent but not exceeding 30per cent was earned by 25per cent. Present this information in the form of frequency table with intervals as follows:

Exceeding 0per cent but not exceeding 5per cent
 Exceeding 5per cent but not exceeding 10per cent
 Exceeding 10per cent but not exceeding 15per cent
 Exceeding 15per cent but not exceeding 20per cent
 Exceeding 20per cent but not exceeding 25per cent
 Exceeding 25per cent but not exceeding 30per cent

Use $\frac{N}{4}$, $\frac{2N}{4}$, $\frac{3N}{4}$ as ranks of lower, middle and upper quartiles, respectively.

Find the rate of return around which there is maximum concentration of units.

Solution: (A) Median = $\left(\frac{N}{2}\right)^{th}$ item = $\frac{500}{2} = 250^{th}$ item

Upper Quartile $Q_3 = \left(\frac{3N}{4}\right)^{th}$ item = $\frac{3 \times 500}{4} = 375^{th}$ item

Returns	Return Percentage	Units (f)	Cumulative Frequency (c.f)
0 — 5	5% of 500	25	25
5 — 10	15% of 500	75	100
10 — 15		150	250
15 — 20		125	375
20 — 25		0	375
25 — 30	25% of 500	125	500

The maximum concentration of units is around 10-15 percent rate of return.

(b) Discuss briefly the various components of time series.

Answer. See Q.3(b)

Or

Does correlation imply causation between two variables?

5

Solution: No, the correlation does not always imply causation between two variables. There may be a spurious correlation between two variables which is otherwise caused by a third independent factor. For example, in emergency situation such as war both the price and the demand of few products increases, but it does not mean that the correlation between price and demand is positive. This temporary situation is caused by a third factor, i.e., war.

Q.5. (a) Given below is a trend equation:

$$Y_c = 372 + 288X$$

(Origin 2006, X unit = 1 year, Y unit = Annual sales)

Convert the above equation:

(i) To monthly trend equation with January, 2007 as origin and estimate sales for March, 2007.

(ii) To quarterly trend equation with first quarter, 2007 as origin and estimate sales for third quarter of 2007.

9

Solution: (i) The required equation is

$$Y = \frac{372}{12} + \frac{288}{144}(X + 6.5) = 31 + 2(X + 6.5)$$

Sales for March 2007: $Y = 31 + 2(2 + 6.5) = 31 + 17 = 48$

Sales for third quarter 2007: $Y = \frac{372}{4} + \frac{288}{16}(2 + 2.5) = 93 + 18 \times 4.5 = 93 + 81 = 174$

Working notes:

(a) Shift the origin from 2006 to January 2007.

No. of months from July 01, 2006 to January 15, 2007, becomes $6 + 0.5 = 6.5$

(b) Shift the origin from 2006 to first quarter of 2007.

No. of quarters = 2 (in 2006) + 0.5 (in 2007) = 2.5

Or

(a) 100 students appeared for an examination. The results of those who failed are given below:

Marks	5	10	15	20	25	30	Total
No. of Students	4	6	8	7	3	2	30

If the average marks of all students were 68.6, find out average marks of those who passed. 3

(b) A computer while calculating the coefficient of correlation between two variables X and Y from 25 pairs of observations obtained the following results:

$$N = 25, \Sigma X = 125, \Sigma X^2 = 650, \Sigma Y = 100, \Sigma Y^2 = 460, \Sigma XY = 508.$$

It was however, discovered at the time of checking that it had copied down two pairs as:

n	n		n	n
6	14	While correct values were	8	12
8	6		6	8

Obtain the correct value of correlation coefficient. 6

Solution: (a) Total marks of all the students = $68.6 \times 100 = 6860$

Total marks of the students who failed

$$= 5 \times 4 + 10 \times 6 + 15 \times 8 + 20 \times 7 + 25 \times 3 + 30 \times 2 \\ = 20 + 60 + 120 + 140 + 75 + 60 = 475$$

$$\text{Average marks of the students who passed} = \frac{6860 - 475}{100 - 30} = \frac{6385}{70} = 91.2$$

(b) Correct $\Sigma X = 125 - 6 - 8 + 8 + 6 = 125$

$$\text{Correct } \Sigma Y = 100 - 14 - 6 + 12 + 8 = 100$$

$$\text{Correct } \Sigma X^2 = 650 - (6)^2 - (8)^2 + (8)^2 + (6)^2 = 650$$

$$\text{Correct } \Sigma Y^2 = 460 - (14)^2 - (6)^2 + (12)^2 + (8)^2 = 436$$

$$\text{Correct } \Sigma XY = 508 - 6 \times 14 - 8 \times 6 + 8 \times 12 + 6 \times 8 = 520$$

$$\begin{aligned} \text{Correct, } r &= \frac{n \Sigma XY - (\Sigma X)(\Sigma Y)}{\sqrt{n \Sigma X^2 - (\Sigma X)^2} \sqrt{n \Sigma Y^2 - (\Sigma Y)^2}} \\ &= \frac{25 \times 520 - 125 \times 100}{\sqrt{25 \times 650 - (125)^2} \sqrt{25 \times 436 - (100)^2}} \\ &= \frac{13,000 - 12,500}{\sqrt{(16,250 - 15,625)} \sqrt{(10,900 - 10,000)}} \\ &= \frac{500}{\sqrt{625} \sqrt{900}} = \frac{500}{25 \times 30} = \frac{2}{3} = 0.67 \end{aligned}$$

(b) If X is a Poisson variable such that:

$$P(X = 2) = 9P(X = 4) + 90P(X = 6)$$

Find mean and variance of X .

Also state the condition under which Poisson distribution is used. 6

Solution: Given

$$\begin{aligned}\frac{e^{-m}m^2}{2!} &= 9\frac{e^{-m}m^4}{4!} + 90\frac{e^{-m}m^6}{6!} \\ &= 9\frac{e^{-m}m^4}{4 \times 3 \times 2 \times 1} + 90\frac{e^{-m}m^6}{6 \times 5 \times 4 \times 3 \times 2 \times 1} \\ 1 &= \frac{9}{12}m^2 + \frac{m^4}{4} = \frac{3}{4}m^2 + \frac{m^4}{4}\end{aligned}$$

$$\frac{3m^2 + m^4}{4} = 1$$

$$m^4 + 3m^2 - 4 = 0$$

$$m^4 + 4m^2 - m^2 - 4 = 0$$

$$m^2(m^2 + 4) - 1(m^2 + 4) = 0$$

$$(m^2 - 1)(m^2 + 4) = 0 \text{ i.e. } m^2 = 1 \text{ or } m^2 = -4$$

As m^2 cannot be negative, $m^2 = -4$ is rejected. Hence $m^2 = 1$ or $m = \pm 1$

Since m is variance, and cannot be -ve, $m = 1 = \text{Mean} = \text{Variance}$

Or

With the help of given data, find:

(i) Value of middle 50per cent item

(ii) Value of exactly 50per cent item

(iii) The value of P_{40} and D_6

(iv) Graphically with the help of ogive curve, the values of Q_1 , Q_3 , median, P_{40} and D_6 :

Class Interval	10—14	15—19	20—24	25—29	30—34	35—39	
Frequencies	5	10	15	20	10	5 = 65(=N)	6

Solution: Changing: to exclusive type, we get

Class interval	Frequency	Cumulative Frequency (c.f.)	
9.5—14.5	5	5	
14.5—19.5	10	15	
19.5—24.5	15	30	$\leftarrow Q_1 [(N/4)^{\text{th}} = 16.25^{\text{th}} \text{ item}]$
24.5—29.5	20	50	$\leftarrow Q_2 [(N/2)^{\text{th}} = 32.5^{\text{th}} \text{ item}]$
29.5—34.5	10	60	$\leftarrow Q_3 [(3N/4)^{\text{th}} = 48.75^{\text{th}} \text{ item}]$
34.5—39.5	5	65	

$$P_{40} = \left(\frac{40N}{100} \right)^{\text{th}} = \frac{40 \times 65}{100} = 26^{\text{th}} \text{ frequency}$$

$$D_6 = \left(\frac{6N}{100} \right)^{\text{th}} = \frac{6 \times 65}{10} = 39^{\text{th}} \text{ frequency}$$

Middle 50per cent values = $Q_3 - Q_1$ and Exactly 50per cent item = Median

$$Q_1 = \left(\frac{65}{4} \right)^{\text{th}} = 16.25^{\text{th}} \text{ value lies in the class } 19.5 - 24.5$$

$$Q_2 = \left(\frac{65}{2} \right)^{\text{th}} = 32.5^{\text{th}} \text{ value lies in the class } 24.5 - 29.5$$

$$Q_3 = \left(\frac{3}{4} \times 65 \right)^{th} = 48.75^{th} \text{ value lies in the class } 24.5 - 29.5$$

$P_{40} = 26^{th}$ value lies in the class 19.5 – 24.5

$D_6 = 39^{th}$ value lies in the class 24.5 – 29.5

Thus $Q_1 = 19.5 + \frac{16.25 - 15}{10} \times 5 = 19.5 + \frac{1.25}{2} = 20.125$

$$Q_2 = 24.5 + \frac{32.50 - 30}{20} \times 5 = 24.5 + \frac{2.50}{4} = 25.12$$

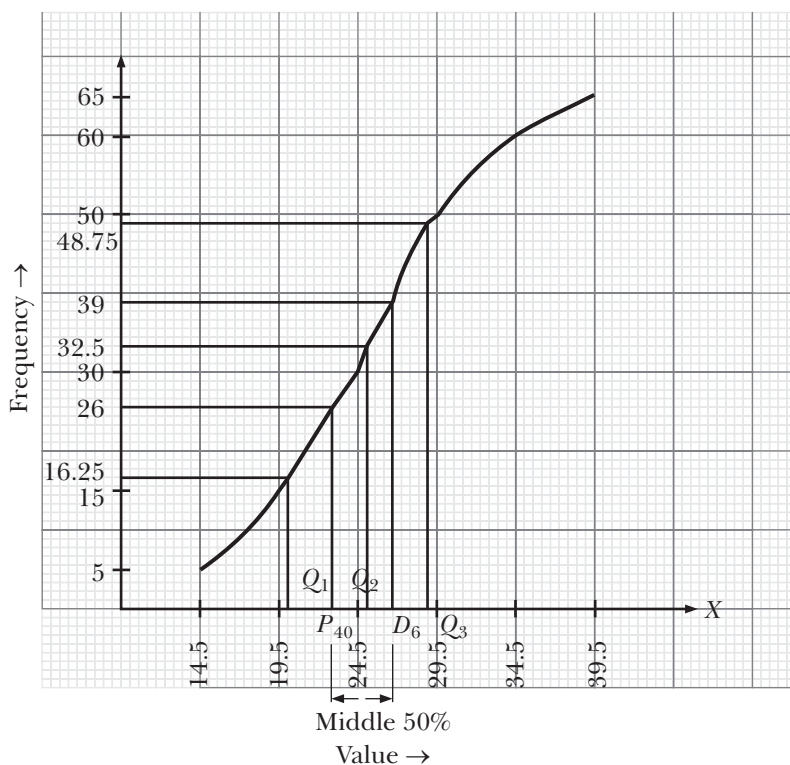
$$Q_3 = 24.5 + \frac{48.75 - 30}{20} \times 5 = 24.5 + \frac{18.75}{4} = 29.187$$

(i) Value of middle 50 per cent item = $Q_3 - Q_1 = 29.187 - 20.125 = 9.062$

(ii) Value of exactly 50 per cent item = $Q_2 = 25.12$

(iii) $P_{40} = 19.5 + \frac{26 - 15}{15} \times 5 = 19.5 + \frac{11}{15} \times 5 = 23.167$

$$D_6 = 24.5 + \frac{39 - 30}{20} \times 5 = 24.5 + \frac{9}{4} = 26.75$$



BUSINESS STATISTICS
B.Com. (Hons.) First Year
Paper IV—2009

Time : 3 hours

Maximum Marks : 75

Note: The maximum marks printed on the Question Paper are applicable for the candidates registered with the School of Open Learning for B.Com. (Hons.). These marks will, however, be scaled down proportionately in respect of the students of regular colleges, at the time of posting of awards for compilation of result.

Attempt **all** the questions.

Q. 1. (a) Comment on the following:

- (i) For a binomial distribution, Mean = 7 and Variance is 11.
- (ii) The G.M. and A.M. of a distribution are 27 and 30 respectively. Then H.M. is 26.
- (iii) The Standard Deviation of a symmetrical distribution is 3 and the value of the Fourth Moment about Mean is 243 in order that the distribution be Mesokurtic.
- (iv) If the coefficient of Q.D. = 0.6 and Q.D. = 15, then $Q_1 = 10$ and $Q_3 = 40$.
- (v) A and B are two events such that $P(A) = \frac{1}{4}$, $P(B) = \frac{1}{3}$ and $P(A \cup B) = \frac{1}{2}$

$$\text{Hence } n\left(\frac{n}{n}\right) = \frac{1}{5}.$$

$$2 \times 5 = 10$$

Solution: (a) (i) Given

$$\text{Mean} = np = 7, \quad \dots(i)$$

$$\text{Variance} = npq = 11 \quad \dots(ii)$$

Dividing (ii) by (i), we get $\frac{npq}{np} = \frac{11}{7}$, i.e. $q = \frac{11}{7} > 1$ which is not feasible because the probability of failure cannot be more than 1.

(ii) The statement is correct because A.M. > G.M. > H.M., i.e. $30 > 27 > 26$

(iii) For a mesokurtic curve, $\frac{\mu_4}{(\mu_2)^2}$ should be equal to 3, which is true in this case That is, $\frac{243}{(3)^4} = 3$, hence the statement is true.

(iv) Coefficient of quartile deviation is given by:

$$\frac{Q_3 - Q_1}{Q_3 + Q_1} = 0.6 \quad \dots(i)$$

and $\text{Quartile deviation} = \frac{Q_3 - Q_1}{2} = 15$, i.e. $Q_3 - Q_1 = 30 \quad \dots(ii)$

Substituting the value of $Q_3 - Q_1$ in (i), we get

$$\frac{30}{Q_3 + Q_1} = 0.6, \text{ i.e. } Q_3 + Q_1 = \frac{30}{0.6} = 50 \quad \dots(iii)$$

Adding (ii) and (iii), we get, $2Q_3 = 80$, i.e. $Q_3 = 40$. Substituting the value of Q_3 in (iii), we get $Q_1 = 10$. Hence the given statement is true.

(v) Since $P(A \cup B) = P(A) + P(B) - P(A \cap B)$, and $P(A \cap B) = P(A) P(B|A)$

therefore $\frac{1}{2} = \frac{1}{4} + \frac{1}{3} - P(A \cap B)$, and $\frac{1}{12} = \frac{1}{4} P(B|A)$

$$P(A \cap B) = \frac{1}{4} + \frac{1}{3} - \frac{1}{2} = \frac{1}{12}, \text{ and } P(B|A) = \frac{4}{12} = \frac{1}{3}$$

Hence the given statement is wrong.

(b) Define Dispersion.

The mean and the Standard Deviation of a series of 100 items were found to be 60 and 10 respectively. While calculating two items were wrongly taken as 5 and 45 instead of 30 and 20. Calculate corrected variance and corrected coefficient of variation. 5

Solution: Dispersion is the degree of the variation (or scatter) of the variables about a central value. Dispersion measures the extent to which the value of variables vary from some central value. Measures of dispersion shows the range within which the values of a variable lie.

$$\text{Wrong total of 100 items} = 100 \times 60 = 6000$$

$$\text{Correct total of 100 items} = 6000 - 5 - 45 + 30 + 20 = 6000$$

$$\text{Correct mean, } \bar{x} = \frac{6000}{100} = 60, \text{ i.e. the mean remains unchanged}$$

$$\text{Wrong, S.D. (=10)} = \sqrt{\frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2}$$

$$\text{or } 100 = \frac{\sum x^2}{100} - (60)^2 \quad \text{i.e. } \sum x^2 = (100 + 3,600)100 = 3,70,000$$

$$\begin{aligned} \text{Correct } \sum x^2 &= 3,70,000 - (5)^2 - (45)^2 + (30)^2 + (20)^2 \\ &= 3,70,000 - 25 - 2025 + 900 + 400 = 3,69,250 \end{aligned}$$

$$\text{Correct S.D.} = \sqrt{\frac{369250}{100} - (60)^2} = 9.618$$

$$\text{Correct Variance (S.D.)}^2 = (9.618)^2 = 92.505$$

$$\text{Coefficient of variation} = \frac{\text{S.D.}}{\text{Mean}} \times 100 = \frac{9.618}{60} \times 100 = 16\%$$

Q. 2. (a) What are the different methods of measuring trend? Explain the methods of eliminating trend in a time series. Which one do you consider better? 5

Solution: (a) Different methods of measuring trend are:

- (i) Freehand (or graphical) method
- (ii) Method of semi-averages
- (iii) Method of moving averages
- (iv) Method of least squares

Trend can be eliminated by:

- (i) Method of moving averages
- (ii) Least squares method (i.e. Ratio-to-trend method)

Least squares method is preferred as it is mathematical in nature

(b) 100 car radios are inspected as they come off the production line and the number of defects per set are recorded below:

<i>No. of Defects</i>	<i>No. of Sets</i>
0	79
1	18
2	2
3	1
4	0
	<u>100</u>

Fit a Poisson distribution to the above data and calculate theoretical frequencies of 0, 1, 2, 3 and 4 defects. 5

Solution: Mean of the Poisson distribution is given by:

$$\lambda(=m) = \frac{0 \times 79 + 1 \times 18 + 2 \times 2 + 3 \times 1 + 4 \times 0}{79 + 18 + 2 + 1 + 0} = \frac{25}{100} = 0.25$$

Thus $e^{-m} = e^{-0.25} = 0.7788$ (From table)

$$\text{Frequency of 0 defect} = 100 \frac{e^{-m}(m)^0}{0!} = 100 \times e^{-m} = 100 \times 0.7788 = 78$$

$$\text{Frequency of 1 defect} = 100 \frac{e^{-m}(m)^1}{1!} = 100m e^{-m} = 100 \times 0.25 \times 0.7788 = 19.47 = 20 \text{ (approx.)}$$

$$\text{Frequency of 2 defects} = 100 \frac{m^2}{2!} e^{-m} = 100 \times \frac{(0.25)^2}{2} \times 0.7788 = 2.43 = 2 \text{ approx.}$$

$$\text{Frequency of 3 defects} = 100 \frac{m^3}{3!} e^{-m} = 100 \times \frac{(0.25)^3}{6} \times 0.7788 = 0.20 = 0 \text{ approx.}$$

$$\text{Frequency of 4 defects} = 100 \frac{m^4}{4!} e^{-m} = 100 \times \frac{(0.25)^4}{24} \times 0.7788 = 0.0126 = 0 \text{ approx.}$$

(c) Define skewness.

From a moderately skewed distribution of retail prices for Men's Shirts it is found that the mean price is ₹ 200 and median price is ₹ 170. If the coefficient of variation is 20 per cent, find the Pearson co-efficient of skewness of the distribution. 5

Solution: When a series is not symmetrical, the distribution is said to be asymmetrical or skewed. The degree of skewness indicates the cluster of deviations around a measure of central tendency. In an asymmetrical distribution, the deviations around an average are not equal. Measures of skewness shows the shape of the curve of values in any data set.

$$\text{Coefficient of variation} = \frac{\text{S.D.}}{\text{Mean}} \times 100 \quad \text{or} \quad 20 = \frac{\text{S.D.}}{200} \times 100, \text{ i.e. S.D.} = 40$$

$$\begin{aligned} \text{Pearson's coefficient of skewness} &= \frac{3(\text{Mean} - \text{Median})}{\text{S.D.}} \\ &= \frac{3(200 - 170)}{40} = \frac{90}{40} = 2.25 \end{aligned}$$

Or

Define Geometric Mean and Harmonic Mean. A train runs for 30 minutes at a speed of 40 miles an hour and then because of repairs of the track, the train runs for 10 minutes at a speed of 8 miles an hour, after which it resumes its previous speed and runs for 18 minutes except for a period of 2 minutes when it had to run over a bridge with a speed of 30 miles per hour. Using a suitable average, calculate the average speed of the train. 5

Answer: Geometric Mean is defined as the n^{th} root of the product of N items (or values). If there are two items, we take the square root; if there are three items, the cube root; and so on.

$$\text{G.M.} = \sqrt[n]{x_1 \times x_2 \times \dots \times x_n}$$

where $x_1, x_2, \dots, x_n$, etc. refer to the various items of the series.

The harmonic mean is based on the reciprocals of the numbers' averaged. It is defined as the reciprocal of the arithmetic mean of the reciprocal of the individual observation, That is

$$\text{H.M.} = \frac{N}{\left(\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_n} \right)}$$

Total distance travelled = Speed $\times$ Time

$$\begin{aligned} &= \frac{40}{60} \times 30 + \frac{8}{60} \times 10 + \frac{40}{60} \times 18 + \frac{30}{60} \times 2 \\ &= 20 + 1.33 + 12 + 1 = 34.33 \text{ miles} \end{aligned}$$

Total time taken = 30 + 10 + 18 + 2 = 60 minutes

Thus,
$$\text{Average speed} = \frac{34.33}{60} \times 60 = 34.33 \text{ miles/hr.}$$

Q. 3. (a) Distinguish between Karl Pearson's coefficient of correlation and Spearman's Rank Correlation. Explain with the help of an example when Spearman Rank Correlation results to +1, -1 and between -1 to +1. 5

Solution: (a) Pearson's coefficient of correlation is a linear correlation coefficient and is based on quantitative measurement but Rank correlation coefficient might be based on some qualitative factor, such as taste, appearance, intelligence or some other factor.

$$\text{Spearman's rank correlation coefficient (R)} = 1 - \frac{6\sum D^2}{N(N^2 - 1)}$$

where D is the difference between the corresponding ranks. The value of Spearman's rank correlation co-efficient ranges between +1 and -1. If $\sum D^2 = 0$, $R = 1$, i.e. there is a perfect positive correlation between the variables. If the ranks assigned are exactly inverse i.e. an individual getting the highest score from one person gets the lowest score from the other person and so on, then the correlation is -1, that is, there is perfect negative correlation between the variables. For example:

Rank I :	3	8	2	4	6	7	5
Rank II :	3	8	2	4	6	7	5
D :	0	0	0	0	0	0	0

In this case the correlation is perfect positive. Again if

Ranks R_1	Ranks R_2	$D = R_1 - R_2$	D^2
3	7	-4	16
8	2	6	36
2	8	-6	36
4	6	-2	4
6	4	2	4
7	3	4	16
4	4	0	0
			112

$$R = 1 - \frac{6\Sigma D^2}{N(N^2 - 1)} = 1 - \frac{6 \times 112}{7(49 - 1)} = 1 - \frac{6 \times 112}{7 \times 48} = -1$$

In this case, rank correlation is perfect negative. However, in other cases rank correlation coefficient will always lie between +1 and -1.

(b) For the following data verify that five-yearly weighted moving average trend values with weights 1, 2, 2, 2, 1 respectively is equivalent to four-yearly centred moving average trend values **5**

Year	: 1995	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005
Sales	: 5	3	7	6	4	8	9	10	8	9	9

(₹ in Lakh)

Solution:

Year	Sales (in lakhs)	five-years' Weighted Moving Total	five-years' Weighted Moving Average
1995	5	—	
1996	3	—	
1997	7	$1 \times 5 + 2 \times 3 + 2 \times 7 + 2 \times 6 + 1 \times 4 = 41$	$\frac{41}{8} = 5.125$
1998	6	$1 \times 3 + 2 \times 7 + 2 \times 6 + 2 \times 4 + 1 \times 8 = 45$	$\frac{45}{8} = 5.625$
1999	4	$1 \times 7 + 2 \times 6 + 2 \times 4 + 2 \times 8 + 1 \times 9 = 52$	$\frac{52}{8} = 6.500$
2000	8	$1 \times 6 + 2 \times 4 + 2 \times 8 + 2 \times 9 + 1 \times 10 = 58$	$\frac{58}{8} = 7.250$
2001	9	$1 \times 4 + 2 \times 8 + 2 \times 9 + 2 \times 10 + 1 \times 8 = 66$	$\frac{66}{8} = 8.250$
2002	10	$1 \times 8 + 2 \times 9 + 2 \times 10 + 2 \times 8 + 1 \times 9 = 71$	$\frac{71}{8} = 8.875$
2003	8	$1 \times 9 + 2 \times 10 + 2 \times 8 + 2 \times 9 + 1 \times 9 = 72$	$\frac{72}{8} = 9.000$
2004	9	—	
2005	9	—	

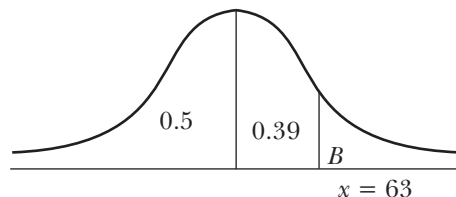
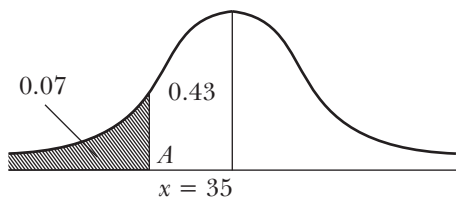
4-yearly centred moving average would be as under:

Year (1)	Sales (2)	four-yearly Moving (Total of 2 column)	two-yearly Moving Total of Column (3)	four-years' Centred Moving Average
1995	5	—		—
1996	3	—		—
		21		
1997	7	20	41	$\frac{41}{8} = 5.125$
1998	6	25	45	$\frac{45}{8} = 5.625$
1999	4	27	52	$\frac{52}{8} = 6.500$
2000	8	31	58	$\frac{58}{8} = 7.250$
2001	9	35	66	$\frac{66}{8} = 8.250$
2002	10	36	71	$\frac{71}{8} = 8.875$
2003	8	36	72	$\frac{72}{8} = 9.000$
2004	9			
2005	9	—		

Both the series are same.

(c) In a distribution exactly normal 7 per cent of the items are under 35 and 89 per cent are under 63. What are the Mean and standard Deviation of the distribution? 5

Solution: At point A, $z = -1.475$ and at point B, $z = 1.23$ (From table). Now



$$z = \frac{x - \mu}{\sigma}$$

$$-1.475 = \frac{35 - \mu}{\sigma}, \text{ i.e. } -1.475 \sigma = 35 - \mu \quad \dots(i)$$

or $1.23 = \frac{63 - \mu}{\sigma}$, i.e. $1.23 \sigma = 63 - \mu$... (ii)

Subtracting (i) from (ii), we get $2.705 \sigma = 28$ or $\sigma = 10.35$

Or

Determine the values of Median and Mode of the following distribution graphically. Verify the results by actual calculations. After verifying, calculate the value of Mean and sketch a curve indicating the general shape of the distribution and comment: 5

Size	Frequency
10—19	11
20—29	19
30—39	21
40—49	16
50—59	10
60—69	8
70—79	6
80—89	3
90—99	1

Solution: Converting given class intervals into exclusive class intervals is as under

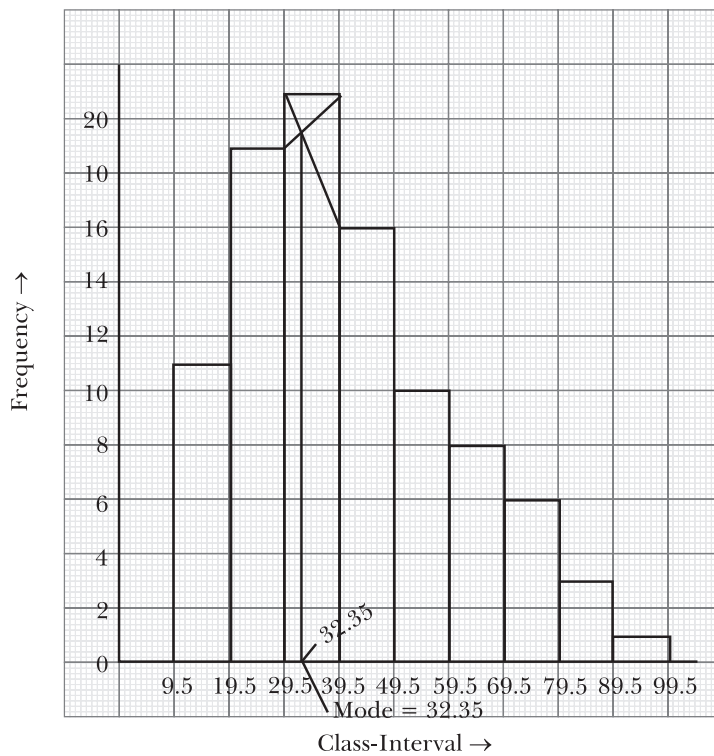
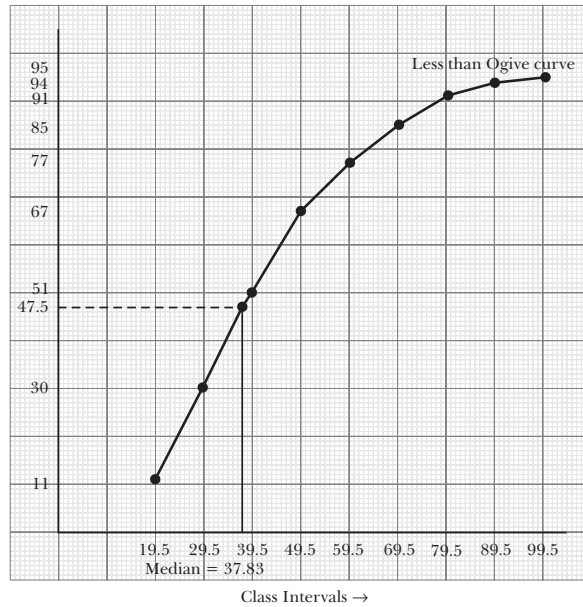
Class interval	Mid-value	Frequency (f)	Cumulative Frequency (c.f.)
9.5—19.5	14.5	11	11
19.5—29.5	24.5	19	30
29.5—39.5	34.5	21 $\leftarrow M_0$	51 $\leftarrow Me$
39.5—49.5	44.5	16	67
49.5—59.5	54.5	10	77
59.5—69.5	64.5	8	85
69.5—79.5	74.5	6	91
79.5—89.5	84.5	3	94
89.5—99.5	94.5	1	95

$$\text{Median class} = \left(\frac{95}{2} \right) \text{th value} = 47.5 \text{th value}$$

$$\begin{aligned} \text{Median} &= l + \frac{\frac{N}{2} - c.f.}{f} \times h \\ &= 29.5 + \frac{47.5 - 30}{21} \times 10 = 29.5 + \frac{17.5}{21} \times 10 = 37.833 \end{aligned}$$

$$\text{Mode} = l + \frac{f_m - f_{m-1}}{2f_m - f_{m+1} - f_{m-1}} \times h$$

$$= 29.5 + \frac{21 - 19}{42 - 16 - 19} \times 10 = 29.5 + \frac{2 \times 10}{7} = 32.35$$



Q. 4. (a) Explain the mathematical properties of Standard Deviation. Why is Standard Deviation sed more than Mean Deviation? 5

Answer: Mathematical properties of Standard Deviation

- (i) Σd^2 or Σfd^2 is the minimum as the deviations are taken from the actual mean.
- (ii) If the standard deviations of a number of groups are known, their combined S.D. can be computed.
- (iii) In a normal distribution
 - $x \pm \sigma$ covers 68.26 per cent area
 - $x \pm 2\sigma$ covers 95.45 per cent area
 - $x \pm 3\sigma$ covers 99.73 per cent area

Standard Deviation is regarded as a better measure of dispersion as compared to mean deviation due to the following reasons:

- (i) In the calculation of mean deviation, the algebraic signs are ignored which is mathematically not correct, whereas in the calculation of standard deviation, these signs are considered.
- (ii) Standard deviation is amenable to algebraic treatment, i.e., we can find the combined standard deviation of two or more series but it is not possible to compute combined mean deviation.
- (iii) The standard deviation is used in the analysis of statistical series such as in correlation, skewness, sampling or the analysis of variance.

(b) The following table gives age (in years) of husbands and wives for 50 newly married couples:

<i>Age of Wives (Y)</i>	<i>Age of Husbands (X)</i>			<i>Total</i>
	20—25	25—30	30—35	
16—20	9	14	—	23
20—24	6	11	3	20
24—28	—	—	7	07
Total	15	25	10	50

Find: (i) Regression line of husbands on wives.

(ii) The age of husband when wife is 20 years old.

Solution: Let age of the husband be denoted by X and that of wife by Y .

$$\text{Mean of } X = A + \frac{\Sigma fd_x}{\Sigma f} \times h = 27.5 + \left(\frac{-5}{50} \times 5 \right) = 27.5 - 0.5 = 27$$

$$\text{Mean of } Y = A + \frac{\Sigma fd_y}{\Sigma f} \times h = 22 + \left(\frac{-16}{50} \times 4 \right) = 22 - 1.28 = 20.7$$

$$\begin{aligned} b_{yx} &= \frac{N \Sigma fd_x d_y - \Sigma fd_x \times \Sigma fd_y}{N \Sigma fd_x^2 - (\Sigma fd_x)^2} \times \frac{h_y}{h_x} = \frac{50 \times 16 - (-5 \times -16)}{50 \times 25 - (-5)^2} \times \frac{4}{5} \\ &= \frac{800 - 80}{1250 - 25} \times \frac{4}{5} = \frac{720}{1225} \times \frac{4}{5} = 0.47 \end{aligned}$$

$$b_{xy} = \frac{N \Sigma fd_x d_y - \Sigma fd_x \times \Sigma fd_y}{N \Sigma fd_y^2 - (\Sigma fd_y)^2} \times \frac{h_x}{h_y} = \frac{50 \times 16 - (-5 \times -16)}{50 \times 30 - (-16)^2} \times \frac{5}{4} = 0.72$$

Computations for Regression Equations

Y m.v. dy X m.v. dx				20-25	25-30	30-35	Total			
				22.5	27.5	32.5				
				1	0	1				
16-20	18	4	1	9	0			fd_y	fd_y^2	fd_xd_y
				9	14		23	23	+23	+9
20-24	22	0	0	0	0	0				
				6	11	3	20	0	0	0
24-28	26	+4	+1			7				
						7	7	+7	+7	+7
Total f				15	25	10	50	Σfd_y	Σfd_y^2	Σfd_xd_y
								16	+30	+16
fd_x				15	0	+10	Σfd_x			
							5			
fd_x^2				+15	0	+10	Σfd_x^2			
							+25			
fd_xd_y				9	0	7	Σfd_xd_y			
							+16			

Regression equation of Y on X

$$Y - \bar{Y} = b_{yx}(X - \bar{X}) \quad \text{or} \quad Y - 20.7 = 0.47(X - 27)$$

$$Y = 0.47X + 8.01 \quad \dots(i)$$

Regression equation of X on Y

$$(X - \bar{X}) = b_{xy}(Y - \bar{Y}) \quad \text{or} \quad X - 27 = 0.72(Y - 20.7)$$

$$X = 0.72Y + 12.1 \quad \dots(ii)$$

The age of husband (value of X) when age of wife $Y = 20$ is given by

$$X = 0.72Y + 12.1 = 0.72 \times 20 + 12.1 = 26.5 \text{ years}$$

(c) Two sets of Indices one with 1976 as base and the other with 1984 as base given below:

Year	Index A	Year	Index B
1976	100	1984	100
1977	110	1985	105
1978	120	1986	90
1979	190	1987	95
1980	300	1988	102
1981	330	1989	110
1982	360	1990	96
1983	390	—	—
1984	400	—	—

You are required to splice the Index (B) to Index (A). Then also shift the base to 1986.

Solution:

<i>Year</i>	<i>Index A</i>	<i>Year</i>	<i>Index B</i>
1976	100	1984	$100 \times 4 = 400$
1977	110	1985	$105 \times 4 = 420$
1978	120	1986	$90 \times 4 = 360$
1979	190	1987	$95 \times 4 = 380$
1980	300	1988	$102 \times 4 = 408$
1981	330	1989	$110 \times 4 = 440$
1982	360	1990	$96 \times 4 = 384$
1983	390	—	—
1984	400	—	—

If 1986 is taken as base year, then the index for 1986 will be taken as 100. So dividing all the indices by 3.6 as $[360/3.6 = 100]$

The new index numbers will be

<i>Year</i>	<i>Index Number</i>	<i>Year</i>	<i>Index Number</i>
1976	$\frac{100}{3.6} = 27.77$	1984	$\frac{400}{3.6} = 111.1$
1977	$\frac{110}{3.6} = 30.5$	1985	$\frac{420}{3.6} = 116.60$
1978	$\frac{120}{3.6} = 33.3$	1986	$\frac{360}{3.6} = 100$
1979	$\frac{190}{3.6} = 52.77$	1987	$\frac{380}{3.6} = 105.5$
1980	$\frac{300}{3.6} = 83.3$	1988	$\frac{408}{3.6} = 113.3$
1981	$\frac{330}{3.6} = 91.66$	1989	$\frac{440}{3.6} = 122.2$
1982	$\frac{360}{3.6} = 100$	1990	$\frac{384}{3.6} = 106.66$
1983	$\frac{390}{3.6} = 108.3$		

Q. 5. (a) Distinguish between (any two):

- (i) **Time Reversal and Factor Reversal tests of adequacy of Index Number formulae.**
- (ii) **$Y = a + bx$ and $Y = ab^x$ for obtaining Trend values in an analysis of time series.**
- (iii) **Priori and Posteriori definitions of probability.**

$2\frac{1}{2} + 2\frac{1}{2}$

Answer: (a) (i) **Time Reversal Test:** It is a test which helps in determining whether a given formula will work both ways in time, *forward* and *backward*. This test implies that if the time subscript in any index number formulae be interchanged, then the resulting index number should be the reciprocal of original index number. Symbolically, the product $I_{01} \times I_{10}$ should be equal to unity [omitting the factor 100 from both the indices] where I_{01} is the price index number with 1 as current year and 0 as base year and I_{10} is the index number with 1 as base year and 0 as the current year.

Fisher's Ideal index number formula satisfies this test.

Factor Reversal Test: According to this test the product of a price index and the quantity index should be equal to the corresponding value index, i.e., true value ratio, omitting the factor 100 from both the indices, i.e.

$$P_{01} \times Q_{01} = \frac{\Sigma P_1 Q_1}{\Sigma P_0 Q_0}$$

where P_{01} is the price index with Q_{01} as the quantity index with 1 as current year and 0 as base year.

This test indicates that the change in price multiplied by the change in quantity should be equal to the total change in value.

- (ii) $y = a + bx$ is the equation for linear trend whereas $y = ab^x$ is the equation for obtaining exponential trend.
- (iii) Priori probability is the probability of an event occurring before the sample evidence relevant to the event has been observed.

Posteriori probability is the probability of an event modified in the light of an experimental evidence. Posterior probabilities are generally determined with the help of Baye's theorem.

(b) A manufacturing company is faced with the problem of choosing from four products. The potential demand for each product may turn out to be good, satisfactory and poor. The probabilities estimated for each type of demand are given below:

Product	Good	Satisfactory	Poor
A	0.60	0.20	0.20
B	0.75	0.15	0.10
C	0.60	0.25	0.15
D	0.50	0.20	0.30

The estimated profit or loss under different states of demand in respect of each product may be taken as

Product	Profit/Loss Under Different States of Demand (₹)		
A	40,000	10,000	1,100
B	40,000	20,000	(-) 7,000
C	50,000	15,000	(-) 8,000
D	40,000	18,000	15,000

Prepare the expected value table and advise the company about the choice of the product to manufacture.

5

(b) Solution:

Demand	Product A		Product B		Product C		Product D	
	Probability	Profit	Probability	Profit	Probability	Profit	Probability	Profit
Good	0.6	40,000	0.75	40,000	0.6	50,000	0.5	40,000
Satisfactory	0.2	10,000	0.15	20,000	0.25	15,000	0.2	18,000
Poor	0.2	1,100	0.1	-7,000	0.15	-8,000	0.3	15,000

$$\begin{aligned}\text{Expected value for } A &= 40,000 \times 0.6 + 10,000 \times 0.2 + 1,100 \times 0.2 \\ &= 24,000 + 2,000 + 220 = ₹ 26,220\end{aligned}$$

$$\begin{aligned}\text{Expected value for } B &= 40,000 \times 0.75 + 20,000 \times 0.15 - 7,000 \times 0.10 \\ &= 30,000 + 3,000 - 700 = ₹ 32,300\end{aligned}$$

$$\begin{aligned}\text{Expected value for } C &= 50,000 \times 0.60 + 15,000 \times 0.25 - 8,000 \times 0.15 \\ &= 30,000 + 3,750 - 1,200 = ₹ 32,550\end{aligned}$$

$$\begin{aligned}\text{Expected value for } D &= 40,000 \times 0.5 + 18,000 \times 0.2 + 15,000 \times 0.3 \\ &= 20,000 + 3,600 + 4,500 = ₹ 28,100\end{aligned}$$

Since the profit is maximum for product C, the company should manufacture C.

(c) The first four moments of a distribution about the value 4 are -1.5, 17, -30 and 108 respectively. Calculate:

(i) Moments about Mean

(ii) Skewness

(iii) Show whether the distribution is Leptokurtic or Platykurtic

5

Solution. First moment about the arbitrary point 4 is:

$$\begin{aligned}\frac{\Sigma(x-4)}{n} &= -1.5 \text{ or } \frac{\Sigma x}{n} - 4 = -1.5 \text{ or } \frac{\Sigma x}{n} = 2.5 (= \bar{x}) \\ \frac{\Sigma(x-\bar{x})^2}{2} &= \frac{\Sigma(x-4+4-\bar{x})^2}{n} = \frac{\Sigma[(x-4)+(4-\bar{x})]^2}{n} \\ &= \frac{\Sigma[(x-4)^2 + (4-\bar{x})^2 + 2(x-4)(4-\bar{x})]}{n} \\ &= \frac{\Sigma(x-4)^2}{n} + \frac{\Sigma(4-\bar{x})^2}{n} + 2(4-\bar{x}) \frac{\Sigma(x-4)}{n} \\ &= 17 + 2.25 + 2(4-2.5)(-1.5) = 14.75 (= \sigma^2) \\ &= \text{Second moment about the mean}\end{aligned}$$

$$\therefore \mu_2 (= \text{SD}) = \sqrt{14.75} = 3.84$$

$$\begin{aligned}\frac{\Sigma(x-\bar{x})^3}{2} &= \frac{\Sigma(x-4+4-\bar{x})^3}{n} = \frac{\Sigma[(x-4)+(4-\bar{x})]^3}{n} = \frac{\Sigma[(x-4)+1.5]^3}{n} \\ &= \frac{\Sigma[(x-4)^3 + (1.5)^3 + 3 \times 1.5(x-4)^2 + 3 \times 1.5 \times 1.5 \times (x-4)]}{n} \\ &= \frac{\Sigma(x-4)^3}{n} + 3.375 + 4.5 \frac{\Sigma(x-4)^2}{n} + 6.75 \frac{\Sigma(x-4)}{n} \\ &= -30 + 3.375 + 4.5(17) + 6.75(-1.5) \\ &= ₹ 39.75 (= \mu_3) \text{ Third, moment about the mean}\end{aligned}$$

$$\text{Skewness } \frac{\mu_3}{\mu_2 \cdot (\mu_2)^{1/2}} = \frac{39.75}{14.75 \times 3.84} = 0.702$$

$$\begin{aligned}
\frac{\Sigma(x - \bar{x})^4}{n} &= \frac{\Sigma(x - 4 + 4 - \bar{x})}{n} = \frac{\Sigma[(x - 4) + (4 - \bar{x})]^4}{n} = \frac{\Sigma[(x - 4) + 1.5]^4}{n} \\
&= \Sigma \left[{}^4C_0 \frac{(x - 4)^0}{n} (1.5)^4 + {}^4C_1 \frac{(x - 4)}{n} (1.5)^3 + {}^4C_2 \frac{(x - 4)^2}{n} (1.5)^2 \right. \\
&\quad \left. + {}^4C_3 \frac{(x - 4)^3}{n} (1.5) + {}^4C_4 \frac{(x - 4)^4}{n} \right] \\
&= \frac{\Sigma(1.5)^4}{n} + 4(1.5)^3 \Sigma \frac{x - 4}{n} + 6 \times (1.5)^2 \Sigma \frac{(x - 4)^2}{n} \\
&\quad + 4(1.5) \frac{\Sigma(x - 4)^3}{n} + \frac{\Sigma(x - 4)^4}{n} \\
&= (1.5)^4 + 13.5(-1.5) + 13.5 \times 17 + 6(-30) + 108 \\
&= 5.0625 - 20.25 + 229.5 - 180 + 108 \\
&= 142.3125 (= \mu_4) \text{ Fourth moment about the mean} \\
\text{Kurtosis} &= \frac{\mu_4}{(\mu_2)^2} = \frac{142.3125}{(14.75)^2} = 0.654 < 3, \text{ Distribution is platykurtic.}
\end{aligned}$$

BUSINESS STATISTICS
B.Com. (Hons.) First Year
Paper IV—2010

Time : 3 hours

Maximum Marks : 75

Note: The maximum marks printed on the Question Paper are applicable for the candidates registered with the School of Open Learning for B.Com.(Hons.). These marks will, however, be scaled down proportionately in respect of the students of regular colleges, at the time of posting of awards for compilation of result.

Attempt all the questions.

Use of simple calculator is allowed.

Logarithmic tables and graphs will be supplied on demand.

Q. 1. (a) Comment on any three of the following:

6

(i) In a factory outlet, a unit of work is completed by A in 10 minutes, by B in 15 minutes, by C in 16 minutes and by D in 20 minutes. What is the average number of units of work completed per minute?

(i) What is relative frequency approach of probability?

(iii) Standard deviation may be negative or positive.

(iv) State the tests that are used to determine the presence of skewness in a distribution.

Solution: (i) Work completed by A in one minute = $\frac{1}{10}$

Work completed by B in one minute = $\frac{1}{15}$

Work completed by C in one minute = $\frac{1}{16}$

Work completed by D in one minute = $\frac{1}{20}$

$$\begin{aligned}\text{Total work done by A, B, C and D in one minute} &= \frac{1}{10} + \frac{1}{15} + \frac{1}{16} + \frac{1}{20} \\ &= \frac{24 + 16 + 15 + 12}{240} = \frac{67}{240}\end{aligned}$$

$$\text{Average amount of work done in one minute} = \frac{67}{240 \times 4} = 0.0698$$

(ii) As per this approach, the probability is defined as the proportion of times an event occurs in the long run when the conditions are stable.

(iii) Standard deviation is always positive. It is never negative. It is the square root of the Arithmetic Mean of the squared deviations which are always positive.

(iv) 1. In a symmetrical distribution: Mean = Median = Mode but in a skewed distribution: Mean, Median, Mode do not coincide.

2. In a skewed distribution, the two quartiles are not equidistant from the Median, i.e. $Q_3 - M \neq M - Q_1$.

3. Skewed distribution when plotted on a graph paper will not give a symmetrical bell-shaped curve.
4. The sum of the positive deviations from the Median is not equal to the sum of the negative deviations.
5. Frequencies are not equally distributed at various points which are equidistant from Mode on either side.

(b) The first four central moments of a distribution are 0, 16, -36 and 120, respectively. Comment upon the Skewness and Kurtosis of the distribution. 5

Solution: Skewness = $\frac{\mu_3}{\mu_2^{3/2}} = \frac{-36}{(16)^{3/2}} = \frac{-36}{64} = \frac{-9}{16}$ skewness is negative

Kurtosis = $\frac{\mu_4}{(\mu_2)^2} = \frac{120}{(16)^2} = \frac{120}{256} < 3$, Distribution is Platykurtic

Or

Write a short note on BSE Sensex. 4

Answer: Bombay Stock Exchange (BSE): Sensitive Index Number of equity prices is the most widely used and accepted equity price index in the country. It comprises of 30 scrips representing a sample of large, well established and financially sound companies. Popularly known as Sensex, the Index has been serving in a large measure. The purpose of qualifying the price movements as also the sensitivity of the market in an effective manner. It is a weighted average of prices of the 30 selected stocks where the weight is the market capitalisation of individual stocks. These stocks constitute a large chunk of both volume and value of shares traded on the Stock Exchange.

(c) Are the following statements True or False? Give reasons:

- (i) If the coefficient of correlation between two variables X and Y is 0.8, then coefficient of correlation between $-X$ and $-Y$ is -0.8.
- (ii) If the coefficient of correlation between X and Y is perfect, the two lines of regression of X on Y and Y on X are reversible. 5

Solution: (i) The Statement is False because the coefficient of correlation between $-X$ and $-Y$ will also be +0.8.

If Mean of X is $\bar{x}$, then Mean of $-X$ will be $-\bar{x}$, Also coefficient of correlation between $-X$ and $-Y$ will be

$$\begin{aligned}
 &= \frac{[-X - (-\bar{X})][Y - (-\bar{Y})]}{\sigma_x \sigma_y} \quad [\text{The SD of } X \text{ values and } -X \text{ values will remain the same}] \\
 &= \frac{(X - \bar{X})(Y - \bar{Y})}{\sigma_x \sigma_y} = \text{Coefficient of correlation between } X \text{ and } Y
 \end{aligned}$$

(ii) True. The two regression lines are

$$\begin{aligned}
 Y - \bar{Y} &= b_{yx}(X - \bar{X}) = r \frac{\sigma_y}{\sigma_x}(X - \bar{X}) \\
 Y - \bar{Y} &= b_{xy}(Y - \bar{Y}) = r \frac{\sigma_x}{\sigma_y}(Y - \bar{Y})
 \end{aligned}$$

If $r = \pm 1$, the two regression lines become the same, i.e., reversible.

Q. 2. (a) The coefficient of correlation between ages of husbands and wives in a community was found to be + 0.8, the average of the husbands' age was 25 years and that of wives' age was 22 years. Their standard deviations were 4 and 5 respectively. Find with the help of regression equations:

(i) The expected age of the husband when wife's age is 18 years.

(ii) The expected age of the wife when husband's age is 32 years.

5

Solution: (a) Let X denote the age of the husband and Y denote the age of the wife. Then the regression of X on Y is:

$$(X - \bar{X}) = r \frac{\sigma_x}{\sigma_y} (Y - \bar{Y}) \quad \text{or} \quad X - 25 = 0.8 \left(\frac{4}{5} \right) (Y - 22)$$

When $Y = 18$, we have

$$X - 25 = 0.8 \left(\frac{4}{5} \right) (18 - 22) = -2.56$$

$$X = 25 - 2.56 = 22.44 \text{ years}$$

The regression equation of Y on X

$$Y - \bar{Y} = b_{yx} (X - \bar{X}) = r \frac{\sigma_y}{\sigma_x} (X - \bar{X})$$

When $X = 32$, we have

$$Y - 22 = 0.8 \left(\frac{5}{4} \right) (32 - 25)$$

$$Y = 22 + 1(7) = 29 \text{ years}$$

(b) The trend equation for quarterly sales of a firm is estimated to be as: $Y = 20 + 2X$, where Y is sales per quarter in millions of rupees, unit of X is one quarter and the origin is middle of the first quarter (Jan-March) of 2005. The seasonal indices of sales for the four quarters are given below:

Quarter :	I	II	III	IV
Seasonal Indices :	120	105	85	90

Estimate the actual sales for each quarter of 2010.

5

Solution: For first quarter of 2005, $X = 0$
 For second quarter of 2005, $X = 1$
 For third quarter of 2005, $X = 2$ and so on.
 For first quarter of 2010, $X = 20$
 For second quarter of 2010, $X = 21$
 For third quarter of 2010, $X = 22$
 For fourth quarter of 2010, $X = 23$

Quarter of 2010	X	Trend Values (T) $Y_t = 20 + 2X$	Seasonal (S) Effect (Seasonal Index)	Estimated Sales $T \times S$
1	20	$20 + 40 = 60$	1.20	$60 \times 1.20 = 71$
2	21	$20 + 42 = 62$	1.05	$62 \times 1.05 = 65.1$
3	22	$20 + 44 = 64$	0.85	$64 \times 0.85 = 54.4$
4	23	$20 + 46 = 66$	0.90	$66 \times 0.90 = 59.4$

(c) A textile worker in a city earns ₹ 3,655 per month. The cost of living index for a particular month is 145. Find the amount of money he spends on house rent and clothing with the help of information given below: 5

Group	Expenditure	Group Index
Food	1400	180
Clothing	?	150
House Rent	?	100
Fuel & Lighting	555	110
Miscellaneous	640	91

Solution: Let the expenditure on clothing and house rent be ₹ X and ₹ Y respectively,

$$1400 + X + Y + 555 + 640 = 3655$$

$$X + Y = 3655 - 1400 - 555 - 640 = 1060 \quad \dots(i)$$

$$145 = \frac{1400 \times 180 + 150X + 100Y + 110 \times 555 + 640 \times 91}{3655}$$

$$145 \times 3655 = 2,52,000 + 150X + 100Y + 61,050 + 58,240$$

$$529975 = 3,71,290 + 150X + 100Y$$

$$150X + 100Y = 5,29,975 - 3,71,290 = 1,58,685$$

$$30X + 20Y = 31737 \quad (\text{Dividing on both sides by common factor 5}) \quad \dots(ii)$$

Multiplying (i) by 20 and subtracting from (ii), we get

$$10X = 10537, \text{ i.e. } X = 1053.7 \text{ and then } Y = 1060 - 1053.7 = 6.3$$

So worker spends ₹ 1053.70 on clothing and ₹ 6.30 on house rent.

Or

(a) Explain the properties of Regression Coefficients. 5

Answer: Properties of regression coefficients:

- Both the regression coefficients must be of the same sign either both positive or both negative.
- The product of the two regression coefficients is equal to r^2 which lies between 0 and 1. Therefore both the regression coefficients cannot be more than 1. One of the regression coefficients should be more than 1 and the other less than 1 such that their product is positive and less than 1.

(b) Given below is the information relating to a bivariate distribution: Regression equation Y on X : $Y_c = 20 + 0.4X$, Mean of $X = 30$, and correlation coefficient between X and $Y = 0.8$.

Find regression equation of X on Y . 5

Solution: Given correlation coefficient between X and Y , $r = 0.8$. Also

$$b_{yx} = 0.4 = r \frac{\sigma_y}{\sigma_x} \text{ or } 0.4 = 0.8 \frac{\sigma_y}{\sigma_x}$$

$$\frac{\sigma_y}{\sigma_x} = \frac{1}{2} \text{ or } \frac{\sigma_x}{\sigma_y} = 2$$

Regression equation of X on Y is:

$$\begin{aligned} X - \bar{X} &= b_{xy} (y - \bar{y}) = r \frac{\sigma_x}{\sigma_y} (Y - \bar{Y}) \\ &= 0.8 \times 2(Y - \bar{Y}) = 1.6(Y - \bar{Y}) \end{aligned} \quad \dots(i)$$

Putting $X = 30$ in $Y_c = 20 + 0.4X$, we get

$$Y_c = 20 + 0.4 \times 30 = 32 (= \bar{Y})$$

From (i), we get

$$X - 30 = 1.6 (Y - 32), \text{ i.e. } X = -21.2 + 1.6Y$$

(c) A box of 100 gaskets contains 10 gaskets with type A defect, 5 gaskets with type B defect and 2 gaskets with both types of defect. Find the probabilities that:

(i) a gasket to be drawn has a type B defect under the condition that it has a type A defect.

(ii) a gasket to be drawn has no type of defect under the condition that it has no type A defect. 5

Solution: Given $P(A) = \frac{10}{100} = 0.1$ $P(B) = \frac{5}{100} = 0.05$ and $P(A \cap B) = \frac{2}{100} = 0.02$. Thus

$$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{0.02}{0.1} = \frac{2}{10} = 0.2$$

$$\begin{aligned} P(\bar{B}|\bar{A}) &= \frac{P(\bar{A} \cap \bar{B})}{P(\bar{A})} = \frac{1 - P(A \cap B)}{1 - P(A)} \\ &= \frac{1 - [P(A) + P(B) - P(A \cap B)]}{1 - P(A)} \\ &= \frac{1 - [0.1 + 0.05 - 0.02]}{1 - 0.1} = \frac{1 - 0.13}{0.9} = \frac{0.87}{0.9} = 0.966 \end{aligned}$$

Q. 3. (a) Write the mathematical properties of standard deviation. 5

Solution:

- (i) We can find the combined S.D. of two or more groups.
- (ii) We can find coefficient of variation which is used to calculate the uniformity of two distributions which are in different units.
- (iii) The sum of squares of the deviations of the values from their mean is minimum.
- (iv) The standard deviation enables us to determine with a great deal of accuracy where the values of a frequency distribution are located. For example, in a symmetric distribution

$\bar{x} \pm 1 \sigma$ covers 61.27 per cent of the items.

$\bar{x} \pm 2 \sigma$ covers 95.45 per cent of the items.

$\bar{x} \pm 3 \sigma$ covers 99.73 per cent of the items.

(b) Find the coefficient of correlation between age and playing habits of the following students: 5

No. of students	:	2500	2000	1500	1200	1000	800
Regular players	:	2250	1200	1050	480	250	120
Age of players	:	16	17	18	19	20	21

Solution:

Age (X)	Proportion of Players (Y)	$x = \frac{(X - 18.5)}{0.5}$	$y = Y - 0.5$	xy	x^2	y^2
16	0.9	-5	0.4	-2	25	0.16
17	0.6	-3	0.1	-0.3	9	0.01
18	0.7	-1	0.2	-0.2	1	0.04
19	0.4	1	-0.1	-0.1	1	0.01
20	0.25	3	-0.25	-0.75	9	0.0625
21	0.15	5	-0.35	-1.75	25	0.1225
111	3.00			-5.1	70	0.405

$$\bar{X} = \frac{\Sigma X}{N} = \frac{111}{6} = 18.5 \text{ and } \bar{Y} = \frac{\Sigma Y}{N} = \frac{3.00}{6} = 0.5$$

$$\text{Coefficient of Correlation, } r = \frac{\Sigma xy}{\sqrt{x^2} \sqrt{y^2}} = \frac{-5.1}{\sqrt{70} \sqrt{0.405}} = \frac{-5.1}{5.324} = 0.9579$$

(c) Under an employment promotion programme, it is proposed to allow sale of newspapers on the buses during off peak hours. The vendor can purchase the newspaper at a special rate of ₹ 1.25 per copy and sell it for a price of ₹ 1.60. Any unsold copies are treated a dead loss. A vendor has estimated the following probability distribution for the number of copies demanded during 150 days:

Number of copies	:	15	16	17	18	19	20	
Number of days	:	102	5	35	40	25	15	= 150

How many copies the vendor should buy for maximum gain? Also calculate EVPI.

5

Solution: Profit on each copy sold = ₹(1.60 – 1.25) = ₹ 0.35

Loss on each unsold copy = ₹ 1.25

Demand	Probability of Demand	Stock					
		15	16	17	18	19	20
15	$\frac{10}{150} = 0.067$	5.25	4.0	2.75	1.5	0.25	-1.00
16	$\frac{25}{150} = 0.167$	5.25	5.6	4.35	3.1	1.6	0.35
17	$\frac{35}{150} = 0.233$	5.25	5.6	5.95	4.7	3.45	2.20
18	$\frac{40}{150} = 0.266$	5.25	5.6	5.95	6.3	5.05	3.80
19	$\frac{25}{150} = 0.167$	5.25	5.6	5.95	6.3	6.65	5.40
20	$\frac{15}{150} = 0.100$	5.25	5.6	5.95	6.3	6.65	7.00

The Expected profit likely to occur at various levels of stock is as follows:

Stock	Expected Profit
15	5.25
16	$4 \times 0.067 + 5.6 \times 0.933 = 5.5$
17	$2.75 \times 0.067 + 4.35 \times 0.167 + 5.95 \times 0.766 = 5.468$
18	Further the profit will reduce

Hence the vender should stock only 16 copies

$$\begin{aligned} \text{EVPI} &= 0.067 \times 5.25 + 0.167 \times 5.6 + 0.233 \times 5.95 + 0.266 \times 6.3 + 0.167 \times 6.65 + 0.1 \times 7 \\ &= 0.352 + 0.935 + 1.39 + 1.68 + 1.111 + 0.7 = 6.168 \end{aligned}$$

$$\text{Value of perfect information} = 6.168 - 5.468 = ₹ 0.7$$

Or

(a) Write a short note on Bayes' Theorem.

5

Answer: The basic formula for conditional probability under dependence

$$P(A/B) = \frac{P(A \cap B)}{P(B)} \text{ is called Bayes' theorem.}$$

Bayes' theorem is an important statistical method which is used in evaluating new information as well as in revising prior estimates of the probability in the light of that information. Bayes' theorem may be viewed as a means of transforming our Prior Probability of an event into Posterior Probability of that event. If properly used, Bayes' theorem makes it unnecessary to collect huge data over a long period in order to make good decisions on the basis of probabilities.

(b) A computer while calculating the coefficient of correlation between the variables X and Y derived the following results:

$$N = 30, \Sigma X = 120, \Sigma X^2 = 600, \Sigma Y = 90, \Sigma Y^2 = 250, \Sigma XY = 335.$$

It was later discovered that the computer copied down two pairs of observation as:

$$X : \quad 8 \quad 12$$

$$Y : \quad 10 \quad 7$$

whereas the correct values were:

$$X : \quad 8 \quad 10$$

$$Y : \quad 12 \quad 8$$

Compute the correct coefficient of correlation between X and Y.

5

Solution: Correct $\Sigma X = 120 - 8 - 12 + 8 + 10 = 118$

$$\text{Correct } \Sigma Y = 90 - 10 - 7 + 12 + 8 = 93$$

$$\text{Correct } \Sigma XY = 335 - 80 - 84 + 96 + 80 = 347$$

$$\text{Correct } \Sigma X^2 = 600 - 64 - 144 + 64 + 100 = 556$$

$$\text{Correct } \Sigma Y^2 = 250 - 100 - 49 + 144 + 64 = 309$$

Correct coefficient of correlation

$$\begin{aligned} r &= \frac{N \Sigma XY - (\Sigma X)(\Sigma Y)}{\sqrt{N \Sigma X^2 - (\Sigma X)^2} \sqrt{N \Sigma Y^2 - (\Sigma Y)^2}} \\ &= \frac{30 \times 347 - 118 \times 93}{\sqrt{30 \times 556 - (118)^2} \sqrt{30 \times 309 - (93)^2}} \end{aligned}$$

$$= \frac{10,410 - 10,974}{\sqrt{16,680 - 13,924} \sqrt{9,270 - 8,649}}$$

$$= -\frac{564}{52.5 \times 24.92} = -0.431$$

(c) The sales of a company for the year 2003 to 2009 are given below:

<i>Years</i>	:	2003	2004	2005	2006	2007	2008	2009
<i>Sales (in millions)</i>	:	32	47	65	92	190	132	275

Estimate sales for the year 2012 with the help of exponential curve ($Y = LB^X$).

5

Solution:

<i>Years (X)</i>	<i>x = X - 2006</i>	<i>Sales (Y)</i>	<i>log Y</i>	<i>x log Y</i>	<i>x²</i>
2003	-3	32	1.505	-4.515	9
2004	-2	47	1.672	-3.344	4
2005	-1	65	1.813	-1.813	1
2006	0	92	1.964	0	0
2007	1	190	2.28	2.28	1
2008	2	132	2.12	4.24	4
2009	3	275	2.44	7.32	9
	0		13.794	4.168	28

The required equations are

$$\Sigma \log Y = n \log A + \log B \Sigma x \quad \text{or } 13.794 = 7 \log A + 0 \quad \dots(i)$$

$$\Sigma x \log Y = \log A \Sigma x + \log B \Sigma x^2 \quad \text{or } 4.168 = 0 + 28 \log B \quad \dots(ii)$$

From (i), $\log A = \frac{13.794}{7} = 1.97$ or $A = 93.325$

From (ii), $\log B = \frac{4.168}{28} = 0.149$ or $B = 1.409$

Then required equation is $Y = 93.325 (1.409)^{(X-2006)}$

Sales for the year 2012, are $Y = 93.325(1.409)^6 = 730.237$

Q. 4. (a) If 5per cent students appearing in an examination fail, using Poisson distribution find the probability that out of 100 students appearing in the examination:

7

(i) none failed

(ii) 5 students failed

(iii) maximum 3 failed

Solution: $m (= \lambda) = \frac{5}{100} \times 100 = 5$

(i) $P(X=0) = e^{-m} = 0.00698$

(ii) $P(X=5) = \frac{e^{-m} m^r}{r!} = \frac{e^{-m} (5)^r}{5!} = \frac{e^{-m} (5)^5}{120} = 0.1817$

(iii) $P(X=0) + P(X=1) + P(X=2) + P(X=3) = e^{-m} + m e^{-m} + \frac{m^2 e^{-m}}{2!} + \frac{m^3 e^{-m}}{3!}$

$$\begin{aligned}
 &= e^{-m} \left[1 + m + \frac{m^2}{2!} + \frac{m^3}{3!} \right] = e^{-m} \left[1 + 5 + \frac{25}{2} + \frac{125}{6} \right] \\
 &= 0.00698[6 + 12.5 + 20.833] = 0.00698 (39.333) = 0.2745
 \end{aligned}$$

(b) The production of a commodity during 2005-2010 is given below:

8

Fit the second degree parabola curve and estimate the production for the year 2014:

<i>Year</i>	: 2005	2006	2007	2008	2009	2010
<i>Prod. (000 tons.)</i>	: 10	12	15	16	18	21

Solution:

<i>Year (X)</i>	$x = \frac{X - 2007 \cdot 5}{0.5}$	<i>Prod. (Y)</i>	x^2	x^3	x^4	xY	x^2Y
2005	-5	10	25	-125	625	-50	250
2006	-3	12	9	-27	81	-36	108
2007	-1	15	1	-1	1	-15	15
2008	1	16	1	1	1	16	16
2009	3	18	9	27	81	54	162
2010	5	21	25	125	625	105	525
	<u>0</u>	<u>92</u>	<u>70</u>	<u>0</u>	<u>1414</u>	<u>74</u>	<u>1076</u>

The three normal equations are as follows:

$$\Sigma Y = na + b\Sigma x + c\Sigma x^2 \quad \text{or } 92 = 6a + 70c \quad \dots(i)$$

$$\Sigma xY = a\Sigma x + b\Sigma x^2 + c\Sigma x^3 \quad \text{or } 74 = 70b \quad \dots(ii)$$

$$\Sigma x^2Y = a\Sigma x^2 + b\Sigma x^3 + c\Sigma x^4 \quad \text{or } 1076 = 70a + 1414c \quad \dots(iii)$$

From (ii), $b = \frac{74}{70} = 1.057$

Multiplying (i) by 70 and (iii) by 6, we get

$$6440 = 420a + 4900c \quad \dots(iv)$$

$$6456 = 420a + 8484c \quad \dots(v)$$

Subtracting (iv) from (v), we get

$$16 = 3584c, \text{ i.e. } c = 0.00446$$

From (iii) $1076 = 70a + 6.3$ or $70a = 1076 - 6.3 = 1069.7$, i.e. $a = 15.28$

The required equation is: $Y = 15.28 + 1.057x + 0.00446x^2$

When $X = 2014$

$$\begin{aligned}
 &= 15.28 + 1.057 \left[\frac{X - 2007.5}{0.5} \right] + 0.00446 \left[\frac{X - 2007.5}{0.5} \right]^2
 \end{aligned}$$

$$\begin{aligned}
 Y &= 15.28 + 1.057 \left[\frac{2014 - 2007.5}{0.5} \right] + 0.00446 \left[\frac{2014 - 2007.5}{0.5} \right]^2 \\
 &= 15.28 + 13.74 + 0.75 = 29.77
 \end{aligned}$$

Or

(a) A project yields an average cash-flow of ₹550 lakh and standard deviation cash flow of ₹110. Calculate the following probabilities assuming the normal distribution:

- (i) Cash flow will be more than ₹ 675 lakh.
- (ii) Cash flow will be less than ₹ 450 lakh.
- (iii) Cash flow will be between ₹ 425 lakh and ₹ 750 lakh.

Solution: (i) $Z = \frac{X - \mu}{\sigma} = \frac{675 - 550}{110} = \frac{125}{110} = 1.14$

Area covered between $Z = 0$ and $Z = 1.14$ is 0.3729.
Therefore the required probability
 $= 0.5 - 0.3729 = 0.1271$

(ii) $Z = \frac{X - \mu}{\sigma} = \frac{450 - 550}{110} = -\frac{100}{110} = -0.91$

Area covered between $Z = 0$ and $Z = -0.91$ is 0.3986.
Thus the required probability
 $= 0.5 - 0.3986 = 0.1014$

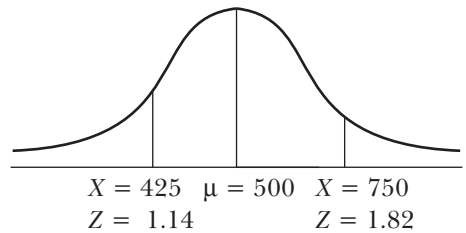
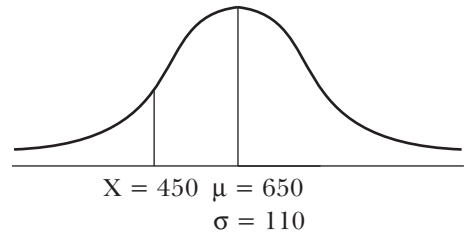
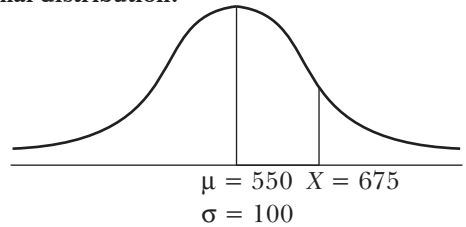
(iii) $Z_1 = \frac{425 - 550}{110} = -1.14,$

$Z_2 = \frac{750 - 550}{110} = 1.82$

Area covered between $Z = 0$ and $Z_1 = 1.14$ is 0.3729

Area covered between $Z = 0$ and $Z_2 = 1.82$ is 0.4656

Thus, the required probability $= 0.3729 + 0.4656 = 0.8385$



(b) From the following quarterly prices of Coke for six years, calculate seasonal variations by the method of ratio-to-moving average: 8

Quarters	2003	2004	2005	2006	2007	2008
Jan. –March	30	35	31	31	34	36
April–June	26	28	29	31	36	32
July –Sept.	22	22	28	25	26	27
Oct. –Dec.	31	36	32	35	33	38

Solution: Calculations of Ratio to Moving Averages (Multiplication Model) are shown below:

<i>Year</i>	<i>Quarter</i>	<i>Price</i>	<i>Years' Moving Total</i>	<i>Four Years' Moving Average</i>	<i>Given Figures as %age of Moving Average = $\frac{(3)}{(5)} \times 100$ (Trend Elimination Value)</i>
(1)	(2)	(3)	(4)	(5)	
2003	Qtr. I	30			
	II	26	109		
	III	22	114	$\frac{109 + 114}{8} = 27.875$	$\frac{22}{27.875} \times 100 = 78.9$
	IV	31	116	28.75	107.82
2004	Qtr. I	35	116	29	120.69
	II	28	121	29.625	94.5
	III	22	117	29.75	73.95
	IV	36	118	29.375	122.55
2005	Qtr. I	31	124	30.25	102.48
	II	29	120	30.5	95.08
	III	28	120	30	93.33
	IV	32	122	30.25	105.78
2006	Qtr. I	31	119	30.125	102.9
	II	31	122	30.125	102.9
	III	25	125	30.875	80.97
	IV	35	130	31.875	109.8
2007	Qtr. I	34	131	32.625	104.2
	II	36	129	32.5	110.77

(Continued)

2008	Qtr. I	III	26	131	32.5	80
		IV	33		32.25	102.3
	II		36	127	31.875	112.94
	III	II	32	128	32.625	98
	IV	III	27	133		
		IV	38			

Calculations of Seasonal Index are shown below:

<i>Year</i>	<i>1st quarter</i>	<i>2nd quarter</i>	<i>3rd quarter</i>	<i>4th quarter</i>
2003	—	—	78.90	107.82
2004	120.69	94.50	73.95	122.55
2005	102.48	95.08	93.33	105.78
2006	102.90	102.90	80.97	109.80
2007	104.20	110.77	80.00	102.30
2008	112.94	98.00	—	—
Total	543.21	501.25	407.15	548.25
Average	108.64	100.25	81.43	109.65 = 399.97
Adjusted Seasonal Index	$\frac{108.64}{399.97} \times 400$ = 108.65	$\frac{100.25}{399.97} \times 400$ = 100.26	$\frac{81.43}{399.97} \times 400$ = 81.44	$\frac{109.65}{399.97} \times 400$ = 109.66

Q. 5. (a) If a machine is correctly set-up, it produces 90per cent acceptable items. If it is incorrectly set-up, it produces 40per cent acceptable items. Experience shows that 80per cent of the set-ups are correctly done.

(i) If after a certain set-up, out of the first two items produced, first is found to be acceptable and second unacceptable, what is the probability that the machine is correctly set-up?

(ii) If the machine produced first two items as acceptable, what is the probability that the machine is correctly set-up?

7

Solution: (i) Let B_1 be the event that the machine is correctly set up and B_2 the event that the machine is incorrectly set up. A be the event that the first item is acceptable and second unacceptable. Given that

$$\begin{aligned}
 P(B_1) &= 0.8, & P(B_2) &= 0.2 \\
 P(A|B_1) &= 0.9 \times 0.1 = 0.9 & P(A|B_2) &= 0.4 \times 0.6 = 0.24 \\
 P(B_1|A) &= \frac{P(B_1 \cap A)}{P(A)} = \frac{P(B_1) \times P(A|B_1)}{P(B_1) \times P(A|B_1) + P(B_2) \times P(A|B_2)} \\
 &= \frac{0.8 \times 0.9 \times 0.1}{0.8 \times 0.9 \times 0.1 + 0.2 \times 0.4 \times 0.6} \\
 &= \frac{0.072}{0.072 + 0.048} = \frac{0.072}{0.120} = 0.6
 \end{aligned}$$

(ii) Let A be the event that the first two items are acceptable

Then $P(A/B_1) = 0.8 \times 0.9 \times 0.9$, $P(A/B_2) = 0.2 \times 0.4 \times 0.4$

$$P(B_1|A) = \frac{0.8 \times 0.9 \times 0.9}{0.8 \times 0.9 \times 0.9 + 0.2 \times 0.4 \times 0.4} = \frac{0.648}{0.648 + 0.032} = \frac{0.648}{0.680} = 0.953$$

(b) In a partially destroyed laboratory record, the following results are only legible:

Variance of $X = 9$, Regression equations: $8X - 10Y = -66$, and $10X - 4.5Y = 53.5$. Find

(i) the coefficient of correlation between X and Y .

(ii) the standard deviation of Y .

(iii) the standard error of estimate between X and Y and Y and X .

8

Solution: The regression equations may be rewritten as:

$$Y = \frac{8}{10}X + 6.6, \text{ and } X = \frac{4.5}{10}Y + 5.3$$

Thus $b_{yx} = 8/10 = 0.8$ and $b_{xy} = 4.5/10 = 0.45$

(i) Correlation coefficient between X and Y , $r = \sqrt{b_{yx} \times b_{xy}} = \sqrt{0.8 \times 0.45} = 0.60$

(ii) $b_{yx} = r \frac{\sigma_y}{\sigma_x}$ or $0.8 = 0.6$, i.e. $\sigma_y = 0.8 \times 3 / 0.6 = 4$

(iii) $SE_{YX} = \sigma_y \sqrt{1 - r^2} = 4 \sqrt{1 - 0.36} = 4 \times 0.8 = 3.2$

$$SE_{XY} = \sigma_x \sqrt{1 - r^2} = 3 \sqrt{1 - 0.36} = 3 \times 0.8 = 2.4$$

Or

(a) A toy company is planning to introduce few changes in the toys under three categories; complete change, partial change and minimum change. Further, the company has three levels of product acceptance. Management will make its decision on the basis of expected profits from the first year of production. The relevant information are given below:

Anticipated 1st Year Profits (₹ '000)

Product acceptance	Complete change	Partial change	Minimum change
Good	80	70	50
Fair	50	45	40
Poor	(25)	(10)	0

You are asked to take an optimal decision on the basis of the following decision criteria:

- (i) Minimax (ii) Maximax (iii) Maximin
(iv) Laplace (v) Hurwicz

assuming the pessimism index as 0.35.

8

Solution: (i)

	Good	Fair	Poor	Max. Profit	Min. Profit
Complete change	80	50	(25)	80	25
Partial change	70	45	(10)	70	10
Minimum change	50	40	0	50	0

(ii) & (iii) Under maximax and maximin complete change should be planned.

(iv) In Laplace criteria equal probability is assigned to all the outcomes.

Expected profit for complete change = $(80 + 50 + 25)/3 = 51.67$

Expected profit for partial change = $(70 + 45 + 10)/3 = 41.67$

Expected profit for minimum change = $(50 + 40 + 0)/3 = 30$

(v) Hurwicz criteria.

Expected profit for complete change = $25 \times 0.35 + 80 \times 0.65 = 60.75$

Expected profit for partial change = $10 \times 0.35 + 70 \times 0.65 = 49$

Expected profit for minimum change = $0 \times 0.35 + 50 \times 0.65 = 32.5$

Opportunity Loss Table

	<i>Good</i>	<i>Fair</i>	<i>Poor</i>	<i>Max. Loss</i>
Complete change	0	0	0	0
Partial change	10	5	15	15
Minimum change	30	10	25	30

As per this criteria also complete change should be preferred.

(b) The first four moments of a distribution about the value 5 of the variables are 2, 20, 40 and 50. Calculate the mean, variance, β_1 and β_2 and comment upon the nature of distribution. 7

Solution:

$$\mu_2 = \mu'_2 - (\mu'_1)^2 = 20 - (2)^2 = 16$$

$$\mu_3 = \mu'_3 - 3\mu'_1 \mu'_2 + 2(\mu'_1)^3 = 40 - 3 \times 2 \times 20 + 2 \times (2)^3 = 40 - 120 + 16 = -64$$

$$\begin{aligned} \mu_4 &= \mu'_4 - 4\mu'_1 \mu'_3 + 6(\mu'_1)^2 \mu'_2 - 3(\mu'_1)^4 = 50 - 4 \times 2 \times 40 + 6(2)^2 \times 20 - 3(2)^4 \\ &= 50 - 320 + 480 - 48 = 162 \end{aligned}$$

$$\text{Also, Mean } (= \mu_1) = \frac{\Sigma(x - 5)}{n} = 2 \text{ (Given) or } \frac{\Sigma x}{n} - 5 = 2, \text{ i.e. } \mu_1 = \frac{\Sigma x}{n} = 7$$

$$\text{Variance } (= \mu_2) = 16$$

$$\beta_1 = \frac{\mu_3}{(\mu_2)^{3/2}} = \frac{-64}{(16)^{3/2}} = -\frac{64}{(4)^3} = -1$$

$$\beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{162}{(16)^2} = \frac{162}{256} = 0.63$$